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# ARJUNA

*for*

# JEE

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# CHEMISTRY

**Full Course Study Material** XI

## SOLUTIONS

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# PREFACE

A highly skilled professional team of Physics Wallah (PW) works arduously to ensure that the students receive the best content for the **JEE** exam.

From the beginning, the whole content team comprising Associate faculties, DTP operators, Proofreaders and others are involved in shaping the material to their best knowledge and experience to produce powerful content for the students.

Faculties have adopted a new style of presenting the content in easy-to-understand language and have provided the team with their guidance and supervision throughout the creation of this Study Material.

Physics Wallah (PW) strongly believes in conceptual and fun-based learning. PW provides highly exam-oriented content to bring quality and clarity to the students.

A plethora of **JEE Study Material** is available in the market but PW professionals are continuously working to provide the supreme Study Material for our **JEE** students.

This Study Material adopts a multi-faceted approach to master and understanding the concepts by having a rich diversity of questions asked in the examination and equip the students with the knowledge for the competitive exam.

The main objective of the study material is to provide a large number of quality problems with varying cognitive levels to facilitate the teaching-learning of concepts that are presented through the book.

It has become popular among aspirants because of its easy-to-understand language and clear illustrations with diagrams, mnemonics, flow charts and tables.

Students can benefit themselves by attempting the exercise given in this Study Material for the self-assessment and also mastering the basic techniques of problem-solving.

The questions are strictly designed in accordance with the exam relevant topics that helps to develop examination temperament within the students.

Mastering the Physics Wallah (PW) study material curated by the PW team, the students can easily qualify for the exam with a Top Rank in the **JEE**.

In each chapter, for better understanding, questions have been classified according to the latest syllabus of **JEE**

- The nature and diversity of the questions help students to ace the examination
- Quality questions to strengthen the concept of the topic at the zenith level

## BOOK FEATURES

- Detailed theory with Train Your Brain and Concept Application Questions
- Short Notes & Solved Miscellaneous Examples
- Topicwise, Learning Plus & Advance Level MCQs covered in exercises
- Match the Column, Comprehension Based, Numerical Type and Multicorrect questions covered for complete practice.
- JEE Past Year Questions
- Elaborated Solutions

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CHAPTER

# 1 Some Basic Concepts of Chemistry

## Exercise-1 (Topicwise)

**1.** (*a*) All chemical compounds are found to have constant composition irrespective of their method of preparation and sources.

**2.** (*c*) Nitrogen equivalent weight is variable.

**3.** (*a*) $\frac{1}{12}$ of C – 12

**4.** (*b*) In chemical scale, the relative mass of the isotopic mixture of oxygen atoms ($O^{16}$, $O^{17}$, $O^{18}$) is assumed to be equal to 16.00.

**5.** (*b*) 1 Mole of $CH_4$ contains 4 mole of hydrogen atom *i.e.* 4g atom of hydrogen.

**6.** (*a*) $\because$ 5.8L of gas has mass = 7.5 gm

$\therefore$ 22.4L of gas has mass = $\frac{7.5}{5.8}\times 22.4 = 28.96$

So molecular weight = 29

So, molecular formula of compound is NO

**7.** (*d*) $\because$ 17*gm* $NH_3$ contains $6 \times 10^{23}$ molecules of $NH_3$

$\therefore$ 4.25*gm* $NH_3$ contains = $\frac{6\times10^{23}}{17}\times 4.25$

$\therefore$ No. of atoms = $\frac{6\times10^{23}\times4.25}{17}\times 4 = 6\times10^{23}$

**8.** (*a*) $\because$ 1L of gas at S.T.P. weight 1.16*g*

$\therefore$ 22.4 L of gas at S.T.P. weight = $22.4\times1.16$

$= 25.984 \approx 26$

This molecular weight indicates that given compound is $C_2H_2$.

**9.** (*a*) $6 \times 10^{23}$ molecules has mass = 18 gm

1 molecules has mass = $\frac{18}{6\times10^{23}} = 3\times10^{-23}$ gm

$= 3\times 10^{-26}$.

**10.** (*a*) 14 *gm* $N^{3-}$ ions have = $8N_A$ valence electrons

4.2 *gm* of $N^{3-}$ ions have = $\frac{8N_A\times4.2}{14} = 2.4N_A$

**11.** (*b*) $\because$ 22400 *ml* at NTP has $6.023 \times 10^{23}$ molecule

$\therefore$ 1 *ml* at NTP has = $\frac{6.023\times10^{23}}{22400}$

$= 0.0002688 \times 10^{23} = 2.69 \times 10^{19}$.

**12.** (*b*) $\because$ 22400*cc* of gas at STP has molecules

$\therefore$ $1.12 \times 10^{-7}$ of gas at STP has

$\frac{6\times10^{23}\times1.12\times10^{-7}}{22400} = .03\times10^{14} = 3\times10^{12}$.

**13.** (*a*) $\because$ 2.24*L* of gas has mass = 4.4*gm*

$\therefore$ 22.4*L* of gas has mass = $\frac{4.4}{2.24}\times22.4 = 44$

So given gas is $CO_2$ because $CO_2$ has molecular mass = 44.

**14.** (*a*) 44*g* of $CO_2$ has $2 \times 6 \times 10^{23}$ atoms of oxygen

4.4*g* of $CO_2$ has = $\frac{12\times10^{23}}{44}\times4.4$

$= 1.2 \times 10^{23}$ atoms.

**15.** (*c*) $\because$ 100*gm* $CaCO_3 = 6.023 \times 10^{23}$molecules

$\therefore$ 10*gm* = $CaCO_3$ = $\frac{6.023\times10^{23}}{100}\times10$

$= 6.023 \times 10^{22}$ molecule

1 molecule of $CaCO_3$= 50 protons

$6.023 \times 10^{22}$ molecule of $CaCO_3 = 50 \times 6.023 \times 10^{22}$

$= 3.0115 \times 10^{24}$

**16.** (*c*) According to Avogadro's hypothesis equal volumes of all gases under similar conditions of temperature and pressure contains equal no. of molecules.

**17.** (*d*) d = $\frac{M}{V}$ density, M = mass, V = volume)

Since *d* = 1

So, M = V

18*gm* = 18*ml*

18*ml* = $N_A$ molecules ($N_A$ = avogadro's no.)

1000*ml* = $\frac{N_A}{18}\times1000 = 55.555\ N_A$.

**18.** (*b*) 2*gm* of oxygen contains atom = $\frac{2}{16} = \frac{1}{8}$ mole

also 4*g* of sulphur = $\frac{4}{32} = \frac{1}{8}$ mole.

**19.** (*a*) 100 *gm* caffeine has 28.9 *gm* nitrogen

194 *gm* caffeine has = $\frac{28.9}{100}\times194 = 56.06$ gm

$\therefore$ No. of atoms in caffeine = $\frac{56.06}{14} \approx 4$.

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**20.** (a) $\because$ 40gm *NaOH* contains 16gm of oxygen

$\therefore$ 100gm of *NaOH* contains $\frac{16}{40}\times 100 = 40\%$ oxygen.

**21.** (d) C = 24 gm, H = 4 gm, O = 32 gm

So, Molecular formula = $C_2H_4O_2$

So, Empirical formula = $CH_2O$

(Simplest formula).

**22.** (c) In $Fe(CNS)_3.3H_2O$

$$\% \text{ of } H_2O = \frac{3\times 18}{284}\times 100 = 19\%.$$

**23.** (b)

| Element | At.wt | Mole | Ratio | Empirical formula |
|---|---|---|---|---|
| C = 86% | 12 | 7.1 | 1 | $CH_2$ |
| H = 14% | 1 | 14 | 2 | Belongs to alkene |

**24.** (b)

| Element | % | At.wt. | Ratio | |
|---|---|---|---|---|
| X | 50 | 10 | 5 | 2 |
| Y | 50 | 20 | 2.5 | 1 |

Simplest formula = $X_2Y$

**25.** (c) $N = \frac{W(gm)\times 1000}{V\times Eq.wt.}$

1500*ml* of 0.1*N HCl* = 150*ml* (*N*)

$$1 = \frac{W(gm)\times 1000}{150\times 40}, W(gm) = \frac{150\times 40}{1000} = 6gm.$$

**26.** (c)

| | $4NH_{3(g)}$ | + $5O_{2(g)}$ | → $4NO_{(g)}$ | + $6H_2O_{(g)}$ |
|---|---|---|---|---|
| t = 0 | 1 | 1 | 0 | 0 |
| t = t | 1 – 4x | 1 – 5x | 4x | 6x |

Oxygen is limiting reagent

So, $X = \frac{1}{5} = 0.2$ all oxygen consumed

Left $NH_3 = 1 - 4\times 0.2 = 0.2$.

**27.** (c) $\because$ 100*gm Hb* contain = 0.33*gm Fe*

$\therefore$ 67200*gm Hb* – = $\frac{67200\times 0.33}{100}$ gm Fe.

*gm* atom of *Fe* = $\frac{672\times 0.33}{56} = 4.$

**28.** (d) $\underset{27gm}{Al} + H_2O + NaOH \rightarrow NaAlO_2 + \frac{3}{2}H_2$

$\frac{3}{2}\times 22.4 = 33.6L$

**29.** (c) 0.1 M $AgNO_3$ will react with 0.1 M NaCl to form 0.1 M $NaNO_3$. But as the volume is doubled, conc. of $NO_3^- = \frac{0.1}{2} = 0.05M$

**30.** (b) $Mg^{2+} \equiv H_2$

$n = \frac{12gm}{24gm} = \frac{1}{2}$ mole of $H_2$

**31.** (c) $\underset{100g}{CaCO_3} + \underset{2N}{2HCl} \rightarrow CaCl_2 + \underset{44g}{CO_2} + H_2O$

100 g $CaCO_3$ with 2 N HCl gives 44 g $CO_2$

100 g with $CaCO_3$ with 1 N HCl gives 22 g

**32.** (b) $I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$

moles of $I_2 = \frac{5}{127}$

1 mole $I_2$ reacts with 10 moles of $HNO_3$

$\frac{5}{127}$ moles of $I_2$ reacts with $\rightarrow 10\times\frac{5}{127}$

$= 0.393$

Wt. of $HNO_3 = 0.393 \times 63$

$= 24.8g$

**33.** (d) $\underset{1\text{ mole}}{KMnO_4} + \underset{16\text{ mole}}{16HCl} \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$

2 mole of $KMnO_4$ produces 5 moles of $Cl_2$

1 mole of $KMnO_4$ will produce $\rightarrow \frac{5}{2}$ mole of $Cl_2$

2.5 moles × 22.4

= 56 litre of $Cl_2$

**34.** (a) $M + X_2 \rightarrow MX_2$

$3MX_2 + X_2 \rightarrow M_2X_8$

$\frac{3}{4}$moles $\quad$ $\frac{1}{4}$moles

$$\underset{\frac{1}{4}\text{moles}}{M_3X_8} + 4N_2CO_3 \longrightarrow \underset{(2\text{moles})}{\underset{206g}{8NX}} + 4CO_2 + M_3O_4$$

8 moles of NX produced by $\rightarrow$ 1 moles $M_3X_8$

so, 2 moles of NX will be produced by $\rightarrow \frac{1}{4}$ mole $M_3X_8$

$\therefore \frac{1}{4}$ mole $M_3X_8$ is produced by $\rightarrow 3\times\frac{1}{4}$ mole $MX_2$

$\therefore \frac{3}{4}$ moles of $MX_2$ is produced by $\rightarrow \frac{3}{4}$ moles of M

wt. of M = $\frac{3}{4}\times 56 = 42g$

**35.** (d) $\underset{2\text{moles}}{2Al} + \underset{\frac{3}{2}\text{moles}}{\frac{3}{2}O_2} \longrightarrow Al_2O_3$

2 moles Al reacts with $\rightarrow \frac{3}{2}$ moles of $O_2$

$\therefore$ 1 mole Al reacts with $\rightarrow \frac{3}{4}$ moles of $O_2$

wt. of $O_2 = \frac{3}{4}\times 32 = 24g$

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**36.** (*d*) Balanced eq$^{n}$.

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

0.5 moles 0.2 moles

(L.R)

2 moles $Na_3PO_4$ react with → 1 mole of $BaCl_2$

∴ 0.2 mole $Na_3PO_4$ react with → 0.1 mole

**37.** (*b*) $2KClO_3 \rightarrow 2KCl + 3O_2$

3 moles of $O_2$ is produced by = 2 moles of $KClO_3$

$\therefore \frac{1}{3}$ mole of $O_2$ is produced by = $\frac{2}{3}\times\frac{1}{2}=\frac{1}{3}$ moles of $KClO_3$

**38.** (*a*) $H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$

0.5 0.2 ?

(L.R)

1 mole of $Ca(OH)_2$ produce → 1 mole $CaSO_4$

∴ 0.2 mole of $Ca(OH)_2$ will produce → 0.2 mole $CaSO_4$

**39.** (*d*) $H_3PO_4$ is tribasic so N = 3 M = 3 × 1 = 3

**40.** (*c*) Number of moles of NaOH

$$=\frac{\text{given mass of NaOH}}{\text{molar mass}}=\frac{10^{-3}}{40}$$

$$\text{Molarity}=\frac{\text{number of moles of solute}}{\text{volume of solution in mL}}\times 1000$$

$$\text{Molarity}=\frac{\frac{10^{-3}}{40}\times 1000}{250}=10^{-4}\,M$$

**41.** (*c*) $NaOH + HCl \rightarrow NaCl + H_2O$

1 mole of HCl is neutralized by 1 mole (40 g) of NaOH.

100 ml of 0.1N HCl corresponds to 0.01 moles.

Thus, 0.01 mole of HCl s neutralized by 0.01 mole (or 0.4 g) of NaOH.

**42.** (*b*) For hydrated oxalic acid, $(COOH)_2, 2H_2O$, molar mass is 126 g/mol.

$$\text{Equivalent weight}=\frac{M}{2}=63$$

Meq. of oxalic acid = 500 × 0.1 = 50

$$\frac{w}{E\times 1000}=50$$

$$w=\frac{126}{2}\times 50\times 1000=3.15\,g$$

Hence, mass of oxalic acid required is 3.15 g.

**43.** (*c*) $M_{H_2SO_4}$ = 18M = 18 mole of $H_2SO_4$ in 1 litre solution density = 1.8 g/mL (given)

so, Mass of Sol$^{n}$ = 1.8 × 1000 = 1800 g

Mass of Solute = 18 × 98 = 1760

∴ Mass of solvent = 1800 – 1760

= 36 g

$$\text{Molality}=\frac{18}{36(g)}\times 1000=500\,\text{molal}$$

**44.** (*c*) $\%\ w/w=\frac{110\times 10^{-3}}{100}=0.11\%$

**45.** (*a*) $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

0.02 0.05

L.R

1 mole of $CaCO_3$ produce → 1 mole of $CO_2$

So 0.02 mole $CaCO_3$ will produce → 0.02 moles of $CO_2$

Volume of $CO_2$ evolved = 0.02 × 22.4

= 0.448 L

**46.** (*c*) $N_1V_1 = N_2V_2$

Dibasic acid NaOH

Here,

$$N_1=\frac{w}{E}=\frac{0.16}{E}$$

$V_1 = 1000$

$$N_1=\frac{1}{10}$$

$V_2$ = 25 mL

$$\frac{0.16}{E}\times 1000=\frac{1}{10}\times 25$$

$$\frac{160}{E}=\frac{25}{10}$$

$$E=\frac{1600}{25}=64$$

M = 2 × E = 2 × 64 = 128 g

**47.** (*a*) One molal soltuion is one that contains one mole of a solute in 100g of the solvent.

**48.** (*a*) $(COOH)_2.2H_2O$ oxalic acid dihydrate

$$\text{Molarity}=\frac{6.3}{126}\times\frac{1}{250}\times 1000=0.2\,M$$

No. of equivalents of oxalic acid = No. of equivalent of NaOH

$N_1V_1 = N_2V_2$

$0.2 \times 2 \times 10 = 0.1 \times V_2$

$0.4 \times 10 = 0.1 \times V_2$

$V_2$ = 40 mL

**49.** (*c*) Methyl orange indicator gives the end point corresponding to 100% of sodium carbonate.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

$20 \times x = 10 \times 0.1 + 5 \times 0.2 \times 2$

x = 0.15 M

**50.** (*a*) 2.5 mL of 0.1 M $H_2SO_4$ = 2.5 mL of 0.2 $NH_2SO_4$

$= \frac{1}{2}$ $Na_2CO_3$ present in 10 mL of mixture

So,

5 mL of 0.2 N $H_2SO_4$ = $Na_2CO_3$ present in 10 mL of mixture

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$\equiv$ 5 mL of 0.2 N $Na_2CO_3$

$\equiv \frac{0.2\times53}{1000}\times5 = 0.053\,g$

Amount of $Na_2CO_3 = \frac{0.053}{10}\times1000 = 5.3$ g/L of mixture

Between first and second end points,

= 2.5 mL of 0.2 M $H_2SO_4$ used

= 2.5 mL of 0.4 N $H_2SO_4$ used

= 5 mL of 0.2 N $H_2SO_4$ used

$\equiv \frac{1}{2}$ $Na_2CO_3$ + $NaHCO_3$ present in 10 mL of mixture

(5 – 2.5) mL 0.2 N $H_2SO_4$

$\equiv$ $NaHCO_3$ present in 10 mL of mixture

$\equiv$ 2.5 mL 0.2N $NaHCO_3$

$\equiv \frac{0.2\times84}{1000}\times2.5 = 0.042\,g$

Amount of $NaHCO_3 = \frac{0.042}{10}\times1000$

= 4.20 g/L of mixture.

**51.** (*a*) 24 $dm^3 \rightarrow$ 1 mole of $O_2$ gas.

$3dm^3 \rightarrow \frac{3}{24} = \frac{1}{8}$ mole of $O_2$ gas

$2H_2O_2 \rightarrow 2H_2O + O_2$

$n_{H_2O_2} = 2\times n_{O_2} = 2\times\frac{1}{8} = \frac{1}{4}$ moles

$\therefore X \times 0.1 = 0.25$

$X = 2.5$

## Exercise-2 (Learning Plus)

**1.** (*c*) In $Ca_3(PO_4)_2$

$\frac{\text{mole of Ca atom}}{\text{mole of O atom}} = \frac{3}{8}$

Mole of 'O' atom = $\frac{8}{3}$ (mole of Ca atom)

Mole of 'Ca' atom = 3

**2.** (*a*) $H_2SO_4$ $\qquad$ $A\ell_2(SO_4)_3$

$\Downarrow$ $\qquad$ $\Downarrow$

32 $\qquad$ $3 \times 32$

so total molecular mass = 98

$\frac{1}{3}$ $(A\ell_2(SO_4)_3)$

$\Downarrow$

$\frac{1}{3}\times 342$

$\Downarrow$

114

$\frac{98}{114} = 0.86$

**3.** (*c*) (*a*) $n = \frac{10\times1}{18} = 0.55$

(*b*) $n = 0.1 \times 5 = 0.5$

(*c*) $n = \frac{12}{48} \times 3 = 0.75$

(*d*) $n = \frac{N}{NA} = 0.2\times2 = 0.4$

**4.** (*a*) $3.9854 \times 10^{-23}$ g of element = 1atom.

1 g of element = $\frac{1}{3.9854\times10^{-23}}$

$= 2.509 \times 10^{22}$

**5.** (*c*) (*a*) $n = \frac{12}{12} = 1$

(*b*) $n = \frac{8}{16} = 0.5$

(*c*) $n = \frac{32}{32} = 1$

(*d*) $n = \frac{24}{24} = 1$

**6.** (*b*) Moles of $Mg_3(PO_4)_2 = \frac{1}{8}\times0.25 = 3.125 \times 10^{-2}$

**7.** (*a*)

| | C | H | O |
|---|---|---|---|
| Mass | 24 | 8 | 32 |
| Moles | $\frac{24}{12}$ | $\frac{8}{1}$ | $\frac{32}{16}$ |
| Ratio | 2 | 8 | 2 |
| Simple integer ratio | 1 | 4 | 1 |

Hence empirical formula is $CH_4O$

**8.** (*a*)

| X | Y |
|---|---|
| $\frac{75.8}{75}$ | $\frac{24.2}{16}$ |
| 1.01 | $1.5 \times 2$ |
| 2 | 3 |

**9.** (*a*) $C_xH_y + \left(x+\frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$

$a\left(x+\frac{y}{4}\right)a \qquad ax \qquad \frac{ay}{2}$

$a + \left(x+\frac{y}{4}\right)a = 600$ ...(i)

$ax + a\frac{y}{2} = 700$ ...(ii)

(i) divide by (ii)

$7 + x = 5y/4$

$x < 5$

put the value

if $x = 3$ $\qquad$ $10 = 5y/4$ $\qquad$ $y = 8$

Ans. is $C_3H_8$

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**10.** (*d*) Mole fraction of $H_2O = 1 - 0.25 = 0.75$

$$\frac{x_{C_2H_5OH}}{X_{C_2H_5OH} + X_{H_2O}} = \frac{n_{C_2H_5OH}}{n_{C_2H_5OH} + n_{H_2O}} \text{ or wt. \%}$$

$$= \frac{0.25 \times 46}{0.25 \times 46 + 0.75 \times 18} \times 100 = 46\%.$$

**11.** (*c*) $CO_2 = 132\ g = \frac{132}{44}$ mole = 3 mole

$H_2O = 54\ g = \frac{54}{18}$ mole = 3 mole

$\Rightarrow$ C atoms = 3 mole

H atoms = 6 mole

**12.** (*a*) $Fe_2O_3 = \frac{2 \times 56}{3 \times 16} = \frac{7}{3}$

$FeO = \frac{56}{16} = \frac{7}{2}$

$\therefore$ $Fe_2O_3 : FeO = \frac{7}{3} \times \frac{7}{2} = 3 : 2$

**13.** (*a*) Amount of butter $= \frac{2 \times 10^{-3}}{5.5 \times 10^{-6}} = 363.6$ gm

**14.** (*b*) According to question, atomic ratio of metal and oxide is 2:3

$\therefore$ formula $= M_2O_3$

$$\% \text{ of Oxygen} = \frac{\text{Mass of Oxygen}}{\text{Mol. mass of oxide}} = \frac{30}{100}$$

$$= \frac{16 \times 3}{2M + (16 \times 3)} = \frac{30}{100}$$

or $M = \frac{160 - 48}{2}$

$= 56g$

**15.** (*a*) 2 moles of $SO_2$ will react with 1 mole of $O_2$ to form 4 moles of $SO_3$. 8 moles of $SO_2$ will react with 4 moles of $O_2$ to form 8 moles of $SO_3$.

Hence, $10 - 8 = 2$ moles of $SO_2$

and $15 - 4 = 11$ moles of $O_2$

did not enter into the combination respectively.

**16.** (*b*) $C_6H_5OH + 7O_2 \longrightarrow 6CO_2 + 3H_2O(\ell)$

30 ml

$6 \times 30 = 180$ ml of $CO_2$ is produced

Volume used initially

$= 30 + 210 = 240$

(for $C_6H_5OH$) (for $O_2$)

change in volume $= 240 - 180 = 60$ ml

**17.** (*b*) C = 84/12 = 7 mole

$H_2 = 12\ g = 6$ mole

$O_2 = 56/22.4 = 5/2$ mole

$12C + 11H_2 + 11/2\ O_2 \longrightarrow C_{12}H_{22}O_{11}$

L.R. $= O_2$

11/2 mole $O_2$ produce 1 mole sucrose

5/2 mole $O_2$ will for 5/11 mole sucrose

mass of sucrose = 5/11 × (mol. mass)

$= 5/11 \times 342$

$= 155.45$ g

**18.** (*a*) Let volume is V ml

$H_2SO_4 + 2\ NaOH \longrightarrow Na_2SO_4 + 2H_2O$

mole 0.2 V 40 x 0.1

moles of $H_2SO_4$ remains $= 0.2\ V - \frac{40 \times 0.1}{2}$

$$\frac{0.2V - \frac{40 \times 0.1}{2}}{V + 40} = \frac{6}{55}$$ V = 70 mL

**19.** (*a*) 1 Mol of x will give $= \frac{5}{2} = 2.5$ mol of w

But % yield $= \frac{1.25}{2.5} \times 100 = \mathbf{50\%}$

**20.** (*a*) Explanation : $2\ Ag + S \rightarrow Ag_2S$

$2 \times 108$ g of Ag reacts with 32 g of sulphur

10 g of Ag reacts with $\frac{32}{216} \times 10 = \frac{320}{216} > 1$ g

It means 'S' is limiting reagent

32 g of S reacts to form $216 + 32 = 248$ g of $Ag_2S$

1 g of S reacts to form $= \frac{248}{32} = 7.75$ g

Alternately

$n_{eq}$ of Ag $= \frac{10}{108} = 0.0925 n_{eq}$ of S $= \frac{1}{16} = 0.0625$

($n_{eq}$ = number of equivalents)

Since $n_{eq}$ of S is less than $n_{eq}$ of Ag

$\Rightarrow$ 0.0625 eq of Ag will react with 0.0625 eq of S to form 0.0625 eq of $Ag_2S$

Hence, amount of $Ag_2S = n_{eq} \times$ Eq. wt. of $Ag_2S$

$= 0.0626 \times 124 = 7.75$ g

**21.** (*a*) Molality $= \frac{X_B}{X_A \times M_A} \times 1000$

$m_B = 75$ m

$m = \frac{M \times 1000}{d \times 1000 - M \times M_1}$

$M = 30$

**22.** (*a*) NaOH $= \frac{125\text{ml} \times 1 \times \frac{8}{100}}{40}$ mole = 0.25 mole

$HCl = \frac{125 \times \frac{10}{100}}{36.5} = 0.34$ mole

HCl > NaOH

Acidic

**23.** (*d*) $M = \frac{\%\text{ by weight} \times 10 \times d}{Mw_2} = \frac{36.5 \times 10 \times 1.2}{36.5} = 12$ M

$m = \frac{36.5 \times 1000}{36.5 \times (100 - 36.5)} = \frac{1000}{63.5} = 15.7$ m

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**24.** (*b*) 1000 mL solution contain 2 mole of ethanol or 1000 × 1.025 g solution contain 2 mole of ethanol

wt. of solvent = 1000 × 1.025 – 2 × 46

$$m = \frac{2}{1000\times1.025-2\times46}\times1000$$

$$m = \frac{2}{933}\times1000 = 2.143$$

**25.** (*c*) Molarity $= \frac{6.02\times10^{22}}{6.02\times10^{23}}\times\frac{1}{1/2} = 0.2$

**26.** (*b*) Let, $n_{H_2O} = n_{NaCl} = n$

$$m = \frac{\text{Mole of solute}}{\text{wt. of solvent (kg)}} = \frac{n}{n\times18}\times1000$$

$$= \frac{1}{18}\times1000 = 55.55 \text{ m.}$$

**27.** (*b*)

(I) 50g of 40% (w/w) NaOH for 100 gm solution wich means 60 gm of $H_2O$ and 40 gm of NaOH and for 50 g solution has 20 gm NaOH.

(II) 50 ml of 50% (w/v) NaOH ($d_{sol.}$ = 1.2 g/ml) and for 100 ml of solution has 50 gm NaOH

$\therefore$ for 50 ml solution

$\Rightarrow$ 25 gm of NaOH.

(III) 50 g of 15M NaOH ($d_{sol.}$ = 1 g/ml)

$d_{sol.}$ = 1 g/ml , for 50 gm solution $\Rightarrow$ 50 ml solution

moles of NaOH = M × V

= 15 × 0.05 = 0.75

Mass of NaOH = 0.75 × 40

= 30 gm

So mass in decreasing order is III > II > I.

**28.** (*a*) Mole fraction of A i.e. $X_A = \frac{n_A}{\text{Total moles}}$

So $X_{H_2O} = \frac{n_{H_2O}}{\text{Total moles}}$

Now $\frac{X_A}{X_{H_2O}} = \frac{n_A}{n_{H_2O}}$

and molality $= \frac{n_A\times1000}{n_{H_2O}\times18} = \frac{X_A\times1000}{X_{H_2O}\times18}$

$= \frac{0.2\times1000}{0.8\times18} =$ **13.9 Ans.**

**29.** (*c*) Molarity $= \frac{98\times10\times1.84}{98} = 18.4$ M

$\left\{\therefore M = \frac{(\%\,w/w)\times(d)\times10}{\text{Mol. mass of solute}}\right\}$ (d in g/ml.)

**30.** (*b*) Weight of KOH = 2.8 gram

Volume of solution = 100 ml

$$M = \frac{2.8\times1000}{56\times100} = \frac{28}{56} = 0.5 \text{ M}$$

**31.** (*c*) Molarity of $Cl^-$ = 3 (molarity of $FeCl_3$)

$$= 3\left(\frac{M}{30}\right) = \frac{M}{10}$$

**32.** (*a*) $M_1V + M_2V_2$

$= M_R[V_1 + V_2]$

$1\times500 + 1\times500$

$= M_R[500 + 500]$

$M_R = 1$

**33.** (*b*) Mole = M × V

$100\times10^{-3} = 0.8\times V$

$V = 0.125$

**34.** (*d*) Moles of $Cl^-$ in 100 ml of solution

$$= \frac{2}{58.5}+\frac{4}{111}\times2+\frac{6}{53.5} = 0.2184$$

Molarity of $Cl^- = \frac{0.2184}{100}\times1000 = 2.184.$

**35.** (*d*) Conc. of cation $= \frac{400+300+200}{400} = \frac{900}{400} = 9/4$

Conc. of anion $= \frac{200+300+400}{400} = \frac{900}{400} = 9/4$

$\therefore$ Ratio of the conc. = 1

**36.** (*b*) No. of moles of $CO_2$ in 4.4g $CO_2 = \frac{4.4}{44} = 0.1$ mole

No. of moles of $H_2$ gas in 2.24 liters at STP = $\frac{2.24}{22.4} = 0.1$ mole

So total moles of $H_2$ and $CO_2$ gas are = (0.1 + 0.1)

= 0.2 moles

Total No. of molecules in the container = $0.2\times6.022\times10^{23}$

$= 1.2046\times10^{23}$

**37.** (*b*) If percentage of A = 75.08, then percentage of B = (100 – 75.08) = 24.92

Ratio in which A and B atoms are combining can be calculated as:

$$A : B = \frac{75.08}{75} : \frac{24.92}{16} = 1:1.5 = 2:3$$

$\therefore$ Formula of the compound is $A_2B_3$

**38.** (*c*) $CaCO_3 \rightarrow CaO + CO_2$

10g of $CaCO_3$ = 0.1 mole of $CaCO_3$

0.1 mole of $CaCO_3$ will produce $\rightarrow$ 0.1 mole of CaO

0.1 mole of CaO = 5.6g of CaO

$$\%\text{ yield} = \frac{\text{Actual}}{\text{Theortical}}\times100$$

$$= \frac{5}{5.6}\times100 = 89\%$$

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**39.** (*b*) $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

Moles of $CO_2 = \frac{4.4}{44} = 0.1$ moles

Then moles of $NaHCO_3$ will be 0.2 moles

wt. of $NaHCO_3 = 0.2 \times 84 = 16.8$ g

$\%\text{ purity} = \frac{16.8}{33.6} \times 100 = 50\%$

**40.** (*c*) The atomic masses of nitrogen and oxygen are 14 and 16g/mol respectively.

$n_N = \frac{28}{14} = 2$ moles

$n_O = \frac{80}{16} = 5$ moles

The mole ratio N : O = 2 : 5

The formula of the compound is $N_2O_5$.

**41.** (*a*) No. of $e^-$ in 1 molecule of $CO_2 = 6 + 2(8) = 22$

**42.** (*c*)

| | $O_2$ | $N_2$ |
|---|---|---|
| Weight ratio | 1 | 4 |
| Moles ratio | $\frac{1}{32}$ | $\frac{4}{28} = \frac{1}{7}$ |
| Molecules ratio | $\frac{1}{37}N_A$ | $\frac{1}{7}N_A$ |

Molecules ratio = 7 : 32

**43.** (*a*) $Ag_2CO_3(s) \longrightarrow 2Ag(s) + CO_2(g) + \frac{1}{2}O_2(g)$

Molecular weight of $Ag_2CO_3 = 276$ g

And, molecular weight of Ag = 2 × 108 = 216 g

276 g of $Ag_2CO_3$ gives 216 g of Ag.

Therefore, 2.76g of $Ag_2CO_3$ on heating will give

$\frac{216}{276} \times 2.76 = 216$g Ag as residuce.

**44.** (*c*) Let the metal be M with atomic weight be X.

So, $3M + N_2 \rightarrow M_3N_2$

12g 14.8g

$3x \rightarrow 3x + 28$

So,

$\frac{12}{3x} = \frac{14.8}{3x+28}$

$\frac{12}{14.8} = \frac{3x}{3x+28}$

$X = 40$

**45.** (*c*) $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$

1 mole of sulfuric acid will produce 1 mole (22.4 L at STP) of $CO_2$

100 mL (0.1 L) of 0.1 M $H_2SO_4$ coresponds 0.01 mole

Thus 0.01 mole of $H_2SO_4$ will produce 0.01 moles of $CO_2$

0.01 moles of $CO_2$ = 0.224 L of $CO_2$

**46.** (*c*) To form one molecule of $X_2Y_3$, two atoms of X and three atoms of Y will be needed

It means 36 × 2 = 72 amu and 24 × 3 = 72 amu of Y willl be needed to form 144 amu of $X_2Y_3$.

**47.** (*a*) Moles of sugar = $\frac{1.71}{342} = \frac{1}{200}$ moles

No. of molecules = $\frac{1}{200} \times N_A$

No. of C-atoms = $12 \times \frac{1}{200} \times N_A = 3.6 \times 10^{22}$

**48.** (*a*) The ratio among C : H : N is 38 : 8 : 16 : 45.2

The molar ratio is $\frac{38.8}{12} : \frac{16}{1} : \frac{45.2}{14}$

= 3.23 : 16 : 3.22

= 1 : 5 : 1

Therefore the empirical formula of the compound would be $[CH_3NH_2]$

**49.** (*b*) When 100g of Ethylene polymerises entirely to polyethene, the weight of polyethene formed as →

$n\,CH_2 = CH_3 \rightarrow (CH_2 - CH_2)_n$ is 100g

Because entire mass of ethylene is converted to polyethylene as there is no addition, elimination or condensation of simple molecules.

**50.** (*c*) Given,

Mass % = 98% i.e 98 g of $H_2SO_4$ is present in 100g of the solution

density = 1.84 g/cc = 1.84 g/mL

Hence volume = $\frac{100}{1.84}\text{mL} = \frac{1}{18.4}\text{L}$

given mass of $H_2SO_4$ = 98 g

$n_{H_2SO_4} = \frac{98}{98} = 1$

$\text{Molarity} = \frac{1}{\frac{1}{18.4}} = 18.4\,\text{M}$

## Exercise-3 (JEE Advanced Level)

**1.** (*a,b,c*)

Mole of $NH_3 = 1.7 = 0.1$

Mole H atom = 0.3

Total atoms = $0.4 \times 6.02 \times 10^{23} = 2.408 \times 10^{23}$

$\%\text{ H} = \frac{3 \times 1}{17} \times 100 = 17.65\%$

**2.** (*b,c,d*)

| | C | + | $O_2$ | ⟶ | $CO_2$ |
|---|---|---|---|---|---|
| Mass | 27 | | 88 | | |
| Moles | $\frac{27}{12}$ | | $\frac{88}{32}$ | | |

C is limiting reagent

Moles of $CO_2$ produced = moles of C = $\frac{27}{12} = 2.25$

$\therefore$ Volume of $CO_2$ at STP = $2.25 \times 2.4 = 50.4$ L

Ratio of C and O in $CO_2$ = 12 : 32 = 3 : 8

Moles of unreacted $O_2 = 2.75 - 2.25 = 0.5$

$\therefore$ Volume of unreacted $O_2$ at STP = $0.5 \times 22.4 = 11.2$ L

**3.** (*a*,*c*)

$0.5 \times n = \frac{216}{108}$ = mol of Ag

$n = 4$

M.wt = 58 + [165]n g/mol = 718 g/mol

**4.** (*a*,*c*)

Convert all the wt. in mole and use limiting reagent concept ind out the mole produced of $NH_3$.

In (*a*) & (*c*) it comes equal to 10 moles

**5.** (*a*,*b*,*c*)

(M wt. of $Na_2CO_3$ = 106 M wt. of HCl = 36.5, M wt. of NaCl = 58.5)

Moles of $Na_2CO_3 = \frac{106}{106} = 1.0$ mol

Moles of $HCl = \frac{109.5}{36.5} = 3.0$ mol

(*a*) Since for 1 mol of $Na_2CO_3$, 2 mol of HCl is required.

So, HCl is in excess (3 – 2) = 1.0 mol

Therefore, $Na_2CO_3$ is the limiting quantity.

(*b*) Weight of NaCl formed = (1.0 mol $Na_2CO_3$)

$\left(\frac{2 \text{ mol NaCl}}{\text{mol } Na_2CO_3}\right)\left(\frac{58.5\text{g NaCl}}{\text{mol NaCl}}\right) = 1 \times 58.5 = 117.0$ g NaCl

(*c*) 1 mol of $Na_2CO_3$ = 1 mol of $CO_2$ = 22.4 L at NTP

**6.** (*a*,*c*)

$CaCl_2\ CaCO_3 \rightarrow CaO\ \frac{1.12}{56} = 0.02$ mole CaO

$\therefore$ Moles of $CaCl_2$ = 0.02 Mole

Mass of $CaCl_2 = 0.02 \times 111 = 2.22$ g

$\therefore$ % of $CaCl_2 = \frac{2.22}{4.44} \times 100 = 50\ \%$

**7.** (*b*,*d*)

| | $3A$ | + | $2B$ | $\longrightarrow$ | $A_3B_2$ |
|---|---|---|---|---|---|
| Initial mole | 3 | | 3 | | 0 |
| Final mole | 0 | | 3–2 | | 1 |
| | $A_3B_2$ | + | $2C$ | $\longrightarrow$ | $A_3B_2C_2$ |
| Initial mole | 1 | | 1 | | 0 |
| Final mole | $1-\frac{1}{2}$ | | 0 | | $\frac{1}{2}$ |

**8.** (*a*,*c*)

M wt. of $CaCO_3$ = 100, M wt. of $Na_2CO_3$ = 106

M wt. of $HNO_3$ = 63 g $mol^{-1}$

$Na_2CO_3 + CaCO_3 \longrightarrow CaCO_3 + 2NaCl$

(*a*) moles of $CaCO_3 = \frac{10}{100} = 0.1$ mol

moles of $Na_2CO_3$ = moles of $CaCO_3 \equiv 2 \times$ moles of NaCl

Weight of $Na_2CO_3 = 0.1 \times 106 = 10.6$ g

(*c*) % purity $Na_2CO_3 = \frac{10.6}{21.2} = 100 = 50\%$

(*d*) moles of NaCl = $2 \times 0.1 = 0.2$ mol

**9.** (*a*,*b*)

$C_3H_8 + O_2 \rightarrow 100$ mL

**Case I:**

Let assume $C_3H_8$ is L.R.

| | $C_3H_8$ | + | $5O_2$ | $\rightarrow$ | $3CO_2$ | + | $4H_2O$ |
|---|---|---|---|---|---|---|---|
| initial | $V_1$ | | $V_2$ | | | | |
| final | 0 | | $V_2 - 5V_1$ | | $3V_1$ | | |

Vol. contraction = $V_i - V_f$

$= (V_1 + V_2) - (V_2 - 5V_1 + 3V_1)$

$45 = 3V_1$

$V_1 = 15$ mL ($C_3H_8$)

$V_2 = 85$ mL ($O_2$)

**Case II:**

Let assume $O_2$ is L.R.

| | $C_3H_8$ | + | $5O_2$ | $\rightarrow$ | $3CO_2$ | + | $4H_2OH$ |
|---|---|---|---|---|---|---|---|
| initial | $V_1$ | | $V_2$ | | | | |
| final | $V_1 - \frac{V_2}{5}$ | | 0 | | $\frac{3}{5}V_2$ | | |

Vol contraction = $V_i + V_f$

$(V_1 + V_2) - \left(V_1 - \frac{V_2}{5} + \frac{3}{5}V_2\right)$

$45 = \frac{3}{5}V_2$

$V_2 = 75$ mL

$V_1 = 25$ mL

**10.** (*b*,*d*)

Molality of $Cl^- = \frac{2 \times 1000 \times 2}{(1000 \times 1.09) - 190} = 4.44$

**11.** (*a*,*b*)

Mol. wt. of HCOOH is 46.

Mass of HCOOH in 46 g of 70% $\left(\frac{w}{v}\right)$ HCOOH ($d_{solution}$ = 1.40 g/mL) is = (46/1.4) × 0.7 = 23 gm.

Mass of HCOOH in 50 g of 10 M HCOOH($d_{solution}$ = 1 g/mL) is = 50 × 10 × 46/1000 = 23 gm.

**12.** (*a*,*c*,*d*)

Molecular weight of KI, $(NH_4)_2SO_4$, $CuSO_4.5H_2O$ and $Al^{3+}$ are 166, 132, 250 and 27 g/mol respectively.

(*a*) For KI, Molarity = $\frac{166 \times 100}{166 \times 100} = 1.0$M

(*b*) For $(NH_4)_2SO_4$, Molarity = $\frac{33 \times 1000}{132 \times 200} = 1.25$M

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(c) For $CuSO_4.5H_2O$, Molarity $\frac{25\times1000}{250\times100}=1.0M$

(d) For $Al^{3+}$, Molarity $=\frac{27\times10^{-3}\times1000}{27\times1}=1.0M$

**13.** (a,b,d)

(a) Molarity of second solution is $=\frac{10\times d\times x}{M}=1\ M$

(b) Volume = 100 + 100 = 200 mL

(d) Mass of $H_2SO_4=\frac{200\times1}{1000}\times98=19.6$ g.

**14.** (c) 11.2 g of $N_2$ $\frac{11.2}{28}=0.4$ mole

$\therefore$ air = 0.5 mole $\Rightarrow 0.5\times22.4=11.2$ L

**15.** (b) 1 mole of air $\Rightarrow$ 0.8 mole of $N_2 = 08\times28$ g $N_2$

$\Rightarrow$ 0.2 mole of $O_2 = 0.2\times2$ g $O_2$

$\therefore$ % w/w $O_2=\frac{w_{O_2}\times100}{w_{O_2}+w_{N_2}}=\frac{0.2\times32\times100}{0.2\times32+0.8\times28}$

$=22.2\%$

**16.** (b) Density of air at NTP

1 mole of air = 0.8 mole $N_2$ + 0.2 mole $O_2$

$=0.8\times28+0.2\times32=28.8$ g = 22.4 Ltr volume.

$D=\frac{m}{V}=28.8=1.2857$ g/L $=\frac{288}{22.4}=1.2857$g/L

**17.** (a) $\%O=60\times\frac{16}{18}=53.33\%$

**18.** (b)

| Element | Symbol | Percentage of element | At. mass of element | Relative no. of atoms = Percentage / at. mass | Simplest atomic ratio | Simplest whole no. atomic ratio |
|---|---|---|---|---|---|---|
| Carbon | C | 40 | 12 | $\frac{40}{12}=3.33$ | $\frac{3.33}{3.33}=1$ | 1 |
| Hydrogen | H | 6.67 | 1 | $\frac{6.67}{1}=6.67$ | $\frac{6.67}{3.33}=2$ | 2 |
| Oxygen | O | 53.33 | 16 | $\frac{53.33}{16}=3.33$ | $\frac{3.33}{3.33}=1$ | 1 |

So emperical formula is $CH_2O$

**19.** (b) For molecular formula

$n=\frac{\text{Molecular mass}}{\text{Empirical formula mass}}=\frac{178}{30}=6$

So, molecular formula is $C_6H_{12}O_6$

**20.** A → r; B → p; C → q

$$\%\text{ of Y}=\frac{89\times3}{(89\times3)+(5\times27)+(12\times16)}\times100$$

$$=\frac{267\times100}{594}=44.95\%$$

$$\%\text{ Al}=\frac{5\times27}{594}\times100=22.73$$

$$\%\text{ O}=\frac{12\times16}{594}\times100=32.32\%$$

**21.** A → p,s; B → s; C → p,q; D → r

(a) Molarity of cation $=\frac{M_1V_1+M_2V_2}{V_1+V_2}$

$=\frac{0.2\times100+0.1\times400}{500}=\frac{0.6}{5}=0.12$

Molarity of $Cl^-=\frac{3(0.2)100+0.1\times400}{500}=\frac{0.6+0.4}{5}=0.2$

(b) Molarity of cation $=\frac{50\times0.4+0}{100}=0.2$

Molarity of $Cl^-=\frac{0.4\times50+0}{100}=0.2$

(c) Molarity of cation $=\frac{2(0.2)30+0}{100}=0.12$

Molarity of $SO_4^{2-}=\frac{30\times0.2}{100}=0.06$

(d) 24.5 g $H_2SO_4$ in 100 mL solution

Molarity $=\frac{\frac{25.4}{98}}{0.1}=2.5$

$\therefore$ Concentration of cation = 2 × 2.5 M

Concentration of $SO_4^{2-}=2.5$ M.

**22.** (d) A → q,s; B → p,s; C → p,q,r; D → q,r

(a) C : H : O $=\frac{51.17}{12}:\frac{13.04}{1}:\frac{34.78}{16}$

$=4:12:2$ or $2:6:1$

$\therefore$ Empirical formula = $C_2H_6O$ & molar mass = 46 g/mol

$\therefore$ Mol formula = $C_2H_6O$

$C_2H_6O + 3O_2 \longrightarrow 2CO_2 + 3H_2O$

1 mole → 44.8 L at STP

0.25 mole → (11.2 L at STP)

(*b*) Mass of C in organic compound = mass of C in $CO_2$

$= \frac{0.44}{44} \times 12 = 0.12$ g

Mass of H in organic compound = Mass of H in $H_2O$

$= \frac{0.18}{18} \times 2 = 0.02$ g

$\therefore$ Mass of O in organic compound = 0.3 – (0.12 + 0.02) = 0.16 g

$\therefore$ C : H : O $= \frac{0.12}{12} : \frac{0.02}{1} : \frac{0.16}{16} = 0.01 : 0.02 : 0.01$

$= 1 : 2 : 1$

$\therefore$ Empirical formula = $CH_2O$, but it contains 2 O atom per molecule

$\therefore$ Molecular formula = $C_2H_4O_2$

1 mole of $C_2H_4O_2$ contains $4\ N_A$ hydrogen atoms.

$C_2H_4O_2 + 2O_2 \longrightarrow 2CO_2 + 2H_2O$

1 mole 44.8 L

0.25 mole 11.2 L

(*c*) C : H = 42.857 : 57.143

= 3 : x (given)

On solving, x = 4 $\therefore$ molecular formula = $C_3H_4$

1 mole of $C_3H_4$ contains $4N_A$ hydrogen atoms.

Empirical formula is same as molecular formula

$C_3H_4 + 4O_2 \longrightarrow 3CO_2 + 2H_2O$

$n_{CO_2} > n_{H_2O}$

(*d*) C : H $= \frac{10.5}{12} : \frac{1}{1} = \frac{7}{8} : 1 = 7 : 8$

Empirical formula = $C_7H_8$

Mol wt. = 2 × VD = 2 × 46 = 92

$\therefore$ Mol formula = Empirical formula = $C_7H_8$

$C_7H_8 + 9O_2 \longrightarrow 7CO_2 + 4H_2O$

$n_{CO_2} > n_{H_2O}$

**23.** [+8]

$x + 4(-2) = 0$

$x = +8$

**24.** [–3]

$x + 4(+1) = +1$

$= -3$

**25.** [500]

$M = \frac{18 \text{ moles of solute}}{(1 \text{ lit. solution})}$

Mass of solution = 1000 × 1.8 = 1800 g.

Mass of solute = 18 × 98 = 1764.

Mass of solvent = 1800 – 1764) g = 36 g.

Hence molality $= \frac{\text{moles of solute}}{\text{mass of solvent}} \times 1000 = \frac{18}{36} \times 1000$

= 500.

**26.** [40]

$\because$ 40gm NaOH contain 16 gm of oxygen

$\therefore$ 100gm of NaOH contains $\frac{16}{40} \times 100 = 40\%$ oxygen.

**27.** [6g]

$N = \frac{W(gm) \times 1000}{V \times Eq.wt.}$

1500ml of 0.1N HCl = 150ml (N)

$1 = \frac{W(gm) \times 1000}{150 \times 40}, W(gm) = \frac{150 \times 40}{1000} = 6gm.$

**28.** [1]

$2Al + KClO_3 \rightarrow Al_2O_3 + KCl$

(excess) 1 mole ($KClO_3$ is L.R.)

From mole-mole analysis

$\frac{n_{KClO_3}}{1} = \frac{n_{Al_2O_3}}{1}$

$\Rightarrow n_{Al_2O_3} = 1$ mole

**29.** [22g]

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

100g 2N 44g

100 g $CaCO_3$ with 2 N HCl gives 44 g $CO_2$

100 g $CaCO_3$ with 1 N HCl gives 22 g $CO_2$

**30.** [1]

Conc. of cation $= \frac{400 + 300 + 200}{400}$

conc. of anion $= \frac{200 + 300 + 400}{400}$

$\therefore$ ratio of the conc. = 1

**31.** [27g]

Let wg water in added to 16 g $CH_3OH$

molality $= \frac{16 \times 1000}{W \times 32} = \frac{500}{W}$

$\frac{500}{W} = \frac{x_A \times 1000}{(1 - x_A)m_B}$

$= \frac{0.25 \times 1000}{0.75 \times 18}$

W = 27 gm

## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*b*) Given chemical eq$^n$

$M_2CO_3 + 2HCl \rightarrow 2MCl + H_2O + CO_2$

1 gm 0.01186 mol

$\Rightarrow$ from the balanced chemical eq$^n$

$\frac{1}{M} = 0.01186 \Rightarrow M = 84.3$ gm/mol

**2.** (*c*) Mass in body of a healthy human adult has:

Oxygen = 61.4%, carbon = 22.9 %

Hydrogen = 10.0% and Nitrogen = 2.6%

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Total weight of person = 75kg

Mass due to $^1H$ is = $75 \times \frac{10}{100} = 7.5$ kg

$^1H$ atoms are replaced by $^2H$ atoms.

So mass gain by person = 7.5 kg

3. (*c*) $\frac{12x}{y} = \frac{6}{1}$

$\boxed{2x = y}$ for $C_xH_yO_z$

$$C_xH_y(g) + \left(x + \frac{y}{4}\right)O_{2(g)} \rightarrow xCO_{2(g)} + \frac{y}{2}H_2O(\ell)$$

no. of oxygen atom in $C_xH_yO_z = z$

no. of oxygen atom required for $C_xH_y$ combustion is $\left(x + \frac{y}{4}\right) \times 2 = \left(2x + \frac{y}{2}\right)$

So $z = \frac{1}{2}\left(2x + \frac{y}{2}\right)$

$z = x + \frac{y}{4}$

$z = x + \frac{2x}{4} = \frac{3x}{2}$

$x : 2x : \frac{3x}{2}$

$2x : 4x : 3x$

$2 : 4 : 3$

Hence $C_2H_4O_3$

4. (*d*) $2NaHCO_3 + H_2C_2O \rightarrow Na_2C_2O + 2CO_2 + 2H_2O$

Let mass of $NaHCO_3$ be x mg

$n_{CO_2} = \frac{0.25}{25000} = 10^{-5}$

$w_{NaHCO_3} = 84 \times 10^{-5}$ g

$\% = \frac{84 \times 10^{-5}}{10^{-2}} \times 100 = 8.4\%$

5. (*a*) More polluted water has high biological oxygen demand.

6. (*c*) $2C_{57}H_{110}O_6$ (s) + 163 $O_2$ (g) → 114 $CO_2$ (g) + 110 $H_2O$ (l)

$\frac{\text{Moles of } C_{57}H_{110}O_6}{2} = \frac{\text{Moles of } H_2O}{110}$

$\frac{445}{\frac{890}{2}} = \frac{\frac{\text{mass of } H_2O}{18}}{110}$

Mass of $H_2O$ = 495 g

7. (*a*) Eq. of $(COOH)^2$ = Eq. of NaOH

$50 \times 0.5 \times 2 = 25 \times M \times 1$

Mass of NaOH in 50 mL = $\frac{50 \times 2}{1000} \times 40 = 4$g

8. (*c*) Molality of $Na^+ = \left(\frac{w}{M} \times \frac{1000}{W}\right) \times 2$ ($Na_2SO_4$ contains two $Na^+$ ions)

$= \left[\left(\frac{92}{23} \times \frac{1000}{1000}\right)\right] \times 2 = 8$

9. (*d*) As 1L solution have $10^{-3}$ mol $CaSO_4$

Eq. of $CaSO_4$ = eq. of $CaCO_3$

In 1L solution

$n_{CaSO_4} \times$ v.f. = $n_{CaCO_3} \times$ v.f.

$10^{-3} \times 2 = n_{CaCO_3} \times 2$

$n_{CaCO_3} = 10^{-3}$ mol in 1L

$\therefore$ $w_{CaCO_3} = 100 \times 10^{-3}$ g in 1L solution

$\therefore$ hardness in terms of $CaCO_3$

$= \frac{w_{CaCO_3}}{w_{Total}} \times 10^6 = \frac{100 \times 10^{-3}\text{g}}{1000\text{g}} \times 10^6 = 100$ ppm

10. (*c*) $0.1 = \frac{n_{C_{12}H_{22}O_{11}}}{2}$

$n_{C_{12}H_{22}O_{11}} = 0.2$

Wt. of $C_{12}H_{22}O_{11} = 0.2 \times 342 = 68.4$

11. (*c*) Moles of NaOH = $\frac{8}{40} = 0.2$

Moles of $H_2O = \frac{18}{18} = 1$

Moles fraction of NaOH = $\frac{0.2}{1.2} = 0.167$

Molality = $\frac{8}{40} \times \frac{1000}{18} = 11.11$

12. (*c*) Volume strength = 11.35 × M = 11.35 (STP)

13. (*b*) $n_{eq}.CaCO_3 = N_{eq}Ca(HCO_3)_2 + n_{eq}Mg(HCO_3)_2$

Or, $\frac{W}{100} \times 2 = \frac{0.81}{162} \times 2 + \frac{0.73}{146} \times 2$

$\therefore$ w = 1.0

$\therefore$ Hardness = $\frac{1.0}{100} \times 10^6 = 10000$ppm

Correct option : (b)

14. (*c*) $N_2(g) + 3H_2 \rightarrow 2NH_3$

1 mole $N_2$ required 3 moles of $H_2$.

(*a*) $\frac{14}{28}$ $\quad$ $\frac{4}{2}$

0.5 mole $\quad$ 2 mole

$N_2$ is L.R.

(*b*) $\frac{28}{28}$ $\quad$ $\frac{6}{2}$

1 mole $\quad$ 3 mole

No L.R.

(*c*) $\frac{56}{28}$ $\quad$ $\frac{10}{2}$

2 mol. $\quad$ 5 mole

$\therefore$ 2mole $N_2$ required 6 moles of

So, $H_2$ is L.R.

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**15.** (*d*) In $CH_4$

$$\% \text{ by mole of carbon} = \frac{1 \text{ mol atom}}{5 \text{ mol atom}} \times 100 = 20\%$$

**16.** (*d*) The maximum prescribed concentration of Cu in drinking water is 3 ppm.

**17.** (*b*) Volume strength = 11.2 × molarity = 11.2

$\Rightarrow$ molarity = 1 M

$\Rightarrow$ strength = 34 g/L

$$\Rightarrow \% \text{ ww} = \frac{34}{1000} \times 100 = 3.4\%$$

**18.** (*b*) $C_xH_y + \left(X + \frac{Y}{4}\right)O_2 \rightarrow xCO_2 + \frac{Y}{2}H_2O$

$\left(\frac{25}{M}\right)X \times \frac{25}{M}$ $\quad$ $\frac{y}{2} \times \frac{25}{M}$

$= 2$ $\quad$ $= 0.5$

C $\quad x \times \frac{25}{M} = 2$

H $\quad y \times \frac{25}{M} = 1$

$C_{2y}H_y \equiv 24_y \text{ gm C} + y \text{ gm H}$

or

24 : 1 ratio by mass

**19.** (*c*) $\frac{W}{W}\% = 20$

100 gm solution has 20 gm KI

80 gm solvent has 20 gm KI

$$M\frac{\frac{20}{166}}{\frac{80}{1000}} = \frac{20 \times 1000}{166 \times 80} = 1.506 = 1.51 \text{ mol/kg}$$

**20.** (*c*) $5[M_A + 2M_B] = 125$

$M_A + 2M_B = 25$ $\quad$ ...(i)

$2M_A + 2M_B = 30$ $\quad$ ...(ii)

from eq. (i) & (ii)

$M_A = 5$

$M_B = 10$

**21.** (*c*) $X_{solvent} = 0.8$

If $n_T = 1$

$n_{Solvent} = 0.8$

$$\text{molality} = \frac{2}{\frac{0.8 \times 18}{1000}} = 13.88$$

**22.** (*d*) $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$

10 $\quad 10\left(x + \frac{y}{4}\right)$ $\quad 10x$

By given data, $10\left(x + \frac{y}{4}\right) = 55$ $\quad$ ...(i)

$10x = 40$ $\quad$ ...(ii)

$\therefore$ $x = 4$, $y = 6$ $\Rightarrow$ $C_4H_6$

**23.** (*c*) $C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(\ell)$

Each 1 g of $C_3H_8$ requires 3.63 g of $O_2$

$P_4(s) + 5O_2g \longrightarrow P_4O_{10}(s)$

Each 1 g of $P_4$ requires 1.29 g of $O_2$

$4Fe(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s)$

Each 1 g of Fe requires 0.428 g of $O_2$

$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$

Each 1 g of Mg requires 0.66 g of $O_2$ therefore least amount of $O_2$ is required in option (iii).

**24.** (*a*) Refer Notes

**25.** [26.80 to 27.00]

$[Co(NH_3)_6]Cl_3$ + $3AgNO_3$ → $3AgCl$

0.3g $\quad$ v ml, 0.125 M

$$\frac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3} \quad \text{or,}$$

$$V = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 \text{ ml}$$

**26.** [2130 g]

mol of $NaClO_3$ = mol of $O_2$

$$\text{mol of } O_2 = \frac{PV}{RT} = \frac{1 \times 492}{0.082 \times 300} = 20 \text{ mol}$$

mass of $NaClO_3 = 20 \times 106.5 = 2130$ g

**27.** [100.00 ]

$10^{-3}$ molar $MgSO_4 \equiv 10^{-3}$ moles of $MgSO_4$ present in 1 L solutions.

$n_{CaCO_3} = n_{MgSO_4}$

$$ppm_{(\text{in term of } CaCO_3)} = \frac{10^{-3} \times 100}{1000} \times 10^6$$

$ppm_{(\text{in term of } CaCO_3)} = 100$ ppm

**28.** [25.00]

Mass of $6.023 \times 10^{22}$ molecules of a substance = 10 g

Mass of $6.023 \times 10^{23}$ molecules of the substance = 100 g

$$\text{Molar mass of the substance} = \frac{5}{100 \times 2} = 2.5 \times 10^{-2}$$

$$= 25 \times 10^{-3}$$

**29.** (*a*) Volume strength = 5.6

V.S = 11.2 × Molarity

$$\text{Molarity (M)} = \frac{\text{V.S.}}{11.2} = \frac{5.6}{11.2} = 0.5$$

0.5 M → 0.5 mole in 1L

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Mass percentage $= \frac{0.5\times34}{1000\times d}\times100$

$= \frac{17\times100}{1000\times1} = 1.7$

**30.** [47] $X_{C_6H_{12}O_6} = 0.1$

Let total mole is 1 mole then mole of glucose will be 0.1 and mole of water will be 0.9

so, mass % of water = (0.9 × 18)/(0.1 × 180 + 0.9 × 18) × 100

$= 47.36$

$= 47$

**31.** [100]

Molarity of $H_2O_2$ solution = 8.9 M

Volume strength of $H_2O_2$ solution

= 8.9 × 11.2 = 100 V

**32.** [3400]

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

Number of moles of $N_2 = \frac{2.8\times10^3}{28} = 100$

Number of moles of $H_2 = \frac{1000}{2} = 500$

Number of moles of $NH_3$ produced = 200

Mass of $NH_3$ produced = 200 × 17 = 3400 gm

**33.** [18]

$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$

$C_4H_{10}(g) + \frac{13}{2}O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

No. of moles of $O_2$ required to oxidise 1 mole of propane and 2 moles of butane $= 5 + 2\times\frac{13}{2} = 18$

**34.** (*c*) Average molar mass $= \frac{35\times3+37\times1}{4} = 35.5$

$\Rightarrow {}^{35}Cl : {}^{37}Cl = 3 : 1$

**35.** (*b*) To express $C_2$ in terms of mole fraction $x_2$

| | 1st component | 2nd component |
|---|---|---|
| mole | $n_1$ | $n_2$ |
| m.w | $M_2$ | $M_2$ |
| mass | $n_1M_1$ | $n_2M_2$ |

mass of solution $= n_1M_1 + n_2M_2$

mole fraction $x_2 = \frac{n_2}{n_1+n_2}$

$n_1 = \frac{n_2(1-x_2)}{x_2}$

Mass of solution $= n_1M_1 + n_2M_2$

$= \frac{n_2M_1(1-x_2)}{x_2} + n_2M_2$

$= \frac{n_2}{x_2}[M_2x_2 - x_2M_1 + M_1]$

Volume of solution

$= \frac{n_2[M_2x_2 - x_2M_1 + M_1]}{1000dx_2}$ Litre

$C_2 = \frac{1000n_2dx_2}{n_2[M_2x_2 - x_2M_1 + M_1]}$

$$\boxed{C_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}}$$

**36.** [64]

Given, mass of $CuSO_4\cdot5H_2O$ = 80 g

The concentration of copper sulphate solution is $x\times10^{-3}$ mol/L.

Molarity $= \frac{\text{Number of moles of solute}}{\text{Volume of solution(L)}}$ ...(*i*)

Molar mass of $CuSO_4\cdot5H_2O = 63.54+32+16\times4$

$= 5\times18 = 249.54$ g/mol

Number of moles of solute

$= \frac{\text{Weight of solute}}{\text{Molecular mass of solvent}}$

$= \frac{90g}{249.54g/mol} = 0.32$ mol

Volume of solution = 5L

From Eq. (*i*),

Molarity $= \frac{0.3205}{5} = 64.11\times10^{-3}$ mol/L

$\therefore$ x = 64.11

or $x \approx 64$

Hence, **Ans.** is 64.

**37.** [2] Given, mass of Na = 8 g

Molar mass of Na = 23gmol$^{-1}$

$\frac{8g}{23g} = \frac{\text{Number of atoms}}{6.022\times10^{23}}$

Number of atoms $= \frac{8\times6.022}{23}\times10^{23}$

Number of atoms = $2.09\times10^{23}$

$x \geq 2$

Hence, **Ans.** is 2

**38.** [13]

$Na_2O + H_2O \rightarrow 2NaOH$

$\frac{20}{62}$ moles

Moles of NaOH formed $= \frac{20}{62}\times2$

**39.** [20] $[H_2C_2O_4, 2H_2O] = \frac{\text{weight/M.M}}{V(L)}$

$\rightleftharpoons \approx\times10^{-2} = \frac{6.3\times125}{250/1000}$

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**40.** [2592 g and 2376 g] $C_{15}H_{30} + \frac{45}{2}O_2 \rightarrow 15CO_2 + 15H_2O$

Mass of the fuel = $0.756 \times 1000 = 756$ g

Number of moles of fuel = $\frac{756}{210}$

Weight of oxygen = $\frac{756}{210} \times \frac{45}{2} \times 32 = 2592$g

Weight of $CO_2 = \frac{756}{210} \times 15 \times 44 = 2376$g

**41.** [46]

| $N_2$ | + | $3H_2$ | → | $2NH_3$ |
|---|---|---|---|---|
| (56L) | | (in excess) | | (0 L) |
| (−10L) | | (−30L) | | (+20L) |
| 46 L | | | | 20 L |

**42.** [24] $ppm = \frac{W_{Mg}}{V_{soln}} \times 10^6 = 48$

$\Rightarrow W_{Mg} = \frac{48 \times 2 \times 1000}{10^6} = 96 \times 10^{-3}$g

No. of Mg atoms = $\frac{96 \times 10^{-3}}{24} \times 6.02 \times 10^{23}$

$= 24.08 \times 10^{20}$

**43.** [54] $n_{HNO_3} = 0.4$ mol

Remaining $n_{HNO_3} = 0.4 - \frac{11.5}{63} = 0.2175$

$\therefore$ Molarity = $\frac{0.2175}{400} \times 1000$

$= 0.5437$ mol/L $= 54 \times 10^{-2}$ mol/lit

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**44.** [5] The balance chemical equation is

$3Br_2 + 3Na_2CO_3 \longrightarrow 5NaBr + NaBrO_3 + 3CO_2$

**45.** (*c*) Mole = $\frac{120}{60} = 2$

mass of solution = 1120 g

$V = \frac{1120}{1.15 \times 1000} = \frac{112}{115}$L

$M = \frac{2 \times 115}{112} = 2.05$ mol/litre

**46.** [8 mL.]

29.2% (w/w) HCl has density = 1.25 g/ml

Now, mole of HCl required in 0.4 M HCl = $0.4 \times 0.2$ mole = 0.08 mole

if v mol of orginal HCl solution is taken

then volume of solutio = 1.25 v

mass of HCl = (1.25 v × 0.292)

mole of HCl = $\frac{1.25v \times 0.292}{36.5} = 0.08$

so, v = $\frac{36.5 \times 0.08}{0.29 \times 1.25}$ mol = 8 mL

**47.** [8] Given 3.2 M solution

$\therefore$ moles of solution = 3.2 mol

Consider 1 L Solution.

$\therefore$ volume of solvent = 1 L

$P_{solvent} = 0.4$ g.mL$^{-1}$

$\therefore m_{solvent} = P \times V = 400$ g

$\therefore$ molality = $\frac{3.2 \text{ mol}}{0.4 \text{ kg}} = 8$ molal

**48.** [9] Given, molality = Molarity

And assuming no volume change in forming solution

Density of solvent = 1 gml$^{-1}$

And density of solution (given) = 2 gml$^{-1}$

Implies, solvent and solute are present in equal qualities

$\therefore \chi_{solute} = 0.1 = \frac{\frac{1}{M_{solute}}}{\frac{1}{M_{solvent}} + \frac{1}{M_{solute}}} \rightleftharpoons \frac{M_{solute}}{m_{solvent}} = 9$

**49.** [6] $8KMnO_4 + 3Na_2S_2O_3 + H_2O \rightarrow 8MnO_2 + 3Na_2SO_4 + 3K_2SO_4 + 2KOH$

No. of sulphur containing products is 3 + 3 = 6

**50.** [2992]

$\underset{\substack{1584g \\ =12mol}}{(NH_4)_2SO_4} + Ca(OH)_2 \rightarrow \underset{\substack{gypsum(M=172) \\ 12mol}}{CaSO_4.2H_2O} + \underset{24mole}{2NH_3}$

$\underset{952g=4mol}{NiCl_2.6H_2O} + \underset{24mol}{6NH_3} \rightarrow \underset{\substack{(M=232) \\ 4mol}}{[Ni(NH_3)_6]Cl_2} + 6H_2O$

Total mass = $12 \times 172 + 4 \times 232 = 2992$ g

**51.** [2.98 or 2.99]

$X_{urea} = 0.05 = \frac{n}{n+50}$

$19n = 50$

$n = 2.6315$

$V_{sol} = \frac{(2.6315 \times 60 + 900)}{1.2} = 881.5789$ ml

Molarity = $\frac{2.6315 \times 1000}{881.5789} = 2.9849$

Molarity = 2.98M

**52.** [6.15]

$2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$

Moles of Al takes = $\frac{5.4}{27} = 0.2$

moles of $H_2SO_4$ taken = $\frac{50 \times 5.0}{1000} = 0.25$

As $\frac{0.2}{2} > \frac{0.25}{3}$, $H_2SO_4$ is limiting reagent

Now, moles of $H_2$ formed = $\frac{3}{3} \times 0.25 = 0.25$

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$\therefore$ Volume of $H_2$ gas formed $= \frac{nRT}{P}$

$= \frac{0.25 \times 0.082 \times 300}{1} = 6.15L$

**53.** (*b,c*)

(*a*) Compound 3 if compound is

$P_3O_4 \Rightarrow \frac{3P}{4Q} = \frac{40}{60} = \frac{2}{3}$

$\frac{P}{Q} = \frac{8}{9}$

Compound 2

$P_3O_5 \Rightarrow \frac{3P}{5Q} = \frac{44.8}{55.6}$

$\frac{P}{Q} = \frac{5}{3}0.8$

$\frac{4}{3} = \frac{8}{6}$

Option '*a*' is wrong

(*b*) Compound 3

$P\frac{40}{20} = 2 \quad 1.5 \quad 3$

formula = $P_3Q_2$

$Q\frac{60}{45} = \frac{4}{3} \quad 1 \quad 2$

(*c*) Compound 2 $\Rightarrow \frac{P}{Q} = \frac{44.4}{55.6} = 0.8$

Compound 1 $\Rightarrow \frac{5P}{4Q} = \frac{5}{4} \Rightarrow \frac{P}{Q} = 1$

(*d*) Compound 1

$P\frac{50}{70}1$

$Q\frac{50}{35}2$

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CHAPTER

# 2 Atomic Structure

## Exercise-1 (Topicwise)

**1.** (*a*) Volume fraction = $\frac{\text{Volume of nucleus}}{\text{Total volume of atom}}$

$$= \frac{\left(\frac{4}{3}\right)\pi\left(10^{-13}\right)^3}{\left(\frac{4}{3}\right)\pi\left(10^{-8}\right)^3}$$

$= 10^{-15}$

**2.** (*c*) $r = RoA^{1/3}$

r = radius of nucleus

Ro = Constant = 1.25

A = Atomic mass of element

$\therefore r = 1.25 \times (64)^{1/3} = 5\text{fm}$

**3.** (*b*) Since the charge and mass of an electron is fixed, the ratio of e/m for a cathode ray is fixed.

**4.** (*d*) Cathode rays have both charge and mass.

**5.** (*c*) Proton is ionized hydrogen atom i.e. it contains only protons and no neutrons or electrons.

**6.** (*d*) Rutherford observed that the positively charged α-particles were repelled and deflected by the positive charge in the atom. Rutherford named this positively charged portion of atom as nucleus.

**7.** (*c*) Most of the space in an atom is empty.

**8.** (*d*) e/m for neutron = $\frac{0}{1} = 0\,v$

e/m for alpha particles = $\frac{2}{4} = 0.5$

e/m for proton = $\frac{1}{1} = 1$

e/m for electron = $\frac{1}{\left(\frac{1}{1850}\right)} = 1850$

**9.** (*a*) Nucleons are protons and neutrons.

**10.** (*d*) As neutrons are neutral particles, the scientists were unable to observe neutrons.

**11.** (*d*) Atomic number represents the number of protons is an atom.

**12.** (*c*) e/m for nitrogen = $\frac{14}{28} = 0.5$

e/m for oxygen = $\frac{16}{32} = 0.5$

e/m for hydrogen = $\frac{2}{2} = 1$

e/m for helium = $\frac{2}{4} = 0.5$

**13.** (*a*) $\nu = \frac{c}{\lambda} = \frac{3\times10^8}{\lambda}$

$\lambda = \frac{c}{\nu} = \frac{3\times10^8}{400\times10^6} = 0.75$

**14.** (*a*) I.E. of one mole Na atom = $\frac{hc \times Na}{\lambda}$

$$= \frac{6.6\times10^{-34}\times3\times10^8\times6\times10^{23}}{242\times10^{-9}}$$

$= 494.65$ kJ/mol

**15.** (*d*) Energy emitted by a bulb = Power × Time

$= 40 \times 20 = 800$ J

Energy of photous emitted by a bulb = $\frac{80}{100}\times8000$

$= 640$ J

Energy of photon = $\frac{hc}{\lambda} = \frac{12400\times1.6\times10^{-19}\times10^{-10}}{620\times10^{-9}}$

$= 2 \times 10^{21}$ photons

**16.** (*c*) $h\nu = h\nu_0 + \text{K.E.}$

K.E. varies from O to any positive number.

Hence, $h\nu \geq h\nu_0$

$\frac{hc}{\lambda} \geq \frac{hc}{\lambda_0}$

$\Rightarrow \lambda \leq \lambda_0$

**17.** (*c*) Absorbed energy = $h\nu = 6.626 \times 10^{-34} \times 6 \times 10^{14}$

$= 2.49$ eV

$2.49 = 2 + \text{KE}$

$\Rightarrow \text{KE} = 0.49$ eV

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**18.** (*c*) $E = \frac{nhc}{\lambda}$

$\Rightarrow n = \frac{E\lambda}{hc} = \frac{600 \times 331.3 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^{8}}$

$n = 1 \times 10^{21}$

**19.** (*d*) Photoelectric effect can be explained by assuming that light consists of quanta.

**20.** (*c*) $h\nu_1 = h\nu_0 + KE_1$ .....(i)

$2h\nu_1 = h\nu_0 + KE_2$ .....(ii)

Solving (i) and (ii), we get

$KE_2 = 2KE_1 + h\nu_0$

**21.** (*a*) Wavelength of 1st photon = $\lambda_1$

Wavelength of 2nd photon = $\lambda_2$

E(total) = $E_1 + E_2$ = Emitted Energy

$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$

$\frac{1}{300} = \frac{1}{400} + \frac{1}{\lambda_2}$

$\lambda_2 = 759$ nm

**22.** (*b*) For H-atom, $E_1 = -13.6 \times \frac{1^2}{1^2} = -13.6$ eV/atom

For $He^+$ ion, $E_2 = -13.6 \times \frac{(2)^2}{(2)^2} = -13.6$ eV/atom

For $Li^{2+}$ ion, $E_3 = -13.6 \times \frac{(3)^2}{(3)^2} = -13.6$ eV/atom

For $Be^{3+}$ ion, $E_4 = -13.6 \times \frac{(4)^2}{(4)^2} = -13.6$ eV/atom

**23.** (*a*) Velocity = $\frac{2\pi Ze^2}{nh}$

$v \times \frac{Z}{n}$

Velocity of $H_2^{\oplus} \propto 1$

Velocity of $Li^{2+} \propto \frac{3}{3} = 1$

**24.** (*a*) radius $\propto \frac{n^2}{Z}$

$r_{n2} - r_{n1} = 24 \times r_0$

Since,

$n_2^2 - n_1^2 = 24$

$n_2 = 7, n_1 = 5$ or $n_2 = 5, n_1 = 1$.

**25.** (*b*) $S_1$: PE $\propto \frac{1}{r}$

$S_2$: KE $\propto \frac{1}{n^2}$

$S_3$: PE = 2 × T.E.

$PE = 2\left(-13.6 \times \frac{2^2}{n^2}\right)$

P.E. $\propto \frac{-1}{n^2}$

$S_4$: $E = -13.6 \times \frac{(2)^2}{(2)^2}$

$= -13.6$ eV

**26.** (*a*) $En = -R_H \times \frac{Z^2}{n^2}$

Energy of orbit becomes less negative as n increases.

**27.** (*b*) $r = \frac{0.0529 \times n^2}{Z}$

radius of 1st orbit of H-atom = $\frac{0.0529 \times 1^2}{3}$
= 0.0529 Å

radius of 1st orbit of $Li^{2+}$ ion = $\frac{0.0529 \times 1}{3}$

$= \frac{0.0529}{3}$ Å

**28.** (*b*) $L = \frac{nh}{2\pi} = \frac{5h}{2\pi} = \frac{2.5h}{\pi}$

**29.** (*a*) Amount of energy $\propto \frac{1}{n_1^2} - \frac{1}{n_2^2}$

The difference is maximum from $n_1 = 1$ to $n_2 = 2$

**30.** (*b*) $E_n = \frac{-313.6}{n^2}$ ....(i)

$E_n = -34.84$ ....(ii)

From (i) and (ii)

$-34.84 = \frac{-313.6}{n^2}$

$n^2 = \frac{313.6}{34.84} = 9$

$n = 3$

**31.** (*b*) $V \propto \frac{Z}{n}$ ; $V_1 \propto \frac{1}{1}$

In 3rd orbit, n = 3, Z = 1

$V_3 \propto \frac{1}{3}$

**32.** (*a*) $r = \frac{n^2h^2}{4\pi^2 mkze^2}$

Since, n = 1 and Z = 1 for both hydrogen and deuterium nucleus, the ratio of the radius is 1 : 1.

**33.** (*d*) Energy given to electron = 1.5×13.6 eV
= 20.4 eV

Excess energy = 20.4 – 13.6
= 6.8 eV

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34. (c) Bohr's theory is not valid for multielectron atom.

35. (c) Radius of $n^{th}$ orbit = $\frac{n^2 r_1}{Z}$

$r_1$ = radius of first orbit of H-atom

$r_2 = \frac{(2)^2 r_1}{1} = 4r_1$

$r_3 = \frac{(3)^2 r_1}{1} = 9r_1$

$r4 = \frac{(4)^2 r_1}{1} = 16r_1$

36. (c) Number of lines in the infrared region

= Total emitted lines - [Number of lines in visible region + Number of lines in U.V.region]

$n_1 = 6 \; ; \; n_2 = 1$

Total emitted lines = $\frac{(n_1 - n_2)(n_1 - n_2 + 1)}{2}$

= 15

All the lines falling onto n = 2 are in the visible bond, the number of lines of the visible region = 4

All the lines falling onto n = 1 are in the U.V region, the number of lines in the ultraviolet region = 5

∴ Number of lines in infrared region = 15 – 4 – 5

= 6

37. (b) Total number of lines in lyman series is given by n – 1

38. (d) Number of spectral lines = $\frac{n(n-1)}{2}$

$= \frac{5(5-1)}{2}$

= 10

39. (d) The transition 3 → 2 will correspond to red line as it has lowest energy.

40. (d) $\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Balmer : $n_1 = 2 \; ; \; n_2 = 3$

Paschen : $n_1 = 3 \; ; \; n_2 = \infty$

$V_B - V_P = R(3)^2\left[\frac{1}{4} - \frac{1}{9}\right] - R(3)^2\left[\frac{1}{9} - \frac{1}{\infty}\right]$

$= \frac{R}{4}$

41. (c) Lyman series corresponds to transition to n = 1 level.

42. (c) Paschen series corresponds to transition to n = 3 level.

43. (c) Series limit of Lyman series, n = 1

$\frac{1}{\lambda} = R\left(\frac{1}{n^2}\right) = R$

$\lambda = \frac{4}{R} = 4 \times 912\text{Å}$

44. (a) $\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{(\infty)^2}\right]$

$\frac{1}{\lambda} = 109678$

$\lambda = 911.7 \times 10^{-8}$ cm

45. (b) $\frac{1}{\lambda} = Z^2R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

In both the cases, transition states are same.

$\frac{1}{\lambda} \propto Z^2$

$\therefore \nu \propto Z^2$

$\Rightarrow \frac{\nu_1}{\nu_2} = \frac{Z_1^2}{Z_2^2}$

Let $\nu'$ be the frequency of line emitted by $He^+$

$\frac{\nu_0}{\nu'} = \left(\frac{1}{2}\right)^2$

$\nu' = 4\nu_0$

46. (c) Paschen series of hydrogen atom lies in the visible region of electromagnetic readiation.

47. (d) $\bar{\nu} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)\text{cm}^{-1}$

$\bar{\nu} = R_H\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$\bar{\nu} = \frac{5R_H}{36}\text{cm}^{-1}$

48. (d) $\frac{1}{\lambda} = R \times (1)^2\left[\frac{1}{(1)^2} - \frac{1}{(3)2}\right]$

$\frac{1}{\lambda} = \frac{8R}{9}$

$\lambda = \frac{9}{8R}$

49. (b) $\frac{r_1}{r_3} = \frac{(1)^2}{(3)^2}$

$r_3 = 9r_1 = 9x$

According to De Broglie;

$mvr_3 = \frac{3h}{2\pi} \; ; \; \frac{h}{mv} = \frac{2\pi r_3}{3}$

$\lambda = \frac{2\pi(9x)}{3}$

$6\pi x$

50. (b) For $Li^{2+}$ ion, $\frac{Z}{n} = \frac{3}{2}$

For $C^{5+}$ ion, $\frac{Z}{n} = \frac{6}{4} = \frac{3}{2}$

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51. (c) $\lambda = \frac{h}{p}$

52. (d) $\lambda = \frac{h}{m \times v}$

$m = \frac{h}{\lambda \times v}$

$m = \frac{6.62 \times 10^{-34}}{6.62 \times 10^{-35} \times 100}$

$m = 0.1$ kg

53. (a) $\lambda = \frac{h}{\sqrt{2mE}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}}$

$\frac{3}{5} = \sqrt{\frac{E_2}{E_1}}$

$\frac{E_1}{E_2} = \frac{25}{9}$

54. (d) $\Delta n = \frac{h}{4\pi \times \frac{1}{5} \times 0.01}$

$= 2.62 \times 10^{-32}$ m

55. (b) $\frac{r_1}{r_3} = \frac{(1)^2}{(3)^2}$

$r_3 = 9r_1 = 9$ times

According to De Broglies;

$mvr_3 = \frac{3h}{2\pi}$ ; $\frac{h}{mv} = \frac{2\pi r_3}{3}$

$\lambda = \frac{2\pi(9x)}{3}$

$6\pi$ times

56. (b) $\lambda = \frac{h}{mv}$

molecule with smallest mass will have maximum De broglie wavelength.

57. (a) $\sqrt{n(n+2)} = 24$

$\Rightarrow n = 4$ (No. of umpaired electrons)

$Z = 26$

$X^{n+} \rightarrow X = Fe, n = 2$

58. (b) $Fe^{2+} \rightarrow [Ar]3d^6 4s^0$

4 unpaired electorns

59. (b) The number of electrons with $l = 1$ is 12.

The number of electrons with $l = 2$ is 5

60. (b) $|L| = \sqrt{l(l+1)}\frac{h}{2\pi}$

$= \sqrt{0(0+0)}\frac{h}{2\pi} = 0$

61. (c) $Cl^- \rightarrow [Ne]3s^2 3p^6$

for 3p orbital, $l = 1$ and m is any one of -1, 0, 1.

62. (c) No. of radial nodes = $n - l - 1$

$1 = 3 - l - 1$

$l = 1$

$|L| = \sqrt{l(l+1)}\frac{h}{2\pi} = \sqrt{l(l+1)}\frac{h}{2\pi}$

$= \sqrt{2}\frac{h}{2\pi}$

63. (a) For 1st orbital, the distance at which maximum radial probability occurs is $r_0$.

64. (c) The maximum probability of finding an electron in the $d_{xy}$ orbital is at aan angle of 45° from the x and y-axis

65. (c) $3p_y$ orbital has ZX modal plane.

66. (c) Total no. of nodes = $n - l$

$= 3 - 1 = 2$

Radial nodes = $3 - l - 1 = 1$

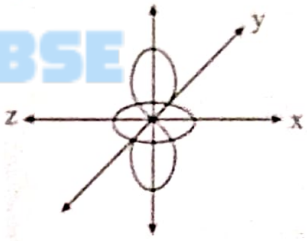
Angual nodes = $l = 1$

67. (b) According to schrodinger model, the nature of an electron in an atom is only as a wave.

68. (d) (A) $3d_{x^2-y^2}$ orbital has lobes in x and y axis.

(B) $3d_{z^2}$ is a mixed orbital.

69. (d) $d_{z^2}$ has dough nut shape.




70. (b) Spin quantum number was discovered before schrodinger even gave his equation.

## Exercise-2 (Learning Plus)

1. (a) $E = K.E + w$

$E = h\nu = \frac{hc}{\lambda}$

$\frac{hc}{\lambda} = 1 + w$ ...(i)

when light of wavelength $\frac{\lambda}{3}$ is incident

$\frac{hc}{\lambda/3} = 5 + w$ ...(ii)

$\frac{3hc}{\lambda} = 5 + w$

put the value of $\frac{hc}{\lambda}$ in equation (ii) we get

$3(1 + w) = 5 + w$

$2w = 2$

$\boxed{w = 1eV}$

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**2.** (*c*) $r_1 = R$

$r_2 = 4R$

we know radius $\propto n^2$

$\therefore r_2 = 4r_1 \Rightarrow n_2 = 2r_1$

$\boxed{v \propto \frac{1}{n}}$

$v_1 = 2v_2$

$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{2\pi r}{v}$

$\therefore \frac{T_1}{T_2} = \frac{2\pi r_1}{v_1} \times \frac{v_2}{2\pi r_2} = \frac{r_1 \times 0.5 v_1}{v_1 \times 4r_1} \Rightarrow \frac{1}{8}$

**3.** (*d*) $\text{P.E.} = \frac{-ze^2}{4\pi\varepsilon_o r}$

for Be; $z = 4$

$\text{P.E.} = -\frac{4e^2}{4\pi\varepsilon_o r} = -\frac{e^2}{\pi\varepsilon_o r}$

**4.** (*c*) Some wavelength means energy equal.

$\text{Energy} = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) z^2$ z for Li = 3, He = 2

$\left(\frac{1}{2^2} - \frac{1}{4^2}\right) \times 2^2 = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \times 3^2$

$\left(\frac{1}{4} - \frac{1}{16}\right) \times 4 = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \times 9$

$\left(\frac{4-1}{16}\right) \times 4 = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \times 9$

$\frac{3 \times 4}{16 \times 9} = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{12} = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$n_1 = 3, n_2 = 6$ then $\frac{1}{12} = \left(\frac{1}{9} - \frac{1}{36}\right)$

**5.** (*a*)

$hv_1 = 13.6\left(\frac{1}{1} - \frac{1}{\infty}\right)$

$v_1 = C_o \left(\text{where } C_o = \frac{13.6}{h}\right)$

Then,

$hv_2 = 13.6 \times \left(\frac{1}{1} - \frac{1}{2}\right)$

$v_2 = \frac{v_1}{2}$

Also, $hv_3 = 13.6$

$v_3 = \frac{v_1}{2}$

$\Rightarrow \boxed{v_2 + v_3 = v_1}$

$\boxed{v_1 - v_2 = v_3}$

**6.** (*b*) For shortest wove length of lymon series

$n_1 = 1, n_2 = \infty$

$\therefore \frac{1}{x} = R\left[\frac{1}{1^2} - \frac{1}{\infty}\right]$

So, $x = 1/R$ or $R = 1/x$

for first line of balmer

$n_1 = 2, n_2 = 3$

$\frac{1}{\lambda} = R(1^2)\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$

$\frac{1}{\lambda} = R\left[\frac{5}{36}\right] \qquad R = \frac{1}{x}$

$\lambda = \frac{36x}{5}$

**7.** (*a*) Balmer series corresponds to transition upto second energy level.

So, total no. of lines = 2

$4 \rightarrow 2$ and $3 \rightarrow 2$

Hence, $n = 4$

$\text{Energy} = \frac{-13.6}{n^2} = \frac{13.6}{4^2} = 0.85$

The kinetic energy of the ejected photelections

$= 13 - 0.85 = 12.15$ eV

**8.** (*b*) Acc. to question

$M_y = M_x \times \frac{25}{1000} = 0.25\, M_x$

and $V_y = V_x \times \frac{75}{1000} = 0.75\, V_x$

De-broglic $\lambda = \frac{h}{mV}$

$\lambda_x = 1\text{Å} = \frac{h}{M_x \times V_x}$

$\lambda_y = \frac{h}{0.25 M_x \times 0.75 V_x} = 5.33 \frac{h}{M_x \times V_x}$

$\lambda_y = 5.33\,\text{Å}$

**9.** (*b*) $\lambda = \frac{h}{mv}$

$\frac{\lambda_1}{\lambda_2} = \frac{m_2 v_2}{m_1 v_1}$

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$$\frac{v_1}{v_2}=\frac{m_2\lambda_2}{m_1\lambda_1}$$

$$\frac{v_1}{v_2}=\frac{4m_\alpha\lambda_\alpha}{m_p\lambda_p}$$

$$\frac{\lambda_p}{\lambda_\alpha}=\frac{1}{2}\Rightarrow\frac{\lambda_\alpha}{\lambda_p}=\frac{2}{1}$$

$$\Rightarrow\frac{\lambda_p}{\lambda_\alpha}=4\times2=8:1$$

**10.** (*d*) by de-broglie hypothsis $mvr=\frac{nh}{2\pi}$

$$2\pi r=n\left(\frac{h}{mv}\right)=n\lambda$$

**11.** (*c*) In case of $dx^2-y^2$ orbital electron density is lies on x and y axis.

So, probability of finding electron is maximum along x and y-axis.

**12.** (*b*) Magnetic moment = 2.83 = $\sqrt{n(n+2)}$

So no. of unpaired electron = 2 = n

no. of unpaired is in $Ni^{2+}$ = 2.

**13.** (*d*) The expresion for calculating the total number of electrons in any energy level is $\sum_{l=0}^{l=n-1}2(2l+1)$

**14.** (*d*) The spin quantum numbers +1/2 and –1/2 represents two quantum mechanical spin states which have no classical analogue.

**15.** (*a*) $\lambda=\frac{h}{\sqrt{2mK.E.}}$

Mass of electron is significantly lower than the mass of a proton or alpha particle K.E. order 16 E = electron, 4 E = Proton, E = alpha particle.

$\lambda_e>\lambda_p=\lambda_\alpha$

(mass of alpha particle is 4 times mass of proton)

**16.** (*c*) $r_1=R$

$r_2=4R$

we know radius $\propto n^2$

$\therefore r_2=4r_1\Rightarrow n_2=2r_1$

$$\boxed{v\propto\frac{1}{n}}$$

$v_1=2v_2$

$$\text{Time}=\frac{\text{Distance}}{\text{Speed}}=\frac{2\pi r}{v}$$

$$\therefore\frac{T_1}{T_2}=\frac{2\pi r_1}{v_1}\times\frac{v_2}{2\pi r_2}=\frac{r_1\times0.5v_1}{v_1\times4r_1}\Rightarrow\frac{1}{8}$$

**17.** (*c*) Total nodes = n – 1

Angular node = *l*

radial nodes = n – *l* – 1

for 3p, n = 3; *l* = 1

angular node = 1, radial node = 1

Total nodes = 2.

**18.** (*b*) angular momentum

$$L_n=\frac{nh}{2\pi}$$

$$\Rightarrow L_{n+1}-L_n=(n+1)\frac{h}{2\pi}-\frac{nh}{2\pi}$$

$$\Rightarrow\frac{nh}{2\pi}+\frac{h}{2\pi}-\frac{nh}{2\pi}$$

$$\frac{h}{2\pi}$$

**19.** (*b*) Potential digfernece = 6V

$$\frac{1}{2}mv^2=e=P.D$$

$$\therefore V=\sqrt{\frac{2e6}{m}}=\sqrt{\frac{2\times1.6\times10^{-19}\times6}{4\times1.67\times10^{-27}}}=2.4\times10^4$$

As de broglie wave length λ is given by

$$\lambda=\frac{h}{mv}$$

$$\Rightarrow\frac{6.6\times10^{-34}}{4\times1.67\times10^{-27}\times2.4\times10^4}$$

= 4.15 pm

**20.** (*c*) Energy of photon $\Delta E=2.55eV=2.55\times1.6\times10^{-19}J$

$$\Delta E=\frac{hc}{\lambda}\quad;\quad\lambda=\frac{hc}{\Delta E}$$

$$\lambda=\frac{6.6\times10^{-34}\times3\times10^8}{2.55\times1.6\times10^{-19}}=4861\text{Å}$$

Corresponds to the second line of balmer series $n_1=2$ to $n_2=4$

$$\text{So, }\Delta\lambda=\frac{h}{mv'}-\frac{h}{mv}\Rightarrow\frac{6.6\times10^{-34}}{9.1\times10^{-31}}\left(\frac{2}{2.19\times10}\right)$$

$\Delta\lambda=6.64$Å

**21.** (*d*) Principal quantum number (n)

n = 3

Now, magnetic quantum number (m) given the releative position of the electrons in orbitals.

m = –l to +l

l = |m|

l = |m| = 2 = l

**22.** (*b*) Ground state configuration of Boron = $1s^22s^22p^1$

Excited state configuration of Boron = $1s^22s^12p^2$

**23.** (*d*) Most probable radius = $\frac{a_o}{Z}$

$a_o=52.9$ z for helium = 2

$$r_{mp}=\frac{52.9}{2}=26.45\,\text{pm}$$

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$= \frac{429\times10^3}{6.023\times10^{23}} = 7.12\times10^{-19}$ J / molecule

$E_{photon} = \frac{hc}{\lambda}$

$= \frac{6.626\times10^{-34}\times3\times10^8}{270\times10^{-9}} = 7.36\times10^{-19}$ J

Energy converted into kinetic Energy = Energy left after bond dissociation

$\Rightarrow (7.36 - 7.12)\times10^{-19}$ J

$\Rightarrow 0.24\times10^{-19}$ J

% energy converted into K.E

$\Rightarrow \frac{0.24\times10^{-19}}{7.36\times10^{-19}}\times100 = 3.26\%$

**25.** (*d*) Orbital angular momentum

$= \sqrt{l(l+1)}\frac{h}{2\pi}$ ...(i)

$\Rightarrow \sqrt{5}\frac{h}{2\pi}$ ...(ii)

Comparing both

$\sqrt{l(l+1)}\frac{h}{2\pi} = \sqrt{5}\frac{h}{2\pi}$

$\sqrt{l(l+1)} = \sqrt{20}$

$l(l+1) = 20$

$l = 4$

g subshell

$\Rightarrow$ 9 orbitals possible.

**26.** (*a*) Ne = 10 es

$F^-$ = 10 $e^-$s , $Na^+$ = 10$e^-$s , $Mg^{2+}$ = 10$e^-$s , $Cl^-$ = 18$e^-$s

**27.** (*b*) Pauli's exclusion principle :- two electrons can't have the same values for all four grantum numbers.

**28.** (*c*) Total number of electrons in shell = $2n^2$

$\Rightarrow 2\times(3)^2 = 2\times9 = 18$

**29.** (*c*) $l$ = 0, 1, 2, 3,..... = n – 1 to 0

$l$ = |m|

4f orbital = n = 4 $l$ = 3

m = 3, 2, 1

**30.** (*c*) In any subshell, the maximum number of electrons having same value of spin quantum number is $2l + 1$

**31.** (*b*) For $l$ = 0 (s) = m = 0

For $l$ = 1 (p) = m = –1, 0, 1

For $l$ = 2 (d) = m = –2, –1, 0, 1, 2

For $l$ = 3 (f) = m = –3, –2, –1, 0, 1, 2, 3

total m value = 1 + 3 + 5 + 7 = 16

**32.** (*d*) The electronic configuration of an atom / ion can be defined by Aufbau's principle, Hund's rule of maximum multiplicity and pauli's exclusion principle.

lower n have lower

| | $n + l$ |
|---|---|
| 4s | 4 + 0 = 4 |
| 4p | 4 + 1 = 5 |
| 3d | 3 + 2 = 5 |

**34.** (*b*) Aufbau's principle: electrons are filled in order of energy level. If we compare two level ns have less energy So, ns will be filled

**35.** (*b*) Cu $\Rightarrow$ 29 electrons

$Cu^-$ $\Rightarrow$ 28 electrons

in K shell = $2\times1^2 = 2$

in L shell = $2\times2^2 = 8$

in M shell = $2\times3^2 = 18$

So in outer most shell 18 $e^-$s.

**36.** (*d*) A strong argument for the particle nature of cathode rays is they are deflected by electric and magnetic fields. The deflection through electric and magnetic fields proves that cathode rays are strems of minute particles having a negative charge.

**37.** (*b*) Let a,b,c be the relative abundances of three isotopes having mass numbers M(M + 1) and (M + 2) respectively

$\frac{aM + b(M+1) + c(M+2)}{a+b+c} = M + 0.5$

$aM + bM + b + cM + 2c = aM + bM + cM + 0.5a + 0.5b + 0.5c$

$b + 2c = 0.5a + 0.5b + 0.5c$

$0.5b + 1.5c = 0.5a$

$b + 3c = a$

This is true for the option (b)

$\Rightarrow a : b : c$

$\Rightarrow 4 : 1 : 1$

**38.** (*c*) The statement (*c*) is not correct. The energy of electron in an atomic orbitla of a multi-electron atom depends on the principle quantum number as well as azimuthal quantum number.

**39.** (*b*) At high pressure the mean free path of electron is low the atoms of gas are closely packed at high pressure when the gas pressure is low, the atoms are not closely spaced, the electrons get enough time to accelerate. As a result they collide with another atom, they gain sufficient energy from electric field to ionize the atoms. So gas conducts at low pressure.

**40.** (*c*) The uncertainity in the position of electron is

$\Delta x = \frac{h}{4\pi\Delta p}$

$= \frac{6.626\times10^{-34}}{4\times3.14\times1.0\times10^{-5}}$

$\Delta x = 5.27\times10^{-30}$ m

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# Exercise-3
# (JEE Advanced Level)

**1.** (*a,c*) $r_n = \frac{n^2}{z}\left(0.529\,\text{Å}\right)$, radius of first Bohr's orbits = 0.529 Å = $r_1$

For $He^{2+}$, z = 2, n = 2

$r_{He^{2+}} = \frac{4}{2}r_1$ or $r_{He^{2+}} = 2r_1$

For $r_{O^{7+}}$, z = 8, n = 4

$r_{O^{7+}} = r_1 \times \frac{16}{8}$ or $r_{O^{7+}} = 2r_1$

**2.** (*a,b,c* )

Maximum no. of electron in the subshell is given by = 2 (2*l* + 1)

**3.** (*a,b,c,d* )

Ψ is a mathematical function $|\Psi|^2$ denotes the probability density $R^2(r)_{max.}$ at nucleus.

**4.** (*b,c*)

Cathode rays when strikes to fluorescent material, produces fluorescence.

Cathode rays behaves as negatively charge in magnetic field or electric field.

**5.** (*a,b,c,d*)

Total no. of spherical nodes = n – *l* – 1

Total angular nodes = *l*

no. of maxima in an orbital = n – *l*

**6.** (*a,c,d*)

$E = \frac{-13.6}{n^2}$ eV (For excited state)

$n = 2, E_2 = \frac{-13.6}{4}\text{eV} = -3.4\text{eV}$

$n = 3, E_3 = \frac{-13.6}{9}\text{eV} = -1.51\text{eV}$

$n = 4, E_4 = \frac{-13.6}{16}\text{eV} = -0.85\text{eV}$

**7.** (*b,c,d*)

(*b*) We know, 2 × K.E = – P.E.

(T.E.) or E = K.E. + P.E or E = K.E – 2 K.E.

or E = – K.E.

∴ E : K.E. : P.E. = – K.E. : K.E. : –2 K.E.

= 1 : 1 : 2

(*c*) $\nu = 6 \times 10^{14}$ Hz.

$c = \nu\lambda$ or $\lambda = \frac{c}{\nu} = \frac{3\times10^8\,\text{m/s}}{6\times10^{14}\,\text{s}^{-1}}$

or λ = 500 nm.

(*d*) $\lambda = \frac{h}{\sqrt{2m\,K.E}}$, K.E. are same for all species.

$\lambda \propto \frac{1}{\sqrt{m}}$, m = molecular mass

$\lambda_H : \lambda_{He} : \lambda_{CH_4} = \frac{1}{\sqrt{1}} : \frac{1}{\sqrt{4}} : \frac{1}{\sqrt{16}} = 4 : 2 : 1$

**8.** (*a,b,c*)

We know, $\lambda \propto \frac{1}{p}$ and K.E. $\propto p^2$

or, K.E. $\propto \frac{1}{\lambda^2}$

$\therefore \frac{T_A}{T_B} = \frac{\lambda_B^2}{\lambda_A^2}$

Given, $\lambda_B = 2\lambda_A$

$\therefore \frac{T_A}{T_B} = \frac{2^2}{1^2} = 4$ or $T_A = 4\,T_B$

$T_B = T_A - 1.50$ eV or $T_B = 0.5$ eV

or $T_A = 2.00$ eV

Work function is given as for metal (A) is = 4.25 eV – $T_A$ or (4.25 – 2) eV or 2.25 eV

Similarly work function is given as for metal B = 4.20 eV – $T_B$ = (4.20 – 0.50) eV

= 3.70 eV

**9.** (*a,b,c*) Na (Z = 11) $1s^2\,2s^2\,2p^6\,3s^1$

**10.** (*a*) For node = $(12 - 8\sigma + \sigma^2) = 0$

or $(\sigma - 6)(\sigma - 2) = 0$

or σ = 6 and σ = 2

$\frac{2Zr}{na_0} = 6$ and $\frac{2Zr}{na_o} = 2$

$\underset{\text{Radial node}}{\frac{Zr}{na_0} = 3}$ $\underset{\text{Angular node}}{\frac{Zr}{na_o} = 1}$

**11.** (*b*) Radial node = n – *l* – 1 = 3 and Angular node = *l* = 1

∴ n = 5, *l* = 1 i.e., P- subshell.

Hence, 5p orbital.

**12.** (*c*) $p_Z$ - orbital

**13.** (*d*) Radial node occurs where probability of finding $e^-$ is zero

∴ $\Psi^2 = 0$ or $\Psi = 0$

$(1 - \sigma)(12 - 8\sigma + \sigma^2) = 0$

$(1 - \sigma)(\sigma - 6)(\sigma - 2) = 0$

or σ = 1, σ = 6, σ = 2,

For max. distance.

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$$\sigma = \frac{2Zr}{na_o} = 6 \text{ or } \frac{2Zr}{5a_o} = 6$$

$$\text{or } r = \frac{15a_o}{Z}$$

**14.** (*a*) $E_n = -13.6\frac{Z^2}{n^2}\text{eV}$ i.e., $E_n \propto Z^2, E_n \propto \frac{1}{n^2}$

and $E_n = \frac{hc}{\lambda}$ or $E_n = h\nu$ or $E_n \propto \nu$

$$T \propto \frac{1}{\nu}$$

**15.** (*c*) Values of m varies from $+l$ to $-l$ (including 0)

So, for $l = 2$, values of m have $-2, -1, 0, 1, 2$, i.e., 5

**16.** (*a*) $E_n = -13.6\frac{Z^2}{n^2}\text{eV}$

$$r_n = 0.529\frac{n^2}{Z}\text{A}^\circ$$

$$mvr = \frac{nh}{2\pi} \text{ or } v = \frac{nh}{2\pi mr}$$

$$\text{or } v = \frac{nh}{2\pi m \times 0.529\frac{n^2}{Z}\text{Å}} \text{ or } v \propto \frac{Z}{n}$$

**17.** (*b*) For Lyman's series = $n_h$ to $n_l = 1$, lies in UV region.

For Paschen series = $n_h$ to $n_l = 3$ lies in IR.

Last line of Balmer series lies in visible region.

**18.** (*c*) $mvr = \frac{nh}{2\pi} = \sqrt{\ell(\ell+1)}\ h$

Magnetic moment $= \sqrt{n(n+2)}$

spin angular momentum $= \sqrt{s(s+1)}\ h$

**19.** (*d*) For 3s orbital it has two nodes in (R vs. r graph)

Similarly for 4s it has 3

Similarly for 3p it has 1

**20.** (*a,c*) In a vacuum, the speed of all electromagnetic radiations is equal to the speed of light. Hence, option (*a*) is correct.

$He^+$ and H both are monoelectronic species so it will have similar spectrum as of hydrogen atom.

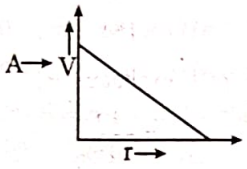
**21.** (*b,c,d*)

$V \alpha \frac{z}{n}$ $\quad\therefore V \alpha \frac{1}{n}$

$r \alpha \frac{n^2}{z}$ $\quad\therefore r \alpha n^2$

P. E. $\alpha -\frac{1}{r}$

K.E. $\alpha \frac{1}{r}$




**22.** (*c,d*) $E \propto \nu$ and $E_n \propto \frac{1}{n^2}$

Also, $\nu \propto \frac{1}{\lambda}$

**23.** (*a,c*)

Max. number of different photons emitted is

$4[(4 \to 3 \to 1 \text{ and } 4 \to 2 \to 1) \text{ or } (4 \to 3 \to 2 \to 1 \text{ and } 4 \to 1)]$.

Minimum number of different photons emitted is

$1(4 \to 1 \text{ and } 4 \to 1)$.

**24.** (*a,b,c*)

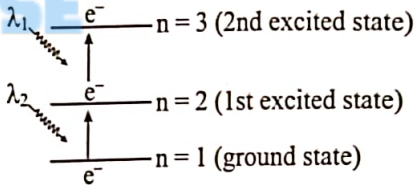
$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2mqV}}$$

When v, KE and V are same, as m increasing, decreases. $\lambda_e > \lambda_p > \lambda_\alpha$ (if v, KE and V are same).

**25.** (*b,c*)

If intensity or no. of photons falling per unit area increased then photocurrent will increase in surface area also causes increases in no. of photons.

**26.** (*b*) $\frac{1/\lambda_1}{1/\lambda_2} = \frac{R_H \times 1^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]}{R_H \times 1^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right]}$




$$\frac{\lambda_2}{\lambda_1} = \frac{1}{x} = \frac{5}{27}$$

$$x = \frac{27}{5}$$

As, $\lambda = \frac{h}{mc}$

for photons

$$\therefore \frac{(mc)_1}{(mc)_2} = \frac{h/\lambda_1}{h/\lambda_2}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{5}{27} = y$$

$$\frac{E_1}{E_2} = \frac{hc/\lambda_1}{hc/\lambda_2} = \frac{\lambda_2}{\lambda_1} = z = \frac{5}{27}$$

**27.** (*c,d*)

For s orbital $\theta$ & $\phi$ are zero $\quad\therefore$ independent

s-orbital spherical $\quad\therefore$ no angular component

2p, $n = 2$

$l = 2 \to$ angular dependence

$m = 0$

**28.** $(a,b,c)$

$n = 4, m = 2$

Value of $\ell = 0$ to $(n-1)$ but $m = 2$. $\therefore \ell = 2$ or 3 only

Value of s may be $+1/2$ or $-1/2$.

**29.** $(a,b,c)$

$(a)$ $_{24}Cr : [Ar]3d^5 4s^1$

$(b)$ $m = -\ell$ to $+\ell$ through zero.

$(c)$ $_{47}Ag : 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 4d^{10}$.

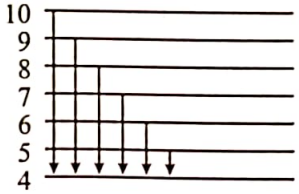
Since only one unpaired electron is present.

$\therefore$ 23 electrons have spin of one type and 24 of the opposite type.

**30.** [14] Radius $= 0.529 \dfrac{n^2}{Z}$ Å $\Rightarrow 10 \times 10^{-9}$ m $= 0.529\, n^2$ Å

So, $n^2 = 189$ or, $n \approx 14$

**31.** [6]




First Excited level = 2

$\therefore$ ninth level = 10

Total line = 6

**32.** [6] infrared lines = total lines – visible lines – UV lines

$$= \frac{6(6-1)}{2} - 4 - 5 = 15 - 9 = 6.$$

(visible lines = 4 $6 \to 2, 5 \to 2, 4 \to 2, 3 \to 2$)

(UV lines = 5 $6 \to 1, 5 \to 1, 4 \to 1, 3 \to 1, 2 \to 1$)

**33.** [5] For 3d →

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |

**34.** [26] $\sqrt{n(n+2)} = \sqrt{35}$

$\therefore n = 5$

$x^{3+} \to 4s^0\, 3d^5$

$x \to 4s^2\, 3d^6$

i.e., $_{26}Fe$

**35.** (4) $E = \dfrac{hc}{\lambda} = \dfrac{6.6\times10^{-34}\times3\times10^{8}}{300\times10^{-9}}$ J

or, E = 4.125 eV

Work function of the given metal must be less than eenrgy of incident photon. i.e., Li, Na, K, Mg.

**36.** (5) Let excited state of $He^+$ is $n_2$. Then according to question.

$$\frac{1}{\lambda_2} = R_H Z^2 \left[\frac{1}{1^2} - \frac{1}{n_1^2}\right]$$

Putting all values, we get

$n_1 = 2$

Now, for $\lambda_1$, $n_1 = 2$ and $n_2 = n$

$$\frac{1}{\lambda_1} = R_H Z^2 \left[\frac{1}{2^2} - \frac{1}{n^2}\right] \Rightarrow n = 5$$

**37.** [20] For closest approach distance

$$\frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r} = 3.6\times10^{6}\times1.6\times10^{-19}$$

$$\text{or, } \frac{9\times10^{9}\times\left(Z\times1.6\times10^{-19}\right)\times\left(2\times1.6\times10^{-19}\right)}{1.6\times10^{-14}} = 3.6\times10^{-13}$$

or, Z = 20

**38.** [3] Let, $\lambda_1$ is the wavelength of 1st line in Balmer series

$$\frac{1}{\lambda_1} = Z^2 R_H \left[\frac{1}{2^2} - \frac{1}{3^2}\right] \text{or} \lambda_1 = \frac{36}{5R_H Z^2}$$

Let, wavelength of 1st line in lyman series $\lambda_2$.

$$\frac{1}{\lambda_2} = Z^2 R_H \left(\frac{1}{1^2} - \frac{1}{2^2}\right) \text{ or } \lambda_2 = \frac{4}{3R_H Z^2}$$

According to question,

$$\lambda_1 - \lambda_2 = 59.3\times10^{-9}\,\text{m} = \left[\frac{36}{5} - \frac{4}{3}\right]\frac{1}{R_H Z^2}$$

or Z = 3

**39.** [4] $K.E._1 = E_{Photon} - B.E._{n=1}$

$K.E._2 = E_{Photon} - B.E._{n=n}$

$K.E._2 - K.E._1 = B.E._{n=1} - B.E._{n=n}$

$$= 13.6\, Z^2 \left[\frac{1}{1^2} - \frac{1}{n^2}\right] = 12.75 \text{ (given)}$$

$\therefore n^2 = 16$ or $n = 4$

**40.** $\left[\dfrac{5R}{9}\right] mvr = \dfrac{nh}{2\pi}$

$$3.1652\times10^{-34} = \frac{n\times6.6\times10^{-34}}{2\times3.14} \text{ or } n = 3$$

$$v = R_H q^2 \left[\frac{1}{2^2} - \frac{1}{3^2}\right] \Rightarrow \frac{5R}{9}$$

**41.** [1240] $E = \dfrac{hc}{\lambda} = \dfrac{6.6\times10^{-34}\times3\times10^{8}}{960\times10^{-10}} = 2.0625\times10^{-18}$ J

For one mol of photon

$E = 6.023 \times 10^{23} \times 2.0625 \times 10^{-21}$ kJ

= 1240 kJ/mol

**42.** [31] $w_o = 40$ eV or $w_o = 40 \times 1.6 \times 10^{-14}$J $= 6.4 \times 10^{-18}$J.

Given, v = 0 or K.E. = 0 and $\lambda = x$

$h\nu = w_o + K.E.$

or $\dfrac{hc}{\lambda} = w_o$

or $x = \dfrac{6.625\times10^{-34}\times3\times10^{8}}{6.4\times10^{-18}} = 31.05$ nm

**43.** [12]

1st ionization potential or B.E. of that nucleus i.e., from $n = 1$ to $n = \infty$.

$E_\infty - E_1 = 16$ V or $E_1 = -16$ eV

For 1st excitation i.e., $n = 1$ to $n = 2$.

$$E_2 - E_1 = -\frac{16}{2^2} - (-16) = 12\,\text{eV}$$

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**44.** [5] No. of spectral lines

$$\frac{(n_2-n_1)(n_2-n_1+1)}{2}=3$$

or $n_2^2-5n_2+6=6$

or $n_2(n_2-5)=0$ or $n_2=5$.

**45.** [12] K.E. of emitted $e^- = \frac{150}{(12.016)^2}=13.6\times\frac{11}{144}$

B.E. of 2nd E.S. of $H=\frac{13.6}{n^2}=\frac{13.6}{9}$

Total energy = B.E. + K.E.

$$13.6Z^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)=\frac{13.6}{9}+13.6\times\frac{11}{144}$$

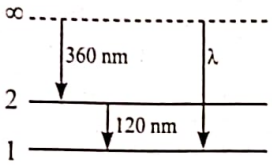
$$3^2\left[\frac{1}{6^2}-\frac{1}{n_2^2}\right]=\frac{1}{9}+\frac{11}{144} \text{ or } n_2=12$$

**46.** [–435.2 eV] $T.E.=\frac{-kZe^2}{r}$ or $r=\frac{n^2h^2}{4\pi^2mkZe^2}$

$$\text{or } T.E.=-\frac{kZe^2}{\frac{n^2h^2}{4\pi^2mkZe^3}} \text{ or } \frac{T.E._2}{T.E.}=\frac{m_2e_2^4}{m_1e_2^4}=2^5=32$$

$T.E._2 = -13.6 \times 32$ eV $= -435.2$ eV

**47.** [90 nm] $\frac{1}{\lambda}=\frac{1}{120}+\frac{1}{360}\Rightarrow\lambda=90\text{nm}$




**48.** [91 nm] $E=\frac{hc}{\lambda}$ or $\lambda=\frac{hc}{E}$

$$\lambda=\frac{6.626\times10^{-34}\times3\times10^8}{2.17\times10^{-18}}=91\text{nm}$$

**49.** [973.5 Å] $\Delta E=13.6\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]=\frac{hc}{\lambda}$

$n_1=4, n_2=1$ putting this values in above equation

$\lambda = 973.5$Å

**50.** [$5 \times 10^{-26}$] $\Delta x\cdot\Delta v\geq\frac{h}{4\pi m}$ or $\Delta x\geq\frac{h}{4\pi m\Delta v}$

$$\Delta x\geq\frac{6.626\times10^{-34}}{4\pi\times10^{-3}\times\frac{3.13}{\pi}\times10^{-3}} \text{ or } \Delta x\geq5\times10^{-26}\text{ m}$$

**51.** [$3.63 \times 10^6\ m^{-1}$] Energy of accelerated electron

$= 1.6 \times 10^{-19} \times 4.5$ or $7.2 \times 10^{-19}$ J

The above energy completely converted in light.

$$\frac{hc}{\lambda}=7.2\times10^{-19}$$

$$\bar{\nu}=\frac{1}{\lambda}=\frac{7.2\times10^{-19}}{6.6\times10^{-34}\times3\times10^8}=3.63\times10^6\text{ m}^{-1}$$

**52.** [300303] General equation of $\Psi(r)$

$$=4pk_1e^{-r/k_2}\times r^{\ell}\left(\text{Polynomial}\right)^{n-\ell-1}$$

On comparing

$n-l-1=2$ , $l=0$

$\therefore n=3$ i.e., 3s-orbital

$\therefore A=3, B=0, C=0$

For D ⇒ 8s to 8p there are 3 orbitals which are 6d, 5f, 7p

For F = 3 (on solving) $r^2-5r+6=0$.

## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*d*) De –Broglie equation

$$\lambda=\frac{h}{mv} \quad ...(i)$$

$$\because KE=\frac{1}{2}mv^2$$

$$\therefore mv=\sqrt{2m.KE}$$

$KE = eV$

$mv=\sqrt{2meV}$

Put in equation (i)

$$\frac{h}{\lambda}=\sqrt{2meV}$$

**2.** (*d*) Radius of $n^{th}$ Bohr orbit in H-atom $= 0.53\ n^2$ Å

Radius of IInd Bohr orbit $= 0.53 \times (2)^2 = 2.12$ Å

**3.** (*d*) 2s feels maximum attraction from the nucleus due to more $Z_{eff}$

**4.** (*c*) (A) $mvr=\frac{nh}{2\pi}$

As n increases, angular momentum increases.

(C) Angular momentum $=\frac{nh}{2\pi}$

**5.** (*d*) $\lambda=\frac{h}{mv}$

According to Einstein's theory of photoelectric effect:

$hv = hv_0 + KE$

$$hv=hv_0+\frac{1}{2}mv^2$$

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$$2h(v - v_0) = mv^2$$

$$\frac{2h(v-v_0)}{m} = v^2$$

$$v \propto (v - v_0)^{1/2}$$

$$\lambda \propto \frac{h}{m(v-v_0)^{1/2}}$$

$$\lambda \propto \frac{1}{(v-v_0)^{1/2}}$$

**6.** (*a*) [i] 900 nm = 9000 Å

It is in far infra red region hence paschen.

**7.** (*d*) For emission line

$n_f < n_i$

$$\therefore \bar{v} = RZ^2\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right] = R\left[\frac{1}{8^2} - \frac{1}{n^2}\right]$$

$$\text{or, } \bar{v} = R_H\left(\frac{1}{64} - \frac{1}{n^2}\right)$$

$$= \frac{R_H}{64} - \frac{R_H}{n^2}$$

$$\bar{v} = -R_H\left(\frac{1}{n^2}\right) + \frac{R_H}{64}$$

$\therefore\ y = mx + c$

Slope $= -R_H$

**8.** (*c*) The isotopes are:

$${}_1^1H, {}_1^2H \text{ and } {}_1^3H \equiv P, D, T$$

**9.** (*c*) $h\nu = \phi + h\nu^\circ$

$$\frac{1}{2}mv^2 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$$\phi = \frac{6.626\times10^{-34}\times3\times10^8}{4000\times10^{-10}} - \frac{1}{2}\times9\times10^{-31}\times(6\times10^5)^2$$

$$\phi = 3.35\times10^{-19}\,J \Rightarrow \phi \simeq 2.1\ eV$$

**10.** (*c*) The electron configuration is [Xe] $4f^{14}\ 5d^1\ 6s^2$

**11.** (*c*) $$E = -13.6\frac{Z^2}{n^2}eV$$

$$E_{He}^{+} = -13.6\frac{4}{9} = -6.04$$

**12.** (*d*) $2\pi r = n\lambda$

$$\lambda = \frac{2\pi r}{n} = \frac{2\pi n^2 a_0}{n\times Z} = 2\pi\frac{n}{Z}a_0$$

$$\lambda = 1.5\ \pi a_0$$

$$\therefore\ 2\pi\frac{n}{Z}a_0 = 1.5\pi a_0$$

$$\therefore\ \frac{n}{Z} = \frac{1.5}{2} = 0.75$$

**13.** (*a*) 3 and 1

**14.** (*c*) K.E = $h\nu - h\nu_0$, so (*c*) is not correct.

**15.** (*b*) According to $(n + \ell)$ rule : $3p < 3d < 4p < 4d$
Correct option : (*b*)

**16.** (*b*) For isoelectronic species the size is compared by nuclear charge.
Correct option: (*b*)

**17.** (*b*) K = 2, 8, 8, 1

After removal of one electron, second electron we have to remove from another shell, hence there is large difference between first and second ionization energies.

**18.** (*d*) For Lyman

$$\bar{v}_{max} = R_H\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = R_H$$

$$\bar{v}_{min} = R_H\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3}{4}R_H$$

$$\Delta\bar{v}_{Lyman} = \frac{R_H}{4}$$

For Balmer

$$\bar{v}_{max} = R_H\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R_H}{4}$$

$$\bar{v}_{min} = R_H\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5}{36}R_H$$

$$\Delta\bar{v}_{Balmer} = \frac{R_H}{4} - \frac{5R_H}{36} = \frac{4R_H}{36} = \frac{R_H}{9}$$

$$\frac{\Delta\bar{v}_{Lyman}}{\Delta\bar{v}_{Balmer}} = \frac{R_{H/4}}{R_{H/9}} = \frac{9}{4}$$

**19.** (*d*) $h\nu - \phi = KE$

$$\Rightarrow \left(\frac{hc}{\lambda}\right)_{incident} = KE + \phi$$

$$\left(\frac{hc}{\lambda}\right)_{incident} \simeq KE$$

$$KE = \frac{p^2}{2m} = \frac{hc}{\lambda_{incident}} = \frac{hc}{\lambda} \qquad ...(i)$$

$$\Rightarrow \frac{p^2\times(1.5)^2}{2m} = \frac{hc}{\lambda'} \qquad ...(ii)$$

divide (i) and (ii)

$$(1.5)^2 = \frac{\lambda}{\lambda'}$$

$$\Rightarrow \lambda' = \frac{4\lambda}{9}$$

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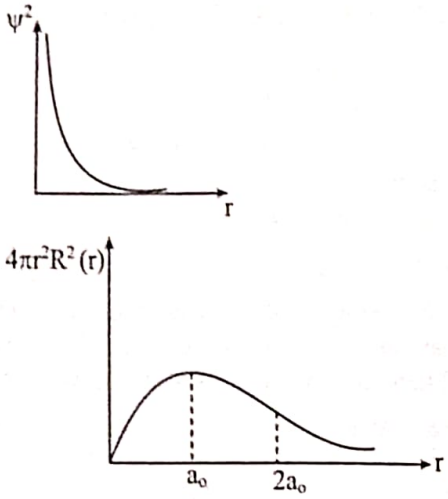
**20.** (*d*)




**21.** (*b*) $P(x) = 4\pi x^2 \times [\psi x]^2$

Probability will be maximum at A and C.

**22.** (*d*) $N^{-3} \rightarrow 1s^2\ 2s^2\ 2p^6$

$Li^{+} \rightarrow 1s^2$

$Mg^{+2} \rightarrow 1s^2\ 2s^2\ 2p^6$

$O^{-2} \rightarrow 1s^2\ 2s^2\ 2p^6$

$F^{-} \rightarrow 1s^2\ 2s^2\ 2p^6$

$Na^{+} \rightarrow 1s^2\ 2s^2\ 2p^6$

$N^{-3}$, $O^{-2}$, $F^{-}$ and $Na^{+}$ are isoelectronic

**23.** (*d*) Graph of $|\psi|^2$ v/s r touches r axis at 1 point so it has one radial node and since at r = 0, it has some value so it should be for 's' orbital.

$\therefore\ n - \ell - 1 = 1$ where $\ell = 0 \Rightarrow n - 1 = 1$

$\therefore\ n = 2 \Rightarrow$ '2s' orbital

**24.** (*b*)

$$\frac{\frac{1}{\lambda_2} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)Z^2}{\frac{1}{\lambda_2} = R_n\left(\frac{1}{m_1^2} - \frac{1}{m_2^2}\right)Z^2}$$

as for shortest wavelengths both $n_2$ and $m_2$ are $\infty$

$$\therefore\ \frac{\lambda_1}{\lambda_2} = \frac{9}{1} = \frac{m_1^2}{n_1^2}$$

Now if $m_1 = 3$ & $n_1 = 1$ it will justify the statement hence Lyman and Paschen (*b*) is correct.

**25.** (*d*) Balmer series lies in visible region.

**26.** (*b*) No. of orbitals = 1 + 3 + 5 + 7 + 9 = 25

**27.** (*b*) I. As wavelength decreases, the spacing between the lines ($\Delta\lambda$) becomes smaller and smaller.

II. For balmer series, $n_1 = 2$

III. $n_2 = 3$ to $n_1 = 1$ will produce the longest wavelength.

**28.** (*c*)

| | $^{1}_{1}H$ | | $^{2}_{1}H(D)$ | | $^{3}_{1}H(T)$ | |
|---|---|---|---|---|---|---|
| Number of neutrons | 0 | + | 1 | + | 2 | = 3 |

**29.** (*c*) $r = \frac{a_0 n^2}{Z}$

For $Li^{2+}$ $r = \frac{a_0(2)^2}{3} = \frac{4a_0}{3}$

**30.** (*d*) $2\pi r = n\lambda$

$$2\pi \times \frac{n^2}{Z} a_0 = n\lambda$$

$$2\pi \times \frac{4^2}{1} a_0 = 4\lambda$$

$$\lambda = 8\pi a_0$$

**31.** (222)

w, work function of sodium metal = $4.41 \times 10^{-19}$ J

$\lambda$, wavelength of incident light = 300 nm = $3 \times 10^{-7}$ m

According to Photoelectric effect,

$$\frac{hc}{\lambda} = w + KE$$

$$\frac{6.63\times10^{-34}\times3\times10^{8}}{3\times10^{-7}} = 4.41 \times 10^{-19} + KE$$

$$6.63 \times 10^{-19} = 4.41 \times 10^{-19} + KE$$

$$KE = 2.22 \times 10^{-19}\ J = 222 \times 10^{-21}\ J$$

**32.** (*d*) Cs has least work function value because of le electronegativity thats why it is used in photoelectr cells.

**33.** (*a*) 2 subshells are associated with n = 4 and m = –2.

**34.** (*b*) Explanation of variation of internal energy of Ar wi temperature (Straight line and $U \propto T$) is not a direc manifestation of the quantum nature of atoms. While explanation of absorption spectrum, natur of emission of radiation from hot bodies (blac body radiation) and photoelectric effect are direc manifestation of the quantum nature of atoms.

**35.** (*d*) Under hypothetical situation, the value of *l* is greate than n which varies from 0 to n + 1

for n = 1, *l* = 0, 1, 2

n = 2, *l* = 0, 1, 2, 3

Atomic number (Z) = 9

$1s^2\ 1p^6\ 1d^1$

Atomic number 6

$1s^2\ 1p^4$

Atomic number 8

$1s^2 1p^6$

Atomic number 13

$1s^2\ 1p^6\ 1d^5$

Here atomic number of first noble gas will be 18.

**36.** [142]

$Pt|H_2\ (1\ bar)|HCl(pH = 1)|AgCl|Ag$

$$\frac{1}{2}H_2 + AgCl \rightarrow H^{+} + Ag + Cl^{-}$$

$$E_{cell} = E^0_{cell} - \frac{0.06}{1}\log[H^+][Cl^-]$$

$= 0.22 - 0.06 \log 10^{-2} = 0.34$ V

Work function of Na metal = 2.3 eV

KE of photoelectron = 0.34 eV

Energy of incident radiation = 2.3 + 0.34 = 2.64 eV

Also energy of incident radiation for K metal = 2.64 eV

Work function of K metal = 2.25 eV

KE of photoelectrons = 2.64 – 2.25 = 0.39 eV

$\therefore E_{cell} = 0.39\ V = 0.22 - 0.06 \log [H^+]^2$

$= 0.22 + 0.12 \times pH$

$pH \approx 1.42 = 142\ 10^{-2}$

**37.** (*d*) Shortest wavelength → Max energy ($\infty \rightarrow 1$)

(Lyman series)

$$\frac{1}{\lambda_1} = R_H(1)^2\left[\frac{1}{1} - 0\right]$$

$$\frac{1}{\lambda_1} = R_H \Rightarrow R_H = \frac{1}{\lambda_1}$$

For Balmer series,

$$\frac{1}{\lambda_1} = R_H(2)^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$\Rightarrow \frac{1}{\lambda_1} = R_H(4)\left(\frac{9-4}{36}\right)$$

$$\frac{1}{\lambda_1} = \frac{5R_H}{9} \Rightarrow \lambda = \frac{9}{5R_H} = \frac{9\lambda_1}{5}$$

**38.** (*b*) $\psi^2$ (probability density) can be zero for 3p orbital other than infinite distance. It has one radial node.

**39.** (*c*) $r_n = a_0\frac{n^2}{Z}$

$$r_n \propto \frac{n^2}{Z}$$

$$\therefore \frac{\Delta R_1}{\Delta R_2} = \frac{Z_{He^+}}{Z_{Li^{2+}}} = \frac{2}{3}$$

**40.** (*b*) $Gd^{3+}$ (Z = 64) = [Xe] $4f^7$

$m = \sqrt{n(n+2)} = \sqrt{7(7+2)} = 7.9$ B.M.

**41.** (*a*) Filling of electrons in orbitals in any period takes place as: $ns, \underset{\text{if possible}}{(n-2)f}\ \underset{\text{if possible}}{(n-1)d}\ np$

$\therefore$ for sixth period n = 6,

orbitals that are filled are 6s, 4f, 5d and 6p

**42.** [2]

Energy of photon is given as $E = \frac{Nh}{\lambda}$ ...(*i*)

where, E = energy of photon (50W),

n = number of photon

h = Planck's constant ($6.63\times10^{-34}$ Js)

c = speed of light ($3\times10^8$ m/s)

$\lambda$ = wavelength of light ($795\times10^{-9}$ m)

E = 50W = 50J = energy of photon

$$50\,J = \frac{n\times6.63\times10^{-3}\,J_{s\times3}\times10^{-}m/s}{795\times10^{-9}\,m}$$

$$\Rightarrow n = \frac{50\times195\times10^{-9}}{6.63\times10^{-\frac{3}{3}}\times3\times10^3}$$

$= 1998.49\times10^{17} = 1.998\times10^{20}$

$\approx 2 \times 10^{20}$

$\therefore x = 2$

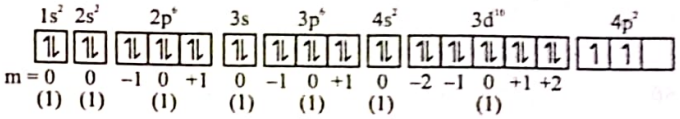
**43.** [0]

$Zn^+ \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$

Outermost electron is in 4 s subshell

m = 0

**44.** [7]




Completely filled orbital with $m_l = 0$ are

$= 1+1+1+1+1+1+1 = 7$

**45.** [50] Energy emitted in 0.1sec. $= 0.1\text{sec.}\times10^{-3}\frac{J}{s} = 10^{-4}$ J

If 'n' photons of $\lambda = 1000$nm are emitted, then

$$10^{-4} = n\times\frac{hc}{\lambda}$$

$$\Rightarrow 10^{-4} = \frac{n\times6.63\times10^{-34}\times3\times10^8}{1000\times10^{-9}}$$

$$\Rightarrow n = 5.02\times10^{14} = 50.2\times10^{13} \Rightarrow 50$$

**46.** [2] $\lambda = \frac{h}{\sqrt{2meV}}$

$$\frac{\lambda_{Li}}{\lambda_p} = \sqrt{\frac{m_p(e)(V)}{m_{Li}(3e)(V)}};\ m_{Li} = 8.3m_p$$

$$\frac{\lambda_{Li}}{\lambda_p} = \sqrt{\frac{1}{8.3\times3}} = \frac{1}{5} = 0.2 = 2\times10^{-1}$$

$\boxed{X = 2}$

**47.** (*b*) (I) KE = –TE

$$KE = 13.6 \times \frac{Z^2}{n^2}\ eV$$

$$KE \propto \frac{Z^2}{n^2}$$

(II) $V = 2.188 \times 10^6 \times \frac{Z}{n}$ m/s

$V_n \propto Z$

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(III) $F = \frac{V}{2\pi r}$

$F \propto \frac{Z^2}{n^2}\left[\because r \propto \frac{Z^2}{n^2} \text{ and } V \propto \frac{Z}{n}\right]$

(IV) $F \propto \frac{Z^2}{r^2} \propto \frac{Z^3}{n^4}$

**48.** [180]

Total energy required = $N_A \times \frac{hc}{\lambda}$

$= \frac{6.02\times10^{23}\times6.63\times10^{-34}\times3\times10^{8}}{663\times10^{-9}}$

= 180.4 kJ/mol

**49.** (*d*) Angular nodes = $l = 2$

Radial nodes = $n - l - 1$

$= 5 - 2 - 1 = 2$

**50.** (*d*) Tritium is radioactive and emits low energy $\beta^-$ particles.

**51.** [1.17] $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

$\Delta x = \frac{h}{4\pi m \Delta V}$

$= \frac{6.63\times10^{-34}}{4\times3.14\times10\times10^{-3}\times90\times0.05}$

$= 1.17 \times 10^{-33}$ m

**52.** [9] $h\nu = h\nu_o + KE$

$= \frac{6.63\times10^{-34}\times3\times10^{8}}{248\times10^{-9}\times1.6\times10^{-19}} = 3.0 + KE$

KE = 2.0 eV

$\lambda = \frac{h}{\sqrt{2mKE}} = \frac{6.6\times10^{-34}}{2\times9.1\times10^{-31}\times2\times10^{-19}\times1.6}$

$= 8.68 \times 10^{-10}$ m = 9A°

**53.** [25] No. of orbitals = 1 + 3 + 5 + 7 + 9 = 25

**54.** [0] $n = 4, m_e = -3 \Rightarrow l = 3$

Radial nodes = $n - l - 1$

$= 4 - 3 - 1 = 0$

**55.** (*c*) $l = 0$

Radial nodes = $n - 0 - l = 2$

$\Rightarrow n = 3$

$\therefore$ 3s

**56.** [0] $Ga^+ \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

Valence electrons $\rightarrow 4s^2$

So, azimuthal quantum number, $l = 0$

**57.** [6] $\lambda = \frac{h}{\sqrt{2meV}}$

$= \frac{6.634\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1.6\times10^{-19}\times40\times10^{3}}}$

$= 6 \times 10^{-12}$

X = 6

**58.** [12] $V \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

Number of electrons in p-orbital = 12

**59.** [2] $E = \frac{nhc}{\lambda}$

$1000 = \frac{n\times6.626\times10^{-34}\times3\times10^{8}}{400\times10^{-3}}$

$n = 20.122 \times 10^{20}$ photons incident on metal surface in 10 seconds

$n = 2.0.122 \times 10^{20}$ photons incident on metal surface in 1 second

Number of electrons ejected = Number of incident photons

X = 2

**60.** [58] Uncertainity in speed of electron = $\frac{0.02\times5\times10^{6}}{100}$

$= 10^3$ ms$^{-1}$

$m \times \Delta v \times \Delta x = \frac{h}{4\pi}$

$\Delta x = \frac{6.6\times10^{-34}}{4\times3.14\times9.1\times10^{-31}\times10^{3}}$

$= 58 \times 10^{-9}$ m

x = 58

**61.** (*a*) Rutherford model doesn't say anything about the electronic structure of atom.

Bohr's model gave exact formula for simultaneous calculations of speed and distance of electron from the nucleus, which was stated impossible by Heisenberg.

**62.** (*a*) As in Thomson model, protons are diffused, X-particles deviate by small angles and due to repulsion from protons, their speed decreases.

**63.** (*b*) $V = 2.188 \times 10^6 \times \frac{Z}{n}$ m/s

$V \propto \frac{Z}{n}$

**64.** [5] $\frac{hc}{\lambda} = h\nu_o + \frac{1}{2}m_e V^2$

$\frac{6.63\times10^{-34}\times3\times10^{8}}{500\times10^{-9}}$

$= 6.63 \times 10^{-34} \times 4.3 \times 10^{14} + \frac{1}{2} \times 9 \times 10^{-31} \times V^2$

$V = 5 \times 10^5$ ms$^{-1}$

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**65.** [3155] $KE = \frac{1}{2} mv^2$ ...(i)

$mvr = \frac{nh}{2\pi}$

from both,

$(mv)^2 = \frac{n^2h^2}{4\pi^2r^2}$

$mv^2 = \frac{1}{m} \times \frac{n^2h^2}{4\pi^2r^2}$ ...(ii)

put eqn (ii) in (i),

$KE = \frac{1}{2} \times \frac{1}{m} \times \frac{n^2h^2}{4\pi^2r^2}$

Now, $n = 2$; $r_1 = a_o$; $r_2 = a_o(2)^2$; $r_2 = 4a_o$

$KE = \frac{1}{2} \times \frac{(2)^2h^2}{4\pi^2(4a_o)^2 \times m}$

$= \frac{h^2}{32\pi^2 \times ma_o^2} = \frac{h^2}{315.5 \times ma_o^2}$

**66.** [4] $Np \rightarrow [Rn]\ 5f^4 6d^1 7s^2$

$\therefore$ It has 4 electrons in the f-subshell of the anti penultimate shell.

**67.** [50] Power = 1mW

$= 10^{-3}$J in 1 second

$= 10^{-4}$J in 0.1 second

$\therefore$ Energy $= \frac{nhc}{\lambda}$

$10^{-4} = \frac{n \times 6.63 \times 10^{-34} \times 3 \times 10^8}{1000 \times 10^{-9}}$

$n = 50.2 \times 10^{13}$

$X = 50$

**68.** (*b*) $Eu \rightarrow [Xe]\ 4f^7 6s^2$

$Eu^{2+} \rightarrow [Xe]\ 4f^7$

**69.** (*d*) $Fe \rightarrow [Ar]\ 3d^6 4s^2$

$Fe^{3+} \rightarrow [Ar]\ 3d^5$

**70.** (*b*) According to $(n + l)$ rule.

**71.** (*b*) $r = 0.529 \times \frac{n^2}{Z} A°$

$r_3 = 0.529 \times \frac{(3)^2}{1}$

$r_4 = 0.529 \times \frac{(4)^2}{1}$

$\frac{r_4}{r_3} = \frac{(4)^2}{(3)^2} = \frac{16}{9}$

$r_4 = \frac{16r_3}{9}$

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**72.** (*c*) Values of $l$ lies between 0 to n – 1

**73.** [1758] $\lambda = \frac{h}{mv}$

$\lambda_e = \lambda_N$

$\frac{h}{m_e V_e} = \frac{h}{m_N V_N}$

$V_e = \frac{m_N}{m_e} . V_N$

$V_e = \frac{1.6 \times 10^{-27}}{9.1 \times 10^{-31}} V_N$

$\therefore X = 1758.24$

**74.** [4] No. of neutrous = 26

No. of electrons = 25

% of extra neutrons than electrons $= \left(\frac{26-25}{25}\right) \times 100$

$= 4$

**75.** [22] According to Heisenberg uncertainty principle,

$\Delta x \times \Delta p = \frac{h}{4\pi}$

$\Rightarrow 10^{-7} \times m . \Delta V = \frac{6.62 \times 10^{-34}}{4 \times 3.14}$

$\Rightarrow 10^{-7} \times m \times 2.4 \times 10^{-24} = \frac{6.62 \times 10^{-34}}{4 \times 3.14}$

$m = 0.2196 \times 10^{-3}$ Kg

$m = 21.96 \times 10^{-5}$ Kg

**76.** (*a*) The orbitals with same n and $l$ value but with different m value are degenerate.

**77.** (*c*) $mv\,(4a_0) = \frac{h}{\pi}$

$so, v = \frac{h}{4m\pi a_0}$

$so, KE = \frac{1}{2} mv^2 = \frac{1}{2} m . \frac{h^2}{16m^2\pi^2 a_0^2} = \frac{h^2}{32m\pi^2 a_0^2}$

**78.** [8] $^{63}_{29}Cu + ^1_1H \rightarrow 6^1_0n + ^4_2\alpha + 2^1_1H + X$

$64 = 6 + 4 + 2 + A \Rightarrow A = 52$

$29 + 1 = 30 = 0 + 2 + 2 + z \Rightarrow z = 26$

Element X should be iron in group 8.

**79.** (*a,b*)

$_4Be^9 + X \longrightarrow {}_4Be^8 + Y$

If X is $_0\gamma^0$ then Y is $_0n^1$

If X is $_1P^1$ then Y is $_1D^2$

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**80.** [6] $n = 4, \quad m_\ell = 1, -1$

Hence $\ell$ can be = 3,2,1

i.e. $H_f$ ; 2 orbitals

$H_d$ ; 2 orbitals

$H_p$ ; 2 orbitals

Hence total of 6 orbitals, and we want $m_s = -\frac{1}{2}$, that is only one kind of spin. So, 6 electrons.

**81.** [3] Energy order of orbitals of H is decided by only principle quantum number (n) while energy order of $H^-$ is decided by $(n + \ell)$ rule:

Electronic configuration of '$H^-$' is - $1s^2$ its Energy order is decided by $n+\ell$ rule.

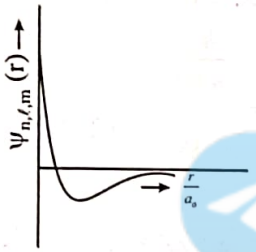
$H^- = 1s^2 2s^0 2p^0$

Its 2nd excited state is 2p and degeneracy of 2p is '3'

**82.** (*d*) The qualitative sketch of the dependence of P on r is the radial probability curve.

**83.** (*c*) s-orbital is non directional so wave function will be independent of cos θ.

**84.** (*a*) For 2s orbital, no. of radial nodes = $n - \ell - 1 = 1$




**85.** (*d*) For 1s orbital, Ψ should be independent of θ, also it does not contain any radial node.

**86.** (*a,c*)

$-3.6 = \frac{-13.6 \times 4}{n^2}$

$n = 4$

$\ell = 2$

$m = 0$

Angular nodes = $\ell = 2$

Radial nodes = $(n - \ell - 1) = 1$

$n\ell$ = 4d state

**87.** (*c*) $r = 0.529 \times \frac{n^2}{z} \Rightarrow r \propto n^2 \Rightarrow$ (I) (T)

$mvr = \frac{nh}{2\pi} \Rightarrow (mrv) \propto n \Rightarrow$ (II) (S)

$KE = +13.6 \times \frac{z^2}{n^2} \Rightarrow KE \propto n^{-2} \Rightarrow$ (III) (P)

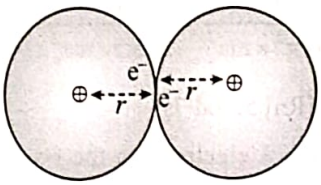
$PE = -2 \times 13.6 \times \frac{z^2}{n^2} \Rightarrow PE \propto n^{-2} \Rightarrow$ (IV) (P)

**88.** (*d*) Refer to Q 87.

**89.** [–5246.49]

At $d = d_0$, nucleus-nucleus & electron-electron repulsion is absent.

Hence potential energy will be calculated for 2 H atoms (P.E. due to attraction of proton & electron)




$$P.E. = \underset{\text{(Bohr radius)}}{\frac{-Kq_1q_2}{r}} = \frac{(9\times10^9)(1.6\times10^{-19})^2}{0.529\times10^{-10}} = -4.355\times10^{-21}\text{ kJ}$$

For 1 mol = $-4.355 \times 10^{-21} \times 6.023 \times 10^{23}$

$= -2623.249$ kJ/mol

For 2 H atoms = –5246.49 kJ/mol

**90.** [30] $\lambda = \frac{h}{m(\Delta v)}$

$$330\times10^{-9} = \frac{6.6\times10^{-34}}{\left(\frac{4\times10^{-3}}{6\times10^{23}}\right)\times\Delta v}$$

$$\Delta v = \frac{6.6\times6\times10^{23}\times10^{-34}}{4\times10^{-3}\times330\times10^{-9}} = 30 \text{ cm/s}$$

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CHAPTER

# 3 States of Matter

## Exercise-1 (Topicwise)

**1.** (*c*) Boyle's law is $V \propto \frac{1}{P}$ at constant *T*.

**2.** (*d*) $PV = nRT = K$ (Constant) (Boyle's law)

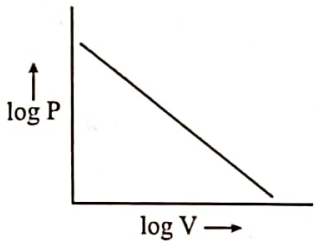
Taking log both side

$\log P + \log V = \log k$

$\log P = -\log V + \text{constant}$

$y = mx + c$

so




**3.** (*a*) At constant T, $P_1V_1 = P_2V_2$

$1 \times 20 = P_2 \times 50 \Rightarrow P_2 = 20/50$

**4.** (*a*) $P_1 = 5$ atm $V_1 = 10$ L

$P_2 = 1$ atm $V_2 = ?$

$P_1V_1 = P_2V_2$ or $\frac{P_1}{P_2} = \frac{V_2}{V_1}$

$\frac{5}{1} = \frac{V_2}{10}$

$V_2 = 50$ L

**5.** (*d*) According to Boyle's Law, $PV$ = constant, at constant temperature either P increases or V increases both (*b*) & (*c*) are correct.

**6.** (*c*) $V \propto T$

**7.** (*a*) $P = \frac{nRT}{V} = \frac{2 \times 0.0821 \times 546}{44.8l}$ atm = 2 atm

**8.** (*b*) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2}$

**9.** (*b*) Molecular weight = *V.d.* × 2 = 11.2 × 2 = 22.4

Volume of 22.4 *gm* Substance of NTP = 22.4 *litre*

1 *gm* substance at NTP = $\frac{22.4}{22.4}$ litre

11.2 *gm* substance of NTP = 11.2 *litre*

**10.** (*b*) $\frac{\text{M.wt. of } O_2}{\text{M.wt. of } SO_2} \Rightarrow \frac{M_1}{M_2} \Rightarrow \frac{32}{64} = \frac{1}{2}$

The weight of oxygen will be $\frac{1}{2}$ that of $SO_2$

**11.** (*d*) The value of universal gas constant would depend only on the units of measurement rather than temperature and pressure

**12.** (*b*) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\frac{P \times 5}{300} = \frac{2P \times V_2}{600}$

$\boxed{V_2 = 5L}$

**13.** (*a*) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\frac{1 \times 1}{273} = \frac{2 \times V_2}{300}$

$\boxed{V_2 = \frac{300}{2 \times 273} L}$

**14.** (*c*) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\frac{760 \times 273}{273} = \frac{600 \times V_2}{300}$

$\boxed{V_2 = 380\,mL}$

**15.** (*d*) Moles $= \frac{5.6}{22.4} = \frac{1}{4}$

$\frac{\text{Given mass}}{\text{molar mass}} = \frac{1}{4} \Rightarrow$ molar mass = 40g

$\eta = \frac{\text{moleculor mass}}{\text{empirical mass}} = \frac{40}{20} = 2$

molecular formula = $(HF)_2 = H_2F_2$

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**16.** (*b*) moles of $CO_2 = \frac{4400}{44} = 100$

$$100 = \frac{\text{volume}}{24.5\text{L}}$$

volume = $100 \times 24.5L$

**17.** (*c*) $P = \frac{dRT}{M}$

$d_1T_1 = d_2T_2$

$d \times 300 = 0.75d \times T$

$T = 400K$

**18.** (*c*) moles of $CO_2(n_1) = \frac{44}{44} = 1\text{mol}$

moles of $H_2(n_2) = \frac{44}{2} = 22\text{mol}$

$\frac{P_1}{n_1} = \frac{P_2}{n_2} \Rightarrow \frac{1}{1} = \frac{P_2}{22}$

$P_2 = 22\text{atm}$

**19.** (*d*) $P_{total} = P_1 + P_2 + P_3 + \ldots\ldots$

There should be non reacting gases, behave ideally and temperature of the component gases remain constant

**20.** (*c*) From Daltons partial pressure law

$$P_1 = \frac{1000}{1500} \times 600 + \frac{500}{1500} \times 800$$

$$= 400 + \frac{800}{3}$$

$$= \frac{2000}{3} \text{ torr.}$$

$V_1 = 1500$ mL

$P_2 = ?$

$V_2 = 2\text{L} = 2000$ mL

From $P \propto \frac{1}{V}$

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

$$\Rightarrow \frac{2000/3}{P_2} = \frac{2000}{1500}$$

$$\Rightarrow P_2 = \frac{1500}{3} = 500 \text{ torr}$$

**21.** (*c*) $N_2 + 3H_2 \rightarrow 2NH_3$

Initial 1 3

final – – 2

ratio = $\frac{4}{2} = \frac{2}{1}$

**22.** (*a*) Rate of effusion $\propto$ Area of cross section

$$\therefore \frac{r_A}{r_B} = \frac{A_A}{A_B} = \frac{\pi r^2}{r^2} = \frac{\pi}{1}$$

**23.** (*a*) $P = P_A + P_B + P_C$

**24.** (*d*) $r \propto \frac{1}{\sqrt{M}}$

**25.** (*b*) $\frac{d_1}{d_2} = \frac{1}{16}; \frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{16} = \frac{4}{1}$

**26.** (*b*) $NH_4Cl$ ring will first formed near the HCl bottle because rate of diffusion of $NH_3$ is more than that of HCl because $M_{NH_3} : M_{HCl} = 17 : 36.5$. SO $NH_3$ will reach first to the HCl bottle & will react there with HCl to form $NH_4Cl$ ring.

**27.** (*d*) $\frac{t_{(gas)}}{t_{(O_2)}} = \sqrt{\frac{M_{(gas)}}{M_{(O_2)}}}$

$$\Rightarrow \frac{40}{16} = \sqrt{\frac{M_{(gas)}}{32}}$$

$$\Rightarrow \frac{5}{2} = \sqrt{\frac{M_{(gas)}}{32}} \quad \Rightarrow \frac{25}{4} = \frac{M_{(gas)}}{32}$$

$\Rightarrow M_{(gas)} = 200$ g/mol.

**28.** (*a*) $\frac{\text{Rate}_1}{\text{Rate}_2} = \sqrt{\frac{(\text{molar mass})_2}{(\text{molar mass})_1}}$

$$\frac{50}{40} = \frac{\sqrt{(\text{molar mass})_2}}{8}$$

$\boxed{m_2 = 100}$

**29.** (*a*) $KE = \frac{3}{2}RT = \frac{3}{2}PV$

$\therefore P = \frac{2}{3}\frac{E}{V}$ for unit volume ($V = 1$) $\Rightarrow P = \frac{2}{3}E$

**30.** (*d*) The most accurate form in which velocity can be used in these calculations.

**31.** (*c*) $d \propto M \Rightarrow \frac{d_1}{d_2} = \frac{M_1}{M_2}; \frac{3d}{d} = \frac{M}{M_2} \Rightarrow M_2 = \frac{M}{3}$

**32.** (*a*) $V_{rms} = \sqrt{\frac{3RT}{M}}, V_{av} = \sqrt{\frac{8RT}{\pi M}}; \frac{V_{rms}}{V_{av}} = \sqrt{\frac{3\pi}{8}}$

$$= \sqrt{\frac{66}{56}} \Rightarrow \frac{1.086}{1}$$

**33.** (*d*) most probable velocity : mean velocity : $V_{rms}$

$$= \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}} = \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$$

**34.** (*a*) When average speed of molecule is increased due to increase in temperature then the change in momentum during collision between wall of container and molecules of gas also increases.

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35. (*d*) $\frac{U_1}{U_2}=\sqrt{\frac{m_2}{m_1}\cdot\frac{T_1}{T_2}}$ $\because T_1=T_2$

$\frac{U_1^2}{U_2^2}=\frac{m_2}{m_1}$ $\therefore m_1U_1^2=m_2U_2^2$

36. (*d*) Kinetic energy of the gas molecule remains the same.

$K.E=\frac{3}{2}kT$ $[K.E\propto T]$

37. (*a*) At constant volume, on increasing tmperature → Average speed of molecule increased due to which the pressure of gas also increase.

38. (*c*) The number of mole of the gas increases is the incorrect statment as mass doesn't change with temperature.

39. (*a*) Speed $=\sqrt{\frac{3RT}{M}}$

Gas which has maximum value of $\frac{T}{M}$ has highest speed.

40. (*a*) $Z<1$ shows that the gas has greater compressibility at intermediate pressure and all reactive forces are dominant, higher value of 'a' and lower value of 'b' above Boyle's temperature it is not possible because $Z>1$.

41. (*b*) Because molecules of real gases have intermolecular forces of attraction so the effective impact on the wall of container is diminished. Pressure of real gas is reduced by $\frac{a}{v^2}$ factor hence behaviour of real gas deviate from ideal behaviour.

42. (*c*) $Z=\frac{PV}{RT}$ $\because$ for ideal gas $PV=RT$ so $Z=1$

43. (*d*) $\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT$

44. (*b*) At low temperature and high pressure.

45. (*c*) Value of constant *a* is greater than other for $NH_3$ that's why $NH_3$ can be most easily liquefied.

46. (*b*) If $Z<1$ then molar volume is less than 22.4 L.

47. (*b*) Critical temperature is defined as the temperature above which a gas cannot be liquified.

48. (*a*) $\left(p+\frac{a}{v^2}\right)$ term represent the intermolecular force in non-ideal gas equation.

## Exercise-2 (Learning Plus)

1. (*c*) $\frac{V_1}{V_2}=\frac{T_1}{T_2};\frac{3}{2.7}=\frac{320}{T_2}\Rightarrow T_2=288K=15^\circ C$

2. (*d*) n,T → constant

PV = constant

3. (*d*) n → constant

v → fixed

$\therefore$ P & T→ constant

4. (*b*) V = 2 litre

$\frac{v_1}{v_2}=\frac{T_1}{T_2}$

$\Rightarrow \frac{2}{4}=\frac{273}{T(inK)}\Rightarrow T=546\ K\Rightarrow T=273^\circ C$

5. (*b*) n = constant

no of molecules = constant

same number of molecules

6. (*c*) $\frac{10}{V_2}=\frac{273}{373}\Rightarrow V_2=13.66$ L.

7. (*c*) $P=CRT;\ T=\frac{P}{RC}=\frac{1\times12}{1\times1}=12$ K.

8. (*c*)

A B

0.5 $dm^3$ 1 $dm^3$

$PV=nRT$

$PV=\frac{W}{M}RT$

$\Rightarrow P\times M=\rho RT$

$\frac{P_A\times M_A=\rho_A RT}{P_B\times M_B=\rho_B RT}$

$\frac{P_A}{P_B}\times\frac{1/2M_B}{M_B}=\frac{3}{1.5}$

$\frac{P_A}{P_B}=4$

9. (*b*) $\rho\propto\frac{P}{T}$

10. (*d*) $\frac{p_{H_2}}{p_{C_2H_6}}=\frac{n_{H_2}}{n_{C_2H_6}}=\frac{30}{2}=\frac{15}{1}$.

$X_{H_2}=15/16,\ P_{H_2}=X_{H_2}.X_{PT}\Rightarrow\frac{P_{H_2}}{P_T}=\frac{15}{16}$

11. (*d*) Weight of $H_2$ = 20 g in 100 g mixture;

Weight of $O_2$ = 80 g

$\therefore$ Moles of $H_2=\frac{20}{2}=10$; $\therefore$ Moles of $O_2=\frac{80}{32}=\frac{5}{2}$

$\therefore$ Total moles $=10+\frac{5}{2}=\frac{25}{2}$

$\therefore$ $P'_{H_2}=(P_T)$ x (mole fraction of $H_2$) $=1\times\frac{10}{25/2}$

= 0.8 bar

**12.** (*d*) $r \propto \frac{1}{\sqrt{M}}$

**13.** (*b*) $\frac{20}{60} \times \frac{30}{V} = \sqrt{\frac{32}{64}}$

$\Rightarrow V = 14.14\ l$

**14.** (*c*) $r \propto \frac{1}{\sqrt{M}}$

So $NH_3$ diffuses with faster rate.

**15.** (*b*) $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

$\frac{x/5}{x/t_2} = \sqrt{\frac{M_2}{2}} \Rightarrow \frac{t_2}{5} = \sqrt{\frac{M_2}{2}}$

when $t_2 = 20$

$\frac{20}{5} = \sqrt{\frac{M_2}{2}}$

$M_2 = 32.$

**16.** (*d*) Given $\frac{r_A}{r_B} = \frac{16}{3}; \frac{w_A}{w_B} = \frac{2}{3}$

We have $\frac{r_A}{r_B} = \frac{n_A}{n_B}\sqrt{\frac{M_B}{M_A}}$

$\frac{16}{3} = \frac{w_A}{M_A}\frac{M_B}{w_B}\sqrt{\frac{M_B}{M_A}}$

$\frac{16}{3} = \frac{2}{3}\left(\frac{M_B}{M_A}\right)^{3/2} \Rightarrow \left(\frac{M_B}{M_A}\right)^{3/2} = 8$

$\Rightarrow \frac{M_B}{M_A} = 4 \quad \therefore$ mole ratio $= \frac{8}{3}$

**17.** (*a*) $\frac{r_{N_2}}{r_{H_2}} = \sqrt{\frac{(MM)_{H_2}}{(MM)_{N_2}}}$

$= \sqrt{\frac{2}{28}} = \sqrt{\frac{1}{14}}$

$1 : \sqrt{14} : \sqrt{7}$

**18.** (*d*) $C_3H_6O + O_2 \rightarrow 3CO_2$

40 (exess)

— $40 \times 3 = 120$ ml.

**19.** (*b*) Urms $= \sqrt{\frac{3RT}{M}}$

$\frac{T_1}{M_1} = \frac{T_2}{M_2}$

$\frac{T_1}{32} = \frac{300}{20}$

$T_1 = 480$ K

**20.** (*b*) $v \propto \sqrt{T}$

**21.** (*c*) $\frac{(V_{rms})_1}{(V_{rms})_2} = \sqrt{\frac{T_1 M_2}{M_1 T_2}} = \sqrt{\frac{50 \times 32}{2 \times 800}} = 1$

**22.** (*a*) K.E. $= 3/2$ nRT

$n_1T_1 = n_2T_2$

$T_1 = \frac{0.4 \times 400}{0.3} \Rightarrow T_1 = 533$ K

**23.** (*c*) $U_{rms} = \sqrt{\frac{3RT}{M}} = \frac{5 \times 10^4}{10 \times 10^4}$

$= \frac{\sqrt{\frac{3RT_1}{M}}}{\sqrt{\frac{3RT_2}{M}}} = \frac{1}{4} = \frac{T_1}{T_2}$

$T_2 = 4T_1$

$T_2 = 4$ times $T_1$

**24.** (*a*) $T_C = \frac{8a}{27Rb}$

$T_B = \frac{a}{Rb}$

$T_i = 2T_B$

$T_C < T_B < T_i$

**25.** (*c*) In vander waals equation, the term $\frac{n^2a}{v^2}$ represent the intermolecular forces.

**26.** (*b*) Boiling point $\propto$ a

**27.** (*a*) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT.$

**28.** (*a*) Vander waal's equation is

$\left(p + \frac{n^2a}{v^2}\right)(v - nb) = nRT$

For non-zero values of force of attraction

$\left(P + \frac{n^2a}{V^2}\right)V = nRT$

$$\boxed{PV = nRT - \frac{n^2a}{V}}$$

**29.** (*c*) $PV = Pb + RT$

$\frac{PV}{RT} = 1 + \frac{Pb}{RT}$

**30.** (*a*) $\left(P + \frac{a}{V^2}\right)(V) = RT$

$PV + \frac{a}{V} = RT$

$\frac{PV}{RT} = 1 - \frac{a}{VRT}$

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**31.** (*a*) $b = \left(\frac{4}{3}\pi r^3\right)(4N)$

$24\times10^{-3}\ L\ mol^{-1} = \left(\frac{4}{3}\times3.14\times8^3\right)\left(4\times6.022\times10^{23}\right)$

$r^3 = 0.37\times10^{-27}\ dm^3$

$r = 1.355\mathring{A}$

**32.** (*c*) The gas which has the largest size has the highest value of the vander waals constant '*b*'. $O_3$ has largest size among also option.

**33.** (*b*) Rate of effusien $\propto \frac{1}{\sqrt{\text{molar mass}}}$

$0.525 = \sqrt{\frac{16}{m}}$

$m = 58\ g$

$C_4H_{10}$ is the answer

**34.** (*d*) $N_2 + 3H_2 \rightarrow 2NH_3$

$N_2$ is the limting reagent

After the completion of the reaction , 2 mol $NH_3$ formed

5*L* water added which makes volume increases to 20*L*

$PV = nRT$

$P\times20 = 2\times0.0821\times300$

$$\boxed{P = \frac{1\times0.0821\times300}{10}}$$

**35.** (*c*)

| | $3O_2$ | $\rightarrow 2O_3$ |
|---|---|---|
| $t=0$ | $20ml$ | $0$ |
| $t=t$ | $20-3x$ | $2x$ |

$20-3x+2x = 17ml$

$x = 3ml$

volume of ozone $= 2x = 2(3) = 6\ ml$

common oil absorb 6 *ml* ozone which result in contraction.

**36.** (*b*) More the value of '*a*' constant means more the ability of a gas to get quilified and '*b*' for $N_2$ is greater than that of $NH_3$

**37.** (*b*) $U_{rms} \propto \sqrt{T}$

$\frac{U_{rms_1}}{U_{rms_2}} = \sqrt{\frac{1200}{300}}$

$U_{rms_2} = \frac{1}{2}U_{rms_1}$.

**38.** (*b*) $PV = nRT$

If $P,V,R$ = constant then $\frac{n_1}{n_2} = \frac{T_1}{T_2}$

Let $n_1 = 1$ and $n_2 = x$

$\frac{1}{x} = \frac{400}{300} \rightarrow x = \frac{3}{4}$

So has escoped out will $= 1-\frac{3}{4} = \frac{1}{4}$

**39.** (*c*) $PV = nRT$

Initally,

$PV = \frac{4}{m}\times R\times T$ ....(*i*)

At higher temperature,

$PV = \frac{8.2}{m}\times R\times(T+50)$ ....(*ii*)

equate both equation (*i*) and (*ii*)

$4T = 3.2\ T + 160$

$T = 200\ K$

**40.** (*a*) $PV = nRT$

$\frac{7.6\times10^{-6}}{760}\times\frac{1}{1000} = n\times0.0821\times250$

$n = 4.87\times10^{-13}$ moles

No of molecules

$= 6.022\times10^{23}\times4.87\times10^{-13} = 2.93\times10^{11}$ molecules

**41.** (*c*) $P = \frac{d}{m}RT$

$m = \frac{2.64\times0.0821\times583}{(775/760)}$

$m = 124\ g/\text{mol}$

No. of $P$ atom $= \frac{124}{31} = 4$

Molecular formula is $P_4$

**42.** (*c*) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$T_2 = \frac{14.9\times300}{12} = 372.5K = 99.5°C$

**43.** (*b*) $d_1T_1 = d_2T_2$

$d_1T_1 = d_2(1.1T_1)$

$d_1 = 1.1d_2$

$\frac{d_1}{d_2} = \frac{1.1}{1}$

% decrease in density $= \frac{0.1}{1.1}\times100 = 9.1\%$

**44.** (*a*) At constant pressure and temperature

$\frac{V}{n} = \text{constant}$

$\frac{V_1}{n_1} = \frac{V_2}{n_2}$

$\frac{4/3\pi r_1^3}{n_1} = \frac{4/3\pi r_2^3}{n_2}$

$r_2 = r_1\left[\frac{n_2}{n_1}\right]^{1/3}$

$r_2 = \frac{10}{2}\left[\frac{1}{8}\right]^{1/3}$

$r_2 = 5\text{cm}$

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**45.** (*b*) For $H_2O$ vapour

$$PV = \frac{V}{m}RT$$

$$P \times 50 = \frac{1.2}{18} \times 0.0821 \times 300$$

$$P = 0.03284 \text{ atm} = 24.96 \text{ mm}$$

**46.** (*a*) *PV* curve for hydrogen gas lie above then *PV* curve for ideal gas.

**47.** (*a*) $P(V_m - n) = RT$

$$\frac{PV_m}{RT} = \frac{Pb}{RT} + 1$$

$$Z = \frac{Pb}{RT} + 1$$

$$Z = \frac{1}{9} + 1$$

$$Z = 10/9$$

$$PVm - Pb = RT$$

$$P(10b) - Pb = RT$$

$$9Pb = RT$$

$$\frac{Pb}{RT} = \frac{1}{9}$$

**48.** (*a*) $\frac{n}{v} = \frac{\rho}{m}$

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

$$\left(P + \frac{a\rho^2}{m^2}\right)\left(1 - \frac{\rho}{m}b\right) = \frac{\rho}{m}RT$$

$$\left[P + \frac{3.6(2.2)^2}{(44)^2}\right]\left(1 - \frac{2.2 \times 0.05}{44}\right) = \frac{2.2}{44} \times 0.0821 \times 300$$

$$P = 1.23 \text{ atm}$$

**49.** (*b*) At moderote temprature, $Z = 1$

$$Z = 1 + 0.35P - \frac{168}{T}P$$

$$0.35 = \frac{168}{T}$$

$$T = 480\ K$$

## Exercise-3 (JEE Advanced Level)

**1.** (*a,b*)

Suppose the cylinder will burst at $T_2$K, when $V_1 = V_2$

$$T_2 = \frac{P_2T_1}{P_1} = \frac{14.9 \times 300}{12} = 372.5 \text{ K}$$

**2.** (*a,c*)

(*a*) $PV = nRT$

$P \times 8.21 = 2 \times 0.0821 \times 300$

$P = 6$ atm.

(*b*) Open container

$P_T = P_{gas} + P_{atm}$

$$P_{gas} = \frac{2 \times 0.821 \times 400}{8.21} = 8 \text{ atm}$$

So, $P_T > 8$ atm.

(*c*) Closed non-rigid

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{8.21}{300} = \frac{V_2}{600}$$

$V_2 = 16.42$

(*d*) $\frac{P_1}{T_1} = \frac{P_2}{T_2} = \frac{6}{300} = \frac{P_2}{150}$

$P_2 = 3$ atm, $P_1 = 6$ atm.

**3.** (*a,d*)

| 300 K | 400 K |
|---|---|
| mole $H_2$ | mole $H_2$ |
| 16.42 | 8.211 |

(*a*) I. P. = 3 atm by $P = \frac{nRT}{V}$

(*b*) Pressure just after openin doesn't changes.

(*d*) Pressure becomes same after some time.

**4.** (*b,d*)

$$r \propto \frac{1}{\sqrt{M}}$$

**5.** (*b,d*)

Given $\frac{r_A}{r_B} = \frac{16}{3}; \frac{w_A}{w_B} = \frac{2}{3}$

We have $\frac{r_A}{r_B} = \frac{n_A}{n_B}\sqrt{\frac{M_B}{M_A}}$

$$\frac{16}{3} = \frac{w_A}{M_A}\frac{M_B}{w_B}\sqrt{\frac{M_B}{M_A}}$$

$$\frac{16}{3} = \frac{2}{3}\left(\frac{M_B}{M_A}\right)^{3/2} \Rightarrow \left(\frac{M_B}{M_A}\right)^{3/2} = 8$$

$$\Rightarrow \frac{M_B}{M_A} = 4$$

$$\therefore \text{ Mole ratio} = \frac{8}{3}$$

**6.** (*a,b,c,d*)

With increase in temperature, most probable velocity increases & fraction of molecules with velocity equal to M.P. velocity decreases. Total no. of molecules remain same.

**7.** (*d*) According to Gay Lussac's law

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{1}{200} = \frac{P_2}{821k}$$

$P_2 = 4.1$ atm

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**8.** (*a,b,c,d*)

Vanderwaals constant '*a*' represent the magnitude of interamolecular attractive forces and vanderwaal constant '*b*' also represent the co-volume.

Unit of '*b*' is L $mol^{-1}$

**9.** (*a,d*) Average speed, $\mu_{av} = \sqrt{\frac{8RT}{\pi M}}$

$\mu_{av} \propto T$

$\mu_{av} \propto \frac{1}{M}$

**10.** (*c,d*) According to kinetic theory of gas

$K.E. \propto T$

Pressure of gas is due to collisions of moleucles against the side of container and we assumed that there is no force of attraction exist between molecules.

**11.** (*a,b*) At temperature *T*,

Mean velocity of $A = V_{rms}$ of *B*

$$\sqrt{\frac{8RT}{\pi M_A}} = \sqrt{\frac{3RT}{M_B}}$$

$$\frac{M_B}{M_A} = \frac{3\pi}{8}$$

$M_B > M_A$

If mean velocity of both *A* and *B* can be equal

$$\sqrt{\frac{8RT_A}{\pi M_A}} = \sqrt{\frac{8RT_B}{\pi M_B}}$$

$$\frac{M_B}{M_A} = \frac{T_B}{T_A}$$

$T_B > T_A$ [$M_B > M_A$]

option *B* is correct

**12.** (*a,d*)

(*a*) $PV = \frac{wRT}{m}$

If mass of gas is same

*PV* depends only on ratio of (*T*/*m*) which is equal for both gases

(*b*) Pressure also depend upon density rather than temp and mass.

(*c*) K.E. is diffecent for diff gases at diff. temperature.

(*d*) $V_{rms} = \sqrt{\frac{3RT}{m}}$

This proprety has same mangitude as (T/m) is same for both gasses.

**13.** (*a,b,d*)

At critical point, the gases experience force of attraction and they can be liquified.

At high pressure and low temperature, there is force of attraction between molecules which can't be neglected.

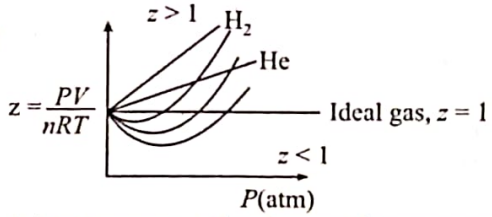
**14.** (*a,b,c*)

Vapour may be condensed to liquid by pressure.

if $T < T_C$, A gas can be liquified by applying pressure.

At $T_C$, equilibrium exist between liquid and vapour statge.

**15.** (*a,c*)




At high pressure, all real gas are less compressible than ideal gas

for $H_2$ and He, Z > 1 and for other gas Z < 1 at low pressure.

**16.** (*b,c*) Molar mass of Neon > Malor mass of He

$U_{av} = \sqrt{\frac{8RT}{\pi M}}$

$U_{av} \propto \sqrt{\frac{1}{M}}$

He has high $U_{avg}$ and strike the wall more frequently.

**17.** (*a,d*) Speed $\propto \sqrt{\frac{T}{m}}$

Option (*a*) and (*d*) are correct

Higher the temperature means Higher the value of *c*

Lower the molar mass means Higher the value of *c*.

**18.** (*a,d*) Gas can be liquified f $T < T_C$

or if $T = T_C$ then high pressure ($P > P_c$) is needed to liquify the gas.

**19.** (*a,b,c,d*)

(*a*) At critical condition, the substance exist as three phase equilibrium due to which physical state cannot be defined.

(*b*) At critical condition, due to equilibrium, their physical properties become identical.

(*c*) At critical condtition, ther is no separation between liquid and vapour state due to equilibrium.

(*d*) Gases which can be easily liquified ∝ Boyle's temp.

**20.** (*c*) $P = \frac{RT}{V_m}$

Gases behave idealy at or above $T_B$ (Byle's Temperature)

$T = \frac{a}{Rb}$

so, $P = \frac{R\left(\frac{a}{Rb}\right)}{V_m} = \frac{a}{V_m.b}$

**21.** (*a,b,d*)

As we know $P_c = \frac{a}{27b^2}$ , $T_c = \frac{8a}{27Rb}$ , $V_c = 3b$

22. (d) $Z = z\sigma^2 \bar{u} N^*$

$N^* \propto \frac{1}{V}$

$x \propto \frac{1}{V} = A$

$y \propto \frac{1}{2V} = \frac{A}{2}$

Ratio = 2 : 1

None of these

23. (c) Total no. of collisions per unit volume $\propto \frac{(N^*)^2}{V}$

Ratio = $1 : \sqrt{2}$

24. (a) $H_2 \longrightarrow 2H$

$\frac{n_x}{n_y} = \frac{n_{He}}{n_H} = \frac{x/4}{x/1} = \frac{1}{4}$

$\frac{V_x}{V_y} = \frac{1}{2} \Rightarrow \frac{N_x^x}{N_y^x} = \frac{1}{4} \times 2 = \frac{1}{4}$

$\frac{\sigma_x}{\sigma_y} = 2$ (because $H_2$ become H)

$\frac{U_{avg}(x)}{U_{avg}(y)} = \sqrt{\frac{(T/M)_x}{(T/M)_y}} = \sqrt{\frac{T/4}{2T/1}} = \frac{1}{2\sqrt{2}}$

$\frac{Z_{11x}}{Z_{11y}} = \left(\frac{N_x}{N_y}\right)^2 \times \left(\frac{U_{ave}(x)}{U_{ave}(y)}\right) \times \left(\frac{\sigma_x}{\sigma_y}\right)^2$

$= \frac{1}{4} \times \frac{1}{2\sqrt{2}} \times 4 \frac{1}{2\sqrt{2}}$

$Z_{11y} = 2\sqrt{2}$ ; $Z_{11x} = 2\sqrt{2}$ A

25. (c) $w = -pv$

$w = 0 \times (4)$

$w = 0$

26. (d) $n_{N_2} = \frac{0.82 \times 10}{0.082 \times 300} = 0.33$

$n_{He} = \frac{4.1 \times 6}{0.052 \times 300} = 1$

$n_{Total} = 1.33$

$K.E = n_T \times \frac{3}{2} \times R \times T$

$= 1.33 \times \frac{3}{2} \times 8.314 \times 300$

$= 4982.4\ J$

27. (d) $X_{N_2} = \frac{0.33}{1.33} = 0.248$

$P_{N_2} = x_{N_2} \cdot P_T = 0.248 \times 1 = 0.248\ atm$

$P_{N_2} \approx 0.25$ atm

28. (b) $n_t = 1 + 0.33 = 1.33$

$p = \frac{1.33 \times 0.082 \times 300}{16} = 2.05$

$P = 2.05$ atm

29. (a) A → s; B → q,s; C → r; D → p,r

(A) $PV = nRT$

At constant temperature

$PV = K$ (T = constant)

Higher the value of PV, higher the temperature.

So, $T_3 > T_2 > T_1$

Since, $P_1 = P_2 = P_3$

So, $V \propto T \Rightarrow V_3 > V_2 > V_1$

$d = \frac{PM}{RT}$

Since, $P_1 = P_2 = P_3$

$d \propto \frac{1}{T} \Rightarrow d_1 > d_2 > d_3$

(B) From Graph,

$V_3 > V_2 > V_1$ and $T_1 = T_2 = T_3$

Higher the volume, lesser the pressure because temperature is same for all.

$P_1 > P_2 > P_3$

$d = \frac{PM}{RT}$

Since, $T_1 = T_2 = T_3$

So, $d \propto P \Rightarrow d_1 > d_2 > d_3$

(C) From the graph,

$P_3 > P_2 > P_1$ and $T_1 = T_2 = T_3$

Higher the pressure, lesser the volume because temperature is same for all.

$V_1 > V_2 > V_3$

$d = \frac{PM}{RT}$

Since, $T_1 = T_2 = T_3$

So, $d \propto P \Rightarrow d_3 > d_2 > d_1$

(D) From the graph,

$d_3 > d_2 > d_1$ and $P_1 = P_2 = P_3$

$d = \frac{PM}{RT} \Rightarrow d \propto \frac{1}{T}$

So, $T_1 > T_2 > T_3$

$PV = nRT$

Since, $P_1 = P_2 = P_3$

$V \propto T$

So, $V_1 > V_2 > V_3$

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**30.** (*b,d*)

$$P_C = \frac{a}{27b^2}; T_C^2 = \frac{64\,a^2}{27\times 27R^2b^2}$$

$$V_C = 3b; \frac{T_C^2}{P_C} = \frac{64\,a^2}{27\times 27R^2b^2}\times\frac{27b^2}{a}$$

$$T_C = \frac{8a}{27Rb}, a = \frac{27\,R^2T_C^2}{64\,P_C}$$

**31.** (*a,c,d*)

For gaeous state, Thermal energy is always greater than the molecular attraction.

**32.** (*a,b*)

(*a*) At Boyle's temperture a real gas behave as ideal irresp. of pressure.

(*b*) At critical condition a real gas behave as ideal.

**33.** (*a*) A → p,q; B → r,s; C → p,q; D → r,s

(A) At ciritcal temperature, gas on liquify and deviate from ideal gas equation.

(B) At or above Boyle's Temperature, gas behaves like idealy.

(C) $Z < 1$ → Means deviates from ideal gas equation ($Z = 1$) and gas can be liquifed.

(D) At high temperature and low pressure, gas starts follow ideal gas equation and we assume vanderwalls constant '*a*' is negligible.

**34.** (*b*) A → p,r; B → q,r,s; C → p,r; D → q,s

(A) Diffiusion $\propto \frac{1}{\sqrt{m}}$ K.E. increase with increase in temperature due to which diffusion of gas increases.

(B) $Z < 1$, $V_{ideal} < V_{real}$ and $Z$ increase with increase in temperature.

(C) $V_{rms} = \sqrt{\frac{3RT}{M}}$

(D) Liquification of gas takes place easily when attractive force dominotes and $V_{ideal} > V_{real}$.

**35.** (*c*) $A \to r; B \to s; C \to q; D \to p$

(A) As we know, $Z$ for ideal gas is always unity

(B) for real gas, at low $P$, $\left(P+\frac{a}{V_m^2}\right)V_m = RT$

$$\boxed{\frac{PV_m}{RT} = 1 - \frac{a}{RTV}}$$

(C) For real gas, at high $P$

$P(v - nb) = nRT$

$$\frac{PV_M}{RT} = 1 + \frac{Pb}{RT}$$

(D) $Z$ for critical state

$$V_C = 3b \quad T_c = \frac{8a}{27Rb} \quad P_c = \frac{a}{27b^2}$$

$$Z = \frac{PV}{RT} = 3/8$$

**36.** (*d*) A → q,r; B → s; C → p; D → q

Translational K.E. per gram

$= \frac{3}{2}\frac{RT}{m} \Rightarrow$ Depends upon ratio of (T/m)

Translational K.E. per mole

$= \frac{3}{2}RT \Rightarrow$ Depends upon only temperature.

**37.** (*c*) A → s; B → r; C → q; D → p

(A) $PV = nRT$, $\frac{P}{V} = y$ *axis*, $P = x$-*axis*

$v = \frac{x}{y}$

$n, R, T$ = constant

$PV$ = constant

$x.\frac{x}{y}$ = constant

$\frac{P}{V}$ vs $P \to y = x^2$

(B) $\frac{P}{V}$ vs $V \Rightarrow \frac{P}{V} = \frac{nRT}{V} \Rightarrow \frac{P}{V} \propto \frac{1}{V}$

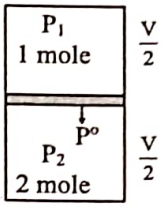
(C) $\frac{V}{p} = y$ axis, $\frac{1}{p^2}$ = axis → $y/x = PV$ = constant

so, $\boxed{y = kx}$

(D) $\frac{p}{v} = y$ axis $\log P = x$ axis $\to P = 10^x$

$v = \frac{10^x}{y} \to k10^{2x} = y$

**38.** [90]



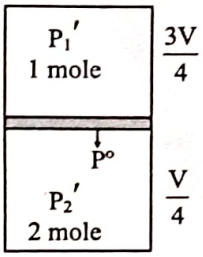

P° is the pressure due to weight of piston

at 300K

$P_1 + P° = P_2$

$$P° = \frac{2\times R\times 300}{\frac{V}{2}} - \frac{1\times R\times 300}{\frac{V}{2}}$$

at temperature T

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$$P_T' = P' = P_{H_2}' \quad P' = \frac{2 \times R \times T}{\frac{V}{4}} - \frac{1 \times R \times T}{\frac{3V}{4}}$$

$$\frac{2 \times R \times 300}{\frac{V}{2}} - \frac{1 \times R \times 300}{\frac{V}{2}} = \frac{2RT}{\frac{V}{4}} - \frac{1 \times R \times T}{\frac{3V}{4}}$$

Solving T = 90 K

39. [1]

$V_1 = 500$ ml

$V_2 = 1800$ ml

$P_{total} = 3.2$ atm

$T_2 = 360$ K

$T_1 = 320$ K

V.P = 0.6 atm

V.P = 0.2 atm

$P_{gas} = 3.2 - 0.2 = 3$ atm

For $H_2O$(vap.)

For gas :

$$\frac{0.2 \times 500}{320} = \frac{P_2 \times 1800}{360}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{3 \times 500}{320} = \frac{P_2 \times 1800}{360}$$

$$P_2 = \frac{20}{320} = \frac{2}{32} \text{ atm } (< \text{V.P.})$$

$$P_2 = \frac{6 \times 500}{320 \times 10} = \frac{30}{320} \text{ atm}$$

$$P_{total} = \frac{30}{32} + \frac{2}{32} = 1 \text{ atm}$$

40. [4] $\left(\sqrt{\frac{3RT}{M}}\right)_X = \left(\sqrt{\frac{2RT}{M}}\right)_Y$

$$\sqrt{\frac{3R400}{40}} = \sqrt{\frac{2R60}{M}}$$

$$\sqrt{30} = \sqrt{\frac{4 \times 30}{M}}$$

M = 4

41. [72]

$$\frac{1}{6} = \sqrt{\frac{2}{x}} \text{ (Where X is molecular weight of gas)}$$

$$\frac{1}{36} = \frac{2}{x}$$

x = 72

42. [4] Rate of effusion $\propto \frac{1}{\sqrt{M}}$

For Hydrogen ($H_2$) and Oxygen ($O_2$)

$$\frac{Rate^4_{H_2}}{Rate\ O_2} \propto \left(\frac{\sqrt{32}}{\sqrt{2}}\right)^4$$

$Rate^4 \propto 256 \Rightarrow x = 256$

For Helium (He) and Oxygen ($O_2$)

$$\frac{Rate^4_{He}}{Rate\ O_2} \propto \left(\frac{\sqrt{32}}{\sqrt{4}}\right)^4$$

$Rate^4 \propto 64 \Rightarrow y = 64$

so, $\frac{x}{y} = \frac{4}{1}$

43. [10] $P_{Final} = 1 + \frac{40}{76} = \frac{116}{76}$ atm

Final Height = 19cm

Final Volume = $(19 \times A)$cm$^3$

$P_iV_i = P_fV_f$

$1 \times H_i(A) = \frac{116}{76} \times 19(A)$

$H_i = 29$ cm

Hence, length by which the mercury column shorts down is = (29 – 19) cm = 10 cm

44. [5] At STP, the amount of $_{36}Ar$ found 1.911 mm$^3$ per kg of meteorite.

1 mm$^3$ = $10^{-6}$ litre

1.911 mm$^3$ = $1.911 \times 10^{-6}$ L

As we know

22.4 L = 1 mole

$1.911 \times 10^{-6}$ L = $\left(\frac{1}{22.4} \times 1.911 \times 10^{-6}\right)$ mol $_{36}Ar$

No. of $_{36}Ar$ atoms = $\left(\frac{6 \times 10^{23} \times 1.911 \times 10^{-6}}{22.4}\right)$ atoms

$$\frac{\text{No of atoms } {}_{36}Ar}{10^{16}} = \left(\frac{6 \times 10^{23} \times 1.911 \times 10^{-6}}{22.4 \times 10^{16}}\right) = 5$$

45. [3] Density (at 0°C and 1 atm) = 1.25 g/L

Volume = 3L

Amount of gas (in g) = 1.25 × 3 = 3.75 g

Let us say, xg of gas goes out P decrease by 0.8 atm

So, $\frac{P_1}{P_2} = \frac{n_1}{n_2}$

$\frac{P_1}{P_2} = \frac{w_1}{w_2}$

$\frac{1}{0.2} = \frac{3.75}{375 - x} \Rightarrow x = 3g$

Hence, 3g gas oges out

46. [3] $P_2' = \frac{P_1T_2}{T_1} - P_2' - 2$

$P_2' = \frac{P_1T_2}{2T_1} - 1$

$P_2' = \frac{4 \times 600}{2 \times 300} - 1$

$= 4 - 1$

$P_2' = 3$atm

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# Exercise-4 (Past Year Questions)

## JEE MAIN

1. (c)

$$\frac{P_i(V)}{RT_1}+\frac{P_i(V)}{RT_1}=\frac{P_f(V)}{RT_1}+\frac{P_f(V)}{RT_2}$$

$$\frac{2P_i}{T_1}=P_f\left(\frac{1}{T_1}+\frac{1}{T_2}\right)$$

$$P_f=\frac{2P_i}{T_1}\left(\frac{T_1T_2}{T_1+T_2}\right)$$

$$P_f=\frac{2P_iT_2}{T_1+T_2}$$

2. (b) $Z = PV/nRT$

$$P=\frac{ZnRT}{V}$$

at constant T and mol $P \propto \frac{Z}{V}$

$$\frac{P_A}{P_B}=\frac{Z_A}{A_B}=\frac{V_B}{V_A}=\left(\frac{3}{2}\right)\times\left(\frac{1}{2}\right)=\frac{3}{2}$$

$\therefore 2P_A = 3P_B$

3. (d) $PV = nRT$

$200 \times 10 = (0.5 + x)R \times 1000$

On solving $x=\frac{4-R}{2R}$

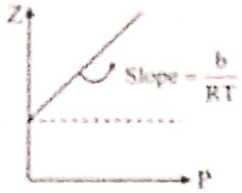
4. (a) $T_c=\frac{8a}{27Rb}$

Greater value of $\frac{a}{b}$ $\Rightarrow$ higher is $T_c$

| Gas | $a/b$ |
|---|---|
| Ar | $\frac{1.3}{3.2}=0.406$ |
| Ne | $\frac{0.2}{1.7}=0.118$ |
| Kr | $\frac{5.1}{1}=5.1$ |
| Xe | $\frac{4.1}{5}=0.82$ |

$\therefore$ $T_C$ has order : Kr > Xe > Ar > Ne

5. (c)




As $b\uparrow \Rightarrow$ slope$\uparrow$

Hence, Xe, will have highest slope

6. (c)
- Gas A and C have same value of 'b'but different value of 'a' so gas having higher value of 'a' have more force of attraction so molecules will be more closer hence occupy less volume.
- Gas B and D have same value of 'a' but different value of 'b' so gas having lesser value of 'b' will be more compressible.

7. (d) $V_{mp}=\sqrt{\frac{2RT}{M}} \Rightarrow V_{mp} \propto \sqrt{\frac{2RT}{M}}$

For $N_2, O_2, H_2$

$$\sqrt{\frac{300}{28}}<\sqrt{\frac{400}{32}}<\sqrt{\frac{300}{2}}$$

$V_{mp}$ of $N_2(300K) < V_{mp}$ of $O_2(400K) < V_{mp}$ of $H_2(300K)$

8. (b) Radon is a radioactive substance. This is not present in atmosphere.

9. (d) $C_{RMS} > C_{Avg} > C_{MPS}$

10. (b) $PM = dRT$

$\Rightarrow d \propto \frac{1}{T}$

$\Rightarrow d \propto P$

11. (d) Pressure due to 1 mole of $H_2$ gas = 2 atm

$\therefore$ Pressure due to 1 mol of $H_2$ + 1 mol of He + 1 mol of $O_2$ = 6 atm.

12. [750]

At constant temperature and number of moles,

$P_1V_1 = P_2V_2$

$P_1 = 48 \times 10^{-3}$ bar; $V_1=\frac{4}{3}\pi(3)^3$

$V_2=\frac{4}{3}\pi(12)^3$

$$P_2=\frac{P_1V_1}{V_2}=\frac{48\times10^{-3}\times(3)^3}{(12)^3}$$

$$=\frac{48\times10^{-3}}{64}=7.5\times10^{-4}=750\times10^{-6}\text{ bar}$$

13. [2]

Weight of empty LPG cylinder = 14.8 kg

Weight of full LPG cylinder = 29 kg

$\therefore$ Weight of gas = 29 – 14.8 = 14.2 kg

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If weight of full LPG cylinder = 23 kg

Then weight of gas used = 29 – 23 = 6 kg at ambient temperature.

From ideal gas equation, $pV = nRT$

$$\text{or } pV = \frac{\text{Weight of solute}}{\text{Molecular mass of solvent}} \times RT$$

$$\text{or } pV = \frac{W}{M} \times RT$$

Applying ideal gas to LPG cylinder when gas is full,

$PV = nRC$

$$3.47\,\text{atm} \times \bar{V} = \frac{14.2\,\text{kg}}{M} \times RT \quad \ldots(i)$$

Applying ideal gas to LPG cylinder when gas is reduced to 23kg at ambient temperature,

$pV = nRT$

$$p \times V = \frac{8.2\,\text{kg}}{M2} \times RT \quad \ldots(ii)$$

Divide Eq. (*i*) by (*ii*)

$$\frac{3.47}{p} = \frac{14.2}{8.2}$$

$$\Rightarrow p = \frac{3.47 \times 41}{71} = 2.003\,\text{atm}$$

**14.** [6]

Moles of $NH_4HS$ initially taken $= \frac{5.1\,\text{g}}{51\,\text{g/mol}}$

$= 0.1$ mol

Volume of vessel = 2*l*

$$NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$$

| | | | |
|---|---|---|---|
| $t = 0$ | 0.1 mol | | |
| $t = \infty$ | $1(1 - 0.2)$ | $0.1 \times 0.2$ | $0.1 \times 0.2$ |

$\Rightarrow$ Partial pressure of each component

$$P = \frac{nRT}{V} = \frac{0.1 \times 0.2 \times 0.002 \times 300}{2}$$

$= 0.246$ atm

$$\Rightarrow k_p = P_{NH_3} \times P_{H_2S} = (0.246)^2 = 0.060516$$

$= 6.05 \times 10^{-2}$

$\Rightarrow 6$

**15.** [927]

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{300 \times 10^3}{300} = \frac{1.2 \times 10^6}{T_2}$$

$\Rightarrow T_2 = 1200$ K

$\Rightarrow T_2 = 927°C$

**16.** [26]

$$n(CH_4) = \frac{PV}{RT}$$

$$= \frac{1 \times 4 \times 10^3 \times 1000}{0.083 \times 300}$$

Weight of $CH_4$

$$= \frac{40 \times 16 \times 10^5}{0.033 \times 300}\,\text{gm}$$

$= 25.7 \times 10^5$ gm

**17.** [45] Using $PV = nRT$

$$n_1T_1 = n_2T_2$$

$$\frac{3}{\text{Molar mass of A}} \times 300 = \frac{0.2}{2} \times 200$$

$\therefore$ Molar mass of A = 45 g $mol^{-1}$

**18.** [32] Using $\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{30}{300} = \frac{P_2}{318}$

$\Rightarrow P_2 = 32$ atm

**19.** [1655]

Using $PV = nRT$

$$2 \times 10^6 \times 2 \times 10^{-3} = \frac{11}{44} \times 8.3 \times T$$

$\therefore T = 1927.710$ K $= 1654.56°C$

**20.** [2] $n_{H_2} = \frac{40}{2} = 20$ and $n_{O_2} = \frac{60}{32} = \frac{15}{8}$

$$P_{H_2} = \left(\frac{20}{20 + \frac{15}{8}}\right) \times 2.2$$

$= 2.0114$ bar $= 2$ bar

**21.** [25] $Z = 1 + \frac{Pb}{RT}$ (For real gas under high pressure)

$$2 = 1 + \frac{Pb}{RT} \Rightarrow b = \frac{RT}{P}$$

$$\Rightarrow b = \frac{0.083 \times 298}{99} = 0.25 = 25 \times 10^{-2}\,\text{L mol}^{-1}$$

## JEE ADVANCED

**22.** [5] $\lambda = \frac{h}{\sqrt{2m(KE)}}$ $\quad KE \propto T$

$$\frac{\lambda_{He}}{\lambda_{Ne}} = \sqrt{\frac{m_{Ne}KE_{Ne}}{m_{He}KE_{He}}} = \sqrt{\frac{20 \times 1000}{4 \times 200}} = 5.$$

**23.** (*c*) According to Grham's law, if all conditions are identical,

$$r \propto \frac{1}{\sqrt{M}}$$

As in this question, all conditions are identical for X and Y, it will be followed

Hence $\frac{r_x}{r_y} = \sqrt{\frac{M_y}{M_x}}$

$$\frac{d}{24 - d} = \sqrt{\frac{40}{10}} \Rightarrow \frac{d}{24 - d} = 2$$

$d = 48 - 2d$

$3d = 48$

$d = 16$ cm.

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**24.** (*d*) The general formula of mean free path ($\lambda$) is

$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P}$$ (d = diameter of molecule, p = pressure inside the vessel).

$\because$ d & p are same for both gases, ideally their $\lambda$ are same. Hence it must be the higher drift speed of X due to which it is seeing more collisions per second, with the inert gas in comparison to gas Y. So X see comparably more resistance from noble gas than Y and hence covers lesser distance than that predicted by Graham's Law.

**25.** [9] Initial moles of gases = 1

$$^{238}_{92}U \longrightarrow {}^{206}_{82}Pb + 8{}^{4}_{2}He + 6{}_{-1}e^{\circ}$$

Inital moles 1 moles

Moles after decompostion 8 mole

Total gaseous moles after decompostion = 8 + 1 = 9 moles

Ratio of pressures $\frac{P_f}{P_i} = \frac{n_f}{n_i} = 9$

**26.** (*c*) $P(V-b) = RT$

$\Rightarrow PV - Pb = RT$

$\Rightarrow \frac{PV}{RT} = \frac{Pb}{RT} + 1$

$\Rightarrow Z = 1 + \frac{Pb}{RT}$

Hence $Z > 1$ at all pressures.

This means , repulsive tendencies will be dominant when interatomic distance are small.

This means, interatomic potential is never negative but becomes positive at small interatomic distances.

**27.** [4 times]

Given diffusion coefficient is proportional to mean free path ($\lambda$) and mean speed ($V_{mean}$)

And absolute T is increased by 4 times

And average $K_E \propto T \Rightarrow \frac{1}{2}mV^2 \propto T$

When T increased by 4 times $\Rightarrow$ $V_{mean}$ increases by 2 times (i)

Also mean free path, $\lambda$

$$= \frac{KT}{\sqrt{2}\pi d^2 P} \Rightarrow \lambda \propto \frac{T}{P}$$

Increasing T 4 times and P 2 times,

$\lambda$ increases 2 times (ii)

from (i) and (ii) implies, diffusion coefficient increases 4 times

**28.** [2.22]

$P_1 = 5 \quad P_2 = 1$

$v_1 = 1 \quad v_2 = 3$

$T_1 = 400 T_2 = 300$

$n_1 = \frac{5}{400R} \quad n_2 = \frac{3}{300R}$

Let volume be $(v + x) \quad v = (3 - x) \quad 15 - 5x = 4 + 4x$

$\frac{P_A}{T_A} = \frac{P_B}{T_B}$

$\frac{n_{b_1} \times R}{v_{b_1}} = \frac{n_{b_2} \times R}{v_{b_2}}$

$\Rightarrow \frac{5}{400(4+x)} = \frac{3}{300R(3-x)}$

$\Rightarrow 5(3-x) = 4 + 4x \quad \Rightarrow x = \frac{11}{9}$

$v = 1 + x = 1 + \frac{11}{9} = \left(\frac{20}{9}\right) = 2.22$

**29.** (*a,b,c*)

$U_{rms} = \sqrt{\frac{3RT}{M}}$

$E_{avg} = \frac{3}{2}kT$

**30.** (*b*) Graph represents symmetrical distribution of speed and hence, the most probable and the average speed should be same. But the root mean square speed must be greater than the average speed.

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CHAPTER

# 4 Chemical Thermodynamics and Energetics

## Exercise-1 (Topicwise)

**1.** (*c*) In isolated system neither exchange of matter nor exchange of energy is possible with surroundings.

**2.** (*c*) Work is not a state function as during a process its value depends on the path followed. The value of enthalpy, internal energy and entropy depends on the state and not on the path followed to get that state, hence these are state functions.

**3.** (*b*) Sliding of legs on roof surface results in mechanical energy loss.

**4.** (*c*) Critical density, viscosity, specific heat capacity, molar heat capacity, kinetic energy, specific gravity, dielectric constant, pH.

**5.** (*c*) PV = constant (Boyle's Law)
PV = nRT (Ideal gas equation)
PV/T = Constant
W = –q
ΔV = 0
ΔU = q + W = 0

**6.** (*a*) In an ideal gas intermolecular forces are absent.

**7.** (*b*) The process starts and ends at the same point.

**8.** (*c*) In a well stoppered thermoflask, no exchange of energy or matter can take place.

**9.** (*a*) A → B, Volume is constant
B → C, Slope (V/T) of line is constant.
C → A, Temperature is constant.

**10.** (*c*) A → B: Isochoric
B → C: Isothermal
C → D: Isochoric
D → A: Isothermal

**11.** (*c*) $\Delta H = nC_p\Delta T = +ve$ [$\because \Delta T = -ve$]
$W = -P(V_2 - V_1) = -ve$ [$\because (V_2 - V_1)$ is +ve]

**12.** (*d*) q = 0; because system is thermally insulated.
W > 0; because work is done on the system by stirring.

**13.** (*b*) (i) Volume = constant
W = –PdV = 0
(ii) Pressure = constant
$W = -P(2V_1 - V_1) = -PV_1$

**14.** (*b*) $W = -2.303\ nRT \log \frac{V_2}{V_1}$

$= -2.303 \times 1 \times 8.314 \times 10^7 \times 298 \log \frac{20}{10}$

$= -298 \times 10^7 \times 8.314 \times 2.303 \log 2.$

**15.** (*c*) $W_{ABCA}$ = Area enclosed in figure ABCA
$= 1/2 \times (3V_1 - V_1)(6P_1 - P_1) = 5P_1V_1$

**16.** (*c*) Heat is released during freezing.

**17.** (*a*) $C_P - C_V = nR$
$\therefore C_P - C_V = 2R$

**18.** (*a*) ΔU = q + W
= 600 – 300 = 300 J

**19.** (*a*) ΔV = 0 for isochoric process
$\therefore q_v = \Delta E$

**20.** (*d*) $dU = nC_v dT$
dT = 0
∴ dU = 0
W = –q

**21.** (*b*) W = – PdV = – 1 × dV = – 101.33 × ΔV J
∵ ΔU = q + W
300 = 500 + (–101.3 × ΔV)
⇒ ΔV = 2L

**22.** (*b*) Heat is evolved; so $q_{sys}$ = –ve
W = –ΔnRT
W = –(–1) RT
W = RT
ΔU = q + W = –ve

**23.** (*d*) The temperature of the system decreases in adiabatic expansion.

**24.** (*a*) $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$(300)(V_1)^{1/3} = T_2(8V_1)^{1/3}$

$T_2 = 150K$

$W = \frac{nR}{\gamma - 1}[T_2 - T_1]$

$W = \frac{1 \times 2}{1.33 - 1}(150 - 300)$

W = –900 Cal

**25.** (*b*) $\gamma = \frac{C_P}{C_V}$ increases with atomicity

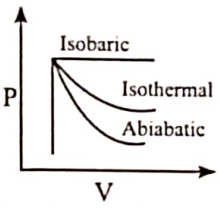
Atomicity of B > Atomicity of A

| | Ar | He | $H_2$ | $O_2$ |
|---|---|---|---|---|
| Atomicity | 1 | 1 | 2 | 2 |

Atomicity of $H_2$ > Atomicity of He

$\gamma$ of $H_2$ > $\gamma$ of He

**26.** (*a*)




Work done is area under the curve.

**27.** (*b*) $\Delta H = \Delta U + \Delta n_g RT$

$= \Delta U + \Delta PV$

$= 40 + [(4)(5) - (1)(3)] = 57$ L-atm

**28.** (*d*) H and U are functions of temperature.

$\Delta U = nC_v dT$

$\Delta H = nC_p dT$

**29.** (*d*) $W = -P_{ext}(V_2 - V_1)$

$= -1(2) = -2 \times 101.3$ J $= -202.6$

$q = \Delta H = 202.6$ J

$\Delta U = q + W$

$= 202.6 - 202.6 = 0$

**30.** (*b*) $W = -P(V_{final} - V_{initial})$

$= +ve$ [$\because V_{final} > V_{initial}$]

**31.** (*b*) When water freezes, it gives up heat to the surrounding. This heat transfer is $n\Delta H_{fusion}$.

**32.** (*c*) Randomness of gaseous phase is maximum.

**33.** (*c*) $\Delta S_{reaction} = \Sigma\Delta S_{products} - \Sigma\Delta S_{reactants}$

Entropy of gases is higher than entropy of solids or liquid.

**34.** (*b*) Number of particles increases, number of ways of energy distribution increases, thus entropy increases.

**35.** (*d*) More the negative value of $\Delta n_g$, more will be the negative change in entropy.

(*a*) $\Delta n_g = -1$ (*b*) $\Delta n_g = -1$ (c) $\Delta n_g = 0$ (d) $\Delta n_g = -2$

**36.** (*b*) $\Delta n_g = +1 \Rightarrow$ Entropy change is positive.

Decomposition reactions are endothermic.

**37.** (*c*) In this reaction $NaNO_3$ changes from solid to its aqueous ions $Na^+$ and $NO_3^-$. Thus, entropy is greater than zero.

**38.** (*b*) $\Delta n_g > 0$, hence entropy change will be maximum.

**39.** (*a*) Eontropy of universe is continuously increasing.

**40.** (*b*) For an isolated system,

$\Delta S = 0$

$S_p - S_r = 0$

$S_p = S_r$

**41.** (*c*) $\Delta S = nC_{p,m}\ln\left(\frac{T_2}{T_1}\right)$

$\Delta S = 2 \times \frac{5}{2} R \times \ln\left(\frac{600}{300}\right)$

$\Delta S = 5R\ln 2$

**42.** (*c*) $\Delta S = nC_{v,m}\ln\left(\frac{T_2}{T_1}\right)$

$C_{p,m} - C_{v,m} = R$

$\frac{5}{2}R - C_{v,m} = R$

$C_{v,m} = \frac{3}{2}R$

$\Delta S = 2 \times \frac{3}{2} R \times \ln\left(\frac{600}{300}\right)$

$= 3R\ln 2$

**43.** (*c*) $\Delta S = nC_{v,m}\ln\left(\frac{T_2}{T_1}\right)$

$= 2 \times \frac{3}{2} R \times \ln\left(\frac{573}{273}\right)$

$= 3R\ln\left(\frac{573}{273}\right)$

**44.** (*c*) Total entropy change $= \frac{\Delta H}{T_2} - \frac{\Delta H}{T_1}$

$= \frac{-401.7}{273} - \left[\frac{-401.7}{368}\right] = -0.38$ J/K

**45.** (*c*) $4Fe_3O_4 + O_2 \longrightarrow 6Fe_2O_3$

$\Delta S^\circ_{reaction} = 6\Delta S^\circ(Fe_2O_3) - \Delta S^\circ(O_2) - 4\Delta S^\circ(Fe_3O_4)$

$-266 = 6 \times 87 - 205 - 4 \times \Delta S^\circ(Fe_3O_4)$

$\Delta S^\circ(Fe_3O_4) = 145.75$ J $K^{-1}$

**46.** (*d*) Gibbs free energy in an extensive property.

**47.** (*b*) When $\Delta H = -ve$ and $\Delta S = +ve$ then reaction is spontaneous.

**48.** (*d*) At equilibrium, $\Delta G = 0$

$-2.303$ RT log k $= 0$

$\Rightarrow k = 1$

**49.** (*a*) Curd forms from milk because of the chemical reaction between the lactic acid, bacteria and casein protein.

**50.** (*d*) $\Delta G = nRT\ln\left(\frac{P_2}{P_1}\right)$

$= nRT\ln(1) = 0$

**51.** (*b*) $\Delta G = \Delta H - T\Delta S$ must be less than zero

$\Delta H - T\Delta S < 0$

$\Delta H = -ve$

For $\Delta G < 0$, $\Delta S$ can be negative but the numerical value must be less than $\frac{X}{298}$ Cal $K^{-1}$ $mol^{-1}$.

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**52.** (*d*) $\Delta G = \Delta H - T\Delta S < 0$ [for spontaneity]

$-33 - T(-58) < 0$

$T < \frac{33}{58}$

**53.** (*a*) $\Delta G = -RT \times \ln k$

$k = \frac{P_B}{P_A} = \frac{P}{4P} = \frac{1}{4}$

$\Delta G^\circ = -RT \ln \left(\frac{1}{4}\right)$

$= RT \ln (4)$

**54.** (*b*) $2Fe(s) + \frac{3}{2} O_2(g) \longrightarrow Fe_2O_3(s)$; $\Delta G = -177$ k Cal

$4Fe_2O_3(s) + Fe(s) \longrightarrow 3Fe_3O_4(s)$; $\Delta G = -19$ k Cal

$\Delta G$ for $3Fe(s) + 2O_2(g) \longrightarrow Fe_3O_4(s)$ can be obtained by taking $[2 + 4(1)] \times \frac{1}{3}$

Hence, we get $\Delta G_f = [19 + 4 \times (-177)] \times \frac{1}{3}$

$= -242.3$ k Cal for 1 mol. $Fe_3O_4$

**55.** (*c*) For spontaneity, $\Delta G < 0$

$\Delta H - T\Delta S < 0$

$-76.6 - T(226 \times 10^{-3}) < 0$

$T < 338.93$ k

$T < 66°C$

**56.** (*c*) Free energy change for a reversible reaction at equilibrium is zero.

**57.** (*a*) $\Delta n_g = 1 - 1 - 1 = -1$

$\Delta U = \Delta H^\circ - \Delta n_g RT$

$= -4600 - (-1 \times 2 \times 500)$

$= -3600$ Cal $= -3.6$ k Cal

**58.** (*d*) $W = -\Delta n_g RT = -PdV$

For I and II, $\Delta n_g$ = –ve

$\Rightarrow$ W = +ve (work done on the system)

For III, $\Delta n_g$ = +ve

$\Rightarrow$ W = –ve (work done by the system)

For IV, dV = –ve

$\Rightarrow$ W = +ve (work done on the system)

**59.** (*d*) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$; $\Delta H = -185$ k J

2mole 2mole

$\Rightarrow 2H_2(g) + 2Cl_2(g) \longrightarrow 4HCl(g)$; $\Delta H = -370$ k J

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta n_g = 0$

$\Rightarrow \Delta H = \Delta U = -370$ k J

**60.** (*b*) $NH_4Cl(s) \xrightarrow{\text{Sublime}} NH_3(g) + HCl(g)$

Internal energy becomes more negative if the system expands.

**61.** (*d*) $\Delta H = \Delta E + \Delta n_g RT$

(a) $\Delta n_g = 2 - 1 - 1 = 0$

(b) $\Delta n_g = 0 - 0 = 0$

(c) $\Delta n_g = 1 - 1 = 0$

(d) $\Delta n_g = 2 - 3 - 1 = -2 \neq 0$

In (d), $\Delta H \neq \Delta E$

**62.** (*c*) $3C(s) + 3H_2(g) + O_2(g) \longrightarrow CH_3COOCH_3(l)$

$\Delta n_g = 0 - (4) = -4$

$\Delta H = \Delta U + \Delta n_g RT$

$-442.9 = \Delta U + 4 \times 8.314 \times 10^{-3} \times 298$

$\Rightarrow \Delta U = -452.82$ k J/mol

**63.** (*b*) The negative sign of $\Delta H$ indicates that this reaction is exothermic.

**64.** (*d*) (i) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H_1 = -94.3$ k Cal mol$^{-1}$

(ii) $CO(g) + \frac{1}{2} O_2(g) \longrightarrow CO_2(g)$; $\Delta H_2 = -67.4$ k Cal mol$^{-1}$

(iii) $O_2(g) \longrightarrow 2O(g)$; $\Delta H_3 = 117.4$ k Cal mol$^{-1}$

(iv) $CO(g) \longrightarrow C(g) + O(g)$; $\Delta H_4 = 230.6$ k Cal mol$^{-1}$

$C(s) \longrightarrow C(g)$; can be obtained by

Target equation = Eq. (i) – Eq. (ii) $-\frac{1}{2}$ Eq (iii) + Eq. (iv)

So, enthalpy change will be

$\Delta H = \Delta H_1 - \Delta H_2 - \frac{1}{2}\Delta H_3 + \Delta H_4$

$= -94.3 + 67.4 - 58.7 + 230.6 = 145$ k Cal mol$^{-1}$

**65.** (*a*) $\Delta H$ represents heat of reaction.

**66.** (*d*) Eq. (i) $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$; $\Delta H = \Delta H_1^\circ$

Eq. (ii) $Br_2(g) \longrightarrow Br_2(l)$; $\Delta H = \Delta H_2^\circ$

eq. (i) – eq. (ii)

$H_2(g) + Br_2(l) \longrightarrow 2HBr(g)$

$\Delta H = \Delta H_1^\circ - \Delta H_2^\circ$

For 1 mol of HBr, $\Delta H = \frac{\Delta H_1^\circ - \Delta H_2^\circ}{2}$

**67.** $C(\text{diamond}) + O_2 \longrightarrow CO_2(g)$; $\Delta H = -94.3$ k cal/mol

$C(\text{graphite}) + O_2 \longrightarrow CO_2(g)$; $\Delta H = -97.6$ k cal/mol

eq. (i) – eq. (ii)

$\Delta H = -94.3 - (-97.6) = 3.3$ k cal mol$^{-1}$

**68.** (*a*) Heat of combustion of 1mole of sucrose = 1350 kCal

Heat of combustion of 342g of sucrose = 1350 kCal

Heat of combustion of 17.1g of sucrose

$= \frac{1350}{342} \times 17.1 = 67.5$ kCal

**69.** (*a*) Given,

$\Delta H_1 = -298.2$ k J mol$^{-1}$; $\Delta H_2 = -98.7$ k J mol$^{-1}$

$\Delta H_3 = -130.2$ k J mol$^{-1}$; $\Delta H_4 = -287.3$ k J mol$^{-1}$

By adding all the equation, we get

$\Delta H$ of formation of $H_2SO_4$ at 298 K = –814.4 k J mol$^{-1}$

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**70.** (*a*) $C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$; $\Delta H = -5644$ k J

$n_{O_2} = \frac{5644 \times 3}{1411} = 12$

$V_{O_2} = 12 \times 22.4 = 268.8$ L

**71.** (*a*) Calorific value for ethane

$= \frac{-341.1}{30} = -11.37$ k Cal $g^{-1}$

Calorific value for ethyne $= \frac{-310}{26} = 11.92$ k Cal $g^{-1}$

**72.** (*d*) $CH_4 \longrightarrow C + 4H$, $\Delta H = 320$

$E_{C-H} = 80$ cal

$C_2H_6 \longrightarrow 2C + 6H$, $\Delta H = 360$

$\therefore 360 = E_{C-C} + 6E_{C-H}$

$\therefore E_{C-C} = 360 - 6 \times 80 = 120$ cal

**73.** (*b*) Aim: $\frac{1}{2}H_2 + \frac{1}{2}Cl_2 \longrightarrow HCl$

$\Delta H = \Sigma B.E._{(products)} - \Sigma B.E._{Reactants}$

$= B.E.(HCl) - \left[\frac{1}{2}B.E.(H_2) + \frac{1}{2}B.E.(Cl_2)\right]$

$= -103 - \left[\frac{1}{2}(-104) + \frac{1}{2}(-58)\right]$

$= -103 - (-52 - 29) = -22$ k cal

**74.** (*a*) $\frac{1}{2}A_2 + \frac{1}{2}B_2 \longrightarrow AB$; $\Delta H = -100$ k J

Let bond enthalpy of $A_2$ and AB be X and that of $B_2$ will be $\frac{X}{2}$.

$\left(\frac{X}{2} + \frac{X}{4}\right) - X = -100$

$X = 400$ k J

**75.** (*a*) $\Delta H_{diss} = \Delta H_{hyd} + (\Delta H_{diss})_{hyd}$

$-25 = -30 + (\Delta H_{diss})_{hyd}$

$\Rightarrow (\Delta H_{diss})_{hyd} = 5$ Cal $mol^{-1}$

**76.** (*a*) Since process is exothermic, there heat is evolved due to which temperature of water increases.

**77.** (*c*) $C_2H_2(g) + H_2(s) \longrightarrow C_2H_4(g)$

$\Delta H = \Delta U + P\Delta V$

$-0.5 = \Delta U + (-1.5)(0.05)$

$\Delta U = 0.25$ k J

**78.** (*a*) $\Delta H_f LiCl = \Delta H_{sub} + \Delta H_{I.P} + \frac{\Delta H_{diss}}{2} + \Delta H_{EA}$

$-400 = 160 + 520 + \frac{244}{2} - 365 + I.E$

$I.E = -837$ k J $mol^{-1}$

**79.** (*c*) Heat of neutralization of a strong acid and stong base is nearly –13.7 k Cal.

**80.** (*d*) Methanoic acid is a weak acid and therefore, energy will be required for its dissociation.

**81.** (*a*) $\Delta H = -13.7 + 0.7$

$\Delta H = -13$ k cal

## Exercise-2 (Learning Plus)

**1.** (*d*) Boiling point, pH, presure, temperature and density are intensive properties.

Mass, volume and entropy are extensive properties.

**2.** (*b*) For IV volume is constant → Isochoric

For I presure is constant → Isobaric

**3.** (*b*) Isothermal process → Temperature constant → $\Delta T = 0$

Isobaric process → Pressure constant → $\Delta P = 0$

Adiabatic process → Heat constant → $q = 0$

Isochoric process → Volume constant → $\Delta V = 0$

**4.** (*d*) We know in P-V graph process will be either isothermal or Adiabatic but pressure will be different. But here given process is starting from some point and ending at same point so it is not possible.

**5.** (*d*) A → B Isobaric process

B → C Isothermal process

C → D Isobaric process

$W_{Isobaric} = -Pext\,(\Delta V) = -Pext\,(V_f - V_i)$

$W_{Isobaric} = -pv\ln\left(\frac{V_2}{V_1}\right)$

$W_{Total} = W_{A\to B} + W_{B\to C} + W_{C\to D}$

$\Rightarrow -P_o(2V_o - V_o) + (-P_oV_o)\ln\frac{4V_o}{2V_o} + \left(\frac{-P_o}{2}\right)(2V_o - 4V_o)$

$\Rightarrow -P_oV_o - P_oV_o\ln 2 + P_oV_o$

$W_{total} \Rightarrow -P_oV_o\ln 2$

**6.** (*a*) $W = P\Delta V$

Volume increases by 1L = $V_2 = 51$L

$V_1 = 50$L

$p = 1$ atm

$W = P\Delta V = 1\text{atm} \times [51L - 50L]$

$W \Rightarrow 1$ atm L

**7.** (*c*) For cyclic process : $\Delta U = 0 \Rightarrow q = -W$

Isothermal process: $\Delta U = 0 \Rightarrow q = -W$

Adiabatic process: $q = 0 \Rightarrow \Delta U = -W$

Isochoric process: $q = 0 \Rightarrow \Delta U = q$

In Adiabatic process $q = 0 \neq \Delta E$

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8. (c) Given

$P_{ext} = 2$ atm $P_1 = 5$ atm $P_2 = 2$ atm

$T_1 = 300$ K $T_2 = T$, $n = 1$, $C_V = \frac{5}{2}R$

For an adiabatic process

$w = nC_V\,dT = -P\Delta V$

$\therefore nC_V(T_2 - T_1) = -P_{ext}(V_2 - V_1)$

$$\Rightarrow 1\times\frac{5}{2}R(T-300) = -2\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

$$\Rightarrow 1\times\frac{5}{2}R(T-300) = -2\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

$$\Rightarrow \frac{5}{2}R(T-300) = -2R\left(\frac{T}{2} - \frac{300}{5}\right)$$

$\Rightarrow 5T - 1500 = -2T + 240$

$\Rightarrow 7T = 1740 \Rightarrow T = 248.57$ K

$\therefore T_2 = T = 248.57$ K.

9. (b) As we know that,

$\Delta H = \Delta U + P\Delta V$

for adiabatic process,

$$\Delta H = \frac{f}{2}nR\Delta T + nR\Delta T = nC_P\Delta T$$

$\therefore \Delta H \propto \Delta T$ ....(i)

Now the work done by the gas is maximum in reversible adiabatic process than irreversible adiabatic process.

Now, $P\Delta V = W$

$\Rightarrow \Delta T \propto W$ ....(ii)

from equation (i) and (ii), we have

$\Delta H_1 < \Delta H_2$

10. (b) For the state 1

$PV = nRT$

$1 \times 10 = n \times (0.082)(300)$

$n = 0.4$

For the state 2

$PV = nRT$

$4 \times 5 = 0.4 \times (0.082)(T_2)$

$T_2 = 600$K

$\Delta U = q + W$

$= 50 \times (\Delta T) + (-P_{ext})\Delta V$

$= 50(600 - 300) - (1)(4 - 1)$

$\Delta U = 15000 - 300 \Rightarrow 14700$J

$\Delta H = \Delta U + \Delta PV$

$\Rightarrow 14700 + (5 \times 4 - 10)$ atm L

$\Rightarrow 14700 + 10 \times 100$J

$\Rightarrow 15700$ J

$\Delta H \Rightarrow 15.7$ kJ

11. (c) HgO(s) HgS(s)

Entropy also depends upon total molar mass.

Molar mass of HgS > HgO.

HgO has lower entropy than Hgs.

12. (c) $S_1 \rightarrow AlCl_3 \rightarrow Al^{3+} + 3cl-$

$Al^{3+} \rightarrow$ high charge density

So water molecule Attracts $Al^{3+}$ So Arragement is highly order. So entropy decreases.

$S_2 \rightarrow$ Adsorption leads to exothermic process $\Delta H = -Ve$

$S_3 \rightarrow$ Higher the molar mass Higher entrophy $D_2$ has higher entropy than $H_2$.

Statement 2 and 3 are correct.

13. (c) Isoentropic process is Adiabatic and reversible process.

14. (b) At absolute zero (At OK) the entropy of a perfectly crystalline substance is taken as zero.

15. (c) for a chemical reaction:-

at equilibrium Gibbs free energy change is zero.

But for every chemical reaction at equilibrium stardard Gibbs energy of reaction is not zero.

for a Spontaneous reaction $\Delta G$ is negative.

Spontaneity is related to change in entropy of universe.

Statement 2 and 3 are correct.

16. (d) $\Delta G = \Delta H - T\Delta S$

$\Rightarrow -6000 - 300 \times 0.18 \Rightarrow -6054$ kJ/mol

So, for 34.2 gm (0.1 mol) of sucrose maximum available energy $\Delta G = 605.4$kJ

17. (c) $\Delta_r H = \Delta H_{Reactants} - \Delta H_{products}$

$\Delta_r H = -110.5 - (-266.3) = 155.8$ kJ/mol

Similarly, $\Delta rs = S_{Reactants} - S_{products}$

$\Delta_r s = (27.28 + 197.6) - (57.49 + 5.74)$

$\Rightarrow 161.65$ J/mol k

Therefore at $\Delta_r G = O$, $\Delta_r H = T\Delta_r S$

$$T = \Delta_r H/\Delta_r S = \frac{155.8\times1000\,J\,mol}{161.65} = 963.94k$$

$T = 964$ K

18. (a) $\Delta_r H = \Delta H_{product} - H_{Reactat} = H_B - H_A$

$\Rightarrow 15 - 40 = -25$ kJ

19. (c) Given $\Delta H° = -92.2$ KJ

$\Delta C_P = 2C_P(NH_3, g) - C_P(N_2, g) - 3C_P(H_2)$

$\Rightarrow 2 \times 35.1 - 29.1 - 3 \times 28.8$

$\Rightarrow 70.2 - 29.1 - 86.4 = -45.3$ J/K

$$\Delta H°_{100°C} = \Delta H°_{25°C} + C_P\Delta T = -92.2 - \frac{45.3\times75}{1000}$$

$\Delta H°_{100°C} = -92.2 - 1.3975 = -95.6$ kJ/mole

Then, reaction at→ 100°C as compared to that at 25°C will be more exothermic.

**20.** (*c*) The enthalpy of formation of pure elements is zero. Therefore,

$\Delta H_f^\circ(X_2) = 0$

Let the enthalpy of formation of $AX_2$ (g) and $BX_2$ (g) be 4X and 3X respectively.

$\Delta H = \Delta H_{Products} - \Delta H_{Reactants}$

$\Delta H = \left[\Delta H_f^\circ(AX_2) - 2\Delta H_f^\circ(BX_2)\right] - \left[\Delta H_f^\circ(AB_2)\right]$

$\Delta H = [4x - (2 \times 3x)] - (+30)\text{kJ mol}^{-1} = -270 \text{ kJ mol}^{-1}$

$\Rightarrow 4x - 6x - 30 = -270 \text{ kJ mol}^{-1}$

$2x = 240 \text{ kJ mol}^{-1}$

$x = 120 \text{ kJ mol}^{-1}$

Therefore, $\Delta H_f^\circ(AX_2) = 4x = 4 \times 120 = 480 \text{ kJmol}^{-1}$

$\Delta H_f^\circ(BX_2) = -6x = 6 \times 120 = -720 \text{ kJmol}^{-1}$

$K_P = K_C (RT)^{\Delta n}$

$\Delta n = n_{AX_2} + n_{BX_2} - n_{X_2}$

$(2 + 1) - 3 = 0$

Therefore, $K_P = K_C (Rt)^o$

$K_P = K_C$

**21.** (*b*) $2\,CO_2 + 3H_2O \rightarrow C_2H_5OH + O_2 \quad \Delta H_1 = 66$ kcal ..(1)

$CH_3OCH_3\,(g) + O_2 \rightarrow 2CO_2 + 3H_2O \; \Delta H_2 = -348$ kcal ...(2)

$\Delta H(H_2O) = -68$ kcal

$\Delta H(O_2) = -94$ kcal

$\Delta H$ of reaction 2

$\Rightarrow -348 - (3(-68) + 2(-94)) \Rightarrow 44$ kcal

by adding equation (1) and (2)

$CH_3OCH_3 \rightarrow C_2H_5OH$

$\Delta H = +66 - 44 = 22$ kcal

$\Delta E = 21.408$ kcal/mol.

**22.** (*d*) $\Delta H_f = \Delta H_S + \text{I.E} + \frac{\Delta H_d}{2} + \text{E. A} + \text{L}$

$L = -388.6 - 81.2 - \frac{243}{2} - 375.7 + 348.3$

$L = -618.7$ KJ/mol.

**23.** (*a*) Enthalpy of neutralization is the energy released when one equivalent of acid reacts with one equivalent of loose to produce salt and water as products

$\therefore$ from question:

$-12250x - 13000\,(1-x) = -12500$ cal/mol

Let x be the HCl acid dissolved in weak base AOH

$\therefore 750x - 13000 = -12500$

$x = \frac{500}{750} = \frac{2}{3}$

$(1-x) = \frac{1}{3}$

Ratio of acid in AOH & BOH = 2:1

**24.** (*b*) 40g of NaOH $= \frac{40}{40} = 1$mol NaOH

60g $CH_3COOH = \frac{60}{60} = 1$mol $CH_3COOH$

Since acetic acid is a weak acid, some of the neat is utilised to ionize it. So, enthalpy of neutralisation of 1 mol of NaOH by 1 mol $CH_3COOH$ is less than 57.1 kJ

Enthalpy of neutralisation of a strong acid by a strong acid by a strong base is always 57.1 kJ

**25.** (*b*) HCN is a weaker acid than $CH_3COOH$ from $\Delta H$ ionization value

$\text{pka} \propto \frac{1}{\text{Strength of acid}}$

Since HCN is weaker acid

$\therefore$ pka (HCN) > pka ($CH_3COOH$)

**26.** (*c*) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$

$\Delta_f H = -285.8 = -\frac{571.6}{2}$

$H_2O(l) \rightarrow H_2O\,(g) + 40.6$

$\therefore H_2O\,(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g) - 245.2\text{kJ / mol}$

Let the average OH bond energy in $H_2O$ be x

$\therefore -245.2 = (435 + 244.8) - (2x)$

$\Rightarrow 2x = 435 + 244.8 + 245.2$

$\Rightarrow x = \frac{925}{2} = 462.5\,\text{kJ / mol}$

**27.** (*b*) When double bond of one molecule of $CH_2 = CH_2$ breaks then addition of two $-CH_2-$ groups takes place, that is two new single bonds are formed.

So, for each C = C two new C – C bonds are formed. Therefore, in the complete polymer of ethylene there are 2 single bonds per mole of ethylene.

$\Delta H$ polymerisation $= \Delta H_{(CH_2 = CH_2)} - 2\Delta H_{(-CH_2 - CH_2-)}$

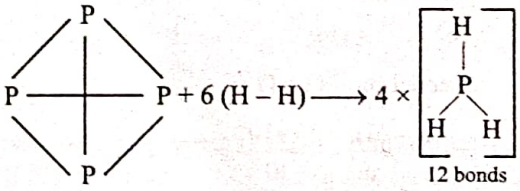
$2\Delta H_{(-CH_2 - CH_2-)} = 600 - (-100) = 700 \text{ kJ mol}^{-1}$

$\Delta H_{(-CH_2 - CH_2-)} = \frac{700}{2} = 350\text{kJmol}^{-1}$

**28.** (*c*) $P_4(s) \rightarrow 4P(g) \Rightarrow 53$ kcal/mol

$H_2(g) \rightarrow 2H(g) \Rightarrow 104.2$ kcal/mol

In $P_4(s)$ 6 bonds are present.

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We know

$\Delta H_f$ (reaction) = OH(reactant) OH(product)

$\Rightarrow 5.5 = (53.2 \times 6) + (104.2 \times 6) - 12x$

$5.5 = 319.2 + 625.2 - 12x$

$x \approx 77$

**29.** (*c*) Since formation reaction is the reaction involving formation of 1 mole of that species from its constituent elements while decomposition reaction is breaking down of 1 mole of any species into its constituent speicies. i.e for water

formation reaction :-

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \quad \Delta_f H = -286 kJmol^{-1}$$

Decomposition reaction:-

$$H_2O \rightarrow H_2 + \frac{1}{2}O_2$$

Hence the formation reaction and decomposition reaction are reverse reaction of each other.

Hence, enthalpy of decomposition reaction will be negative of enthalpy of formation reaction:-

$\Delta H_d(H_2O) = -\Delta H_f(H_2O) = -(286) = 286\ kJmol^{-1}$

**30.** (*d*) $P_{(yellow)} \rightarrow P_{(Red)}$

$\Delta H$= heat of combustion of $P_{(yellow)}$ – heat of combustion of $P_{(Red)}$

$\Rightarrow -9.91kJ - (-8.78\ kJ)$

$\Rightarrow -1.13\ KJ$

**31.** (*c*) $Cl_2(g) \rightarrow 2Cl(g) \quad \Delta H_1 = 243.3\ kJ/mol$

$I_2(g) \rightarrow 2I(g) \quad \Delta H_2 = 151\ kJ/mol$

$ICl(g) \rightarrow I(g) + Cl(g) \quad \Delta H_3 = 211.3\ kJ/mol$

$I_2(g) \rightarrow I_2(g) \quad \Delta H_4 = 62.8\ kJ/mol$

Required equation :

$$\frac{1}{2}I_2(S) + \frac{1}{2}Cl_2(g) \rightarrow ICl(g) \quad \Delta H = ?$$

$$\Delta H = \frac{62.8 + 151 + 242.3}{2} - 211.3$$

$\Delta H = 16.75\ kJ/mol.$

**32.** (*a*) $\frac{1}{2}x_2 + \frac{3}{2}y_2 \rightleftharpoons xy_3 \quad \Delta H = 30kJ$

$S_{x_2} = 60\ S_{y_2} = 40\ S_{xy_3} = 50$

$\therefore \Delta S_{rxn} = 50 - \frac{2}{2} \times 40 - \frac{1}{2} \times 60$

$\Rightarrow 50 - 60 - 30$

$\Delta S_{rxn} = -40J/K$

at equation, $\Delta G = O$

$$\Rightarrow T = \frac{\Delta H}{\Delta S} = \frac{30 \times 10^3}{40} = 750K$$

**33.** (*d*) $\Delta H = 30.558\ kJ/mol \quad \Delta S = 0.066\ kJK^{-1}mol^{-1}$

$p = 1\ atm$

$\Delta G = \Delta H - T\Delta S$

$O = 30.558 - 0.066 \times T$

$$T = \frac{30.558}{0.066}$$

$\therefore T = 463K \Rightarrow$

as temperature decreases $\Delta G$ becomes positive which means that reaction becomes Non-spontaneous.

**34.** (*b*) As we know that melting of ice below 273K is a non-spontaneous process and the value of $\Delta G$ for a non-spontaneous process is greater than 0.

Hence $\Delta G > 0$.

**35.** (*c*) A better gas welder is one which possesses high calorific value i.e., heat produced by 1g of Fuel.

Calorific value for ethane:-

$$\left(\frac{341.1}{30}\right) \Rightarrow 11.37 kcal/g \quad (\text{mol wt.} = 30)$$

Calorific value for ethyne:-

$$\left(\frac{310}{26}\right) = 11.92 kcal/g \quad (\text{mol wt} = 26)$$

Thus ethyne is better gas welder.

**36.** (*c*) 1 L of each rise in temperature 3°C.

**37.** (*d*) $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)\ \Delta H = -110kJ$ Exothermic Reaction

$C(s) + H_2O(g) \rightarrow CO + H_2(g)\ \Delta H = 132kJ$ Endothermic Reaction

Total heat Evolved I = Total heat Evolved in II

$n_1 \times 110 = n_2 \times 132$

$$\therefore \frac{n_1}{n_2} = \frac{132}{110} = 1.2$$

$$\frac{n_1}{n_2} \cong 1 = \frac{1}{1}$$

**38.** (*d*) Rise in temperature will be highest if the extent of neutralization reaction is higher due to which the heat evolved will be greater in quantity

g eq = eq = volume × normality

normality = molarity × n-factor

g eq of acid = g eq of bass

$V \times 0.5 \times 2 = (100 - V) \times 1 \times 1$

$V = 100 - V$

$2V = 100$

$V = 50$

$\therefore$ Volume of $H_2SO_4$ = V = 50 ml.

Volume of KOH = 100 – V = 100 – 50 = 50 ml.

50:50

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**39.** (*c*) $\Delta rG^\circ = \sum \Delta_f G^\circ(\text{product}) - \sum \Delta_f G^\circ \text{reactats}$

$\Rightarrow 2(\Delta_f G^\circ)_{H_2} - O$

$\Rightarrow 2 \times 1.3\,\text{kJ mol}^{-1} = 2.6\,\text{kJ mol}^{-1}$

**40.** (*c*) For cyclic process : $\Delta U = 0 \Rightarrow q = -W$

Isothermal process: $\Delta U = 0 \Rightarrow q = -W$

Adiabatic process: $q = 0 \Rightarrow \Delta U = -W$

Isochoric process: $q = 0 \Rightarrow \Delta U = q$

In Adiabatic process $q = 0 \neq \Delta E$

## Exercise-3 (JEE Advanced Level)

**1.** (*a,b,d*)

Color, kinetic energy per mole and specific volume are intensive properties because it does not depend on mass.

**2.** (*a,b,c*)

The boundary may be real or imaginary which is separated to the system and surrounding perfectly isolated system can't be achieved. In reversible steps energy change in each step can be reversed.

**3.** (*b,d*)

$W_{irrv} > W_{rev}$

$\Delta E_{rev} = \Delta E_{irr}$ (Because $\Delta E$ is a state function)

**4.** (*a,b,c*)

The isothermal expansion means a gas is expanding from initial volume to final volume at constant temperature. The internal energy and enthalpy depend upon the temperature only in case of ideal gas.

**5.** (*b,c,d*)

For adiabatic process, $PV^\gamma$ = constant

$$\frac{dP}{dV} = -\gamma \frac{p}{V}$$

From graph, slope (B) > slope (A)

$\gamma_B > \gamma_A$

For mono-atomic gas, $\gamma = 1.67$ (He, Ar)

For diatomic and linear polyatomic gas, $\gamma = 1.4$ ($H_2$, $N_2$ and $CO_2$)

For non-linear polyatomic, $\gamma = 1.33$

**6.** (*b,d*) $PV^\gamma$ = constant (for reversible adiabatic process)

$\Delta H = C_p \Delta T$

$\Delta T$ changes enthalpy changes.

**7.** (*a,b,c,d*)

Sublimation of a solid requires heat, it means system is absorbed heat ($q > 0$). At low temperatures internal energy increases while at high temperature internal energy decreases. $\Delta H > 0$

**8.** (*b,d*) Entropy is a measure of the disorder or randomness of the system. The greater the randomness in a system, greater is its entropy. The randomness is greater in liquid state as compared to solid state So the entropy increases when ice melts into water. Similarly, in vaporization of Camphor, it converts from solid to gas which increases randomness or entropy.

**9.** (*a,b*) Whenever a substance is transforming from a liquid state to a solid state, the change in enthalpy ($\Delta H$) is negative. This process is commonly known as the freezing and results in the molecules within the substance becoming more ordered. In Other words randomness decreases.

**10.** (*a,b,d*)

Molar entropy is greater for heavier gas under identical conditions.

**11.** (*a,b,d*)

In adiabatic process, no heat exchange between the system and the surrounding.

$\therefore \Delta S_{surr} = 0$

For adiabatic reversible process, $\Delta S_{univ} = 0$

$\Delta S_{sys.} + \Delta S_{surr} = 0$

$\Rightarrow \Delta S_{sys} = 0$

$\therefore$ isoentropic process

For irreversible adiabatic compression, $(\Delta S_{sys.})_{rev.} > 0$

For irreversible isothermal compression,

$$\Delta S_{surr} = \frac{-Q_{irr}}{T} > 0$$

**12.** (*a,b,c*)

The normal boiling point of a liquid X is 400K. At boiling point temperature equilibrium exist between liquid to vapor.

Thus, $\Delta G = 0$

At 400K and 2 atm pressure, the liquid does not boil so that process is non-spontaneous and hence, $\Delta G$ = +ve.

At 400K and 0.1 atm pressure, the liquid boil spontaneously and thus $\Delta G$ = –ve.

**13.** (*a,b,c,d*)

For an isothermal expansion of ideal gas.

Isothermal means temperature is constant during whole process

$\Delta H = 0$

$\Delta E = 0$

$\Delta G = \Delta H - T\Delta S$

$= 0 - T\Delta S$

$\Delta G = T\Delta S$

**14.** (*b,c,d*)

Electrolysis of water is the decomposition of water ($H_2O$) into oxygen ($O_2$) and hydrogen ($H_2$) due to an

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electric current being passed through the water. Graphite is more stable than diamond. Hence, energy is required for graphite to convert into diamond. Dehydrogenation of ethane to ethylene is endothermic reaction because energy is required for removing hydrogen from ethane molecule.

**15.** *(a,b,d)*

Heat of reaction depends on physical state of the reactants for example, some reactant may react only in certain physical state, Also heat may need to be supplied to change the state of state, like transforming it from liquid to gas.

Since, enthalpy is also a state function so the final product are obtained from reactant according to Kirchoff's law, $\Delta H_2 = \Delta H_1 + C_p \Delta T$. Kirchoff's law describes the enthalpy of a reaction's variation with temperature changes.

To study heat of reaction, whether the reaction is carried out at constant pressure or at constant volume, because heat is a path function, but when it is studied at constant pressure or volume it become state function.

**16.** *(a,b,c)*

The elemental form of carbon is graphite and diamond.

$$\underset{(\text{graphite})}{C} + O_2(g) \rightarrow CO_2(g)$$

**17.** *(a,b)* $C(s) + O_2(g) \rightarrow CO_2(g) \quad \Delta H_1 = -394 kJ ...(i)$

$$O_2(g) + 2H_2(g) \rightarrow 2H_2O(l) \qquad \Delta H_2 = -574 kJ ...(ii)$$

$$CO_2(g) + 2H_2O(\ell) \rightarrow CH_4(g) + 2O_2(g) \; \Delta H_3 = +892 kJ \; ...(iii)$$

On adding (i), (ii), (iii) equations we get,

$$C(s) + 2H_2(g) \rightarrow CH_4(g) \Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

$\Delta H = -394 - 574 + 892 = -76$ kJ.

$= -18.13$ kcal.

**18.** *(c)* $HNO_3 + LiOH \rightarrow LiNO_3 + H_2O \; \Delta H = +57.32$ kJ.

**19.** *(a,b,c)*

The formation of $CO_2$ is in two steps

$$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$$

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$$

$\therefore H_f^\circ \neq \Delta H^\circ_{reaction}$

Similarly for $N_2O_3$ and $CH_2Cl_2$ formation.

**20.** *(b,c,d)*

Molar heat of formation is the quantity of heat evolved or absorbed when one mole of substance is formed from its constituent elements under given conditions of temperature and pressure.

**21.** *(a,b,c)* Fact Based

**22.** *(b)* U is a state function so that $\Delta U_{A \rightarrow B}$ is not depending on path.

$\Delta U_{A \rightarrow B} = Q + W$ (1st law)

$\Delta U_{A \rightarrow B} = (80 - 30)$ J $= 50$ J

$\Delta U_{A \rightarrow B} = Q_{A \rightarrow D \rightarrow B} + W_{A \rightarrow D \rightarrow B}$

$50J = Q_{ADB} + (-10J)$

$Q_{ADB} = (60J)$

**23.** *(a)* $\Delta U_{BA} = Q_{curve} + W$

$-50$ J $= Q + (+20J)$

$Q = -70J$ ; heat is liberated

**24.** *(d)* $\Delta E_{A \rightarrow D} = Q + W$

$-40J = q_{AD} + (-10)$ (Given in question number 22)

$q_{AD} = -30J$

**25.** *(a)* $\Delta S = 2.303 \, n \, C_V \log\left(\frac{T_2}{T_1}\right) + 2.303 nR \log\left(\frac{V_2}{V_1}\right)$

At same temperature, $T_1 = T_2$

$\Delta S = 2.303 \times 2 mol \times 8.314 \, JK^{-1} \, mol^{-1} \log (50/5)$

$\Delta S = 38.23 \, JK^{-1}$.

**26.** *(b)* Isobaric process, $P_1 = P_2$

$$\Delta S = 2.303 \, nCp \left(\frac{T_2}{T_1}\right)$$

$$23.03 \, JK^{-1} = 2.303 \times 1 \, mol \times C_p \log\left(\frac{600}{300}\right)$$

$10JK^{-1} = 1 \, mol \times C_p \times \log 2$

$$C_p = \frac{10}{\log 2} JK^{-1}mol^{-1}$$

$C_p - C_v = R$

$$\frac{10}{\log 2} - C_V = R \Rightarrow C_V = -R + \frac{10}{\log 2}$$

**27.** *(d)* $\Delta G = -$ve (For spontaneous process)

$\Delta H - T\Delta S < 0$

$30 \, kJ - T \times 0.07 < 0$

$30 < T \times 0.07$

$$T > \frac{30}{7} \times 100 \Rightarrow T > 428.6K$$

For non-spontaneous $\Rightarrow T < 428.6$ K.

**28.** *(b)* $CaCO_3 \xrightarrow{\Delta} CaO(s) + CO_2(g)$

$\Delta G = \Delta H - T\Delta S$

Heat is supplied to $CaCO_3$ for decomposition into CaO and $CO_2$. Therefore, it is endothermic in nature.

Entropy also increases, therefore, $\Delta S = +$ve (small)

At high temperature $T\Delta S$ term overweigh and it become more $-$ve so that $\Delta G = -$ve at high temperature.

**29.** *(a)* $\Delta G = \Delta H - T\Delta S$

$\Delta G = 2.1 \times 1000 - 298 \times 20$

$= 2100 - 5560 =$ -ve

Spontaneous

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**30.** (*c*) $\Delta G = \Delta H - T\Delta S < 0$

$100 - T \times 0.05 < 0$

$100 < T \times 0.05$

$T > \frac{100}{0.05}$

$T > 2000$ K.

**31.** (*d*) $\Delta G = \Delta H - T\Delta S$

$0 = -40 \times 1000 - 400\,\Delta S$

$-40000 = 400\,\Delta S$

$\Delta S = -100$ cal $K^{-1}$.

**32.** (*a*) $\Delta G = \Delta H - T\Delta S$

$= -15 \times 1000 + 300 \times 7.2$

$= -12.84$ kcal. $mol^{-1}$.

$= -ve$.

Spontaneous

**33.** (*a*) $\Delta H^\circ_{neu} = \Delta H^\circ_{ion} + \Delta_r H^\circ$

$-49.86$ kJ $mol^{-1} = \Delta H^\circ_{ion} + (-55.84$ kJ $mol^{-1})$

$\Delta H^\circ_{ion} = +5.98$ kJ $mol^{-1}$

**34.** (*b*) Diacidic base produce twice $OH^-_{ions}$

$\Delta H^\circ = 2 \times (-55.84$ kJ $mol^{-1})$

$= -111.68$ kJ

**35.** (*b*) $NaOH \rightarrow V_1$ volume (mL)

$H_2A \rightarrow V_2$ volume (mL)

At complete neutralization, $(meq)_{Acid} = (meq)_{Base}$

$2 \times 0.05\, V_2 = 0.1 \times V_1$

$\frac{V_1}{V_2} = 1$

$V_1 + V_2 = 100$

$2V_1 = 100$

$V_1 = 50$

$V_2 = 50$

$V_1 : V_2 = 50 : 50$

**36.** (*a*) Reversible isothermal expansion of an ideal gas work done, $W = -2.303\, nRT \log\left(\frac{V_2}{V_1}\right)$

$\Delta H = 0$ (for ideal gas) in case of reversible isothermal expansion.

**37.** (*b*) A process carried out infinitesimally slowly, then process is known as reversible.

**38.** (*c*) $(\Delta G)_{T,P} \leq 0 \Rightarrow$ process is in equilibrium when it is equal to zero and spontaneous when it is negative. Therefore, $\Delta G \leq 0$, system is unable to do useful work.

**39.** (*a*) Reversible adiabatic, $\Delta S_{surr} = 0$ because there is no heat exchange between system and surrounding.

$MgCO_3(s) \xrightarrow{\Delta} MgO(s) + CO_2(g)\,\Delta S = +ve$

$\Delta S_{surr} < 0$

$\Delta S_{sys.} > 0$

**40.** (*d*) $C(s, graphite) + O_2(g) \rightarrow CO_2(g)$, this step is known as combustion, formation.

$CH_4 \rightarrow C(g) + 4H(g)$. This step is known as atomisation.

**41.** Faced Based

**42.** [7] Extensive property are those properties which depends on mass.

Free energy → extensive.

Vapour pressure → Intensive

Mole → extensive.

Kinetic energy → extensive

Entropy → extensive

Internal energy → extensive

Enthalpy → extensive

Specific heat capacity → Intensive

n → intensive

Total heat capacity → extensive

**43.** [8] Work and heat are path functions others are state functions.

$\therefore 8$

**44.** (*c*) $n = 2$ $\qquad$ $n = 2$

$T = 300K \xrightarrow[\text{at constant pressure}]{\text{expansion}} T' = ?$

$V = 20$ L $\qquad$ $V' = 40$L

$P = \frac{nRT}{V}$ $\qquad$ $P = \frac{nRT'}{V'}$

$\frac{300}{20} = \frac{T'}{40}$

$T' = 600$ K

$T' = 600K \xrightarrow[\text{adiabatic expansion}]{\text{Reversible}} T'' = ?$ (final temperature)

$V' = 40$ L $\qquad$ $V'' = 110$ L

$TV^{\gamma-1} = \text{constant}$

$600 \times (40)^{\gamma-1} = T\,(110)^{\gamma-1}$ $\quad (\because T'' = T)$

$6 \times \left(\frac{40}{110}\right)^{\gamma-1} = \frac{T}{100}$

$\Rightarrow \frac{T}{100} = 3$

**45.** [–4] For reversible process, $\int dW = -p\int_{V_1}^{V_2} dV$

$w = -k\int_{V_1}^{V_2} V^3 dV$

$= -\frac{k}{4}\left[v_2^4 - v_1^4\right]$

$= -\frac{1}{5\times4}\left(3^4 - 1^4\right)$

$= \frac{-1}{5\times4}\left(3^4 - 1^4\right)$

$= -4$ L atm

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**46.** [14] Work is done by the system during expansion,

$$W = -2.303\ nRT\ \log\left(\frac{P_1}{P_2}\right)$$

$$= -2.303 P_1V_1 \log\left(\frac{P_1}{P_2}\right)$$

$$= -2.303 \times 7 \times 1 \log\left(\frac{7}{1}\right)$$

$$= -13.6\ L\text{–}atm$$

$\therefore$ work is done on atmosphere = + 13.6L atm

**47.** [10] 1st law $\Delta U = Q + W$

$60\ J = +50\ J + W$

$W = 10\ J$

**48.** [24] Heat absorbs = 20 Ca = 20 × 4.2J = 84 J

Heat reject = 60 J

Work done = (84 – 60) = 24 J

**49.** [80] Work done in an adiabatic compression,

$$W = nCv(T_2 - T_1)$$

$$W = 2mol \times \frac{3R}{2}(T_2 - 30K)$$

$W = 8.314\ (T_2 - 30)\ J$

$P_1^{1-\gamma} T_1^{\gamma} = P_2^{1-\gamma} T_2^{\gamma}$

$1^{1-\gamma}.(30)^{\gamma} = 2^{1-\gamma}.T_2^{\gamma}$

$(30)^{\lambda} = 2^{1-\gamma}.\ T_2^{\gamma}$

$$\frac{60^{\gamma}}{2} = T_2^{\gamma}$$

$T_2 = 39.6\ K$

$W = 8.314\ (39.6 - 30)\ J$

$= 8.314 \times 9.6$

$= 79.9\ J$

**50.** Fact Based

**51.** [43] $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$

$= -29.8 - 298\ (-0.1)$

$= -29.8 - 29.8$

$= -59.6 \times 1000$

$\Delta G^{\circ} = -RT\ lm\ keq$

$-59600 = -2 \times 298 \times 2.303 \times \log_{10} K_{eq.}$

$\log_{10} K_{eq.} = 43.42$

**52.** Fact Based

**53.** [8] $A + 3B \rightarrow 2C$

$\Delta H^{\circ}_{rxn} = 2 \times (-10) - 3\ (\times 0) - 3(\times 0) = -20 \times 1000$

$\Delta S^{\circ}_{rxn} = 2 \times (45) - 3\ (30) - 40 = -40$

$\Delta G^{\circ}_{rxn} = \Delta H^{\circ}_{rxn} - T\Delta S^{\circ}_{rxn}$

$= -20000 - 300\ (-40) = -8000 = -8\ kcal$

**54.** [8] $H_2(g) \rightarrow 2H(g)$

$\Delta S^{\circ}_{rxn} = 2 \times 50 - 60 = 40\ JK^{-1}$

$\Delta G^{\circ}_{rxn} = 21.6 \times 1000\ JK^{-1}\ mol^{-1}$

$\Delta G^{\circ} = \Delta H^{\circ} - 300 \times 40$

$21600 = \Delta H^{\circ} - 300 \times 40$

$\Delta H^{\circ} = 21600 + 12000$

$= 33600\ J\ mol^{-1}$

$= 33.6\ kJ\ mol^{-1}$

$= 8\ kCa\ mol$

**55.** [–50] $X \rightleftharpoons Y$ $\qquad \Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$

$\Delta H^{\circ} = 25\ kJ$ $\qquad 40.2 = 25 - 300 \times \Delta S^{\circ}$

$\Delta G^{\circ} = -RT \times 2.303 \log_{10} K_{eq}$ $\quad \Delta S^{\circ} = -50.67\ kJ$

$= -8.314 \times 300 \times 2.303 \times \log_{10} 10^{-7}$

$= 40.2\ kJ$

**56.** [40] At equilibrium, $\Delta G = 0$

$\Delta G = \Delta H - T\Delta S$

$0 = -30 \times 1000 - 750 \times \Delta S$

$$\Delta S = -\frac{30,000}{750} = -40$$

$$S_{NH_3} - \frac{1}{2}S_{N_2} - \frac{3}{2}S_{H_2} = -40$$

$$50 - \frac{1}{2} \times 60 - \frac{3}{2}S_{H_2} = -40$$

$$20 - \frac{3}{2}S_{H_2} = -40$$

$$-\frac{3}{2}S_{H_2} = -60$$

$$S_{H_2} = 40$$

**57.** [9] $C_6H_{12} + 9O_2 \rightarrow 6CO_2 + 6H_2O$ $\quad \Delta H^{\circ}_1 = -3920$

$$C_6H_{10} + \frac{17}{2}O_2 + 6CO_2 + 5H_2O \quad \Delta H_2O = -3717$$

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \qquad \Delta H^{\circ}_3 = -241$$

$\Delta H^{\circ}_{Hydrogen} = \Delta H^{\circ}_2 + \Delta H_3{}^{\circ} - \Delta H_1{}^{\circ} = -3717 - 241 + 3920$

$= -38\ kJ\ mol^{-1}$

$= -9\ kcal\ mol^{-1}$

**58.** [3] moles of glucose, $n = \frac{0.36g}{180g\,mol^{-1}} = 0.002$ moles

$$= \frac{600 \times 10}{0.002} = 3 \times 10^6\ J/mol$$

$= 3\ MJ/mol$

**59.** Fact Based

**60.** [–86] $\Delta_f H^{\circ} = \Delta_{sub} H^{\circ} + \frac{1}{2}\Delta_{diss} H^{\circ} + \Delta_i H^{\circ} + \Delta_{eg} H^{\circ} + \Delta_{lattice} H^{\circ}$

$$= 137 + \frac{1}{2} \times 144 + 496 + (-325) + (-742)$$

$= -362\ kJ\ mol^{-1}$

$= -86\ kcal\ mol^{-1}$

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**61.** [8] Standard enthalpy of formation is zero for all elements in their standard states.

$Br_2(l)$, C(graphite), $Cl_2(g)$, $F_2(g)$, $N_2(g)$, $P_{Black}$, $P_{red}$, $S_{Monoclinic}$

**62.** [53] $C_3H_6 + \frac{9}{2}O_2 \longrightarrow 3CO_2 + 3H_2O,\ \Delta H_1^\circ = -2091$

$3CO_2 + 3H_2O \longrightarrow C_3H_6 + \frac{9}{2}O_2\ \Delta H_1^\circ = +2091$ ...(i)

$3\times\left(C + O_2 \longrightarrow CO_2 \quad \Delta H_2^\circ = -393.5\right)$ ...(ii)

$3\times\left(\frac{1}{2}O_2 + H_2 \longrightarrow H_2O \quad \Delta H_3^\circ = -285.8\right)$ ...(iii)

$3C + 3H_2 \rightarrow C_3H_6$

$\Delta H_4^\circ = \Delta H_1^\circ + 3\Delta H_2^\circ + 3\Delta H_3^\circ$

$\Delta H_4^\circ = 53.1\ kJmol^{-1}$

**63.** [–87] $\Delta H = \Delta U + P\Delta V$

$-92 = \Delta U + 40\ bar \times (-1.25)\ L$

$-92\ kJ = \Delta U + (-50\ L\text{-bar})$

$-92kJ = \Delta U - 50 \times 101.26J$

$\Delta U = (-92 + 5.06)\ kJ$

$\Delta U = -87kJ$

**64.** [–40] $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O\ \Delta H_{coml} = -337.2$

$C_2H_2 + \frac{5}{2}O_2 \longrightarrow 2CO_2 + 2H_2O\ \Delta H_{coml} = -310.1$

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O\ \Delta H_f = -67.8$

$\Delta H_{Hydr} = \Delta H^1_{coml} + \Delta H_f - \Delta H_{comf} = -310.1 + (-67.8) + 337.2$

$\Delta H_{Hyd} = -40.7\ kcal\ mol^{-1}$

$\Delta H = \Delta U + P\Delta V$

$-40.7 = \Delta U + \Delta ng\ RT$

$-40.7 = \Delta U + (-1)\times 2 \times \frac{350}{1000}$

$\Delta U = -40.k\ Cal\ mol^{-1}$

**65.** [78] $-20 + 6 \times 99 + (C - C)$ bond ethalpy $= 170 \times 2 + 6 \times 52$

$(C - C) = 78$ kcal

**66.** [99.5] $\Delta H = \Delta U + \Delta(PV)$

$\Delta H = \Delta U + P_2V_2 - P_1V_1$ ...(i)

$P_1 = 1atm\ P_2 = 1001\ atm$

$V_1 = 1L\ V_2 = 0.99L$

Let $P = a + bV$

On finiding $a = 100001$, $b = -10^5$

$$W = -\int_{V_1}^{V_2}\left(100001 - 10^5 V\right)dV$$

$W = 501J$

$\Delta U = W = 501\ J$

In eqn (1) by using $\Delta U = 501\ J$

$\Delta U = 501 + (1001 \times 0.99 - 1 \times 1) \times 100$

$= 995005$

$\Delta H = 99.5\ kJ$

**67.** [100] $\Delta H = \Delta U + P\Delta V$

$|\Delta H - \Delta U| = P\Delta V$

$|\Delta H - \Delta U| = P|-V_{graphicte} + V_{diamond}|$

$$= 500\times10^3\ bar\left|-\frac{12g\ mol^{-1}}{2g\ cm^{-3}} + \frac{12g\ mol^{-1}}{3g\ cm^{-3}}\right|$$

$= 500 \times 10^3\ bar \times 2 \times 10^{-3}\ L\ mol^{-1} = 1000\ L\ bar\ mol^{-1}$

$= 100\ kJ\ mol^{-1}$

$(\because\ 1\ L\ bar = 100\ J)$

**68.** [–1200] for adiabatic revesible process, $T^\gamma P^{1-\gamma}$ = constant

$$T_2 = T_1\left(\frac{P_1}{P_2}\right)\left(\frac{1-4}{4}\right) = 600K$$

$$W_1 = W_{rev} = nC_v\,(T_2 - T_1) = 2\times\frac{3}{2}R\times300$$

$W_1 = 900R$

$$W_2 = -2.303nRT\ \log\frac{P_1}{P_2} = -2\times R\times600\times\ln\left(\frac{4\sqrt{2}}{1}\right)$$

$W_2 = 600R$

$$W_3 = -P\Delta V = -nRT = -\left(200R\times\frac{5}{2}\times0.7\right)$$

$W_3 = -2100R$

$W_{Total} = 900\ R - 2100\ R + 600\ R$

$= -600\ R = -1200\ Cal$

**69.** [1020] In 1st Case, $W_1 = -P\Delta V = -P - 0 \quad (\because \Delta V = 0)$

$W_1 = 0$

In 2nd Case, $W_2 = -2.303\ nRT\ \log\frac{P_1}{P_2}$

$$P_1 = \frac{2\times300}{V_1}$$

$$P_2 = \frac{P_1\times T_2}{T_1} = \frac{12000}{V_1}$$

$$W_2 = -2.303\times1\times2\times300\times\log\left(\frac{1}{2}\right) = +420$$

In 3rd case, $W_3 = -P_2 \times \Delta V$

$$= -\left(\frac{1200}{V_1}\times\frac{V_1}{2}\right) = 600\,cal$$

$W_{Total} = W_1 + W_2 + W_3$

$= 0 + 420 + 600$

$= 1.02\ k\ Cal$

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**70.** [32] For revensibly adiabatic process

$TV^{\gamma-1}$ = contant

$$\frac{T_1}{T_2}=\left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\frac{(u_{rms})_1}{(u_{rms})_2}=\sqrt{\frac{T_1}{T_2}}=\left(\frac{V_2}{V_1}\right)^{\frac{\gamma-1}{2}}=\left(\frac{V_2}{V_1}\right)^{\frac{\frac{7}{5}-1}{2}}=2^5=32$$

**71.** [–2880] $f_{av}=\frac{f_An_A+f_Bn_B}{n_A+n_B}=\frac{1\times6+2\times3}{1+2}=4$

$$\gamma_{avg}=1+\frac{2}{f_{avg}}=1+\frac{2}{4}=\frac{3}{2}$$

$TV^{\gamma_{avg}-1}$ = constant

$$T_2=T_1\left(\frac{V_1}{V_2}\right)^{\frac{3}{2}-1}=320\left(\frac{1}{4}\right)^{\frac{1}{2}}$$

$T_2 = 160K$

$\Delta U = n_T C_V \Delta T$

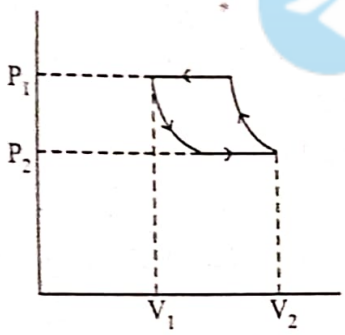
$= 3 \times 2R \times (160 - 320)$

$= -960\ R$

$\Delta H = n_T C_p \Delta T$

$= 3 \times 3R \times (-160) = -2880$ Cal

**72.** [14]




$$W_{net}=P_2(V_2-V_1)\ln\left(\frac{P_1}{P_2}\right)$$

$= P_2 (8 - 1)$

$= 2 \times 7$ L-bar

$= 14$ L-bar

**73.** [20] $\Delta G = -RT \ln K$

$\Delta G_{300} = -8.314 \times 300 \times \ln 2 = -1746$ Jmol$^{-1}$

$\Delta G_{400} = -8.314 \times 400 \times \ln 8 = -6983$ Jmol$^{-1}$

$-1746 = \Delta H - 300\ \Delta S$ ...(i)

$-6983 = \Delta H - 373 \times \Delta S$ ...(ii)

from (i) & (ii), $\Delta S = \frac{\Delta H + 1746}{300}$

$$-6983=\Delta H-\frac{373}{300}(\Delta H+1746)$$

$\Delta H = 20$ kJ mol$^{-1}$

**74.** [16] $\Delta S_{system} = nR \ln\left(\frac{P_1}{P_2}\right)$

$$=10\times8.314\times\ln\left(\frac{2}{1}\right)=56$$

$$\Delta S_{surr}=\frac{-Q}{T}$$

$\Delta U = Q + W$

$W = -P_{ext}(V_2 - V_1)$

$$q=-W=nRTP_{ext}\left(\frac{1}{P_2}-\frac{1}{P_1}\right)$$

$$q=10\times8\times T\left(1-\frac{1}{2}\right)\Rightarrow\frac{-Q}{T}=-40$$

$\Delta S_{system} + \Delta s_{surr} = 56 - 40 = 16$J

**75.** [–17.8]

$$2A(s)\rightleftharpoons B+2C+3D$$

At equilibrium (1 – 2x) x 2x 3x

$K_p = p_B^1 \cdot p_C^2 \cdot p_D^3$

$= (2)^1 . (4)^2 . (6)^3$

$= 32 \times 216$

$= 6912$

$\Delta G^o = -RT \ln kp = -2 \times 1000 \times \ln 6912 = -17.68$ kcal

**76.** [10] $dS = n\bar{C}_p \frac{dT}{T}$

$$\Delta S=1\times\int_{300}^{600}\left(10+10^{-2}T\right)\frac{dT}{T}$$

$= 10$ Jk$^{-1}$ mol$^{-1}$

**77.** [5] $\Delta H_{rxn} = 2\Delta H_A + \Delta H_B - 3\Delta H_C$

$= 2 \times (-100) + (-60) - 3\ (-285)]$

$= 595$ kJ mol$^{-1}$

$\Delta G = \Delta H - T\Delta S$

$= 595 - 300 \times 2$

$= -5$ kJ mol$^{-1}$

$|\Delta G| = 5$ kJ mol$^{-1}$

**78.** [22] $CH_3OCH_3 + 3O_2 \rightarrow 2CO_2 + 3H_2O - 348 = \Delta H^\circ_1$

$$2C+3H_2+\frac{1}{2}O_2\rightarrow C_2H_5OH\qquad\Delta H_2^0=-66$$

$C + O_2 \rightarrow CO_2 \quad \Delta H_3^\circ = -94$

$$H_2+\frac{1}{2}O_2\rightarrow H_2O\qquad\Delta H_4^\circ=-68$$

$C_2H_5OH \rightarrow CH_3OCH_3 \quad \Delta H^\circ = ?$

$\Delta H^\circ = -\Delta H_2^\circ - \Delta H_1^\circ + 2\Delta H_3^\circ + 3\Delta H_4^\circ$

$= -(-66) - (-348) + 2 \times (-94) + 3 \times (-68)$

$= 22$ kcal mol$^{-1}$

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**79.** [461] $\Delta H_{r\times n} = 2 \times \Delta_f H_f^\circ [H(g)] + \Delta_f H^\circ [O(g)] - \Delta_f H^\circ [H_2O(g)]$
$= (2 \times 218 + 249.2 + 241.8)$ kJ mol$^{-1}$.
$= 927$ kJ mol$^{-1}$

Average bond enthalpy of OH $= \frac{\Delta H_{r\times n}}{2} = \frac{927}{2}$

$= 463.5$ kJ mol$^{-1}$
$U = H - nRT$
$= 463.5 - \left(\frac{1\times 8.314\times 298}{1000}\right)$
$U = 461$ kJ mol$^{-1}$.

**80.** (i) [342]
$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$ $\Delta_f H^\circ = -83$ kJ mol$^{-1}$
$2 \times (C(s) \rightarrow C(g))$ $(\Delta_{sub}H^\circ = 719$ kJ mol$^{-1}) \times 2$
$3 \times (H_2(g) \rightarrow 2H(g))$ $(\Delta_{diss}H^\circ = 435$ kJ mol$^{-1}) \times 3$
Adding (2), (3) equation and subtract from (i) equation.
$2C(g) + 6H(g) \rightarrow C_2H_6(g)$ $\Delta H = -2826$ kJ mol$^{-1}$
$\Delta H = 0 - (6\,(Eh) + (C\text{-}C))$
$-2826 = -6 \times 414 + -(C\text{-}C)$
$-(C\text{-}C) = -342$
BE (C–C) = 342 kJ mol$^{-1}$

(ii) [891]
$CH_3CN(g) \rightarrow 2C$ (graphite) $+ 3/2\ H_2(g) + 1/2\ N_2(g)$
$\Delta H = -87$
$2C$ (graphite) $\rightarrow 2C(g)$ $\Delta H_1 = 2 \times 719$ kJ mol$^{-1}$

$\frac{3}{2}H_2(g) \rightarrow 3H(0)$ $\Delta H_2 = \frac{3}{2} \times 435$ kJ mol$^{-1}$

$\frac{1}{2}N_2(g) \rightarrow N(g)$ $\Delta H_3 = \frac{1}{2} \times 945$ kJ mol$^{-1}$

On adding all equation, we get
$CH_3CN(g) \rightarrow 2C(g) + 3H(g) + N(g)$
$\Delta H = \Delta H' + \Delta H_1 + \Delta H_2 + \Delta H_3$
$\Delta H = 2477$ kJ mol$^{-1}$
$2477 = BE_{C-C} + 3BE_{C-n} + BE_{C\equiv N}$
$BE_{C\equiv N} = 891$ kJ mol$^{-1}$

**81.** [26.5] $2C + 3H_2 \rightarrow C_2H_6$
$\Delta H_{rxn} = +\Delta H_{react} + \Delta H_{pro}$
$-2840 = 0 - (6\,(C\text{–}H) + 2(C\text{–}C))$
$-2840 = -(410 \times 6 + 2 \times (C\text{–}C))$
$2\,(C\text{–}C) = -380$
BE (C–C) = 380 kJ
$2C + 2H_2 \rightarrow C_2H_4$
$-2275 = 0 - (4(C\text{–}H) + (C = C))$
BE (C = C) = 635 kJ
In benzene, $= 6\,(C\text{-}C) + 3\,(C = C) + 6\,(C\text{–}H)$
$= 6 \times 380 + 3 \times 635 + 6\,(410)$
$= 6645$ kJ
Resonance energy = Actual – Theoretical
$= 6645 - 5530$
$= 1115$ kJ mol$^{-1}$
$= 26.5$ k Cal mol$^{-1}$

**82.** (–137) $CH_3OH\,(l) \rightarrow CH_3OH(g)$ $\Delta H_1 = 35.5$ kJ mol$^{-1}$
$CH_3OH(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
$\Delta H_{vap} = 3H_{CH} + H_{C\text{-}O} + H_{O\text{=}O} - (2H_{C\text{=}O}$ – Resonance energy $+ 4\ H_{OH})$
$= 3046 - 3137 = -91$ kJ mol$^{-1}$.
Latent heat of vaporisation of water is 40.6 kJ mol$^{-1}$
2 moles of water, $\Delta H_3 = 2 \times 40.6 = 81.2$ kJ mol$^{-1}$
$\Delta H = \Delta H_{vap} + \Delta H_2 + (-\Delta H_3)$
$= 35.5 - 91 - 81.2$
$= -136.7$ kJ mol$^{-1}$.
$\Delta H = -137$ kJ mol$^{-1}$.

## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*d*) As the heat is absorbed
$\therefore q = +2085$
Now for reversible isothermal process,
$q = -w$
$\therefore w = -q = -2085$

**2.** (*a*) $\Delta E = -1364.47$ kJ mol$^{-1}$
$T = 298$ k, $\Delta ng = -1$
$\because \Delta H = \Delta E + P\Delta V$
but $P\Delta V = \Delta ngRT$
$\Delta H = -1364.47 + [(-1) \times 0.0083 \times 298)$
$\Delta H = -1366.95$ kJ mol$^{-1}$.

**3.** (*b*) For methane
$CH_4(g) \rightarrow C(g) + 4H(g)$
So, $4 \times E_{C\text{-}H} = 360$ kJ mol
$E_{C\text{-}H} = 90$ kJ mol
For ethane,
$C_2H_6(g) \rightarrow 2(g) + 6H(g)$
$E_{C\text{-}C} + 6 \times E_{C\text{-}H} = 620$ kJ/mol
$E_{C\text{-}C} = 80$ kJ/mol

Therefore, $N_A\frac{hc}{\lambda} = 80\times 1000$ J

$\lambda = \frac{6.02\times 10^{23} \times 6.62\times 10^{-34} \times 3\times 10^8}{80000}$

$\lambda = 1.49 \times 10^3$ nm

**4.** (*b*) $\Delta G^\circ_{r\times n} = -RT \ln k_p$ ...(i)
$\Delta G^\circ_{r\times n} = 2\Delta G^\circ_F(NO_2) - 2FG^\circ_F(NO)$ ...(ii)
$2\Delta G^\circ_F(NO_2) - 2\Delta G^\circ_F(NO) = -RT\ln k_p$
$\Delta G^\circ_F(NO_2) = 0.5 \times (2\Delta G^\circ_F(NO) - RT \ln k_p)$
$\Delta G^\circ_F(NO_2) = 0.5 \times (2\Delta G^\circ_F(NO) - RT \ln k_p)$
$\Delta G^\circ_F(NO_2) = 0.5 \times [2 \times 86600 - R(298) \ln 1.6 \times 10^{12})]$ J/mol$^{-1}$

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5. (c) $C(s) + O_2(g) \rightarrow CO_2(g)$ ; $\Delta H = -393.5$ kJ/mol.

$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$; $\Delta H = -283.5$ kJ/mol.

$C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$; $\Delta H = -393.5 + 283.5$ kJ/mol $= -110$ kJ/mol.

6. (c) $C(s) + O_2(g) \rightarrow CO_2(g)$ ; $\Delta H = -393.5$ kJ/mol.

$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$; $\Delta H = -283.5$ kJ/mol.

$C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$; $\Delta H = -393.5 + 283.5$ kJ/mol $= -110$ kJ/mol.

7. (c) $CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g)$; $\Delta rH^\circ = +890.3$ kJ mol$^{-1}$

$-393.5 \quad -285.8 \quad ? \quad 0$

$\Delta_r H^\circ = \sum(\Delta_f H^\circ)_{Products} - \sum(\Delta_f H^\circ)_{reactants}$

$890.3 = [1 \times (\Delta_f) H^\circ)_{CH_4} + 2 \times 0] - [1 \times (-393.5) + 2(-285.8)]$

$(\Delta_f)H^\circ)_{CH_4} = 890.3 - 965.1 = -74.8$ kJ/mol

8. (c) From 1st law :

$\Delta U = q + w$

For adiabatic process:

$q = 0$

$\therefore \Delta U = w$

Work involve in adiabatic process is at the expense of change in internal energy of the system.

9. (c) $C_6H_5(\ell) + \frac{15}{2} O(g) \rightarrow 6CO_2(g) + 3H_2O(\ell)$

$\Delta ng = 6 - \frac{15}{2} = -\frac{3}{2}$

$\Delta U = -3263.9$ kj mol.

$\Delta H = \Delta U + \Delta ngRT$

$\Delta H = -3263.9 - \frac{3}{2} \times 8.314 \times 298$

$\Delta H = -3265.9$ kJ/mol.

10. (a) $C_p$ does not changes with change in pressure

11. (c) (A) $q + w = \Delta U$ = state function

(B) $Q \Rightarrow$ path function

(C) $H - TS = G \Rightarrow$ state function

12. (c) $q = P\Delta V$

$q = 16$

$C_P = 24$

$C_p = \frac{q_P}{\Delta T} \Rightarrow \Delta T = \frac{16}{24} K = \frac{2}{3} K$

13. (a) $w = -nRT \ln \frac{v_2}{v_2}$

$w = -nRT \ln \frac{v_b}{v_i}$

$|w| = nRT \ln \frac{v_b}{v_i}$

$|w| = nRT(\ln v_b - \ln v_i)$

$y = mx - c$

Slope of curve 2 is more than curve 1 and intercept of curve 2 is more negative than curve.

14. (a) As $PV = nRT$, where n & T are constant

$PV = k = nRT$ ...(i)

isothermal expansion $PVm = k$ (graph $\rightarrow$ c)

$P = \frac{k}{V_m} \rightarrow$ (Graph A)

15. (c) In order to be spontaneous $\Delta G^\circ$ should be –ve

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$0 = 491.1 \times 10^3 - T \times 198$

$T = \frac{491100}{198} = 2480$

If temp is above 2480 K, the reaction will be spontaneous.

16. (a) At the thermal equilibrium,

final temperature $T_f = \frac{T_1 + T_2}{2}$

$\Rightarrow$ for the 1st block, $\Delta S_I = C_P \ln \frac{T_f}{T_1}$

$\Rightarrow$ for the 2nd block, $\Delta S_{II} = C_P \ln \frac{T_f}{T_2}$

When brought in contact with each other,

$\Delta S = \Delta S_1 + \Delta S_{II} = C_P \ln \frac{T_f}{T_1} + C_P \ln \frac{T_f}{T_2}$

$= C_P \ln \left( \frac{T_f}{T_1} \times \frac{T_f}{T_2} \right) = C_P \ln \left[ \frac{T_f^2}{T_1 T_2} \right]$

$= C_P \ln \left[ \frac{\left( \frac{T_1 + T_2}{2} \right)}{T_1 T_2} \right] = C_P \ln \left[ \frac{(T_1 + T_2)^2}{4 T_1 T_2} \right]$

17. (a) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Delta G^\circ = A - BT$

In endothermic reaction $\Delta H = +$ve. Hence, A = +ve

18. (d) $H_2O(s) \rightarrow H_2O(l) \rightarrow H_2O(l) \rightarrow H_2O(g) \rightarrow H_2O(g)$

1 kg 1 kg 1 kg

at 273 K at 273 K at 373 K at 373 K at 383 K

$\Delta s = \Delta s_1 + \Delta s_2 + \Delta s_3 + \Delta s_4$

$= \frac{334}{273} + 4.2 \ell n \frac{373}{273} + \frac{2491}{373} + 2 \ell n \frac{383}{373} = 9.267 kJKg^{-1}K^{-1}$

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**19.** (*c*) $\Delta G = \Delta H - TDS < 0$

$$T = \frac{\Delta H}{\Delta H} = \frac{200}{40} = 5K$$

**20.** (*d*) Four moles of gas

$(N_2 + 3H_2) \rightarrow 2NH_3$ reacts to give only two moles of gas the hence entropy decreases.

**21.** (*c*) $\Delta G^\circ = -2.303RT \log K_{eq}$

$= -2.303 \times 8.314 \times 298 \log 10^{-14}$

$= -2.303 \times 8.314 \times 298 \times -14$

$= 79,881.87$

$\simeq 80$ KJ mol$^{-1}$

**22.** (*b*) $\Delta G = \Delta H - T\Delta S$

for spontaneous process at all temp. $\Delta G < 0$ and it is possible when $\Delta H < 0$ and $\Delta S > 0$.

**23.** (*c*) For cyclic process : $\Delta U = 0 \Rightarrow q = -w$

For isothermal process : $\Delta U = 0 \Rightarrow q = -w$

For adiabatic process : $q = 0 \Rightarrow \Delta U = W$

For isochoric process : $w = 0 \Rightarrow \Delta U = q$

**24.** (*d*) $\Delta H = n\int_{T_1}^{T_2} C_{p,m}dT = 3\times \int_{300}^{1000} (23+0.01T)dT$

$$= 3[\ 23(1000-300) + \frac{0.01}{2}(1000^2 - 300^2)]$$

$61950\ J \approx 62\ kJ$

**25.** (*a*) $n = 5; T_i = 100\ K; T_f = 200\ K;$

$C_V = 28$ J/mol K; Ideal gas

$\Delta U = nC_V\Delta T$

$= 5$ mol $\times$ 28 J/mol K $\times$ (200 – 100) K

$= 14,000\ J = 14$ kj

$\Rightarrow C_p = C_v + R = (28 + 8)$ J/mol K

$= 36$ J/mol K

$\Rightarrow \Delta H = nC_p\Delta T = 5$ mol $\times$ 36 J/mol K $\times$ 100 K

$= 18000\ J = 18$ kJ

$\Delta H = \Delta U + \Delta(PV)$

$\Rightarrow \Delta(PV) = \Delta H - \Delta U = (18 - 14)$ kJ $= 4$ kJ

**26.** (*a*) $\Delta U = q + w$

$q = -2$kJ, $W = 10$ kJ

$\Delta U = 8$kJ

**27.** (*c*) $W = -P_{ext}(V_2 - V_1)$

$= -1$ bar $\times$ (10 – 1)lit

$= -9$ bar $-$lit $= -900$ J

$= -0.9$ kJ

**28.** (*d*) $\Delta G = 120 - \frac{3}{8}T$

At equilibrium $\Delta G = 0$

$\therefore T = 320$ K

$X \rightleftharpoons Y$, if $T > 320$ K, then $\Delta G$ becomes negative. Thus, the reaction will proceed in the forward direction and the amount of Y will be higher than its amount on equilibrium.

Here only at $T = 350$ K is greater than 320 K.

**29.** (*a*) (ii) + (iii) = (i)

$Y + z = x$

**30.** (*d*) $I_{2(s)} \rightarrow I_{2(g)}$ : $\Delta H_1 = 24$ cal/g at 200° C

$\Delta H_2 = \Delta H_1 + \Delta C_{Prxn}(T_2 - T_1)$

$= 24 + (0.031 - 0.055) \times 50 = 24 - 1.2$

$= 22.8$ cal/g

**31.** (*c*) $C_7H_{16}(\ell) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(\ell)$

$\Delta n_g = n_p - n_r = 7 - 11 = -4$

$\because \Delta H = \Delta U + \Delta n_g RT$

$\therefore \Delta H - \Delta U = -4$ RT

**32.** [48.00]

$$|W| = \frac{1}{2}(6+10)\times 6 = 48J$$

**33.** [6.25 to 6.25]

$\Delta U = nC_v\Delta T$

$5000 = 4 \times C_V$ ( (500 – 300)

$C_v = 6.25$ JK$^{-1}$ mol$^{-1}$

**34.** [–2.70 to –2.70]

$\Delta H = \Delta U + \Delta$ ngRT $= 2.1 \times 10^3 + 2(2)(300)$

$= 2100 + 1200 = 3300$ cal

$\Delta G = \Delta H - T\Delta S = 3300 - (300)(20) = 3300 - 6000$

$= -2700$ cals $= -2.7$ kcal

**35.** (*c*) Fact based

**36.** [–192.50 or –85.00]

$C_2H_6(g) + 3.5O_2(g) \rightarrow 2CO_2(g) + 3H_2O(I)$

From the given data

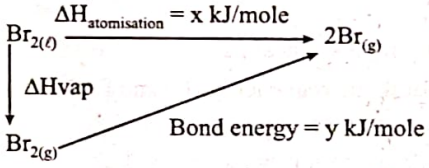
$2 \times (-286) + 3 \times (-393.5) - (-1560) = -572 - 1180.5 + 1560 = -192.50$kJ/mol.

or

$C_2H_6(g) + 3.5O_2(g) \rightarrow 2CO_2(g) + 3H_2O(I)$

$2 \times (-393.5) + 3 \times (-286) - (-1560) = -85$ kJ/mol\

**37.** (*d*)




$\Delta H_{atomisation} = \Delta H_{vap}$ + Bond energy

Hence $x > y$

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**38.** [–326400]

$C_2H_5OH\ (l) + 3O_2(g) \rightarrow 2CO_2\ (g) + 3\ H_2O(l)$

$\Delta H = -327$ kcal; $\Delta n_g = -1$

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta U = -327 + 2 \times 10^{-3} \times 300$

$= -326.4$ kcal

$= -326400$ cal

**39.** (*d*) $\Delta_{Sol}H^\circ = \Delta_{lattice}H^\circ + \Delta_{Hyd}.H^\circ$

$4 = 788 + \Delta_{Hyd}.H^\circ$

$\Delta_{Hyd}.H^\circ = -784$ kJ / mol

**40.** [–13537.57]

$2A(g) \rightarrow A_2(g)$

$\Delta U^\circ = -20$ kJ/mol and $\Delta S^\circ = -30$ J/K mol

$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$

$= -20000 - 1 \times 8.314 \times 298 = -22477.57$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$= -22477.57 + 298 \times 30 = -13537.57$

**41.** (*a*) $\Delta G^\circ = -RT \ln K$

$T_1 = 25^\circ C\ K_1 = 10$

$\Delta G^\circ$ at $T_1 = -8.314 \times 298 \times 2.303 \times \log 10$

$= -5.71$ kJ/mol

$\Delta G^\circ$ at $T_2 = -8.314 \times 373 \times 2.303 \times \log(100)$

$-14.29$kJ/mol

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$-5.71 = \Delta H^\circ - 298\ (\Delta S^\circ)$

$-14.29 = \Delta H^\circ - 373\ (\Delta S^\circ)$

$\Delta H^\circ = 28.4$ kJ/mol

**42.** [189494]

$H_2O(l) \rightarrow H_2O(g)$

$\Delta H_{vap} = \Delta U + \Delta n_g RT$

$\Delta U = \Delta H_{vap} - RT$

$= 41{,}000 - 8.314 \times 373$ J mol$^{-1}$

$= 37898.878$

For 5 moles, $\Delta U = 37898.878 \times 5$ J

$= 189494$ J

**43.** (*a*) $P_{ext} = 0$

$w = -P_{ext}\ \Delta V$

$= 0$

**44.** (*b*) For 1 mole of ideal gas.

- Both internal energy (U) and Enthalpy (H) depends on temperature
- Compressibility factor Z = 1
- $C_{P,m} - C_{V,m} = R$

and $dU = C_{V,m}dT$

**45.** [15] Moles(n) = 5, T = 293 K

Process → Isothermal → Irreversible

$W = -P_{ext}\Delta v$

$$= -4.3 \times \left(\frac{5 \times 293R}{1.3} - \frac{5 \times 293R}{2.1}\right) = -15.34 \text{ kJ}$$

$\Delta v = 0$, $W = -Q = 15.34$ kJ/mol

$Q = 15.34$ kJ/mol

So, the answer is 15.

**46.** [50]

Mass organic compound = 1.6 gm

Mass of AgBr = 1.88 gm

$$\text{Moles of Br} = \text{Moles of AgBr} = \frac{1.88}{188} = 0.01$$

Mass of Br = $0.01 \times 80 = 0.80$ gm

$$\%\text{ of Br} = \frac{0.80 \times 100}{1.60} = 50\%$$

**47.** [31]

$\Delta H - \Delta U = \Delta n_g RT$

$\Delta n_g = 1 - 0 = 1$

$\Delta H - \Delta U = 1 \times 8.31 \times 373$

$= 30.99 \times 10^2$ J/mol $= 31 \times 10^2$ J/mol

**48.** [–741.5]

For given reaction

$\Delta n = 1 + 1 - 1.5 = 0.5$

$\Delta H = \Delta U + \Delta_{ng} RT$

$= -742.7 + 0.5 \times 8.314 \times 10^{-3} \times 298$

$= -741.5$ kJ mol$^{-1}$

**49.** [–2218]

$$\underset{50g}{Fe} + 2HCl \rightarrow FeCl_2 + H_2(g)$$

$$n = \frac{50}{55.85}$$

$W = -P_{ext}\Delta V$, $PV = nRT$

$W = -\Delta n_g RT$

$W = -1 \times 8.314 \times 298$

$$W \text{ for } \frac{50}{55.85} \text{ mole} = -8.314 \times 298 \times \frac{50}{55.25}$$

$= -2218.059$ J

**50.** [101]

$\Delta_{sub}H = \Delta_{Fus}H + D_{Vap}H$

$\Delta_{sub}H = 98.2 + 2.8$

$\Delta_{sub}H = 101$ kJ/mol

**51.** [200]

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Delta G < 0$ (spontaneous)

$\Delta H^\circ - T\Delta S^\circ < 0$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ}$$

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$$T > \frac{80000}{2T}$$

$2T^2 > 80000$

$T > 200$ (minimum T for spontaneous)

**52.** (*b*) $\Delta G = \Delta G^\circ + RT \ln K$

At equilibrium,

$\Delta G = \Delta G^\circ + RT \ln K = 0$

$\Delta G^\circ = -RT \ln K$

$-RT \ln K < 0$

$RT \ln K > 0$

$K > 1$

**53.** [1380]

$\Delta G^\circ = -RT \ln K_P$

$= -R(300)\ln(100)$

$\Delta G^\circ = -1380R$

$x = 1380$

**54.** [336]

$\Delta G = -43.4$ kJ/mol

$\Delta H = 51.4$ kJ/mol

$\Delta G = \Delta H - T\Delta S$

$-49400 = 51400 - 300\,\Delta S$

$$\Delta S = \frac{+100800}{300}$$

$\Delta S = 336\ JK^{-1}\ mol^{-1}$

**55.** [5]

Given, $\Delta H = -57.8\ kJ\ mol^{-1}$

$\Delta S = -176\ JK^{-1}\ mol^{-1}$

$T = 298$ K

Using Gibb's free energy relation

$\Delta G = \Delta H - T\Delta S$

Where, = change in Gibb's free eergy

$\Delta H$ = Change in enthalpy

T = Temperature

$\Delta S$ = Change in entropy

$$\Delta G = 57.8 KJ/mol - \left[298K \times \left(-176 Jk^{-1}\,mol^{-1}\right)\right]$$

$$= 57.8 KJ/mol - \left(298 \times \frac{-175}{1000} kJ\right) [\because 1kJ = 1000J]$$

$= -5.352$ kJ/mol

$|\Delta G| = 5.352$

**56.** [964]

**57.** (*c*) $\Delta G^\circ = -RT \ln K$ ...(i)

$\Delta G = \Delta H^\circ - T\Delta S^\circ$ ...(ii)

From (i) and (ii)

$-(\Delta H^\circ + T\Delta S^\circ) = RT \ln K$

$$\ln K = \frac{-(\Delta H^\circ - T\Delta S^\circ)}{RT}$$

**58.** (*a*) Fact based

**59.** [964]

$$T_{rain} = \left(\frac{\Delta H^0}{\Delta S^0}\right)$$

$$\Delta H^0_{rem} = \left[\Delta_f H^0(Fe) + \Delta_f H^0(CO)\right] -$$

$$= \left[\Delta_f H^0(FeO) + \Delta_f H^0\left(C_{(graphite)}\right)\right]$$

$= [0 - 110.5] - [-266.3 + 0]$

$= 155.8$ kJ/mol

$$\left[\Delta S^0(Fe') + \Delta S^0\left(C_{(graphite)}\right)\right]$$

$= [27.28 + 197.6] - [57.49 + 5.74]$

$= 161.65\ J/mol-K$

$$T_{rain} = \frac{155.8 \times 10^3\ J/mol}{161.65\ J/mol - K} = 963.8K$$

$\simeq$ 964k (Nearest integer)

**60.** [855]

$3CH \equiv CH(g) \rightleftharpoons C_6H_6(\ell)$

$\Delta G^\circ = \Delta G^\circ_F(C_6H_6) - 3\Delta G^\circ_F(CH \equiv CH)$

$= -1.24 \times 10^5 - 3(-2.04 \times 10^5)$

$= 4.88 \times 10^5$ J/mol

$\Delta G^\circ = -RT \ln K = -2.303RT \log K$

$$\log K = \frac{-4.88 \times 10^5}{2.303 \times 8.314 \times 298}$$

$|\log K| = 85.5$

$x \times 10^{-1} = 85.5$

$x = 855$

**61.** [309]

$S(s) + 3F_2(g) \rightarrow SF_6(g);\ \Delta H = -1100$ kJ ...(i)

$S(s) \rightarrow S(g);\ \Delta H = 275$ kJ ...(ii)

$\frac{1}{2}F_2 \rightarrow F(g);\ \Delta H = 80$ kJ ...(iii)

Equation (ii) + 6 equation (iii) – equation (i)

$SF_6 \rightarrow S(g) + 6F(g);\ \Delta H = 1855$ kJ

Thus average bond energy for S – F bond

$$= \frac{1855}{6} = 309.16 kJ$$

**62.** [336]

$\Delta G = -43.4$ kJ/mol

$\Delta H = 51.4$ kJ/mol

$\Delta G = \Delta H - T\Delta S$

$-49400 = 51400 - 300\,\Delta S$

$$\Delta S = \frac{+100800}{300}$$

$\Delta S = 336\ JK^{-1}\ mol^{-1}$

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63. [238]

$3CaO + 2Al \rightarrow 3Ca + Al_2O_3$

Al and Ca are in its standard state so enthalpy of formation of these is zero.

$\therefore \Delta H_f^\circ = \Delta H_f^\circ(Al_2O_3) - 3\Delta H_f^\circ(CaO)$

$\Delta H_f^\circ = -1675 - 3(-625)$

$\Delta H_f^\circ = 230$ kJ

64. [21] Heat of combustion per mol for 12 g $= -2.48 \times 10^2$ kJ

For 1g of graphite:

$$\text{Heat of combustion} = \frac{-2.48\times10^2}{12} = -2066\,kJ = -21$$

65. [131]

$C_2H_6 \rightarrow C_2H_4 + H_2$

$\Delta H = [(E_{C-C} + 6E_{C-H}) - (E_{C=C} + 4E_{C-H} + E_{H-H})]$

$\Delta H = [(347 + 6 \times 414) - (611 + 4 \times 414 + 433)]$

$= 2831 - 2700$

$= 131$ kJ/mol

Hence, the reaction enthalpy $\Delta_r H$ is 131 kJ mol$^{-1}$.

66. [5]

Given, $\Delta H = -57.8$ kJ mol$^{-1}$

$\Delta S = -176$ JK$^{-1}$ mol$^{-1}$

$T = 298$ K

Using Gibb's free energy relation

$\Delta G = \Delta H - T\Delta S$

Where, = change in Gibb's free eergy

$\Delta H$ = Change in enthalpy

T = Temperature

$\Delta S$ = Change in entropy

$$\Delta G = 57.8\text{KJ/mol} - \left[298\text{K} \times \left(-176\text{Jk}^{-1}\,\text{mol}^{-1}\right)\right]$$

$$= 57.8\text{KJ/mol} - \left(298 \times \frac{-175}{1000}kJ\right)[\because 1\text{kJ} = 1000\text{J}]$$

$= -5.352$ kJ/mol

$|\Delta G| = 5.352$

67. [82] Milli moles of HCl $= 200 \times 0.2 = 40$

$\Rightarrow$ Milli moles of NaOH $= 300 \times 0.1 = 30$

$$\Rightarrow \text{Heat released} = \left(\frac{30}{1000} \times 57.1 \times 1000\right) = 1713\,J$$

$\Rightarrow$ Mass of solution = 500 mL × 1gm/mL = 500gm

$$\Rightarrow \Delta T = \frac{Q}{m \times 0} = \frac{1713\,J}{500g \times 4.18\frac{J}{3-K}} = 0.8196\,K$$

$= 81.96 \times 10^{-2}$ K

68. [38] $H_2O(\ell) \rightleftharpoons H_2O,\ \Delta H_{vap} = 41\text{kJ/mol}$

$\Delta H = \Delta E + \Delta n_g RT$

$\Delta E$ is internal energy change

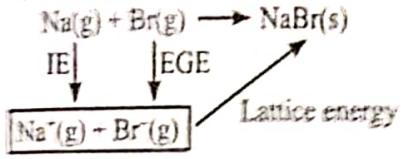
$\Delta n_g$ is change in gaseous moles $= 1 - 0 = 1$

$41 = \Delta E + 1 \times 8.3 \times 10^{-3} \times 373$

$\Delta E \approx (41 - 3) = 38$ kJ mol$^{-1}$.

69. [–5576]

The total enthalpy change is the sum of the enthalpy changes of each reaction.




the enthalpy of reaction of :

NaBr = IE + EGE + lattice energy

$\Delta_r H = 495.8 - 325 - 728.4$

$= -557.6$ kJ/mol

$= -5576 \times 10^{-1}$ kJ/mol

70. [131]

$C_2H_6 \rightarrow C_2H_4 + H_2$

$\Delta H = [(E_{C-C} + 6E_{C-H}) - (E_{C=C} + 4E_{C-H} + E_{H-H})]$

$\Delta H = [(347 + 6 \times 414) - (611 + 4 \times 414 + 433)]$

$= 2831 - 2700$

$= 131$ kJ/mol

Hence, the reaction enthalpy $\Delta_r H$ is 131 kJ mol$^{-1}$.

71. [82]

Millimoles of $H_2SO_4 = 400 \times 0.2 = 80$

meq of NaOH $= 600 \times 0.1 = 60$

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

$t = 0 \quad 80 \quad 60 \quad --$

$50 \quad - \quad 30-$

30 m moles of product of formed.

$H^+ + OH^- \rightarrow H_2O,\ \Delta H = -57.1\ |kJ|$ mol.

$$\text{Moles of } H^+ \text{ \& } OH^- \text{ neutralised} = \frac{60}{1000}$$

$$\therefore \Delta H = \frac{60}{1000} \times (-57.1) = 3426\,J/mol$$

Total volume = 1 L, mass = 1000 g

$mS\Delta T = \Delta H$

$1000 \times 4.18 \times \Delta T = 3426$

$\Delta T = 0.8196 = 81.9 \times 10^2\,K = 82 \times 10^{-2}\,K$

x = 82

72. [718]

$\Delta_F H^\circ_{KCl} = \Delta_{Sub} H^\circ_{(K)} + \Delta_{ionisation} H^\circ_{(K)}$

$+ \Delta_{EGE} H^\circ_{(Cl)} = \Delta_{lattice} H^\circ_{(KCl)} + \frac{1}{2}\Delta_{bond} H^\circ_{(Cl_2)}$

$-436.7 = 89.2 + 419 + (-348.6) + \Delta_{lattice} H^\circ_{(KCl)}$

$+ \frac{1}{2}(243.0)$

$\Delta_{lattice} H^\circ_{(KCl)} = -717.8$ kJ/mol

73. [38] $H_2O(\ell) \rightleftharpoons H_2O,\ \Delta H_{vap} = 41\text{kJ/mol}$

$\Delta H = \Delta E + \Delta n_g RT$

$\Delta E$ is internal energy change

$\Delta n_g$ is change in gaseous moles $= 1 - 0 = 1$

$41 = \Delta E + 1 \times 8.3 \times 10^{-3} \times 373$

$\Delta E \approx (41 - 3) = 38$ kJ mol$^{-1}$.

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**74.** [667]

Molar mass of glucose ($C_6H_{12}O_6$) = 180 g/mol

moles of glucose required per day = $\frac{1000\,kJ/day}{2700\,kJ/mol}$

$= \frac{100}{27}$ mol/day

Gram of glucose required by the body in a day

$= \frac{100}{27} \times 180\left(\frac{g}{day}\right) = 666.67\,g/day \approx 667\,g/day$

**75.** (*b*) Fact based

**76.** [300]

$\Delta G = \Delta H - T\Delta S = 0$ at equilibrium

$\Rightarrow -165 \times 10^3 - T \times (-505) = 0$

$T = 300$ K

**77.** (*a*) $\Delta H = \Delta U - P\Delta V$

if U + PV (by defintion)

$\Delta H = \Delta U + P\Delta V$

**78.** [195]

$N_2O$ moles = $\frac{22}{44} = \frac{1}{20}$

$\Delta H = nC_P\Delta T = \frac{1}{20} \times 100(-40) = -200\,J$

$\Delta U = q_P + W$

$W = -P_{ext}\Delta V$

$W = -1\frac{(167.75 - 217.1)}{1000} \times 101.35$

$W = +5$ J

$\Delta U = -200 + 5 = -195$ J

**79.** [0] For free expansion:

$P_{ext} = 0$, $W = 0$

$q = 0$, $\Delta U = 0$

**80.** [8630]

$n = 5$ mol, $T = 300$ K

$v_1 = 10$L, $v_2 = 20$ L

$W = -nRT\ln\frac{v_2}{v_1}$

$= -5 \times 8.3 \times 300 \times \ln\frac{20}{10} = -8630.38\,J$

**81.** [38] $H_2O(l) \rightarrow H_2O(g)$

$n = \frac{36}{18} = 2\,mol$

$\Delta U = \Delta H - \Delta_n gRT$

$41.1 - \frac{1 \times 8.31 \times 373}{1000}\,kJ/mol = 38\,kJ/mol$

**82.** [2] $C_{P_1m} = C_{v_1m} + R$

$\Rightarrow C_{v_1m} = 20.785 - 8.314 = 12{,}471\ Jk^{-1}\ mol^{-1}$

$\Delta U = nCv_1m\Delta T$

$n = \frac{5000}{12.471 \times 200} = \frac{25}{12.471} \approx 2$

**83.** [3] Internal energy, volume enthalpy are state variable.

**84.** [57]

$\Delta H = \Delta U + \Delta_{ng}RT$

$\Delta H = -59.6 + 1 \times 8.314 \times 300 \times 10^{-3}$

$\Delta H = -57.10$

**85.** (*c*) $\Delta H = \sum\Delta H_{combustion}(\text{Reactant}) - \sum\Delta H_{combustion}(\text{product})$

$= 3 \times (-1300) - (-3268) = -632$ kJ/mol

**86.** (*c*) $C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

$\Delta_C H(C_2H_6) = 2\Delta_F H(CO_2)(g) + 3\Delta_F H(H_2O)(l) - \Delta_F H(C_2H_6(g))$

$-1560 = 2(-394) + 3(-286) - \Delta_F H(C_2H_6)(g)$

$\Delta_F H(C_2H_6) = -86$ kJ/mole.

**87.** [104]

The desired reaction is:

$3C\ (\text{graphite}) + 4H_2(g) \rightarrow C_3H_8(g)$

$\Delta H_F = 3\Delta H_2 + 4\Delta H_3 + \Delta H_1$

$= 3(-393.5) + 4(-285.8) - (-2220) = -103.7$ kJ/mol

$|\Delta H_F| = 104$ kJ/mol.

**88.** [117]

Given data is for 1 moles and asked for 5 moles

So value is $23.4 \times 5 = 117$ kJ

**89.** [600]

$Mg(s) + \frac{1}{2}O_2(g) \rightarrow MgO(s)$

$\Delta H = \Delta U + \Delta_{ng}RT$

$-601.70 \times 10^3 = \Delta U - \frac{1}{2} \times 8.3 \times 300$

$-601.7$ kJ $= \Delta U - 1.245$ kJ

$\Delta U = -600.455$ kJ $\approx 600$

**90.** [–727]

$\Delta U = -726$ kJ/mol

$\Delta_{ng} = 1 - \frac{3}{2} = -\frac{1}{2}$

$\Delta H = \Delta U + \Delta_{ng}RT$

$= -726 - \frac{1}{2} \times \frac{8.3 \times 300}{1000} = -727.245$

**91.** [2] $\Delta H_{ionisation}$ of $CH_3COOH$

$= [-57.3 - (-55.3)] = 2$ kJ/mol

**92.** [35]

Let Xg is burnt

moles = $\frac{X}{280}$

Heat released by $\frac{X}{280}$ moles = $2.5 \times 0.45$

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Heat released by 1 mole = $\frac{2.5\times0.45\times280}{X}$ kJ

$\Delta H = \Delta U + \Delta_{ng}T$

$\Delta H = \Delta U$

$9 = \frac{2.5\times280\times0.45}{X}$

$X = 35$ g

**93.** (*d*) $C(s) + O_2(g) \rightarrow CO_2(g) + 400$ kJ ...(i)

1 g mole

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + 100$ kJ ...(ii)

$0.6 \times 1000 = 600$ gm

$600\times\frac{60}{100}$ (pure carbon)

$= 360\,gm = \frac{360}{12} = 30\,mole$

Carbon converted into $CO_2 = \left(30 - 30\times\frac{60}{100}\right) = 12$ mole

& carbon converted in CO = $30\times\frac{60}{100} = 18$ mole

Energy generated = $18 \times 100 = 1800$ kJ

Energy generated during 1st reaction = $12 \times 400 = 4800$

Total = 4800 + 1800 = 6600 kJ

**94.** [54]

$HNO_3$ NaOH

600 mL × 0.2 M 400 ml × 0.1 M

= 120 mmol = 400 mmol

$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$

120 40

80 0 40 mmol

$\Delta_r H = 40\ mmol \times (57\times10^3)\frac{J}{mol}$

$= 40\times10^{-3}\ mol\times57\times10^3\frac{J}{mol} = 2280\,J$

$mS\Delta T = 2280$

$1000\,mL\times\frac{1\,gm}{mL}\times4.2\times\Delta T = 2280$

$\Delta T = 542.86 \times 10^{-3}$

$\Delta T = 54.286 \times 10^{-2}\,K$

$\Delta T = 54.286 \times 10^{-2}\,^\circ C$

**95.** [200]

$C(s) + O_2(g) \rightarrow CO_2(g);\ \Delta H = -X$ kJ/mol

$Q = C\Delta T = 20\ kJ \times 2$

40 kJ heat is released for 2.4 g of C

for 1 mole 'C': $Q = \frac{40}{2.4}\times12$

$= \frac{400}{24}\times12 = 200\,kJ/mol$

$Q = \Delta E = \Delta H = 200$ kJ ($\because \Delta_{ng} = 0$)

$X = 200$

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**96.** [2] Work done along dashed path

$|-W| = \sum P\Delta V$

$= 4 \times 1.5 + 1 \times 1 + \frac{2}{3} \times 2.5 = 8.65$ atm

Work done along solid path

$W = nRT\ln\frac{V_2}{V_1} = P_1V_1\ln\frac{V_2}{V_1}$

$= 2\times2.3\log\frac{5.5}{0.5}$

$= 2\times2.3\log11 = 4.79$

$= \frac{Wd}{Ws} = \frac{8.65}{4.79} = 1.80 \approx 2$

**97.** (A) $CO_2(s) \rightarrow CO_2(g)$

It is phase transition. The process is endothermic (sublime)

Gas is produced, so entropy increase.

(B) On heating $CaCO_3$ decomposed. So, process is endothermic. The entropy increases as gaseous product is formed.

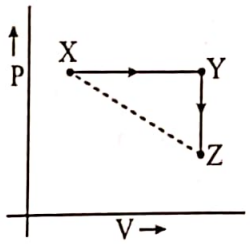
(C) $2H \rightarrow H_2(g)$

Entropy decrease as number of gaseous particles decrease.

(D) It is phase transition. White and red P are allotropes since red P is polymeric form

So $\Delta S$ is –ve.

**98.** (*a,c*)



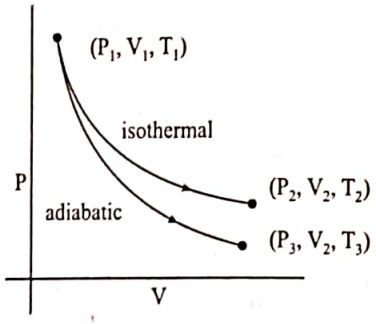

(*a*) $\Delta S_{x\rightarrow z} = \Delta S_{x-y} + \Delta S_{y\rightarrow z}$ (Correct)

(*b*) $W_{x\rightarrow y} = W_{x-y} + W_{y\rightarrow z}$ (Incorrect)

(*c*) $W_{x\rightarrow y\rightarrow z} = W_{x-y}$ (Correct)

(*d*) $\Delta S_{x\rightarrow y\rightarrow z} = \Delta S_{x-y}$ (Incorrect)

**99.** (*a,d*)




(*a*) $T_1 = T_2$ (due to isothermal)

(*b*) $T_3 > T_1$ (incorrect) cooling will take place in adiabatic expansion)

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(*c*) $W_{isothermal} > W_{adiabatic}$ {with sign, this is incorrect}

(*d*) $\Delta U_{isothermal} = 0 > \Delta U_{adiabatic} = -$ ve

So, answer is (*a*,*d*)

**100.** (*d*) $2C(s) + H_2(g) \longrightarrow C_2H_2(g)\ \Delta H = 225$ KJ/mol

$2C(g) + 2H(g) \longrightarrow C_2H_2(g)$ : $-(2\varepsilon_{C-H} + \varepsilon_{C=C})$

$2C(g) + 2H(g)$ : 1410, 330

$\therefore \Delta H = +1410 + 330 - (350 \times 2) - \varepsilon_{C\equiv C} = +225$

$\therefore \varepsilon_{C\equiv C} = 1740 - 700 - 225 = +815$ KJ/mol.

**101.** (*c*) $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O()$

$\Delta_C H = 6 \times \Delta_f H(CO_2) + 6\,\Delta_f H(H_2O) - \Delta_f H(C_6H_{12}O_6) - 6\Delta_f(O_2,g)$

$= 6 \times (-400 - 300) - (-1300) - 0$

$= -4200 + 1300$

$= -2900$ KJ/ mol

For one gram of glucose, enthalpy of combustion

$= -\frac{2900}{180} = -16.11$ KJ/g.

**102.** (*a*,*b*,*c*)

Since the vessel is thermally insulated so

$q = 0$

$p_{ext} = 0$ , so $w = 0$

so $\Delta U = 0$ (ideal gas)

Hence $\Delta T = 0$

$\Rightarrow \Delta T = 0$

$\Rightarrow T_2 = T_1$

$\Rightarrow P_2V_2 = P_1V_1$

The process is however adiabatic irriversible.

So we cannot apply $P_2V_2^{\gamma} = P_1V_1^{\gamma}$

**103.** (*b*) For $H_2O(\ell) \rightarrow H_2O(g)$ at T = 100°C, 1atm

equilibrium exists. $\therefore \Delta G = 0, \Delta H - T\Delta S = 0$

$\Delta H = T\Delta S > 0$ for system, since evaporation is endothermic

$\therefore (\Delta S)_{system} > 0$, also $(\Delta S)_{surrounding} = \frac{q_{surr}}{T_{surr}}$

Heat gained by system = heat lost by surroundings

$\therefore q_{surr.} < 0 \therefore (\Delta S)_{surr.} < 0$

**104.** (*a*) Let the heat capacity of insulated beaker be C.

Mass of aqueous content in expt. 1 = (100 + 100) × 1

= 200 g

$\Rightarrow$ Total heat capacity = (C + 200 × 4.2) J/K

Moles of acid, base neutralised in expt. 1 = 0.1 × 1 = 0.1

$\Rightarrow$ Heat released in expt 1 = 0.1 × 57 = 5.7 KJ

$\Rightarrow 5.7 \times 1000 = (C + 200 \times 4.2) \times \Delta T.$

$5.7 \times 1000 = (C + 200 + 4.2) \times 5.7$

$\Rightarrow (C + 200 \times 4.2) = 1000$

In second experiment, $n_{CH_3COOH} = 0.2$, $n_{NaOH} = 0.1$

Total mass of aqueous content = 200 g

$\Rightarrow$ Total heat capacity = (C + 200 × 4.2) = 1000

$\Rightarrow$ Heat released = 1000 × 5.6 = 5600 J.

Overall, only 0.1 mol of $CH_3COOH$ undergo neutralization.

$\Rightarrow \Delta H_{neutralization}$ of $CH_3COOH = \frac{-5600}{0.1}$

$= -56000$ J/mol

$= -56$ KJ/mol.

$\Rightarrow \Delta H_{ionization}$ of $CH_3COOH = 57 - 56 = 1$ KJ/mol

**105.** (*b*) Final solution contain 0.1 mole of $CH_3COOH$ and $CH_3COONa$ each.

Hence it is a buffer solution.

$$pH = pK_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$= 5 - \log 2 + \log\frac{0.1}{0.1} = 4.7$$

**106.** (*a*) -R,T (*b*)-P,Q,S (*c*)-P,Q,S (*d*)-P,Q,S,T

(*a*) $H_2O(\ell) \rightarrow H_2O(s)$ at 273 K. & 1 atm

$\Delta H = -$ ve $= q$

$\Delta S_{sys} < 0$, $\Delta G = 0$.

$w \neq 0$ (as water expands on freezing), $\Delta U \neq 0$

(*b*) Free expansion of ideal gas.

$q = 0$ $w = 0$

$\Delta U = 0$ $\Delta S_{sys} > 0$

$\Delta G < 0$

(*c*) Mixing of equal volume of ideal gases at constant pressure & temp in an isolated container

$q = 0$, $w = 0$,

$\Delta U = 0$, $\Delta S_{sys} > 0$

$\Delta G < 0$

(*d*) $H_2(g)$ 300 K $\xrightarrow[\text{Heating, 1atm}]{\text{Reversible}}$ 600 K

$\xrightarrow[\text{Cooling, 1atm}]{\text{Reversible}}$ 300 K.

$q = 0, w = 0, \Delta U = 0, \Delta G = 0, \Delta S_{sys} = 0$

**107.** (*c*) $\because$ Process is done against const. External P, process will be irreversible.

$$\therefore \text{“}S_{sure} = \frac{-P_{ext}(V_2 - V_1)}{T}$$

$$= \frac{-3 \times (2-1) \times 101.3}{300} = -1.013 J/K$$

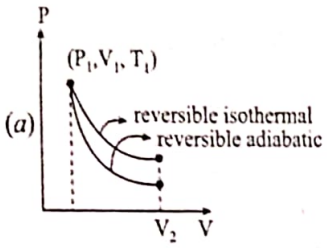
**108.** (*a,b*)

$$\Delta S_{Surr} = \frac{-\Delta H}{T_{Surr}}$$

For endothermic, if $T_{surr.}$ increases, $\Delta S_{surr}$ will increases.

For exothermic, if $T_{surr.}$ increases, $\Delta S_{surr}$ will decreases.

**109.** (*a,c,d*)




Area under curve in reversible isothermal is more. So, more work will be done by gas.

(*b*) $T_1 = T_2 \Rightarrow \Delta U = nC_V\Delta T = 0$

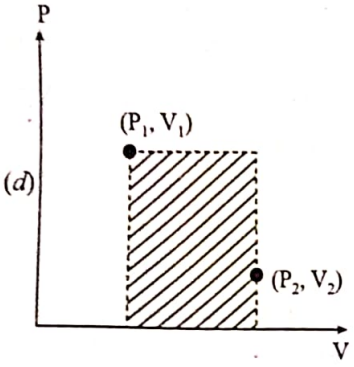
In reversible adiabatic expansion, $T_2 < T_1$

$\therefore \Delta T = -ve \therefore \Delta U = -ve$

(*c*) In Free expansion, $P_{ext} = 0 \therefore W = 0$

If carried out isothermally ($\Delta U = 0$) $\Rightarrow q = 0$ (Adiabatic); From I law

If carried out adiabatically ($q = 0$) $\Rightarrow \Delta U = 0$ (isothermal); From I law




During irreversible compression, maximum work is done on the gas (corresponding to shaded area)

**110.** (*c*) $dG = VdP - SdT$

At 298 K, $SdT = 0$

$\therefore dG = VdP$

$$\int_1^P dG = \int_1^P VdP \therefore G - G^\circ = V(P-1)$$

[$\because$ Solids involved $\therefore$ V almost constant]

$\therefore \Delta_r G = [G^\circ_{diamond} + V_d(P-1)] - [G^\circ_{graphite} + V_g(P-1)]$

$0 = 2.9 \times 10^3 + (P-1)\,10^5\,(-2 \times 10^{-6})$

$\therefore P = 14501$ bat

**111.** (*a,c*)

$A \rightleftharpoons P$

given $T_2 > T_1$

$$\frac{\ln k_1}{\ln k_2} > \frac{T_2}{T_1}$$

$T_1 \ln k_1 > T_2 \ln k_2$

$-\Delta G_1^\circ > -\Delta G_2^\circ$

$(-\Delta H^\circ + T_1\Delta S^\circ) > (-\Delta H^\circ + T_2\Delta S)$

$T_1\Delta S^\circ > T_2\Delta S^\circ$

$\Delta S^\circ < 0$

**112.** [−14.6]

From the given data

$2Cu(s) + H_2O(g) \rightleftharpoons Cu_2O(s) + H_2(g)$

$\Delta G^\circ = 1{,}00{,}000$

$\Delta G^\circ = 1{,}00{,}000 = -RT \ln k_p$

$$k_P = \frac{P_{H_2}}{P_{H_2O}}(P_{H_2O(g)}) = 0.01\,bar$$

$$100000 = -RT \ln \frac{P_{H_2}}{P_{H_2O}}$$

$P_{H_2} = -14.6$

**113.** (*b,c*)

AC → Isochoric

AB → Isothermal

BC → Isobaric

# $q_{AC} = \Delta U_{BC} = nC_V(T_2 - T_1)$

$$W_{AB} = nRT_1 \ln\left(\frac{V_2}{V_1}\right)$$ A (wrong)

$q_{BC} = \Delta H_{AC} = nC_P(T_2 - T_1)$

$W_{BC} = -P_2(V_1 - V_2)$ B (correct)

$nC_P(T_1 - T_2) < nC_V(T_1 - T_2)$ C (correct)

$\Delta H_{CA} < \Delta U_{CA}$

D (wrong)

**114.** (*a,b*)

Enthalpy of formation is defined as enthalpy change for formation of 1 mole of substance from its elements, present in their natural most stable form.

**115.** (*a,b,c*)

For 1 mole Vander Waal's gas

$$P = \frac{RT}{V-b} - \frac{a}{V^2}$$

If $P_{ext}$=P, means process is reversible. For Vanderwaal gas, expression is correct for all reversible process.

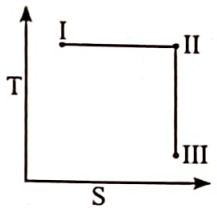
**116.** [935.00] $SnO_{2(S)} + C_{(S)} \rightarrow Sn_{(S)} + CO_{2(g)}$

$\Delta H°_{rxn}$=[–394]–[–581] =187 kJ/mole

$\Delta S°_{rxn}$=[52+210]–[56+6]

= 200J/k–mole

**117.** (*a,b,d*)




$$(I) \xrightarrow[\text{Isothermal expansion}]{\text{Reversible}} (II) \xrightarrow[\text{Expansion}]{\text{Rev. Adiabatic}} (III)$$

Isothermal process is isoenthalpic while reversible adiabatic process is isoentropic

**118.** [10] 1st process is adiabatic since entropy is constant.

$W_1 = \Delta U$

$\Delta U = 450R - 2250R = -1800R$ ...(i)

In 2nd process

$$W_2 = -2.303\ nRT \log \frac{v_3}{v_2}$$ ...(ii)

$$= -nRT\ In \frac{v_3}{v_2}$$

U = nCvT for process II

$$450R = 1 \times \frac{5}{2} RT$$

$$T = \frac{450 \times 2}{5} = 180k$$

$W_1 = W_2$

$$-1800R = -1 \times R \times 100 \ln \frac{v_3}{v_2}$$

$$\ln \frac{v_3}{v_2} = \frac{1800}{180} = 10$$

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CHAPTER

# 5 Equilibrium

## Exercise-1 (Topicwise)

1. (*c*) When rate of forward reaction is equal to rate of backward reaction the reaction is said to be in equilibrium.

2. (*b*) At equilibrium, the rate of forward & backward reaction become equal.

3. (*d*) $[HI] = \frac{64\text{ gm}}{128 \times 2\text{ litre}} = 0.25$

4. (*b*) As conc. of reactant increases, rate of reaction increases.

5. (*d*) Fact Based

6. (*a*) As we increase the concentration of substance, then speed of the reaction increases.

7. (*d*) Equilibrium constant for the reaction

$$3A + 2B \rightleftharpoons C \text{ is } K = \frac{[C]}{[A]^3[B]^2}$$

8. (*d*) Suppose 1 mole of A and B each taken then 0.8 mole/litre of C and D each formed remaining concentration of A and B will be (1 - 0.8) = 0.2 mole/litre each.

$$Kc = \frac{[C][D]}{[A][B]} = \frac{0.8 \times 0.8}{0.2 \times 0.2} = 16.0$$

9. (*c*)

| | A | + B | ⇌ C | + D |
|---|---|---|---|---|
| Initial conc. | 4 | 4 | 0 | 0 |
| After T time conc. | (4–2) | (4–2) | 2 | 2 |

$$\text{Equilibrium constant} = \frac{[C][D]}{[A][B]} = \frac{2 \times 2}{2 \times 2} = 1$$

10. (*a*) For reaction $A + 2B \rightleftharpoons C$

$$K = \frac{[C]}{[A][B]^2} = \frac{0.216}{0.06 \times 0.12 \times 0.12} = 250$$

11. (*b*) $A + 2B \rightleftharpoons C + 3D$

$$K = \frac{[P_C][P_D]^3}{[P_A][P_B]^2} = \frac{0.30 \times 0.50 \times 0.50 \times 0.50}{0.20 \times 0.10 \times 0.10} = 18.75$$

12. (*c*) $A + 2B \rightleftharpoons C$

$$K_C = \frac{[C]}{[A][B]^2}$$

13. (*a*)

| | $PCl_5 \rightleftharpoons$ | $PCl_3 +$ | $Cl_2$ |
|---|---|---|---|
| Moles | 2 | 0 | 0 |
| At eq$^m$ | $\frac{2\times60}{100}$ | $\frac{2\times40}{100}$ | $\frac{2\times40}{100}$ |

Volume of container = 2 litre.

$$K_c = \frac{\frac{2\times40}{100\times2} \times \frac{2\times40}{100\times2}}{\frac{2\times60}{100\times2}} = 0.266$$

14. (*d*) Unit of $K_p = (atm)^{\Delta n}$

Unit of $K_c = (mole/litre)^{\Delta n}$

$= [mole / litre]^0 = 1$ (unitless)

15. (*b*) For $A + B \rightleftharpoons C + D$

$$K = \frac{[C][D]}{[A][B]} = \frac{0.4 \times 1}{0.5 \times 0.8} = 1$$

16. (*a*) $K_p = \frac{[P_{CO}]^2[P_{O_2}]}{[P_{CO_2}]^2} = \frac{[0.4]^2 \times [0.2]}{[0.6]^2} = 0.0888$

17. (*b*) $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)\uparrow, K_{P_1} = 8 \times 10^{-2}$

$C(s) + CO_2(g) \rightleftharpoons 2CO(g); K_{P_2} = 2$

$$K_{P_3} = \frac{[P_{CO}]^2}{[P_{CO_2}]}; \; P_{CO} = \sqrt{[Kp_1 \times Kp_2]}$$

$$P_{CO} = \sqrt{[8 \times 10^{-2} \times 2]} = \sqrt{16 \times 10^{-2}} = 4 \times 10^{-1} = 0.4$$

18. (*a*)

| | $H_2(g)$ + | $CO_2(g) \rightleftharpoons$ | $CO(g) +$ | $H_2O(l)$ |
|---|---|---|---|---|
| Initial conc. | 1 | 1 | 0 | 0 |
| At equili | (1- x) | (1 - x) | x | x |

$$K_p = \frac{P_{CO} \cdot P_{H_2O}}{P_{H_2} \cdot P_{CO_2}} = \frac{x.x}{(1-x)(1-x)} = \frac{x^2}{(1-x)^2}$$

19. (*c*) $K_p = \frac{[P_{co}]^2}{[P_{co_2}]} = \frac{4 \times 4}{2} = 8$

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**20.** (*b*) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$K_p = P_{CO_2}$

Solid molecule does not have partial pressure so in calculation of $K_p$ only $P_{CO_2}$ is applicable.

**21.** (*a*) $n_p = n_r$ then $K_p = K_c$

where $n_p$ = no. of moles of product in gas

$n_r$ = no. of moles of reactant in gas

**22.** (*b*) $K_p = K_c (RT)^{\Delta n}$

**23.** (*c*) $K_p = K_c (RT)^{-2}$, $\Delta n = 2$

**24.** (*b*) $K_p = K_c (RT)^{\Delta n}$ When $\Delta n = 2 - (3 + 1) = -2$

i.e. negative = $K_p < K_c$

**25.** (*c*) $K_p = K_c (RT)^{\Delta n}$; $\Delta n = 2 - 2 = 0$

**26.** (*c*) $\Delta n = 2 - 1 = 1$

$K_p = K_c (RT)$

**27.** (*a*) $K_p = K_c (RT)^{\Delta n}$

$\Delta n = 3 - 2 = 1$; $k_p > k_c$

**28.** (*a*) $K_p = K_c (RT)^{\Delta n}$; $\Delta n = 2 - 4 = -2$

$K_p = 6 \times 10^{-2} \times (0.0812 \times 773)^{-2}$

$$K_p = \frac{6 \times 10^{-2}}{(0.0812 \times 773)^2} = 1.5 \times 10^{-5}$$

**29.** (*a*) Those reaction which have more values of K proceeds towards compeletion.

**30.** (*c*) For this reaction, there is no change in equilirium constant by change of volume (as $\Delta n = 0$).

**31.** (*d*) $K_f$ is a characteristic constant for the given reaction its only change with change in temp.

**32.** (*c*) $K_1 = \dfrac{[SO_3]}{[SO_2][O_2]^{1/2}}$ and $K_2 = \dfrac{[SO_2]^2[O_2]}{[SO_3]^2}$; $K_2 = \dfrac{1}{K_1^2}$

**33.** (*d*) $K_{C_1}$ for $H_2 + I_2 \rightleftharpoons 2HI$ is 50

$K_{C_2}$ for $2HI \rightleftharpoons H_2 + I_2$

$$K_{c_2} = \frac{1}{K_{c_1}} = \frac{1}{50} = 0.02$$

**34.** (*b*) $K' = K^n$; Hence $n = \dfrac{1}{2}$

$\therefore K' = K^{1/2} = \sqrt{K}$

**35.** (*c*) $2NO_2 \rightleftharpoons 2NO + O_2$; $K = 1.6 \times 10^{-12}$

$$K = \frac{[NO]^2[O_2]}{[NO_2]^2} \quad \text{...(i)}$$

$$NO + \frac{1}{2}O_2 \rightleftharpoons NO_2$$

$$K' = \frac{[NO_2]}{[NO][O_2]^{1/2}} \quad \text{...(ii)}$$

On multiplying (i) and (ii)

$$K \times K' = \frac{[NO]^2[O_2]}{[NO_2]^2} \times \frac{[NO_2]}{[NO][O_2]^{1/2}} = \frac{[NO][O_2]^{1/2}}{[NO_2]} = \frac{1}{K'}$$

$$K \times K' = \frac{1}{K}\ ;\ K = \frac{1}{K'^2}\ ;\ K' = \frac{1}{\sqrt{K}}$$

**36.** (*d*) $K_1$ for $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$K_2$ for $NH_3 \rightleftharpoons \dfrac{1}{2}N_2 + \dfrac{3}{2}H_2$

$$K_1 \times K_2 = \frac{[NH_3]^2}{[N_2][H_2]^3} \times \frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]}$$

$$K_1 \times K_2 = \frac{1}{K_2}\ ;\ K_2 = \frac{1}{\sqrt{K_1}}$$

**37.** (*b*) $2.303 \log \dfrac{K_2}{K_1} = \dfrac{\Delta H}{R}\dfrac{[T_2 - T_1]}{T_1 \times T_2}$

$\Delta H$ = + ve for the reaction

**38.** (*d*) $NH_3 \rightleftharpoons \dfrac{1}{2}N_2 + \dfrac{3}{2}H_2$

$$K_c = \frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]} \text{ and } \frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$$

$$K_d = \frac{[NH_3]}{[N_2]^{1/2}[H_2]^{3/2}}$$

So for dissociation = $\dfrac{1}{K_C}$

**39.** (*c*) $K_1 = \dfrac{[H_2S]}{[H_2][S_2]^{1/2}}$ ; $K_2 = \dfrac{[HBr]^2}{[H_2][Br_2]}$

$$K_3 = \frac{[HBr]^2 \times [S_2]^{1/2}}{[Br_2] \times [H_2S]}\ ;\ \frac{K_2}{K_1} = K_3$$

**40.** (*b*)

| | $2SO_2$ | + | $O_2$ | → | $2SO_3$ |
|---|---|---|---|---|---|
| at eq. | 5–3 | | 5–1.5 | | 3 |

total moles = 2 + 3.5 + 3 = 8.5

**41.** (*c*) $K_c = \dfrac{[NO_2]^2}{[N_2O_4]} = \dfrac{(0.1)^2}{0.05} = 0.2$

**42.** (*a*)

| | $2HI$ | $\rightleftharpoons$ | $H_2$ | + | $I_2$ |
|---|---|---|---|---|---|
| t = 0 | 3.2 | | | | |
| At eq$^m$ | $3.2 - 3.2\,\alpha$ | | $\dfrac{3.2\alpha}{2}$ | | $\dfrac{3.2\alpha}{2}$ |

$$\alpha = \frac{22}{100} = 0.22$$

So, $n_{HI} = 3.2 - 3.2 \times 0.22 = 2.496$

**43.** (*a*) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

For this reaction $\Delta n = 2 - 1 = 1$

value of $\Delta n$ is positive so the dissociation of $PCl_5$ take forward by decrease in pressure & by increase in pressure the dissociation of $PCl_5$ decrease.

44. (b) $\frac{\text{Normal molecular weight}}{\text{experimental molecular wt.}} = 1+\alpha$

$NH_4Cl \rightleftharpoons NH_3 + HCl$

for complete dissociation

$\because \alpha = 1$

$\therefore$ Experimental Molecular wt $= \frac{\text{nor. mol. wt.}}{2}$

45. (b) $K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left[\frac{20}{100}\right]\times\left[\frac{20}{100}\right]}{\left[\frac{80}{100}\right]}$

$= \frac{0.2\times0.2}{0.8} = \frac{0.04}{0.8} = 0.05$

46. (d) $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

a — 0.5atm

a − x — 0.5+x — x

Total pressure = 0.5 + 2x = 0.84

i.e., x=0.17

$K_p = P_{NH_3} . P_{H_2S} = (0.67).(0.17) = 0.1139$

47. (b) $Kp = (P_{NH_3})(P_{H_2S})$

$P_{NH_3} = 0.5 + x$

$0.11 = (0.5 + x)(x)$

$0.11 = 0.5x + x^2$

$x^2 + 0.5x - 0.11 = 0$

$x = 0.165$

48. (a) For this reaction $\Delta n$ is negative & $\Delta H$ is negative so it take forward by decrease in temperature.

49. (c) According to Le-chatelier principle when concentration of reactant increases, the equilibrium shift in favour of forward reaction.

50. (d) We know that

P become $\frac{1}{2}$P & V become 2V so,

$\frac{1}{2}$ P × 2V = PV = nRT

So there is no effect in equation

51. (d) Equilibrium constant depends upon temperature.

52. (a) According to Le-Chatelier principle exothermic reaction is forwarded by low temperure, in forward direction number of moles is less, hence pressure is high.

53. (d) Equilibrium constant changes only with temperature, not with pressure and the concentration of either reactant or product.

54. (a) $N_2 + O_2 \rightleftharpoons 2NO$; $Q$ cal

The above reaction is endothermic so for higher production of $NO$, and the temperature should be high.

55. (d) According to Le-chatelier's principle.

56. (c) At low pressure, reaction proceeds where volume is increasing. This is the favourable condition for the reaction. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$.

57. (a) T & P.

58. (b) At equilibrium $\Delta G = 0$.

59. (d) $\underset{acid}{HClO_4} + \underset{base}{H_2O} \rightleftharpoons \underset{C.A.}{H_3O^+} + \underset{C.B.}{ClO_4^-}$

60. (b) Those substance which accept the proton are called Bronsted base and which donate the proton are called Bronsted acid.

$HCO_3^- + H^+ \rightleftharpoons H_2CO_3$ Bronsted base.

$HCO_3^- \rightleftharpoons H^+ + CO_3^{--}$ Bronsted acid.

61. (c) Because it is an electron pair acceptor as central atom have a vacant d-orbital.

62. (b) The basic character of hyride decreases down the group

63. (c) $H_3PO_4 \underset{}{\overset{Step-1}{\rightleftharpoons}} H^+ + H_2PO_4^-$

$H_2PO_4^- \underset{}{\overset{Step-2}{\rightleftharpoons}} H^+ + HPO_4^{2-}$

$HPO_4^{2-} \underset{}{\overset{Step-3}{\rightleftharpoons}} H^+ + PO_4^{3-}$

64. (d) Water is a neutral, protic solvent and act as both acid as well as Base.

65. (b) According to Bronsted principle $HNO_3$ is acid so they give $H^+$ in aqueous solution and form $NO_3^-$.

66. (c) $BF_3$ have vaccant orbital, so it can accept lone pair of $e^-$, but can not donate proton.

67. (d) $K_w = [H^+] \times [OH^-] = mol^2L^{-2}$

68. (b) For pure water $[H^+]=[OH^-]$, $\therefore K_w = 10^{-12}$

69. (a) Because pure water has a 7 pH.

70. (b) $CH_3COOH + HF \rightleftharpoons CH_3COOH_2^+ + F^-$

$HF$ gives $H^+$ to the $CH_3COOH$. So it is a conjugate base of $F^-$

71. (c) $\underset{acid}{H_2SO_4} \rightleftharpoons H^+ + \underset{C.B.}{HSO_4^-}$

72. (c) $\underset{\text{Conjugate acid}}{H_2PO_4^-} \rightleftharpoons H^+ + \underset{\text{Base}}{HPO_4^{2-}}$

73. (a) $\underset{\text{Base}}{NH_2^-} + H^+ \rightleftharpoons \underset{\text{C.A.}}{NH_3}$

74. (d) Conjugate acid is obtained from the base by means. $Cl^-$ is a weak base

75. (c) pH = 4 means; $[H^+] = 10^{-4}$ mol

76. (d) $10^{-3}$N KOH will give $[OH^-] = 10^{-3}$M

pOH=3

$\because$ pH + pOH = 14, pH = 14 − 3 = 11

77. (b) $pH + pOH = pK_w = 14$

78. (a) $[OH^-] = 10^{-2}$ M; pOH = 2

pH + pOH = 14; pH = 14 − pOH

pH = 14 − 2 = 12

79. (b) $\frac{N}{100}$ = 0.01 N HCl; $[H^+] = 10^{-2}$ M; pH=2

[OH] = $10^{-2}$ M for NaOH

pH + pOH = 14; pH = 14 − 2; pH = 12

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**80.** (*c*) $H_2O \rightleftharpoons [H^+]\,[OH^-]$

$HCl \rightleftharpoons [H^+]\,[Cl^-]$

Total $[H^+] = [H^+]_{H_2O} + [H^+]_{HCl} = 10^{-7} + 10^{-8}$

$= 10^{-7}\,[1 + 10^{-1}]$

$[H^+] = 10^{-7} \times \frac{11}{10}$

$pH = -\log [H^+] = -\log\left(10^{-7} + \frac{11}{10}\right)$; $pH = 6.958$

**81.** (*c*) $H_2SO_4$ ionized in two step

1st $H_2SO_4 \rightarrow H^+ + HSO_4^-$

2nd $HSO_4^- \rightarrow H^+ + SO_4^{2-}$

**82.** (*b*) $[H^+] = 2 \times 10^{-2}$ M

$\because$ $pH = -\log [2 \times 10^{-2}]$;

pH = 1.7 i.e. in between 1 and 2.

**83.** (*b*) [NaOH(aq)] = $10^{-3}$, pOH = 3, pH =11

[HCl(aq)] = $10^{-3}$, pH = 3

[NaCl(aq)] = neutral, pH = 7

**84.** (*b*) $10^{-10}$ M $HCl$ = $10^{-10}$ M $[H^+]$. But pH ≠ 10 because solution is acidic. This is because $H^+$ from $H_2O$ ($10^{-7}$M) cannot be neglected.

Total $[H^+] = 10^{-7} + 10^{-10}$

$= 10^{-7}\,[1+10^{-3}] = 10^{-7}\,(1.001)$

That is why pH = 7 (slightly less than 7)

**85.** (*b*) Because the degree of dissociation is inversely proportional to the concentration of the electrolyte.

**86.** (*b*) Electrolytes are those substances which dissociates in water to give ions.

**87.** (*a*) $K = \frac{\alpha^2 C}{1-\alpha}; \alpha = \frac{0.01}{100}; \therefore K = \alpha^2 C = \left[\frac{0.01}{100}\right]^2 \times 1$

$= 1 \times 10^{-8}$.

**88.** (*a*) NaCl, being a salt, is a strong electrolyte.

**89.** (*c*) According to the Ostwald's dilution formula $\alpha^2 = \frac{K(1-\alpha)}{C}$. But for weak electrolytes α is very small. So that (1–α) can be neglected. So that $\alpha = \sqrt{\frac{K_a}{C}}$.

**90.** (*b*) $\alpha \propto$ dilution of solution

**91.** (*d*) Mathematical form of Ostwald's dilution law.

**92.** (*c*) $[CN^-] = \sqrt{KaC} = \sqrt{4\times10^{-10}\times10^{-1}} = \sqrt{4\times10\times10^{-12}}$

$= 6.3 \times 10^{-6}$

**93.** (*d*) $[H^+] = \alpha.C = \frac{2}{100} \times .02; [H^+] = 4\times10^{-4}$ M

$pH = -\log[H^+] = 4 - \log 4; pH = 3.3979$

**94.** (*b*) pH = 3, $[H^+] = 10^{-3}$ M $\because [H^+] = \sqrt{K \times c}$

$[10^{-3}]^2 = K \times c$ ; $\frac{[10^{-6}]}{0.1} = K = 10^{-5}$

**95.** (*c*) For $NH_4OH$.

$[OH^-] = C.\,\alpha$ ; $C = \frac{1}{10}$ M, $\alpha = 0.2$

$[OH^-] = \frac{1}{10} \times 0.2 = 2 \times 10^{-2}$ M

$pOH = -\log [OH^-] = \log [2 \times 10^{-2}]$; pOH = 1.7

pH = 14 – pOH = 14 – 1.7 = 12.30.

**96.** (*b*) $NH_4CN$ is a salt of weak acid and weak base and thus for it.

$K_H = \frac{K_w}{K_a \times K_b}$

**97.** (*b*) Alkaline,

$CH_3COONa + H_2O \rightleftharpoons \underset{\text{Weak acid}}{CH_3COOH} + \underset{\text{Strong base}}{NaOH}$

**98.** (*c*) Because salt of S.A + S.B. does not undergo hydrolysis.

**99.** (*c*) $Fe^{3+}$ions are hydrolysed to develop acidic nature

**100.** (*c*) 0.001 M of NaOH means $[OH^-]$ = .001

$= 10^{-3}$ M ⇒ pOH = 3

pH + pOH = 14 ⇒ pH = 14 – 3 = 11

**101.** (*c*) $pK_a = -\log K_a$, $pK_b = -\log K_b$

$pH = -\frac{1}{2}[\log K_a + \log K_w - \log K_b]$

$= -\frac{1}{2}[-5 + \log(1\times10^{-14}) - (-5)]$

$= -\frac{1}{2}[-5 - 14 + 5] = -\frac{1}{2}(-14) = 7$

**102.** (*a*) $MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2\,HCl$

**103.** (*b*) It contains replacable H atom.

**104.** (*a*) An acid buffer solution consists of solution of weak acid with strong base of its salt.

**105.** (*b*) [Salt]= 0.1 M, [Acid]= 0.1 M

$K_a = 1.8 \times 10^{-5}$; $pH = -\log K_a + \log \frac{[Salt]}{[Acid]}$

$= -\log(1.8 \times 10^{-5}) + \log \frac{0.1}{0.1} = -(\log 1.8 \times 10^{-5})$

pH = 4.7

**106.** (*c*) $CH_3COONH_4$ is a simple buffer and called salt of weak acid.

**107.** (*c*) A strong acid is not used to make a buffer

@PROCHEM

108. (*d*) $NH_4OH$ is a weak acid and $NH_4Cl$ is a strong base salt.

109. (*a*) $pH = pK_a + \log \frac{[Salt]}{[Acid]} = (-\log 2 \times 10^{-5}) + \log \frac{10 \times 1}{50 \times 2} = 4.$

110. (*c*) $BaSO_4 \rightleftharpoons Ba^{2+} + SO_4^{2-}$

$-$ S S

$K_{SP} = [Ba^{2+}][SO_4^{2-}]$

$K_{SP} = 10^{-9}, [Ba^{2+}] = 0.01$

$[SO_4^{2-}] = \frac{K_{SP}}{[Ba^{2+}]} = \frac{10^{-9}}{10^{-2}} = 10^{-7}$

111. (*b*) $CaF_2 \rightleftharpoons \underset{(S)}{Ca^{++}} + \underset{(2S)}{2F^-}; K_{sp} = 4S^3$

112. (*d*) $K_{sp}$ for $CaF_2 = 4s^3 = 4 \times [2 \times 10^{-4}]^3 = 3.2 \times 10^{-11}$.

## Exercise-2 (Learning Plus)

1. (*a*)

| | A | + | B | $\rightleftharpoons$ | C | + | D |
|---|---|---|---|---|---|---|---|
| t = 0 | 3n | | n | | 0 | | 0 |
| t = teq. | (3n–x) | | (n–x) | | x | | x |

$(n-x) = x ; n = 2x \Rightarrow x = \frac{n}{2}$

2. (*b*) Given: Volume = 20.0 lit

0.50 mole of each gas that is $H_2$ & $I_2$

| Now, | $H_2$ | + | $I_2$ | $\rightleftharpoons$ | 2HI |
|---|---|---|---|---|---|
| intially | 0.5 | | 0.5 | | 0 |
| at eq. | 0.5–x | | 0.5–x | | 2x |

$K_{eq} = \frac{[HI]^2}{[H_2][I_2]}$

$49 = \frac{(2x)^2}{(0.5-x)^2}$ (as $K_{eq} = 49$)

$7 = \frac{2x}{0.5-x}$

$2x = 3.5 - 7x$

$9x = 3.5$

$x = 0.39$

$[HI] = \frac{2x}{20} = \frac{x}{10} \Rightarrow \frac{.39}{10} = 0.039$

3. (*a*) Given: Equilibrium constant, $K_C = 2.5$

We know:

$\frac{K_f}{K_b} = 2.5$

$K_f = 2.5 \times 3.2 \times 10^{-2} = 8 \times 10^{-2}$

4. (*b*) $C_2H_5OH(\ell) + CH_3COOH(\ell) \rightleftharpoons CH_3COOC_2H_5(\ell) + H_2O(\ell)$

a a 0 0

a – 0.33a a – 0.33a 0.33a 0.33a

$K_C = \frac{(0.33a) \times (0.33a)}{(0.67a) \times (0.67a)} = K_C = 1/4.$

5. (*a*) Given: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

At equilibrium

conc. of $N_2O_4 = 4.8 \times 10^{-2}$ mol/lit

$[NO_2] = 1.2 \times 10^{-2}$ mol/lit

$\therefore K_C = \frac{(1.2 \times 10^{-2})^2}{(4.8 \times 10^{-2})} = 3 \times 10^{-3}$

6. (*b*) Given: $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$; $K_C = 4 \times 10^{-4}$

Now, $NO(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$

$K'_C = ?$

$K_C' \frac{1}{[K_C]^{1/2}} \Rightarrow \frac{1}{(4 \times 10^{-4})^{1/2}} \Rightarrow \frac{1}{2 \times 10^{-2}} \Rightarrow 50$

7. (*b*) Given:

(i) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

at equ$^m$ 0.01 0.01 0.01

$K_{c_1} = \frac{[PCl_3] \times [Cl_2]}{[PCl_5]} \Rightarrow \frac{0.01 \times 0.01}{0.01} = 0.01$

(ii) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

at equ$^m$ 0.01 0.01 0.01

$K_{c_2} = \frac{0.01 \times 0.01}{(0.01)^2} = 1$

(iii) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

at equ$^m$ 0.01 0.01 0.01

$K_{c_3} = \frac{(0.01)^2}{(0.01)(0.01)^3} = \frac{1}{10^{-4}} = 10^4$

Extent of reaction $K_C \rightarrow$ (iii) > (ii) > (i)

8. (*c*) Given: reaction

$A(g) + 2B(g) \rightleftharpoons C(g)$

a 4 0

(at equill.) a–x 4–2x x

as x = 1

$n_{A(g)} = a - 1, \quad n_{B(g)} = 2, \quad n_{C(g)} = 1$

$[A] = \frac{a-1}{5} \quad [B] = \frac{2}{5}, [C] = \frac{1}{5}$

$K_C = \frac{(1/5)}{\left(\frac{2}{5}\right)^2\left(\frac{a-1}{5}\right)}; K_c = \frac{25}{4(a-1)}$ ...(i)

as $K_C = 0.25$ ...(ii)

from (i) & (ii)

$0.25 = \frac{25}{4(a-1)}$

$\Rightarrow 4(a-1) = 100$

$a - 1 = 25 \Rightarrow a = 26$

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**9.** (*b*) Given 1 mole $N_2$, 3 mole $H_2$

total pressure = 4 atm

$$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$$

| | | | |
|---|---|---|---|
| initially | 1 | 3 | |
| at equilibrium | 1–x | 3–3x | 2x |

$P_{total} = 1 - x + 3 - 3x + 2x$

$= 4 - 2x$ ...(i)

an given in question

$P_{total}$ fall to 3 atm

$\therefore P_{total} = 3$ ...(ii)

from (i) & (ii)

$3 = 4 - 2x \Rightarrow \boxed{x = \frac{1}{2}}$

$P_{N_2} = \frac{1}{2}$ ; $P_{H_2} = 3 - \frac{3}{2} = \frac{3}{2}$; $P_{NH_3}(eq.) = 2 \times \frac{1}{2} = 1$

(for formation of $NH_3$) $K_p = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3} = \frac{(1)^2}{\frac{1}{2} \times \left(\frac{3}{2}\right)^3}$

$= \frac{1}{0.5 \times (1.5)^3}$

$K_P$ for dissociation of $NH_3 \Rightarrow (0.5) \times (1.5)^3$

**10.** (*a*) $2H_2O\,(g) + 2Cl_2\,(g) \rightleftharpoons 4HCl\,(g) + O_2\,(g)$

$K_p = 0.03$, $T = 427^\circ C = 700$ K

$K_p = K_C(RT)^1$

$K_C = \frac{K_P}{RT} = \left[\frac{0.03}{0.082 \times 700}\right]$

$K_C = 5.23 \times 10^{-4}$

**11.** (*b*) Using $K_P = K_C\,(RT)^{\Delta n_g}$

so, $\log \frac{K_P}{K_C} = \Delta n_g \log RT$

so, $\Delta n_g = -1$, this is for (b).

**12.** (*d*) Given: $N_2O_4(g) \rightleftharpoons 2NO_2\,(g)$

$K_p = K_C$ ...(i)

$K_p = K_C\,(RT)^{\Delta n_g}$

$\Delta n_g = 2 - 1 \Rightarrow 1$

$K_P = K_C \times RT$

$RT = 1$ $T = \frac{1}{R}$

$\Rightarrow \frac{1}{0.082} \Rightarrow 12.19$ K

$T = 12.19$ K

**13.** (*c*) Given: $CO(g) + \frac{1}{2}\,O_2\,(g) \rightleftharpoons CO_2(g)$

$\Delta n_g = 1 - \left(\frac{1}{2} + 1\right)$, $\Delta n_g = -\frac{1}{2}$

as we know

$K_p = K_C\,(RT)^{\Delta n_g} \Rightarrow \frac{K_p}{K_c} = (RT)^{-1/2}$

**14.** (*b*) Given: Rection as

$$H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$$

Here $K_P = K_C$

| | | | | |
|---|---|---|---|---|
| initially | 0.25 | 0.25 | 0 | 0 |
| at eq. | 0.25–x | 0.25–x | x | x |

$K_C = \frac{[CO][H_2O]}{[H_2][CO_2]} \Rightarrow \frac{x^2}{(0.25 - x)^2}$

$\Rightarrow 0.16 = \frac{x^2}{(0.25 - x)^2}$

[as $K_P = K_C = 0.16$]

$\Delta n_g = 0$

$\frac{x}{(0.25 - x)} = 0.4$

$x = 0.1 - 0.4x$

$x = \frac{0.1}{1.4} = \frac{1}{14}$

% mole of CO = $\frac{(1/14)}{\text{total mole}} \times 100$

total mole = 0.5 mole

% mole of CO = $\frac{(1/14)}{0.5} \times 100$

$= 14.28\%$

**15.** (*c*) Given: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $K_p = 50$

$K_P = K_C\,(RT)^{\Delta n_g}$

now,

$\Delta n_g = 2 - 2 = 0$

$\therefore K_p = K_C$

$K_C = 50$

**16.** (*a*) $PCl_5$ dissociation in a closed container

$PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)$

| | | |
|---|---|---|
| 1 | 0 | 0 |
| 1 – α | α | α |

Total mole = $1 - \alpha + \alpha + \alpha$

$\Rightarrow 1 + \alpha$

$\chi_{PCl_5} = \frac{\alpha}{1 + \alpha}$

$\therefore$ (partial pressure) $P_{PCl_5} = P\left[\frac{\alpha}{1 + \alpha}\right]$

**17.** (*a*) $\alpha = \frac{D - d}{d} \Rightarrow \alpha = \frac{D}{d} - 1; \alpha + 1 = \frac{D}{d}$

$1 + \alpha = \frac{D}{d}$

$y = mx + c$, By compairing we get

$m = 1$, $c = 0$

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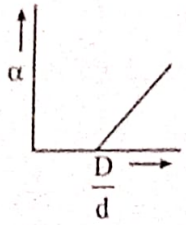
**18.** (*b*) $\alpha = \frac{D-d}{(n-1)d}; \alpha = \frac{D-d}{d}$

$\alpha = \left[\frac{D}{d} - 1\right]; \left(\frac{D}{d}\right) = \alpha + 1$

$\alpha = \left(\frac{D}{d}\right) - 1.$

Comparing with $y = mx + c$

$m = +ve, c = -ve$




**19.** (*d*) Given: Reacion

$2HI\ (g \rightleftharpoons H_2(g) + I_2(g)$

| | | |
|---|---|---|
| 1 | 0 | 0 |
| $1-\alpha$ | $\frac{\alpha}{2}$ | $\frac{\alpha}{2}$ |
| $C(1-\alpha)$ | $\frac{C\alpha}{2}$ | $\frac{C\alpha}{2}$ |

(where $\alpha$ is degree of dissociation)

as $K_C = K_p = \frac{\alpha^2}{4(1-\alpha)^2}$

$\sqrt{K_p} = \frac{\alpha}{2(1-\alpha)}, 2\sqrt{Kp} = \frac{\alpha}{1-\alpha}$

$\frac{1-\alpha}{\alpha} = \frac{1}{2\sqrt{K_p}} \Rightarrow \frac{1}{\alpha} - 1 = \frac{1}{2\sqrt{K_p}}, \frac{1}{\alpha} = 1 + \frac{1}{2\sqrt{Kp}}$

$\frac{1}{\alpha} = \frac{2\sqrt{Kp}+1}{2\sqrt{Kp}} \Rightarrow \alpha = \frac{2\sqrt{Kp}}{2\sqrt{Kp}+1}$

**20.** (*a*) $(VD)_{mix} = \frac{M_{mix}}{2} = \frac{M}{2(1+\alpha)}$

so, as $\alpha$ increases $(VD)_{mix}$ decreases.

**21.** (*a*) Given: V.D of $N_2O_4 = 30$

$N_2O_4 \rightleftharpoons 2NO_2$

| | |
|---|---|
| 1 | 0 |
| $1-\alpha$ | $2\alpha$ |

Total mole $= 1 + 2\alpha - \alpha \Rightarrow 1 + \alpha$

Mass conservation

(wt. of $N_2O_4$ at $t = 0$) = (wt. of mixture.)

$1 \times (92) = (1 + \alpha)\, 2 \times 30$

$\Rightarrow \frac{92}{60} = 1 + \alpha, \alpha = \frac{32}{60} \Rightarrow \frac{8}{15}$

$\alpha\% = \frac{8}{15} \times 100 = 53.3\%$

**22.** (*a*) Given:

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$

| | | | |
|---|---|---|---|
| 1 | 1 | 0 | 0 |
| $1-\alpha$ | $1-\alpha$ | $\alpha$ | $\alpha$ |

$K = \frac{[C][D]}{[B][A]}, K = \frac{\alpha^2}{(1-\alpha)^2}$

$\sqrt{K} = \frac{\alpha}{1-\alpha} \quad \frac{1-\alpha}{\alpha} = \frac{1}{\sqrt{K}}$

$\frac{1}{\alpha} = \frac{1+\sqrt{K}}{\sqrt{K}}$

Now,

$\alpha = \frac{\sqrt{K}}{1+\sqrt{K}}$

**23.** (*a*) Given:

| | $A(g) \rightleftharpoons$ | $B(g)$ | $+ C(g)$ | $+ D(g)$ |
|---|---|---|---|---|
| initially | 1 | 0 | 0 | 0 |
| at eq. | $1-x$ | x | x | x |

Total mole $= 1 + 2x$

Apply mass conservation

(Wt. of A at $t = 0$) = (wt of mixture at eq.)

$1 \times M_{wt.} = [1 + 2x]\, 2 \times 50$

$M_{wt.} = (1 + 0.2) \times 100$ (as $x = 0.1$)

$M_{wt.} = 120$

**24.** (*c*) Given: $N_2O_4$ dissociated

$N_2O_4 \rightleftharpoons 2NO_2$

| | | |
|---|---|---|
| | 1 | 0 |
| at eq. | $(1-x)$ | $2x$ |

Total mole $= 1 - x + 2x = 1 + x$

Mass conservation

(wt. of $N_2O_4$ at $t = 0$) = (wt. of $NO_2$ at eq.)

$1 \times 92 = (1 + x) \times 82$

$\frac{92}{82} - 1 = x$

$x = \frac{10}{82}$

$x\% = \frac{10}{82} \times 100 = 12.2\%$

**25.** (*b*) We know that $d = \frac{D}{1+(n-1)\alpha}$

where n is number of mole of gases produced from one mole of a gas.

**26.** (*b*) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$\log\frac{K_2}{K_1} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

$\log\frac{50}{66.9} = \frac{\Delta H}{2.303R}\left[\frac{1}{623} - \frac{1}{721}\right]$

After calculation negative value of $\Delta H$ is obtained.

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27. (*a*) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$; $\Delta n_g = 0$
If we change the volume the ultimate effect will be due to change in pressure.
$\Rightarrow$ If $\Delta n_g = 0$
no effect of change in volume.

28. (*d*) On adding any reactant equilibrium shifts in forward direction, so amount of product increases.

29. (*b*) On adding inert gas at constant pressure effect on equilibrium will be similar to as if volume of container has been increased.

30. (*b*) For any physical equilibrium on increasing pressure equilibrium shifts in the direction of higher density.

31. (*c*) Solubility of gas is directly proportional to the pressure of gas above liquid.

32. (*a*) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$
Volume of container is constant
so no effect of addition of inert gas.

33. (*d*) Exothermic
$\Rightarrow$ low yield at high temperature
$\Delta n_g < 0$
$\Rightarrow$ low yield at low pressure

34. (*a*) $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$
$\Delta n_g = 2 - 1 \Rightarrow 1$
$\Delta n_g > 0$
$\Rightarrow$ so on increasing the pressure equilibrium shifted left (or backward direction)
as pressure increases volume decreases.

35. (*a*) $\ln k_p = \frac{-\Delta H}{RT} + \ln A$
For exothermic $\Delta H < 0$
so, slope +ve

36. (*c*) Amphiprotic: can accept and Release $H^+$
Only $H_2PO_4^-$ & $HCO_3^-$

37. (*d*) Fact based

38. (*a*) $[OH^-] = 10^{-7}$ in pure water.
So as temperature increases $K_w$ increases $\Rightarrow$ $[OH^-]$ decreases.

39. (*c*) Factual.

40. (*b*) In this solution, source of $OH^-$ is water
$\therefore$ $C\alpha = [OH^-]$, C(molarity) of $H_2O = 55.55$
$\alpha = 1.8 \times 10^{-11}$ M
% ionisation $= 1.8 \times 10^{-9}$ M

41. (*d*) $[H^+]_1 = 10^{-2}$; $[H^+]_2 = 10^{-6}$; Factor $= 10^4$

42. (*c*) $[H^+] = 0.016$ M
$[H^+][OH^-] = 10^{-14} \Rightarrow [OH^-] = 6.25 \times 10^{-13}$ M

43. (*a*)

| Inital | Final |
|---|---|
| pH = 12 | pH = 11 |
| $[H^+] = 10^{-12}$ M | $[H^+] = 10^{-11}$ M |
| $[OH^-] = 10^{-2}$ M | $[OH^-] = 10^{-3}$ M |

Inital No. of mole of $OH^- = 10^{-2}$ Final No. of mole of $OH^- = 10^{-3}$
So no. of mole of $OH^-$ removed = [.01 – 0.001] = **0.009**

44. (*d*)

| | HCl c × v | NaOH c × v | $[H^+]_{remaing}$ | pH |
|---|---|---|---|---|
| (*a*) | $\frac{1}{10} \times 100$ | $\frac{1}{10} \times 100$ | neutral | 7 |
| (*b*) | $\frac{1}{10} \times 55$ | $\frac{1}{10} \times 45$ | $\frac{1}{100}$ | 2 |
| (*c*) | $\frac{1}{10} \times 10$ | $\frac{1}{10} \times 90$ | Basic as $OH^-$ are more | $P^H > 7$ |
| (*d*) | $\frac{1}{5} \times 75$ | $\frac{1}{5} \times 25$ | $\frac{10}{100}$ | 1 |

45. (*c*) $CH_3COOH \rightleftharpoons CH_3COO^{\ominus} + H^+$
0.01
$0.01 - 0.01\alpha$ $0.01\alpha$ $0.01\alpha$
$[HCl^-] = 0.01$
$[H^+]_T = 0.01 + 0.01\alpha$
as $\alpha$ is small, so
$0.01 + 0.01\alpha = 0.01$
$K_d = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$
$K_d = \frac{(0.01\alpha)(0.01)}{0.01 - 0.01\alpha}$
as $\alpha$ is small so
$0.01 - 0.01\alpha \approx 0.01$
$1.69 \times 10^{-5} = \frac{(0.01\alpha)(0.01)}{0.01 - 0.01\alpha}$
on solving we get
$\alpha = 1.69 \times 10^{-3}$

46. (*c*) $pH = pK_a + \log\left[\frac{salt}{acid}\right]$
50% ionised $\Rightarrow$ [Salt] = [Acid]
$\Rightarrow$ pH = pKa = 4.5 $\Rightarrow$ pOH = 9.5

47. (*a*) $M_{eq}$ of HCl = 25 × 0.5 = 12.5
$M_{eq}$ of NaOH = 10 × 0.5 = 5
$M_{eq}$ of HCl remaining 12.5 – 5 = 7.5
$[H^+] = 0.15$
$pH = -\log(0.15)$
$= 0.8239$

48. (*a*) $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$
$P^{OH} = 3$, $[OH^-] = 10^{-3}$, % dissociation $= \frac{10^{-3}}{0.05} \times 100 = 2\%$

49. (*d*) As concentration of solution decreases, degree of dissociation of weak electrolyte increases.

50. (*c*) $HF \longrightarrow H^+ + F^-$
$pK_w = pK_a + pK_b$ [For conjugate Acid-Base]
$\Rightarrow$ $pK_a = 14 - 10.87 = 3.17$
$K_a = 6.76 \times 10^{-4}$

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**51.** (*b*) $HA \longrightarrow H^+ + A^-$

$1-x \quad x \quad x$

$x = 1\% = 0.01$

$\Rightarrow [H^+] = 0.01$

$\Rightarrow pH = 2$

**52.** (*b*) 0.1 N HCl, $V_T = 100$

$$[H^+] = \frac{0.1}{100} = 10^{-3}$$

$P^H = -\log[H^+]$

$P^H = -\log[10^{-3}]$

$P^H = 3$

**53.** (*b*) % Hydrolysis does not depend on the conc. in case of "Weak acid + weak base : Salt"

**54.** (*a*) $\%\ h = \sqrt{\frac{K_h}{C}}$, $K_h = \frac{K_w}{K_a}$

$$\%\ h = \sqrt{\frac{K_w}{K_a \cdot C}} = \sqrt{\frac{10^{-14} \times 80}{1.3 \times 10^{-9} \times 1}}$$

$\%\ h = 2.48\%$

**55.** (*d*) $K_2CO_3$ (Salt of WA + SB)

**56.** (*c*) Salt of weak acid & weak base

$pH = [pK_w + pK_a - pK_b]$

$= [14 + 4.8 - 4.78]$

$pH = 7.01$

**57.** (*b*) Solution of HCl & $NH_4Cl$ will be acidic, solution of NaCl neutral whereas solution of NaCN will be basic.

**58.** (*a*) NaCl + HCl : Not the Buffer and Solution is acidic due to HCl.

$\Rightarrow pH < 7.$

**59.** (*d*) m. equivalent of KOH = 8

m. equivalent of HCOOH = 16

Remaining m. eq. after reaction (HCOOH) = 8

Formed m. eq. (HCOOK) = 8

$\Rightarrow$ Acidic Buffer

$$P^H = P^{ka} + \log\frac{[salt]}{[acid]}$$

As [salt] = [acid]

$P^H = P^{ka} = 3.7$

$P^{OH} = 10.3$

**60.** (*b*) $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$

$t = 0 \quad 20\ m \quad 20\ m$

$t = eq \quad\quad\quad 20\ m$

So, $[CH_3COO^-] = \frac{20}{200} = 0.1\ M$

$$P^H = \frac{1}{2}[14 + 4.74 + \log(0.1)]$$

$P^H = 8.87$

**61.** (*c*) $pH = 8.7 - P^{ka}$

Basic Buffer

$$\Rightarrow pOH = pK_b + \log\left[\frac{salt}{base}\right]$$

If volume of salt = V ml

$$\Rightarrow 5.3 = 4.7 + \log\left[\frac{0.2 \times V}{10}\right]$$

$\Rightarrow V = 200ml$

**62.** (*d*) Because it is an acidic buffer.

**63.** (*c*) $K_a = 5 \times 10^{-10} \Rightarrow pK_a = 9.3$

$$pH = pK_b + \log\left[\frac{salt}{acid}\right]$$

$$9 = 9.3 + \log\frac{5 \times V}{10 \times 2}$$

$V = 8$ ml

**64.** (*a*) $HA + NaOH \rightarrow NaA + H_2O$

$10 \quad x \quad 10$

$10-x \quad - \quad 10+x$

$$P^H = P^{ka} + \log\frac{[salt]}{[acid]}$$

$$5.5 = 5 + \log\left[\frac{10+x}{10-x}\right] \quad 5 = \log\left[\frac{10+x}{10-x}\right]$$

on solving

$x = 5.192$ mmol

wt of NaOH = $40 \times 5.192 = 0.208$ g

**65.** (*b*) $CH_3COOH + NaOH \rightarrow CH_3COO^- Na + H_2O$

$40 \quad 20$

$20 \quad - \quad 20 \quad 20$

$P^H$ for acidic buffer

$$P^H = P^{ka} + \log\frac{[salt]}{[acid]}$$

$$= -\log(1.8 \times 10^{-5}) + \log\left[\frac{20/200}{20/200}\right]$$

$P^H = 4.74$

$[H^+] = 10^{-P^H} = 10^{-4.74} \Rightarrow [H^+] = 1.81 \times 10^{-5}$

**66.** (*d*) $HgSO_4 \rightarrow Hg^{2+} + SO_4^{2-}$

$- \quad S \quad S$

$Ksp = S^2$

$\Rightarrow 6.4 \times 10^{-5} = S^2$

$\Rightarrow S = 8 \times 10^{-3}$ mole/L

$S = 8 \times 10^{-3} \times 10^3$ mole/$m^3$

$\Rightarrow S = 8$ mole/$m^3$

**67.** (*b*) For ppt $Q_{sp} > K_{sp}$

$CaF_2 \longrightarrow Ca^{2+} + 2F^-$

$Q_{sp} = [Ca^{2+}][F^-]^2$

(*a*) $Q_{sp} = 12.5 \times 10^{-14}$ (*b*) $Q_{sp} = 12.5 \times 10^{-10}$

(*c*) $Q_{sp} = 12.5 \times 10^{-13}$ (*d*) $Q_{sp} = 12.5 \times 10^{-15}$

Only (*b*) option will get precipitate.

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**68.** (*d*) WA Vs SB end point > 7, Phenolphthalein

**69.** (*a*) Ppt. occurs only when

$K_{ip} > K_{sp}$

$K_{ip} = [Ag^+]\,[Cl^-]$

$$= \frac{10^{-4} \times 10^{-4}}{4} = \frac{10^{-8}}{4} = 2.5 \times 10^{-9}$$

**70.** (*c*) $Q_{sp}(AgBr) = [Ag^+]\,[Br^-]$

$\Rightarrow Q_{sp} = 2 \times 10^{-16}$

$Q_{sp} < K_{sp}$

$\Rightarrow$ No precipitation

$[Ag^+] = 10^{-7}$ M

**71.** (*c*) $HIn \rightleftharpoons H^+ + In^-$

$$P^H = P^{ka} + \log\left[\frac{\text{Ionised}}{\text{Unionised}}\right]$$

$$6 = 5 + \log\left[\frac{\text{Ionised}}{\text{Unionised}}\right]$$

$$1 = \log\left[\frac{\text{Ionised}}{\text{Unionised}}\right]$$

$$\frac{\text{Ionised}}{\text{Unionised}} = \frac{10}{1}$$

Ionised = 10, Unionised = 1

$$\frac{\text{Ionised}}{\text{Ionised} + \text{Unionised}} = \frac{10}{10+1} = \frac{10}{11}$$

**72.** (*a*) $MX_4 \rightarrow M^{4+} + 4X^-$

$\quad - \qquad S \qquad 4S$

$K_{sp} = [S][4S]^4$

$$\Rightarrow K_{sp} = 256S^5 \Rightarrow S = \left(\frac{K_{sp}}{256}\right)^{1/5}$$

**73.** (*d*) $Li_3Na_3(AlF_6)_2 \rightarrow 3Li^+ + 3Na^+ + 2[AlF_6]^{3-}$

$\qquad\qquad\qquad\qquad 3s \qquad 3s \qquad 2s$

$K_{sp} = (3s)^3\,(3s)^3\,(2s)^2 = 2916\ s^8$.

**74.** (*c*) $CaF_2 \rightarrow Ca^{2+} + 2F^-$

$\quad - \qquad S \qquad 2S$

On adding 0.1 M NaF

$\qquad\qquad S \qquad 2S + 0.1$

As S is small so $2S + 0.1 \approx 0.1$

$Ksp = [Ca^{2+}]\,[F^-]^2$

$3.4 \times 10^{-11} = S \times (0.1)^2$

$S = 3.4 \times 10^{-9}$

**75.** (*a*) $Ag_2CO_3 \rightarrow 2Ag^+ + CO_3^{-2}$

$\qquad\qquad\qquad 2s \qquad s$

Here $[Ag^+] = 2s = 2 \times 10^{-4}$ M $\Rightarrow s = 1 \times 10^{-4}$

$\therefore\ K_{SP} = 4s^3 = 4\,(1 \times 10^{-4})^3 = 4 \times 10^{-12}$

**76.** (*c*) $Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$

$\qquad\qquad\qquad 2 \times 0.01 \qquad 0.01$

$Ag_2CO_3 \rightarrow 2Ag^+ + CO_3^{2-}$

$\qquad\qquad\qquad 2x \qquad (x+0.01)$

$x + 0.01 \approx 0.01$

$K_{sp} = 4x^2\,(0.01)$

$K_{sp} = 4 \times 10^{-12} = 4x^2\,(0.01) = x = 10^{-5}$

**77.** (*c*) order of solubility: Complex formation > Pure water> Common ion effect.

So, $s_1 > s_3 > s_2 > s_4$

**78.** (*a*) $K_{SP}(BaCrO_4) = 2.4 \times 10^{-10}\ M^2$

$[CrO_4^{-2}] = 6 \times 10^{-4}$

$K_{SP}(BaCrO_4) = [Ba^{+2}]\,[CrO_4^{-2}] = 2.4 \times 10^{-10}$

$\Rightarrow [Ba^{2+}] \times 6 \times 10^{-4} = 2.4 \times 10^{-10}$

$[Ba^{2+}] = 4 \times 10^{-7}$ M Ans.

**79.** (*a*) $pH = 4 \Rightarrow [H^+] = 10^{-4}$ M $\Rightarrow [OH^-] = 10^{-10}$ M

$Al(OH)_3 \rightarrow Al^{+3} + 3\,OH^-$

$K_{sp}(Al(OH)_3) = [Al^{+3}]\,[OH^-]^3$

$[Al^{3+}]\,[OH^-]^3 = 1 \times 10^{-33}$

$[Al^{3+}]\,(10^{-10})^3 = 1 \times 10^{-33} \Rightarrow [Al^{+3}] = 10^{-3}$ M

**80.** (*d*) Higher the concentration of $H^+$, higher is the solubility of $Fe(OH)_3$.

solubility of $Fe(OH)_3$ is maximum in 0.1 M $H_2SO_4$

**81.** (*a*) order of solubility: Complex formation > Pure water> Common ion effect.

AgBr form complex $[Ag(NH_3)_2]^+$ in $NH_3$, so solubility is maximum in $NH_3$ (aq).

## Exercise-3 (JEE Advanced Level)

**1.** (*a,c*)

$$AB \quad \rightleftharpoons \quad A^+ + \quad B^- \qquad K_1$$

$$\qquad\quad y \qquad\quad y-x$$

$$AB \quad + \quad B^- \quad \rightleftharpoons \quad AB_2^- \qquad K_2$$

$$\qquad\quad y-x \qquad\quad x$$

$$\Rightarrow K_1 = (y-x)y,\ K_2 = \frac{x}{y-x}$$

$$\Rightarrow \frac{k_1}{k_2} = \frac{y}{x}(y-x)^2$$

$[B^-] = y - x$

**2.** (*a,c*)

(*a*) An irreversible reaction goes to almost completion.

(*b*) At equilibrium, the rate of forward reaction becomes equal to that of backward reaction

**3.** (*a,b,c*)

**4.** (*c,d*) Based on theory

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5. (*b,c*) Based on theory

6. (*c,d*)

Addition of solids have no effect on equilibrium and temperature favours endothermic direction while increasing pressure will shift equilibrium in backward direction as $\Delta n_g$ is +ve.

7. (*b,c*)

Exothermic $\Rightarrow$ low temp

$\therefore$ forward reaction favoured

High temp backward

8. (*c,d*)

As given:

$HgO(s)+4I^-(aq)+H_2O(\ell) \rightleftharpoons HgI_4^{2-}(aq)+2OH^-(aq)$,

$$K = \frac{[HgI_4^{2-}][OH^-]^2}{[I^-]^4}$$

as, $OH^\ominus \uparrow$ $HgI^{2-} \downarrow$

$H_2O(\ell) \uparrow$ $OH^\ominus \uparrow$

9. (*c,d*)

Introduction of inert gas at constant Pressure have effect of decrease in Pressure & addition of reactant amount always take place forward reactions.

$\Rightarrow$ equilibrium shifted forward

10. (*c,d*)

As few moles of CO(g) are introduced into the vessel second equilibrium shifts backward, decreasing the concentration of $Cl_2$. So, first equilibrium will go forward.

11. (*a,c*)

Addition of solids have no effect on equilibrium and temperature favours endothermic direction while increasing pressure will shift equilibrium in backward direction as $\Delta n_g$ is +ve.

12. (*a,b,c*)

Addition of inert gas at constant volume has no effect on equilibrium concentrations.

13. (*b,d*)

$pK_a(H_3O^+) = -1.74 = pK_b$ of $OH^-$

$pK_a + pK_b = 14$ only for conjugate acid base pair.

$\alpha = 1.8 \times 10^{-9}$ or $1.8 \times 10^{-7}$ % for $H_2O$.

14. (*a,b,c*)

(*a*) $pK_w = -\log K_w = -\log(1 \times 10^{-12}) = 12$.

(*b*) $K_w = [H^+][OH^-] = 10^{-12}$.

$[H^+] = [OH^-]$

$\Rightarrow [H^+]^2 = 10^{-12}$ ; $[H^+] = 10^{-6}$ ; $pH = -\log[H^+]$

$= -\log 10^{-6} = 6$.

(*c*) $H_2O$ is neutral because $[H^+] = [OH^-]$ at 373 K even when pH = 6.

(*d*) is not correct at 373 K. Water cannot become acidic.

15. (*b,c*)

(*a*) pH of $10^{-8}$ m sol. of HCl is 6.97

(consider the $H^+$ from $H_2O$ also)

(*b*) $H_2PO_4^- \xrightarrow{-H^+} HPO_4^{2-}$

(*c*) $Kw = [H^+][OH^-]$

On $\uparrow$ temp $[H^+][OH^-]$ both $\uparrow \Rightarrow Kw \uparrow$

(*d*) $HA + NaOH \rightarrow NaA + H_2O$

C C

C–C/2 C–C/2 C/2 C/2

$$pH = pKa + \log\frac{salt}{acid}$$

pH = pKa as [salt] = [acid]

16. (*b,c,d*)

(*b*), (*c*) and (*d*) are the mixture of conjugate acid-base pairs.

17. (*a,d*)

(*a*) $[H^+] = 10^{-2}$ and $[OH^-] = 10^{-2}$

$H^+ + OH^- \longrightarrow H_2O$

This leads complete neutralisation

$$\text{so, pH} = \frac{2+12}{2} = 7$$

(*d*) $[H^+] = 10^{-5}$ and $[OH^-] = 10^{-5}$

$[H^+] + [OH^-] \longrightarrow H_2O$

This leads complete neutralisation

$$\text{so, pH} = \frac{5+9}{2} = 7$$

18. (*a*) $HCl \rightarrow H^+ + Cl^-$ $Cl_2HCCOOH \rightleftharpoons Cl_2HCCOO^- + H^+$

0.09 0.09

– 0.09 0.09 $0.09 - x$ $x$ $x$

$[H^+]_T = [H^+]_{HCl} + [H^+]_{Cl_2HCCOOH}$

$= 0.09 + x = 0.09 + x$

At pH = 1; $pH = -\log[H^+] \Rightarrow [H^+] = 10^{-1} = 0.1$

Now

$0.09 + x = 0.1 \Rightarrow x = 0.01$

$$Ka(CHCl_2COOH) = \frac{[CHCl_2COO^-][H^+]}{[CHCl_2COOH]}$$

$$= \frac{0.1 \times 0.1}{(0.09 - 0.01)} = 1.25 \times 10^{-2}$$

19. (*a,c,d*)

$$[H^+]_{CH_3COOH} = \sqrt{Ka.c} = \sqrt{1.8 \times 10^{-5} \times 0.1}$$

$$= 1.34 \times 10^{-3} M$$

The solution which have $[H^+]$ range $1.34 \times 10^{-3}$ will not change pH.

Sol. A, C, D provide. $[H^+]$ in this range.

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**20.** (*a, b, c*)

Total $[H^+] = \sqrt{K_{a1}C_1 + K_{a2}C_2}$

$$= \sqrt{\left(3.6\times10^{-4}\times\frac{0.5}{500}\times100\right)+\left(8\times10^{-4}\times\frac{0.1}{500}\times400\right)}$$

$\therefore [H^+] = 10^{-2}$ M.

For $HN_3$, $[N_3^-] = \frac{K_a[HN_3]}{[H^+]} = \frac{3.6\times10^{-4}\times0.5}{0.01\times500}\times100$

$= \mathbf{3.6 \times 10^{-3}}$ **M.**

For HOCN, $[OCN^-]$

$= \frac{K_a[HOCN]}{[H^+]} = \frac{8\times10^{-4}\times0.1\times400}{0.01\times500}$ **6.4 × 10$^{-3}$ M.**

**21.** (*a,c*)

As $K_a$ of acid increases, $K_b$ of its conjugate base decreases.

**22.** (*b,c*)

On the basis of ostwald dilution law, number of $H^+$ ions will increase but increase in volume will be more. Therefore, $[H^+]$ decreases, pH increases.

**23.** (*b,c,d*)

$\%h = \sqrt{\frac{kw}{ka.C}}$, from given formula

(*a*) depends on concentration

(*b*) $h \propto \frac{1}{\sqrt{C}}$ with dilution C decrease so h increase

(*c*) $h \propto \frac{1}{K_h}$

(*d*) $h \propto K_w$ & $K_w \propto T$

**24.** (*a,d*)

(*a*) $\frac{[H_3AsO_4]}{[H_2AsO_4^-]} = \frac{[H^+]}{K_1} = \frac{10^{-8}}{2.5\times10^{-4}} = \frac{1}{25000}$

$\therefore [H_3AsO_4] << [H_3AsO_4^-]$.

(*b*) $\frac{[H_2AsO_4^-]}{[HAsO_4^{2-}]} = \frac{[H^+]}{K_2} = \frac{10^{-8}}{5\times10^{-8}} = \frac{1}{5}$

(*c*) $\frac{[AsO_4^{3-}]}{[HAsO_4^{2-}]} = \frac{K_3}{[H^+]} = \frac{2\times10^{-13}}{10^{-8}} = \frac{1}{50000}$

$\therefore [AsO_4^{3-}] << [HAsO_4^{2-}]$.

**25.** (*a,b,c*)

(*a*) $H_2CO_3 + OH^- \longrightarrow HCO_3^- + H_2O$

moles 1.5 1

0.5 – 1 1

so it is a buffer solution.

(*b*) $H_2CO_3 + OH^- \longrightarrow HCO_3^- + H_2O$

moles 1.5 2

– 0.5 1.5 1.5

$HCO_3^- + OH^- \longrightarrow CO_3^{2-} + H_2O$

moles 1.5 0.5

1 – 0.5

so it is a buffer solution.

(*c*) $NH_4OH + H^+ \longrightarrow NH_4^+ + H_2O$

moles 5 4

1 – 4

so it is a buffer solution.

**26.** (*a,b*)

(*a*) A buffer solution is a solution which contains weak acid and its conjugate base. it is acidic buffer.

(*b*) Basic buffer contains weak base and its conjugate acid.

(*c*) is wrong because it does not show change in pH on adding small amount of acid or base.

(*d*) is wrong; all the above statements are not correct.

**27.** (*a,d*)

$HIn \longrightarrow H^+ + In^-$

$K_{in} = \frac{[H^+][In^-]}{[HIn]} \Rightarrow \frac{[In^-]}{[HIn]} = \frac{K_{In}}{[H^+]} = \frac{10^{-9}}{10^{-9.6}} = 10^{0.6} = 4$

$\therefore$ % of $[In^-]$ in solution $= \frac{4}{5}\times100 = 80\%$

so pink colour will be visible.

**28.** (*b,c*) $BOH + HCl \longrightarrow BCl + H_2O$

$pOH = p^{kb} + \log\frac{Salt}{Base}$

(*a*) as $\frac{1}{5}$ neutralised so,

$pH = 9, \Rightarrow p^{OH} = 5$

$5 = p^{Kb} + \log\frac{1/5}{4/5}$

$\Rightarrow$ Kb = 2.5 × 10$^{-6}$

(*b*) At equivalence point

$pH = \frac{1}{2}\left[p^{kw} - p^{K_D} - \log c\right]$

$4.5 = \frac{1}{2}[14 - 5.6 - \log c] \Rightarrow C = 0.25M$

(*c*) $C_{HCl} = \frac{\eta_{HCl}}{v_f} = 0.25 = \frac{0.5\times v_L}{v_L + 0.1} \Rightarrow v_L = 0.1L = 100mL$

(*d*) at equivalence point $n_{WB} = n_{HCl}$

M/45 = 0.5 × 0.1

M = 2.25g

% weight of base $= \frac{2.25}{2.5}\times100 = 90\%$

**29.** (*c, d*)

In $AgNO_3$ solution, the solubility of AgCN will decrease as compared to pure water because of common ion effect of $Ag^+$ ion.

In $NH_3$ solution and buffer of pH = 5, the solubility of AgCN will increase due to complex formation in case of $NH_3$ solution and hydrolysis of $CN^-$ ions in case of buffer of pH = 5.

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**30.** (*a,b*)

$K_{sp} = 1.1 \times 10^{-11} = (1.4 \times 10^{-4})^{x+y}\, x^x \,.\, y^y$

so we have $x + y = 3$ (by comparing values)

so, $x^x \,.\, y^y = \dfrac{1.1\times10^{-11}}{1.4\times1.4\times1.4\times10^{-12}} = \dfrac{110}{1.96\times1.4} = 4$

Hence $x = 1, y = 2$ or $y = 1, x = 2$

**31.** (*b,d*) Factual

**32.** (*a,b*)

$H_2SO_4$ is a strong acid and it completely dissociated in $H_2O$. Hence its and are determined in $CH_3COOH$.

Also $H_2SO_4$ is neutral while $HSO_4^-$ because the – ve charge has more affinity towards $H^+$ ion. Hence choices (a) and (b) are correct while (c) is incorrect. As $H_2SO_4$ is completely dissociated in water so a 0.01 M $H_2SO_4$ is completely dissociated and its $[H^+]$ ion conc. will be equal to 0.02 M.

Hence choice (d) will be incorrect.

**33.** (*b*)

| | $CO_2(g)$ | + C(s) | ⇌ | $2CO(g)$ |
|---|---|---|---|---|
| t= 0 | 0.1 | | | |
| $t = t_{eq.}$ | $0.1 - x$ | | | $2x$ |

$$\frac{(0.1-x)44 + (2x \times 28)}{0.1+x} = 36 \quad \Rightarrow x = \frac{1}{30}$$

Total moles $= 0.1 + \dfrac{1}{30} = \dfrac{2}{15} = 0.1 + x$

**34.** (*a*) Moles of $CO_2$ at eq. $= 0.1 - x = \dfrac{1}{15}$

Moles of CO at eq. $= 2x = \dfrac{2}{30} = \dfrac{1}{15}$

$$P_{CO} = \frac{0.082 \times 900}{15 \times 0.82} = 6 \text{ atm}$$

$$P_{CO_2} = \frac{1}{15} \times \frac{0.082}{0.82} \times 900 = 6 \text{ atm}$$

$$\therefore K_p = \frac{6 \times 6}{6} = 6 \text{ atm}$$

**35.** (*b*) $n_c = 0.1$

| | C(s) | + $CO_2$ (g) | ⇌ | 2CO (g) |
|---|---|---|---|---|
| t = 0 | 0.1 | n | | |
| $t = t_{eq.}$ | – | $n - 0.1$ | | 0.2 |

$$P_{CO} = \frac{0.2 \times 0.082 \times 900}{0.82} = 18 \text{ atm}$$

$$K_P = 6 = \frac{(18)^2}{P_{CO_2}};\ P_{CO_2} = 54 \text{ atm}$$

$$\therefore \frac{(n-0.1)\,(0.082)\,(900)}{0.82} = 54$$

$$n = 0.7$$

**36.** (*d*) Solublity of gas is favourable at high pressure and this process is exothermic hence solubility will be more at low temperature.

**37.** (*c*) Since density of gold decreases after melting therefore it is favourable at low pressure and high temperature.

**38.** (*a,b*)

Number of moles will remain unchanged but due decreased volume pressure will get increased and the concentrations.

**39.** (*c*) Given as:

| | $H_2(g)$ + | $I_2(g)$ ⇌ | $2\,HI(g)$ |
|---|---|---|---|
| initally | 1 | 3 | |
| At equilbrium | 1–x/2 | 3–x/2 | $x$ |

Addition of 2 mole $H_2$.

$$\Rightarrow \frac{(x)^2}{\left(1-\frac{x}{2}\right)\left(3-\frac{x}{2}\right)} = \frac{(2x)^2}{(3-x)(3-x)}$$

$$\Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$$

$$x = 1.5$$

**40.** (*c*) $K_C = \dfrac{(2x)^2}{(3-x)(3-x)} = \dfrac{9}{\left(\frac{9}{4}\right)} = 4$

$K_C = K_p$ because $\Delta n_g = 0$

$K_C = K_p = 4$

**41.** (*b*) Given :

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

| | 2 | 1.5 | |
|---|---|---|---|
| at equ. | 2–2x | 1.5–x | 2x |

$$\text{conc.} = \frac{2-2x}{5}\ \frac{1.5-x}{5}\ \frac{2x}{5}$$

$SO_2 \longrightarrow SO_3$ (nf of $SO_2 = 2$)

m.eq of $KMnO_4$ = m.eq of $SO_2$

$0.4\times5 = (2.2x) \times 2$

$2 = 2(1-x) \times 2$

$$\Rightarrow 1 - x = 1/2 \Rightarrow x = \frac{1}{2}$$

$$[SO_2] = \frac{1}{5};\ [O_2] = \frac{1}{5}$$

$$[SO_3] = \frac{1}{5}$$

$K_C = 5$

**42.** (*d*) Based on theory

**43.** (*a*) Since $K_a$ of $CH_3COOH$ & $K_b$ of $NH_4OH$ are same so degree of hydrolysis of $CH_3COO^-$ & $NH_4^+$ are exactly same.

**44.** (*c*) Consider equal behaviour of acid & base

**45.** (*a*) For $SrF_2$ in pure water

$4s_1^3 = K_{sp}$

For $SrF_2$ in 0.1 M NaF solution

$s_2 (0.1)^2 = K_{sp}$

$$\Rightarrow 4s_1^3 = s_2\,(0.01)$$

$$\Rightarrow 4s_1^3 = s_1 \times \frac{256}{10^6}(0.01)$$

$$\Rightarrow s_1 = 8 \times 10^{-4}\ \text{M}$$

$$\therefore K_{sp} = 4s_1^3 = \mathbf{2.048 \times 10^{-9}}$$

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**46.** (*c*) $[Sr^{2+}]_i = 0.0011 = 11 \times 10^{-4}$ M

$[Sr^{2+}]_f = 2 \times 10^{-4}$ M

$\therefore$ $[Sr^{2+}]$ precipitated $= (11 - 2) \times 10^{-4}$ M

$= 9 \times 10^{-4}$ M

$[F^-]$ needed for this precipitation $= 2 \times 9 \times 10^{-4}$

$= 18 \times 10^{-4}$ M

Also, $[Sr^{2+}]_f [F^-]^2_f = K_{sp} = 2.048 \times 10^{-9}$

But, $[Sr^{2+}]_f = 2 \times 10^{-4}$ M

$\therefore$ $[F^-]_f = 3.2 \times 10^{-3}$ M

$\therefore$ Total $[F^-]$ needed $= 3.2 \times 10^{-3} + 18 \times 10^{-4}$

$= 5 \times 10^{-3}$ M

$\therefore$ NaF needed for 100 ml solution $= \dfrac{5 \times 10^{-3} \times 42}{10}$

$=$ **0.021 g**

**47.** (*b*) $SrF_2 \longrightarrow Sr^{2+} + 2F^-$

s 2s

$F^-$ will react wth $H^+$ to produce HF

$F^- + H^+ \rightleftharpoons HF$

$K = \dfrac{1}{K_a} = \dfrac{[HF]}{[H^+][F^-]} = \dfrac{7}{10^{-5}}$

$\therefore$ $[HF] = 7 \times 10^5 [F^-] [H^+]$ ($\because [H^+] = 10^{-5}$ ; pH=5)

$= 7 \times 10^5 [F^-] \times 10^{-5} = 7 [F^-]$

Here $[F^-] + [HF] = 2s$

$\therefore$ $[F^-] = \dfrac{s}{4}$

$K_{sp} = s\left(\dfrac{s}{4}\right)^2 = 2.048 \times 10^{-9}$

$\therefore$ $s = 3.2 \times 10^{-3}$ mol/L

**48.** (*b*) A→q,s; B→p; C→p; D→r

(*a*) $\Delta n_g = 2 - 4 = -2$ and $K_p = K_C (RT)^{\Delta n_g}$

(*b*) $\Delta n_g = 2 - 1 = 1$ and $K_p = K_C (RT)^{\Delta n_g}$

(*c*) $\Delta n_g = 2 - 1 = 1$ and $K_p = K_C (RT)^{\Delta n_g}$

(*d*) $K_p$ is not defined as all are in liquid state.

**49.** (*a*) A→p; B→q; C→s; D→r

$K_p = K_C (RT)^{\Delta n_g}$

(*a*) $\Delta ng = -2$ (*b*) $\Delta ng = 0$

(*c*) $\Delta ng = -1/2$ (*d*) $\Delta ng = 1/2$

**50.** (*c*) A → p,t; B→q,r; C→q,s

(*a*) $\Delta n_g = 0$

(*b*) $\Delta n_g = -1$

(*c*) $\Delta n_g = 2$

if reaction is exothermic then T ↑se then reaction goes forward.

$\Delta n_g = 0$, no effect on pressure

$\Delta n_g < 0$, P high then reaction goes forward.

$\Delta n_g > 0$, P low then reaction goes forward.

**51.** (*d*) A→s, B→s, C→q

(*a*) $[H^+] = \dfrac{200 \times 1.225 \times \frac{25}{100} \times 2}{98}$

$= 1.25$

$[OH^-] = 0.525 \times \dfrac{800}{1000} \times 3 = 1.26$

Remaining $[OH^-] = 10^{-2}$

$\Rightarrow$ pOH = 2

$\Rightarrow$ pH = 12

Indicator having range 11.4 to 13.

(*b*) $pH = 11 - \log 2 + \log \dfrac{0.8}{0.1}$

$= 11 + 2 \log 2$

$= 11.6$

Indicator having range 11.4 to 13.

(*c*) HCl m.eq = 5

NaOH m.eq = 13

↓

Remaining NaOH = 8

m.eq of HA = 10

$HA + OH^- \longrightarrow A^- + H_2O$

10 8

2 08

$pH = P^{Ka} + \log \dfrac{[A^-]}{[HA]}$

$= 5 + \log 4$

$\Rightarrow 5.6$

Indicator having range 4.6 to 6.4

**52.** [50]

Given:– $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $K_p = 0.5$

$K_p = K_C (RT)^{\Delta n_g}$

now,

$\Delta n_g = 2 - 2 = 0$

$\therefore$ $K_p = K_C$

$K_C = 50$

**53.** [64]

For this reaction there is no change in equilibrium constant by change of volume.

**54.** [16]

$K_p$ remain constant on change in volume.

**55.** [120]

Given:

| | $A(g) \rightleftharpoons$ | $B(g)$ | $+ C(g)$ | $+ D(g)$ |
|---|---|---|---|---|
| initially | 1 | 0 | 0 | 0 |
| at eq. | 1–x | x | x | x |

Total mole = 1 + 2x

Apply mass conservation

(Wt. of A at t = 0) = (wt of mixture at eq.)

$1 \times M_{wt.} = [1 + 2x]\ 2 \times 50$

$M_{w.} = (1 + 0.2) \times 100$ (as x = 0.1)

$M_{wt.} = 120$

@PROCHEM

**56.** [1]

If $\Delta G^\circ = 0$

$\Delta G^\circ = -2.303\ RT \log K_p$

$\log K_p = 0$

$(\because \log 1 = 0)$

$K_p = 1.$

**57.** [4.0%]

$$\%h = \sqrt{\frac{kw}{ka.c}}$$

$$= \sqrt{\frac{10^{-14}}{6.2\times10^{-10}\times10^{-2}}}$$

$\% h = 0.04$

Percent hydrolysis = 4%

**58.** [10]

$$\frac{\alpha_2}{\alpha_1} = \sqrt{\frac{K_a.C_2}{K_a.C_1}} = \sqrt{\frac{C_2}{C_1}} = 10$$

**59.** [9] $K_b = 10^{-5}$, $pK_b = 5$

$BOH + HCl \rightarrow BCl + H_2O$

At half neutralization, 50% of the base is converted to its salt, with strong acid HCl, it forms a basic buffer.

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pOH = 5 + \log\frac{[salt]}{[base]}$$

$[salt] = [base]$

$pOH = 5$

$pH = 14 - pOH = 9$

**60.** [3]

$$H_3PO_4 \underset{}{\overset{\text{Step-1}}{\rightleftharpoons}} H^+ + H_2PO_4^-$$

$$H_2PO_4^- \underset{}{\overset{\text{Step-2}}{\rightleftharpoons}} H^+ + HPO_4^{--}$$

$$HPO_4^{--} \underset{}{\overset{\text{Step-3}}{\rightleftharpoons}} H^+ + HP_4^{---}$$

## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*b*)

| | A | + | B | $\rightleftharpoons$ | C | + | D |
|---|---|---|---|---|---|---|---|
| t = 0 | 1 | | 1 | | 1 | | 1 |
| $t_{eq}$ | 1–x | | 1–x | | 1 + x | | 1 + x |

$$K_{eq} = \frac{(1+x)^2}{(1-x)^2} = 100 \quad \Rightarrow \quad \frac{1+x}{1-x} = 10$$

$\Rightarrow 1 + x = 10 - 10x \quad \Rightarrow \quad 11x = 9$

$\Rightarrow x = \frac{9}{11}$

$\Rightarrow [D] = 1 + \frac{9}{11} \quad \Rightarrow [D] = 1.818$

**2.** (*d*) From thermodynamics

$$\ln k = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

for exothermic reaction,

$\Delta H = -ve$

$$\text{slope} = \frac{-\Delta H^\circ}{R} = +ve$$

So from graph, line should be A & B.

**3.** (*a*) $CH_3COOK + H_2O \rightarrow \underset{\text{Weak acid}}{CH_3COOH} + \underset{\text{Strong base}}{KOH}$

Hence nature of solution is basic.

**4.** (*b*) Methyl orange shows Red (pinkish) color in Ac medium & yellow color in basic medium since orig solution is basic so

initial color ⇒ yellow & Titrated with acid so

Final color ⇒ pinkish (red)

**5.** (*a*) $H_2S \rightleftharpoons HS^- + H^+ \qquad K_1 = 10 \times 10^{-7}$

$$Ka_1 = \frac{[H^+][HS]}{[H_2S]}$$

and $HS^- \rightleftharpoons S^{2-} + H^+ \qquad K_2 = 1.2 \times 10^{-13}$

$$Ka_2 = \frac{[H^+][S^{2-}]}{[HS^-]}$$

$\therefore H_2S \rightleftharpoons S^{2-} + 2H^+ \quad K_{overall} = \times K_1 \times K_2 = 1.2 \times 1$

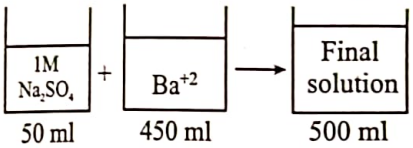
So,

$$[S^{2-}] = \frac{1.2\times10^{-20}\times[H_2S]}{[H^+]^2} = \frac{1.2\times10^{-20}\times10^{-1}}{4\times10^{-2}}$$

$= 3 \times 1^{-20}$ M

*All the $[H^+]$ will come from strong acid [HCl] only

**6.** (*b*)




Concentration of $SO_4^{-2}$ in $Ba^{+2}$ solution

$M_1V_1 = M_2V_2$

$1 \times 50 = M_2 \times 500$

$$M_2 = \frac{1}{10}$$

for just precipitaion

$I.P = K_{sp}$

$$\left[Ba^{+2}\right]\left[SO_4^{-2}\right] = K_{sp}(BaSO_4)$$

$$\left[Ba^{+2}\right]\times\frac{1}{10} = 10^{-10}$$

$[Ba^{+2}] = 10^{-9}$ M in 500 ml solution

For calculation of $[Ba^{+2}]$ in original solution (450 ml)

$M_1 \times 450 = 10^{-9} \times 500$

$$M_1 = \frac{500}{450}\times10^{-9} = 1.11\times10^{-9}\text{M}$$

[$M_1$ = molarity of $Ba^{+2}$ in original solution (450 ml)]

@PROCBSE

**7.** (*d*) Moles of $NH_4SH = \frac{5.1}{51} = 0.1$ mol

$NH_4SH(s) \rightleftharpoons NH_3(g) + H_2S(g)$

| | | | |
|---|---|---|---|
| 0.1 | – | – | Initially |
| $0.1(1-\alpha)$ | $0.1\alpha$ | $0.1\alpha$ | At eq. |

given $\alpha = 30\% = 0.3$

Therefor number of moles equilibrium due

$= 0.1 \times 0.3 + 0.1 \times 0.3$

$= 0.03 + 0.03 = 0.06$

as we know $PV = nRT$

At equlibrium,

$P_{total} \times 3L = 0.06 \times .082 \times 600$

$P_{total} = 0.984$ atm

At equilibrium,

$P_{NH_3} = P_{H_2S} = \frac{P_{total}}{2} = 4.492$

$K_p = P_{NH_3} \times P_{H_2S} = 0.492 \times 0.492$

$K_p = 0.242\ atm^2$

**8.** (*d*) $\Delta G^\circ = 120 - \frac{3}{8} = 315$

$= 1.875$

X is maximum

**9.** (*a*)

| | | | | |
|---|---|---|---|---|
| | $2NH_3$ | $\rightleftharpoons N_2$ | + | $3H_2$ |
| t = 0 | $P_1$ | – | | – |
| t = eq | $P_{NH_3}$ | $\frac{P_1}{2}$ | | $\frac{3P_1}{2}$ |

$P = P_{total} = \frac{P_1}{2} + \frac{3P_1}{2} = 2P_1$

$\therefore\ P_1 = \frac{P}{2}$

$k_{eq} = \frac{1}{k_p} = \frac{\left(\frac{P_1}{2}\right)\left(\frac{3P_1}{2}\right)^3}{(P_{NH_3})^2}$

$P_{NH_3} = \frac{(K_P)^{\frac{1}{2}} 3^{\frac{3}{2}} P^2}{16}$

**10.** (*c*) $A_2(g) + B_2(g) \overset{K_1}{\rightleftharpoons} 2AB(g)$ ...(i)

$6AB(g) \overset{K_2}{\rightleftharpoons} 3A_2(g) + 3B_2(g)$ ...(ii)

Reaction (ii) = –3 × reaction (i)

$\therefore K_2 = \left(\frac{1}{K_1}\right)^3 \Rightarrow K_2 = K_1^{-3}$

**11.** (*a*)

| | | | | | | |
|---|---|---|---|---|---|---|
| | A | + | 2B | $\rightleftharpoons$ | 2C | + D |
| Initially conc | a | | 1.5a | | | |
| at eq. | a – x | | 1.5a – 2x | | 2x | x |

at equilibrium, a – x = 1.5 a – 2x

or a = 2x

$k_C = \frac{(2x)^2 x}{(a-x)(1.5a-2x)^2} = \frac{4x^2.x}{(x)(x)^2} = 4$

**12.** (*b*) $A(s) \rightleftharpoons \underset{P_1}{B(g)} + \underset{P_1+P_2}{C(g)}$ $k_{P_1} = x\ atm^2$

$D(s) \rightleftharpoons \underset{P_1+P_2}{C(g)} + \underset{P_2}{E(g)}$ $k_{P_2} = x\ atm^2$

$x = k_{P_1} = P_1(P_1 + P_2)$ ... (i)

$y = k_{P_2} = P_2(P_1 + P_2)$ ... (ii)

$k_{P_1} + k_{P_2} = (P_1 + P_2)^2$

on addition equation (i) and (ii)

$x + y = (P_1 + P_2)^2$

or $P_1 + P_2 = \sqrt{x+y}$

or $P_{Total} = P_B + P_C + P_E = 2(P_1 + P_2) = 2\sqrt{x+y}$

**13.** (*b*) $K_p = K_c (RT)^{\Delta ng}$

**14.** (*c*) $S(s) + O_2(g) \rightleftharpoons SO_2(g)$ $K_1 = 10^{52}$ ... (i)

$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$ $K_2 = 10^{129}$ ... (ii)

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $K_3 = x$

multiplying equation (i) by (ii) ;

$2SO(s) + 2O_2(g) \rightleftharpoons 2SO_2(g)$ $K'_1 = 10^{104}$ ... (iii)

$\Rightarrow$ Substracting (iii) from (ii); we get

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ ;

$K_{eq} = 10^{(129-104)} = 10^{25}$

**15.** (*a*) $\Delta G^\circ = -RT\ln K$

if $K < 1 \Rightarrow \Delta G^\circ > 0$

**16.** (*d*) if $\Delta n_g \neq 0$

$Kp \neq Kc$

**17.** (*a*) $Kc = 1.7 \times 10^{16}$ i.e. reaction goes to completion. catalyst won't affects the equilibrium it only affect rate of reaction.

**18.** (*a*) The pH of NaOH is more than 7 and during the titration it decreases.

**19.** (*b*)

(*a*) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2$

$400 \times .1 = 40$ $400 \times .1 = 40$

20 0

$\therefore [H^+] = \frac{20 \times 20}{800} = \frac{1}{20} \Rightarrow pH = -\log\left(\frac{1}{20}\right)$

$\therefore$ pH = 1.3

(*b*) $\log\left(\frac{Kw_2}{Kw_1}\right) = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

(*c*) $Ka = 10^{-5}$, $pH = 5 \Rightarrow [H^+] = 10^{-5}$

$Ka = \frac{c\alpha^2}{(1-\alpha)} \Rightarrow Ka = \frac{[H^+].\alpha}{(1-\alpha)}$

$10^{-5} = \frac{10^{-5} \times \alpha}{(1-\alpha)}$ or $\alpha = 50\%$

@PROCBSE

**20.** (*a*) $Ca(OH)_2 + Na_2SO_4 \rightarrow CaSO_4 + 2NaOH$

10 mmol 14 mmole 14 mmol 28 mmol

$w_{CaSO_4} = 14 \times 10^{-3} \times 136 = 1.9$ gm

$$[OH^-] = \frac{28}{100} = 0.28M$$

**21.** (*c*) $8 \times 10^{-12} = (2S' + 0.1)^2 S'$

or $S' = 8 \times 10^{-10}$ M

**22.** (*b*) $\underset{2m.m}{H_2SO_4} + \underset{6m.m}{2NH_4OH} \rightarrow \underset{2m.m}{(NH_4)_2SO_4} + 2H_2O$

$$pOH = 4.7 + \log\frac{4}{2} = 5$$

$pH = 14 - 5 = 9$

**23.** (*a*) Rainwater becomes acidic because gases present in the environment are dissolved so its pH will be less than 7. pH of rainwater is approximate 5.6.

**24.** (*d*) $Zr_3(PO_4)_4(S) \rightleftharpoons 3Zr^{4+}(aq.) + 4PO_4^{3-}(aq.)$

3S M 4S M

$K_{sp} = [Zr^{4+}]^3 [PO_4^{3-}]^4 = (3S)^3 . (4S)^4 = 6912\ S^7$

$$\therefore S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$$

Correct option: (*d*)

**25.** (*d*) $K_{sp} = 4(s)^3$

$= 4 \times (1.84 \times 10^{-5})^3$

$Cd(OH)_2 \rightleftharpoons Cd^{+2} + 2OH^-$

$S' \qquad S' \qquad (10^{-2} + S') = 10^{-2}$

$S' \times (10^{-2})^2 = 4 \times (1.84 \times 10^{-5})^3$

$S' = 4 \times (1.84)^3 \times 10^{-11}$

$(S') = 2.491 \times 10^{-10}$ M

**26.** (*d*) $Al(OH)_3 \rightleftharpoons Al^{+3} + 3OH^-$

$S' \qquad 0.2 + 3(S') \cong 0.2$

$S' \times (0.2)^3 = k_{sp} = 2.4 \times 10^{-24}$

$(S') = 3 \times 10^{-22}$ M

**27.** (*b*) For the salt of strong acid and weak base

$$[H^+] = \sqrt{\frac{K_w \times C}{K_b}}$$

$$[H^+] = \sqrt{\frac{10^{-14} \times 2 \times 10^{-2}}{10^{-5}}}$$

$$-\log[H^+] = 6 - \frac{1}{2}\log 20$$

$\therefore pH = 5.35$

**28.** (*d*) on increasing temperature pH decreases and the reaction endothermic

**29.** (*b*) $2H_2(g) + 2NO(g) \rightleftharpoons N_2(g) + 2H_2O(g)$

$$K_c = \frac{k_f}{k_b} = \frac{[N_2][H_2O]^2}{[H_2]^2[NO]^2}$$

Given for forward reaction, rate $= k_f[NO]^2[H_2]^2$

for reverse reaction rate $= k_b[N_2][H_2O]^2$

**30.** (*d*) $A \rightarrow B$, $K_C = \frac{12}{6} = 2$

**31.** (*d*) $A \rightleftharpoons B + C$ .....(i) $K_{eq}^{(1)}$

$B + C \rightleftharpoons P$ .....(ii) $K_{eq}^{(2)}$

$A \rightleftharpoons P$

$K_{eq}$ (overall) $= K_{eq}^{(1)} K_{eq}^{(2)}$

**32.** (*b*) At equilirium,

rate of forward reaction = Rate of backward reaction

**33.** [16.00]

| | X | + | Y | $\rightleftharpoons$ | 2Z |
|---|---|---|---|---|---|
| t = 0 | 1 | | 1.5 | | 0.5 |
| t = eq | 1–0.25 | | 1.5–0.25 | | 0.5+0.5 |

$$\therefore K_{eq} = \frac{(1)^2}{0.75 \times 1.25} = \frac{x}{15}$$

$$\Rightarrow x = \frac{15}{0.75 \times 1.25} = 16$$

**34.** (*b*) $\because$ Given reaction is endothermic

$\therefore$ On decreasing temperature backward reation will be favoured. On addng $N_2$, pressure is increased at constant T, and volume would also be constant so no change is observed.

**35.** (*a*) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $K_C$

$3NH_3(g) + N_2(g) \rightleftharpoons 3H_2(g)$ $\frac{1}{K_C}$

For

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \quad \frac{1}{K_C^{1/2}}$$

$$\frac{1}{K_C^{1/2}} = \frac{1}{(64)^{1/2}} = \frac{1}{8}$$

**36.** (*a*) $K_p = K_C(RT)^{\Delta n_g}$

$= K_C(RT)^{1-3/2}$

$= K_C(RT)^{-1/2}$

$\Rightarrow K_C = K_p(RT)^{1/2}$

**37.** (*c*) Catalyst will effect the $R_f$ & $R_b$ at some extent, so $K_{eq}$ won't change.

**38.** [3] $K_C = \frac{1.5 \times 0.5}{0.5 \times 0.5} = 3$

**39.** [10.60 to 10.60]

$$M_{H_2SO_4} \Rightarrow \frac{9.8}{98 \times 100} = 10^{-3}$$

$$M_{NaOH} \Rightarrow \frac{4}{40 \times 100} = 10^{-3}$$

$$= \frac{40 \times 10^{-3} - 10 \times 10^{-3} \times 2}{50} = \frac{20}{50} \times 10^{-3}$$

$$[OH^-] = \frac{2}{5} \times 10^{-3}$$

$pOH = 3.397$

$pH = 10.603$

@PROCBSE

**40.** [5.22 to 5.24]

m mole of acetic acid in 20 mL = 2

m mole of HCl in 20 ml = 1

m mole of NaOH = 2.5

$CH_3COOH + NaOH$ (remaining) $\rightarrow CH_3COONa + water$

| 2 | 3/2 | 0 | 0 |
|---|---|---|---|
| 0.5 | 0 | 3/2 | |

$$pH = pK_a + \log \frac{3/2}{1/2}$$

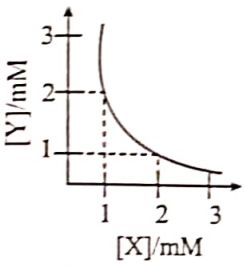
$= 4.75 + \log 3 = 4.75 + 0.4771 = 5.2271 \approx 5.23$

$= 4.74 + 0.48 = 5.22$

**41.** (*b*) The strength of NaOH measured by titrating against oxalic by taking NaOH in the burette and oxalic acid in the conical flask.

$$2NaOH + \begin{matrix}COOH\\|\\COOH\end{matrix} \longrightarrow \begin{matrix}COONa\\|\\COONa\end{matrix} + 2H_2O$$

**42.** (*c*)




$$XY_2(s) \rightarrow \underset{10^{-3}}{X^-(aq)} + \underset{2\times10^{-3}}{2Y^-(aq)}$$

$K_{sp} = [X^-][Y^-]^2$

or $K_{sp} = 10^{-3} \times (2 \times 10^{-3})^2$ or $k_{sp} = 4 \times 10^{-9}\ M^3$

**43.** (*b*) $Q = [Pb^{2-}][Cl^-]^2$

$$= \frac{300\times0.134}{400} \times \left[\frac{100\times0.4}{400}\right]^2$$

$$= \frac{3\times0.134}{4} \times (0.1)^2$$

$= 0.105 \times 10^{-2}$

$= 1.005 \times 10^{-3}$

$\boxed{Q > Ksp}$

**44.** [4] $Cr(OH)_3 \rightarrow Cr^{-3} + 3OH^-$

s 3s

$k_{sp} = s.(3s)^3$

$\Rightarrow 6 \times 10^{-31} = 27.s^4$

$$\Rightarrow s = \left(\frac{6}{27}\times10^{-31}\right)^{1/4}$$

$[OH^-] = 3s$

$$= 3\times\left(\frac{6}{27}\times10^{-31}\right)^{1/4} = (18 \times 10^{-31})^{1/4}M$$

**45.** (*b*) $K_{sp}$ value of CuS is very low = $10^{-36}$ ($36 \times 10^{-36}$) due to low $K_{sp}$ value $Cu^{+2}$ ion gets precipitated very quickly even with very low concentration of $S^{-2}$ ion.

$CuS(s) \rightleftharpoons Cu^{+2} + S^{2-}$

$K_{sp} = [Cu^{+2}]\ [S^{-2}]$

$Cu^{+2} + S^{-2}(s) \overset{K_{eq}}{\rightleftharpoons} CuS(s)$

$$K_{eq} = \frac{1}{K_{sp}} = \frac{1}{3.6\times10^{-36}} = \frac{10^{36}}{3.6}$$

Due to high value of K (equilibrium constant) CuS precipitated quickly.

**46.** (*a*) For titration between HCl and NaOH, pH at equivalence point is found to be 7.

**47.** (*b*)

| | HCl | + $CH_3COONa$ | $\rightarrow CH_3COOH$ | + NaCl |
|---|---|---|---|---|
| Initial | $100 \times 0.1$ = 10 mmol | $200 \times 0.1$ = 20 mmol | | |
| Final moles | 0 | 10 mmol | 10 mmol | 10 mmol |

$CH_3COOH$ and $CH_3OONa$ both are present. Both form acidic buffer.

**48.** [37]

At 30 bar pressure mass of $CO_2$ in 1 kg water = 44gm

At 3 bar pressure mass of $CO_2$ in 1 kg water = 4.4 gm Moles of $CO_2$ in 1 kg water = 0.1

$$\underset{0.1(1-\alpha)}{H_2CO_3} \rightleftharpoons \underset{0.1\alpha}{H^+} + \underset{0.1\alpha}{HCO_3^-}$$

$$Ka_1 = \frac{[H^+][HCO_3^-]}{[H_2CO_3]}$$

$$4\times10^{-7} = \frac{0.1\alpha^2}{1-\alpha} \simeq 0.1\alpha^2$$

$\alpha = 2 \times 10^{-3}$

$[H^+] = 0.1\ \alpha = 2 \times 10^{-4}$ ; pH = 3.7

$= 37 \times 10^{-1}$

**49.** [02.00]

$$AB_2 \rightleftharpoons \underset{s}{A^{2-}}(aq) + \underset{2s}{2B^-}(aq)$$

$K_{sp} = 4s^3 = 3.2 \times 10^{-11}$

$\Rightarrow s^3 = 8 \times 10^{-12}$

$s = 2 \times 10^{-4}$

**50.** (*b*) (*a*) 0.01 M HCl

$[H^+] = 10^{-2}$, $pH = -\log 10^{-2} = 2$

$pOH = 14 - 2 = 12$

(*b*) 0.01 M NaOH

$[OH^-]\ 10^{-2}$, $pOH = -\log [OH]$

$= 2$

(*c*) 0.01 M $CH_3COONa$

$$pH = 7 + \frac{1}{2}[pK_a \log 0.01]$$

$pH > 7 \Rightarrow pOH < 7$

(*d*) 0.01 M NaCl, pH = 7, pOH = 7

Order of pOH value A > D > C > B

@PROCBSE

**51.** [1] $K = K_1 \times K_2 \times K_3 \times K_4$

$= 10^4 \times 1.58 \times 10^3 \times 5 \times 10^2 \times 10^2$

$K = 7.9 \times 10^{11}$

Where K → Equilibrium constant for formation of $[Cu(NH_3)_4]^{2+}$ So equilibrium constant (K) for dissociation of $[CuCNH_3)^{2+}$ is $\frac{1}{K}$

$K' = \frac{1}{K}$

$K' = \frac{1}{7.9 \times 10^{11}} = 1.26 \times 10^{-12}$

So, the value of x = 1.26

**52.** [5]

$\because$ No. of atoms of Cl = no. of molecules of $Cl_2$ i.e $n_{Cl} = n_{Cl_2}$

According Dalton's low of partial pressure.

$P_{Cl} = 0.5$ atm $= P_{Cl_2}$

$\therefore K_P = \frac{(0.5)^2}{0.5} = 0.5$

**53.**[1380] $\Delta G° = -RT \times \log K_{eq}$

$= -R \times 300 \times \log 10^2$

$= -1380\ R$

**54.** [72] $AB_2(G) \rightleftharpoons A(g) + 2B(g)$

| | | | |
|---|---|---|---|
| t = 0 | 1 | – | – |
| t = eq$^m$ | 1–α | α | 2α |

$\therefore$ No. of motes at equilibrium

$= 1 + 2\alpha = \frac{PV}{RT} = 1.93$

$\therefore \alpha = 0.465$

$\therefore P_{AB_2} = \frac{1-\alpha}{1+2\alpha} \times P_T \approx 0.46$ atm

$P_A = \frac{\alpha}{1+2\alpha} \times P_T \approx 0.46$ atm

$P_B = \frac{2\alpha}{1+2\alpha} \times P_T \approx 0.91$ atm

$K_B = \frac{P_A \cdot (P_B)^2}{P_{AB_2}}$

$= \frac{0.46 \times (0.91)^2}{0.53} = 72 \times 10^{-2}$

**55.**[354] $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$\Delta n_g = 2 - 1 = 1$

$K_p = K_C(RT)^{\Delta n_g}$

$6000.1 = 20.4\ (0.0831 \times T)1$

$t = \frac{600.1}{20.4 \times 0.0831}$

**56.**[1.66] $\Delta G° = -RT \ln K_P$

$25200 = -2.3 \times 8.3 \times 400 \log (KP)$

$K_p = 10^{-33} = 10^{-3} \times 0.501$

$K_p = 5.01 \times 10^{-4}$ bar$^{-1}$

$= 5.01 \times 10^{-5} \times 10^{-4}$ Pa$^{-1}$

Relation of $K_p$ and $K_c$,

$K_p = K_c\ (RT)^{\Delta n_g}$

$\Delta n_g = -1$

$K_p = \frac{K_c}{RT}$

$K_p = \frac{K_c}{8.3 \times 400}$

$K_C = 1.66 \times 10^{-5}$ m$^3$/mole

$= 1.66 \times 10^{-2}$ L/mol

**57.** [172] $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

$K_p = \frac{P_{SO_2}^2}{P_{SO_2}^2 P_{O_2}}$

$= \frac{(43)^2}{(0.53)(45)^2}$

$= 1.7228 kPa^{-1}$

$= 172.28 \times 10^{-2} kPa^{-1}$

$= 172 \times 10^{-2}$ kPa$^{-1}$

**58.** [2] $K_p = K_C(RT)\Delta n_g$

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$\Delta n_g = 1$

$K_C = \frac{Kp}{(RT)} = \frac{47.9}{(0.083 \times 288)}$

$= 2$

**59.** (*a*) Fact based.

**60.** [1400] $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

| | | | |
|---|---|---|---|
| t = 0 | 3 | – | – |
| t = t$_{eq}$ | 3 – x | x | x |

$\frac{x^2}{3-x} = 1.844$

x = 1.60

Moles of $PCl_5$ at equilibrium = 3 – x = 3 – 1.6 = 1.4 or $1400 \times 10^{-3}$ mol

**61.** [16] $A(s) \rightleftharpoons M(s) + \frac{1}{2}O_2(g)$

$K_P = (P_{O_2})^{1/2} = 4$

$P_{O_2} = 16$

@PROCBSE

**62.** [6] $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

Initially: 0.1 mole – –

At equil.:0.1 – 0.02 = 0.08 0.02 mole 0.02 mole

$n_T = 0.02 + 0.02 = 0.04$

$$P_T = \frac{n_T RT}{V} = \frac{0.04 \times 0.082 \times 300}{2} = 0.492 atm$$

$$K_p = \left(\frac{0.492}{2}\right)\left(\frac{0.492}{2}\right) = 6.05 \times 10^{-2}$$

$\therefore$ $x = 6$ (nearest integer)

**63.** [4] $Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+$

0.80 $x$

$5 \times 10^{-8}$ $x - 1.60$ 0.80

$$K_f = \frac{[Ag(NH_3)_2]^+}{[Ag^+][NH_3]^2} = \frac{0.80}{5 \times 10^{-6}(x - 1.60)^2} = 10^8$$

x = 4.1 moles

Number of motes of $NH_3$ required for 2L solution

$= 2 \times 4.1 = 8$

**64.** [50] $[PtCl_4]^{-2} + H_2O \rightleftharpoons [Pt(H_2O)Cl_3]^- + Cl^-$

$$\frac{-d[PtCl_4]^{-2}}{dt} = 4.8 \times 10^{-5}[ptCl_4^{-2}] - 2.4 \times 10^3$$

$[Pt(H_2O)Cl_3]$ [u]

$$\Rightarrow K_{eq} = \frac{k_f}{k_b} = \frac{4.8 \times 10^{-5}}{2.4 \times 10^{-3}} = 0.02$$

**65.** [182]

| | A | + | B | $\rightleftharpoons$ | C | + | D |
|---|---|---|---|---|---|---|---|
| Initial Concentration | 1M | | 1M | | 1M | | 1M |
| At equilibrium | $(1-x)$ | | $(1-x)$ | | $(1+x)$ | | $(1+x)$ |

[QC = 1 and less than KC indicates reaction moves forward]

$$K_C = \frac{[C][D]}{[A][B]}$$

$$100 = \frac{(1+x)(1+x)}{(1-x)(1-x)} \text{ or } x = \frac{9}{11}$$

$$[D] = 1 + x = \frac{20}{11}$$

$[D] = 182 \times 10^{-2}$ M

**66.** [3] $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

$= 2 \times 0.0210 = 0.042$ mmol

$NH4Cl \longrightarrow NH_4^+ + Cl^-$

$= 5 \times 0.0504 = 0.252$ mmol

$$pOH = pK_b + \log \frac{[Salt]}{[Base]}$$

$$pOH = -\log(1.8 \times 10\text{–}5) + \log\left(\frac{0.252}{0.042}\right)$$

$[OH^-] = 10^{-5.51}$

$= 3 \times 10^{-6}$

**67.** [141] $PbI_2(s)$ $Pb^{2+}(aq) + 2I^-(aq)$ $K_{sp} = 8.0 \times 10^{-9}$

$Pb(NO_3)_2 \rightarrow Pb^{2+}(aq) + 2NO_3^-(aq)$

$$[I^-]^2 = \frac{8.0 \times 10^{-9}}{0.1} = 8.0 \times 10^{-8}$$

$[I^-] = 2\sqrt{2} \times 10^{-4}$ M

Solubility of $PbI_2$ in 0.1 M $Pb(NO_3)_2$ solution

$$= \frac{[I^-]}{2} = 141 \times 10^{-6} \text{ M}$$

**68.** (*d*) $AgCN \rightleftharpoons Ag^+ + CN^-$

$CN^- + H_3O^+ \rightleftharpoons HCN + H_2O$

let solubility of AgCN = $x$ molar

$$\frac{[HCN]}{[CN^-]} = 1.6 \times 10^6$$

As each $CN^-$ ion hydrolyses to give one HCN

$x = [Ag^+] = [CN^-] + [HCN]$

$\because$ $[CN^-] << [HCN]$

$\therefore$ $x = [Ag^+] = [HCN]$

$$[CN^-] = \frac{x}{1.6 \times 10^6}$$

$K_{sp} = [Ag^+][CN^-]$

$$2.2 \times 10^{-16} = \frac{x^2}{1.6 \times 10^6}$$

$x = 1.9 \times 10^{-5}$ M

**69.** (*d*) $Ca(OH)_2 \rightleftharpoons \underset{s}{Ca^{+2}} + \underset{2s}{2OH^-}$

$K_{sp} = [Ca^{+2}][OH^-]^2$

$= s \times (2s)^2$

$5.5 \times 10^{-6} = 4s^3$

$s = 1.11 \times 10^{-2}$

**70.** [7.00] $(NH_4)_3$ is a salt of weak base and weak acid.

$$pH = \frac{1}{2}pK_w + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$= 7 + \frac{1}{2}(5.23) - \frac{1}{2}(4.75)$$

= 7.00 (nearest integer)

**71.** [50] $A_2X \rightleftharpoons 2A^+ + X^{2-}$

$2s_1$ $s_1$

$K_{sp} = 4s_1^3$ or $s_1 = 10^{-4}$

$MX \rightleftharpoons M^{2+} + X^{2-}$

$s_2$ $s_2$

$K_{sp} = s_2^2$ or $s_2 = 2 \times 10^{-6}$

$s_1 = s(A_2x)$

$s_2 = s(MX)$

$$\frac{s_1}{s_2} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$$

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72. [01] Kal of $H_2SO_3$ >> $Ka_2$ of $H_2SO_3$

∴ The contribution of $H^+$ from 2nd dissociation of $H_2SO_3$ can be neglected.

$H_2SO_3 \rightleftharpoons H^+ + HSO_3^-$

$c(1-\alpha) \quad c\alpha \quad c\alpha$

$$K_a = \frac{c\alpha^2}{1-\alpha} = 1.7\times10^{-2}$$

$\alpha = 0.156$

$[H^+] = c\alpha = 0.092$

$pH = -\log[H^+]$

$= 1.036$

$= 1$

73. [2] $HCl \rightarrow H^+ + Cl^-$

$0.1M \quad 0.1M$

$HA \rightleftharpoons H^+ + A^- \quad K_a = 2.0\times10^{-6}$

$0.01(1-\alpha) \quad 0.1+0.01\alpha \quad 0.01\alpha = 0.1$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$2\times10^{-6} = \frac{0.1\times0.01\alpha}{0.01(1-\alpha)}$$

$\alpha = 2.0\times10^{-5}$

74. [10] $$pH = pk_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

or $5.74 = 4.74 + \log\frac{x}{1}$

$x = 10$ M

75. [64] $CdSO_4 \rightleftharpoons \underset{s}{Cd^{2+}} + \underset{s}{SO_4^{2-}}$

$K_{sp} = S^2$

$= (8\times10^{-4})^2$

$K_{sp} = 64\times10^{-8}$

$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

$64\times10^{-8} = S'\times0.01$

$$S' = \frac{64\times10^{-8}}{0.01} = 64\times10^{-6}\ mol/L$$

76. (*d*) Cons. of $Cl^- = 0.1\ M = 10^{-1}$ M

Conc. of $CrO_4^{2-} = 0.001\ M = 10^{-3}$ M

$K_{sp}(AgCl) = [Ag^+][Cl^-]$

$$[Ag^+]_{AgCl} = \frac{1.7\times10^{-10}}{10^{-1}} = 1.7\times10^{-9}$$

$K_{sp}(Ag_2CrO_4) = [Ag^+]^2[CrO_4^{2-}]$

$[Ag^+] = \sqrt{19}\times10^{-4}$

∴ AgCl will be precipitated first

77. [1] $Ba(OH)^2 \rightarrow Ba^{2+} + 2OH^-$

$[OH^-] = 2\times0.005 = 0.01$ M

$$[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{10^{-14}}{0.01} = 01.00\times10^{-12}\ M$$

78. [2] $\underset{s}{Zn(OH)_2(s)} \rightleftharpoons \underset{s}{Zn^{2+}} + \underset{(2s+0.1)}{2OH^-}$

$K_{sp} = [Zn^{2+}][OH^-]^2$

$2\times10^{-20} = (s)(2s+0.1)^2$

Neglecting 2s w.r.t. 0.1 gives $s = 2\times10^{-18}$ M

So value of x is 2

79. [6021]

| | HCl | + NaOH | → NaCl | + $H_2O$ |
|---|---|---|---|---|
| Initial (m mol) | 50 × 1 | 30 × 1 | – | – |
| Final (m mol) | 20 | – | 30 | 30 |

$pH = -\log[H^+]$

$$= -\log\left(\frac{20}{80}\right)$$

$= -\log(0.25)$

$= -\log(2.5\times10^{-1})$

$= -\log(2.5) - \log(10^{-1})$

$= -0.3979 + 1$

$= 6021\times10^{-4}$

80. [108] $A_3B_3 \rightleftharpoons \underset{\frac{3x}{M}}{3A^{2+}} + \underset{\frac{2x}{M}}{2B^{3-}}$

$$K_{sp} = \left(\frac{3x}{M}\right)^3\times\left(\frac{2x}{M}\right)^2 = 108\left(\frac{x}{M}\right)^5$$

$a = 108$

81. (*b*) Titration of strong acid and weak base uses methyl orange as indicator.

For titration of weak acid and strong base, phenolphthalein is a suitable indicator.

82. (*d*) $H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-} \quad K_{a_3}$

$H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^- \quad K_{a_3}$

$HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-} \quad K_{a_3}$

$$K_{a_2} = \frac{[H^+]^2[C_2O_4^{2-}]}{[H_2C_2O_4]}$$

$$K_{a_2} = \frac{[H^+][HC_2O_4^{2-}]}{[H_2C_2O_4]}, K_{a_2} = \frac{[H^+][C_2O_4^-]}{[HC_2O_4^-]}$$

$K_{a_1} = K_{a_1}\times K_{a_2}$

83. [2] $Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$

$k = 2\times10^{15}$

$$\frac{1}{2}Cu(s) + Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$$

$$K' = \frac{1}{(K)^{1/2}} = \frac{1}{(2\times10^{15})^{1/2}}$$

$x = 2$

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**84.** [2] $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

| | | | |
|---|---|---|---|
| at intial | 2 | 1 | – |
| at equilibrium | 2 – 0.8 | 0.8 | 0.8 |

Partial pressure of $NO(g) = \frac{1.2}{2.6} \times 1$

Partial pressure of $O_2(g) = \frac{0.6}{2.6}$

Partial pressure of $NO_2(g) = \frac{0.8}{2.6}$

$$K_p = \frac{(P_{NO_2})^2}{(P_{NO})^2 (P_{O_2})} = \frac{0.8 \times 0.8 \times 2.6}{1.2 \times 1.2 \times 0.6}$$

$= 2$

**85.** (*b*) $A(g) \rightleftharpoons B(g) + \frac{1}{2}C(g)$

$t = 0 \quad P_i \quad - \quad -$

$t = t$ (e.q.mcondisions) $\quad P_i - p_i\alpha \quad p_i\alpha \quad \frac{p_i\alpha}{2}$

$\therefore$ P (equilibrium pressure) $= p_i - p_i\alpha + p_i\alpha + \frac{p_i\alpha}{2}$

$$= p_i\left(1 + \frac{\alpha}{2}\right)$$

$$= p_i \frac{p}{\left(1 + \frac{\alpha}{2}\right)}$$

$\therefore$

$$K_p = \frac{\left(p_i \frac{\alpha}{2}\right)^{\frac{1}{2}} \times p_i\alpha}{p_i(1-\alpha)}$$

$$= \frac{p^{\frac{1}{2}}\alpha^{\frac{3}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$$

**86.** [1107] $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

| | | | |
|---|---|---|---|
| Initial moles | 5 | | |
| Equilibrium moles | 5 – x | x | x |

Number of moles of $N_2 = 2$

Equilibrium pressure = 2.46 atm

$$P_{eq} = \frac{(7+x) \times 0.082 \times 600}{200} = 2.46$$

$x = 3$

$$K_P = \frac{\left(\frac{3P}{10}\right)\left(\frac{3P}{10}\right)}{\left(\frac{2P}{10}\right)} = 1107 \times 10^{-3} \text{atm}$$

$\therefore$

**87.** [710] $N_2O_4 \rightleftharpoons 2NO_2$

| | | |
|---|---|---|
| t = 0 | 1mol | |
| t = t | (1 –0.5) mol = 0.5mol | 0.5 × 2 mol 1mol |

$$K_p = \frac{\left(\frac{1}{1.5} \times 1\right)^2}{\left(\frac{0.5}{1.5} \times 1\right)}$$

$= 1.33$

$\Delta G^\circ = -RT\ell nk_p$

$= -8.31 \times 300 \times \ell n(1.33) = -710.45$ J/mol

$= -710$ J/mol.

**88.** [125] $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

| | | | |
|---|---|---|---|
| t = 0 | 2 | | |
| $t = t_{eq}$ | 2 – 0.4 | 0.4 | 0.2 |

$$k_C = \frac{(0.2) \times (0.4)^2}{(1.6)^2}$$

$= 125 \times 10^{-4}$

**89.** (*a*) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

| | | | |
|---|---|---|---|
| t = 0 | 5 | – | – |
| t = t | 5 – n | n | n |

Total moles = 5 – n + n + n

= 5 + n.

For Argon

Total moles = $n_{Ar} + nPCl_5 + nPCl_3 + nPCl_2$

= 4 + 5 + n

= 9 + n

$$K_p = \frac{P_{PCl_3} \cdot P_{Cl_2}}{P_{PCl_5}} PV = nRT$$

$6 \times 100 = (9 + n) \times 0.082 \times 610$

$n = 3$

$$= \frac{\left(\frac{3}{12} \times 6\right) \times \left(\frac{3}{12} \times 6\right)}{\frac{2}{12} \times 6}$$

$$= \frac{27}{12} = \frac{9}{4} = 2.25 \text{ amt}$$

**90.** [29] $H_2O(l) \rightleftharpoons H_2O(g)$

| | | |
|---|---|---|
| $t = t_{eq}$ | $\frac{0.90}{18} - x$ | x |

$PV = nRT$

$$\frac{32}{760} \times V = 082 \times (x) \times 300$$

$$x = \frac{0.90}{18}$$

$$V = .082 \times \frac{0.90}{18} \times \frac{300 \times 760}{32}$$

$= 29.21L \quad = 29L$

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**91.** (c) $\underset{\text{20mL 0.1M}}{NH_4OH} + \underset{\text{40mL 0.05M}}{HCl} \rightarrow NH_4Cl + H_2O$

$\underset{\text{2mmoles}}{\quad} \quad \underset{\text{2mmoles}}{\quad} \quad \underset{\text{2mmoles}}{\quad}$

$\therefore$ In final solution 2 millimoles of $NH_4Cl$ is present.

$\therefore [NH_4Cl] = \frac{1}{30}$ molar

$pH = \frac{1}{2}[pk_w - pk_b - \log C]$

$= \frac{1}{2}[14 - 5 - (-1.48)]$

$= 5.24$

**92.** (c) For basic buffer, $pOH = pK_b + \log\frac{[salt]}{[Base]}$

$pOH = 14 - 8.26 = 5.74$

$5.74 = 4.74 + \log\frac{[NH_4Cl]}{0.2}$

$[NH_4Cl] = 2\ M$

Moles of $NH_4Cl = 2 \times 1 = 2$ moles

Weight of $NH_4Cl = 2 \times 53.5 = 107$ g

**93.** [108] $CaF_2 \underset{s}{\overset{s}{\rightleftharpoons}} \underset{s}{Ca^{2+}} + \underset{2s}{2F^-}$

$K_{sp} = s(2s)^2$

$= 4s^3$

Solubility(s) = 2.34 × 10–3 g/100 mL

$= \frac{2.34\times10^{-3}\times10}{78}(\text{mole/L})^3$

$= 3 \times 10^{-4}$ mole/L

$\therefore K_{sp} = 4 \times (3 \times 10^{-4})^3$

$= 108 \times 10^{-12}$

$= 0.0108 \times 10^{-8}\ (\text{mole/L})^3$

$\therefore x = 0$

**94.** (a) $NH_4OH$ is a weak base and HCl is a strong acid. On additional HCl to $NH_4OH$, pH.

**95.** [27] $K_a = C\alpha^2$ $\quad C = 0.2$ M

$\alpha = \sqrt{\frac{K_a}{C}}$ $\quad K_a = 2 \times 10^{-5}$

$= \sqrt{\frac{2\times10^{-5}}{2\times10^{-1}}}$

$= 10^{-2}$

$[H^+] = C\alpha$

$= 0.2 \times 10^{-2} \quad = 2 \times 10^{-3}$

$pH = 3 - \log 2 = 2.7$

**96.** [282] $PbS(s) \rightleftharpoons \underset{s}{Pb^{2+}}(aq) + \underset{s}{S^{2-}}(aq)$

$K_{sp} = S^2$

$8 \times 10^{-28} = S^2$

$S = 2\sqrt{2} \times 10^{-14}$ mol/L

or $2.82 \times 10^{-14}$ mol/L = $282 \times 10^{-16}$ mol/L

Hence,

$x = 282$

**97.** (b) Molarity of resultant solution is given by

$m_1v_1n_1 + m_2v_2n_2 = mv$

200 mL of 0.01 m HCl + 400 mL of 0.01m $H_2SO_4$

$200 \times 0.01 \times 1 + 400 \times 0.01 \times 2 \quad mx_v$

Molarity = $\frac{10}{600}$ of equivalents.

$[H^+] = \frac{10}{600}$

$pH = -\log[H^+]$

$pH = -\log\left[\frac{10}{600}\right] = 1.778$

**98.** (d) $\underset{\text{(acid)}}{H_2O} + NH_3 \rightleftharpoons OH^- + NH_4^+$

$\underset{\text{(base)}}{H_2O} + H_2S \rightleftharpoons H_3O^+ + HS^-$

Hence, A is false but R is true

**99.** [476] $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

at initially $\underset{\text{mmoles}}{50\times0.1}$ $\underset{\text{mmoles}}{25\times0.1}$

at time t 2.5mmoles 0 2.5mmol

$pH = pKa + \log\left|\frac{(salt)}{(acid)}\right|$

$pH = 4.76 + \log\left|\frac{2.5}{2.5}\right|$

$pH = 4.76$

**100.** [11] $[OH^-] = 0.001 = 10^{-3}$ M

$[H^+][OH^-] = 10^{-14}$

$[H^+] = 10^{-11}$

$pH = -\log[H^+]$

$pH = 11$

**101.** (b) $CH_3CH_2COOH \rightleftharpoons CH_3CH_2COO^- + H^+$

$pH = pKa + \log\frac{[CH_3CH_2COO^-]}{[CH_3CH_2COOH]}$

$4 = -\log 1.3 \times 10{-}5 + \log\frac{[CH_3CH_2COO^-]}{[CH_3CH_2COOH]}$

$\frac{[CH_3CH_2COO^-]}{[CH_3CH_2COOH]} = 0.13$

**102.** (d) Solubility decreases with increasing the concentration of common ion. Therefore, the maximum solubility of AgCl will be in deionized water.

## JEE ADVANCED

**103.** (a,b,d)

(a) $\Delta H_2 - \Delta H_1 = C_{P\,(rxn)}(T_2 - T_1)$ and $C_p$ depends on temperature. Hence enthalpy also depends on temperature.

(b) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ $Kp = (P_{CO_2})_{\text{at equlibrium}}$

For a give reation

$K_{eq.}$ depends only on temperature.

(c) $K_{eq}$ depends only on temperature.

(d) Enthalpy of reaction is independent of the catalyst. Catalyst generaly changes activation energy.

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**104.** (*b*) $X_2(g) \rightleftharpoons 2\times(g)$

$1-\frac{\beta}{2}$ $\quad$ $\beta$

$$K_p = \frac{\left[\frac{\beta}{\left(1+\frac{\beta}{2}\right)}\right]^2 \times 2^2}{\frac{1-\frac{\beta}{2}}{1+\frac{\beta}{2}}\times 2} = \frac{\beta^2}{1-\frac{\beta^2}{4}}\times 2$$

$$K_p = \frac{8\beta^2}{4-\beta^2}$$

**105.** (*c*) $\Delta G^0 = (+)ve \therefore -RT \ln K_p = \Delta G^0 = (+)ve$

$K_P < 1$

Also β cant be 0.7 otherwise $K_P$ will have to be (+) ve

**106.** [8.92 or 8.93]

$Fe^{+2}_{(aq)} + S^{-2}_{(aq)} \rightleftharpoons FeS(s)$

0.03 M $\quad$ 0.1M

(0.03–x) $\quad$ (0.1 – x)

$\simeq y$ $\quad$ $\simeq 0.07$

$K_c >> 10^3 \Rightarrow 0.03 - x \simeq 0 \simeq y$

$\Rightarrow x = 0.03$

$$K_c = 1.6 \times 10^{17} = \frac{1}{y \times 0.07}$$

$$y = \frac{10^{-17}}{1.6\times 0.07} = 8.928 \times 10^{-17} = Y \times 10^{-17}$$

$y = 8.93$

**107.** [0.25]

$$K_{eq.} = \frac{[B]}{[A]}$$

$$K_{1000} = \frac{10}{1} = 10 \text{ and } K_{2000} = \frac{100}{1} = 100$$

$$\text{Now, } \frac{\Delta G^0_{1000}}{\Delta G^0_{2000}} = \frac{\left(-RT\ell nk_{eq}\right)_{1000}}{\left(-RT\ell nk_{eq}\right)_{2000}} = \frac{1000\times \ell n10}{2000\times \ell n100}$$

$= 0.25$

**108.** [4.47]

$$S = \sqrt{K_{sp}\left(\frac{[H^+]}{K_a}+1\right)} = \sqrt{2\times 10^{-10}\left(\frac{10^{-3}}{10^{-8}}+1\right)}$$

$$= \sqrt{2\times 10^{-5}} = 4.47\times 10^{-3}$$

**109.** [0.11]

No. of eq. of oxalic acid = No. of eq. of NaOH

$$\text{or } \frac{5.00\times 0.10}{1000}\times 2 = \frac{9.0\times M}{1000}\times 1$$

$\therefore$ Molarity of NaOH solution $= \frac{1}{9} = 0.11M$

**110.** [2.30 to 3.00]

$B + HA \longrightarrow BH^+ + A^-$

0.1M, V ml

0.1Vm mol $\quad$ 0.1 V m mol $\quad$ 0.1 V 0.1V

$$[BH^+] = \frac{0.1V}{2V} = 0.5M$$

pH at eq. pt = 6 to 6.28

$$pH = 7 - \frac{1}{2}[pK_b + \log 0.05]$$

So $pK_b = 2.30 - 2.80$, Possible

**Solution 2**

at V = ml $\quad$ rx is complete

So V = 3ml $\quad$ is half of eq. pt

at which $\quad$ pH = 11

pOH = (14 – 11) $\quad$ $= pK_b + \log 1$

$pK_b = 3$

**111.** [0.20] For ppt, $[Zn^{+2}][S^{-2}] = K_{sp}$

$$[S^{-2}] = \frac{1.25\times 10^{-22}}{0.05} = 2.5\times 10^{-21} M$$

$H_2S \rightleftharpoons 2H^+ + S^{-2}$

$$K_{Net} = 10^{-21} = \frac{[H^+]^2 \times 2.5\times 10^{-21}}{0.1}$$

$$[H^+]^2 = \frac{1}{25} \quad [H^+]^2 = \frac{1}{5}M = 0.2M$$

**112.** [6] $H_2SO_4 \longrightarrow H^+ + HSO_4^{2-}$ ($K_{a_1}$ is very large)

1M $\quad$ 1M

$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ $K_{a_2} = 1.2 \times 10^{-2}$

$[SO_4^{2-}]$ coming from $Na_2SO_4 = 1.8 \times 10^{-2}$

$$\frac{[SO_4^{2-}][H^+]}{[HSO_4^-]} = \frac{1.8\times 10^{-2}\times 1}{1} > K_{a_2}$$

$\therefore$ Rather than dissociation of $HSO_4^-$ into H and $SO_4^{2-}$ ions, association between already present $H^+$ and $SO_4^{2-}$ will take place.

Assuming 'x' mol/L of $SO_4^{2-}$ and $H^+$ combines to form $HSO_4^-$

$\therefore$ $[SO_4^{2-}] = 1.8 \times 10^{-2} - x$

$[H^+] = 1 - x \approx 1$

$[HSO_4^-] = 1 + x \approx 1$ $\quad$ (assuming $x << 1$)

$$\frac{(1.8\times 10^{-2} - x)}{1} = 1.2\times 10^{-2}$$

$\Rightarrow x = 0.6 \times 10^{-2}$

$[SO_4^{2-}] = 1.2 \times 10^{-2} M$

$PbSO_4(s) \rightleftharpoons Pb^{2+}(aq) + SO_4^{2-}(aq)$

If solubility of $PbSO_4$ = sM

$\therefore$ $[Pb^{2+}] = s$

$[SO_4^{2-}] = s + 1.2 \times 10^{-2} \approx 1.2 \times 10^{-2}$

(assuming $s << 1.2 \times 10^{-2}$)

$\therefore$ $s \times 1.2 \times 10^{-2} = 1.6 \times 10^{-8}$

$$s = \frac{1.6}{1.2}\times 10^{-6} = 1.33\times 10^{-6}$$

On comparing with $X \times 10^{-Y}$

Y = 6ss

@PRODBSE

CHAPTER

# 6 Redox Reaction

## Exercise-1 (Topicwise)

1. (*b*) CO is a neutral molecule So, O.S. of Ni is zero.

2. (*c*) $NH_2OH$

suppose O.S. of N is x.

$x + 2(+1) + (-1) = 0$

$x = -1$

3. (*c*) $CH_4 \Rightarrow x + 4(+1) = 0$

$x = -4$

$CH_3Cl \Rightarrow x + 3(+1) + 1(-1) = 0$

$x = -2$

$CH_2Cl_2 \Rightarrow x + 2(+1) + 2(-1) = 0$

$x = 0$

$CHCl_3 \Rightarrow x + 1(+1) + 3(-1) = 0$

$x = +2$

$CCl_4 \Rightarrow x + 4(-1) = 0$

$x = +4$

4. (*b*) $[Pt(C_2H_4)Cl_3]^{-1}$

suppose O.S. of Pt is x

$x + 0 + 3(-1) = -1$

$x = -1 + 3$

$x = +2$

5. (*a*) HNC

suppose O.S. of C is x.

$(+1) + 1(-3) + x = 0$

$-2 + x = 0$

$x = +2$

6. (*b*) $C_3O_2$

$O = C = C = C = O$

$3(x) + 2(-2) = 0$

$3x = +4$

$x = +4/3$

7. (*a*) $Na_2S_2O_3$

$2(+1) + 2x + 3(-2) = 0$

$2x = +4$

$x = +2$

8. (*b*) $N_3H$

$3(x) + 1 = 0$

$3x = -1$

$x = -1/3$

9. (*a*) Oxidation

$\overset{0}{H_2} + \overset{0}{Br_2} \rightarrow \overset{+1\ -1}{2HBr}$

Reduction

10. (*c*) Reduction

$\overset{0}{Cl_2} + OH^- \rightarrow \overset{-1}{Cl^-} + \overset{+7}{ClO_4^-} H_2O$

Oxidation

Oxidised as well as reduced.

11. (*d*) Reduction

$\overset{0}{Cl_2} + 2OH^- \rightarrow \overset{+1}{ClO^-} + \overset{-1}{Cl^-} + H_2O$

Oxidation

Oxidation

$\overset{+1}{Cu_2O} + 2H^- \rightarrow \overset{0}{Cu} + \overset{+2}{Cu^{2+}} + H_2O$

Reduction

Oxidation

$2\overset{+1}{HCuCl_2} \xrightarrow{\text{disproportionation}} \overset{0}{Cu} + \overset{+2}{Cu^{2+}} + 4Cl^- + 2H^+$

Reduction

12. (*c*) $X_2H_4$

or

$X_2^{-2}H_4 \rightarrow X^{-1/2} + ne^-$

No. of electrons $= 2 \times \{-1/2 - (-2)\}$

$= 3/2 \times 2$

$= 3$

13. (*a*) Reduction

$\overset{-2}{H_2S} + \overset{-1}{H_2O_2} \rightarrow \overset{0}{S} + \overset{-2}{2H_2O}$

Oxidation

$H_2S \Rightarrow$ Reducing agent

$H_2O_2 \Rightarrow$ Oxidising agent

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14. (*b*) $H_2O_2$

In $H_2O_2$, O.S. of oxygen is (–1) which can be increase and decrease.

15. (*c*) Reduction

$\overset{+3}{4H_3PO_3} \rightarrow \overset{-3}{PH_3} + \overset{+5}{3H_3PO_4}$

Oxidation

same element undergoes for oxidation as well as reduction.

16. (*a*) $CaCO_3 \rightarrow CaO + CO_2$ is a decomposition reaction.

17. (*c*) Oxidation

$\overset{+7}{MnO_4^-} + \overset{+4}{SO_3^{2-}} + H^+ \rightarrow \overset{+6}{SO_4^{2-}} + \overset{+2}{Mn^{2+}} + H_2O$

Reduction

18. (*d*) $BaO_2$

$1(+2) + 2(x) = 0$

$2x = -2$

$x = -1$

19. (*c*) Oxidation

$\overset{0}{I_2} + \overset{-1}{KI} \rightarrow \overset{-1/3}{KI_3}$

Reduction

Both Oxidation as well as Reduction.

20. (*a*) $KNO_3 \Rightarrow N = +5$

$HNO_2 \Rightarrow N = +3$

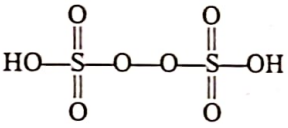
$NH_4Cl \Rightarrow N = -3$

$NaN_3 \Rightarrow N = -1/3$

21. (*a*) $\overset{0}{Br_2} \rightarrow \overset{+5}{BrO_3^-}$

0 to +5

22. (*d*) $H_2S_2O_8$



suppose O.S. of S = x

$2x + 2(+1) + 6(+2) + 2(-1) = 0$

$2x = +12$

$x = +6$

One S has +6 O.S.

So, sum of O.S. of S = 6 + 6 = +12

23. (*a*) $\overset{0}{2Na} + H_2 \rightarrow \overset{+1\ -1}{2NaH}$

Oxidation

Na is oxidised by $H_2$.

24. (*d*) $\overset{+6\ -2}{2CrO_4^{2-}} + \overset{+1}{2H^+} \rightarrow \overset{+6\ -2}{Cr_2O_7^{2-}} + \overset{+1\ -2}{H_2O}$

So there is no element that changes its O.S.

25. (*d*) $H_3AsO_4$

suppose O.S. of As is x

$3(+1) + x + 4(-2) = 0$

$x = +5$

26. (*c*) $MgHxO_3$

$(+2) + 1(+1) + x + 3(-2) = 0$

$x = +3$

27. (*a*) $\overset{}{-O-O-}$

Peroxy linkage

In Peroxy linkage. The O.S. of Oxygen is (–1).

28. (*a*) $\overset{0}{C} + 4\overset{+5}{HNO_3} \rightarrow \overset{+4}{CO_2} + 2H_2O + \overset{+4}{4NO_2}$

Reduction

$HNO_3 \Rightarrow$ Oxidising agent

29. (*c*) $[K_3Fe(CN)_6]$

$3(+1) + x + 6(-1) = 0$

$x = +3$

30. (*d*) $H_2SO_3 \Rightarrow S = +4$

$H_2S \Rightarrow S = -2$

$H_2SO_4 \Rightarrow S = +6$

$H_2S_2O_3 \Rightarrow S = +2$

$H_2S, H_2S_2O_3, H_2SO_3, H_2SO_4$

31. (*d*) $KIO_4$

$(+1) + x + 4(-2) = 0$

$x = +7$

32. (*d*) $XeF_5^-$

$x + 5(-1) = -1$

$x = +4$

33. (*b*) $Fe^{2+} \rightarrow Fe^{3+}e^-$

loses one electron.

34. (*b*) Oxidation

35. (*a*) $\overset{+3}{FeCl_3} + \overset{-2}{H_2S} \rightarrow \overset{+2}{2FeCl_2} + 2HCl + \overset{0}{S}$

Reduction, Oxidation

$FeCl_3 \Rightarrow$ O.A.

$H_2S \Rightarrow$ R.A.

36. (*a*) Reducing Potential

Oxidation

37. (*c*) $\overset{+7}{MnO_4^-} + \overset{+2}{5Fe^{2+}} + 8H^+ \rightarrow \overset{+2}{Mn^{2+}} + \overset{+3}{5Fe^{3+}} + 4H_2O$

Reduction

This is a spontaneous oxidation reduction reaction.

38. (*b*) $x\overset{+6}{Cr_2O_7^{2-}} + yH^+ + 2e^- \rightarrow a\overset{+3}{Cr^{3+}} + bH_2O$

$6e^- + \overset{+6}{Cr_2} \rightarrow 2\overset{+3}{Cr}$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

39. (*a*) $\overset{+7}{MnO_4^-} \rightarrow \overset{+2}{Mn^{2+}}$

So, $5e^- + \overset{+7}{Mn} \rightarrow \overset{+2}{Mn}$

$x = 5$

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40. (a) $\overset{+3}{Cr}(OH)_4^- \rightarrow \overset{+6}{Cr}O_4^{2-}$

So, $\overset{+3}{Cr} \rightarrow \overset{+6}{Cr} + 3e^-$

$x = 3$

41. (c) Balanced reaction is

$4Zn + NO_3^- + 7H_2O \rightarrow 4Zn^{2+} + NH_4^+ + 10OH^-$

42. (d) $\overset{+7}{Mn}O_4^- + H^+ + \overset{+3}{C_2}O_4^{2-} \rightarrow \overset{+2}{Mn}^{2+} + \overset{+4}{C}O_2 + H_2O$

$[5e^- + \overset{+7}{Mn} \rightarrow \overset{+2}{Mn}] \times 2$

$$\frac{[\overset{+3}{C_2} \rightarrow 2\overset{+4}{C} + 2e^-] \times 5}{2MnO_4^- + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2}$$

So, 10 moles of $CO_2$.

43. (c) $MnO_4^- + C_2O_4^{2-} \longrightarrow Mn^{2+} + CO_2$

O.N. = +7, +3, +2, +4

$vf_1 = +5$, $vf_2 = +2$

$n_1vf_1 = n_2vf_2$

$1 \times 5 = n_2 \times 2$

$n_2 = \frac{5}{2}$

44. (d) $Na_3\overset{+3}{As}O_3 \quad \overset{+5}{As}O_4^{3-}$

So, $AsO_4^{3-}$ is Oxidising Product of $Na_3AsO_3$

45. (c) $H_2O + \overset{+4}{S}O_3^{2-} \rightarrow \overset{+6}{S}O_4^{2-} + 2H^+ + x$

So, $\overset{+4}{S} \rightarrow \overset{+6}{S} + 2e^-$

$x = 2e^-$

46. (d) $\overset{+7}{MnO_4^-} + \overset{+2}{Fe}\overset{+3}{C_2}O_4 \longrightarrow \overset{+2}{Mn}^{2+} + \overset{+3}{Fe}^{3+} + \overset{+4}{C}O_2$

v.f.$_1$ = 5, v.f.$_2$ = 1+2 = 3

$n_1vf_1 = n_2vf_2$

$n \times 5 = 1 \times 3$

$n_1 = \frac{3}{5}$

47. (d) $\overset{+4}{Mn}O_2 + 4H^+ + e^- \rightarrow \overset{+3}{Mn}^{3+} + 2H_2O$

$Mn \Rightarrow +4$ to $+3$

48. (c) The total charge of compound should be balanced.

$ABC_2 \Rightarrow +2 + (+5) + 2(-2) = +3$

$B_2(AC_3)_2 \Rightarrow +5 + 2[+2 + (-2) \times 3] = +2$

$A_3(BC_4)_2 \Rightarrow 3(+2) + 2[+5 + 4(-2)] = 0$

$A_3(B_2C)_2 \Rightarrow 3(+2) + 2[4(+5) + (-2)] = 42$

49. (a) $\overset{0}{N_2} \longrightarrow \overset{-3}{NH_3}$ (v.f. = valence factor)

O.N. = 0, O.N. = −3

v.f. = 6

Eq. wt. $= \frac{28}{6}$

50. (d) $\overset{0}{N_2} \rightarrow \overset{-3}{NH_3}$

Eq. wt. $= \frac{\text{mol. wt.}}{\text{Valence factor}}$

V.F. = no. of atoms × change in O.No.

V.F. = 1 × 3 ⇒ 3

Eq. wt. $= \frac{17}{3}$

51. (b) $\overset{0}{I_2} + 2S_2O_3^{2-} \rightarrow 2\overset{-1}{I}^- + S_4O_6^{2-}$

Eq. wt. $= \frac{\text{Mol. wt.}}{\text{Vf}}$

Vf = no. of atoms × change in O.No.

Vf = 2 × 1 ⇒ 2

Eq.wt. $= \frac{M}{2}$

52. (c) $\overset{+5}{BrO_3}^- \rightarrow \overset{-1}{Br}^-$

V.f. = 6

Eq. wt. $= \frac{M}{6}$

53. (c) $\overset{-n_2}{A} + xe^- \rightarrow \overset{-n_1}{A}$

$\boxed{x = n_1 - n_2}$

54. (a) $H_3PO_4 + OH^- \rightarrow H_2PO_4^- + H_2O$

Eq.wt. $= \frac{98}{1} = 98$

$H_3PO_4 + 2OH^- \rightarrow HPO_4^{2-} + 2H_2O$

Eq.wt. $= \frac{98}{2} = 49$

$H_3PO_4 + 3OH^- \rightarrow PO_4^{3-} + 3H_2O$

Eq.wt. $= \frac{98}{3} = 32.67$

55. (c) (A) $\overset{+2}{2}CuSO_4 + 4KI \rightarrow \overset{+1}{2}CuI + I_2 + 2K_2SO_4$

V.f. = 1

Eq. wt. $= \frac{249}{1} = 249$

(B) $\overset{+2}{Cu}^{2+} + 2e^- \rightarrow \overset{0}{Cu}$

V.f. = 2

Eq. wt. $= \frac{249}{2} = 124.5$

56. (b) $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$

0, 8/3

v.f. = 8/3

Eq.wt $= \frac{56}{8} \times 3 = 21$

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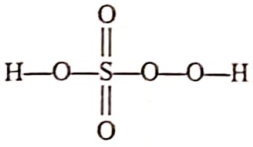
## Exercise-2 (Learning Plus)

**1.** (*d*) $H_2S_2O_7$

$+2 + 2x + (-14) = 0$

$x = +6$

**2.** (*c*)



**3.** (*d*) $F_2O$

**4.** (*c*) $Fe_3O_4 \Rightarrow \overset{+2}{F}eO, \overset{+3}{F}e_2O_3$

**5.** (*a*) $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

$(H_2O_2 \longrightarrow O_2 + 2H^+ + 2e^-) \times 3$

$Cr_2O_7^{2-} + 8H^+ + 3H_2O_2 \longrightarrow 2Cr^{3+} + 7H_2O + 3O_2$

The reaction practically occurs with this stoichiometry.

**6.** (*c*) Let the O.N.of Co be x

O.N. of $NH_3$ is zero

O.N. of Cl is –1

O.N. of Br is –1

Hence, $x + 6(0) - 1 \times 2 - 1 = 0$

$\therefore x = +3$

so, the oxidation number of cobalt in the given complex compound is +3.

**7.** (*c*) In the reaction $P_2O_5 \rightarrow H_4P_2O_7$

The O.N. of P in $P_2O_5$ is $2x + 5(-2) = 0$ or $x = +5$

The O.N. of P in $H_4P_2O_7$ is $4(+1) + 2(x) + 7(-2) = 0$

$2x = 10$ or $x = +5$

Since there is no change in O.N. of P, hence the above reaction is neither oxidation nor reduction.

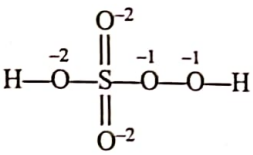
**8.** (*b*) $\overset{+2}{F}e \rightarrow \overset{+3}{F}e, \quad 2\overset{-1}{I^-} \rightarrow \overset{0}{I_2}$

**9.** (*b*) In Phosphorus acid $H_3PO_3$

$3(+1) + x + 3(-2) = 0$

$x = +3$

**10.** (*c*) $H_2SO_5$



$x + 2(+1) + 3(-2) + 2(-1) = 0 \Rightarrow x = +6$

**11.** (*c*) $(N_2H_5)_2SO_4$

$4(x) + 10(+1) - 2 = 0$

$4x + 8 = 0$

$x = -2$

**12.** (*b*) $\overset{+6}{M}o \rightarrow \overset{+3}{M}o$ so, Oxidising agent

**13.** (*a*) $CaOCl_2 \longrightarrow Ca^{2+} + Cl^- + OCl^-$

$Cl^{-1} \Rightarrow -1$ (O.S.)

$OCl^- \Rightarrow +1$ (O.S.)

**14.** (*a*) $\overset{+4}{MnO_2} + 4\overset{-1}{HCl} \rightarrow \overset{+2}{MnCl_2} + \overset{0}{Cl_2} + 2H_2O$

Oxidation

Reduction

Two different elements undergo for Oxdiation and Reduction

**15.** (*b*) $Fe \rightarrow Fe^{2+} + 2e^-$

1 mole Fe $\rightarrow$ 2 mole $e^-$

2 mole $e^- = 2N_A$ electrons

**16.** (*c*) $FeSO_4(NH_4)_2SO_4.6H_2O$

$x + (-2) + 2(+1) + (-2) + 6(0) = 0$

$x = +2$

$\Rightarrow K_4Fe(CN)_6$

$4(+1) + x + 6(-1) = 0$

$x \Rightarrow +2$

$\Rightarrow Fe(CO)_5$

$x + 5(O) = 0$

$x = 0$

$\Rightarrow Fe0.94O$

$0.94(x) + (-2) = 0$

$x = 2/0.94$

**17.** (*c*) Valency factor ratio is inversely related to molar ratio.

(V.f.)HI : (V.f.)$HNO_3 = 1 : 3 = 2 : 6$

$\therefore$ Molar ratio = 6 : 2

**18.** (*a*) $MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$

V.f. = 5 V.f. = 2

$\therefore$ Balanced equation: $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

**19.** (*a*) $\overset{+2}{C}uSO_4 + \overset{0}{F}e \rightarrow \overset{+2}{F}eSO_4 + \overset{0}{C}u$

**20.** (*a*) Fact Based

**21.** (*a*) $2NO_3^- + 6I^- + 8H^+ \rightarrow 2NO + 3I_2 + 4H_2O$

**22.** (*a*) Balance reaction is

$2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \longrightarrow 2MnSO_4 + 5O_2 + 8H_2O + K_2SO_4$

$\therefore$ Sum of stoichiometric coefficients = 2 + 5 + 3 + 2 + 5 + 8 + 1 = 26

**23.** (*a*) $2MnO + 5PbO_2 + 10\,HNO_3 \longrightarrow 2HMnO_4 + 5\,Pb(NO_3)_2 + 4\,H_2O$

**24.** (*d*) Fact Based

**25.** (*d*) $3\overset{0}{F}e + 4H_2O \rightarrow \overset{+8/3}{F}e_3O_4 + 4H_2$

**26.** (*a*) $(NH_4)_2Cr_2O_7 \longrightarrow Cr_2O_3 + N_2 + 4H_2O$

O.S. of N = – 3 O.S. of Cr = +3

O.S. of Cr = + 6 O.S. of N = 0

**27.** (*a*) $\overset{-1.5}{A_2}H_3 \longrightarrow \overset{-0.5}{A_2} + 2e^-$

# Exercise-3
## (JEE Advanced Level)

1. (c,d) In (C) option, Cl goes from +5 to +7 and −1, while in (D) option, Cl goes from 0 to +1 and −1.

2. (b,c) 'Cl' should be in intermediate oxidation state.

3. (a,b,c)
   (a) Oxidation state of S = +6 in both.
   (b) Oxidation state of Cr = +6 in both.
   (c) Oxidation state of P = +5 in both.

4. (b,c,d) $PbO_2$ is an oxide.

5. (b) As $KMnO_4$ reduces, the compound must oxidize. $Fe^{+3}$ cannot oxidize.

6. (b,c) S undergoes increase in oxidation number from −2 to −2.5, while I undergoes decrease in oxidation number from 0 to −1.

7. (a) $n_{eq}\,KMnO_4 = n_{eq}\,FeC_2O_4$

$(FeC_2O_4 \rightarrow Fe^{3+} + CO_2)$

$$\text{or } n \times 5 = 1 \times 3 \Rightarrow n = \frac{3}{5}$$

8. (a,b) m.eq. of $KMnO_4$ = m.eq. of $KHC_2O_4$

$$0.02 \times 100 \times 5 = \frac{x}{M} \times 2 \times 1000 \quad \text{...(i)}$$

m.eq. of $Ca(OH)_2$ = m. eq of $KHC_2O_4$

$$0.05 \times 100 \times 2 = \frac{y}{M} \times 1 \times 1000 \quad (M = \text{Mol. wt. of } KHC_2O_4) \quad \text{...(ii)}$$

Divide (i) and (ii)

$$\frac{0.02 \times 100 \times 5}{0.05 \times 100 \times 2} = \frac{2x}{y} \Rightarrow 1 = \frac{2x}{y} \Rightarrow 2x = y$$

9. (a,c,x)

(a) $6 M_1V_1 = M_2V_2$ [$\because$ For $K_2Cr_2O_7$, Eq. wt. $= \frac{\text{M.wt}}{6}$]

(c) $N_1V_1 = N_2V_2$

(b) and (d) are not possible.

10. (a,b,c,d)

Equivalent of oxidising agent = equivalents of reducing agent

$$Eq_{MnO_4^-} = Eq_{Fe^{2+}}$$

$$n_{MnO_4^-} \times 5 = n_{Fe^{2+}} \times 1$$

$$Eq_{C_2O_4^{2-}} = Eq_{Fe^{2+}}$$

$$n_{C_2O_4^{2-}} \times 6 = n_{Fe^{2+}} \times 1$$

$$Eq_{MnO_4^-} = Eq_{(Mn)}$$

$$n_{MnO_4^-} \times 5 = n_{(Mn)} \times 8$$

$$Eq_{C_2O_4^{2-}} = Eq_{(Mn)}$$

$$n_{C_2O_4^{2-}} \times 6 = n_{(Mn)} \times 8$$

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**11.** (*a,c*)

$n_{eq}$ metal nitrate = $n_{eq}$ metal sulphate

or $\frac{0.5}{E_M+62}=\frac{0.43}{E_M+48}\Rightarrow E_M=38$

**12.** (*a,b,c,d*) For HCl N = M

$$\text{Final molarity}=\frac{V_1\times1+V_2\times0.25}{(V_1+V_2)}=0.75$$

$0.75\,(V_1+V_2)=V_1+V_2\times0.25$

$0.75\,V_1+0.75\,V_2=V_1+V_2\times0.25$

$0.5\,V_2=0.25\,V_1$

$\frac{V_1}{V_2}=2$ (All options are possible)

**13.** (*a,c,d*) milli equivalent of $KMnO_4 = 25\times0.2 = 5$ meq.

(*a*) $Fe^{2+}\longrightarrow Fe^{3+}$

milli equivalent of $FeSO_4 = 25\times0.2\times1 = 0.2\times25 = 5$ (same)

(*b*) $H_3AsO_3\longrightarrow H_3AsO_4$

milli equivalent of $H_3AsO_3 = 2\times50\times0.1 = 10$ (not same)

(*c*) $H_2O_2\longrightarrow 2H^+ + O_2$

milli equivalent of $H_2O_2 = 25\times0.1\times2 = 5$ (same)

(*d*) $Sn^{2+}\longrightarrow Sn^{4+}$

milli equivalent of $SnCl_2 = 25\times0.1\times2 = 5$ (same)

**14.** (*a,b,d*)

$Cu_2S + I_2 \rightarrow Cu^{+2} + SO_4^{-2} + I^-$ ...(i)

$(n_f=10)\quad(n_f=2)$

$CuS + I_2 \rightarrow Cu^{+2} + SO_4^{-2} + I^-$ ...(ii)

$(n_f=8)\quad(n_f=2)$

$I_2 + S_2O_3^{-2} \rightarrow I^- + S_4O_6^{-2}$ ...(iii)

$(n_f=2)\quad(n_f=1)\quad(n_f=2)$

**In reaction (i):**

Meq of $Cu_2S$ = Meq of $I_2$

$0.5\times10\times V = 250\times1\times2$

V = 100 ml (of $Cu_2S$)

**In reaction (ii):**

Meq of CuS = Meq of $I_2$

$0.5\times8\times V_{CuS} = 250\times1\times2$

$V_{CuS}$ = 125 ml

$$\text{Eq.wt. of } I_2=\frac{\text{mol.wt}}{n_f}=\frac{254}{2}=127$$

**15.** (*c*) $10\,e^- + 2MnO_4^- \longrightarrow 2Mn^{2+}$; v.f. = 10

$\therefore$ Eq. mass of $Ba(MnO_4)_2=\frac{M}{10}$

**16.** (*b*) $Fe_{0.9}O + K_2Cr_2O_7 \longrightarrow Fe^{+3} + Cr^{+3}$

$$\text{n factor of } Fe_{0.9}O = 0.9\left(3-\frac{2}{0.9}\right)=0.7$$

$\therefore$ Eq mass $=\frac{M}{0.7}=\frac{10M}{7}$

**17.** (*c*) n factor is 2 for $CaC_2O_4$

Eq. weight $=\frac{M}{2}=\frac{128}{2}=64.$

**18.** (*d*) Let x mole $FeCr_2O_4$ and y mole $Fe_{0.95}O$ present the sample.

Given eqn

$FeCr_2O_4 \rightarrow Fe^{2+} + CrO_4^{2-}$

$Fe^{++} \rightarrow Fe^{+++}$

$\overset{3+}{C}r2O_4^{2-} \rightarrow 2\overset{+5}{C}rO_4^{2-}$

So, Net n factor = 7 ($FeCr_2O_4$)

$Fe_{.95}O \rightarrow Fe^{3+}$

nf = .85 .85$e^-$ approx

Now according to given information

$K_4Fe(CN)_6 + Fe^{3+} \rightarrow KFe[Fe(CN)_6] + 3K^+$

$K_4Fe(CN)_6 + Fe^{2+} \rightarrow K_2Fe[Fe(CN)_6] + 2K^+$

milli moles of $K_4Fe(CN)_6$ reacted with $Fe^{3+}$

$= V_{ml}\times M - V_{ml}\times M$

$= 10\times1 - 7\times.1$ [N = Mxnf]

= milli moles f $Fe^{3+}$

Now,

$x + 9.5y = 9.3\times10^{-3}$

$7x + .85y = .0482$

So, solving $x = 6.5319\times10^{-3}$

$y = 2.9138\times10^{-3}$

wt of x($FeCr_2O_4$) $= 6.5319\times10^{-3}\times224$

= 1.463

$$\text{mole of } FeCr_2O_4=\frac{1.463}{224}=6.5319\times10^{-3}$$

$2FeCr_2O_4 + 8KOH + O_2 \rightarrow 4K_2CrO_4 + 4H_2O$

$6.53\times10^{-3}\qquad 2\times6.53\times10^{-3}+Fe_2O_3$

$\downarrow 4BaSO_4$

$4BaCrO_4 + 4K_2SO_4$

$\therefore$ Wt of $BaCrO_4$ PPt

$= 2\times6.53\times10^{-3}\times253$

(mass of $BaCrO_4$ = 253)

= 3.304g

**19.** (*b*) Mass $Fe_{.95}O$

= mole of FeO × mass $Fe_{.95}O$

$= 2.9138\times10^{-3}\times69.20$

(Calcualted above)

= .2016g

$$\% = \frac{.2016}{2}\times100$$

= 10.08%

**20.** (*b*) Fact Based

**21.** (*c*) $FeCr_2O_4$ $Fe_{0.95}O$

x mol y mol

$FeCr_2O_4 \rightarrow Fe^{+3} + CrO_4^{2-}$

$(n_f = 7)$

$Fe_{0.95}O \rightarrow Fe^{+n}$

$(n_f = 0.85)$

$K_4[Fe(CN)_6] + Fe^{+3} \rightarrow KFe[Fe(CN)_6] + K^+$

$(n_f = 3)$ $(n_f = 3)$

$K_4[Fe(CN)_6] + Fe^{+2} \rightarrow K_2Fe[Fe(CN)_6] + 2K^+$

$(n_f = 2)$ $(n_f = 2)$

m moles of $K_4[Fe(CN)_6]$

reacted with $Fe^{+3} = 10 \times 1 - 0.7 = 9.3$

= m moles of $Fe^{+3}$

$x + 0.95\,y = 9.3 \times 10^{-3}$

$7x + 0.85\,y = 0.0482$

$x = 6.5319 \times 10^{-3}$

$y = 2.9138 \times 10^{-3}$

$w_{Fe_{0.95}O} = 0.2016$ gm

$w_{FeCr_2O_4} = 1.463$

**22.** (*c*) Fact Based

**23.** (*a*) Fact Based

**24.** (*b*) $2MnBr_2 + 17PbO_2 + 30HNO_3 \rightarrow$

$2HMnO_4 + 2Pb(BrO_3)_2 + 15Pb(NO_3)_2 + 14H_2O$

The reaction is balanced by the loss or gain of 34 electrons. Hence,

$$E_{HNO_3} = \frac{30+63}{34} = 55.6$$

**25.** (*a*)-p, q; (*b*)-r, s; (*c*)-r, s; (*d*)-p, q

(*a*) n factor of $Cu_2S = 8$

| $KMnO_4$ | $K_2Cr_2O_7$ |
|---|---|
| $1 \times 5 = x = 8$ | $1 \times 6 = x \times 8$ |
| $x = \frac{5}{8}$ | $x = \frac{6}{8} = \frac{3}{4}$ |

(*b*) Container–I reacts with container–II

n factor of KI = 6

| $KMnO_4$ | $K_2Cr_2O_7$ |
|---|---|
| $1 \times 5 = x \times 6$ | $1 \times 6 = x \times 6$ |
| $x = \frac{5}{6}$ | $x = 1$ |

(*c*) n factor of $NH_4SCN = 6$

| $KMnO_4$ | $K_2Cr_2O_7$ |
|---|---|
| $x = \frac{5}{6}$ | $x = 1$ |

(*d*) n factor of $K_2C_2O_4 \cdot 3H_2C_2O_4 \cdot 3H_2O = 8$

| $KMnO_4$ | $K_2Cr_2O_7$ |
|---|---|
| $x = \frac{5}{8}$ | $x = \frac{3}{4}$ |

**26.** [50%] $Mn^{2+} + KMn^{+7}O_4 \longrightarrow Mn^{3+}$

mole of $Mn^{2+}$ = mole of $KMnO_4 \times 4$

= vol. of $KMnO_4 \times$ molarity $\times 4$

$= 3 \times 0.25 \times 4$

$= 3$

(In acidic medium, n-factor for $KMnO_4$ is 5 so molarity = normality /5)

Mole of $Mn_3O_4 = \frac{1}{3}$ mole of $Mn^{2+}$

= 1 mole

= 229 gm

$\%$ of $Mn_3O_4 = \frac{229}{458} \times 100 = 50\%$

**27.** [80 %] $CuSO_4 + KI \longrightarrow Cu_2I_2 + I_2$

$I_2 + Na_2S_2O_3 \longrightarrow I^- + Na_2S_4O_6$

meq. of $Na_2S_2O_3$ = meq. of $I_2$

meq. of $I_2 = 40 \times 1 = 40$

= meq. of $CuSO_4$ in 25 ml

meq. of $CuSO_4$ in 250 ml = 400 meq.

wt. of $CuSO_4 = 6.4 \times 160 = 64$

$\%$ Purity $= \frac{64}{80} \times 100 = 80\%$

**28.** [5] $Fe_2O_3 \xrightarrow{Zn} Fe^{2+} \rightarrow 30 \times \frac{100}{25}$ ml; 0.01 M

$100 \times N = 120 \times .01 \times n$

$N = 1.2 \times 10^{-2} n$

$\frac{0.48}{160} \times \frac{1000}{100} = 1.2 \times 10^{-2} n$

$n = 5$

**29.** [2] $MnO_4^- \longrightarrow Mn^{2+}$ n-factor = 5

Equivalents of

$MnO_4^- = 1.61 \times 10^{-3} \times 5 = 8.05 \times 10^{-3}$

Equivalents of $A^{n+} = 8.05 \times 10^{-3}$

N –factor of $AO_3^- = 5 - n$

$\therefore (5-n) \times 2.68 \times 10^{-3} = 8.05 \times 10^{-3}$

$5 - n = 3$ $n = 2$

**30.** [27] $CrI_3 + KOH + Cl_2 \Rightarrow K_2CrO_4 + KIO_4 + KCl + H_2O$

$Cr^{3+} + 3I^- + K^+ + OH^- + Cl_2 \Rightarrow 2K^+ + CrO_4^{2-} + K^+ + IO_4^- + K^+ + Cl^- + H_2O$

$(Cr^{3+} \Rightarrow CrO_4^{2-} + 3e^-)$

$3\,(I^- \Rightarrow IO_4^- + 8e^-)$

$3I^- \Rightarrow 3IO_4^- + 24e^-$

$Cr^{3+} + 3I^- \Rightarrow CrO_4^{2-} + IO_4^- + 27e^-$

$Cl_2 + 2e^- \Rightarrow 24^-$

$2\,(Cr^{3+} + 3I^- \Rightarrow CrO_4^{2-} + 3IO_4^- + 27e^-)$

$27\,(Cl_2 + 2e^- \Rightarrow 24^-)$

$2Cr^{3+} + 6I^- + 27Cl_2 + 64\,OH^- \Rightarrow 2\,CrO_4^{2-} + 6IO_4^- + 54Cl^- + 32H_2O$

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# Exercise-4 (Past Year Questions)

## JEE MAIN

**1.** (*a*) In the reaction

$\overset{+4}{XeF_4} + \overset{+1}{O_2}\overset{-1}{F_2} \rightarrow \overset{+6}{XeF_6} + \overset{0}{O_2}$

Xenon undergoes oxidation while oxygen undergoes reduction.

**2.** (*a*) $I_2 + 10HNO_3 \rightarrow 2\,HIO_3 + 10NO_2 \uparrow + 4H_2O$

Iodine in $HIO_3$ has + 5 oxidation state.

**3.** (*d*) Fact based.

**4.** (*b*) $5C_2O_4^{2-} + 2KMnO_4 + H^+ \rightarrow 10CO_2 + 2Mn^{2+} + H_2O$

After balancing 10 gain or loss of electron.

**5.** (*c*) H directly attached to P atom.

**6.** (*b*) $n_{eq.}\ KMnO_4 = n_{eq}\ [FeC_2O_4 + Fe_2(C_2O_4)_3 + FeSO_4]$

or $n \times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$

$\therefore\ n = 2$

Correct option: (*b*)

**7.** (*d*) Correct order of oxidation state of nitrogen in oxides of nitrogen is following

$\overset{+1}{N_2O} < \overset{+2}{NO} < \overset{+3}{N_2O_3} < \overset{+4}{NO_2}$

**8.** (*c*) $\overset{+1}{2\,Cu\,Br} \longrightarrow \overset{+2}{CuBr_2} + \overset{0}{Cu}$

Oxidation

Reduction

**9.** [19]

$\overset{x}{K_2Cr_2O_7} \Rightarrow 2(+1) + 2x + 7 \times (-2) = 0 \Rightarrow x = +6$

$\overset{x}{KMnO_4} \Rightarrow 1\,(+1) + x + 4 \times (-2) = 0 \Rightarrow y = +7$

$\overset{x}{K_2FeO_4} \Rightarrow 2 \times (+1) + z + 4 \times (-2) = 0$

$\Rightarrow z = +6$

So, $(x + y + z) = 6 + 7 + 6 = 19$

**10.** (*a*)

(*a*) The given reaction are as follows :

$\underset{(+6)}{H_2SO_4} + \underset{(+1)}{2NaOH} \longrightarrow \underset{(+1)(+6)}{Na_2SO_4} + 2H_2O$

(No change in oxidation state, so not a redox reaction).

(*b*) $\underset{(+3)}{[Co(H_2O)_6]Cl_3} + 3\ \underset{(+1)}{AgNO_3} \longrightarrow \underset{(+3)}{[Co(H_2O)](NO_3)_3} + 3\underset{(+1)}{AgCl}\downarrow$

(No change in oxidation state, so not a redox reduction).

(*c*) $\underset{(0)}{N_2} + \underset{(0)}{O_2} \xrightarrow{200K} \underset{(+2)(-2)}{2N\ O}$

($N_2$ oxidised from 0 to +2 oxidation state and $O_2$ reduced from 0 to –2 oxidation state).

(*d*) $a\,\underset{(0)}{3O_2} \overset{h\nu}{\rightleftharpoons} \underset{(0)}{2O_3}$

(No change in oxidation state, so not a redox reaction).

**11.** (*c*) $K_2\overset{-2}{O} \Rightarrow x = +1, \quad K\overset{-1/2}{O_2} \Rightarrow x = +1$

$K_2\overset{-1}{O_2} \Rightarrow x = +1$

**12.** [19] $\overset{+1}{K_2}\overset{x}{Cr_2}\overset{-2}{O_7}\ \ 2 + 2x - 14 = 0 \Rightarrow x = +6$

$\overset{+1}{K}\overset{y}{Mn}\overset{-2}{O_4}\ \ 1 + y - 8 = 0 \Rightarrow y = +7$

$\overset{+1}{K_2}\overset{z}{Fe}\overset{-2}{O_4}\ \ 2 + z - 8 = 0 \Rightarrow z = +6$

$x + y + z = 6 + 7 + 6 = 19$

**13.** [6] $[Fe(CN)_5(NOS)]^{4-}$

Let the O.S. of Fe be x

OS of CN = –1

OS of NOS = –1

$\therefore\ x + 5(-1) + (-1) = -4 \Rightarrow x = +2$

$Na_4[FeO_4]$

Let O.S. of Fe be y

$\therefore\ y + 4(-2) = -4 \Rightarrow y = +4$

$[Fe_2(CO)_9]$

Let O.S. of Fe be z

$2(z) + 9(0) = 0 \Rightarrow z = 0$

$x + y + z = 2 + 4 + 0 = 6$

**14.** [10.00] Phosphinic acid is $H_3PO_2$

$NaOH + H_3PO_2 \rightarrow NaH_2PO_2 + H_2O$

Milliequivalents of NaOH = milliequivalents of $H_3PO_2$

$V \times 0.1 = 10 \times 0.1$

$V = 10$ mL

**15.** [10.00] $\text{Normality} = \dfrac{\text{No. of equivalents of solute}}{\text{Volume of Solution (in L)}}$

$0.1 = \dfrac{1.43}{\frac{(106+18x)}{2} \times 0.1}$

$\Rightarrow \dfrac{106+18x}{2} = 143$

$\Rightarrow 18x = 286 - 106 = 180$

$x = 10$

**16.** [19.00] $2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2OH^-$

$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

$\therefore x + y + x' + y' + z' = 19$

**17.** [85] $5H_2O_2 + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$

Moles of $KMnO_4 = \dfrac{0.316}{158} = 2 \times 10^{-3}$

Equivalents of $H_2O_2$ = Equivalent of $KMnO_4$

$= 2 \times 10^{-3} \times 5 = 0.01$

@PROCHEM

Moles of $H_2O_2 = \frac{0.01}{2} = 0.005$

Mass of pure $H_2O_2 = 0.005 \times 34 = 0.170$ gm

Percentage purity $= \frac{0.17}{0.2} \times 100 = 85\%$

**18.** (*d*) Colour of $Fe^{2+}$ is observed green and $Fe^{3+}$ is yellow.

**19.** (*c*) No option contains all species that show disproportionation reaction. So question is bonus.

$MnO_4^-$, $ClO_4^-Cr_2O_7^{2-}$ – Cl, Mn, Cr in these anions are present in highest oxidation state. These will not undergo disproportionation.

**20.** [12]

$16e^{\ominus} + S_8 \longrightarrow 8S^{2-}$

$12H_2O + S_8 \longrightarrow 4S_2O_3^{2-} + 24H^+ + 16e^{\ominus}$

---

$2S_8 + 12H_2O \quad 8S^{2-} + 4S_2O_3^{2-} + 24H^+$

for balancing in basic medium add equal number of OH! that of $H^+$

$2S_8 + 12H_2O + 24OH^{\ominus} \longrightarrow 8S^{2-} + 4S_2O_8^{2-} + 24H_2O$

$2S_8 + 24OH^{\ominus} \longrightarrow 8S^{2-} + 4S_2O_8^{2-} + 12H_2O$

$S_8 + 12OH^{\ominus} \longrightarrow 4S^{2-} + 2S_2O_8^{2-} + 6H_2O$

a = 12]

**21.** [16] Writing the half reaction oxidation half reaction

$MnO_4^- \rightarrow Mn^{2+}$

Balancing oxygen

$MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

Balancing Hydrogen

$8H^+ + MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

Balancing charge

$5e^- + 8H^+ + MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

Reducing half

$C_2O_4^{2-} \rightarrow CO_2$

Balancing charge

$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$

Net equation

$16H^+ + 2MnO_4^- + 5C_2O_4^{2-} \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$

So, c = 16

**22.** [173]

$\overset{+6}{Cr}O_4^{2-} + \overset{+2}{S_2}O_3^{2-} \longrightarrow \overset{+6}{S}O_4^{2-} + \overset{-3}{Cr}(OH)_4^-$

gm equi. of $CrO_4^{2-} = S_2O_3^{2-}$

$0.154 \times 3 \times v = 0.25 \times 40 \times 8$

$v = 173.16 = 173$ ml

**23.** [24]

$n_{eq}\, Fe^{2+} = n_{eq}\, Cr_2O_7^{2+}$

or, $\left(\frac{15 \times M_{Fe^{2+}}}{1000}\right) \times 1 = \left(\frac{20 \times 0.03}{1000}\right) \times 6$

$\therefore = 0.24\ M = 24 \times 10^{-2}\ M$

**24.** (*a*) The oxidation of a mixture of one mole of each of $FeC_2O4$, $Fe_2(C_2O_4)_3FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium with $KMnO_4$ is as follows

$FeC_2O_4 + KMnO_4 \rightarrow Fe^{3+} + CO_2 + Mn^{2+}$ .....

$Fe_2(C_2O_4)_3 + KMnO_4 \rightarrow Fe^{3+} + CO_2 + Mn^{2+}$ .....

$FeSO_4 + KMnO_4 \rightarrow Fe^{3+} + SO_4^{2-} + Mn^{2+}$ .....

Change in oxidation number of Mn is 5. Change oxidation number of Fe in (i), (ii) and (iii) are +3, +6, +1 respectively

$n_{eq} KMnO_4 = n_{eq}[FeC_2O_4 + Fe_2(C_2O_4)_3 + FeSO_4]$

$n \times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$

$n = 2$

**25.** (*d*)

Given, $W_{Ca(HCO_3)_2} = 0.81$ g

$W_{Mg(HCO_3)_2} = 0.73$ g

$W_{Ca(HCO_3)_2} = 162\ gmol^{-1}$

$W_{Mg(HCO_3)_2} = 146\ gmol^{-1}$

$W_{H_2O} = 100$ ml

Now, $n_{eq}(CaCO_3) = n_{eq}[Ca(HCO_3)_2] + n_{eq}[Mg(HCO_3)_2]$

$\frac{W}{100} \times 2 = \frac{0.81}{162} \times 2 + \frac{0.73}{146} \times 2$

$\therefore \frac{W}{100} = 0.005 + 0.005$

$W = 0.01 \times 100 = 1$

Thus, hardness of water sample $= \frac{1}{100} \times 10^6$

$= 10{,}000$ ppm

**26.** (*d*) Upto first end point

gm equi. of $(NaOH + Na_2CO_3) = HCl$

$x + y \times 1 = \frac{1}{10} \times 17.5$

$x + y = 1.75$ ...(i)

Upto second end point

$NaOH + Na_2CO_3 \equiv HCl$

$x + y \times 2 = \frac{1}{10} \times 19$

$x + 2y = 1.9$ ...(ii)

$y = 0.15$

$\%Na_2CO_3 = \frac{0.15 \times 10^{-3}}{0.4} \times 100$

$= 3.975\%$

$= 4\%$

**27.** [6] $V_{NaOH} = 4.4$ ml

eq. of NaOH = eq. of $H_2C_2O_4$

or, $M \times 4.4 \times 1 = 1.25 \times 10 \times 2$

or, $M = 5.68$ M

$\therefore$ Nearest integer answer is 6

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**28.** [50] Most precise volume of HCl = 5 ml at equivalence point

Meq. of $N_2CO_3$ = meq. of HCl.

Let molarity of $Na_2CO_3$

Solution = M, then

$M \times 10 \times 2 = 0.2 \times 5 \times 1$

$M = 0.05$ mol / L

$= 0.05 \times 1000 = 50$ mM

**29.** [3] Acid + Base $\longrightarrow$ Salt + $H_2O$

0.1 M M(OH)2

10 mL 0.05 M

30 mL

At equivalence point

Equivalent of acid = equivalent of base

$0.1 \times 10 \times n = 30 \times 0.05 \times 2$

$n = 3$

**30.** (*a*) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

+6 +7 +4

So, it is disproportionation reaction.

## JEE ADVANCED

**1.** (*d*) The formula of Mohr's salt is $FeSO_4(NH_4)_2SO_4$. It is a mixture of ferrous sulfate and ammonium sulfate. Ferrous sulfate is oxidized by potassium dichromate.

The reaction is as follows:

$K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_5 + 7H_2O$

1 mole of potassium dichromate reacts with 6 moles of ferrous sulphate. Thus, the number of moles of Mohr's salt required per mole of dichromate is 6.

**2.** (*c*) Milli mole of Hypo = $0.25 \times 48$

$= 2 \times$ milli mole of $Cl_2$

milli mole of $Cl_2$

$= \frac{0.25 \times 48}{2} = 6$ milli mole

= milli mole of $Cl_2$ = milli mole of $CaOCl_2$

So, molarity = $\frac{6}{25}M = 0.24$ M

**3.** (*a,b,d*)

$6I^- + ClO_3^- + 6H_2SO_4 \rightarrow Cl^- + 6HSO_4^- + 3I_2 + 3H_2O$

Hence, $I^-$ is oxidised to $I_2$

Coefficient of $HSO_4^- = 6$

and $H_2O$ is one of the product.

Hence (*a*), (*b*), (*d*)

**4.** [8] The balanced chemical equation for the oxidation of the complex is as given below.

$MnO_4^- + 5[Fe(H_2O)(C_2O_4)_2]^{2-} + 8H^+ \rightarrow 5[Fe(H_2O)(C_2O_4)_2]^- + Mn^{2+} + 4H_2O$

Hence, $-\frac{d}{dt}[MnO_4^-] = \frac{1}{8}\frac{d}{dt}[H^+]$

$$\frac{\frac{d}{dt}[H^+]}{\frac{d}{dt}[MnO_4^-]} = 8$$

Hence, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is 8.

**5.** [6]

$8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Mole of $SO_4^{2-}$ formed from 8 moles of $MnO_4^- = 6$.

**6.** (*b*) Correct order:

$\underset{(+5)}{H_3PO_4} > \underset{(+4)}{H_4P_2O_6} > \underset{(+3)}{H_3PO_3} > \underset{(+1)}{H_3PO_2}$

**7.** [126]

$MnCl_2 + K_2S_2O_8 + H_2O \rightarrow KMnO_4 + H_2SO_4 + HCl$

x mole x mole

$C_2O_4^{2-} + MnO_4^- \xrightarrow{H^+} CO_2$

milli equivalents of $C_2O_4^{2-}$ = milli equivalents of $MnO_4^-$

$2 \times \frac{0.225}{90} = x \times 5$

$x = 1 \times [55 + 71]$

$= 126$ mg

CHAPTER

# 7 Classification of Elements and Periodicity in Properties

## Exercise-1 (Topicwise)

1. (*c*) According to Doberiner's Traid rule, in a traid atomic mass of middle element was approximately equal to average atomic mass of other two element

   x $\quad y = \frac{x+z}{2}$

   y

   z $\quad 26 = \frac{10+z}{2}$, $10 + Z = 52$, $Z = 42$

2. (*c*) For k, electronic configuration would be:

   $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1$

   ↑

   So it will belong to d block.

3. (*b*) If n = 7 then it will be 5f or actinide series.

4. (*b*) $15 = [Ne]\ 3s^2\ 3p^3$

   $33 = [Ar]\ 4s^2\ 3d^{10}\ 4p^3$

   $51 = [Kr]\ 5s^2\ 4d^{10}\ 5p^3$

   So all belongs to nitrogen family as outermost configuration is $ns^2\ np^3$ or belongs to $15^{th}$ group.

5. (*a*) Fluorine only shows negative oxidation state whereas other halogen shows negative as well as positive oxidation state.

6. (*c*) $Z_{eff} = z - \sigma$

   Z = atomic no

   S = shielding constant

   If $(Z_{eff})_{45} = z - \sigma$

   $\sigma = 0.85 + 8 + 10 \times 1 + 035 \times 0 = 16.8$

   $Z_{eff} = 19 - 16.8 = 2.20$

   II $(Z_{eff})_{3d} = z - \sigma$

   $\sigma = 0.85 \times 8 + 035 \times 8 + 1 \times 2 = 11.8$

   $(Z_{eff.})_{3d} = 19 - 11.6 = 7.40$

7. (*a*) $Z_{eff.}$ increases on moving left to right in a period.

8. (*a*) The atomic radius is lower due to the high effective nuclear charge.

9. (*c*) As negative charge increases ionic radius increases

   $N^{-3} > O^{-2} > F^-$

10. (*c*) An moving top to down no. of increases. Hence atomic radius increases.

    Mg < Na < K < Rb

11. (*b*) For isoelectronic species more the negative charge more the size.

    $O^{2-} > Mg^{2+} > Al^{3+}$

12. (*c*) The compound in which Mn has least +ve oxidation state will have maximum radius. In MnO, Mn have +2 oxidation state which is least among all other compound.

13. (*a*) Down the group atomic size increases due to increase in no. of shell. but in Pd and Pt this increase is very less due to the poor sheilding of d and f orbital electrons. Thus Pd ≃ Pt.

    Ni < Pd ≃ Pt.

14. (*c*) The atomic radius always decreases from left to right. Hence incorrect option is Y ≃ La

15. (*c*) Cationic radius < atomic radius < anionic radius

    So, K = 1.96Å, F = 0.72Å

16. (*d*) On moving down the group, no. of shell increases. Hence

    N < P

17. (*a*) On moving left to right $Z_{eff}$ increases hence atomic radius decreases.

18. (*d*) On moving top to bottom in a group, the ionic radius increases. Hence $I^- > Br^- > F^-$

    The ionic radius for $O^{-2} > F^-$ due to low effective nuclear charge. Hence

    $I^- > Se^{-2} > Br^- > O^{-2} > F^-$

19. (*b*) After removing 1 $e^-$ from 3p orbital, it will achieve fully filled electronic configuration.

20. (*a*) $r_{Cl^-} = \frac{d}{2} = \frac{10}{2} = 5$ Å

    $r_{H^+} = \frac{d}{2} = \frac{2}{2} = 1$ Å

    $r_{HCl} = r^+ + r^- - 0.9\ (\Delta\ E.N)$

    $= 1 + 5 - 0.9(3 - 2.1)$

    $= 5.919$ Å

@PROCBSE

**21.** (*c*) $Al \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
$Mg \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2$
$O \rightarrow 1s^2\ 2s^2\ 2p^4$
$N \rightarrow 1s^2\ 2s^2\ 2p^3 \rightarrow$ Max no. of unpaired $e^-$
Hence $Mg < Al < O < N$

**22.** (*d*) Lanthanide Contraction is the steady decrease of size of the atom with increase in atomic no. in the lanthanide series.

**23.** (*d*) He has small size and stable configuration.

**24.** (*d*) $Al \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
After removing $3e^-$ Al will have stable configuration hence +3 is most stable and $Al^+$ contain completely filled 3s orbital hence $Al^{+3} > Al^+ > Al^{+2}$.

**25.** (*b*) $Mg \rightarrow Mg^+ + e^-$ $IP_1 = 178$ K cal $mol^-$
$Mg^+ \rightarrow Mg^{+2} + e^-$ $IP_1 = 348$ K cal $mol^-$
$Mg \rightarrow Mg^{+2} + 2e^-$ $IP = IP_1 + IP_2$
$= 348 + 178 = 526$ K cal

**26.** (*b*) Ionisation energy increases from left to right (Li → B) and decrease from top to bottom in periodic table But Be ($1s^2\ 2s^2$) has higher ionisation energy than B ($1s^2\ 2s^2\ 2p^1$) due to fully filled s-orbital.
$Be > B > Li > Na$

**27.** (*d*) $1s^2\ 2s^2\ 2p^6\ 3s^2$ as there is drastic change in I.P from $I.P_2$ to $I.P_3$.

**28.** (*c*) Alkali metals have general electronic configuration of $ns^1$ hence it would be easy to remove only $1e^-$ for removal of second $e^-$ large amount of energy would be given.

**29.** (*a*) Element can be Be as there is large IP difference between $I.P_2$ and $I.P_3$.

**30.** (*c*) The order of I.P is Si > S > P > Cl.
The chlorine has highest I.P value because it has smallest atomic size than others.
P has half filled $e^-$ configuration which leads to stability of the atom so it has higher I.P.

**31.** (*a*) $IE_1 < IE_2 < IE_3$ ..... $< IE_4$

**32.** (*b*) On addition of an electron to Li, energy is released.

**33.** (*a*) 3rd period elements of p block has more electron affinity than 2nd period elements as the incoming $e^-$ will suffer less repulsion.

**34.** (*d*) The most positive $\Delta H_{eg}$ will be for $O^{2-}$ and most negetive $\Delta H_{eg}$ will be for $O^{2+}$

**35.** (*c*) S has heighest electron affinity.

**36.** (*c*) A is N element
B is O element
C is P element
D is S element
electron affinity of 3rd period element in p block is greater than 2nd period element and group 16 has more $e^-$ affinity than 15 group.
So order is D > C > B > A.

**37.** (*d*) Electron affinity order for 16th group is
Te < Se < O < S
S has more electron affinity than oxygen as the incoming $e^-$ will suffer less repulsion than O due to large size of S.

**38.** (*c*) Addition of $e^-$ to isolated atom results in release of energy but as nitrogen has half filled configuration hence energy will be given to add an electron in it.

**39.** (*c*) As $ns^2\ np^3$ half filled configuration hence it will have least electron affinity.

**40.** (*d*) 2nd electron affinity of an element is always endothermic.

**41.** (*a*) 3rd period elements of P block have more electron gain enthalpy than 2nd period as incoming $e^-$ will suffer less repulsion and in a period E.A. increases from left to right due to increase in $Z_{eff}$ Hence
$N < O < F < Cl$

**42.** (*b*) P < S < N < O
In a period, electronegativity increases from left to right and in a group, the electronegativity decreases down the group.

**43.** (*a*) Electron affinity of F is less than Cl.

**44.** (*a*) F atom can't hold additional $e^-$ more tightly than Cl atom as additional $e^-$ will experience repulsion in F-atom due to smaller size.

**45.** (*c*) Hydrogen has more ability to attract shared pair of $e^-$.

**46.** (*b*) Electronegativity decrease down the group
Li > Na > K > Rb > CS

**47.** (*a*) $Na_2O > MgO > Al_2O_3 > SiO_2$
More the metallic character heigher the $e^-$ donating tendency. Therefore lower the I.E. more the basic nature of oxide.

**48.** (*c*) Greater the oxidation state, greater will be the nature of acidic oxide. In $Mn_2O_7$, Mn have +7 oxidation state hence it is most acidic.

**49.** (*d*) All are amphoteric oxide. It reacts with both acid and base to produce respective salts.

## Exercise-2 (Learning Plus)

**1.** (*d*) Terbium (z = 65)
Actinoids are in 2n d f-block element series starting from (z = 90 to z = 103)

**2.** (*c*) Period number is the maximum (n) value

**3.** (*b*) d-subshell can have maximum 10 electrons, so it has 10 groups (d block )

**4.** (*c*) → 1st inner-transition elemental series are lanthanoids in which electrons are filled in 4f-orbitals.

**5.** (*d*) → as it resembles properties of alkali metals; so it can be placed in group.

**6.** (*c*) Cu has atomic No. (47–18) = 29 above Ag (47).
Below Ag has Au with Atomic No. (47 + 32) = 79.

7. (b) In Modern periodic table and long form of periodic table, physical and chemical properties are periodic function of atomic Number

8. (a) for Lanthanoids, this term is used.

9. (a) $\frac{z}{e}$

$N^{-3} \Rightarrow \frac{7}{10} \Rightarrow 0.7$

$O^{-2} \Rightarrow \frac{8}{10} \Rightarrow 0.8$

$F^{-} \Rightarrow \frac{9}{10} \Rightarrow 0.9$

10. (a) $\sigma_{Cu} = x$

$\sigma_{Zn} = x + 0.35$

(as last electron comes in outer 4s) {so electron of 4s is taken}

11. (c) $\sigma$ for d and f = 0.85

12. (a) as diffuseness of s-orbitals increases, their sheilding effect decrease.

13. (c) $\boxed{r \alpha \frac{1}{z}}$ for isoelectronic species

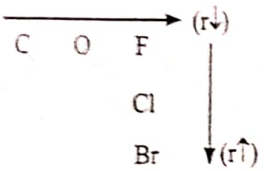
14. (d) $\boxed{r \alpha \frac{1}{z}}$ for isoelectronic species

15. (d) $\boxed{r \alpha \frac{1}{\text{amount of + ve charge}}}$

16. (b) $\boxed{r \alpha \frac{1}{z}}$

17. (c) $\boxed{r \alpha \frac{1}{z}}$

18. (a)




$\therefore \boxed{Br > Cl > C > O > F}$

19. (c) It is because ($z_{eff} = \sigma$) (in 3d series)

20. (d) $M(g) \xrightarrow{IE_1} M^{+}(g) \xrightarrow{IE_2} M^{2+}(g) \xrightarrow{IE_3} M^{3+}(g)$

$I.E_3 > I.E_2 > I.E_1$

(successive I.E. always increases)

21. (b) $M(g) \longrightarrow M^{3+}(g) + 3e;\ \Delta H = x$ kJ

$M(g) \xrightarrow{I.E_1} M^{1+}(g) \xrightarrow{I.E_2} M^{2+}(g)$

$M^{2+}(g) \xrightarrow{I.E_3} M^{3+}(g)$

$I.E_1 + (I.E_2)_M + (I.E)_3 = x \qquad [(I.E_3)]_M = [(I.E_1)]_{Mg}$

$y + (I.E_2)_M + z = x$

$(I.E_2)_M = x - y - z$ (kJ/mol)

22. (c) order of ($I.E_1$) of N, O, Be, B $\Rightarrow$ B < Be < O < N

23. (c) (down the group I.E↓)

24. (c) → Assertion is correct because Be has stable electronic configuration.

→ Reason is false because n + $\ell$ value of 2p > 2s

25. (b) Down the group I.E. decreases.

26. (d) K < Ca ($I.E_1$)

K > Ca ($I.E_2$) {due to stable electronic configuration of $K^{+}$}

27. (c) I.E. α positive charge; (greater than Neutral one)

along the period from ($Be^{+}$ > Be) L to R $\Rightarrow$ I.E↑ (C > Be)

28. (b) Ne > Cl > P > S > Mg > Al

→ Along the period from left to right I.E. increase exceptionally due to stable electronic configuration of Mg and P, they have more I.E. than that of element which is next after to them.

29. (d) $A \xrightarrow{I.E_1} A^{+} \xrightarrow{I.E_2} A^{2+}$

If diff > 16 ev $\Rightarrow$ then lower O.S. is stable

If diff < 11 ev $\Rightarrow$ then higher O.S. is stable

30. (d)

| C | N | O | F |
|---|---|---|---|
| $2s^2 2p^2$ | $2s^2 2p^3$ | $2s^2 2p^4$ | $2s^2 2p^5$ |

$\left[\begin{matrix} C^{+} & N^{+} & O^{+} & F^{+} \\ 2s^2 2p^1 & 2s^2 2p^2 & 2s^2 2p^3 & 2s^2 2p^4 \end{matrix}\right] \Rightarrow$

(I.E.↑) (but exceptianally $O^{+} > F^{+}$)

So 2nd I.E.order $\Rightarrow$ O > F > N > C

31. (a) Tendency to gain electron ↑ $\Rightarrow$ from left to right (exceptionally half filled and full filled has lower tendency to gain electron)

iv > ii > iii > i

32. (c) $O^{-}$ ion will tend to resist the addition of another electron due to electron-electron repulsion.

33. (c) $Be + e^{-} \rightarrow Be^{-}$

Due to stable electronic configuration in Be. We have to give energy, so it is unstable.

34. (b) Adding second electron energy is absorbed due to repulsion.

35. (b) $S^{-} + e^{-} \rightarrow S^{2-}$ (polyanionic species are endothermic in nature)

36. (a) $N + e^{-} \rightarrow N^{-}$ (endothermic step)

And reverse is exothermic due to stable nitrogen electronic configuration.

37. (b) as size of Cl is bigger than F, so it will accomodate electron with least repulsion if we compare it with F.

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38. (*b*) Cl > F > S > O

E.G.E of 3rd period element is greater than 2nd period elements and group 17 has more E.G.E. than 2nd period.

39. (*b*) $Na \xrightarrow{I.E.} Na^+(g) + e^-$ I.E. = 5.1

$Na^+(g) + e^- \longrightarrow Na$ $\Delta H_{eg} = -5.1$

40. (*d*) All statements are correct as per the theory of Electron gain enthalpy.

41. (*b*) As Nitrogen has half filled p subshell, so electron removal is tough. Hence Ionisation energy is high.

42. (*c*) Electron gain enthalpy and ionisation energy are affected by stable electronic configuration.

43. (*c*) Down the group electronegativity decreases.

44. (*c*) Down the group metallic character increases; hence basic strength of metal oxide increases.

45. (*d*) From left to right, Metallic character decreases.

B < Al < Mg < K

46. (*a*) SnO, ZnO, PbO → are amphoteric oxides

47. (*a*) $Cl_2O_7$ → There is maximum oxidation number on Cl; i.e. = 7

Hence it is more acidic

→ as O.N. ↑ ⇒ acidic strength increases.

48. (*b*) The correct order is

Li < B < Be < C

↑

due to stable electronic configuration.

49. (*d*) $CO_2 > SiO_2$

Acidic strength decreases down the group.

## Exercise-3 (JEE Advanced Level)

1. (*c,d*) I[st] group elements show + 1 oxidation state and 2nd group elements show + 2 oxidation state.

Group 18 elements are all noble gases, [He, Ne, Ar, Kr] does not show any oxidation state while Xe shows oxidation states namely II, IV, VI.

2. (*b,d*) Cu shows + 1 and + 2 oxidation state so $Cu_2Cl_2$ exist. Similarly potassium shows + 1 O.S. so general formula with 'O' is $K_2O$.

3. (*a,b,d*)

Zn have only + 2 or zero oxidation state.

4. (*a,b,c,d*)

Due to inert pair effect, Pb, Tℓ, Bi has their lower oxidaton state more stable than higher ones. O has (–2) O.S. move stable than other oxidation states.

5. (*a,b,c,d*)

As positive charge increases by which $Z_{eff.}$ increases, so $Cu^+$, $Cu^{2+}$, $Fe^{3+}$, $Zn^{2+}$ has greater $Z_{eff.}$ than Zn.

6. (*a,b,d*)

(*a*) Size of aluminium and gallium is nearly same due to poor shielding of 3d orbital in Ga.

(*b*) $Te^{2-}$, $I^-$, $Cs^+$, $Ba^{2+}$ are isoelectronic to Xe so elements with move electrons have higher size.

(*d*) Pd and Pt have similar size because of poor shielding of 4f orbital in Pt.

7. (*a,b,c,d*)

$H^-$, $Cl^-$, $Br^-$, $I^-$ has greater radius than F and $H^-$ is very much larger due to very less $Z_{eff.}$ in $H^-$, it's outer electron are free to maximize the radii.

8. (*a,b,d*)

(*a*) As removal of second electron takes place from half filled valance shell, electron configuration of $O^+$ i.e. $2s^22p^3$.

(*b*) Due to higher nuclear charge in phosphorus, $I.E._3$ of P is grearter than aluminium.

(*d*) $I.E._2$ of Cu is greater than Zn because in $Cu^+$ electron is removed from fully filled configuration $[Ar]3d^{10}$.

9. (*a,b,c*)

(A) Mg has $[Ne]3s^2$ configuration so difficult to remove electron from it than Na.

(B) $Mg^{2+}$ – [Ne]

$Al^{2+}$ – $[Ne]3s^1$

Low I.E. required to remove electron from $Al^{2+}$.

So $I.E._3(Mg) > I.E._3(Al)$

(C) $I.E._1(Al) < I.E._1(Mg)$

Mg has $[Ne]3s^2$ electronic configuration so difficult to remove electron from it.

10. (*a,d*) I.E. ∝ |magnitude of + ve charge| ∝ $\frac{1}{\text{Magnitude of } -\text{ve charge}}$

As it is easy to remove electron from a anion (loosely bound electron).

11. (*a,c*) Elements of group 1 and 2 have low I.E. values so they import colour to flame.

12. (*a,c,d*)

(A) More the ionisation energy, then more will be its electronegativity because that atom has high affinity towards electrons.

(C) If ionisation energy is low than we can say it is easier to remove outer electron.

(D) On moving along period electron affinity increases.

13. (*a,b,c,d*)

Electron affinity increases on moving along a period and exact order of electron affinity for group 16 is S > Se > Te > O

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14. *(a,b,d)*

Due to smaller size of 2nd period elements it resists the incoming electron making electron affinity of 2nd period elements greater than 3rd.

15. *(b,c)*

(B) First ionisation enthalpy of nitrogen is higher than its neighbour's due to half filled electronic configuration.

(C) Cl has higher electron gain enthalpy than F due to compact size of F.

16. *(a,b,d)*

(A) Ionisation enthalpy increases along a period so C has highest first ionisation enthalpy.

(B) Atomic size increases down the group and decreases along a period. So Al has largest atomic size.

(D) Metallic character increases down the group so Al has most metallic character.

17. *(a,b,c,d)*

Hydroxides of metals which shows both acidic as well as basic behaviour are amphoteric.

e.g. $Zn(OH)_2$, $Be(OH)_2$, $Al(OH)_3$, $Pb(OH)_2$.

18. *(b)* $I.E._3 > I.E._2$ of an atom because $Z_{eff}$ of an atom in its dipositive ion is more than in mono-positive ion.

19. *(d)* Alkali metals have lowest ionisation energy in period as by loosing only one electron stable noble gas configuration is achieved.

20. *(c)* $I.E._2$ is highest for IIIrd element as after removing Ist electron stable noble gas configuration is achieved so it is difficult to remove the 2nd $e^-$.

21. *(a)* (A) has highest $I.E._1$ by which, this elements is likely to be the noble gas.

22. *(b)* Fully filled and half filled configuration have higher ionisation energy order than its neighbour elements.

So, B < Be < C < O < N

23. *(a)* In Na $I.E._1 << I.E._2$ because $Na^+$ has stable noble gas electronic configuration.

So difference is maximum in case of sodium.

24. *(c)* As phosphorus has stable half filled electronic configuration so $I.E._1$ is highest for it.

Now on moving along a period I.E. increases so sulphur has more I.E. than Mg and Al.

25. *(c)* Group 14 is the carbon family consisting of non-metals (carbon and silicon), metalloid (germanium) and metal (lead and tin).

26. *(b)* Non-metals belong to the p-block of the modern periodic table.

27. *(c)* Metallic character is inversely proportional to electronegativity and Non-metallic character is directly proportional to electronegativity of elements.

28. *(c)* Electronegativity increases along a period and decreases down the group.

29. *(d)* Metallic character decreases on moving along a period so potassium has more metallic character than Mg and B.

30. *(b)* Non-metals have larger size than metals so size of $I^-$, $Br^-$ is greater than $Li^+$ and $Na^+$ and ionic radius increases along a group and decreases along a period.

31. *(c)* On moving left to right in a period, atomic radius decreases due to increase in $Z_{eff}$ and addition of electrons to the same outer most shell.

32. *(a)* La – 57, Cl – 17, K – 19, Cu – 29.

(A) La – 57 is a d-block elements and violet Aufbau principle.

(B) Cl – 17 is a p-block element and is a non-metal.

(C) K – 19 is a s-block element and belong's to 4th period.

(D) Cu – 29 is a d-block and violet Aufbau's principle ($[Ar]_{18}3d^{10}4s^1$).

33. *(c)* Electronic configuration's

$O \rightarrow 1s^2\ 2s^2\ 2p^4$

$O^- \rightarrow 1s^2\ 2s^2\ 2p^5$

$O^{2-} \rightarrow 1s^2\ 2s^2\ 2p^6$

(A) In 'O' electron is gained, So energy is released in making negative electron gain enthalpy.

(B) $(E.G.E.)_2$ is generally positive because the incoming electron faces the repulsion and energy is taken by it to accept an $e^-$.

(C) $Na^-$ has stable configuration and so energy is needed to remove $e^-$ from it.

(D) $Mg^-$ gains $e^-$ easily due to higher $Z_{eff}$, making the process exothermic with negative electron gain enthalpy.

34. *(b)* (A) Iodide ion is the strongest reductant.

(B) Chromium has half filled d-orbital $[Ar]_{18}3d^5 4s^1$

(C) Aurum/Gold is a coinage metal.

(D) Cerium is a lanthanide element.

35. *(d)* (A) Gallium – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$

(B) Vanadium – $2s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$

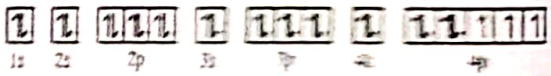
(C) Zinc – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}$

(D) Scandium – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

36. [0015]

$Ni^+ = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^7$

Max. no. of electron having same spin = 15




37. [350]

Lattice enthalpy = 772 kJ/mol

$$\frac{[\Delta H.E.]_{Na^+}}{[\Delta H.E.]_{Cl^-}} = \frac{30}{25} = \frac{6}{5}$$

$$\underset{(x)}{[\Delta HE]_{Na^+}} = \frac{6}{5}\underset{(y)}{[\Delta HE]_{Cl^-}}$$

$$x = \frac{6}{5}y$$

$(\Delta H)_{sol^n}$ = Lattice enthalpy + H.E. of $Na^+$ + H.E. of $Cl^-$

$$2 = 272 + y + \frac{6}{5}y$$

$y = -350$ kJ

**38.** [8] $r_A + r_A = 10\ A°$ ...(1)

$r_B + r_B = 6\ A°$ ...(2)

Add (1) and (2),

$2(r_A + r_B) = 16\ A°$

$r_A + r_B = 8\ A°$

**39.** [30]

Zn has largest size among Ni, Cu and Zn
Because of repulsion of d-electron in zinc.

**40.** [1]

$A(g) + e^- \longrightarrow A^-(g)$ $\Delta H = +$ E.G.E. ...(1)

$A(g) \longrightarrow A^{2+}(g) + 2e^-$ $\Delta H = 1200$ kJ/mol ...(2)

$A^-(g) \longrightarrow A^{2+}(g) + 3e^-$ $\Delta H = 1100$ kJ/mol ...(3)

eq$^n$ (3) = (2) – (1)

1100 = 1200 – E.G.E.

E.G.E. = 100 kJ/mol.

E.G.E. = $1 \times 10^2$ kJ/mol.

So P = 1

**41.** [23]

$A + e^- \longrightarrow A^-$ E.G.E. = –3eV

$$\text{Numer of mole of A} = \frac{10}{30}$$

As 1 mole of A released the ammount energy = $-3 \times 23$ kcal/mol.

Energy released for conversion of $\frac{10}{30}$ mole of gaseous A into $A^-$ ions = $\frac{3\times 23}{30}\times 10$ = –23 kcal/mol.

**42.** [3]

(i) $N^{3-} > P^{3-}$

(ii) $Ca^{2+} > Sr^{2+}$

(iii) $Cl^- > S^{2-}$

**43.** [6]

$Zn^{2+} = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 2p^6\ 3d^{10}\ 4s^0$

All s-electrons i.e. $1s^2, 2s^2, 3s^2$ have $\ell = 0$.

**44.** [3]

Li – Mg, Be – Al, B – Si shows diagonal relationship.

**45.** [7]

The element is nitrogen which belongs to second period and group 15.

**46.** [4]

SnO, PbO, $Al_2O_3$, ZnO are amphoteric in nature.

**47.** [5]

$(Na^+, Li^+)$, (Se, S), $(I^-, Br^-)$, (Ca, Mg), $(S^{2-}, Cl^-)$

are the pair's in which size of first species is greater than second one.

**48.** [4]

He, Be, N, Ne possess a positive value of electron gain enthalpy.

**49.** [4]

(N, O), (Be, B), (Me, Ne), (Mg, Na) are the pair(s) in which I.E. of first element is greater than second element.

**50.** [4.35]

$Zn = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2$

$Z_{eff.}$ = Atomic no. – shielding constant ($\sigma$)

$\sigma = 10 + 0.85 \times 18 + 0.35 \times 1 = 25.65$

$Z_{eff.} = 30 - 25.65 = 4.35$.

**51.** [5]

Na, Mg, P, S, Al have ionisation energy less than chlorine atom.

**52.** [2]

Sudden jump in $I.E._2$ and $I.E._3$ is observed showing valence shell's electron = 2

So group number of element X is 2.

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## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*d*) Sc has highest first ionization energy because:

- Sc: [Ar] $4s^2\ 3d^1$
- Poor screening of d-orbitals.
- I.E. increases left to right in a period.

**2.** (*a*) $O^{2-}$, $F^-$, $Na^+$, $Mg^{2+}$ are isoelectronic species as all of these has same no. of $e^-$ = 10.

**3.** (*b*) $[Og_{118}]\ 8s^2$ is the electronic configuration for Z = 120, (Og is Oganesson).

So, as per the configuration, it will be in II$^{nd}$ Group.

**4.** (*c*) due to lanthanide contraction, atomic and ionic radii decreases.

**5.** (*b*) E.N. of Al is same as that of Be due to presence of Diagonal Relationship.

**6.** (*d*) Correct order of radii : Eu > Ce > Ho > N.

It is because in case of lanthanide, 2 peaks appear at ($_{63}$Eu and $_{70}$Yb) as Eu andYb both provide only $2e^-$ for bonding while other members provide $3e^-$ for bording purpose.

**7.** (*b*) Electronegativity increases from left to right in a period and decreases down the group.

8. (*d*) Inert pair effect gradually increases down the group. Hence stability of lower O.S. increases down the group.

9. (*c*) On moving down a group, Electronegativity decreases and atomic radius increases for representative elements.

10. (*a*) The onter most electronic configuration of Tl is $6s^2$ $6p^1$. The 6s electrons are strongly attracted towards the nucleus due to its more penetrating power and deshielding of d and f-electrons, Hence 6s-electrons do not participate in bonding.

11. (*d*) 2nd electron gain enthalpy of oxygen is positive.

12. (*a*) On moving down, size increases:

order : C < S < Al < Cs.

13. (*d*) 119 symbol is : Ununennium (uue)

14. (*a*) In case of "Be" electron remove from "2s" orbital while in case of "B" electron remove from "2p" orbital. '2s' orbital have greater peretration effect then '2p' orbitals. So, 'Be' have more I.E. than 'B'.

15. (*c*) Atomic number (Z) 15 $\Rightarrow$ P $\rightarrow$ [Ne] $3s^2$ $3p^3$

Phosphorus belongs to 15th group.

no. of valence $e^- = 5$

and valency = 3 in (G.S.)

16. (*a*) As we move down the group, Bond strength decreases there by decreasing the catenation tendency.

Hence; the order expected is

C > Si > Ge $\approx$ Sn

17. (*b*) Ti $\rightarrow$ [Ar] $3d^2$ $4s^2$

Mn $\rightarrow$ [Ar] $3d^5$ $4s^2$

Ni $\rightarrow$ [Ar] $3d^8$ $4s^2$

Zn $\rightarrow$ [Ar] $3d^{10}$ $4s^2$

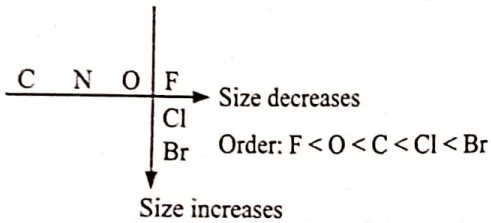
18. (*a*) The element which release more energy upon an $e^-$ gain are [Cl, S and Li]

19. (*d*) Correct order of Ionization energy will be:

Na < Al < Mg < Si

20. (*d*) $_{26}Fe$ = [Ar] $3d^6$ $4s^2$

21. (*b*)




22. (*d*) The electronic configuration of B and Be are:

B $\Rightarrow$ $1s^2$ $2s^2$ $2p^1$

Be $\Rightarrow$ $1s^2$ $2s^2$

(I) 2s - orbital experienced more $Z_{eff}$ than 2p therefore 2p electron could be easily removed than 2s.

(II) In case of B there are total 4 inner core electrons but in Be only $2e^-$ (in $1s^2$) therefore 2p more shielded in B than 2s of Be.

(III) s-orbital has spherical shape through that $Z_{eff}$ work 10% but p-orbital is dumbell shape therefore its $Z_{eff}$ less and penetrating power less than 2s.

(IV) Atomic radius of B less than Be due to more $Z_{eff}$

According to the above explanation only (I), (II) and (III) are correct,

23. (*a*) According to the given date of I.E., this element must belong to group I and thus is monovalent and form hydroxide of the type M(OH).

$MOH + HCl \rightarrow MCl + H_2O$

1 mol 1 mol

$2\,MOH + H_2SO_4 \rightarrow M_2SO_4 + H_2O$

1 mol 1/2 mol

24. (*d*) In periodic table, in a period on moving left to right as atomic number increases the nature of oxide follows the order basic then anphoteric and then acidic i.e., oxide of X is basic, oxide of Y is amphoteric, oxide of Z is acidic.

– Hence, atomic number follows the order:

Z > Y > X

25. (*d*) Atomic radius decreases on moving left to right in periodic table, while other properties given increases (in magnitude) on moving left to right across a period.

26. (*a*) For Unnilennium: Atomic no : 109

Symbol : Une

27. (*b*) $Ar(g) + e^- \rightarrow Ar^-(g)$ (Endo.)

$H(g) + e^- \rightarrow H^-(g)$ (Exo.)

$Na(g) \rightarrow Na^+ + e^-$ (Endo.)

$O^-(g) + e^- \rightarrow O^{-2}(g)$ (Endo.)

– E. G. E. of H(g) is negative while that of Ar(g) is positive due to $ns^2$ $np^6$ configuration.

– 2nd E. G. E. is always positive for an atom.

– Ionisation potential of an atom is positive

28. (*d*) Among Isoelectronic species greater the $z_{eff}$, smaller will be the radius.

Order of $Z_{eff}$ : $O^{2-} < F^- < Na^+ < Mg^{+2}$

Order of Ionic Radii : $O^{2-} > F^- > Na^+ > Mg^{+2}$

29. (*b*) Actinoids contains 14 elements with atomic number 90 to 103. Hence, element with atomic number 101 is actinoids.

Element with atomic number 104 is a d-block element of group 4.

30. (*a*) For isoelectronic species, as the no. of protons increases, size of ions decreases.

$\therefore$ Correct order of ionic Radii :

$Al^{+3} < Mg^{+2} < Na^+ < F^- < O^{2-} < N^{-3}$

31. [101]

Unnilunium

IUPAC Symbol : Unu

At, no. (Z) = 101

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**32.** (*d*) $Mn(25) = [Ar]\ 3d^5\ 4s^2$

$Mn^{3+} = [Ar]\ 3d^4\ 4s^0$

- Ru-belong to 4-d Transition series
- $Co(27) = [Ar]\ 3d^7\ 4s^2$

  $Co^{3+} = [Ar]\ 3d^6\ 4s^0$
- $Fe(26) = [Ar]\ 3d^6\ 4s^2$
- $Fe^{3+} = [Ar]\ 3d^5$

**33.** (*d*) Elements with atomic number 21, 25, 42 and 72 belongs to transition metals.

**34.** (*b*) From left to right in periodic table:

- Metallic character decreases.
- Non-metallic character increases.

– This is due to increase in ionization enthalphy and increase in electron gain enthalphy.

**35.** (*b*) $Mg(Z = 12) = 1s^2 2s^2 2p^6 3s^2 \Rightarrow$ (Fully-filled)

$Al(Z = 13) = 1s^2 2s^2 2p^6 3s^2 3p^1$

$Si(Z = 14) = 1s^2 2s^2 2p^6 3s^2 3p^2$

$P(Z = 15) = 1s^2 2s^2 2p^6 3s^2 3p^3 \Rightarrow$ (Half-filled)

$S(Z = 16) = 1s^2 2s^2 2p^6 3s^2 3p^4$

In general from left to right in a period. I.E. increases due to increase in effective nuclear charge but due to extra stability of half-filled and fully-filled electronic configuration. I.E. of P > S and Mg > Al.

Hence, the order for first I.E. is as follows:

Al < Mg < Si < S < P

**36.** (*b*) The major components in "Gun metal" which is an alloy are as follows:

Cu : 87%

Zn : 3%

Sn : 10%

**37.** (*a*) The given electronic configuration belong to the following elements:

(*a*) $1s^2 2s^2 \Rightarrow$ Be $\Rightarrow$ Fully filled $\Rightarrow$ stable

(*b*) $1s^2 2s^2 2p^4 \Rightarrow$ O

(*c*) $1s^2 2s^2 2p^3 \Rightarrow$ N $\Rightarrow$ Half filled $\Rightarrow$ stable

(*d*) $1s^2 2s^2 pl^1 \Rightarrow$ B

Order for Ionization enthalpy is

B < Be < O < N

Ionization enthalpy depends upon many factors but in given question it can be explained on the basis of half filled and fully filled stable configuration of elements.

Be has more I.E. compared to B due to extra stability (fully filled), whereas N has more I.E. compared to oxygen due to extra stability as it is half-filled.

Point to be noted: "Fully-filled" orbitals are more compared to half-filled orbitals.

Hence,

N $\Rightarrow$ 1402 kJ/mol

O $\Rightarrow$ 1314 kJ/mol

B $\Rightarrow$ 801 kJ/mol

Be $\Rightarrow$ 899 kJ/mol

**38.** (*a*) The correct order of electron gain enthalpy is as follows:

S > Se > Te > O

Due to small size of oxygen atom, it exerts electron-electron repulsion as a result of which incoming electron will not be able to add into the shell and gets repelled by the electron, so more amount of energy is required. Therefore, electron gain enthalpy of oxygen becomes less negative compared to the rest of the elements.

O < S > Se > Te

S > Sc > Te > O

**39.** (*b*) The nature of the oxide depends upon the nature of the elements whether it is a metal, non-metal or a metalloid.

CaO, BaO $\rightarrow$ Basic oxides

$B_2O_3$, $SiO_2 \rightarrow$ Acidic oxides

$N_2O$ Neutral oxide

The nature of oxide also depends upon the oxidation state of the element. Higher the O.S. of elements more acidic is the oxide where as lower O.S. of element, more basic is the compound.

**40.** (*a*) $I.E_1$ of alkali metals is lowest and $I.E_2$ of alkali metals is highest because after losing one electron they acquire noble gas configuration so 'X' is Na.

**41.** (*b*) $X_{33} - 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^3$

Similar configuration to 15th group so its a metalloid.

$Y_{53} - 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^5$

Similar configuration to halogen period so it is a non-metal.

$Z_{83} - 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^6\ 6s^2\ 4f^{14}\ 5d^{10}\ 6p^3$

It's belong to 15th group p-block. As metallic character increases down the group.

It is only metal in group 15.

**42.** (*c*) Due to small size of F there are inter electronic repulsions on accepting electron so $\Delta H_{e.g}$ of Cl is more negative than F.

**43.** (*d*) Li → Na; Be → Mg; B → Al; C → Si

[These three pairs diagonally show similar properties]

**44.** (*b*) $\text{Radius} \propto \frac{1}{\frac{Z}{e}}$

$Al^{3+}$ < $Mg^{2+}$ < $Na^+$ (radius)

0.54Å 0.72Å 1.02Å

**45.** (*c*) Lets first arrange elements asked in choices in periodic table.

Li Ne

Na Mg Cl Ar

Na < Mg < Cl < Ar

X is Ar, Y is Cl, Z is Na

**46.** (*d*)

| | gp 13 | gp 14 |
|---|---|---|
| Periodic-2 | B | C |
| Periodic-3 | Al | Si |
| Periodic-4 | Ga | Ge |
| Periodic-5 | In | Sn |
| | Tl | Pb |

In 5th period:

Orbital filled are 5s, 4d, 5p

E.C. of Sn is $[Kr]4d^{10}\ 5s^2\ 5p^2$

**47.** (*d*) At the time of Mendeleev structure of atom was not known as his basis of classification was atomic mass.

**48.** (*b*)

| | $Na^+$ | $Mg^{2+}$ | $Al^{3+}$ (isoelectric species) |
|---|---|---|---|
| No. of $e^-$ | 10 | 10 | 10 |
| Z | 11 | 12 | 13 |
| $\frac{Z}{e}$ | $\frac{11}{10}$ | $\frac{12}{10}$ | $\frac{13}{10}$ |

More the $\frac{Z}{e}$, then smaller is the radii

$Al^{3+} < Mg^{2+} < Na^+$

$K^+$ is in lower period, higher value of 'n' so large radius.

$Al^{3+} < Mg^{2+} < Na^+ < K^+$

**49.** (*b*)

$N^{3-}$, $O^{2-}$, $F^-$ isoelectronic species

more Z then small size.

So size $N^{3-} > O^{2-} > F^-$

But size of anion is more than its neutral counter part.

$N^{3-} > N$

**50.** (*c*)

I.E. increases from left to right with exception.

Mg > Al (More penetration effect in Mg)

P > S (Half filled p-subshell in P)

**51.** (*a*) Group-16 are chalcogens

**52.** (*b*) Among isoelectronic species more Z then smaller radius as they have same no. of $e^-$s.

$Ca^{2+} < K^+ < Cl^- < S^{2-} < P^{3-}$

**53.** (*b*) Acidic $\rightarrow Cl_2O_7$

Basic $\rightarrow$ $Na_2O$

Amphoteric $\rightarrow$ $As_2O_3$

Neutral $\rightarrow$ NO, $N_2O$

**54.** (*d*)

| Species | $Al^{3+}$ | $Mg^{2+}$ | $Be^{2+}$ | $O^{2-}$ | $Br^-$ | $Cl^-$ | $Li^+$ | $S^{2-}$ | $K^+$ |
|---|---|---|---|---|---|---|---|---|---|
| No. of $e^-$ | 10 | 10 | 2 | 10 | 36 | 18 | 2 | 18 | 18 |

So, $Mg^{2+}$ and $O^{2-}$ are isoelectronic with $Al^{3+}$

**55.** (*b*) Chlorine has the highest $e^-$ gain enthalpy, even greater then F, due to small size and repulsion in florine atom dwon the group, $e^-$ gain enthalpy decreases and along the period it increases

So the correct order is

Po < Te < F < Cl.

**56.** (*d*)

16th group: $\underbrace{O\ S}_{\text{non metal}}$ $\underbrace{Se\ Te}_{\text{metalloid}}$ $\underbrace{Po}_{\text{metal}}$

15th group: N P As Sb $\underbrace{Bi}_{\text{metal}}$

14th group: C Si Ge Sn $\underbrace{Pb}_{\text{metal}}$

Sc is d-block metal.

**57.** (*a*) Ion $O^{2-}$ $Mg^{2+}$

No. of $e^-$ 10 10(isoelectronic)

Z 8 12

$\frac{Z}{e}$ $\frac{8}{10}$ $\frac{12}{10}$

$\frac{Z}{e} \propto$ force of attraction $\propto \frac{1}{\text{size}}$

Thus size of $Mg^{2+} < O^{2-}$.

**58.** (*a*)

| Species | $N^{3-}$ | $O^{2-}$ | $F^-$ | $Na^+$ | $Mg^{2+}$ |
|---|---|---|---|---|---|
| No. of $e^-$ | 10 | 10 | 10 | 10 | 10 |
| Z | 7 | 8 | 9 | 11 | 12 |

size $\propto \frac{1}{z/e}$ thus the order is

$Mg^{2+} < Na^+ < F^- < O^{2-} < N^{3-}$

**59.** (*a*)

| 16th group | EC | Period |
|---|---|---|
| O | $2s^22p^4$ | 2 |
| S | $3s^23p^4$ | 3 |
| Se | $4s^24p^4$ | 4 |
| Te | $5s^24d^{10}5p^4$ | 5 |

Above it

**60.** (*b*)

$Cl_2O_7$ – Acidic

$Na_2O$ – Basic

$Al_2O_3$ – Amphoteric

$N_2O$ – Neutral

**61.** (*c*)

Acidic $\Rightarrow SiO_2.SO_3$

Amphoteric $\Rightarrow Al_2O_3$

Basic $\Rightarrow$ CaO

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**62.** (*b*) $I.E_1$ of O < N thus assertion is true but the reason for this is that because of half-filled 2p subshell in nitrogen which is stable. Thus, reason given is not explaining assertion but reason on its own is correct.

**63.** (*d*) Atomic no. = 103 (86 + 14 + 2)

Un nil tri(ium)

**64.** (*d*) (a) IE of Be is more than B due to stable fully-filled 2s-orbital and penetration effect.

(b) IE of N is more than O due to more stable half-filled 2p-orbitals.

**65.** (*b*) Acidic → $B_2O_3$, $N_2O_5$, $SO_3$ $P_4O_{10}$

Neutral → NO, $N_2O$, CO

**66.** (*d*) Radius of Anionic species is always greater than their parent atomic radius due to decrease in $\frac{Z_{eff}}{c}$. But covalent radius is always less than the atomic radius due to common sharing portion of covlently bonded molecule.

Thus statement-I is wrong and Statement-II is correct

**67.** (*b*) Ga have low M.P. and it is closer to metalloid in periodic table.

**68.** (*d*)

(*a*) Li
Na
K
Size increases
So I.E. ↓ses

K < Na < Li

(*b*) I.E. decreases down the group so correct

(*c*) Atomic number 37 is Rb so outer E.C is $4s^1$, atomic number 38 is Sr so outer E.C is $4s^2$

So, Z = 37 has lesser I.E. than Z = 38, so correct

(*d*) The first I.E. of Zn > I.E. of Ga due to completely filled orbitals in Zn so (d) is incorrect.

**69.** (*b*) I → Mg, II → Al, III → P, IV → S

I.E increases with exception of

P > S (Half filled stability) > Mg > Al (Penetrating power of s > p)

Thus the order is P > S > Mg > Al

**70.** (*a*) In group 1 Li has highest hydration enthalpy which has diagnoal relationship with group II element Mg.

Li — Be

Diagnol relationship

Na — Mg

**71.** (*c*)

(I) Rb and Cs

(III) Ar and Kr are nearly same electron gain enthalpy.

**72.** (*a*) Na > Mg > Be > Si > P

Across a period metallic character decreases.

**73.** (*c*) I.E. increases on going from L → R. with few exceptions such as Mg has more ionization energy than Al because of stable electronic configuration.

Na < Al < Mg < Si

496 737 786

So the value of I.E. of Al must lie in between 496 and 737 i.e. 577.

## JEE ADVANCE

**74.** (*b*) Due to ineffective shielding of d-orbitals in Galium its size will be less than Al.

Thus the order is Ga < Al < In < Tl

**75.** (*c,d*) NO ⇒ Neutral

$B_2O_3$ ⇒ Acidic

CrO ⇒ Basic

All other oxides are amphoteric.

**76.** [9]

| Atomic Number | Ionization Enthalpy (kJ/mol) | | |
|---|---|---|---|
| | $I_1$ | $I_2$ | $I_3$ |
| n | 1681 | 3374 | 6050 |
| n + 1 | 2081 | 3952 | 6122 |
| n + 2 | 496 | 4562 | 6910 |
| n + 3 | 738 | 1451 | 7733 |

By obseving the values of $I_1$, $I_2$ and $I_3$ for atomic number (n+2), it is observed that $I_2 >> I_1$.

This indicates that number of valence shell electrons is 1 and atomic number (n + 2) should be an

alkali metal.

Also for atomic number (n+3), $I_3 >> I_2$. This indicates that it will be an alkaline earth metal which suggests that atomic number (n+1) should be a noble gas and atomic number (n) should belong to halogen family. Since n<10 : Hence n = 9 (F atom)

**Note:** n = 1 (H atom) cannot be the answer because it does not have $I_2$ and $I_3$ values.

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CHAPTER

# 8 Chemical Bonding and Molecular Structure

## Exercise-1 (Topicwise)

1. (*d*) In $BCl_3$ Boron forms 3 bond having $6e^-$ & in $PCl_5$ phosphorous forms 5 bond i.e. having $10e^-$ so both violates octet rule.

2. (*d*) Hydrogen bond is weakest bond.

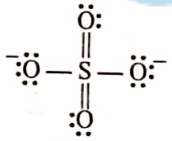
3. (*c*) $\ddot{N} = N = \ddot{N}$

   5-4-2 = −1 ; 5-4 = +1 ; 5-4-2 = −1

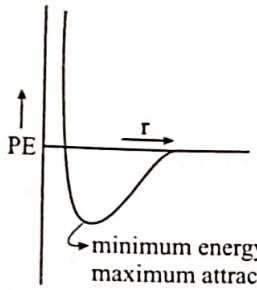
4. (*b*) $SO_4^{2-}$

   Total bond pair = $\frac{8\times5-32}{2} = 4$

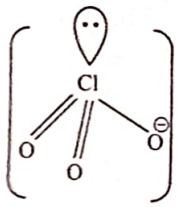
   3 lone pair each oxygen contain

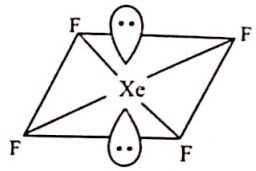


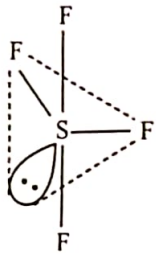
5. (*c*)

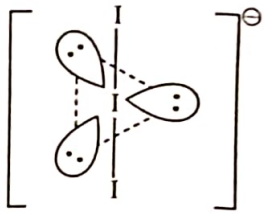



6. (*d*)

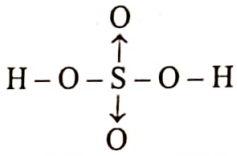



7. (*b*) $H_2SO_4$ has co-ordinate covalent bond.




8. (*d*) $HNO_2$ does not have co-ordinate bond. Structure is H – O – N = O.

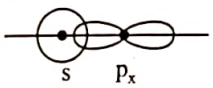
9. (*a*) $CH_3N \equiv C$ contain dative bond.

10. (*d*) $(C_2H_5)_3B \leftarrow :N(CH_3)_3$

    $BF_3 \leftarrow :NH_3$

11. (*d*) In $BF_3$ hybridisation charge from $sp^2$ to $sp^3$ when $NH_3$ donate lone pair acting as Lewis base.

12. (*b*) π-bond is formed by lateral overlapping of unhybridised p-p orbitals.

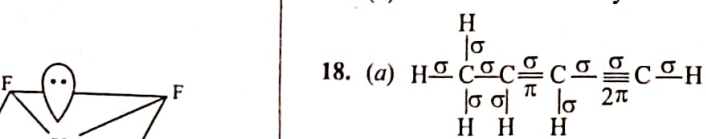
13. (*d*)  σ-bond


14. (*c*) Overlap order is p – p > s – p > s – s.

15. (*b*) $NH_3$ has $sp^3$ hybridised central atom so it is non planar.

16. (*c*) C ≡ C is a multiple bond so it is strongest.

17. (*b*) π-bond are formed by lateral overlap of orbitals.

18. (*a*) 

    σ-bond = 10

    π-bond = 3

19. (*d*) $H_2O$ is not linear because oxygen is $sp^3$ hybridised in $H_2O$

**20.** (*a*) $N \equiv C - C \equiv N$ Linear

**21.** (*b*) In T.B.P. $d_{z^2}$

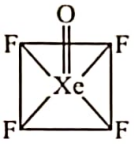
**22.** (*c*) $NH_4^+ \rightarrow sp^3 \rightarrow$ Tetrahedral
$H_3O^- \rightarrow sp^3 \rightarrow$ Pyramidal
$OF_2 \rightarrow sp^3 \rightarrow$ Bent

**23.** (*b*) $d_{x^2-y^2}$, $d_{z^2}$ involved in $d^2sp^3$.

**24.** (*d*)

| Molecule | Hybridisation | Shape |
|---|---|---|
| $SF_4$ | $sp^3d$ with one l.p. | See-Saw |
| $CF_4$ | $sp^3$ zero l.p. | Tetrahedral |
| $XeF_4$ | $sp^3d^2$ two l.p. | Square-planar |

**25.** (*d*)




**26.** (*d*) $Sp^3d^3$

**27.** (*c*) Fluorine does not form any polyhalide as other Halogens because of the absence of d-orbitals in the valence shell of fluorine

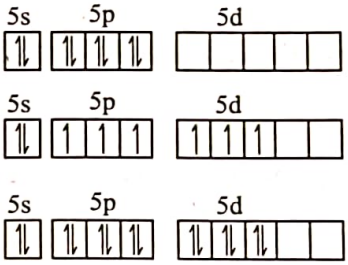
**28.** (*a*) If bond order is high then stronger bond is formed & bond length decreases and bond energy is high.

**29.** (*d*) Choice (*a*), (*b*), (*c*) are the resonance structures of $CO_2$

**30.** (*a*) Bond angle : $sp > sp^2 > sp^3 > sp^3d^2$

**31.** (*b*) Bond order $F_2 = 1$
$O_2 = 2$
$Cl_2 = 1$
$N_2 = 3$
In $Cl_2$ bond length is more than $F_2$ due to size.

**32.** (*b*) $XeF_6$ is distorted Octahedral. It has $sp^3d^3$ hybridisation with lone pair of electron on Xe, so its shape is distorted.

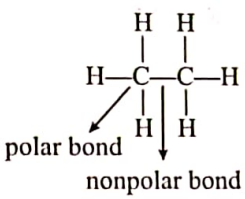



**33.** (*c*) NBEPR THEORY: Larger the size of side atoms larger will be bond angle.

**34.** (*c*) Xe double excitation $XeF_4$ $sp^3d^2$ - hybridization
2 lone pairs + 4 bond pairs = Square planar

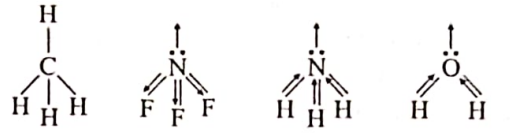
**35.** (*c*) $SO_4^{2-}$ has 32 electrons $CO_3^{2-}$ has 32 electrons; $NO^-$ has 32 electrons.

**36.** (*a*)



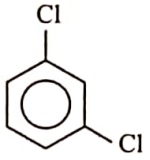

**37.** (*a*) Dipole moment $\propto \cos\theta \propto \frac{1}{\theta}$

**38.** (*a*)

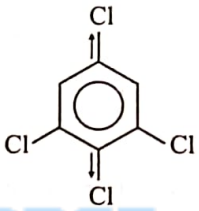



| $\mu = 0$ | Dipole of lp & N–F cancel each other $\mu \neq 0$ | Dipole of lp & bp add bond angle = 107° | Dipole of lp & bp add bond angle = 104.5° |
|---|---|---|---|

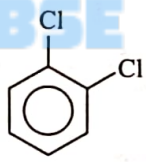
**39.** (*a*) $\mu = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$




$\cos\theta = 120°$ ; $\mu_1 = \mu_2$

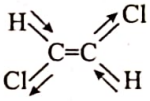



$\cos\theta = 120°$

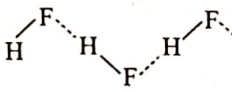



$\cos\theta = 60°$

**40.** (*c*)



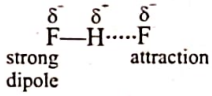

**41.** (*c*)



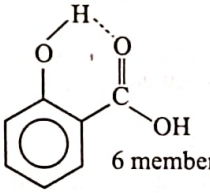

is correct as it will have least repulsion between F.

**42.** (*c*) In HF hydrogen bonding is present.

**43.** (*c*)




**44.** (*a*)




6 membered ring.

**45.** (*c*) Due to intermolecular H-bonding boiling point increases.

**46.** (*b*) $PH_3$ has the lowest boiling point because it does not form Hydrogen bond.

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47. (*b*) Hydrogen bonding increases heat of vaporisation.

48. (*b*) Benzene is non-polar.

49. (*d*) A discrete molecule is a covalent molecule in which the intermolecular forces are really weak, hence the low melting and boiling points of these molecular.

In dry ice intermolecular forces are weak.

50. (*d*) Glycerol has a three OH group hence it is viscous in nature.

51. (*a*) In $BF_3$ molecule back bonding is present hence slightly double bond character is generated.

52. (*c*) Nitrogen does not contain d-orbital so can not expend octet.

53. (*d*) Boron does not have d-orbital.

54. (*a*) Down the group both the lattic energy & hydration energy decreases.

55. (*c*) Valencies of L, Q, P and R is $-2$, $-1$, $+1$ and $+2$ respectively so they will form $P_2L$, RL, PQ and $RQ_2$.

56. (*b*) Due to its small size (fajan's rule)

57. (*d*) Valency of metalis is $+2$ by formula MO so its phosphate would be $M_3(PO_4)_2$ because valency of $[PO_4]$ is $-3$.

58. (*b*) Cation will be formed easily when IE is low.

Anion will be formed easily when EA is high.

59. (*c*) More is charge/size ratio of cation, more is polarisation & more is covalent character.

So $Be^{2+} \longrightarrow Sr^{2+}$ $\frac{\text{charge}}{\text{size}}$ ratio decrease.

60. (*a*) Cs is more electropositive.

61. (*d*) AgI is most covalent and Least soluble.

62. (*d*) Hexane has symmetrical structure so does not have polarity.

63. (*a*) According to Fajan's rule, polarisation of anion is influenced by charge and size of cation more is the charge on cation, more is polarisation of anion.

64. (*b*) Bond orders $O_2^+ = \frac{15-10}{2} = 2.5$

$$O_2 = \frac{16-10}{2} = 2$$

$$O_2^- = \frac{17-10}{2} = 1.5$$

$$O_2^{2-} = \frac{18-10}{2} = 1$$

65. (*c*) $O_2^-$ ($2 \times 8 + 1 = 17$) has odd number of electrons and hence it is paramagnetic. All the remaining molecules/ions, i.e.,

$CN^-$ ($6 + 7 + 1 = 14$) diamagnetic

NO ($7 + 8 = 15$) has odd number of electrons and hence it is paramagnetic.

66. (*a*) Due to s-p mixing.

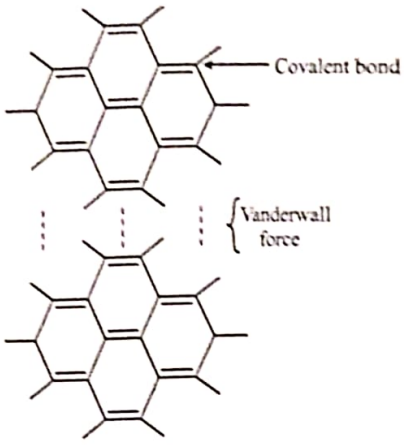
$$\pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi^*_{2p_y} = \pi^*_{2p_x} < \sigma^*_{2p_z}$$

## Exercise-2 (Learning Plus)

1. (*d*) The maximum covalency of representative element is equal to the number of s & p electrons in valence shell.

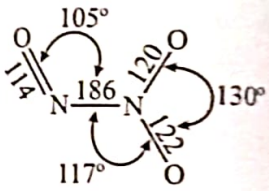
2. (*d*)

3. (*c*)

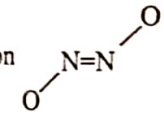



4. (*d*) In $SF_6$, $PCl_5$ and $IF_7$, the valence shell has 12, 10 and 14 electrons. As all contain more than 8 electrons in their valence shell they are example of super octet molecules.

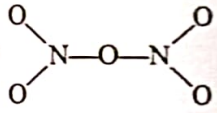
5. (*c*) (*a*) $N_2O_3$ Dinitrogen trioxide



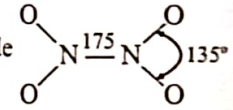

(*b*) $N_2O_2^{2-}$ Hyponitrite ion




(*c*) $N_2O_5$ Dinitrogen pentoxide



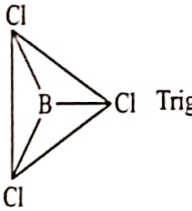

(*d*) $N_2O_4$ Dinitroen tetroxide




6. (*d*) The strength of metallic bonds depends upon the number of mobile electrons per atom. Sodium has only one mobile electron per atom where as iron has 8 mobile electrons per atom.

7. (*d*) (*a*) F–Be–F → Linear

(*b*)




Trigonal planar

(*c*) $\ddot{N}H_3$ Pyramidal

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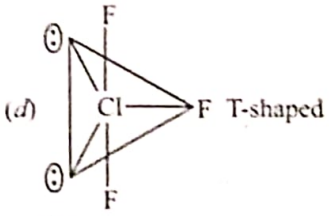
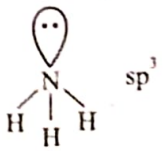
(*d*)  T-shaped

8. (*b*)  $sp^3$

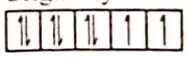
$[PtCl_4]^{2-}$

$\Rightarrow$ Pt is of 3rd transition series

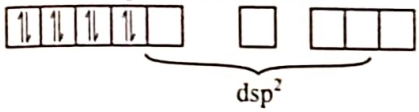
$\Rightarrow$ all ligand will be strong field ligand

$\Rightarrow$ Hybridization $\Rightarrow dsp^2$

Originally

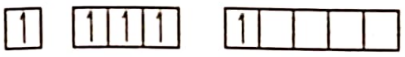


In this compound due to SFL.

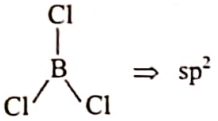


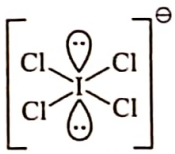
$dsp^2$

$PCl_5$ $\Rightarrow sp^3d$

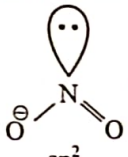


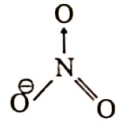
$sp^3d$

 $\Rightarrow sp^2$

9. (*b*) 

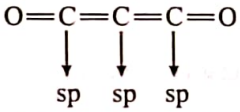
$\Rightarrow$ No of bond pair-lone pair repulsion = 8

10. (*c*) $O=\overset{+}{N}=O$ SP  $sp^2$



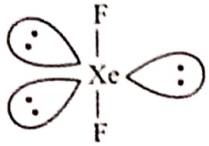
$(NO_2^+ > NO_3^- > NO_2^-)$

(As L.P. $\uparrow$, B.A. $\downarrow$)

11. (*a*) 

sp sp sp

12. (*a*) As the p-character in hybrid orbital increases the size of hybrid orbital increases.

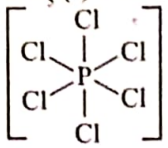
13. (*d*) (*i*) 

$sp^3d$

(*ii*) $[\overset{-}{N} = \overset{+}{N} = N]^{2-}$

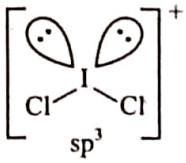
sp

(*iii*) $PCl_5$ (s) exists as $[PCl_4]^+$ and $[PCl_6]^-$,

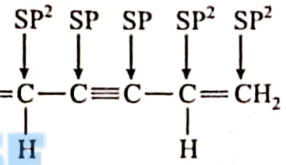


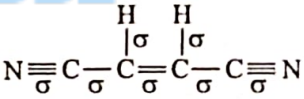
$sp^3d^2$

(*iv*) $ICl_6(\ell) \rightleftharpoons [ICl_2]^+ + [ICl_4]^-$ self ionisation



$sp^3$

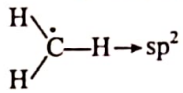
14. (*d*) (*a*) 

(*b*) 

(*c*) In diamond each carbon atom is in $sp^3$ hybridisation.

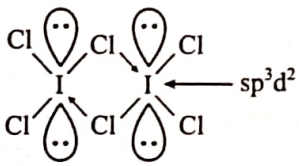
(*d*) O= C = C = C = O

sp sp sp

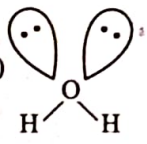
15. (*c*) 

Trigonal Planar

16. (*d*) $ICl_3$ does not exist, but the dimer $I_2Cl_6$ is a bright yellow solid. Its structure is planar.



**Note:** $I_2Cl_4Br_2$ will have the same hybridisation as that of iodine in $I_2Cl_6$. But it exists in 3 different forms.

17. (*b*) (*a*) 

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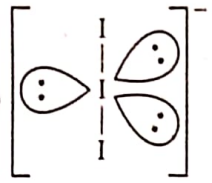
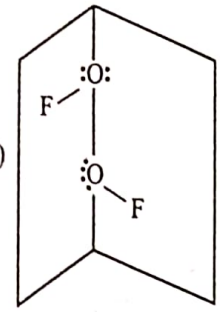
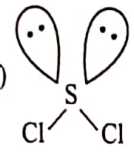
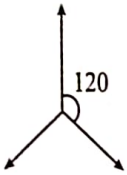
(b) 

Number of bond pairs around I = 2.

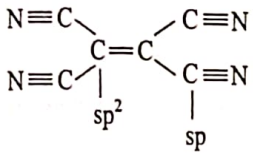
Number of lone pairs around I = 2.

(c) 

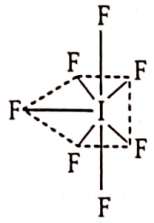
(d) 

18. (c) 

$\mu = 0$ e.g., $BF_3$

19. (c) 

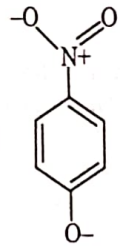
$\Rightarrow$ sp and $sp^2$ hybridized

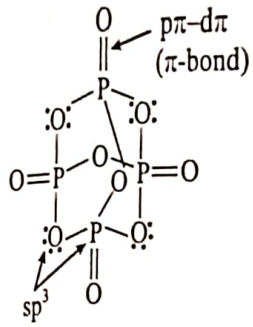
20. (b) 

no. of B.P. =7

21. (d) $\overset{\ominus}{C} \equiv N \leftrightarrow C \equiv \overset{\ominus}{N}$

22. (b) As E.N. of central atom ↑ B.A.↑

23. (a)  most stable.

24. (d) $P_4O_{10}$ 

The P – O bond lengths shows that the bridging bonds on the edges are 1.60 Å but the P = O bonds on the corners are 1.43 Å and this P = O is formed by $p\pi - d\pi$ back bonding. A full p-orbital on the oxygen atom overlaps sideway with an empty d-orbital on the phosphorus atom. The bond angle POP is 127° and there is no P – P bonds.

25. (a) Atomic size arguments can be used for thee species. Larger outer atoms result in larger angles due to steric repulsion.

26. (a) $\Rightarrow$ more s, more EN

27. (a) Polarity order

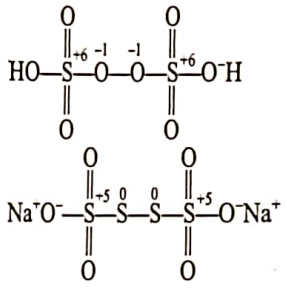
C – F > C – Cl

Distance (Bond) C–Cl > C–F

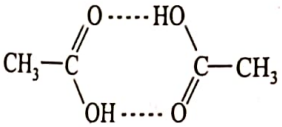
*Here as exception,distance factor is dominant

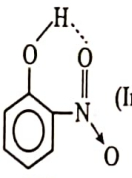
$CH_3$–Cl $CH_3$–F >$CH_3$–Br or >$CH_3$–I

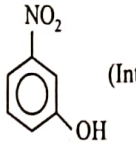
distance factor polarity factor dominant

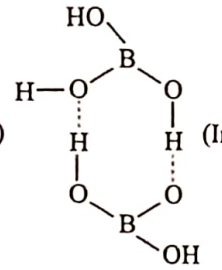
28. (a) 

29. (c)

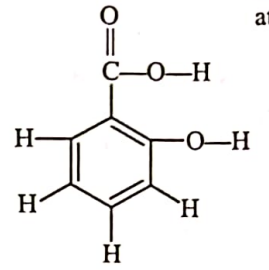
(I)  (Intermolecular H-bonding)

(II)  (Intramolecular H-bonding)

(III)  (Intermolecular H-bonding)

(IV)  (Intermolecular H-bonding)

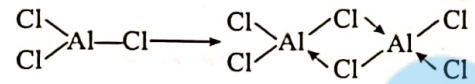
**30.** (*b*) Strength of H-bond $\propto$ Polarity of H-atom / EN of central atom attach to the H-atom

**31.** (*b*) 

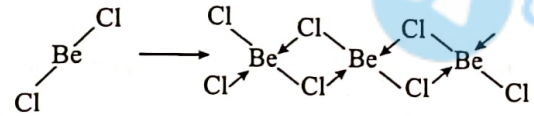
$\sigma$ bond = 16

$\pi$ bond = 4

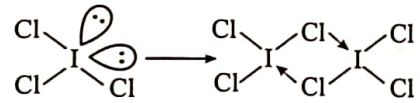
**32.** (*c*)

(*a*) 

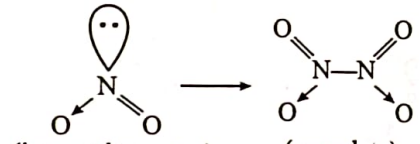
(Incomplete octet) (Complete)

(*b*) 

(Incomplete octet) (Complete octet)

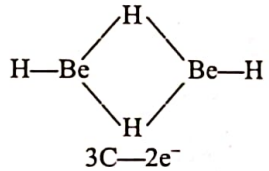
(*c*) 

(complete) (complete)

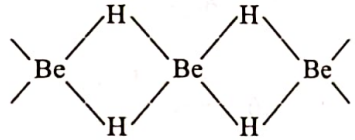
(*d*) 

(incomplete octet) (complete)

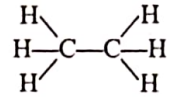
**33.** (*c*)

(*a*) 

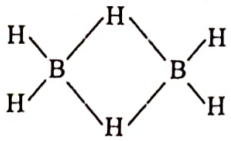
3C—2e⁻

(*b*) 

(3C—2e⁻)

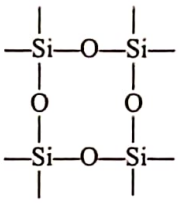
(*c*) 

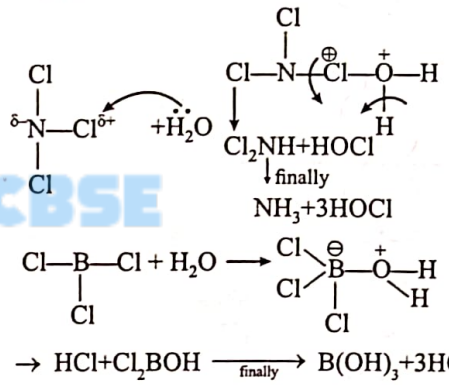
(does not contain 3C–2e⁻)

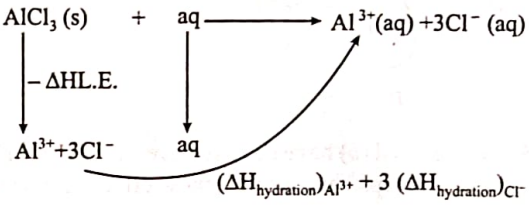
(*d*) 

(3C–2e⁻)

**34.** (*a*) The central atom of $PCl_5$ have vacant d orbital therefore represent extended covalent bonding while in $NCl_5$ N have no vacant d orbital

**35.** (*b*) 

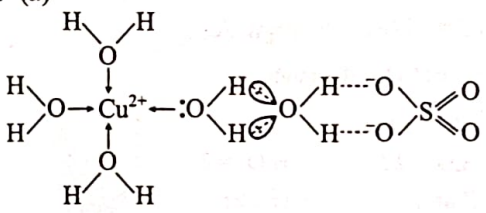
**36.** (*c*) 

**37.** (*c*) 

$\Rightarrow \Delta H_{solution} = -4665 - 3 \times 381 + 5137 < 0$

Hence $AlCl_3$ will dissolve and solution consists of hydrated $Al^{3+}$ and $Cl^-$ ions.

**38.** (*b*) $Sn^{4+}$ has highest polarising power amongst $Na^+$, $Pb^{2+}$, $Sn^{4+}$ and $Al^{3+}$ because of smaller size and higher charge. So $SnCl_4$ is most covalent and thus have least melting point.

**39.** (*d*) 

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**40.** (*a*) $C_3^{4-} \Rightarrow {}^{3-}C - C \equiv C^-$

$= \sigma$ – bond = 2

$\pi$ – bond = 2

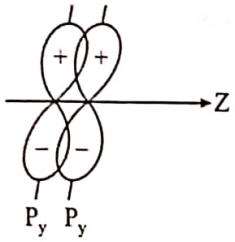
**41.** (*b*) According to Fajan's rule

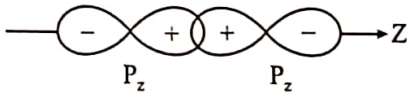
**42.** (*d*) Bond length $\propto \frac{1}{\text{bond order}}$

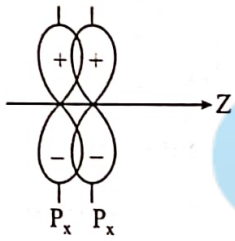
$CO < CO_2 < CO_3^{2-}$

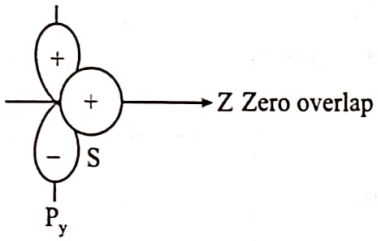
**43.** (*a*) $\overset{\oplus}{\ddot{N}} \equiv \ddot{O}$ $\quad \ddot{C} \equiv \ddot{O}$ $\quad \ddot{C} \equiv \ddot{N}^{\ominus}$ .

**44.** (*d*)

(A) 


(B) 


(C) 


(D) 


**45.** (*c*) (*a*) and (*b*) have negative overlap while (*c*) has positive overlap. Thus (*c*) will show effective overlapping.

**46.** (*b*) (OF)

$$B.O = \frac{N_b - N_a}{2}$$

$$B.O = \frac{10-7}{2} = 1.5$$

**47.** (*a*) (*a*) $N\bar{O}$ $(16\bar{e})$ = paramagnetic

(*b*) $O_2^{2-}$ $(18\bar{e})$ = diamagnetic

(*c*) $(C\bar{N})$ $(14\bar{e})$ = diamagnetic

(*d*) $(14\bar{e})$ CO = diamagnetic

**48.** (*d*) (*a*) $NO \rightarrow \overset{+}{N}O$

B.O = 2.5 (Para) $\quad$ B.O. = 2 (Para)

(*b*) $O_2^- \rightarrow O_2$

unpaired e = 1

B.O = 2.5 (Para) $\quad$ B.O = 2 (Para)

(*c*) $N_2 \rightarrow \overset{\ominus}{N}_2$

B.O = 3 (dia) $\quad$ B.O. = 2.5 (para)

**49.** (*d*) M.O for $C_2 = \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 <$ $\underbrace{\pi 2p_x^2 = \pi 2p_y^2}_{\text{HOMO}} < \underbrace{\sigma 2p_z}_{\text{LUMO}}$

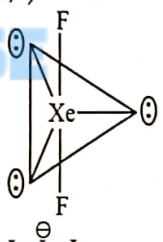
I is important to not that double bond in $C_2$ consists of both pi bonds because of the presence of four electrons in $2\pi$ molecular orbitals $C_2^{2-}$ $[C = C]^{2-}$ .

**50.** (*d*) $O_2^{2+}$ B.O. = 3 $\quad$ $O_2^{2-}$ B.O. = 1

$F_2^{2-}$ B.O. = 0 $\quad$ $H_2^-$ B.O. = 0.5

## Exercise-3 (JEE Advanced Level)

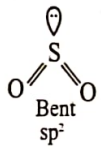
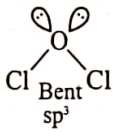
**1.** (*a,b,c,d*)

(*a*) 


$O = \underset{sp}{C} = O$

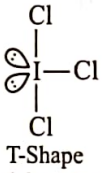
(*b*) $I - \overset{\ominus}{I} - I$ ($sp^3d$, linear) $\quad$ Cl—Hg—Cl (sp linear)

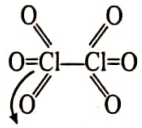
(*c*) $\overset{\ominus}{O}-Cl$ ($sp^3$) $\quad$ $\bar{C} \equiv \overset{+}{O}$ Linear, sp

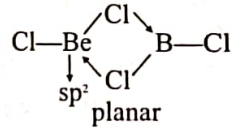
(*d*)  Bent, $sp^2$ $\quad$  Bent, $sp^3$

**2.** (*a,b,d*)

(*a*) $sp^3d$, planer

(*b*)  T-Shape (planer)

(*c*)  $sp^3$, Tetrahedral non planar

(*d*)  $sp^2$ planar

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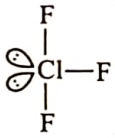
**3.** (*b*) $XeF_6 \longrightarrow [XeF_5]^+F^-$
$sp^3d^2$

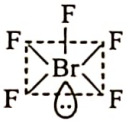
**4.** (*a,c*)

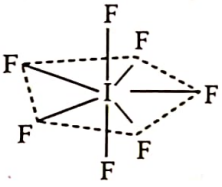
It is the orbital that undergo hybridisation and not the electrons. For example , for orbitals of nitrogen atom ($2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$) belonging to valence shell when hybridise to form four hybrid orbitals, one of which has two electrons (as before) and other three have one electron each. It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

**5.** (*a,b,c,d*)

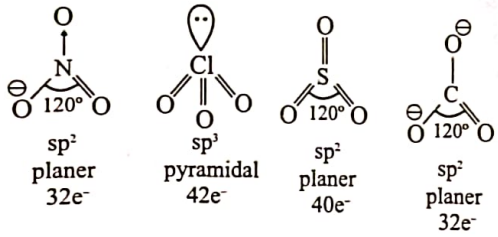
(*a*) I – Cl, linear and polar because of the difference in the electronegativities of iodine and chlorine.

(*b*)  $\mu \neq 0$.

(*c*)  $\mu \neq 0$.

(*d*)  $\mu = 0$.

**6.** (*a*)



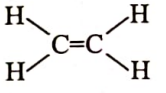

**7.** (*a,b,d*)

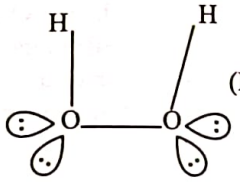
(According to VBT theory)

**8.** (*a,b*)

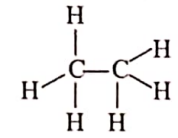
**9.** (*a*) $CH_4 \xrightarrow{\text{Combustion}} CO_2$
$sp^3$ $sp$

**10.** (*a,c*)

(*a*)  no rotation

(*b*)  (Rotation possible)

(*c*) H – C ≡ C – H (no rotation)

(*d*) 
(Rotation possible)

**11.** (*a,b,d*)

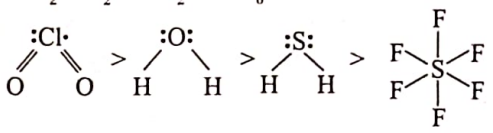
(*a*) According to VSEPR theory as electronegativity of central atom decreases, bond angle decreases. So bond angle of $H_2O > H_2S > H_2Se > H_2Te$

(*b*) $C_2H_2 > C_2H_4 > CH_4 > NH_3$. In $NH_3$ there is bp-lp repulsion so bond angle decreases to 107° from 109.5°.

| | |
|---|---|
| sp | $sp^2$ |
| 180° | 120° |
| $sp^3$ | $sp^3$ |
| 109.5° | 107° |

(*c*) $SF_6 < NH_3 < H_2O < OF_2$ in this case bond angle of $NH_3$ is highest because lp - lp repulsion is absent in it.

(*d*) $ClO_2 > H_2O > H_2S > SF_6$



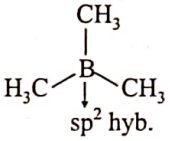

$sp$ ($\approx$117°) $sp^3$ (104.5°) $sp^3$ (92°) $sp^3d^2$ (90°)

**Note:** It is supposed that in $H_2S$ the hybrid orbitals do not participate in bonding but pure p-ayomic orbitals participate in bonding.

**12.** (*b,c*) $\bar{N}{=}\overset{+}{N}{=}O \longleftrightarrow N{\equiv}\overset{+}{N}{-}O^- \longleftrightarrow \bar{N}{-}\overset{+}{N}{\equiv}O^+$

**13.** (*a,b*)

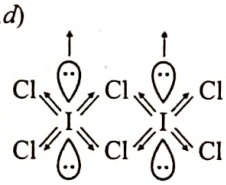
(*a*) Due to the presence of lone pair bond angle decreases.

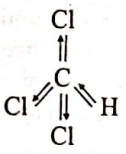
(*b*) 


(*c*) $NH_4Cl$ is an ionic compound and 'N' is in $sp^3$ hybridisation.

(*d*) $S_8$ molecule has 16 electron pairs left behind after the bonding.

**14.** (*a,c,d*)

(*a*) 
Square plane
$\mu = 0$

(*b*)  $\mu \neq 0$

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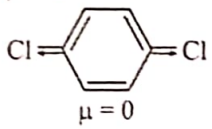
(c) O = C = O

$\mu = 0$

(d) 

$\mu = 0$

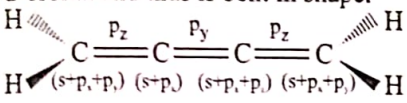
$\Rightarrow$ **Ans.** : (a,c,d)

**15.** (a,c,d)

(a) Ethyne being non-polar is more soluble in acetone.

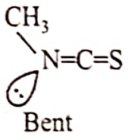
(b) $CD_3F$ is more polar than $CH_3F$. In $CD_3F$, C–D bond is more polar as deuterium is more electropositive than hydrogen.

(c) Silyl isocyanate ($SiH_3NCO$) is linear in shape because of $p\pi$-$d\pi$ delocalisation of lone pair of electrons on nitrogen. While in methyl isocyanate ($CH_3NO$) there is no $p\pi$-$d\pi$ delocaliation of electron as carbon does not have empty d-orbital and thus is bent in shape.

(d) 

In $CH_2=C=C=C=CH_2$, the 2H–atoms on one C–atom lie in a plane perpendicular to the plane in which 2H–atoms on other C–atom lie.

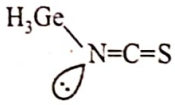
**16.** (b,c,d)

(a) 

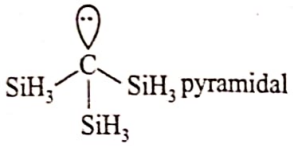
Bent

(b) $H_3Si \leftarrow N = C = S$

Linear due to Back bond

(c) 

Bent as no effective Back Bonding due to large size of Ge

(d) 

$SiH_3$ pyramidal

**17.** (c) The Fluoride ion, by process of hydrogen bonding forms the onion $HF_2^-$.

The compound is written as $K^+[HF_2]^-$.

**18.** (a) Lattice energy is directly proportional to the charge of the ions and inversely proportional to the size of the ions. Thus lattice energy increases as the size of onion decreases.

Lattice energy order: KF > KCl > KBr > KI

**19.** (a,b,c)

(a) With hydrogen sulphur does not undergo $sp^3d^2$ hybridisation because of larger difference in energies between s, p and d-orbitals. Sulphur show +6 oxidation state with highly electronegative elements like O and F.

(b) As fluorine is smaller and more electronegative than oxygen.

(c) $I^-$ being stronger reducing agent reduces $Fe^{3+}$ to $Fe^{2+}$

**20.** (a,d)

$Ba^{2+}$ is larger than $Mg^{2+}$. According to Fajan's rule smaller cation has higher polarising power and therefore, MgO is more covalent than BaO (i.e. BaO is more ionic).

**Melting Points :** BeO = 2500°C; MgO = 2800°C; CaO = 2572°C ; BaO = 1923°C

**21.** (b,c,d)

(a) is incorrect, as ionic compounds have higher melting points as well as higher boiling points due to strong electrostatic force of attraction between the ions. (b), (c) and (d) are properties of ionic compounds.

**22.** (b,c,d)

$\because$ A does not conduct electricity in aq. molten or solid

$\Rightarrow$ it must be non polar covalent comp.

$\Rightarrow$ B only conduct electricity on fused form or in aq. solution hence it must be ionic.

$\Rightarrow$ C only conduct electricity in aq. solution hence

**23.** (a,b,c)

(a) $O_2^{2-}$ (14$\bar{e}$) B.O. = 3

(b) $NO^-$ (14e) B.O. = 3

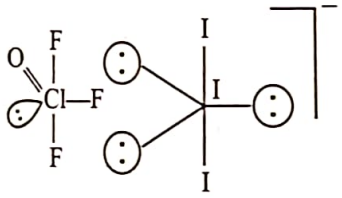
(c) $C\bar{N}$(14e) B.O. = 3

(d) $\overset{\ominus}{C}N$(12$\bar{e}$) B.O. = 2

**24.** (b,d)

| $N_2$ | CO | $C\bar{N}$ | $NO^-$ |
|---|---|---|---|
| (14) | (14e) | (14e) | (14e) |
| ↓ | ↓ | ↓ | ↓ |
| Diamagnetic | Diamagnetic | Diamagnetic | Diamagnetic |
| $N \equiv N$ | $\bar{C} \equiv O^+$ | $\bar{C} \equiv N$ | $N \equiv O^-$ |
| ($\mu \equiv 0$) | $\mu \neq 0$ | $\mu \neq 0$ | $\mu \neq 0$ |
| Linear | Linear | Linear | Linear |

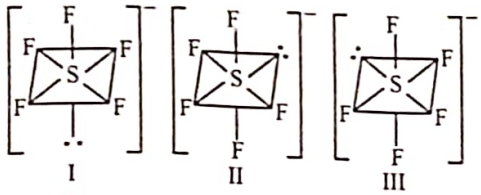
**25.** (c)



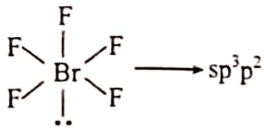
Cl-atom is in $sp^3d$ hybridisation state. Hence geometry is trigonal bi-pyramidal which is similar to $I_3^-$

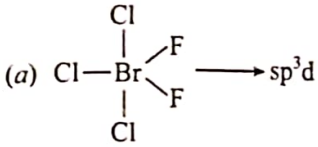
**26.** (d) Number of electrons pairs = 6; number of bond pairs = 5; number of lone pairs = 1. According to VSEPR theory geometry of the molecule is trigonal bipyramidal. As all positions are equivalent the lone pair of electrons can occupy any position in octahedral geometry as given below.

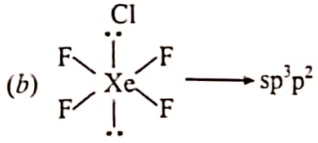
@PROCBSE

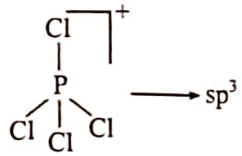


I II III

**27.** (*d*)  $sp^3p^2$

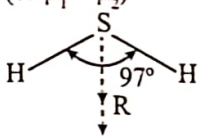
(*a*)  $sp^3d$

(*b*)  $sp^3p^2$

(*c*)  $sp^3$

**28.** (*b*) $\mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$

($\because \mu_1 = \mu_2$)



$\mu_R = \sqrt{2\mu_{S-H}^2 + 2\mu_{S-H}^2 \times \cos 97°}$

$\mu_R = \sqrt{2} \times \mu_{S-H} \times \sqrt{1-0.12}$

$\mu_R = \sqrt{2} \times \mu_{S-H} \times \sqrt{0.88}$

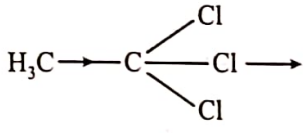
$\mu_{S-H} = \dfrac{1.5}{\sqrt{2}\times 0.94} D$

$(\mu_{S-H})_{Cal} = \dfrac{0.15\times10^{-9}\times1.6\times10^{-19}}{\frac{10}{3}\times10^{-30}}$

$(\mu_{S-H})_{Cal} = 3 \times 0.15 \times 1.6 \times 10\ D$

So % Ionic character $= \dfrac{1.5}{\sqrt{2}\times0.94} \times \left[\dfrac{1}{0.15\times16\times3}\right] \times 100$

$= 16\ \%$

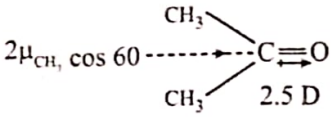
**29.** (*d*) 

I

$\mu_I = 3\ \mu_{C-Cl} \times \cos 70.5 + \mu_{CH3} = 1.5 + 0.4 = 1.9\ D$

$\mu_{II} = 3\ \mu_{C-Cl} \times \cos 70.5 - \mu_{C-H} = 1.5 - 0.4 = 1.1\ D$

$\mu_{III} = 3\ \mu_{C-H} \times \cos 70.5 + \mu_{C-Cl} = 1.9\ D$

**30.** (*a*) 

$\mu_{net} = 2.5 + \mu_{CH_3} = 2.5 + 0.4 = 2.9\ D.$

**31.** (*b*)

| $sp^3d$ | Hydridisation | | Geometry | T.B.P. |
|---|---|---|---|---|
| When | 0 l.p. | 5 b.p. | Shape | T.B.P. |
| When | 1 l.p. | 4 b.p. | Shape | See-Saw |
| When | 2 l.p. | 3 b.p. | Shape | T-Shape |
| When | 3 l.p. | 2 b.p. | Shape | Linear |

**32.** (*b*) $sp^3d^2$ (s, $p_x$, $p_y$, $p_z$, $d_{x^2-y^2}$, $d_{z^2}$)

**33.** (*c*) $sp^3d^2$ ($d_{x^2-y^2}$ and $d_{z^2}$)

**34.** (*b*) A→ p; B→ q; C→ p; D→ s

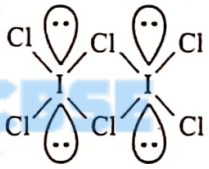
Steric no. of $IO_2F_2^-$ = 4 + 1 = 5, $sp^3d$;

Steric no. of $F_2SeO$ = 3 + 1 = 4, $sp^3$;

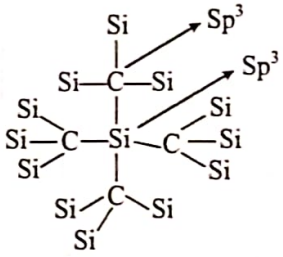
Steric no. of $ClOF_3$ = 4 + 1 = 5, $sp^3d$;

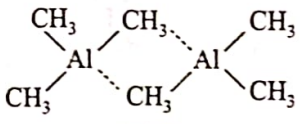
Steric no. of $XeF_5^+$ = 5 + 1 = 6, $sp^3d^2$

**35.** (*a*) A→ r; B→ p; C→ s; D→ q

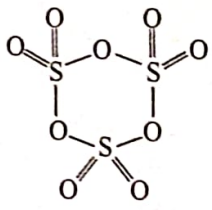
(A) 

$sp^3p^2$

(B) 

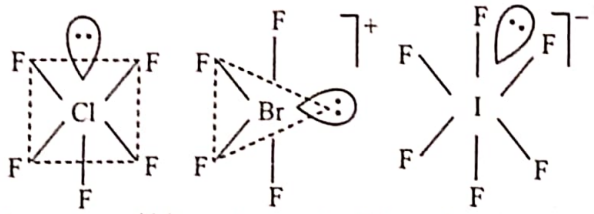
(C) 

3C–2e

(D) 

**36.** (*c*) A→ p,r; B → r; C→ q,s; D→ p,q,r,s

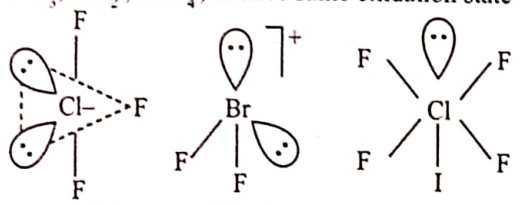
(A) $ClF_5$, $BrF_4^+$, $IF_6^-$; all have same oxidation state (+ 5)



(square pyramidal) (see-saw) (distorted octahedral)

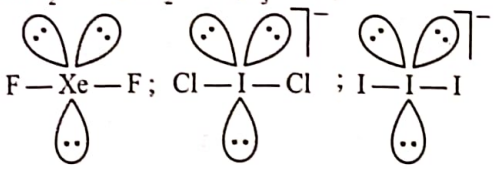
All have one lone pair of electrons each ; but different shapes; $\mu \neq 0$ so polar.

(B) $ClF_3$, $BrF_2^+$, $ICl_4^-$ ;all have same oxidation state (+3);



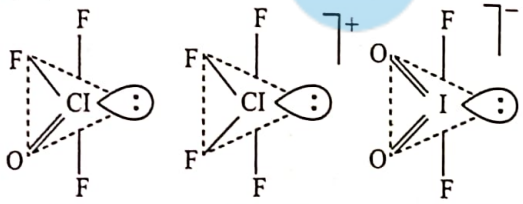
$\mu \neq 0$ (polar) (T-shaped) $\mu \neq 0$ (polar) (bent) $\mu = 0$ (polar) (square planar)

(C) $XeF_2 = +2$ ; $ICl_2^- = +1$ ; $I_3^- = +1$ ;



All have three lone pairs each and same shape but different oxidation state. In all $\mu = 0$; so non - polar

(D) $ClOF_3$ , $ClF_4^+$, $IO_2F_2^-$; All have same oxidation number (+5)

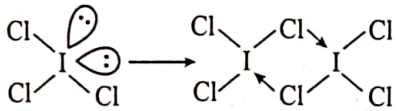


seesaw see - saw see - saw

In all $\mu \neq 0$ , so all polar.

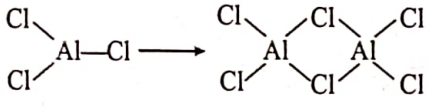
37. (*d*) A→ q, B→ r, C→ s, D→ p

(A)



$sp^3d$ planar $sp^3d^2$ planar

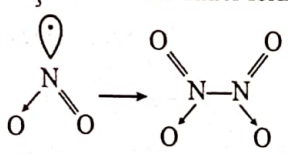
(B)



$sp^2$ planar $sp^3$nonplanar

(C) $AlF_3$ no dimer form

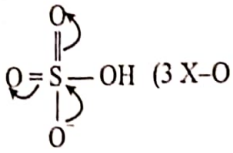
(D)



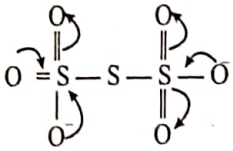
$sp^2$ planar $sp^3$ nonplanar

38. [4.5] Bond orders: $O_2 = 2, O_2^- = 1.5, O_2^{2-} = 1 \Rightarrow 4.5$

39. [36]

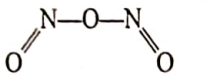


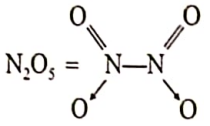
(3 X–O bond length are same)



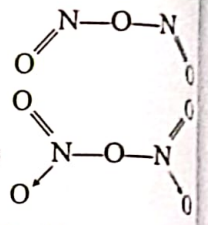
(6–X–O bond length are same)

40. [2] The structure of $N_2O_3$ =





While that of $N_2O_5$



Hence $N_2O_3$ and $N_2O_5$ contain N – O – N bond while in $N_2O_4$ does not.

Hence ans. = 2

## Exercise-4 (Past Year Questions)

### JEE MAIN

1. (*b*) There is extensive intermolecular hydrogen bonding in the condensed phase.

2. (*a*) For $NO_2^+$

$$x = \frac{1}{2}[V + G - C + A]$$
$$= \frac{1}{2}[5 + 0 - 1 + 0]$$
$$= 2 \text{ sp}$$

3. (*a*) In the reaction

$$\overset{+4}{Xe}F_4 + \overset{+1}{O_2}F_2 \rightarrow \overset{+6}{Xe}F_6 + O_2^0$$

Xenon undergoes oxidation while oxygen undergoes reduction.

4. (*b*) NO ⇒ One unpaired electron is present in $\pi^*$ molecular orbital

CO ⇒ No unpaired electron is present.

$O_2$ ⇒ Two unpaired electrons are present in $\pi^*$ molecular orbitals.

$B_2$ ⇒ Two unpaired electrons is present in $\pi$ bonding molecuar orbitals.

**5.** (*c*)

(*a*) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Cr_2O_3 + N_2 + 4H_2O$

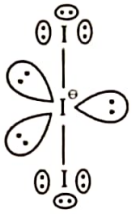
(*b*) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

(*c*) $(NH)_2SO_4 \xrightarrow{\Delta} 2NH_3 + H_2SO_4$

(*d*) $Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$

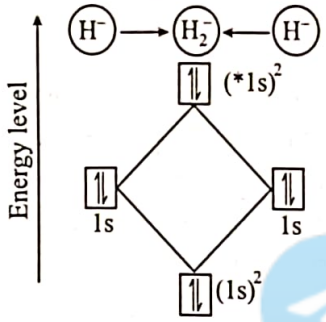
In reaction (3) $NH_3$ is evolved where as in reaction 1,2 and 4 $N_2$ is evolved.

**6.** (*b*)



Total number of lone pairs in $I_3^-$ is 9.

**7.** (*c*) The electronic configuration of $H_2^{2-}$ is $(\sigma 1s^2),(\sigma^* 1s^2)$



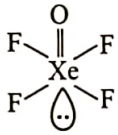

Bond order of $H_2^{2-} = \frac{N_b - N_a}{2} = \frac{2-2}{2} = 0$

Hence $H_2^{2-}$ does not exist, due to zero bond order.

**8.** (*b*) $PH_3$, $P_2$, $B_2H_6$, $H_2SO_4$ all are covalent compounds

KCl is ionic compound.

**9.** (*a*)



**10.** (*d*)

$N_2^{\oplus} \Rightarrow BO = 2.5 \Rightarrow \left[\pi - Bond = 2 \,\&\, \sigma - Bond = \frac{1}{2}\right]$

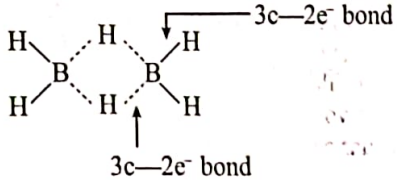
$N_2 \Rightarrow BO = 3.0 \Rightarrow [\pi - Bond = 2 \,\&\, \sigma - Bond = 1]$

$O_2^{\oplus} = B.O. \Rightarrow 2.5 \Rightarrow [\pi - Bond = 1.5 \,\&\, \sigma - Bond = 1]$

$O_2 \Rightarrow B.O. \Rightarrow 2.5 \Rightarrow [\pi - Bond \Rightarrow 1 \,\&\, \sigma - Bond = 1]$

**11.** (*b*) It is due to the smallest atomic size of carbon in the given options. Small size greater pπ-pπ multiple bonds

**12.** (*b*)




**13.** (*a*)

| | | | | | |
|---|---|---|---|---|---|
| | $NO \rightarrow NO^+$ | | | $N_2 \rightarrow N_2^+$ | |
| B.O | 2.5 | 3 | B.O | 3 | 2.5 |
| | Para | Dia | | Dia | Para |
| | $O_2 \rightarrow O_2^+$ | | | $O_2 \rightarrow O_2^{2-}$ | |
| B.O | 2 | 2.5 | B.O | 2 | 1 |
| | Para | Para | | Para | Dia |

**14.** (*b*) Both $Li_2^+$ and $Li_2^-$ have same bond order. But the number of antibonding electrons is less in $Li_2^+$ than in $Li_2^-$

**15.** (*b*) In case of only $C_2$, incoming electron will enter in the bonding molecular orbital which increases the bond order and stability too. Whereas rest of all takes electron in their antibonding molecular orbital which decreases bond order and stability.

**16.** (*c*)

| Chemical species | Hybridisation of central atom |
|---|---|
| $ICl_2^-$ | $sp^3d$ |
| $IF_6^-$ | $sp^3d^3$ |
| $ICl_4^-$ | $sp^3d^2$ |
| $BrF_2^-$ | $sp^3d$ |

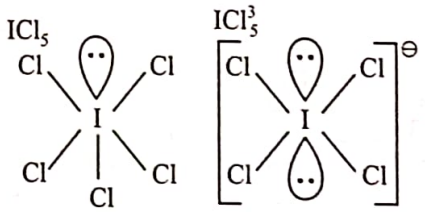
**17.** (*a*)

| Chemical species | Bond order | Magnetic behaviour |
|---|---|---|
| $C_2^{2-}$ | 3 | diamagnetic |
| $N_2^{2-}$ | 2 | paramagnetic |
| $O_2$ | 2 | paramagnetic |
| $O_2^{2-}$ | 1 | diamagnetic |

B.O. $\propto \frac{1}{\text{bond length}}$

**18.** (*c*)

| Chemical species | Hybridisation | Shape |
|---|---|---|
| $ICl_5$ | $sp^3d^2$ | Square pyramidal |
| $ICl_4^-$ | $sp^3d^2$ | Square planar |




**19.** (*c*) In diamod C–C bond have only bond character while in case of graphite and fullerence ($C_{60}$ and $C_{70}$) C–C bond contain double bond character. That's why diamond having maximum C–C bond length.

**20.** (*d*) $O_2$,NO,$B_2$ are paramagnetic according to M.O.T where as CO is diamagnetic.

**21.** (*a*) $BeCl_2$ exist as $(BeCl_2)_n$ polymeric chain in solid form, while $BeCl_2$ exist as dimer $(BeCl_2)_2$ in vapour phase.

**22.** (*c*) HF has highest boiling point among hydrogen halides because it has strongest hydrogen bonding

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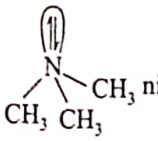
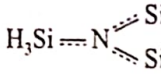
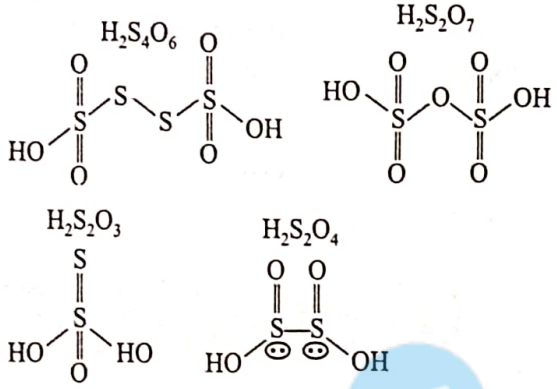
**23.** (*c*) Based on NCERT, statement of limitations of VBT, I & III are correct

**24.** (*d*)

(*a*)  nitrogen is $sp^3$ hybrid and pyramidal no back-bonding i.e. more basic

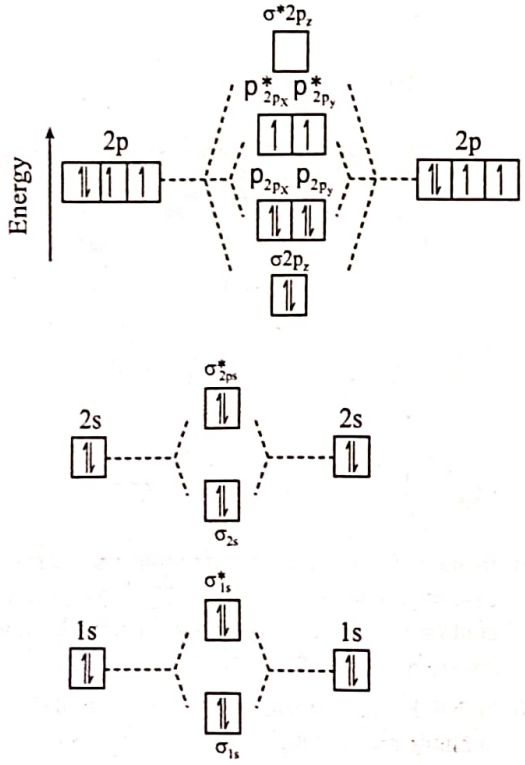
(*b*)  N Nitrogen $sp^2$ hybrid and planar due to back bonding and less basic because lone pair is not available for donation.

**25.** (*b*)



$H_2S_2O_7$ does not contain bond between sulphur atoms.

**26.** (*c*) Molecular orbital diagram of $O_2$ is



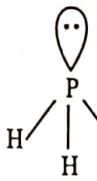
An incoming electron will go in $\pi^*2Px$ orbital.

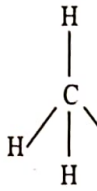
**27.** (*c*)

(*a*) $\overset{\oplus}{M}\overset{\ominus}{H} + HOH \longrightarrow MOH + H_2$

Ionic hydride/saline hydride

(*b*) $4BF_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiF + 3AlF_3$

(*c*)  phosphorous is electron rich hydride due to presence of lone pair

 it is electro precise hydride.

(*d*) HF & $CH_4$ are molecular hydride due to they are covalent molecules.

**28.** (*b*) Relative strength:

Ion-ion > ion-dipole > dipole-dipole

**29.** (*d*) $\mu_{CCl_4} = \mu_{CH_4} = 0$ but $\mu_{CHCl_3} \neq 0$

**30.** (*c*) $CN^-$ is a 14 electron system.

**31.** (*c*) Ethyl acetate is polar molecular so dipole-dipole interaction will be present in it.

**32.** (*b*) Bond energy $\propto \frac{1}{\text{Bond length}}$

**33.** (*a*)

$O_2 = \sigma ls^2\ \sigma^* ls^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$

$O_2^- = \sigma ls^2\ \sigma^* ls^2\ \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_y^2 = \pi^* 2p_y^1$

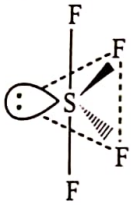
$O_2^+ = \sigma ls^2\ \sigma^* ls^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^0$

**34.** (*b*) $SF_4$ Bond pair = 4

Lone pair = 1

Steric number = 5

Hybridisation - $sp^3d$



Geometry → Trigonal bipyramidal

Shape → See saw

**35.** (*d*) Ion-ion interaction energy $\propto \frac{1}{r}$

Dipole-dipole interaction energy $\propto \frac{1}{r^3}$

London dispersion $\propto \frac{1}{r^6}$

**36.** (*c*)

| Compound | Shape | Hybridization | lp | Polarity |
|---|---|---|---|---|
| $AB_4$ | Tetrahedral | $sp^3$ | 0 | Nonpolar |
| $AB_4$ | Rectangular planar | $sp^3d^2$ | 2 | Nonpolar |
| $AB_4$ | Square pyramidal | $sp^3d$ | 1 | polar |
| $AB_4$ | Square planar | $sp^3d^2$ | 2 | Nonpolar |

**37.** (*a*) Species with minimum bond order will have minimum bond strength.

$NO^-$ has maximum $e^-$ in anti-bonding orbitals hence will have minimum bond strength.

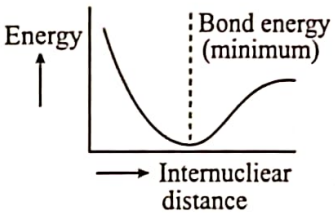
**38.** (*a*) The bond which has greater potential energy (more negative) is considered more stable as it requires more energy to dissociate.

$\because$ A–B bond has most negative potential energy hence it is strongest bond and has maximum bond enthalpy. A–D is longest bond.

**39.** (*b*) Order of bond angle is

$CH_4 > NH_3 > H_2O > H_2S$

**40.** (*c*)




**41.** [173]

$B_2^+ \Rightarrow \sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\pi_{2py}^1 = \pi_{2pz}^0$

It has one unpaired electron.

Spin - only magnetic moment $= \mu$

$= \sqrt{n(n+1)}$

n = Number of unpaired electrons

$= \sqrt{1(1+2)} = \sqrt{3}BM$

$= 1.73BM$

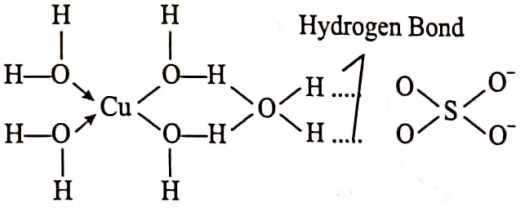
$= 173 \times 10^{-2}BM$

**42.** [0] Molecular orbital configuration of $O_2^{2-}$ is

$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2pz}^2\left(\pi_{2px}^2 = \pi_{2py}^2\right)\left(\pi_{2px}^{*2} = \pi_{2py}^{*2}\right)$

Zero unpaired electron.

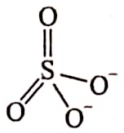
**43.** [1]



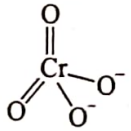

Copper sulphate has a total of five water molecules in its structure. Four of the five water molecules found in the crystal structure of the copper sulphate molecule are coordinated to $Cu^{2+}$, while one is hydrogen linked to $SO_4^{2-}$. One Hydrogen bond $H_2O$ molecule.

**44.** [3] As we know that non-pyramidal species are $SO_3$, $NO_3^-$ and $CO_3^{2-}$

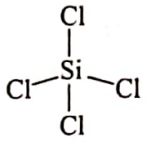
**45.** (*b*) $SO_4^{2-} \Rightarrow$



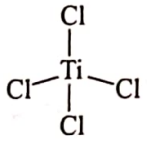
$CrO_4^{2-} \Rightarrow$



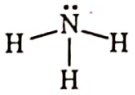
$SiCl_4 \Rightarrow$



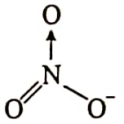
$TiCl_4 \Rightarrow$



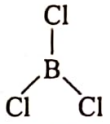
$NH_3 \Rightarrow$



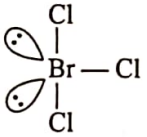
$NO_3^- \Rightarrow$



$BCl_3 \Rightarrow$



$BrCl_3 \Rightarrow$



**46.** (*d*) $[:\ddot{I}—\ddot{I}—\ddot{I}:]^-$ Linear Shape.

**47.** (*a*) Higher the covalent character lower will be melting point.

So LiF > LiCl; MgO > NaCl

**48.** (*c*) $Be_2$ has 8 elections

Bond order of $Be_2$ is zero.

So $Be_2$ does not exist.

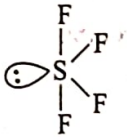
**49.** (*a*) In B2H6 terminal bond angle is greater than bridge bond angle.

$$\text{Bond angle} \propto \text{S-Character} \propto \frac{1}{\text{P-Character}}$$

So, Here in terminal bond more S-character and less P-character.

**50.** (*c*) $SF_4 \Rightarrow$ Hybridisation $\Rightarrow sp^2d$

Geometry $\Rightarrow$ T.B.P Shape $\Rightarrow$ See-Saw



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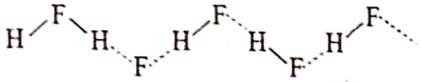
Here 2 bonds are axial and 2 bonds are equitorial.So both bond lengths will be unequal.

**51.** (*a*) A is false but R is true

Assertion is incorrect since in hydrogen bonding. Dipole-dipole interactions are non-covalent but ion-dipole interaction can also result in H-bond formation. Reason is correct since F is most electronegative element & structure is.



Symmetrical H-bonds are present.

**52.** (*a*)

| Molecule | Electrons | Bond order |
|---|---|---|
| $Ne_2$ | 20 | 0 |
| $N_2$ | 14 | 3 |
| $F_2$ | 18 | 1 |
| $O_2$ | 16 | 2 |

A → (iii), B → (iv), C → (i), D → (ii)

**53.** (*d*) $H_2O$ bond angle is 104.5° bond angle is less due to lone pair-lone pair repulsion.

**54.** (*c*) Here 2 lone pairs and 3 bond pairs steric number = 2 + 3 = 5

Geometry is T.B.P.

Shape will be T-Shape ⇒

**55.** (*b*)

| Molecule | Electrons | Bond order |
|---|---|---|
| $C_2^{2-}$ | 14 e⁻s | 3 |
| $N_2^{2-}$ | 16 e⁻s | 2 |
| $O_2^{2-}$ | 18 e⁻s | 1 |

Correct order of bond order.

$O_2^{2-} < N_2^{2-} < C_2^{2-}$

**56.** (*b*) Nitrogen does not have d-orbitals in its valence shell so. It can't form $NCl_5$.

**57.** (*a*) Both statement I and Statement II are correct

Due to presence of double bond alkanes are highly reactive than alkanes.

Therefore, alkanes are less stable than alkanes.

There is more energy required to break double bond than a single bond. So that, strength of double bond is greater than carbon-carbon fingle bond.

**58.** [3] $BeF_2$, $BF_3$ and $CCl_4$ has zero dipole moment.

$H_2O$, $NH_3$ and HCl has non-zero dipole moment.

**59.** (*c*) NO, $O_2$ has last electron in anti-bonding orbitel.

So, here if we remove last electron bond order will increase.

**60.** (*c*) $B_2H_6$ and $BCl_3$ is electron deficient.

**61.** (*b*) $NO_2$ contains odd electron.

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**62.** (*b,c*)

Lone pairs on the central atom in $BrF_5$ = 1

Lone pairs on the central atom in $ClF_3$ = 2

Lone pairs on the central atom in $XeF_4$ = 2

Lone pairs on the central atom in $SF_4$ = 1

**63.** (*a,c*) $C_2^{2-}$ is 14e⁻s species and diamagnetic

$O_2^{2+}$ (B.O. = 3) have shorter bond length than $O_2$ (B.O. = 2)

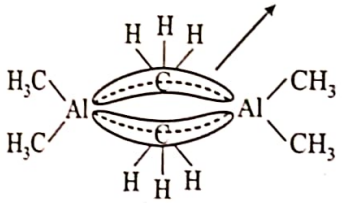
**64.** (*b*) Correct order

$\underset{(+5)}{H_3PO_4} > \underset{(+4)}{H_4P_2O_6} > \underset{(+3)}{H_3PO_3} > \underset{(+1)}{H_3PO_2}$

**65.** (*a,b,d*)

Structure of $Al_2(CH_3)_6$

3 centre-2 electron bond

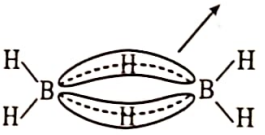


⇒ $BCl_3$ is stronger lewis acid due to small size of boron.

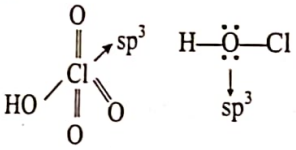
⇒ Structure of $Al_2Cl_6$

Structure of $B_2H_6$

3 centre-2 electron bond



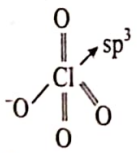
**66.** (*a,c,d*)

(*a*) 

(*b*) $Cl_2 + H_2O \rightarrow HCl + HOCl$

(*c*)

| | | | |
|---|---|---|---|
| Acidic : | $HClO_4$ | > | $H_3^+$ |
| | Acid-1 | | Acid-2 |
| Basic | $ClO_4^-$ | < | $H_2O$ |
| | C.B -1 | | C. B. - 2 |

(*d*) 

Resonance stabilization

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**67.** (*a,d*)

$X_2$ $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 =$

$\pi 2p_y^2, \underbrace{\pi^* 2p_x^2 = \pi^* 2p_y^2}_{\text{(HOMO)}}, \underbrace{\sigma^* 2p_z^0}_{\text{(LUMO)}}$

($F_2$)

Down the group $\pi^*$ to $\sigma^*$ excitation become easy.

**68.** [6]

| Species | | Lone pair on central atom |
|---|---|---|
| $TeBr^{2-}$ | : | 1 |
| $BrF_2^-$ | : | 2 |
| $SNF_3$ | : | 0 |
| $XeF_3^-$ | : | 3 |

Total Number of lone pair = 6

$SNF_3$ :

**69.** [6] ($H_2$, $Cl_2$, $Be_2$, $C_2$, $N_2$, $F_2$)

$H_2$ : $\sigma 1s^2$

$He_2^+$ : $\sigma 1s^2, \sigma^* 1s^2$

$Li_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2$

$Be_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2$

$B_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^1 = \pi 2p_y^1$

$C_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2$

$N_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2 = \pi 2p_z^2$

$O_2^-$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2 = \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^1$

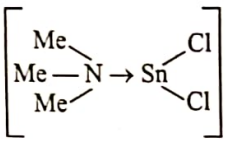
$F_2$ : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_z^2 = \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^2$

**70.** (*a,d*)

$SnCl_2 + Cl^- \rightarrow SnCl_3^-$

**(Q)** **(X)**

$SnCl_2 + Me_3N \rightarrow SnCl_2.NMe_3$



$SnCl_2 + 2CuCl_2 \rightarrow SnCl_4 + 2CuCl$

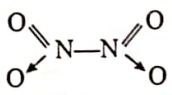
**(Q)** **(Z)**

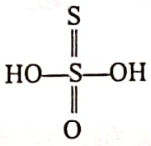
**71.** (*b,d*)

| Polar molecule | Non-polar molecule |
|---|---|
| $CHCl_3$, $SO_2$, $C_6H_5CI$, | $BeCl_2$, $CO_2$, $BCI_3$, $SF_6$ |
| $H_2Se$, $BeF_5$, $O_3$, $XeF_6$, | $NO_2$, $NH_3$, $POCl_3$, $CH_3Cl$ |

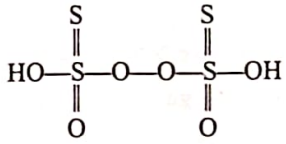
**72.** [4.00]

$N \equiv N \rightarrow O$





$N \equiv N \rightarrow O$



**73.** [19.00]

$XeF_4 + O_2F_2 \rightarrow \underbrace{XeF_6}_{Y} + O_2$

Y has 3 lone pair of electron in each fluorine and one lone pair of electron in xenon.

$6 \times 3 + 1 = 19$

Hence total lone pair of electrons is 19.

**74.** [6]

Here polar molecules in the liquid form will be attracted/ deflected near charged comb.

**Polar molecules** : HF, $H_2O$, $NH_3$, $H_2O_2$, $CHCl_3$, $C_6H_5Cl$ (6-polar molecules)

**Nonpolar molecules** : $O_2$, $CCl_4$, $C_6H_6$

**75.** (*a,d*)

(*a*) σ orbital has a total of two nodal places.

(*d*) $\pi^x$ orbital has one mode in the xy plane the molecular axis.

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CHAPTER

# 9 Hydrogen and Its Compounds

## Exercise-1 (Topicwise)

1. (*b*) $\overset{H}{\boxed{1}} + e^- \rightarrow \overset{H^-}{\boxed{1\downarrow}}$, helium like configuration.
2. (*d*) Hydrogen is an invisible gas.
3. (*a*) ${}^{3}_{1}H \Rightarrow n + p = 3$.
4. (*c*) It can gain or lose it's outermost $e^-$.
5. (*d*) Electrolysis.
6. (*d*) $Ca + H_2 \rightarrow CaH_2$ (Hydrolith)
   $CaH_2 + H_2O \rightarrow Ca(OH)_2 + H_2\uparrow$
7. (*b*) The ionization energy of hydrogen is 1312 kJ $mol^{-1}$
8. (*b*) Alkali metals have very low I.E.
9. (*d*) Hydrogen molecule cannot participate in coordinate bond formation but chlorine molecule can.
10. (*d*) Loss of an electron from hydrogen atom result in a nucleus of very small size as compare to any other atoms or ions.
11. (*a*) $H < H^+ < H^-$, $H^-$ has helium like configuration which is a noble gas, which has highest ionization energy.
12. (*c*) Hydrogen can lose 1 electron or gain 1 electron or it can make a bond with itself or any other atom.
13. (*b*) $CO(g) + H_2O(g) \xrightarrow[\text{Catalyst}]{673K} CO(g) + H_2(g)$.
14. (*d*) In $CH_4$ octate is completed.
15. (*a*) The metals of group 7, 8 and 9 do not form hydride.
16. (*a*) Interstitial hydrides are formed by the compounds of d & f-block elements
17. (*c*) Heavy water is chemically deuterium oxide ($D_2O$).
18. (*b*) Washing soda can remove temporary and permanent hardness of water.
19. (*b*) $D_2O$ used in nuclear reactors as a moderator.
20. (*d*) Temporary handness due to bicarbonates while permanent due to chlorides & sulphates of $Ca^{+2}$ & $Mg^{+2}$.
21. (*a*) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$
22. (*a*) $Na_2Al_2SiO_3.xH_2O$
23. (*b*) $NaZ + CaSO_4 \rightarrow CaZ + Na_2SO_4$
24. (*d*) The density of water is 1 $gcm^{-3}$ at 4°C
    so molarity = $\frac{1000}{18} = 55.5$ M.
25. (*d*) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$
26. (*c*) Produce $H_2$, used in reduction of others.
27. (*c*) ${}_{1}{}^{3}H$
28. (*d*) $2F_2 + 2H_2O \rightarrow 4\,HF + O_2$
29. (*d*) Water obtain from organic ion-exchange resins is free from all ionic impurities.
30. (*b*) $H^{\Theta}$ ion will exchange $H^+$ ion from $H_2O$ molecule.
31. (*a*) $Ca(OH)_2$ (Clark's method)
32. (*c*) Passing it through cation exchange resins ($Mg^{2+}$ and $Ca^{2+}$) replaces $Na^+$ ions.
33. (*b*) Due to production of nascent oxygen.
34. (*b*) $\overset{0}{Cl_2} + H_2O_2 \rightarrow 2H\overset{-1}{Cl} + O_2$
    In this reaction $H_2O_2$ works as reducing agent
35. (*c*) $Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$
36. (*c*) $KO_2 \rightarrow O_2^-$
    $BaO_2 \rightarrow O_2^{-2}$
37. (*c*) Electrolysis of 50% sulphuric acid gives per disulphuric acid ($H_2S_2O_8$) which on distillation yields 30% solution of hydrogen peroxide.
38. (*b*) By the action of cold and dilute sulphuric acid on hydrated barium peroxide
    $BaO_2 \cdot 8H_2O(s) + H_2SO_4(aq) \longrightarrow BaSO_4(s) + H_2O_2(aq) + 8H_2O(l)$
39. (*b*) During the electrolytic oxidation of acidified sulphate solutions at high current density gives peroxodisulphate, which on further hydrolysis gives hydrogen peroxide.
40. (*d*) Fact based.
41. (*a*) The laboratory preparation of $D_2O_2$ is also done by this method
    $K_2S_2O_8(s) + 2D_2O(l) \longrightarrow 2KDSO_4(aq) + D_2O_2(l)$
42. (*c*) $BaO + CO_2 + H_2O \rightarrow BaCO_3 + H_2O_2$
43. (*d*) $H_2O_2$ can be used as (a), (b) and (c).
44. (*b*) Only non-linear structures show dipole moment.

45. (*b*) $Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$

46. (*d*) Conc. $H_2SO_4 + H_2O_2 \rightarrow H_2O + O_2$.

47. (*b*) In Solvey process NaCl, $CO_2$ and $NH_3$ are used.

48. (*d*) $H_2O_2$ exists in two phase gaseous and solid phase with different dihedral angle.

49. (*c*) Higher the no. of hydrogen bonding and molecular mass higher will be boiling point.

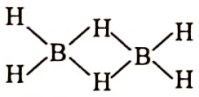
50. (*d*) Strength of hydrogen bond is greater in $H_2O$
$\because$ $H_2O$ has more formal negative charge on the oxygen atom than $H_2O_2$.

## Exercise-2 (Learning Plus)

1. (*d*) Hydrogen does not has lone pair of electron so it can not form coordinate bond with other atoms or molecules.

2. (*c*) The ratio of protium, deuterium and tritium in nature is not certains.

3. (*c*) H– Be – H



4. (*b*) $Na_2Al_2Si_2O_8 \cdot xH_2O + Ca^{2+} \rightarrow CaAl_2Si_2O_8.xH_2O + 2Na^+$

5. (*b*) In bosch process water gas is used for production of $H_2$.

6. (*c*) At Anode
$2H^- \rightarrow H_2 + 2e^-$

7. (*a*) Due to Ca and Mg salts.

8. (*b*) $Ma_3N_2 + 6D_2O \rightarrow 3Mg(OD)_2 + 2ND_3$

9. (*d*) Each oxygen atom attached with four hydrogen, two by covalent bonds and two hydrogen bonds in ice.

10. (*d*) Fact based.

11. (*b*) $Na_2O_2 + 2HCl \rightarrow 2NaCl + H_2O_2$
$Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$

12. (*c*) $H_2O_2$ acts as reducing agent in acidic medium
$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^-$

13. (*c*) It undergoes autoxidation on prolonged standing.

14. (*b*) $CO_2$ is stronger acid than $H_2O_2$ and water is weaker acid than $H_2O_2$.

15. (*c*) 30% (w/v) $H_2O_2$.

16. (*a*) Hydrogen position in the periodic table is not certain and unsatisfactory. Hydrogen exhibits properties similar to both alkali metals (Group (a) and halogens (Group 17).

17. (*b*) Tritium is radioactive and emits low energy beta–particles

18. (*c*) The ionization energy of hydrogen (1312 kJ $mol^{-1}$ is of nearly same order as that of the halogens ($\Delta H$ of F is 1680 kJ $mol^{-1}$.

19. (*a*) Atomic hydrogen is very unstable it readily combines with another hydrogen atom to form hydrogen molecule which is highly stable hence energy is released.

20. (*d*) Dihydrogen gets dissociated into atomic hydrogen by an electric arc and so the hydrogen atoms which are produced are allowed to combine on the surface to be welded to generate the temperature of 4000 K.

21. (*b*) $MgH_2$ is covalent polymeric hydride.

22. (*c*) Molar mass of $D_2O$ and HTO is 20 g/mole

23. (*c*) $H_2$ and $D_2$ are isotopes. Its have same atomic number therefore bond length is same.

24. (*a*) On reaction with water Ionic hydrides will gives basic solution
$NaH + H_2O \longrightarrow NaOH + H_2$

25. (*c*) Boron hydride $BH_3$ is electron deficient because its octet is incomplete.

26. (*c*) d and f block elements form interstitial metal's hydride.

27. (*d*) $2R\text{–}COO\,H + Ca^{2+} \longrightarrow (RCOO)_2Ca + 2H^+$

28. (*a*) Hydrogen peroxide is diamagnetic molecule.

29. (*a*) Volume strength of $H_2O_2 = N \times 5.6$
$= 1.5 \times 5.6 = 8.4$ vol

30. (*a*) $F_2$ is more reactive than $Cl_2$ which is further more reactive than $Br_2$ and least reactive amongst halogens is $I_2$ which requires a catalyst for the reaction.

31. (*b*) Oxygen liberated from 1000 ml of $H_2O_2$ = 20,000 ml.
$\therefore$ Oxygen liberated from 15 ml = 300 ml

32. (*c*) The adsorption of hydrogen by metals is called occlusion.

33. (*b*) Hydrogen is adsorbed by certain metals like Fe, Au, Pt & Pd.

34. (*b*) Deuterium form strong bond with oxygen as compare to hydrogen. So the bond energy of O-H is lesser as compare to O-D bond energy in water.

35. (*b*) $\overset{-1}{H_2}\overset{-1}{O_2} + 2H\underset{\text{Oxidation}}{\overset{-1}{I} \rightarrow \overset{0}{I_2}} + 2\overset{-2}{H_2O}$

$\therefore$ $H_2O_2$ is an oxidising agent.

$HO\underset{\text{Radiation}}{\overset{+1}{Cl} + \overset{-1}{H_2}\overset{-1}{O_2} \rightarrow H_3O^+ + \overset{-1}{Cl} + O_2^0}$

$\therefore$ $H_2O_2$ = Reducing agent.

36. (*a*) In group 6, only chromium forms CrH.

37. (*a*) Pd, Pd adsorbs hydrogen.

38. (*d*) 50% dilute $H_2SO_4$ on electrolysis produces $H_2$.

39. (*c*) $H_2$ and $D_2$ have same bond length.

40. (*a*) $H^- \rightarrow \frac{1}{2}H_2 + e^-$; at anode.

@PBOCBSE

## Exercise-3 (JEE Advanced Level)

1. (*d*) $CaH_2$ (hydrolith)

2. (*d*) Ion-exchange method (Permutit process) is used for removing the permanent hardness of the water.

3. (*c*) Ionic hydride is formed by s-block elements. Which contains $H^-$ ions that's why they are a good reducing agents.

4. (*d*) Vegetable oil + $H_2 \xrightarrow[473K]{Ni}$ Fat
   Liquid Solid

5. (*c*) Ortho and para-hydrogen have the same chemical properties but different physical properties.

6. (*b*) Ionic hydride is formed by s-block elements which are very high electropositive metals.

7. (*d*) s-block elements formed ionic hydride (NaH).

8. (*a,b,d*)
   Ortho & para hydrogen are called as nuclear spin isomers because they are different w.r.t their nuclear spins.

9. (*a,c,d*)
   Bosch process used water gas ($CO_{(g)} + H_2O_{(g)}$).

10. (*a,b,c*)
    Clark's method uses calcium hydroxide to remove the temprary hardness of water.

11. (*a,c,d*)
    (*a*) $Fe + 2H^+(dil) \rightarrow Fe^{2+} + H_2$
    (*c*) $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$
    Sodium zincate
    (*d*) During electrolysis, the following reactions takes place
    At anode : $2Cl^-(aq) \rightarrow Cl_2(g) + 2e-$
    At cathode : $2H_2O(l) + 2e- \rightarrow H_2(g) + 2OH^-(aq)$
    **Overall reaction:** $2Cl^-(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2OH^-(aq)$

12. (*a,b,d*)
    Protium, deuterium and tritium are isotopes. They have only one electron in the shell. The mass number ration is : 1: 2: 3

13. (*a,c*)
    (*a*) $H_2(g) \xrightarrow[\text{Electric arc}]{5000\,K} 2H(g) : \Delta H = 436\,KJ$
    (*c*) Palladium in the powdered state can occlude nearly 1000 times its own volume of hydrogen

14. (*c*) 6% (w/v) of $H_2O_2$.

15. (*a*) Protium is most common isotopes

16. (*d*)
    (*a–c*) Interstitial hydrides are non-stochiometric hydrides and thus deficient in hydrogen. Transition and innertransition elements at elevated temp. absorb hydrogen into the interstices of their lettices to yield metal like hydrides.

17. (*a,b,c*)
    Metallic hydrids are non-stoichiometric hydrides so, theyever deficient of hydrogen and conduct electricity ionic hydrides are crystamine non-volatile and non conducting in solid state.

18. (*a,b,c*)
    Hydrogen can attain stability by using an electron, gaining an electron or forming a bond with itself or another atom.

19. (*a,b,c,d*)
    $LiH + H_2O \rightarrow LiOH + H_2$
    $2Al + qNaOH \rightarrow qNaiv_2 + 3H_2$.
    $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

20. (*c*) Interstitial hydride are non-stoichiometric in nature.

21. (*b*) $SiH_4$ is electron precise hydride (As it's octate is complete).

22. (*d*) $Na + \frac{1}{2}H_2 \rightarrow 2NaH$ . Along the row, electro negativity
    So, hydride becomes more acidic. Metallic hydrides are good conductors of electricity and have high thermal conductivity.

23. (*c*) $_1{}^3H$

24. (*b*) ∵ the earth's gravity field is too small to hold such a light element.

25. (*a*) $2H_2 + O_2 \rightarrow 2\,H_2O$ Highly exothermic occupies very less space and have high thrust.

26. (*c*) Perhydrol '100V' $H_2O_2$ not used to removal of $Ca^{2+}$/ $Mg^{2+}$ from the sample of water.

27. (*d*) permutit is a $Na_2OAl_2O$.

28. (*c*) Zeolite is $Na_2Al_2Si_2O_8.xH_2O$

29. (*a*) $D_2O$ is heavy water, temporary hard water is bicarbonates of Mg and Ca permanent hard water contains sulphates and chlorides of Mg and Ca.

30. (*c*) Calgon are polymetaphosphate of Na. Metallic hydrides are non-stoichiometric compound. $CH_4$ is covalent hydride.

31. (*b*) H-atom can be used in cutting and welding.
    $R\!\!\nearrow + CO + H_2 \xrightarrow{Rh/pph_3} R\!\!\nearrow\!\!CHO + R\!\!\nearrow\!\!\overset{CHO}{\overset{|}{}}$
    (Hydrotormylation) $NaH + H_2O \rightarrow NaOH + H_2$.
    Perhydrol is $H_2O_2$

32. (*c*) $H_2O \xrightarrow{Electrolysis} H_2 + O_2$ $LiAlH_4$ is a reducing agent.
    $\overset{\delta+}{H}-\overset{\delta-}{Cl}$ (polar). $D_2O$ used in aromatic reactor.

33. (*d*) $H_2O_2$ is perhydrol and a propellant sodium hexametaphosphate used in a calgon method.

34. (*a*) LiH is a metallic hydride, conduct heat and electricity. Concept of hybridization is not applicable.

At anode : $H^- \rightarrow \frac{1}{2}H_2 + e^-$

$CH_4(sp^3)$ hybridised.

35. (*c*) perhydrol '100V' $H_2O_2$ not used to removal of $Ca^2/Mg^{2+}$ from the sample of water.

36. [3] $H_2S_2O_8$ on completely hydrolysis gives ....2..... mole of $H_2SO_4$ & .....1..... mole of $H_2O_2$.

37. [4] No. of Peroxy linkage in $H_2S_2O_8$, $CrO_5$ & $H_2TiO_4$ are 1, 2 and 1 respectively.

38. [4] Tritium has one proton, two neutrons, and one electron.

39. [3] Incorrect points

(ii) $H_2 < D_2$ (Bond energy)

(iii) $H_2 < D_2 < T_2$ (Boiling point)

(iv) $H_2 < D_2 < T_2$ (No. of neutrons)

(i) The number of protons in isotopes of hydrogen is the same.

40. [3] Incorrect Points

(i) $CaH_2 < BeH_2$ (Electrical conductance in molten condition)

(ii) $LiH < NaH < CsH$ (Ionic character)

(iii) $H_2 < D_2 < F_2$ (Bond dissociation enthalpy)

(iv) Ionic character is an increase in ionic hydride down the group of s-block elements due to an increase in the size of cation.

41. [9] $[H_9O_4]^+$

## Exercise-4 (Past Year Questions)

### JEE MAIN

1. (*d*) Highly pure hydrogen can be obtained by electrolysis of water. It is the decomposition of water into $O_2$ and $H_2$ gas by passing electric current.

2. (*b*) A reducing agent gets oxidized while reducing the other reactant. Hence, it should release electrons.

3. (*c*) $Na_2O_2$ is peroxide of sodium.

4. (*a*) The oxidation state of Mn in $MnO_4^-$ is +7. $H_2O_2$ can only reduce Mn as it is in maximum O.S.

5. (*c*) Permanent hardness in water cannot be cured by boiling, filtration and distillation.

6. (*c*) $H_2O_2$ can undergo oxidation as well as reduction because the oxidation number of oxygen in $H_2O_2$ is –1.

7. (*c*) $H_2O_2$ oxidises PbS to $PbSO_4$ and itself gets reduced to $H_2O$.

8. (*b*)

(*i*) $[Fe^{+2}(CN)_6]^{4-} + H_2O_2^{-1} + 2H^+ \downarrow [Fe^{+3}(CN)_6]^{3-} + 2H_2O^{-2}$

(*ii*) $[Fe^{+3}(CN)_6]^{3-} + H_2O_2^{-1} + 2OH^- \downarrow [Fe^{+2}(CN)_6]^{4-} + O_2^0 + 2H_2O$

9. (*c*) Temporary hardness is due to soluble $Mg(HCO_3)$ and bicarbonates of calcium and magnesium.

10. (*d*)

I. $H_2$ produces less combustion because it doesn't produce carbon monoxide.

II. Because it has high calorific value.

III. Dihydrogen is stored is tanks of metal alloys.

IV. LPG produces less energy than liquid dihydrogen.

11. (*c*) Temporary hardness is due to soluble $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$.

$Mg(HCO_3)_2 \xrightarrow{\text{Boil}} Mg(OH)_2\downarrow + 2CO_2\uparrow$

$Ca(HCO_3)_2 \xrightarrow{\text{Boil}} CaCO_3\downarrow + H_2O + CO_2\uparrow$

12. (*a*) Temporary hardness is due to soluble $Mg(HCO_3)$ and bicarbonates of calcium and magnesium.

13. (*a*) Mass of $CaSO_4 = 10^{-3} \times 136$

$= 0.136g$

136g of $CaSO_4 \rightarrow$ 100g of $CaCO_3$

0.136g of $CaSO_4 \rightarrow \frac{100}{136} \times 0.136$

$= 0.1g$ of $CaCO_3$

Hardness of water (in terms of $CaCO_3$)

$= \frac{W_{CaCO_3}}{W_{H_2O}} \times 10^6$ ppm

$= \frac{10^{-1}}{1000} \times 10^6 = 100$ ppm

14. (*d*) All are isotopes of hydrogen.

15. (*a*) $I_2$ is least reactive among all the halogens.

16. (*b*) $H_2O_2$ as oxidant:

$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$ (Acidic medium)

$H_2O_2 + 2e^- \rightarrow 2\bar{O}H$ (Basic medium)

$H_2O_2$ as reductant:

$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^-$ (Acidic medium)

$H_2O_2 + 2OH^- \rightarrow 2H_2O + O_2 + 2e^-$ (Basic medium)

17. (*c*) Hydrogen has 3 isotopes and 1 is radioactive amongst them.

18. (*c*) Si has octet configuration.

19. (*b*) NaH is an example of ionic hydride which is also known as saline hydride.

20. (*c*) The synonym for water gas is syn gas.

21. (*c*) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

22. (*a*) Zeolite process uses the mineral zeolite as the exchange resin for cations in the hard water whereas the ion exchange process include several different resins for the ion exchange.

23. (*b*) (*a*) $H_2O_2 \rightarrow 2H_2O + O_2$

(*b*) $KClO_3 \xrightarrow[MnO_2]{\Delta} KCl + \frac{3}{2}O_2$

$Pb(NO_3) \xrightarrow{\Delta} PbO + 2NO_2 + \frac{1}{2}O_2$

$NaNO_3 \xrightarrow{\Delta} NaNO_2 + \frac{1}{2}O_2$

(*c*) 2-Ethyeanthraquinol $\underset{H_2/Pd}{\overset{O_2}{\rightleftharpoons}}$ 2-Ethylanthraquinone ($H_2O_2$)

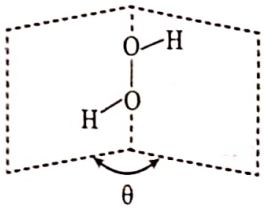
(*d*) Hydrogen peroxide is used for the manufature of sodium perborate.

24. (*d*) $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

25. (*c*) Clark's method is used to remove temporary hardness of water.

26. (*d*)




Open book like structure
- Non planar
- Almost colorless

27. (*b*) In water gas shift reaction, Hydrogen gas is produced economically by the reaction of carbon monoxide with water vapour at 673 K in presence of iron, cromium and copper zinc catalyst.

$CO + H_2O(g) \xrightarrow[\Delta]{Catalyst} CO_2 + H_2(g)$

28. (*b*) To obtain $H_2$ of high purity (> 99.95 %) electrolysis of $Ba(OH)_2$ solution is done using Ni electrodes.

29. (*c*) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$

30. (*b*) (*a*) $PbO_2 + 2H_2O \rightarrow Pb(OH)_4$

(*b*) $Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$ this reaction is possible at room temperature

(*c*) $SnO_2 + 2H_2O \rightarrow Sn(OH)_4$

(*d*) Acidified $BaO_2 \cdot 8H_2O$ gives $H_2O_2$ after evaporation.

31. (*a*) The bond dissociation energy of $D_2$ is greater than $H_2$ and therefore $D_2$ reacts slower than $H_2$.

32. (*d*) $D_2O$ is used for the study of reaction mechanism. Rate of reaction for the cleavage of O–H bond > O–D bond.

33. (*b*) Around 55% of hydrogen goes to ammonia production.

34. (*a*) Sodium hydroxide is generally prepared commercially by electrolysis of sodium chloride in castner Kellner cell.

At cathod: $Na + e^- \xrightarrow{Hg}$ Na–amalgum

Anode: $Cl^- \longrightarrow \frac{1}{2}Cl_2 + e^-$

The Na–amalgam is treated with water to give sodium hydroxide and hydrogen gas:

$2Na\,(amalgam) + H_2O \longrightarrow 2NaOH + H_2 + 2Hg$

35. (*a*) $Mg(HCO_3)_2 \xrightarrow{Boil} Mg(OH)_2 + 2CO_2\uparrow$

$Ca(HCO_3)_2 \xrightarrow{Boil} CaCO_3 + H_2O + CO_2\uparrow$

36. (*d*) They have different neutrons and mass number.

37. (*c*) In option (*a*) and (*c*) reducing action of hydrogen peroxide is shown.

In option (*a*) it is in acidic medium, in option (*b*) it is in basic medium.

## JEE-ADVANCED

38. (*a*) As reducing agent

$KIO_4 + H_2O_2 \longrightarrow KIO_3 + H_2O + O_2$

As oxidising agent

$NH_2OH + 3H_2O_2 \longrightarrow HNO_3 + 4H_2O$

@PROCBSE

CHAPTER

# 10 The s-Block Elements

## Exercise-1 (Topicwise)

1. (*b*) In s-block last electrons comes in the s-orbital.
2. (*d*) $M + O_2 \rightarrow M_2O, M_2O_2, MO_2$
   M = K, Rb, Cs
3. (*c*) Alkali metals are highly reactive metals. They react with
   Alcohol – $2C_2H_5OH + 2K \rightarrow 2C_2H_5OK + H_2$
   Water – $2K + 2H_2O \rightarrow 2KOH + H_2$
   Ammonia – $K + (x + y)NH_3 \rightarrow$
   $$\underset{\text{Ammoniated cation}}{[K(NH_3)_x]^+} + \underset{\text{Ammoniated electron}}{[e(NH_3)_y]^-}$$
   But they do not react with kerosene.
4. (*d*) Smaller size ion in gas form have greater size ion in aq. medium.
5. (*b*) After removal of an electron, the effective nuclear charge per electron increases hence the size decreases.
6. (*c*) K gives violet colour on flame test
7. (*d*)

| Element | – | Li | Na | K | Rb |
|---|---|---|---|---|---|
| Atomic radius (pm) | – | 152 | 186 | 227 | 248 |

8. (*d*) Li forms only oxides
9. (*a*) High IE of Be & Mg.
10. (*b*) H.E. $\propto \frac{\text{charge}}{\text{size}}$
11. (*d*) Down the group reactivity of alkali metal increase.
12. (*b*) Sodium is very reactive towards air and moisture so that it needs to stored under kerosene.
13. (*a*) Cs is the most electropositive element
14. (*b*) Across a period, $Z_{eff}$ increases and size decreases. So, group-I elements are largest in their horizontal periods (except noble gases)
15. (*b*) Alkali metals are strong reducing agent as they loose their valence shell $e^-$ easily.
16. (*c*) Down the group, size increases due to which tendency to lose $e^-$ increases
17. (*c*) Order of their m.p. is Li > Na > k and order of reactivity is opposite to order of m.p.
18. (*c*) Softness of alkali metal increases down the group due to increases in size (close packing of atoms become weak)
19. (*c*) $K_2CO_3$ can not be prepared because $KHCO_3$ is highly soluble and cannot be precipitate out.
20. (*b*) $2Na + 2NH_3 \xrightarrow{\text{heat}} 2NaNH_2 + H_2$
21. (*a*) Greater size of $HCO_3^-$ not stable with $Li^+$
22. (*a*) In polyatomic anions stability inversely depends on polarising power of cation.
23. (*d*) The given compound x must be $CaCO_3$. It can be explained by following reactions,
    $$\underset{(x)}{CaCO_3} \xrightarrow{\Delta} \underset{\text{Residue}}{CaO} + CO_2\uparrow;$$
    $$CaO + H_2O \rightarrow \underset{(y)}{Ca(OH)_2}$$
    $$Ca(OH)_2 + CO_2 + H_2O \rightarrow Ca(HCO_3)$$
    $$Ca(HCO_3)_2 \xrightarrow{\Delta} \underset{(x)}{CaCO_3} + \overset{z}{CO_2}\uparrow + H_2O$$
24. (*c*) According to Fajan's rule RbCl has greatest ionic character due to large ionic size of $Rb^+$ ion. $BeCl_2$ has least ionic (Maximum covalent) due to small size of $Be^{+2}$ ion which has highly polarising.
25. (*a*) $NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$
    $NaHCO_3$ and NaOH pair can't exist in solution as they react with each other.
26. (*c*) Soda ash – $Na_2CO_3$
27. (*b*) In Castner-keller cell wr use: At Anode – Graphite and at cathode – mercury
28. (*d*) $Na \xrightarrow{NH_3} Na^+(NH_3)_x + e^-(NH_3)_y$
29. (*b*) LiCl has more covalent character than other alkyl halide due to which it is more soluble in non-polar solvent (pyridine)
30. (*b*) $Zn\,(s) + 2\,NaOH\,(aq) \longrightarrow \underset{\text{Sodium zincate}}{Na_2\,ZnO_2\,(aq)} + \underset{\text{Hydrogen gas}}{H_2(g)}$
31. (*d*) More the magnitude of $\Delta H_f$ (Heat of formation), more the stability of compound

$\Delta H_f$ LiCl = – 408.8 kJ mol$^{-1}$

$\Delta H_f$ NaCl = – 412.5 kJ mol$^{-1}$

$\Delta H_f$ CsCl = – 433 kJ mol$^{-1}$

$\Delta H_f$ KCl = –436 kJ mol$^{-1}$

**32.** (*b*) $6Li + N_2 \longrightarrow 2Li_3N$

$Li_3N + 3H_2O \longrightarrow 3LiOH + NH_3$

$CuSO_4 + 4NH_3 \longrightarrow [Cu(NH_3)_4]SO_4$

M and B can be Li and $NH_3$

**33.** (*b*) $Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$

Peroxides which have O – O bond give $H_2O_2$ on treatment with dilute acid

**34.** (*a*) $4LiNO_3(s) \longrightarrow 2\,Li_2O(s) + 4\,NO_2(g) + 2O_2(g)$

**35.** (*b*) Calcium gives brick red colour in flame test

**36.** (*c*) Mg element with atomic number 12 belongs to IIA groups and third period.

**37.** (*a*) Hydration energy $\propto \frac{1}{\text{size}}$

**38.** (*a*) (*i*) Small atomic size.

(*ii*) High electronegativity

(*iii*) Absence of *d* orbitals

**39.** (*a*) $3\,Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$

$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$

Ammonia gas was evolved.

**40.** (*d*) Mg does not impart any characteristic color to the flame test where as Ca gives red colour in flame test

**41.** (*a*) Mg is present in chlorophyll

**42.** (*b*) $2\,(CaSO_4.2H_2O) \xrightarrow{120^\circ C} 2\,CaSO_4.H_2O + 3\,H_2O$

Gypsum — Plaster of paris

**43.** (*d*) Solubility of IIA sulphates decreases down to the group.

**44.** (*b*) Basic strength of oxides $\propto \frac{1}{EN}$

**45.** (*a*) M.P. of s-block metal halides $\propto$ L.E.

$F^- > Cl^- > Br^- > I^-$

**46.** (*d*) Amphoteric hydroxide are $Be(OH)_2$ and $Zn(OH)_2$

**47.** (*a*) $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$

As we go down the group I.E. decreases. Hence ionic character increases.

**48.** (*a*) Chlorophyll is complex of Mg.

**49.** (*c*) Radium element is used for treating cancer in cancer radio therapy

**50.** (*b*) $MgCl_2.6H_2O \xrightarrow{\Delta} MgO + 2HCl + 5H_2O$

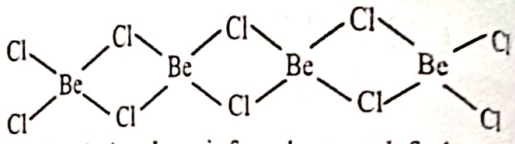
**51.** (*b*) $Be^{2+} \longrightarrow$ smaller size

$F^- \longrightarrow$ smaller size

$\longrightarrow$ High charge density $\downarrow$ Higher Hydration Energy

Greater the hydration energy means greater the solubility

**52.** (*b*) $BeCl_2$ and $MgCl_2$ do not impart any colour to the flame as they have high ionization enthalpies

**53.** (*b*) 

$BeCl_2$ exist in polymeric form due to e$^-$ defecient nat

**54.** (*a*) $BeSO_4 \longrightarrow$ water soluble

$BeO \longrightarrow$ inert on heating

$Be(OH)_2 \longrightarrow$ insoluble in water and soluble in Na

**55.** (*d*) During the Electrolysis of molten anhydrous $CaCl_2$, get calcium at cathode (reduction)

**56.** (*d*) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

Basic oxide — Acidic oxide

**57.** (*c*) Electrolysis of mixture of anhydrous $CaCl_2$ and KC

At anode $2Cl^- \longrightarrow Cl_2 + 2e^-$

At cathode $Ca^{2+} + 2e^- \longrightarrow Ca$

Calcium is obtained at cathode

**58.** (*b*) $Ca(OH)_2$ used as a water softening agent.

$Ca(OH)_2 + Na_2CO_3 \longrightarrow CaCO_3 + NaOH$

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

## Exercise-2 (Learning Plus)

**1.** (*c*) They have weak metallic bond because of one valen electron per atom. So they have low melting points.

**2.** (*a*) $Cs^+$ ion impart violet to Bunsen flame. This is due t the fact that the emitted radiations are of high energy

**3.** (*d*) Li reacts with $N_2$ and $O_2$ of air forming $Li_3N$ and Li$_2$ but Na reacts with only $O_2$ forming $Na_2O$ not with N forming $Na_3N$ (it requires high temperature.)

**4.** (*a*) Down's process involves the electrolysis of fuse sodium chloride containing $CaCl_2$ & KF using Fe a cathode & graphite as anode at about 600°C. $CaCl_2$ & KF are added to decrease fusion temperature (1600°C)

**5.** (*d*) $M + (x + y)\,NH_3 \longrightarrow [M(NH_3)_x]^+ + [e(NH_3)_y]^-$ solution contains unpaired solvated electrons which are responsible for their strongly reducing and highly conducting nature.

**6.** (*a*) $Na + H_2O \rightarrow NaOH + H_2$

A — C — B (Combustible)

$Zn + NaOH \rightarrow Na_2ZnO_2 + H_2$

D Amphoteric

$Zn + dil\ H_2SO_4 \rightarrow ZnSO_4 + H_2$

**7.** (*b*)

(*a*) $2LiNO_3 \xrightarrow{\Delta} Li_2O + 2NO_2\uparrow$ (brown) $+ \frac{1}{2}O_2\uparrow$

(*b*) $KNO_3 \xrightarrow{\Delta} KNO_2 + \frac{1}{2}O_2\uparrow$

(*c*) $Pb(NO_3)_2 \xrightarrow{\Delta} PbO + 2NO_2\uparrow$ (brown) $+ \frac{1}{2}O_2\uparrow$

(*d*) $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2\uparrow$ (brown) $+ O_2\uparrow$

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8. (*a*) According to Fajan rule
solubility in polar solvent $\propto$ Ionic character
solubility in non polar $\propto$ cavalent character solvent

9. (*c*) $Na + Al_2O_3 \rightarrow Na_2O + Al$ (Position in electrochemical series)
$Na_2O + CO_2 \rightarrow Na_2CO_3$

10. (*c*) $BaCO_3$ (white ppt.)
$\uparrow CO_3^{2-}$
$Ba^{2+} + CrO_4^{2-} \rightarrow BaCrO_4(s)$ Yellow ppt
$\downarrow SO_4^{2-}$
$BaSO_4(s)$ (white ppt.)

11. (*b*) $A \longrightarrow \underset{\text{Lime water}}{Ca(OH)_2}$
$\underset{\text{Lime water}}{Ca(OH)_2} + CO_2 \longrightarrow \underset{\text{White ppt (C)}}{CaCO_3} + H_2O$
$Ca(OH)_2 + Na_2CO_3 \rightarrow \underset{C}{CaCO_3} + \underset{B}{NaOH}$

12. (*b*)
(*a*) Ca salt imparts brick red colour to the flame.
(*b*) Sr salt imparts bright crimson colour to the flame.
(*c*) Ba salt imparts apple green colour to the flame.
(*d*) Mg salt does not impart any colour to the flame because of high ionization energy.

13. (*a*) $Mg + 3N_2 \longrightarrow Mg_3N_2$;
$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$.

14. (*c*) $Ca^{2+}$ or $Mg^{2+} + EDTA^{4-}$
$\rightleftharpoons [Ca(EDTA)]^{2-}$ or $[MgEDTA]^{2-}$
Hardness causing ion

15. (*a*) For alkaline earth metals,
(*i*) Solubility of sulphates decreases down the group.
(*ii*) Solubility of hydroxide increases down the group.
BeO and $Be(OH)_2$ are amphoteric; so they are soluble in NaOH forming soluble $[Be(OH)_4]^{2-}$.
Hence metal M is Be.

16. (*a*) $Ba(NO_3)_2 \xrightarrow{\Delta} BaO + 2NO_2 + \frac{1}{2}O_2$;
Alkali metal nitrates give only $O_2$ on heating below 500°C according to following reaction,
$MNO_3 \longrightarrow MNO_2 + \frac{1}{2}O_2$

17. (*a*) $\underset{X}{CaCO_3} \longrightarrow CaO + CO_2$, $CaO + H_2O \longrightarrow \underset{Y}{Ca(OH)_2}$,
$Ca(OH)_2 + 2CO_2 \longrightarrow \underset{Z}{Ca(HCO_3)_2}$,
$Ca(HCO_3)_2 \longrightarrow \underset{X}{CaCO_3} + H_2O + CO_2$

18. (*d*) The solubility of most salt's decreases with increase atomic weight. But trend is reversed with the fluorides and hydroxides.
$BeF_2$, $Be(OH)_2$
$Mg(OH)_2$
$Ca(OH)_2$
$Sr(OH)_2$
$Ba(OH)_2$
$BaF_2$
Ksp ↓ decrease

19. (*a*) It gives a deposit of a carbonate with $CO_2$ of atmosphere, which can jam the tap.
(*b*) LiF has very large lattice energy because of the small size of both the $Li^+$ and $F^-$ ions. Thus its lattice energy exceeds the hydration energy and so in water it is insoluble.
(*c*) $Be^{2+}$ on account of vacant p-orbital and it high polarising power form complex with water as ligands.
$BeCl_2 + 4H_2O \rightleftharpoons [Be(H_2O)_4]^{2+} + 2Cl^-$

20. (*b*) $CaSO_4.\frac{1}{2}H_2O + \frac{3}{2}H_2O \rightarrow CaSO_4.2H_2O$

21. (*b*) Sodium has one electron in valence shell. This requires less energy to remove the valance electron and forms $Na^+$ ion and attains the stable electronic configuration of Ne with a completed octet. The removal of the second electron will break this stable electronic configuration and requires high energy.

22. (*b*) Alkali metals are very reactive so its found in combined state.

23. (*d*) Na – alkali metal
Alkali metals oxide are basic in nature.

24. (*b*) Potassium form superoxide ($KO_2$) when its reacts with oxygen.

25. (*c*) Castner-Kellner cell method use brine solution for manufacturing of NaOH. In this process $Cl_2$ evolved at anode and hydrolysis of Na-Hg gives $H_2$ gas.

26. (*b*) Caustic soda (NaOH) is deliquescent because its absorb moisture.

27. (*d*) It is not used for baking of bread.

28. (*d*) Sodium bicarbonate, ($NaHCO_3$).

29. (*d*) Potassium ions participate in the oxidation of glucose to produce ATP; helps in activation of many nzymes and used in synthesis of proteins.

30. (*b*) $Na^+$ present outside the cell and $K^+$ present in the cell.

31. (*c*) Fact based

32. (*b*) Due to high lattice energy, sulphates and carbonates of alkaline earth metals are insoluble in water.

33. (*d*) NaCl, $NH_3$ and $CO_2$ are used in manufacturing of sodium carbonate.

34. (*c*) For the $BeSO_4$; Hydration enthalpy is higher than lattice enthalpy.

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35. (*c*) Due to presence of OH– ions, solution of $K_2O$ is basic in nature.

36. (*b*) $Ba(NO_3)_2 + Na_2SO_4 \rightarrow BaSO_4 \downarrow + 2Na(NO_3)_2$

37. (*d*) In case of alkaline earth metals, down the group Hydration enthalpy decreae with increase in ion size. So, $Ba(NO_3)_2$ undergoes minimum hydrolysis

38. (*b*) $CaO + 3C \rightarrow CaC_2 + CO$

Quick lime Calcium carbide

39. (*b*) Potassium does not react with nitrogen directly.

$Li + N_2 \rightarrow Li_3N$

$Mg + N_2 \rightarrow Mg_3N_2$

$Ca + N_2 \rightarrow Ca_3N_2$

40. (*c*) Down the group ionization potential decreases.

41. (*b*) $2NaNO_2 + 6Na \rightarrow 4Na_2O + N_2$

42. (*b*) $Na_2O + H_2O \rightarrow NaOH$

(A)

$$Na_2O \xrightarrow[O_2]{400^\circ C} Na_2O_2 \xrightarrow{H_2O} NaOH + O_2$$

(B)

43. (*b*) Due to High hydration in case of AgF and $BeF_2$ they are soluble in water. $CaF_2$ and $MgF_2$ are insoluble in water.

44. (*b*) The loosely bound electron starts oscillating and goes to higher energy level when light falls on its surface. On returning back to lower level it emits light and metal have a lustrous surface.

45. (*b*) In solid $KHCO_3$, $HCO_3^-$ exists as dimer and it involves H– Bonding

46. (*c*) Due to lower melting point and high electripositive character of Na. it react with water more vigrously.

47. (*d*) Due to overvoltage for $H_2$ in case of Hg electrode, Na is formed at cathode.

48. (*b*) $I_3 >> I_2$, hence it have 2 valence electron so metal is Be.

49. (*a*) $4KO_2 + 2CO_2 \rightarrow 2K_2CO_3 + 3O_2$

50. (*b*) Sulphates and chlorides of calcium and Magnesium results in hardness of water.

51. (*a*) Water helps in developing interlocking needle - like crystals of hydrated silicates.

52. (*a*) Calcined gypsum $\rightarrow CaSO_4.\frac{1}{2}H_2O$

Dolomite $\rightarrow CaMg(CO_3)_2$

Marble and Seashells contains $CaCO_3$.

53. (*b*) $Na_2O + H_2O \rightarrow 2NaOH$

$+1\ -2\quad -2\quad +1\ +2$

Oxide ion donates a pair of electron to form OH– ion.

54. (*c*) – 5.1 ev

$Na^+ + e^- \rightarrow Na \quad (-5.1ev)$

55. (*b*) Magnesium prevents rusing or corrosion by provinding cathodic protection.

56. (*d*) In case of alkali metals down the group electropositive character increase and reactivity increase.

In case of Halogen, electronegativity decrease down the group thus reactivity decrease.

57. (*c*) But – 2 – yne does not have any acidic hydrogen.

$H_3C - C \equiv C - CH_3$

58. (*c*) $2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$

## Exercise-3 (JEE Advanced Level)

1. (*b*) When light Falls on the surface of sodium, the free electron starts oscillating at its mean position and get excited to higher levels. On returning Back to lower levels, its emits light in all directions. Due to this sodium has a characteristic metallic lustre.

2. (*d*) Alkali metals have low ionisation energy.

3. (*a*) **S1 :** $Li^+$ being smaller have high polarising power and $I^-$ being larger have high polarisability. So it is most covelent among alkali metal halides according to Fajan's rule.

**S2 :** The $IE_1$ of potussium atom is less then sodium atom.

**S3 :** The presence of transition metals like iron and other impurities catalyses the decompositon of deep blue solution forming amide and liberating $H_2$.

**S4 :** Two opposing tendencies exists. With greater charge and smaller size of cation, lattice energy increases which tends to increase the melting point; while increase in covalent character causes a decrease in melting point. Hence, no unique generalised trend may be stated for melting points.

4. (*b*) The reaction of alkali metals with water becomes increasingly violent on descending the group on account of their decreasing ionisation energies with increasing atomic size. So, the order of reactivity is:

Li < Na < K < Rb.

5. (*c*) Reacts with water being more basic than water.

6. (*c*) Isoelectronic species have same number of electrons.

7. (*b*) All alkali metal hydrides are ionic in nature and react with water according to the reaction ;

$NaH + H_2O \longrightarrow NaOH + H_2$.

8. (*a*) True statement. The CsI, because of bigger cation ($Cs^+$) and bigger anion ($I^-$), has smaller hydration enthalpy. As a result, it does not exceed its lattice energy; so CsI is insoluble in water.

9. (*d*)

(*a*) Lithium show exceptional behavior in reaction directly with nitrogen of air to form the nitride, $Li_3N$.

liberating $CO_2$ gas. So it has the lowest thermal stability.

(*c*) The solubility of the alkali metal hydroxides increases down the group from Li to Cs. This is because of the fact that down the group with increasing size of cation, the lattice energy as well as hydration energy also decreases but the change in lattice energy is more as compare to that of hydration energy.

**10.** (*a*) The reaction Follows as:

$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + H_2O + CaCl_2$

**11.** (*b*) Anhydrous sodium carbonate is stable to heat and does not decompose.

**12.** (*a*) $NaNO_3$ is not used as gun powder because it is hygroscopic in nature and becomes wet by absorbing water molecules from the atmosphere.

**13.** (*d*) Fact based

**14.** (*a*) Dehydration is the process of removal of water from any substance.

**15.** (*c*) Alkali metals showing +1 oxidation state.

**16.** (*c*) In Down's method for the extraction of sodium, the melting point of the electrolyte is lowered by adding both calcium chloride and potassium fluoride.

**17.** (*a,b,d*)

(*a*) The solubility of the alkali metal carbonates increases down the group from Li to Cs. This is because of the fact that down the group with increasing size of cation the lattice energy as well as hydration energy also decrease but the change in lattice energy is more as compare to that of hydration energy. Lithium hydrogencarbonate is not obtained in the solid form while all other elements form solid hydrogencarbonates.

(*b*) $K_2CO_3$ can not be prepared by Solvey process because intermediate formed, $KHCO_3$ is appreciably soluble in water.

(*c*) $Li_2CO_3$ and $MgCO_3$ both are not thermally stable.

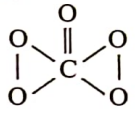
(*d*) $Na_2CO_3.NaHCO_3.2H_2O$ is a mineral called trona.

**18.** (*a,c*)

$KO_2\ K_2O + O_2$

$KO_2 + H_2O\ KOH + O_2$

**19.** (*a,b,c*)



$Na^+.....\bar{O}-\bar{O}.....Na^+$

$Ba^{2+}$ ..... $(O^- — O^-)$

**20.** (*a,d*) Sodium when dissolved in liquid ammonia produces solvated electron which imparts blue colour to the sol.

**21.** (*a,b,c*)

(*a*) The increasing stability of the peroxide or superoxide, as the size of the metal ion increases, is due to the stabilisation of large anions by larger cation through lattice energy effects.

(*b*) Bigger cation stabilises bigger anion and similarly smaller cation stabilises smaller anion through lattice energy effects.

(*c*) The low solubility of LiF is due to its high lattice energy whereas low solubility of CsI is due to smaller hydration energy .

(*d*) NaOH is deliquescent in nature. So, NaOH absorbs moisture.

**22.** (*a,b,c*)

(A) $P_4 + 3NaOH + 3H_2O \xrightarrow{\Delta} PH_3 + 3\ NaH_2PO_2$

(B) $4S + 6NaOH \xrightarrow{\Delta} Na_2S_2O_6 + 2Na_2S + 3H_2O$

(C) $3Cl_2 + 6NaOH \xrightarrow{\Delta} 5\ NaCl + NaClO_3 + 3H_2O$

(D) $2B + 6NaOH \xrightarrow{\Delta} 2Na_3BO_3 + 3H_2$

**23.** (*a,b,d*)

**24.** (*a,c*)

$P_2O_5 + H_2O \rightarrow H_3PO_4$

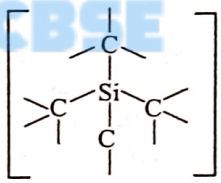
$CaCl_2 + H_2O \rightarrow CaCl_2.2H_2O$

$P_2O_5$ & $CaCl_2$ are dehydrating agent.

**25.** (*c,d*) For solubility

Hydration energy should be greater then lattice energy.

**26.** (*a,b,d*)



**27.** (*a,b,c,d*) Fact Based.

**28.** (*a,b,c,d*) Both become possive on reaction with $HNO_3$ due to formation of oxide layer, their chlorides are lewis axids chlorides exist is polymeric form and hydroxides are soluble in alkali as well as in acid.

**29.** (*b,c*)NaoH, $NaH_2PO_4$ and $NaHCO_3$ can not exist together in the solution as they react with each other.

**30.** (*a,b*) The solubility of a compound depends on the co-relation between lattice enthalpy and hydration enthalpy if the hydration enthalpy is equal to or greater than the lattice enthalpy then the salt is water-soluble.

**31.** (*b,c*)Alkaline earth metal sulphate gives metal oxide and $SO_3$ on heating

$MgSO_4 \rightarrow MgO + SO_3$

$CaSO_4 \rightarrow CaO + SO_3$

Alkali Metal sulphates are thermally stable.

**32.** (*a,d*) Fact based

**33.** (*a,b*) Sodium nitrate decomposes to the nitrite on heating

$2NaNO_3$ (heat) $\rightarrow 2NaNO_2 + O_2$

Sodium forms an ionic hydride NaH, it is a crystalline soft metal.

Sodium carbonate the thermally stable so it doesn't decomposes on heating.

**34.** (*a,b,d*) As $k_2CO_3$ is thermally stable so apart from this all gives $CO_2$ gas.

**35.** (*b,d*) At cathode (iron vessel):

$Na^+ + e^- \rightarrow Na(s)$

at anode (Graphite): $2Cl^- \rightarrow Cl_2 + 2e^-$

By electrolysing aqueous solution of NaCl, hydrogen gas produces.

Sodium is a strong reducing agent Na dissolves in liquid almmonia to give blue solution.

**36.** (*a,b,d*) Ionic mobility in aquous solution increases from Li to Cs thermal stability of carbonate salt also increases. Reactivity towards water increases from Li to Cs.

**37.** (*a,c,d*) Milk of lime : $Ca(OH)_2$ line water is a saturated solution of $Ca(OH)_2$ (aq).

MgO : it is quite stable and fuses at 2800 k hence it is used as a refractory material for lining electric furnancet. The oxides of alkaline earth metals are less basic then those alkali metals.

**38.** (*a,b,c*) The alkali metals from ionic hydrides

Alkali metals have $ns^1$ configeration so they have low potential energy and are highly electropositive metals. Due to this configuration they do not occur in free state in nature and have high affinity towards non metals and form ionic compounds. Also due to higher radius they have low density so they are soft and can be cut with knife.

**39.** (*a,c,d*) Hydration depends upon the size of cation, lower the size higher the hydration So Cs ion have low hydration due to large size. Among the alkali metals Li, Na, K and Rb, lithium has the highest melting point due to the smaller size.

**40** (*d*) As Lithium is less basic than other alkali metals so lithium carbonate decomposes into its oxide on heating unlike other elements.

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$

Also LiCl is covalent in nature so it doesn't provide any ions in molten state and its molten state is non-conductive & due to this nature it dissolves in either. Also Li form $Li_3N$ but none other element from group 1 froms a nitride.

**41.** (*a,c,d*) Metal hydroxides with a highly-charged central metal atom forms amphoteric in addition to aluminium, metal such as zine, tin, lead, and beryllium also forms amphoteric oxides the two elements have same rluctronegativity and their change radius rations.

**42.** (*a,b*) Moving down the group radius increases therm stability of carbonates increases.

Hydration energy is proportional to (1/cation size) so decreases.

**43.** (*a,c,d*)

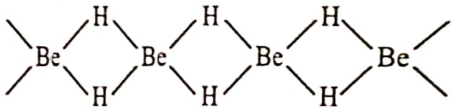
**44.** (*a,b,d*) (*a*) we know stability of carbonates increases doen the group. So, $BeCO_3$ is most unstable of all the carbonate of 2nd group and kept in the atmospher of $CO_2$.

(*b*) $Be + NaOH + H_2O \rightarrow Na_2(Be(OH)_4) + H_2$

(*c*) $BeF_2 + 2NaF \rightarrow Na_2[BeF_4]$

**45.** (*a,b,d*) Mg does not react with NaOH so it can not be used to generate hydrogen gas. All other reagent gives $H_2$ gas

**46.** (*a,b,c,d*) As we more down the group, polarization power of cation decreases so $CaH_2$, $SrH_2$ and $BaH_2$ are ionic and beryllium and magnessium hydrides are corallent. $BeH_2$ contains three centre two-electron bond.




**47.** (*b*) $2CH_3COCH_3 + 6CaCoCl_2 \rightarrow 2CHCl_3 + (CH_3COO)_2Ca + 2Ca(OH)_2$ the product obtained when bleaching powder + 3 $CaCl_2$ is distilled with acetone is $CHCl_3$ or chloroform.

**48.** (*d*) Bleaching powder – $CaOCl_2.2H_2O$

$CaOcl_2.2H_2O \rightarrow Cl_2$

$145g \rightarrow 71$

$$\% \text{ of available chlorine} = \frac{71}{145} \times 100 = 48.9\% = 49\%$$

**49.** (*b*) On long standing, bleaching powder undergoes auto-oxidation which results in the formation of calcium chloride and calcium chloriate

$6CaOCl_2 \rightarrow 5CaCl_2 + 6Ca(ClO_3)_2$

**50.** (*b*)

$$\underset{(A)}{Ca(S)} \xrightarrow{H_2O} \underset{(B)}{H_2(g)} + \underset{(C)}{Ca(OH)_2(aq)}$$

$$\underset{(B)}{Li + H_2} \longrightarrow \underset{(D)}{LiH} \xrightarrow{H_2O} \underset{(\epsilon)}{LiOH} + H_2 \uparrow$$

$$\underset{(C)}{CaCOH_2CaP} \xrightarrow{CO_2,T} \underset{(F)}{CaCO_3} \xrightarrow{CO_2(Excess)} \underset{(G)}{Ca(HCO_3)_2}$$

$$\underset{(F)}{CaCO_3} + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

$$\underset{(F)}{CaCO_3} + 3C \xrightarrow{2000^\circ C} \underset{II}{CaC_2} + CO + CO_2$$

**51.** (*c*) It is $CaHCO_3$, which can not be obtained is solid state and causes temporary hardness.

**52.** (*c*) Fact Based.

**53.** (*a*) Fact Based.

**54.** (*c*) Fact Based.

**55.** (*a*) A → r,s; B → r; C → p,q,r,s; D → q

A. Solvay process is used for the manufacture of $NaHCO_3$ and $Na_2CO_3$; the raw material is NaCl.

B. $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2\uparrow + H_2O\uparrow$.

C. $Na_2O + H_2O \longrightarrow 2NaOH$ ; NaOH being basic turns red litmus blue.

$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$; NaOH being basic turns red litmus blue but $H_2O_2$ being oxidising agent bleaches coloured litmus.

$NaHCO_3 + H_2O \rightleftharpoons NaOH + H_2CO_3$; solution is alkaline and turns red litmus blue.

$Na_2CO_3 + 2H_2O \rightleftharpoons 2NaOH + H_2CO_3$; solution is alkaline and turns red litmus blue.

D. $Na_2O_2 + CO \longrightarrow Na_2CO_3$; $2Na_2O_2 + 2CO \longrightarrow 2Na_2CO_3 + O_2$. Hence, it absorbs CO and $CO_2$ and liberates oxygen.

**56.** (*c*) A → r; B → s; C → q; D → p

A. $3Ca(OH)_2 + 2Cl_2 \xrightarrow[35°C]{\text{below.}} Ca(OCl)_2.CaCl_2.Ca(OH)_2.H_2O + H_2O$
(slaked lime) — bleaching powder

B. $2Ca(OH)_2 + 2Cl_2 \xrightarrow{\text{cold (below 25°C)}} CaCl_2 + Ca(OCl)_2 + 2H_2O.$
(milk of lime) — calcium hypochlorite

C. $6Ca(OH)_2 + 2Cl_2 \xrightarrow{\text{heat (above 35°C)}} 5CaCl_2 + Ca(OCl_3)_2 + 6H_2O$
(slaked lime) — calcium chlorate

D. $2Ca(OH)_2 + 2Cl_2 \xrightarrow{\text{red heat}} 2CaCl_2 + 2H_2O + O_2\uparrow$
calcium chloride

**57.** [6] It is a cyclic structure having six P–O–P linkages.

**58.** [7] Except Be and Sr all are correctly matched.

**59.** [3] Except $Li_2CO_3$, $Na_2CO_3$ all form polymeric chains.
$AlCl_3$ dimeric chain.

**60.** [15] x = $Na_2PbO_2$ y = $Na_2SnO_3$ z = $NaAlO_2$

**61.** [10] Glauber's salt is $Na_2SO_4.10H_2O$

## Exercise-4 (Past Year Questions)

### JEE MAIN

**1.** (*a*) Region 2.
Just above the tip of the Primary flame.
1500°C (2,700° F)

**2.** (*d*) $Li_2O$, $Na_2O_2$, and $KO_2$.

**3.** (*a*) Both form basic carbonates.

**4.** (*b*) $Al + NaOH \rightarrow Al(OH)_3$ (white ppt.)
$Al(OH)_3 + OH^- \rightarrow [Al(OH)_4]^-$ Soluve
$Al(OH)_3 \xrightarrow{\text{heat}} Al_2O_3 + 3H_2O$
$Al_2O_3$ used as an adsorbent in column chromatography

**5.** (*b*) $Rb + O_2 \rightarrow RbO_2$
(excess) — superoxide
X → is $RbO_2$.
$2RbO_2 + 2H_2O \rightarrow 2RbOH + H_2O_2 + O_2$

**6.** (*c*) Be Because of its transparency to X ray

**7.** (*d*) It give a deep blue solution due to formation of ammoniated $e^-$.

**8.** (*c*) Saline hydride. It is an Ionic hydride

**9.** (*d*) $Ba(NO_3)_2$
Due to size of Ba, it doesn't crystallize with water molecule.

**10.** (*b*) Li only element in the group that react with dinitrogen to form nitride.

**11.** (*a*) $Be(OH)_2$.

**12.** (*d*) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$
It depends upon ionic Potential
(Charge/size) I.P. ↑es; H.E. ↑es.

**13.** (*c*) Mg O and $Mg_3N_2$.
$Mg + O_2 \rightarrow MgO$
$3Mg + N_2 \rightarrow Mg_3N_2$

**14.** (*c*) $BeX_2$ {Be has smallest size}

**15.** (*a*) K, 2S feels more attraction from nucleus so It has less energy.

**16.** (*d*) $2LiNO3 \xrightarrow{\Delta} Li_2O + 2NO_2\,(g) + \frac{1}{2}O_2(g)$

**17.** (*b*) $CaSO_4.2H_2O \xrightarrow{393\ K} CaSO_4.\frac{1}{2}H_2O$

**18.** (*a*) Mg and $Mg_3N_2$

$3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$ (B)
$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$
(B)
$NH_3 + CuSO_4 \rightarrow [Cu(NH_3)_4]\,SO_4$ (Blue complex)

**19.** (*d*) A, C and D only.

**20.** (*a*) Li and Mg
$6Li(s) + N_2(g) \xrightarrow{\Delta} 2Li_3N(s)$
$3Li(s) + N_2(g) \xrightarrow{\Delta} Mg_3N(s)$

**21.** (*b*) (i), (ii) and (iv)

**22.** (*d*) $Be(OH)_2 + 2NaOH \rightarrow Na_2BeO_2 + H_2O$
Be is highly soluble in water.

**23.** (*b*) A → q; B → s; C → p; D → r

**24.** (*d*) The solubility of 2nd group element depends on their hydration energy which decreases down the group.
So, the solubility also decreases down the group
Be and Mg are 2nd and 3rd group elements hence the solubility of their sulphate is high.

25. (*a*) Both A and R are correct but R is not the correct explanation of A.

26. (*d*) $CaSO_4.2H_2O$, $CaSO_4$ and $CaSO_4.\frac{1}{2}H_2O$

No. of $H_2O$ 2 0 and 0.5

27. (*b*) $CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow$
slaked lime

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$
water soluble

28. (*d*) A → t; B → r; C → q; D → p

29. (*a*) Baking soda, sodium bicarbonate. It contains Sodium, Oxygen, Hydrogen and Carbon.

30. (*a*) A → p; B → q; C → r; D → s

31. (*a*) Li < K < Na < Rb < Cs.
K is lighter due to unusual tension atomic size of K.

32. (*a*) Mg
Li $\frac{\text{diagonal}}{\text{relationship}}$ Mg.

33. (*b*) Portland cement contains gypsum
$CaSO_4.2H_2O$

## JEE ADVANCED

34. (*a*) $CO_2$, $H_2CO_3$, $HCO_3^-$, $CO_3^{2-}$
When $CO_2$ dissolved in water in water It gives $H_2CO_3$
On ionisation give $HCO_3^-$ and $CO_3^{2-}$ ion.

35. (*a*) $MgSO_4 + Na_2HPO_4 + NH_4OH \rightarrow Mg(NH_4)PO_4 + Na_2SO_4 + H_2O$

36. (*b*) Blue colour is due to ammoniated $e^-$.

37. (*a,b*)
$NH_4NO_3 + NaOH \rightarrow NH_3 + NaNO_3 + H_2O$
$7NaOH + NaNO_3 + 4Zn \rightarrow 4Na_2ZnO_2 + NH_3 + 2H_2O$
$NH_4NO_2 + NaOH \rightarrow NaNO_2 + NH_3 + H_2O$
$3Zn + 5NaON + NaNO_2 \rightarrow 3Na_2ZnO_2 + NH_3$

38. (*b*) $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$
So O.S. of P in $P_4$, $NaH_2PO_2$ and $PH_3$ is 0, +1, –3,

39. (*b*) $NaHSO_4$.
$Na_2S_2O_3 + 4Cl_2 + 5H_2O \rightarrow 2NaHSO_4 + 8HCl$.

40. (*a,b*) $4Na + O_2 \rightarrow 2Na_2O$
$2Na + O_2 \rightarrow Na_2O_2$.

41. (*b*) $2KO_2(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2O_2(aq) + O_2(g)$

@PROCBSE

CHAPTER

# 11 The p-Block Elements-I

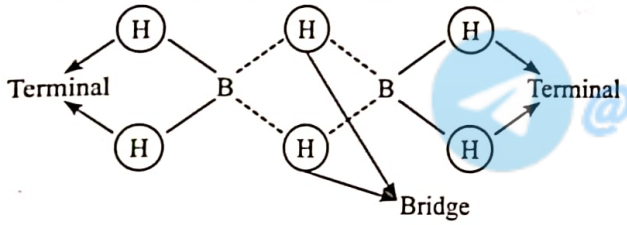
## Exercise-1 (Topicwise)

**1.** (*b*) $TlI_3 \rightleftharpoons Tl^+ + I_3^-$

Tl is in +1 oxidation state because of inert pair effect (*b*) is correct.

**2.** (*c*) As oxygen is divalent so the element R forms oxides of type $R_2O_3$.

**3.** (*b*) 2 bridge hydrogens and 4 terminal hydrogen are present




**4.** (*b*) $H_3BO_3 + 3C_2H_5OH \rightarrow B(OC_2H_5)_3 + 3H_2O$

A mixture of boric acid with ethyl alcohol burns with green edged flame due to the formation of ethyl borate.

**5.** (*a*) Borazine

$(B_3N_3H_6)$

**6.** (*c*) Large sized chlorine atom do not fit in between the small boron atom whereas small sized hydrogen atom get fitted in between boron atoms. So $BCl_3$ does not exist as dimer but $BH_3$ as dimer $(B_2H_6)$.

**7.** (*a*) Electron deficient molecules.

**8.** (*a*) $BF_3 < BCl_3 < BBr_3 < BI_3$

**9.** (*c*) Al is called self-protecting metal because when it comes in contact with the atmosphere it is covered entirely with a layer of aluminium oxide which stage of further corrosion.

**10.** (*a*) Al

**11.** (*b*) 1 : 1 : 2

The formula of alum

$K_2CO_4.Al_2(CO_4)_3\ 2\ 4H_2O$

$2 \rightarrow$ Monovatent cation

$2 \rightarrow$ Trivatent cation

and 4 $SO_4^{2-}$ ion

True Ratio $\rightarrow$ 1 : 1 : 2

**12.** (*c*) Aluminium hydroxide react with excess of NaOH and forms $AlO_2^-$

$Al(OH)_3 + NaOH \rightarrow NaAlO_2 + 2H_2O$

So, the $NaAlO_2$ forms $AlO_2^-$

**13.** (*b*) $[Al(H_2O)_6]^{3+} + 3Cl^-$

$AlCl_3$ is covalent but in water it becomes ionic due to large hydration energy of $Al^{3+}$

$AlCl_3 + H_2O \rightarrow [Al(H_2O)_6]^{3+} + 3Cl^-$

**14.** (*c*) $2Al + 6H_2O \rightarrow 2Al(OH)_3 + 3H_2$

$Al(OH)_3 + NaOH \rightarrow Na^+ + (Al(OH)_4)^-$

**15.** (*a*) $AlCl_3$ can form dimer and exist as $Al_2Cl_6$ aluminium have vacant d-orbital which can accomodate electron from chloride atom

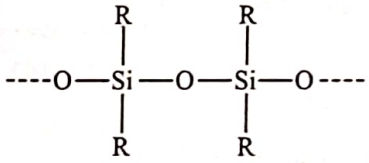
**16.** (*a*) Ge > Sn > Pb

**17.** (*b*) 12 Pentagons and 20 hexagons.

**18.** (*c*) $CO_2$ is acidic oxide and thus more effectively absorbed by an alkali. SO, cold and solid calcium hydroxide is the best absorbing material for the $CO_2$.

**19.** (*a*) Calcium phosphide

**20.** (*c*) Organic polymers containing silicon in them are called as silicons. Silicons have general formula of $R_2SiO$, an empirical formula analogous to Ketone ($R_2CO$).




**21.** (*b*) HF

**22.** (*c*) Lead pipes are not suitable for drinking worm because lead reacts will water containing air to form $Pb(OH)_2$ which poisonous for our health.

**23.** (*b*) Pbs + 4 $H_2O_2$ A $PbSO_4 + 4H_2O$

**24.** (*b*) Due to inert pair effect.

**25.** (*c*) $Pb_3O_4$ is a mixed oxide. It can be represented as 2PbO. $PbO_2$

**26.** (*c*) When tin is treated with conc. nitric acid, it is converted into-meta stannic nitrate ($H_2SnO_2$).

$Sn + \text{Conc. } HNO_3 \rightarrow NO_2 + H_2SnO_2$

**27.** (*c*) White lead is the basic lead carbonate. It is a complex salt containing both carbonate and hydroxide ions. It is also known as hydrate of carussite. It cause lead poisoning.

**28.** (*a*) Litharge is on of the natural mineral formed by lead that is lead (II) oxide, PbO.

**29.** (*a*) No lead in pencils Rather, the core is made up of a non-toxic mineral called graphite. Therefore % of lead in lead pencil is zero.

**30.** (*d*) Lead (II, IV) oxide, red lead or triplumbic tetroxide, is a bright red or orange crystalline or amorphous solid pigment. Chemically, red lead is $Pb_3O_4$ or $2PbO.PbO_2$ it is used in the manufacture lead glass, ruff proof paints and in sindhur.

## Exercise-2 (Learning Plus)

**1.** (*b*) The reluctance of S-subshell electrons to participate in the chemical bonding.

**2.** (*d*) All of these

(*a*) $2BCl_3 + 3H_2 \xrightarrow[W \text{ or } Ta]{\text{Red HOT}} 2B + 6HCl$

(*b*) $2BI_3 \xrightarrow[\Delta]{} 2B + 3I_2$

(*c*) $B_2H_6 \xrightarrow[\Delta]{} 2B + 3H_2\uparrow$

Diborane

**3.** (*d*) Formula of Borax is $Na_2B_4O_7.10H_2O$

**4.** (*a*) $\underset{\text{orthoboric acid}}{\text{conc.}H_3BO_3} \xrightarrow{100^\circ C} \underset{\text{Metaboric acid}}{4HBO_2} \xrightarrow{150^\circ C} \underset{\text{Tetraboric acid}}{H_2B_4O_7}$

(x) (y)

$\xrightarrow{\text{Red Hot}} \underset{\text{boric anhydride}}{2B_2O_3} + H_2O$

**5.** (*d*) Borax bead test

$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\Delta} \underset{\text{sodium Meta Borate}}{Na_2B_4O_7}$

$\xrightarrow{\Delta} 2NaBO_2 + \underset{\text{Glassy make}}{B_2O_3}$

$CuO + B_2O_3 \rightarrow Cu(BO_2)_2$

Blue bead

**6.** (*d*) $Al_2S_3 + 6H_2O \longrightarrow 2Al(OH)_3 + 3H_2S$

Pure

**7.** (*c*) $B(OH)_3 + H_2O \longrightarrow \overset{\oplus}{H}[B(OH)_4]^{\ominus}$

weak mono basic acid

**8.** (*d*) 80% HNO3 + Fe, Co, Ni, Cr, Al, Be → No reaction due to formation of protective layer of oxide.

**9.** (*b*) Acidic

$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

Strong base cation and strong acid anions are always soluble in water as medium for M neutral.

Weak base cation hydrolysis

$Al^{3+} + \overset{+}{H} - \overset{-}{OH}$

$Al(OH)_3 + H^{\oplus}$

In, aq. medium acidic in nature

**10.** (*d*) Alum's

General formula

$M_2(SO_4).N_2(SO_4)_3$

IA ⇒ $Na^+$, $K^+$, $Rb^+$, $Cs^+$, $NH_4^+$ due high P.P.

N ⇒ (+3) O.S. & $Cr^{3+}$, $Fe^{3+}$, $Al^{3+}$

**11.** (*a*) $BCl_3 + H_2O \rightarrow B(OH)_3 + H_2O \rightarrow \overset{\oplus}{H}[B(OH)_4]^{\ominus}$

weak acid due to presence of $H^+$

**12.** (*b*) Group-14 ⇒ +2 and +4 (O.S.)

**13.** (*d*) Methanides

$Be_2C$ and $Al_4C_3$ form methanides.

**14.** (*a*) $K_2CO_3$ and $CO_2$(g)

$2\ HO-C(=O)-C(=O)-OH \xrightarrow{\text{conc.}H_2SO_4} \underset{\text{neutral gas}}{CO} + \underset{\text{acidic gas}}{2CO_2} + H_2O$

$\underset{\text{Base}}{2KOH} + \underset{\text{acidic gas}}{CO_2} \longrightarrow \underset{\text{salt}}{K_2CO_3} + H_2O$

**15.** (*b*) Coal gas is a mixture of $H_2$. Saturated and unsaturated hydrocarbons CO, $CO_2$, $N_2$ and $O_2$.

**16.** (*a*)

| $Al_4C_3$ | $CaC_2$ | $MgC_3$ |
|---|---|---|
| $4 \times 3 + 3x = 0$ | $+2 + 2a = 0$ | $2 \times 2 + 3x = 0$ |
| $x = -4$ | $2x = -2$ | $3x = -4$ |
| | $x = -1$ | $C_3^{-4}$ |
| $C^{-4}$ | $C^{-1}$ or $C_2^{2-}$ | |

**17.** (*a*)

→ Interstitial carbides form only silicon metalloids form covalent carbides.

→ CO and $CN^-$ both are fatal due to complex formation with Fe (III) present in blood.

→ SiC is called carborundum

**18.** (*c*) $Sn^{2+}$ ⇒ (amphoteric)

**19.** (*c*) Due to inability of bromine and iodine to oxidise $Pb^{2+}$ to $Pb^{4+}$

**20.** (*d*) $PbI_4$ due to inert pair effect.

## Exercise-3 (JEE Advanced Level)

**1.** (*a,c,d*)

Boron and silicon halides can be hydrolysed others options are correct.

$BCl_3 + 3H_2O \rightarrow B(OH)_3 + 3HCl$

Boron carbide used in nuclear reactor.

**2.** (*b,c,d*)

$[BF_6]^{3-}$ does not exist since boron does not have any vacant d-orbitals.

**3.** (*a,b,c,d*)

$B_2O_3$ and $B(OH)_3$ are acidic.

$BF_3$ don't get hydrolysed because of strong back bonding.

$BH_3$ get hydrolysed easily

Alane $(AlH_3)_n$ is a polymer.

**4.** (*a,b,c,d*)

$BeF_2$ can accommodate two more $F^-$ ions to form $[BeF_4]^{2-}$

$[B_4O_5(OH)_4]^{2-} + 5H_2O \rightleftharpoons \underset{\text{(weak acid)}}{B(OH)_3} + \underset{\text{(salt)}}{2[B(OH)_4]^-}$

$2[B(OH)_4]^{2-} + 2H_3O^+ \rightarrow 2B(OH)_3 + 4H_2O$

$K_4[Fe(CN)_6] + 6H_2SO_4\,(conc) + 6H_2O \longrightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO\uparrow$

$5O(g) + I_2O_5(s) \rightarrow I_2(s) + 5CO_2(g)$

**5.** (*a,b,d*)

$AlCl_3$ can accept lone pair hence it is a lewis acid.

**6.** (*b,c,d*)

Graphite is less denser than diamond.

**7.** (*c*) $Na_2B_4O_7 + 7H_2O \rightarrow 4B(OH)_3 + 2NaOH$

$MeSiCl_3$ on hydrolysis gives cross linked polymer.

Along the period from left to right acidity increases.

**8.** (*a,b*) $HCOOH \xrightarrow[\text{conc. } H_2SO_4]{\Delta} H_2O + CO(g)$

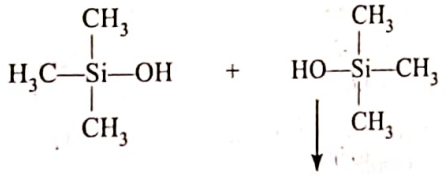
$K_4[Fe(CN)_6] + 6H_2SO_4\,(conc) + 6H_2O \downarrow\Delta \rightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$

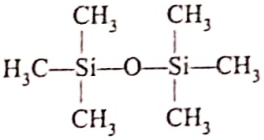
**9.** (*a,c*)

$NaHCO_3$ is basic in nature hence it will give pink colour with phenolpthaline.

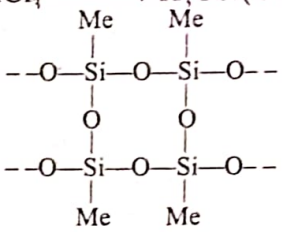
$Mg(NO_3)_2 + NaHCO_3 \rightarrow MgCO_3 + NaNO_3 + H_2O + CO_2$

**10.** (*c*) $(H_3C)_3SiCl \xrightarrow{H_2O} (H_3C)_3Si(OH)$





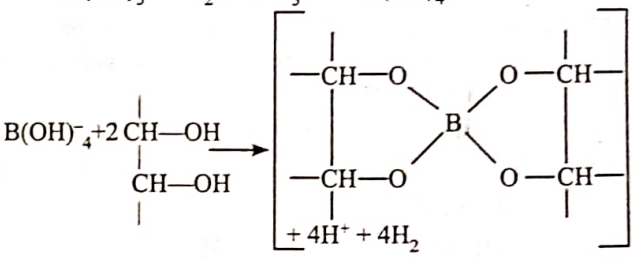
$H_3CSiCl_3 \xrightarrow{H_2O} H_3CSi(OH)_3$



**11.** (*c*) $2Na[BH_4](A) + I_2 \xrightarrow[\text{Solution}]{\text{in Diglyme}} \underset{(B)}{B_2H_6} + H_2 + 2NaI$

$B_2H_6 + 6H_2O \rightarrow 2H_3BO_3(C) + 3H_2$

$B(OH)_3 + 2H_2O \Leftrightarrow H_3O^{\oplus} + B(OH)_4^-$



$B_2H_6(B) + 3O_2 \rightarrow B_2O_3 + 3H_2O$

$CoO + B_2O_3 \rightarrow CO(BO_2)_2$

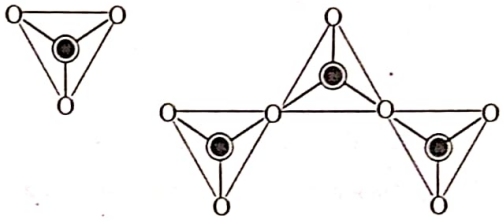
(cobalt metaborate blue colour beed)

$B(OH)_3$ have hydrogen bonding in solid state.

**12.** (*a*) $B(OH)_3$.(aq.) can be titrated against NaOH in presence of phenolpthaline indicator.

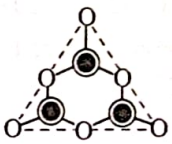
**13.** (*b*) $B_2H_6 + NH_3 \xrightarrow[\text{Low temp.}]{\text{excess } NH_3} B_2H_6.2NH_3$

**14.** (*b*)



$SiO_4^{4-}$ (Td structure) $Si_3O_{10}^{8-}$

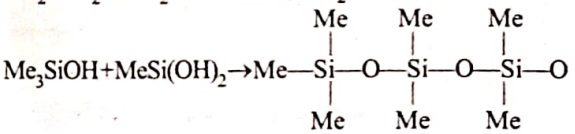
**15.** (*b*) $Si_3O_9^{6-}$



**16.** (*c*) $Ca_2Cu_2Si_3O_{10}.2H_2O$ accomodates $Ca^{2+}$, $Cu^{2+}$ and water molecule in 1 : 1: 1 ratio.

**17.** (*a*) $Me_3SiCl + H_2O \rightarrow Me_3SiOH + HCl$

$Me_2SiCl_2 + 2H_2O \rightarrow MeSi(OH)_2 + HCl$



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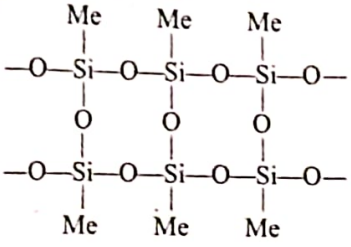
$MeSiCl_3 + 3H_2O \rightarrow MeSi(OH)_3$

$MeSi(OH)_3 \rightarrow$



**19.** (*b*) $CH_3SiCl_3 + H_2O \rightarrow CH_3Si(OH)_3 + HCl$
(Hydrolysis of $CH_3SiCl_3$)
$CH_3Si(OH)_3$ form complex crosslinked polymer
$SnCl_2 + H_2O \rightarrow Sn(OH)Cl$

**20.** (*c*) Barox bead test
$Na_2B_4O_7.10H_2O \rightarrow Na_2B_4O_7 \rightarrow 2NaBO_2 + B_2O_3$
$B_2O_3 + CuSO_4 \rightarrow Cu(BO_2)_2$

**21.** [6] Four 'H' atoms, two on the left and two on the right, known as terminal hydrogens and two boron atoms lie in the same plane.
Total atoms = 4 + 2 = 6

**22.** [4] Compound having higher lattice energy than NaBr are:
LiF, MgO, $Al_2O_3$, $TiO_2$

**23.** [3] $B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 3H_2$
1 mol $B_2H_6$ yields 3 moles $H_2$ gas.

**24.** [4] $C_{60}$ contain 12 five membered ring and 20 six-membered rings. Graphite is dimagnetic in nature and themodynamically more stable than diamond.
Coke is the impure form of carbon.

**25.** [21] B, Al, In, Tl are solid at 40°C while Ga is at 30°C. So,
x = 4
I.E. B > Al > Ga > In > Tl
y = 4
Al → 4
z = 3
$x + 2y + 3z = 4 + 2 \times 4 + 3 \times 3 = 21$

**26.** [26] $C_{60}$ contains 12 five memebered ring ⇒ a = 12
$C_{60}$ contains 20 six-memebered ring ⇒ b = 20
$C_{60}$ contains 30 π-bonds ⇒ c = 30
$2a - 2b + c = 3 \times 12 - 2 \times 20 + 30 = 26$

**27.** [26] $B_{10}C_2H_{12}$ is isostructural and isoelectronic with borate ion of $B_{12}H_{12}^{2-}$
$x + y + z = 12 + 12 + 2 = 24 + 2 = 26$

**28.** [7] Amphoteric nature are those nature which react from both acid as well as base.
PbO, $PbO_2$, SnO, $SnO_2$, $Al_2O_3$, ZnO, BeO are amphoteric nature.

**29.** [3] $Mg + B \rightarrow MgB_2 \xrightarrow{HCl} 3MgCl2 + \underset{\text{Diborane}}{B_2H_6}$
x = 1
y = 2
x + y = 1 + 2 = 3

## Exercise-4 (Past Year Questions)

**1.** (*a*) $4Zn + 10HNO_3(dil) \longrightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$
$Zn + 4HNO_3(conc.) \longrightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

**2.** (*a*) The phosphtous atoms of orthophorus acid $H_3PO_3$ and pyrophosphorus $H_4P_2O_5$ have a formal axidation state of +3

**3.** (*c*) $Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$
Cold & dilute Sodium hypochlorite

**4.** (*c*) $BCl_3$ and $AlCl_3$ has vacant p-orbital to accept lone pair. Hence they will act as lewis acid.

**5.** (*d*) Pb does not show catenation property.

**6.** (*d*) Silicones are polymers with (—$R_3$Si–O—) linkage, they are hydrophobic and biocompatible.
They are also resistant to oxidation and are used in greases.

**7.** (*a*) $SiH_4$ has a complete octet hence, it is not electron deficient.

**8.** (*c*) $B_2H_6 + H_2O \longrightarrow 2H_3BO_3 + 3H_2$
$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$

**9.** (*b*) The amorphous from of silica is called Kiselghur.

**10.** (*a*) All the given statements regarding group-13 are correct
$B_2O_3 \longrightarrow$ Acidic
$Al_2O_3$, $Ga_2O_3 \longrightarrow$ Amphoteric
$In_2O_3$, $Tl_2O_3 \longrightarrow$ Basic

**11.** (*c*) The basic structural unit of feldspar, zeolites, mica and asbestos is $(SiO_4)^{4-}$.

**12.** (*b*) $B_3N_3H_3Cl_3 + LiBH_4 \longrightarrow B_3N_3H_6 + LiCl + BCl_3$
$B_3N_3H_3Cl_3 + 3MeMgBr \longrightarrow$
(A) (C) $B_3N_3H_3(CH_3)_3 + 3MgBrCl$

**13.** (*c*) AgF is highly ionic in nature, hence it will have highest boiling point.

**14.** (*b*) $\underset{(X)}{Na_2SO_3} \xrightarrow{dil.H_2SO_4} \underset{(Y)}{SO_2} \xrightarrow{NaOH} \underset{(X)}{Na_2SO_3} \xrightarrow{SO_2 + H_2O} NaHSO_3$

**15.** (*d*) Black phoshporus has two forms:
α-black & β-black
Red P $\xrightarrow[\text{Sealed tube}]{803\ K}$ α-black P.

**16.** (*a*) CO – Haemoglobin
$SO_2$ – Stiffness to flower buds.
Polychlorinated bipenyls – Carcinogenic
Oxides of Nitrogen – Metabolized by pyrus plants.

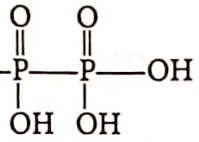
**17.** (*b*) Liquefiction ∝ critical temperature. Hence, $SO_2$ will be absorbed to a large extent than $H_2$.

**18.** (*b*) Acidified sulphate solution gives $H_2S_2O_8$ on electrolytic oxidation.

**19.** (*d*) Electron deficient species are $BCl_3$ and $B_2H_6$.

**20.** (*a*) Cs – Photoelectric effect
Ga – High temperature thermometer
B – Bullet proof vest
Si – Water repellent sprays.

**21.** (*b*) White $P_4$ + alkali $\longrightarrow H_3PO_2$
Red $P_4$ + alkali $\longrightarrow H_4P_2O_6$

Structure of $H_4P_2O_6$ $\longrightarrow$ 

NO P–H bond

**22.** (*b*) $Br_2 + 5F_2 \longrightarrow 2BrF_5$
(Excess)

**23.** (*b*) $P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$
White Phosphorus

**24.** (*b*) $Na_2B_4O_7.10H_2O \xrightarrow{\Delta} Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$
$B_2O_3 + CoO \longrightarrow Co(BO_2)_2$ cabalt metaborate
(Blue beed)

**25.** (*b,d*)

When orthoboric acid react 1,2 dihydroxy or polyhydroxy compounds, they form chelate ring which removes $[B(OH)_4^-]$ and produces a maximum number of $H_3O^{\oplus}$ or $H^{\oplus}$

(*d*) $H_3BO_3$ is weak electrolyte in water as it's pKa is 9.25.

**26.** [2] $B_2H_6 + 6MeOH \longrightarrow 2B(OMe)_3 + 6H_2$
Hence answer is 2 moles of boron containing product.

**27.** (*b*) The hydrolysis of $(H_3C)_2SiCl_2$ produces silane diol which produces linear polymer on condensation. Trimethylchlorosilane can't undergo polymerisation. Hence it is used for chain termination.

**28.** (*a,c,d*)

Crystalline $[B_4O_5(OH)]^{2-}$ have $sp^2$ and $sp^3$ hybridised boron atoms it also contains one terminal hydroxide per boron unit.

**29.** (*b,d*)

$2HNO_3 \xrightarrow[\text{dehydration}]{P_4O_{10}} N_2O_5$ (Diamagnetic in nature)
$N_2O_5 + Na \longrightarrow NaNO_3 + NO_2 \uparrow$
Brown

**30.** (*b*) $2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_2 \uparrow$
Whereas $Na_2O_2$ and $Li_2O_2$ produces $H_2O_2$ and $Pb_3O_4$ do not react with water.

**31.** (*a,b,c*)

$SnCl_2.2H_2O$ is a reducing agent since $Sn^{2+}$ tends to convert $Sn^{4+}$.

(*b*) $SnO_2 + 2KOH\,(aq) + 2H_2O \longrightarrow K_2[Sn(OH)_6]$
(Amphoteric)

(*c*) Filterate of first group contain solution of $PbCL_2$ in HCl which contains $Pb^{2+}$ and $Cl^-$
However in conc. HCl it produces $H_2[PbCl_4]$

(*d*) $Pb_3O_4 + 4HNO_3 \longrightarrow PbO_2 \downarrow + 2Pb(NO_3)_2 + 2H_2O$
($2PbO.PbO_2$) Not a redox reaction

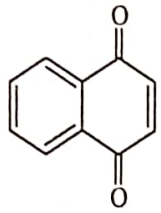
@PROCREE

CHAPTER

# 12 IUPAC Nomenclature

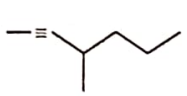
## Exercise-1 (Topicwise)

**1.** (*c*) For molecular formula, we add all atom A of particular elements available in the organic compound and denotes each in subscript.



$C_{10}H_6O_2$

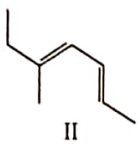
**2.** (*c*) If we see, all the given compounds have molecular formula



I

$C_8H_{14}$

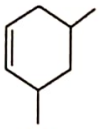
$C_nH_{2n-2}$



II

$C_8H_{14}$

$C_nH_{2n-2}$

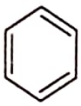


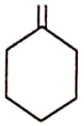
III

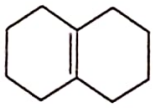
$C_8H_{14}$

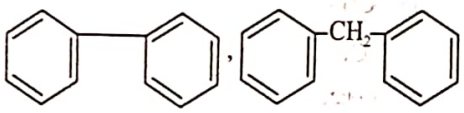
$C_nH_{2n-2}$

It is clear that I, II, III are not identical structure hence correct option will be (C).

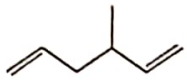
**3.** (*b*)  is aromatic Hydrocarbon.

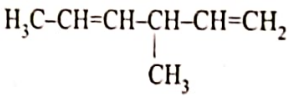
**4.** (*b*)  is unsaturated Hydrocarbon since contain $\pi$–bonds.

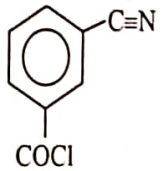
**5.** (*a*)  has only 2° Hydrogen.



**6.** (*c*) Fact Based

**7.** (*d*)  has terminal double bond which the



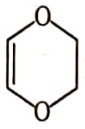
**8.** (*a*) 

No. of $\pi$-bonds = 6

No. of ring = 1

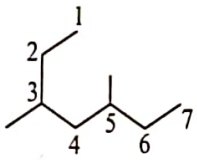
D.O.I = 6 + 1 = 7

**9.** (*c*) Fact based.

**10.** (*c*)  has heteroatom in ring hence it is heterocyclic compound.

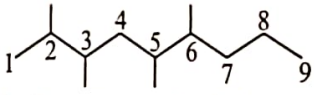
**11.** (*b*) Alkanal has $C_nH_{2n}O$ general formulae.

**12.** (*c*) $H_3CCH_2CHO$ is thrid member of homologus series of aldehyde.

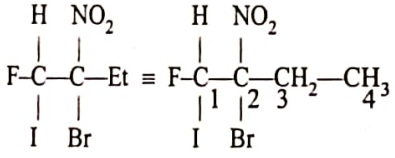
**13.** (*b*) 

3, 5 Dimethylheptane

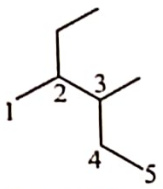
We start numbering of substituent from that end where substituent comes at locast locant.

**14.** (*c*) 

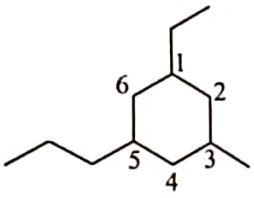
6- ethyl – 2, 3, 5 trimethyl nonane

**15.** (*c*) 

2-Bromo-1-fluoro-1-iodo-2-nitroethane

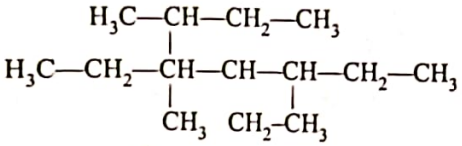
16. (*d*) 

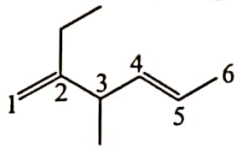
2–ethyl–3–methylpentane.

17. (*a*) 

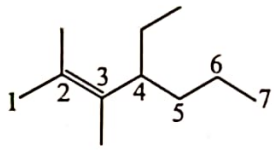
(i) Followed lowest locant rule.

(ii) Followed alphabatical order of substituents.

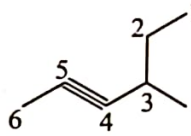
18. (*d*) 

19. (*b*) 

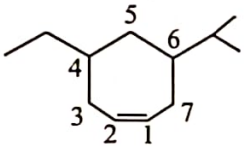
2-ethyl-3-methyl hexa 1, 4 diene

20. (*a*) 

4-ethyl–2, 3 dimethyl–2–heptene

21. (*c*) 

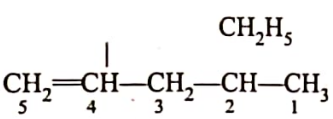
4–methylhexyne–4

22. (*d*) 

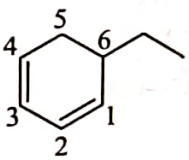
4–ethyl–6-(isopropyl) cycloheptene.

23. (*c*) $\overset{2}{C}H_2=\overset{6}{C}H-\overset{5}{C}H-\overset{4}{C}H_2-\overset{3}{C}H_2-\overset{2}{C}H\equiv\overset{1}{C}H$

Lowest locant rule

24. (*b*) 

s2–Ethylpent-4–ene

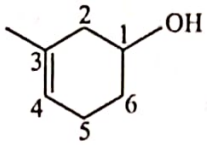
25. (*c*) 

6–Ethenylcyclohexa–1, 3–diene

$Mg^{+} \rightarrow Mg^{+2} + e^{-}$ $IP_1 = 348$ K cal mol$^{-}$

$Mg \rightarrow Mg^{+2} + 2e^{-}$ $IP = IP_1 + IP_2$

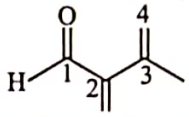
$= 348 + 178 = 526$ K cal

26. (*c*) 

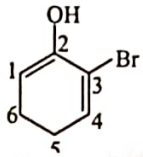
3–methyl cyclohex–3–enol

27. (*c*) $CH_3—NH—\underset{\overset{1}{C}H_3}{\underset{|}{CH}}—CH_2—\overset{2}{\underset{\overset{|}{CH_3}}{\underset{|}{CH}}}—\overset{3}{CH_3}$

2–(N–methylamino)–3–isopropylpropane

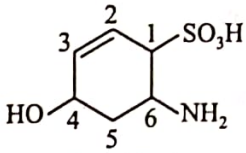
28. (*c*) 

3–Mehtyl–2–methylidene but–3–enal

29. (*b*) 

2 + 3 = 5

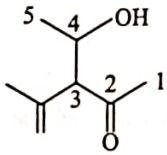
30. (*d*) $-CH > -\overset{\overset{O}{\|}}{C}-R > -OH > -NH_2$

31. (*b*) 

6–Amino–4–hydroxycyclohex–2–ene

32. (*b*) $I-\overset{1}{\underset{\underset{F}{|}}{C}}=\overset{2}{\underset{\underset{Cl}{|}}{C}}-\overset{3}{C}\equiv\overset{4}{C}-OCH_3$

4–methoxy–2–chloro–1–fluoro–1–iodobuten–3–yne

33. (*c*) 

3–(1–Methylethenyl)–4–hydroxy pentan–2–one

34. (*b*) $BrCH_2-\overset{2}{\underset{\underset{CONH_2}{|1}}{CH}}-\overset{3}{CO}-\overset{4}{CH_2}-\overset{5}{CH_2}\overset{6}{CH_3}$

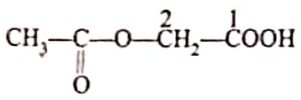
2–(Bromomeththyl)–3–oxohexanamide

35. (*b*) $CH3-\overset{3}{\underset{\underset{CH_2CH_3}{|1}}{CH}}-\overset{2}{CH2}-\overset{1}{CHO}$

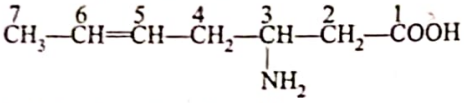
Correct Name: 3–methyl–pentanal

Given Name: 3–ehtyl–butanal in question

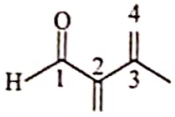
@PROCBSE

36. (*b*) 

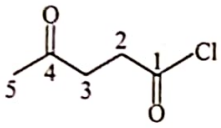
2–Acetoxyethanoic acid

37. (*c*) 

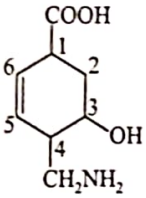
3–Aminohept–5–enoic acid

38. (*c*) 

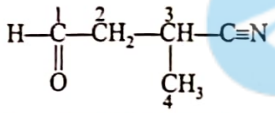
3–Methyl–2–methylenebu–3–enal.

39. (*c*) 

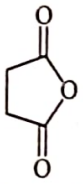
4–oxopentanoyl chloride

40. (*d*) 

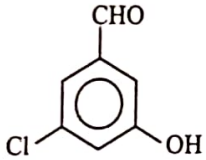
3–Hydroxy–4–aminomethylcyclohex–5–en–1–oic acid

41. (*c*) 

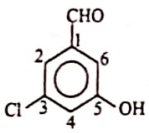
3–carbonitrile–butanal.

42. (*d*) 

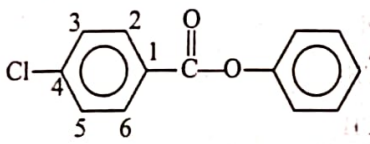
Butanedioic anhydride

43. (*c*) 

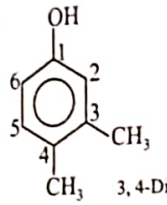
3–chloro–5–hydroxybenzene carbaldehyde

44. (*b*) 

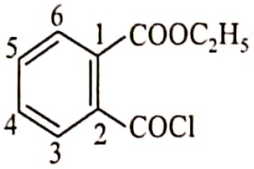
3–cyano–4–formyl benzoic acid

45. (*b*) 

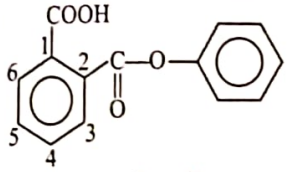
Phenyl–4–chlorobenzoate

46. (*a*) 

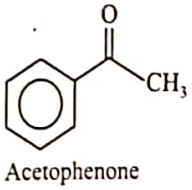
3, 4-Dimethylphenol.

47. (*c*) 

Ethyl–2–(Chlorocarbonyl) benzenecarboxylate.

48. (*a*) 

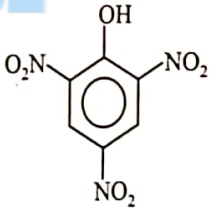
2–phenoxycarbonylbenzen carboxylic acid.

49. (*c*) 

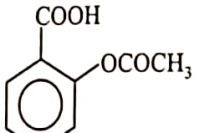
Acetophenone

50. (*c*) $CH_3CH{=}CH{-}COOH$

Crotonic acid

51. (*b*) 

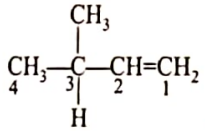
Picric acid

52. (*b*) 

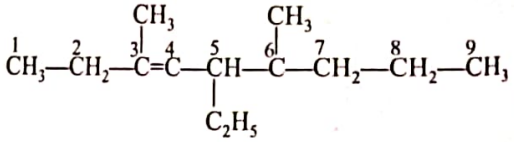
Aspirin

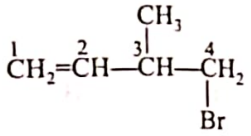
## Exercise-2 (Learning Plus)

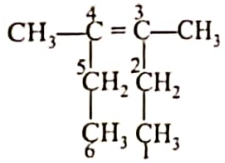
1. (*c*) 3-Methyl-1-butene

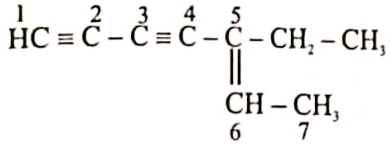


2. (*a*) 5-Ethyl-3, 6-dimethylnon-3-ene

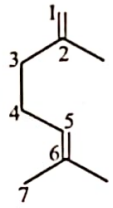


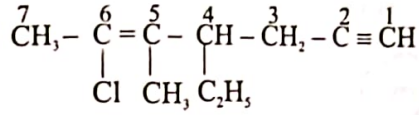
3. (*c*) 

4. (*c*) 

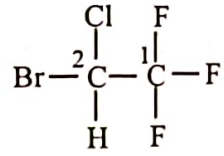
5. (*a*) 

5 Ethyl – 5 – heptene – 1, 3 – diyne

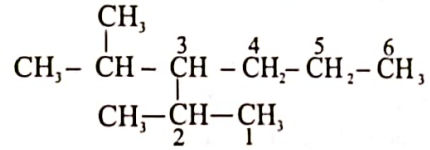
6. (*c*) 

7. (*a*) 

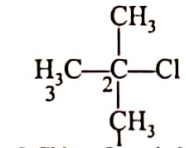
6 – Chloro – 4 – ethyl – 5 – methylhepten – 1 – yne

8. (*c*) 

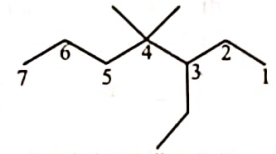
2 – Bromo – 2 – chloro – 1, 1, 1, trifluoroethane

9. (*a*) 

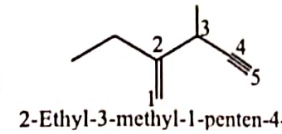
3 – Isopropyl – 4 – methylhexane

10. (*d*) 

2-Chloro-2-methylpropane

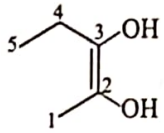
11. (*d*) 

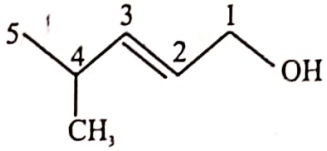
3-Ethyl-4, 4-dimethylheptane

12. (*a*) 

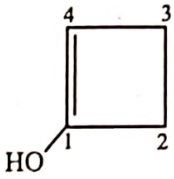
2-Ethyl-3-methyl-1-penten-4-yne

13. (*b*) Pent-2-ene-2, 3-diol

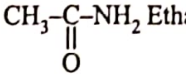


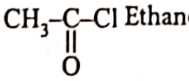
14. (*d*) 

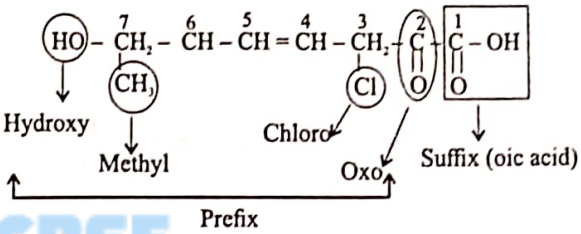
4 – methyl pent – 2 – en – 1 – ol

15. (*a*) 

Cyclobut – 1 – en – 1 – ol

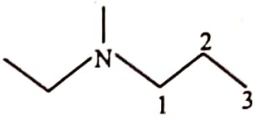
16. (*d*)  Ethanamide

 Ethanoyl chloride

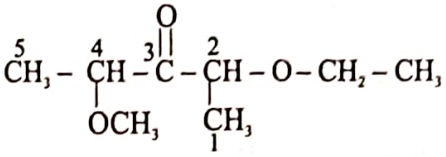
17. (*b*) 

Parent chain → 4 – heptene

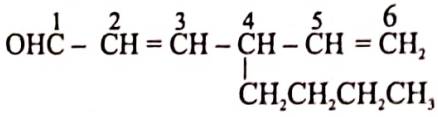
– oic acid, chloro, hydroxy, methyl, oxo, 4 – heptene

18. (*b*) 

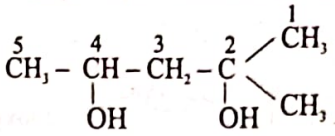
N – Ethyl – N – methylpropan – 1 – amine

19. (*a*) 

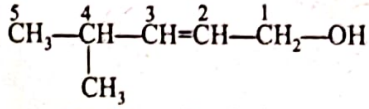
2 – Ethoxy – 4 – methoxy pentan – 3 – one

20. (*a*) 

4 – Butyl – 2, 5 – hexadien –1– al

21. (*c*) 

2 – Methyl – 2, 4 – pentane diol

22. (*d*) 

4-Methyl pent-2-en-1-ol

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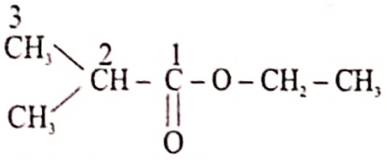
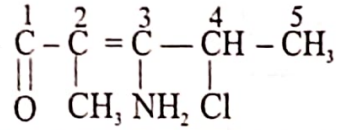
**23.** (*c*) $C_6H_5-\overset{3}{C}H=\overset{2}{C}H-\overset{1}{C}OOH$

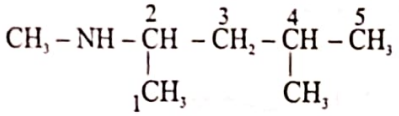
3 – Phenylprop – 2 – enoic acid

**24.** (*c*) 

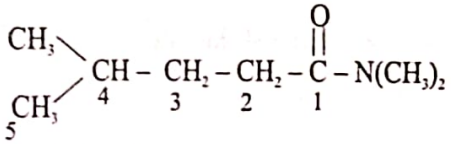
Ethyl – 2 – methylpropanoate

**25.** (*c*) 

3 – amino – 4 – chloro – 2 – methyl – 2 – pentenoic acid

**26.** (*b*) 

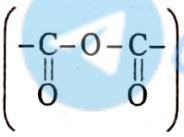
N, 4 – dimethylpentan – 2 – amine

**27.** (*a*) 

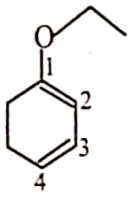
N, N, 4 – trimethylpentanamide

**28.** (*d*) $C_6H_{12}O_3$

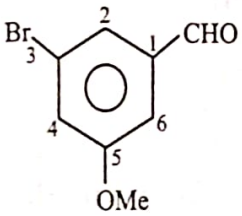
Degree of unsaturation $= 6 + 1 - \frac{12}{2} = 1$

so, it can't be anhydride 

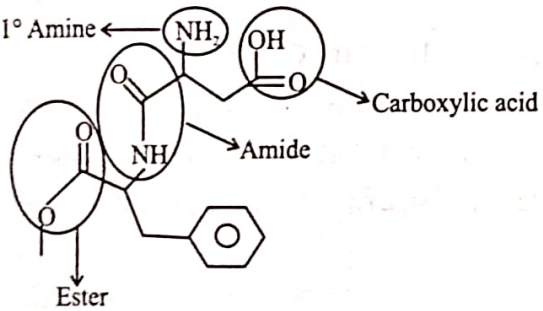
because degree of unsaturation is 2 here.

**29.** (*a*) 

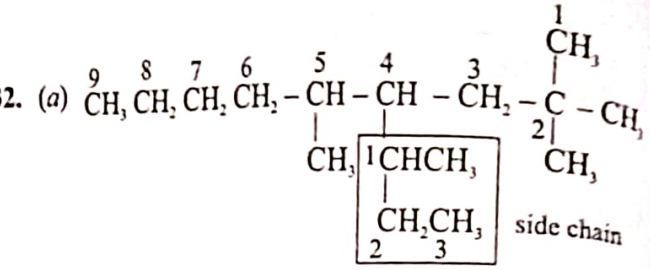
1 – Ethoxy cyclohexa – 1, 3 – diene

**30.** (*d*) 

3 – Bromo – 5 – methoxy benzene carbaldehyde

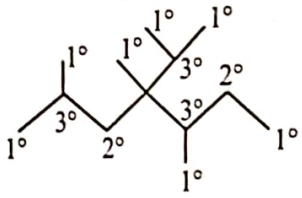
**31.** (*a*) 

No. of functional group = 4

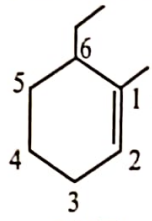
**32.** (*a*) 

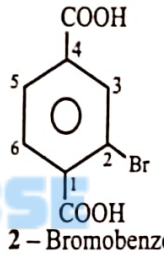
2, 2, 5 – Trimethyl – 4 – (1 – methylpropyl) nonane

**33.** (*b*) 4 – Isopropyl – 2, 4, 5 – trimethylheptane

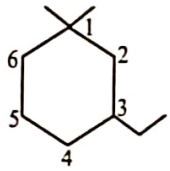


$1° H = 7 \times 3 = 21 \quad 2° H = 2 \times 2 = 4 \quad 3° H = 1 \times 3 = 3$

**34.** (*d*)  Alkene is most preferred.

**35.** (*c*) 

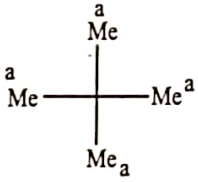
2 – Bromobenzene – 1, 4 – dicarboxylic acid

**36.** (*c*) 

3 – Ethyl – 1, 1 – dimethylcyclohexane

Numbering will start from Ethyl

→ Follow alphabetical order.

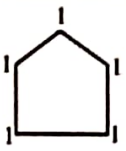
**37.** (*c*) 

Molecular wt. $= 15 \times 4 + 12 = 72$

Only 1 type (a) of hydrogen present so only one monochlorination product will be formed.

**38.** (*b*) $C_5H_{10} \rightarrow DBE = 5 + 1 - \frac{10}{2} = 1$

1 DBE means 1 ring or 1 double bond.



only 1 type of hydrogen so only one product on monochlorination.

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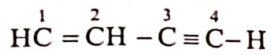
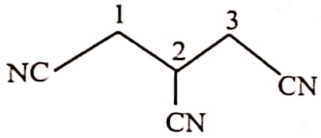
39. (*c*) Vinylacetylene



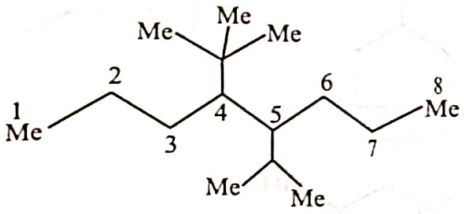
But – 1 – en – 3 – yne

40. (*a*)



Propane – 1, 2, 3 – tricarbonitrile

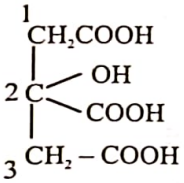
41. (*c*)



4 – t – Butyl – 5 – isopropyloctane

42. (*d*) All members have the same physical properties.

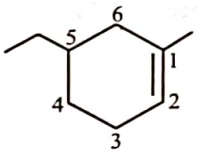
43. (*b*)



2 – Hydroxypropane – 1, 2, 3 – tricarboxylic acid

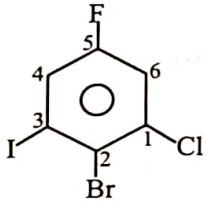
44. (*c*)

45. (*b*)



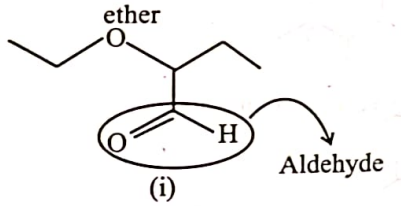
5 – Ethyl – 1 – methylcyclohexane

46. (*b*)

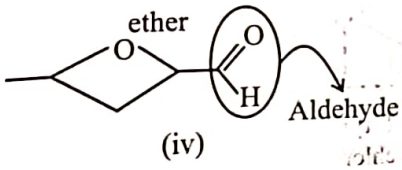


2 – Bromo – 1 – chloro – 5 – fluoro – 3 – iodobenzene

47. (*b*)

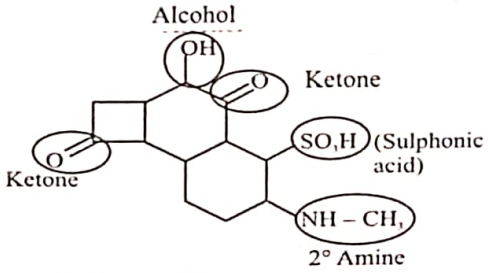


(i)



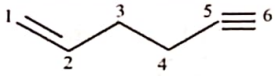
(iv)

48. (*d*)



Alcohol is present here.

49. (*b*)

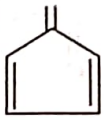


Hex – 1 – en – 5 – yne

## Exercise-3 (JEE Advanced Level)

1. (*a,b,d*)

$C_6H_6$
$C_nH_n$



$C_6H_6$
$C_nH_n$

Both compounds have same general formula.

(*d*)

$C_6H_6$
$C_nH_{2n-6}$

$H-C\equiv C-C\equiv C-H$

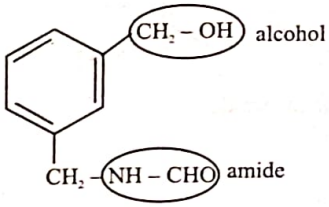
$C_4H_2$
$C_nH_{2n-6}$

(*c*)

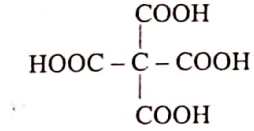
$C_6H_{10}$
$C_nH_{2n-2}$

$C_5H_8$
$C_nH_{2n-2}$

2. (*c,d*)

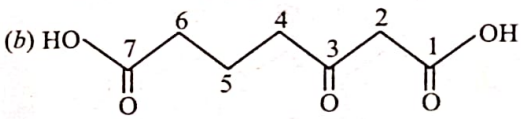


3. (*a,c*) (*a*)



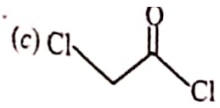
Methane – 1,1,1,1 – tetracarboxylic acid

(*b*)

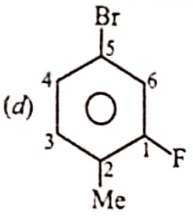


5 – oxo heptane – 1, 7 – dioic acid

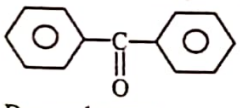
@PROGRSE

(c) 

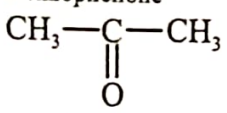
2 – chloroethanoylchloride

(d) 

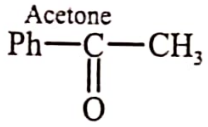
1 – fluoro – 2 – methyl – 5 – bromocyclohexane

**4.** (b,d) (a) 

Benzophenone

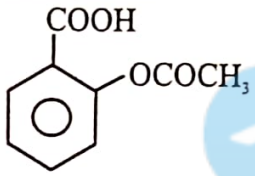
(b) 

Acetone

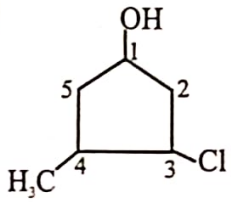


(c)

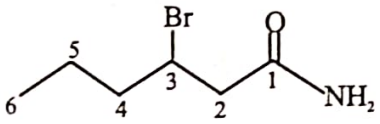
Acetophenone

(d) 

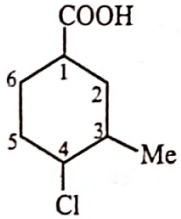
Asprin

**5.** (a,b,d) (a) 

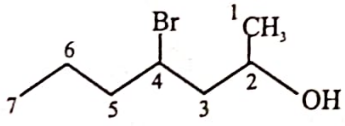
3 – chloro – 4 – methylcyclopentanol

(b) 

3 – bromohexanamide

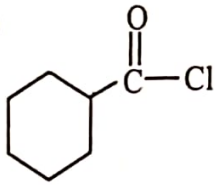
(c) 

4 – chloro – 3 – methylcyclohexane carboxylic acid

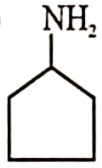
(d) 

4–bromoheptan – 2 – ol

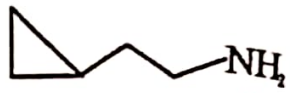
**6.** (b,c) Homologous have same general formula but differ in molecular weight by – CH2 and contain same functional group.

**7.** (a,c,d) 

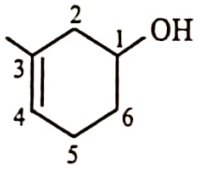
Cyclohexane carbonyl chloride

**8.** (a,b) 

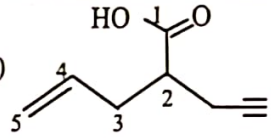
$C_5H_{11}N$



$C_5H_{11}N$

**9.** (c) 

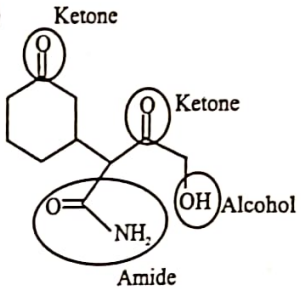
3 – Methylcyclohex – 3 – enol

**10.** (a,b,c) 

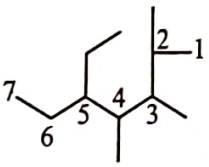
(a) Double bond equivalent = 4

(b) 3 functional group = Alkene, Alkyne, Carboxylic acid

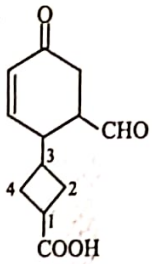
(c) 2 – (prop – 2 – ynyl) pent – 4 – enoic acid

**11.** (a,b,d) 

**12.** (b,c,d) Fact based

**13.** (a) 

5-Ethyl-2,3,4-trimethylheptane

**14.** (b,c,d) 

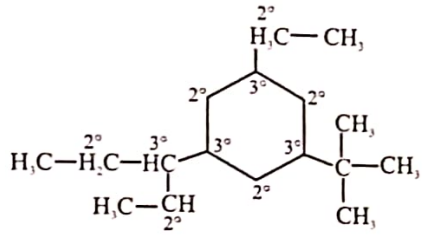
@PROCBST

(*b*) Carbon in parent chain = 4

(*c*) IUPAC name is 3 – (6 – formyl – 4 – oxocyclohex – 2 – enyl) cyclobutane carboxylic acid

(*d*) Number of 2° carbon are 6.

**Sol.15 - 17**



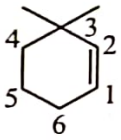
**15.** (*c*) 3° hydrogen = 4

**16.** (*c*) 2° hydrogen = 12

**17.** (*c*) 3 substituents → Ethyl ; 1- ethylpropyl ; 1, 1 - dimethyl ethyl

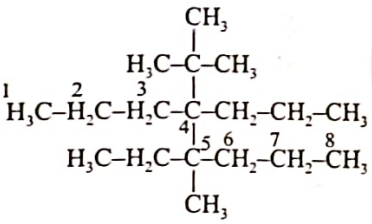
* 1 – (1, 1– Dimethylethyl) – 3 – ethyl – 5 – (1 – ethylpropyl) cyclohexane

**18.** (*c*)



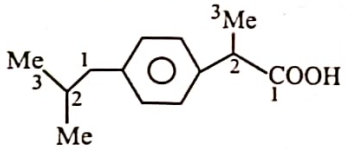
3,3-Dimethylcyclohexene

**19.** (*d*)



**20.** (*b*) –COOH group

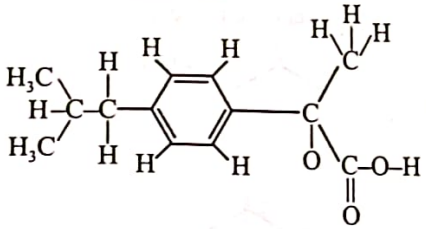
**21.** (*b*)



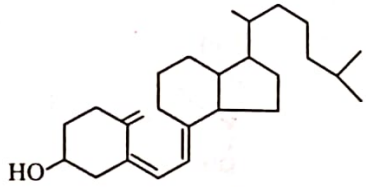
2(p–Isobutylphenyl)propanoic acid

**22.** (*c*) No. of π bonds = 4

**23.** (*d*) No. of σ bonds = 33

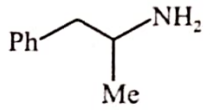


**24.** (*a*) (A)



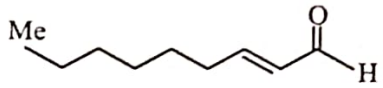
2° Alcohol and Triene

(B)



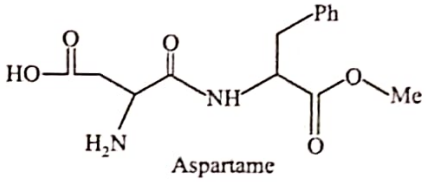
1° Amine

(C)



Aldehyde and ene

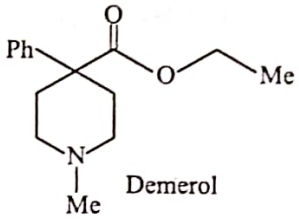
**25.** (*b*) (A)



Aspartame

q, r

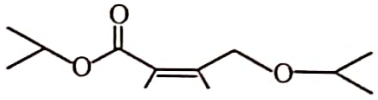
(B)



Demerol

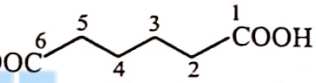
r,s

(C)



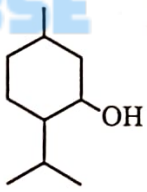
r

**26.** (*d*) (A)



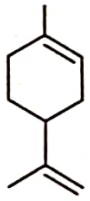
p, s

(B)



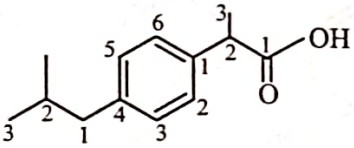
p, s

(C)



p, s

(D)



2-[4-(2-methyl propyl)phenyl]

Propanoic acid

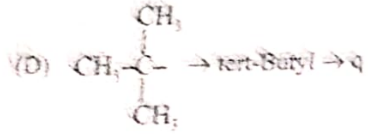
q, r, t

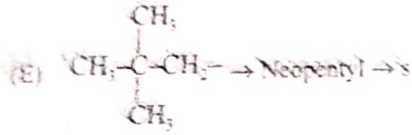
**27.** (*a*) (A) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-$ → Isopropyl → r

(B) $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{|}{CH}}-$ → sec-Butyl → p

(C) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-CH_2-$ → Isobutyl → t

@PROCBSE

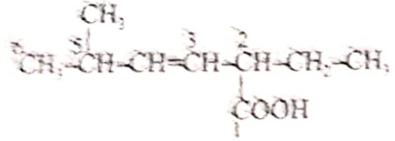
(D)  → tert-Butyl → q

(E)  → Neopentyl → s

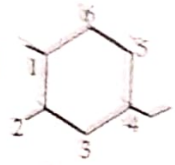
28. (d) Fact based

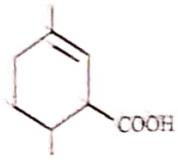
29. [6] Do it yourself

30. [5] Do it yourself

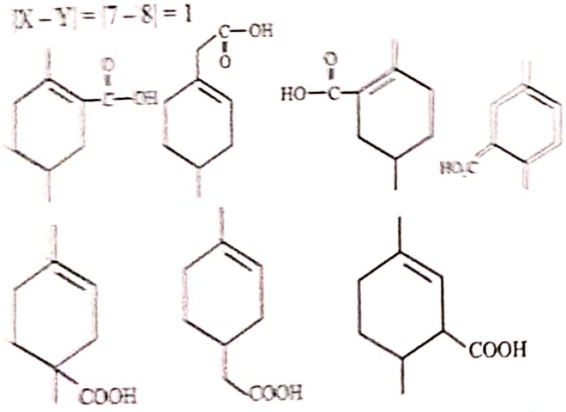
31. [14] 

x = 6
y = 5
z = 3
x + y + z = 14

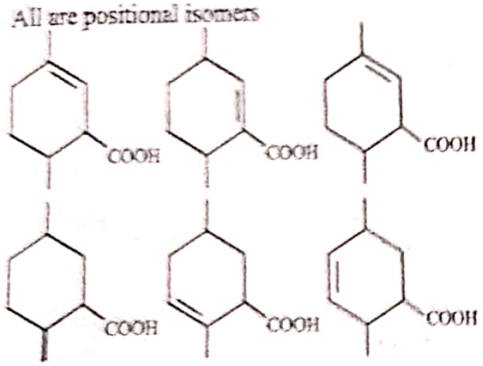
32. [7] 

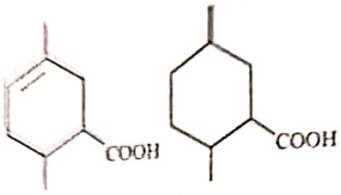
33. [1] 

X = 7
Y = 8
|X – Y| = |7 – 8| = 1



All are positional isomers

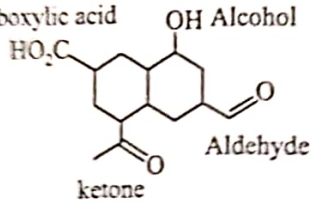




All are positional isomers

34. [8]

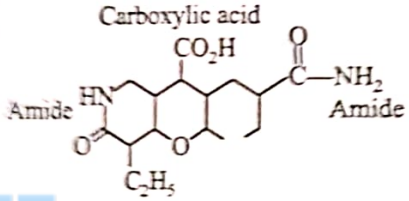
σ = 11
π = 3
σ – π = 8

35. [4] Carboxylic acid  Alcohol

Aldehyde

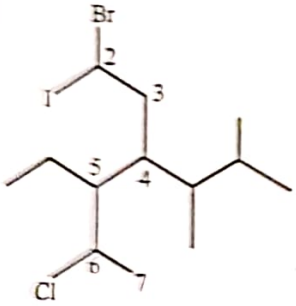
ketone

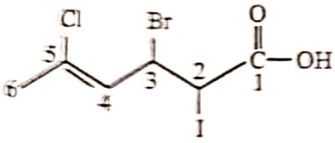
Total 4 different functional groups are present.

36. [5] Carboxylic acid 

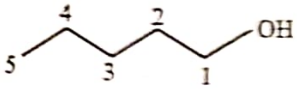
Amide Amide

Total 3 different functional groups are present.

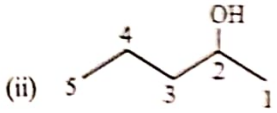
37. [4] 

38. [3] 

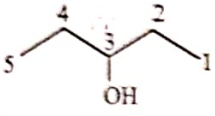
Total 3 substituent are present

39. [0] (i) 

Pentan–1–ol

(ii) 

Pentan–2–ol

(iii) 

Pentan–3–ol

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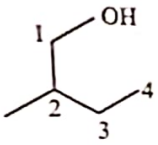
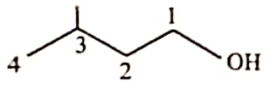
(iv) 

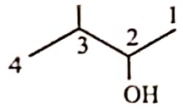
2–Methylbutan–2–ol

(v)

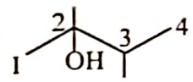
2, 2–Dimethylbutan–1–ol

(vi) 

3–Methylbutan–1–ol

(vii) 

3–Methylbutan–2–ol

(viii) 

sss2–Methylbutan–2–ol

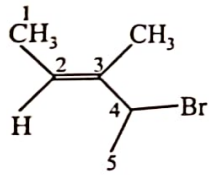
## Exercise-4 (Past Year Questions)

### JEE MAIN

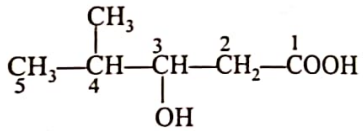
**1.** (*d*) Glycerol which decomposes at its boiling point can be purified by distillation under reduced pressure.

At normal pressure, boiling point of glycerol is 563 K, at this temperature it decomposes.

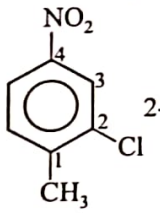
But at reduced pressure of 12 mm Hg, glycerol can be distilled at 453K without decomposition.

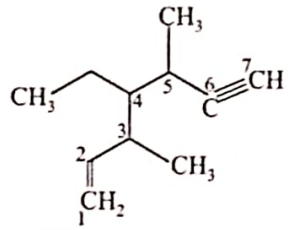
**2.** (*d*) 

4-Bromo-3-methylpent-2- ene

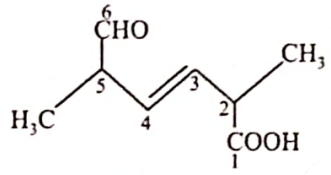
**3.** (*c*) 

3 – Hydroxy– 4 – methylpentanoic acid

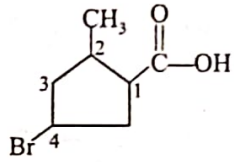
**4.** (*d*)  2-Chloro-1-methyl-4-nitrobenzne

**5.** (*d*) 

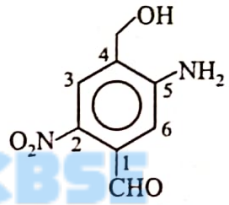
3,5-Dimethyl-4-propylhept-1-en-6-yne

**6.** (*d*) 

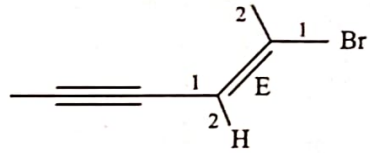
2, 5-Dimethyl-6-oxo-hex-3-enoic acid.

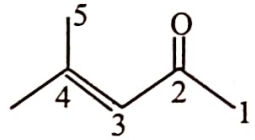
**7.** (*d*) 

4-Bromo-2-methylcyclopentane carboxylic acid.

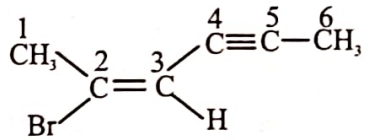
**8.** (*b*) 

5-Amino - 4 - hydroxymethy l - 2-nitrobenzaldehyde

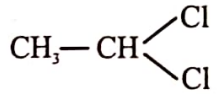
**9.** (*c*) 

**10.** (*d*) 

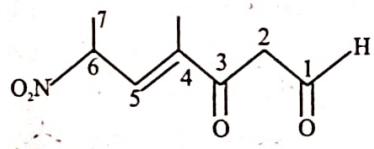
4 – methylpent – 3 – en – 2 – one

**11.** (*c*) 

(2E) – 2 – bromo – hex – 2 – en – 4 – yne

**12.** (*d*)  (ethylidine chloride)

1,1 – dichloroethane

**13.** (*c*) 

4 – Methyl – 6 – nitro – 3 – oxohept – 4 – enal

14. (*b*) Priority order

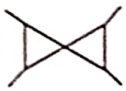
$$-COOH > -SO_3H > -\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}- > -\overset{O}{\overset{\|}{C}}-$$

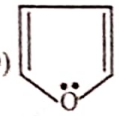
$$OR > -\overset{O}{\overset{\|}{C}}-X > -\overset{O}{\overset{\|}{C}}-NH_2 > -C\equiv N > -N\equiv C > -CHO$$

$$> -\overset{O}{\overset{\|}{C}}- > -OH > -SH > -NH_2$$

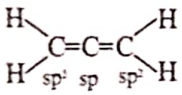
15. (*c*) (A) (r) Non planar Heterocyclic compound

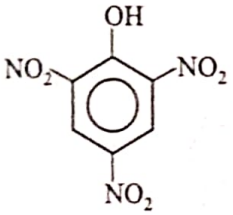
(B) (s) Bicyclo compound

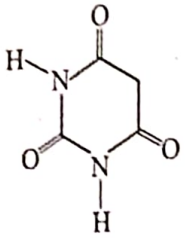
(C)  (p) Spiro compound

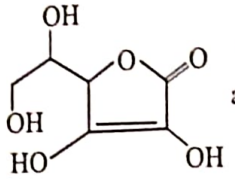
(D)  (q) Aromatic compound

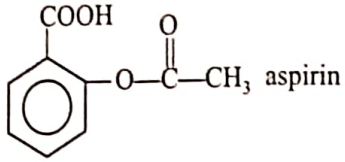
## JEE ADVANCED

16. (*b*) 

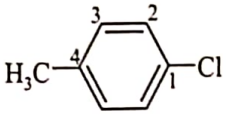
17. (*d*)  picric acid

 barbituric acid

 ascorbic acid;

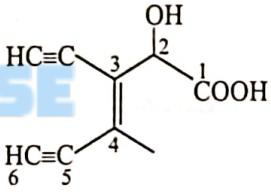
 aspirin

18. (*b,c*)



1-Chloro-4-methylbenzene

4-Chlorotoluene

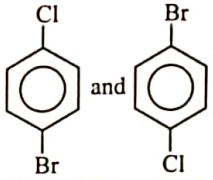
19. (*d*) 

3-Ethynyl-2-hydroxy-4-methyl-hex-3-en-5-ynoic acid

@PROCBSE

# 13 Isomerism

## Exercise-1 (Topicwise)

**1.** (*b*)  and  → Identical
→ 1-Bromo-4-chlorobenzene

**2.** (*d*) $CH_3—CH_2—O—CH_2—CH_3$ (Diethylether)

(*a*) $CH_3—CH_2—CH_2—O—CH_3$

(*b*) $CH_3—CH_2—CH_2—CH_2—OH$

(*c*) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—OH$

(*d*) $CH_3—\overset{\displaystyle O}{\overset{\|}{C}}—CH_2—CH_3$

For isomerism molecular formula must be same.

For isomerism molecular formula must be same.

Molecular formula of diethylether is $C_4H_{10}O$

Molecular formula of butanon is $C_4H_8O$

**3.** (*b*) For position isomerism

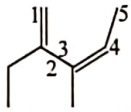
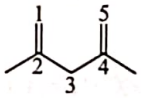
(*a*) Molecular formula = same

(*b*) Parent chain = same

(*c*) Substituent = same

(*d*) Function group = same

(*e*) Position of substitutent/Function group = different

 and 

**4.** (*a*) 2-Ethyl-3-methylpenta-1,3-diene 2,4-Dimethylpenta-1,4-diene

IUPAC name = different

Molecular formula = same

Parent chain = same

Substituent are different

Chain Isomers

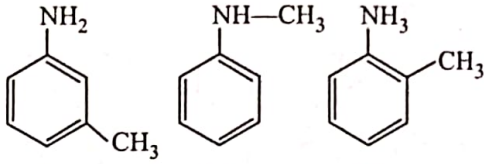
**5.** (*d*) Molecular formula = same

Parent chain = same

Substitutent/Function group = same

Position of substitutent/Function group = different

Position Isomers

**6.** (*d*) 

I II III

1°, 2° and 3° amines are functional isomers

**7.** (*b*) For chain Isomer

Molecular formula (M.F) = same

PC = different

Substitutent = different

Functional group (F.G) = SAME

**8.** (*d*) For isomers molecular formula = same

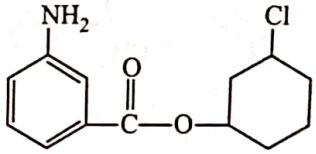
Degree of unsaturation = same

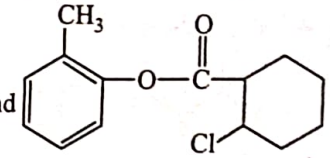
**9.** (*d*) For metamers: Molecular formula (M.F) = same

F.G. = same but polyvalent

Alkyl group = different

All option are correct

**10.** (*b*) 

and 

Molecular formula = same

Functions group = same but polyvalent

Alkyl group = different

Metamers

**11.** (*d*) For metamers

Functional group = bivalent or polyvalent.

**12.** (*b*) Molecular formula = same

Functional group = same but polyvalent

Alkyl group = different

Metamers

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13. [4] $C_4H_9Cl \longrightarrow C_4H_{10}$ (Isomers)

$CH_3-CH_2-CH_2-CH_3$ and $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(2 types of H-atom) — 2; (2 types of H-atom) — 2

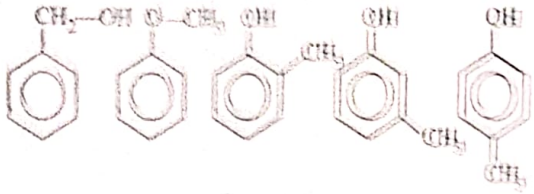
No. of monocholoro product = Types of H-atoms

Total structural isomers = 2 + 2 = 4

14. (a,b,c)

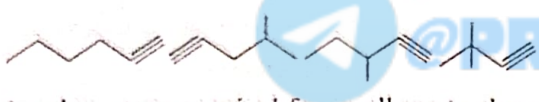
Fact based

15. [3] Formula of compound is not given (correction))



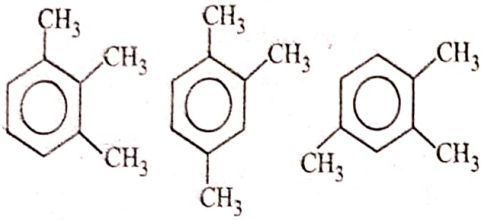
16. (d)

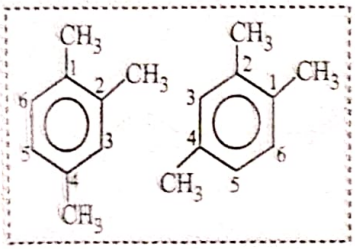
17. (d)



18. (b) 4 carbon atoms required for an alkane to show isomerism.

19. [3]





These both structures are same.

20. (c) $C_4H_9Cl \longrightarrow C_4H_{10}$ (Isomers)

$CH_3-CH_2-CH_2-CH_3$ and $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(2 types of H-atom) — 2; (2 types of H-atom) — 2

No. of monocholoro product = Types of H-atoms

Total structural isomers = 2 + 2 = 4

21. (b) $C_5H_{11}Br \xrightarrow{-HBr} C_5H_{12} \xrightarrow{Cl_2(1)}$

(i) $CH_3-CH_2-CH_2-CH_2-CH_3 \longrightarrow 3$ S.I.

(ii) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3 \longrightarrow 4$ S.I

(iii) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3 \longrightarrow$ 1 S.I. (Total = 8 S.I.)

Only one tertiary alkyl bromide is possible

$CH_3-\underset{\underset{CH_2-CH_3}{|}}{\overset{\overset{Br}{|}}{C}}-CH_2-CH_3$

22. (b)

(i) $CH_3-CH_2-CH_2-CH_2-O-CH_3$ (1-methoxybutane)

(ii) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-O-CH_3$ (1-methoxy-2-methylpropane)

(iii) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-O-CH_3$ (2-methoxybutane)

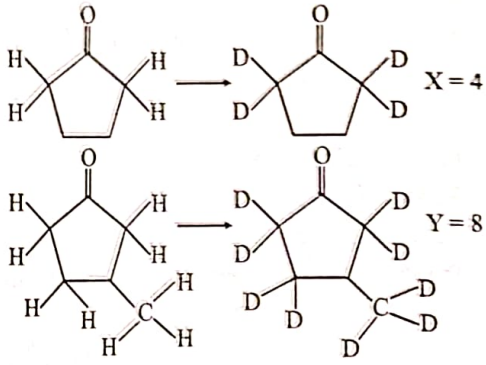
(iv) $CH_3-CH_2-O-CH_2-CH_3$ (2-ethoxyprpane)

(v) $CH_3-\underset{\underset{CH_3}{|}}{CH}-O-CH_2-CH_3$ (2-ethoxypropane)

(vi) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-O-CH_2-CH_3$ (2-ethoxy-2-methylpropane)

23. (b) In (b) option there is no α H-atom

24. (d)



X = 4

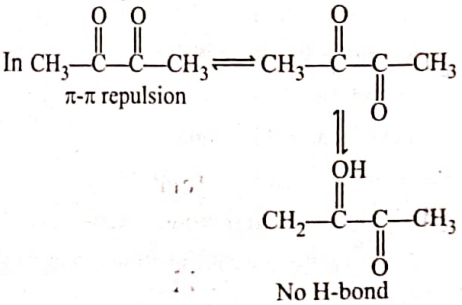


Y = 8

X + Y = 4 + 8 = 12

25. (d) All having enolisable H/D atom.

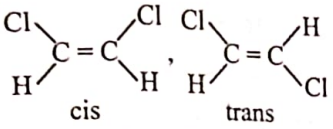
26. (a) % enol content α aromaticity (Aromatic comb) α H-bonding



π-π repulsion

No H-bond

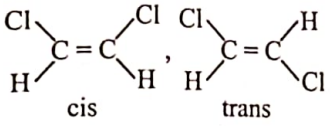
@PROCBSE

**27.** (*c*) 
cis trans

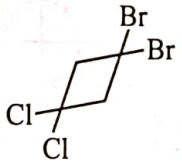
Condition for G.I.

Restriction rotation about double bond

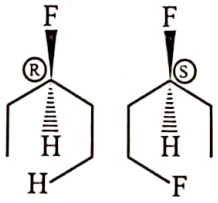
Double bonded C-atom should have different valency.

**28.** (*d*) 
cis trans

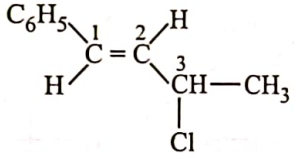
**29.** (*a*) Restricted rotation but the groups attached with C-atom is not different no. G.I.



**30.** (*a*) Ring structure having double bond less than 8C- atom does not show G.I.



**31.** (*b*) Bisubstituted cycloalkane having different position of substitutents.

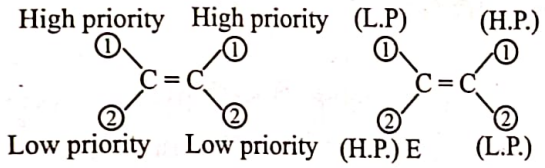
**32.** (*d*) 

**33.** (*d*) Fact based

**34.** (*d*) Fact based

**35.** [2] Fact based

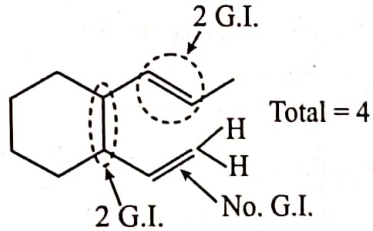
**36.** (*a*) Fact based

**37.** (*c*) 
High priority High priority (L.P) (H.P.)
Low priority Low priority (H.P.) E (L.P.)

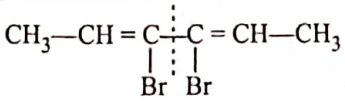
**38.** (*c*) Syn → –OH group and bulky group same side.

i.e. –OH and Et group same side.

**39.** (*a*) In Z configuration higher priority atoms on same side.

**40.** (*b*) 
Total = 4

**41.** (*b*) For symmetrical molecule

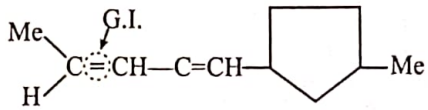


$$G.I. = 2^{n-1} + 2^{P-1} \qquad \left(P = \frac{n}{2} \text{ if } n = \text{even}\right)$$

$$= 2^{n-1} + 2^{P-1}$$

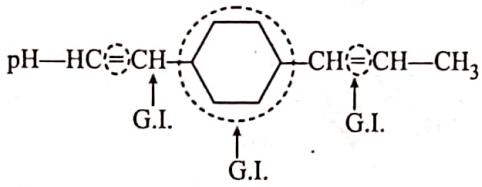
$$= 2^1 + 2^0$$

$$= 2 + 1 = 3$$

**42.** (*b*) 

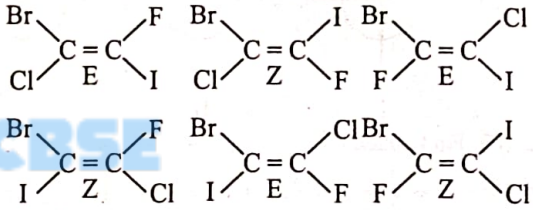
Unsymmetrical molecules

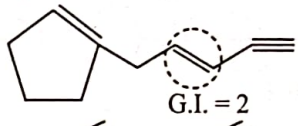
No. of G.I. = $2^n = 2^3 = 8$

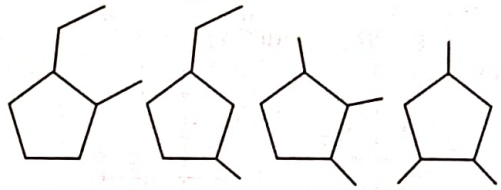
**43.** (*d*) 

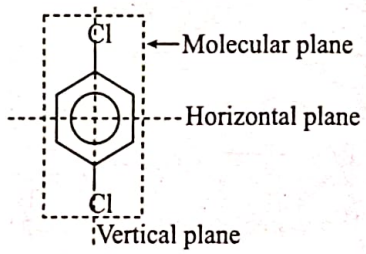
Unsymmetrical molecules

No. of G.I. = $2^n = 2^3 = 8$

**44.** (*d*) 

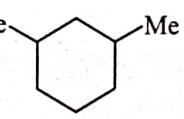
**45.** (*a*) 

**46.** [4] 

**47.** (*b*) 

**48.** (*d*) Fact based

**49.** (*c*) Fact based

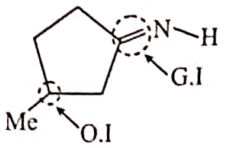
**50.** (*a*) Correction in Y structure 

**51.** (*a*) Fact based

**52.** (*a*)

@PROCBSE

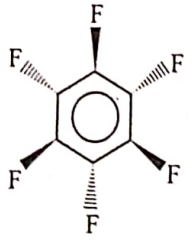
**53.** (*c*)




**54.** (*c*) Fact based

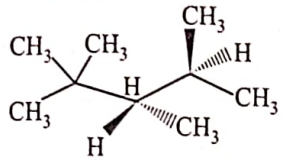
**55.** (*d*) Fact based

**56.** (*c*)

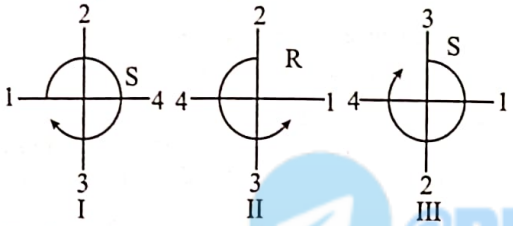



POS + COS

**57.** (*d*)

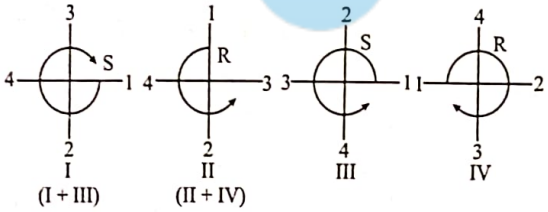



**58.** (*d*)



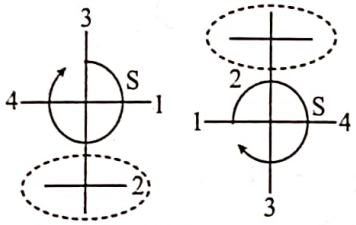

**59.** (*d*) Fact based

**60.** (*c*)

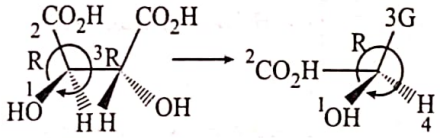



(I + III) (II + IV)

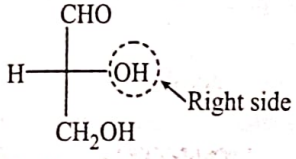
**61.** (*c*)




**62.** (*a*)




**63.** (*a*)




**64.** (*c*) Fact based

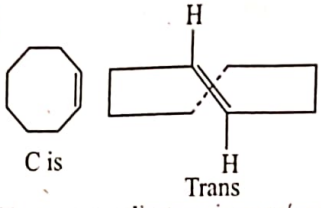
**65.** (*c*) C is and trans Diastereomer

8 members ring containing double bond show G.I.

Less than 8 members ring contain double bond do not show G.I.

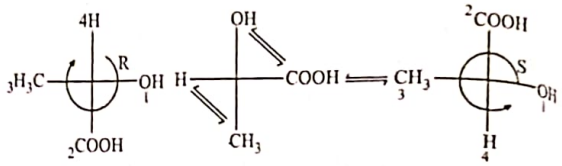
**66.** (*a*) Fact based

**67.** (*a,d*)

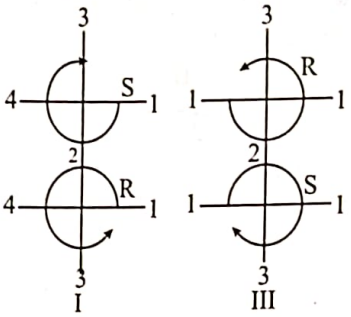



Cis, trans are diastereoisomer/geometrical isomer

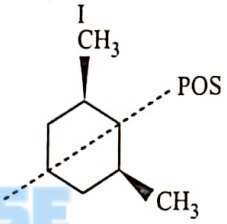
**68.** (*b*)




**69.** (*a*)



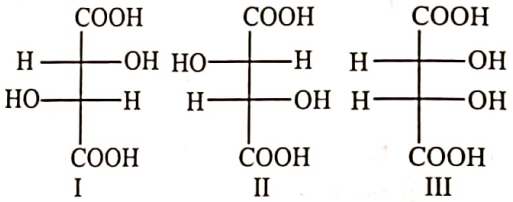

**70.** (*b*)




**71.** (*c*) Fact based

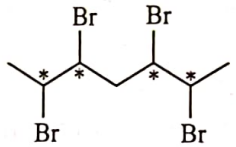
**72.** (*b*) Compound having even number of chiral centre with similar ends steroisomer = O.I. + Meso

$= 2^{n-1} + 2^{n/2-1} = 2^{2-1} + 2^{2/2-1} = 3$

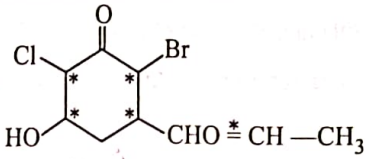



**73.** (*c*) Stereoisomer $= 2^{n-1} + 2^{n/2-1} = 2^{4-1} + 2^{4/2-1}$

$= 2^3 + 2^{2-1} = 8 + 2 = 10$

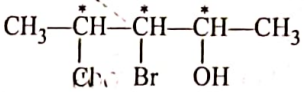



**74.** (*d*) Steroisomer $= 2^n = 2^5 = 32$



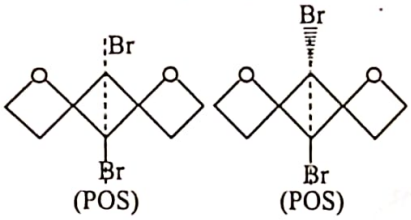

**75.** (*d*) Compound having no mirror image

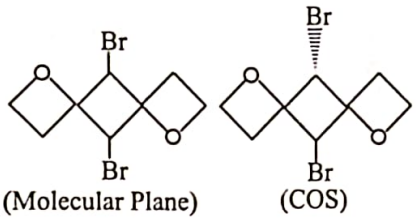
**76.** (*c*) Steroisomer $= 2^n = 2^3 = 8$

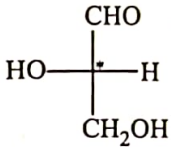



@PROCBSE

**77.** (*a*) All isomer are optically inactive




**78.** (*c*)  ⇒ Optically active

Unsymmetrical compound = $2^n$

n = 3

Total stereoisomers = $2^3 = 8$

**79.** (*a*) Case of Pseudo chiral centre

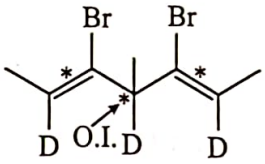
C R T

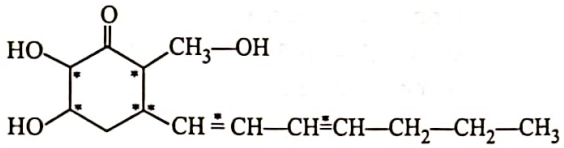
T R C

C S T

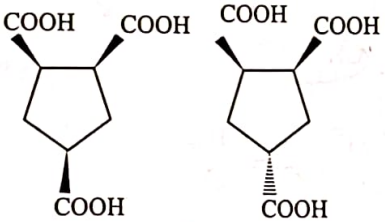
T S C

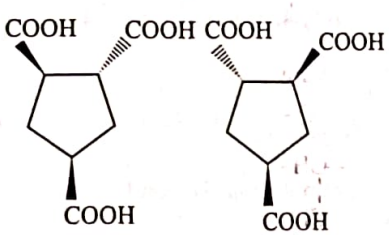
$2^{n-1} = 2^{3-1} = 2^2 = 4$

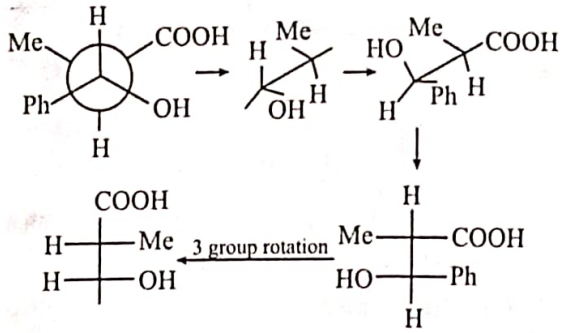


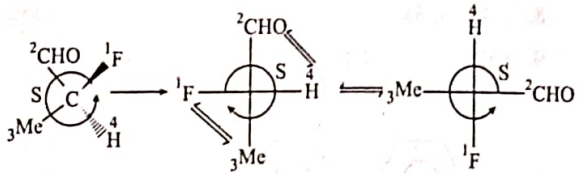

**80.** (*d*) 

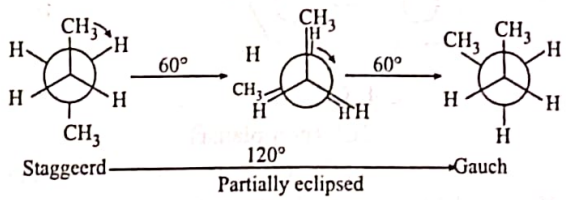
Total Stereoisomer = $2^n = 2^6 = 64$

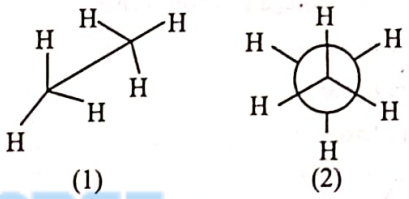
**81.** (*c*) 


**82.** (*c*) 


**83.** (*a*) 

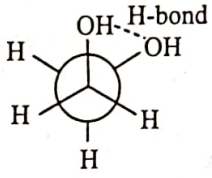

**84.** (*c*) 


**85.** (*c*) 


(1) (2)

Both are staggered conformation of ethane.

**86.** (*c*) Due to intra H-bond in Gauch is more stable than anti in ethylene glycol.




Gauch

**87.** (*a*) Conformer

**88.** (*d*) Due to intra H-bonding.

**89.** (*d*) Flag pole interaction.

**90.** (*d*) PE.(Chair) < PE.(Twist boat) < PE(Boat) < PE(Calf Chair)

**91.** (*c*) Stability = Chair form > Boat form

Equatorial conformer > Axial conformer.

**92.** (*b*) Theory based

**93.** (*d*) Stability equatorial > Axial

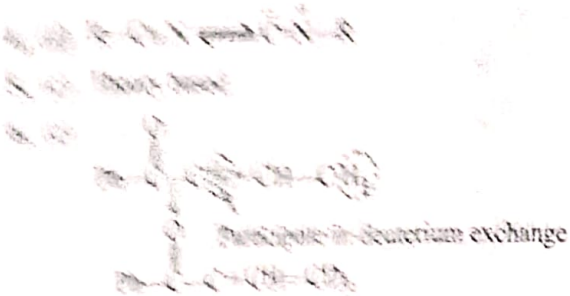
## Exercise-2 (Learning Plus)

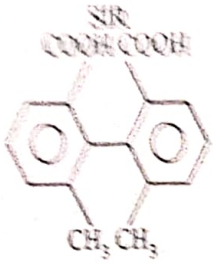
**1.** (*b*) For isomer molecular formula should be same.

**2.** (*c*) Theory based

**3.** (*c*) Theory based

@PROCBSE





SIR (non-planar)

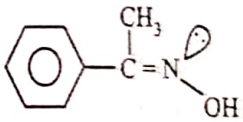
POS and COS = absent

10. (*a*) Due to SIR molecule is non-planar

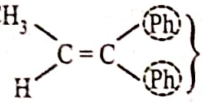
11. (*a*) Theory based

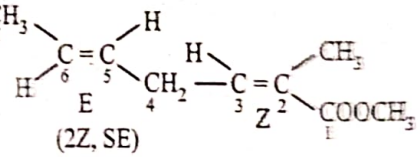
12. (*b*) Theory based

13. (*d*) Theory based

14. (*c*) 

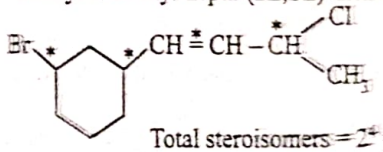
15. (*d*) Theory based

16. (*d*)  same group attached with double bonded c-atom → NO GI

17. (*d*) 

(2Z, SE)

methyl-2-methyl hepta-(2Z,SE)-diene

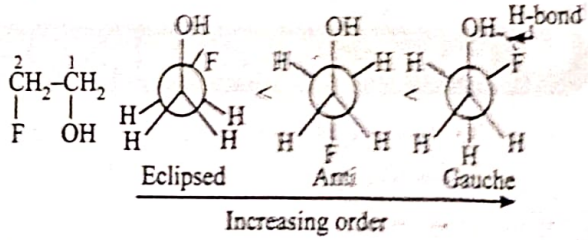
18. (*c*) 

Total steroisomers = $2^4 = 16$

19. (*a,b,c*) Molecular formula = same

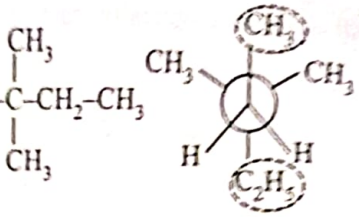
DBE or Du = Same for structural isomers

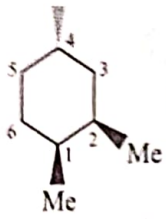
20. (*a*) Theory based

21. (*c*) 

Eclipsed < Anti < Gauche

Increasing order

22. (*b*) 

23. (*a*) 

1e, 2a, 4e e = equitorial

a = axial

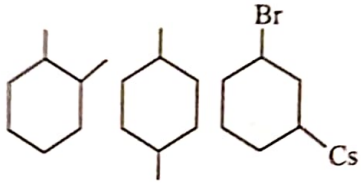
24. (*d*) Theory based

Dipole moment = Cis > trans

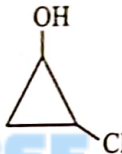
B.P = Cis > Trans (d.p-dp interaction moreno cis)

M.P. = Trans > cis (symmetrical str. of trna)

Stability = trans > cis

25. (*d*) 

All show G.I due to having different valencies at different $sp^3$ atom

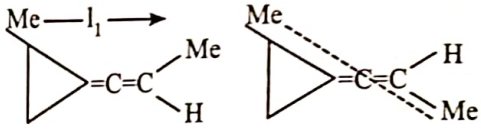
26. (*a*) 

Mol. formula = $C4H_8O$

Mol mass = 72

$\therefore \frac{x}{9} = \frac{72}{9} = 8$

27. (*a*) for G.I → $l_1 \neq l_2$



In all option $l_1 \neq l_2$

28. (*d*) Theory based

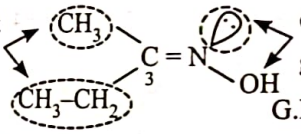
29. (*d*) NO G.I → if Total double bond and ring = even

Terminal end = different

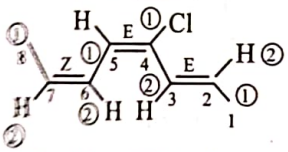
Show G.I → if Total double bond and ring = odd

Terminal = different

30. (*d*) I and III = G.I

31. (*c*) different group  different group

G.I = possible

32. (*c*) 

(2E, 4E, 6Z)-4- chlooc + a- 2, 4, 6-triene

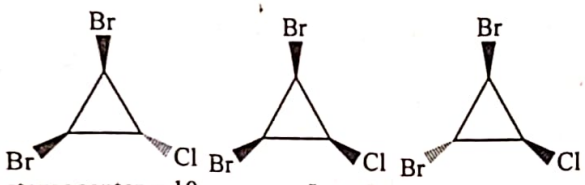
H.P H.P H.P

opposite side same side

33. (*a*) Intramolecular H-bond

@PROCBSE

34. [3]



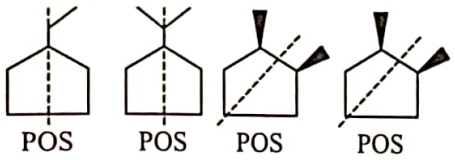

35. (*c*) stereocenter = 10
stereocentre = CC + GC
1 GC = 2
I = 3
II = 4
III = 1
IV = 2
} Total = 10

36. (*b*) Total chiral C-atom = 8

37. (*d*) Total chiral C-atom = 8

38. (*d*) Both $C_3$ + POS

39. (*b*)




40. (*a*) Theory based (Anti form have COS)

41. (*a*) Due to SIR b/w Bromine.

42. (*b*) All having POS but option 'b' having POS + COS

43. (*a*) (A) – POS

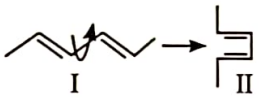
(B) – POS, chiral centre, optically in active but will show optical isomer.

(C) – COS

(D) – $C_2$ axis of symmetry

44. (*d*) 2nd C = 5, 3rd C = R 4th C = S
(2S, 3R, 4S)

45. (*d*)




I and II conformers

46. (*b*) Distereomers having no mirror image + CC
G.I is also diastereomers
R/S configuration → RR + RS
(A) 2S, 4R SS + RS
} Pass
(a) 2S, 4R
(d) 2R 4S
} diastereomers of A

47. (*a,b*) (A) $sp, sp^2$ (B) $C_1$ P-orbital = $C_4$ P-orbital

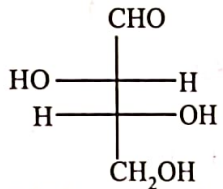
48. (*c*) POS is present = achiral.

49. (*c*) For meso there should be chiral carbon and POS

50. (*a*) Compound (II), (V), (IX), (X) are chiral

51. (*b*) No POS

52. (*d*)




53. (*b*) L,D,L configuration of respective compounds

54. (*b*) $2^n = 2^4 = 16$ (n = chiral centre)

55. (*c*) $2^n = 2^3 = 8$ (n = chiral centre + geometrical centre)

## Exercise-3 (JEE Advanced Level)

1. (*a,c*) Fact based

2. (*a,c*) Fact based

3. (*a,b*) Fact based

4. (*b,c*) Different valency at double bonded c-atom

5. (*c*) Not show G.I

6. (*a,b,c*)
Fact based

7. (*b,d*) Fact based

8. (*a,b,d*)
Fact based

9. (*a,b*) Fact based

10. (*a,c,d*)
In option (b) Allene system with n = even
$L_1 = L_2$ not show G.I

11. (*a,b,c,d*) Fact based

12. (*b,c,d*)
For R/S configure molecule should have at least 1 chiral centre.

13. (*b,c,d*)
a → conformers
b → G.I
c → O.I
d → O.I

14. [3] Enantiomers → non superimposable mirror image
→ different R/S configuration
(*a*) S/R
(*b*) S/R
(*c*) R/S
(*d*) R/R
(*e*) Non superimposable mirror image
(*f*) 2S, 3S/ 2S, 3S
(*g*) nos mirror image i.e. diastereomers

15. [2] (*i*), (*ii*),(v) = diastereomers = 3
(*iii*) = enantiomers
(*iv*) structures isomers having different IUPAC
difference = 3 – 1 = 2

16. (*a*) A-S, B-P, C-S, D-R

17. (*b,c*)
(*iii*) no POS, no COS, optical active

18. [6] (b, c, d, e, f, i) chiral compounds

19. [32] Total steroisomers = $2^n = 2^5 = 32$

20. (*a,d*) a = 2, b = 0 c = 3 d = 2 – even

@PROFBSE

21. (*b,c,d*)
Theory based

22. (*b, d*)
Theory based

23. (*d*) $\alpha_H/\alpha_D$ containing molecules

24. (*b*) In enol form double bonded c-atom having different valency

25. (*a*) % enol ∝ aromaticity
∝ H-bonding

26. (*c*) stability → Anti > gauche > Partial eclipsed > eclipsed of there is intra H-bond gauche > Anti
Anti + Intra H-bond → most stable

27. (*d*) Infinite conformer possible

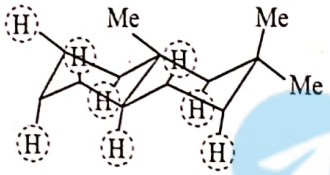
28. (*a*) Theory based

29. (*a*) Theory based

30. (*d*) Theory based

31. (*c*) A-(p,s); B-(q,s,t); C-(r,s,t); D-(q,s,t)

32. (*d*) Theory based

33. [8]



34. [0] (*i*) 2CC (*ii*) 2CC (*iii*) no. CC (*iv*) 2CC
(*v*) 2CC (*vi*) 2CC (*vii*) 4CC (*viii*) 4CC
(*ix*) 4CC (*x*) 2CC

35. [2] Theory based

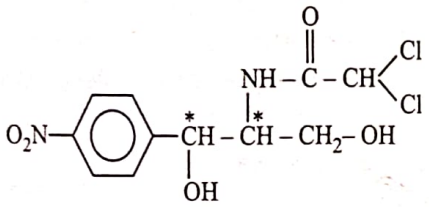
## Exercise-4 (Past Year Questions)

### JEE MAIN

1. (*c*) Angle between H'– C – C – H" is 149°.

2. (*b*) Theory based

3. (*a*) In linear = 4 (Fischer)
In cyclic = 5 (howarth projection)

4. [2]



5. [3] Structure based

6. [5] C-$sp^3$ having four different group are chiral carbon.

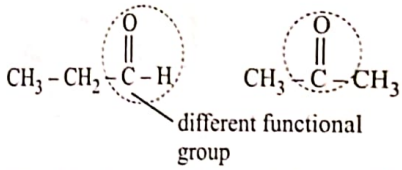
7. (*c,d*) c → G.I. due to ring
d → G.I due to double bonded C–atom.

8. (*d*) Double bonded c-atoms having different valency

9. (*d*) meso comp. ∴ T.S.I = $2^{n-1} = 2^{2-1} = 3$
compounds having chiral center + POS are called me
compounds

10. (*c*) (*i*) Having no chiral c-atom
(*ii*) double bonded c-atom having same valency

11. (*d*) For metamors
(*i*) F.G = same and bivalent or polyvalent
(*ii*) Alky part different
(*iii*) R – C(=O) – R"

12. (*d*) $C_3H_6ODU = \frac{2+2c-H}{2} \Rightarrow \frac{2+2\times3-6}{2} \Rightarrow \frac{2+6-6}{2} =$


[Enol form is not stable so don't consider that structur

13. (*c*) Theory based

14. (*b*) Conformers/Rotamer

15. (*b*) $P.E \propto \frac{1}{\text{stability}}$

16. (*c*) Staggered

17. (*d*) Theory based (i) $sp^3$ c-atom having different group

18. [7] Fact based

19. [60°]

### JEE ADVANCED

20. (*b*) Fact Based

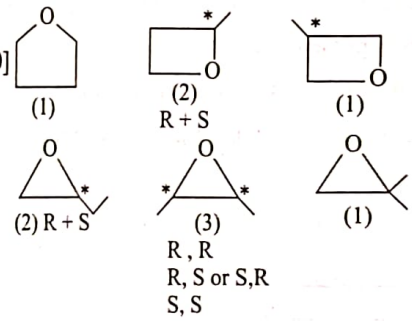
21. (*a*) Fact Based

22. (*a, b, c,*)
M & N → Diasteromers, M & O → Identical
M & P → Enantiomer, M & Q → Diasteromers

23. (*a,c,d*)

24. [3]

25. [7] Theory based

26. [10]


Total = 1+ 2 + 1 + 2+3+1 = 10

27. (*c*) The conformar having same IUPAC name are identical.

28. (*b*) *a* & *b* = meso compound b having intra-moleular H - bond.

29. [2] Theory based

CHAPTER

# 14 General Organic Chemistry

## Exercise-1 (Topicwise)

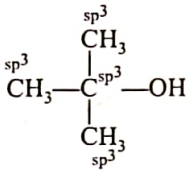
**1.** (*b*)

Benzene

Each carbon atom has 3σ and 1π.

It has 3σ bond so $sp^2$.

**2.** (*c*)



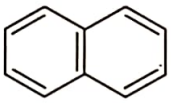
**3.** (*d*)

$$\overset{sp^3}{CH_3}-\overset{sp^2}{CH}=\overset{sp^2}{CH}-\overset{sp^3}{CH_2}-\overset{sp^2}{CH}=\overset{sp^2}{CH}-\overset{sp}{C}=\overset{sp}{CH}$$

**4.** (*c*) *sp* and $sp^2$

$$N\equiv\overset{sp}{\underset{1}{C}}-\overset{sp^2}{\underset{2}{CH}}=\underset{3}{CH_2}$$

**5.** (*c*)

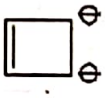


π electrons = 10.

π bonds = 5

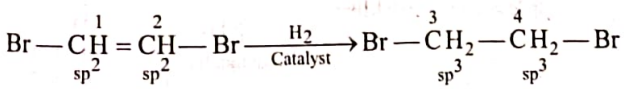
hence electrons are double.

**6.** (*c*)

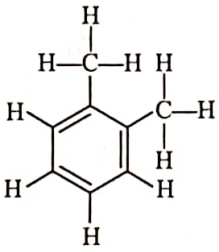


Total $\pi e^- = 6$

**7.** (*a*)



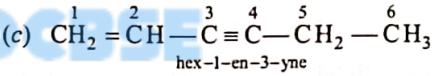
**8.** (*d*)



o-dimethyl benzene
or
o-xylene

It has 18σ bonds and 3π bonds.

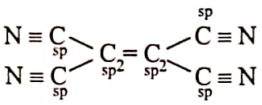
**9.** (*c*)



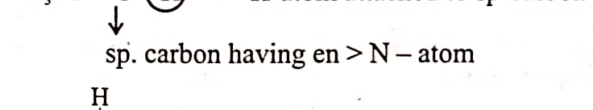
hex-1-en-3-yne

Three π bonds.

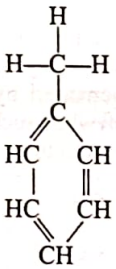
**10.** (*c*)



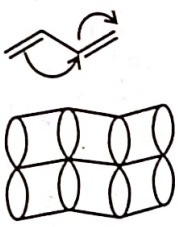
**11.** (*c*)



**12.** (*d*)



**13.** (*b*)



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14. (*a*) $C_2H_5OH$ (H attached to O atom)

15. (*b*) $CH_3—\overset{\overset{\cdot\cdot}{O}H}{\underset{|}{C}}=CH_2$

$9\sigma$, $1\pi$ and 2L.P

16. (*a*) 1,2 - ethandiol

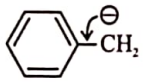
$CH_2(OH)—CH_2(OH)$

17. (*c*) Theory based

18. (*b*) Theory based

19. (*c*) $-NR_2 < -OR < -F$

20. (*b*)

$CH_3C^{\ominus}(CH_3)(CH_3)$ < $ME_2CH^{\ominus}$ < $ME-CH_2^{\ominus}$ < 

3°C⁻ 2°C⁻ 1°C⁻ Involve in resonance

21. (*a*) $-\overset{\oplus}{N}H_3 > -NO_2 > -CN$

22. (*b*) (*a*) Has $6\alpha - H$

(*b*) Has $2\alpha - H$

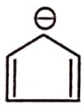
(*c*) Has $9\alpha - H$

(*d*) Resonance

23. (*c*) $-CH_3 \Rightarrow +I$ – effect

24. (*d*) Resonance structure of molecule does not have identical bonding.

25. (*d*)



Envolve in resonance so total

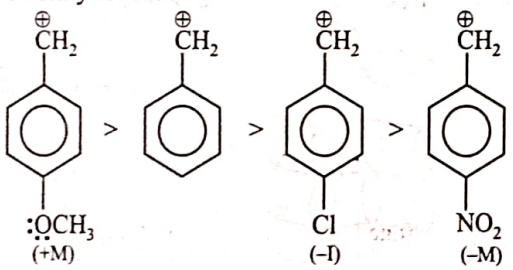
$\pi e^- = 6$ and follow's Huckel's Rule

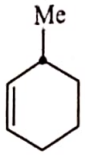
26. (*a*) Triphenyl methyl cation has three benzene resonating ring so it is most stable compound.

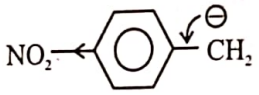
27. (*c*) The octet of all atoms are complete in structures (a) and (b). In structure (d) electron deficiency of positively charged carbon is duly compensated by lone pair electrons of adjacent oxygen atom while such neighbour group support is not available in structure (c).

28. (*a*) $CH_3-\overset{+}{C}H(OCH_3) > CH_3-\overset{+}{C}H(CH_3) > CH_3-\overset{+}{C}H(COCH_3)$

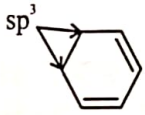
29. (*c*) Stability order :

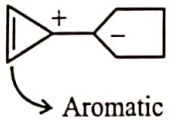



30. (*a*)  (Resonance + hyperconjugation)

31. (*d*) 

32. (*c*) Aromatic compound follows Huckel's rule of aromaticity $[4n + 2]\ \pi e^-$

33. (*b*) 

(c) 

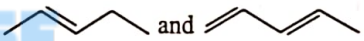
Aromatic

34. (*d*) (I) $\alpha - H \Rightarrow 9$

(II) Resonance (conjugation)

(III) $\alpha - H \Rightarrow 2$

(IV) no $\alpha - H$

35. (*d*)  and 

36. (*a*) Fact Based

37. (*b*) $CH_3-\overset{+}{C}(CH_3)(CH_3)$ Total $9\ \alpha - H$

38. (*b*) More $\alpha$–H

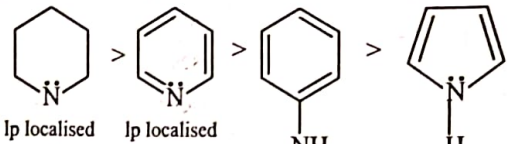
39. (*b*) Acidity x > y > z (–I effect of x support y)

40. (*d*)

$NO_2 \Rightarrow -M, -I$ effect

$NO_2$ increases the stability of anion.

41. (*d*) Basic strength :

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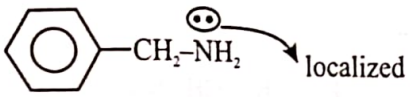
42. (*d*)



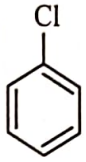

43. (*d*) Aromatic compound follows $(4n+2)\pi e^-$ [Huckel's rule]

44. (*a*) $e^-$ with drawing group decreasing basic strength

45. (*b*) Acidic strength :
$R—COOH > H_2O > R—OH > CH \equiv CH$
stable the conjugate base, stronger the acid

46. (*b*) $H—\ddot{O}—H, :NH_3, R—\ddot{O}—R$ — nucleophiles $SO_3$ has electron deficient centre (a reagent which can accept an electron pair in a reaction, is called an electrophile)

47. (*d*) upcoming group is decided by +M

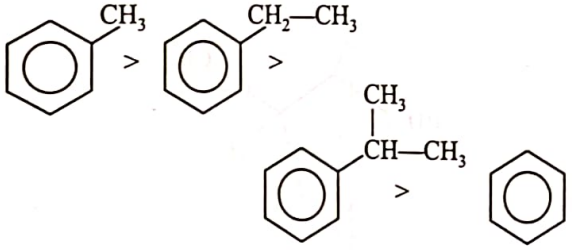



48. (*d*) When methane gas is treated with chlorine in the presence of sunlight, one hydrogen of methane replaced by the chlorine atom and forms methyl chloride. The mechanism involved in this reaction is free radical mechanism. So it is an example of free radical substitution reaction.

49. (*a*) $AlCl_3$ is lewis acid *i.e.*, electron deficient compound. So it is electrophile.

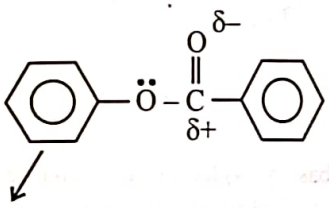
50. (*d*) $CH_3—O^-$ is the strongest nucleophile which is capable of acting as donar of electon pair.

51. (*b*) ESR order :




due to hyper conjugation effect.

52. (*c*)




More $e^-$ density

53. (*c*) Characterstic reaction of aromatic hydrocarbon is ESR.

54. (*c*) $e^-$ donating group increases ESR

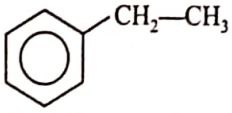
55. (*b*) $e^-$ donating group increases ESR

56. (*c*)

6π electrons are present in compound

57. (*d*) In benzene all C—C bond lengths are same.

58. (*c*)




Benzylic carbon has 2 hydrogen ethyl benzene.

59. (*c*) $R—\ddot{N}H_2$
N has 3σ and 1 L.P.
so, $sp^3$

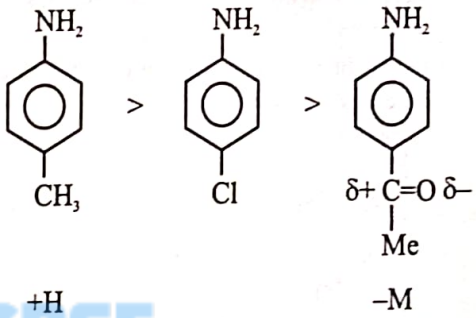
60. (*a*) $e^-$ drawing group decreases basic strength

61. (*a*) Fact based

62. (*a*)

(localised lp)

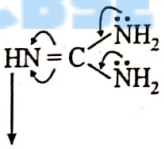
63. (*a*)




+H

–M

64. (*a*)



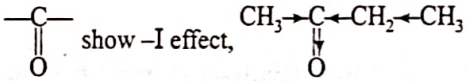

Electron – density will be maximum at this position.

65. (*a*) It is picric acid because it has three $–NO_2$ group are arranged which are ortho and para position

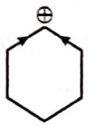
## Exercise-2 (Learning Plus)

1. (*b*) Partial displacement of σ-electrons.

2. (*d*)

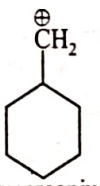



3. (*d*)



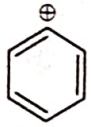

>

(four hyperconjugable H-atom)




(one hyperconjugable H-atom)

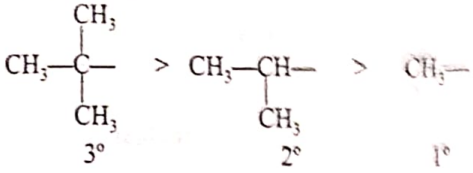
>



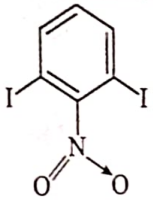

(no delocalisation of positive charge)

(bridgehead carbocation)

4. (*b*) The correct order of these groups were are attached with Benzene ring is–



3° > 2° > 1°

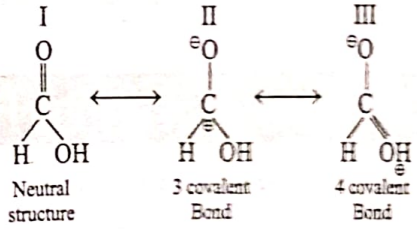
5. (*d*) (*a*) is secondary and (*b*) is primary alcohol
6. (*b*) Due to presence of conjugated system.
7. (*d*) (*a*), (*b*) and (*c*) all are conjugated.
8. (*b*) Do it yourself
9. (*c*) Nitrogen does not have vacant d-orbital, so cannot form five bonds.
10. (*c*) Equivalent resonating structures contribute equally to the resonance hybrid.
11. (*c*) 
12. (*d*) (I) $CH_2 = CH - CH = CH_2$

I, II, III are canonical structures to each other.

(II) $\overset{\ominus}{C}H_2 - CH = CH - \overset{\ominus}{C}H_2$

(III) $\overset{\ominus}{C}H_2 - CH = CH - \overset{\ominus}{C}H_2$

All

13. (*c*) 

Neutral structure | 3 covalent Bond | 4 covalent Bond

14. (*a*)

(I) $CH_2=CH-CH=CH-CH_2 \longleftrightarrow \overset{\ominus}{C}H_2-CH=CH-\overset{\ominus}{C}H-CH_3$

(II) $CH_3-CH=C=CH-CH_3$

Here in str. (I) Resonance occurs But not in II[nd].

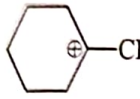
15. (*d*)

Complete octate &

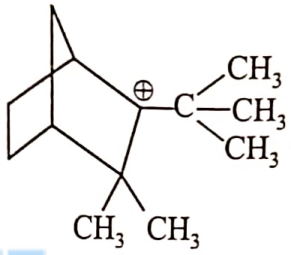
Extended conjugation

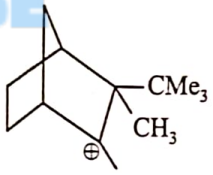
16. (*d*) Do it yourself
17. (*a*) Do it yourself
18. (*d*) $CH_3 - CH_2 - CH = CH_2$ has two α-hydrogen for hyperconjugation.
19. (*d*) Has maximum number of hyperconjugation structure
20. (*b*) (I) & (III) has delocalisation (III has better delocalisation)

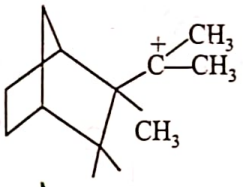
(II) & (IV) has hyperconjugation. (II has better hyperconjugation).

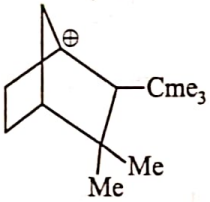
21. (*b*)  3° Carbocation is more stable.
22. (*c*)

hyperconjugation and delocalisation with +I effect.

23. (*b*) (I) 

(II) 

(III) 

(IV) 

Bredt rule

(IV) III > II > I > IV

24. (*d*) Conjugate base of (4) has negative charge on oxygen.
25. (*a*) On the basis of Inductive effect.
26. (*a*) On the basis of Inductive effect.
27. (*b*) $H_3N^{\oplus}-(CH_2)_2-COOH$ has strongest acid due to strong –I effect of $-\overset{\oplus}{N}H_3$.
28. (*a*) On the basis of Inductive effect.

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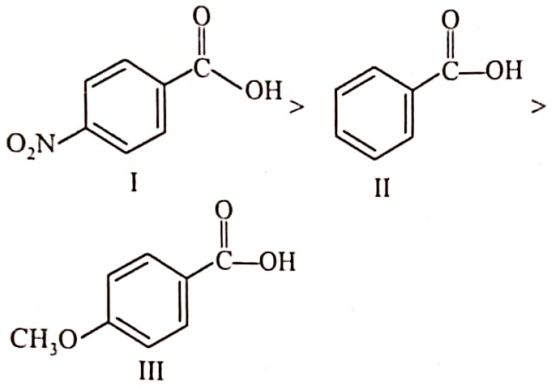
29. (*c*)



I > II > III

(Strong –I, –M effect of $-NO_2$ group) (+M effect of $-OCH_3$ group)

30. (*b*) Conjugate base of (2) has three equivalent resonating structures.

31. (*a*) II → Hyperconjugation III → – M effect
Due to Hyperconjugation basicity increases and due to -M basicity decreases

32. (*a*) Nucleophilicity $\propto \frac{1}{\text{electronegativity}}$ (in a period).

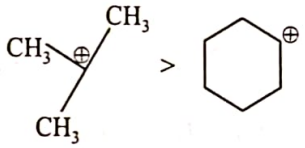
33. (*a*) Nucleophilicity ∝ size (in a group).

34. (*d*) As increases delocalisation of negative charge nucleophilicity decreases.

35. (*b*) Nucleophilicity ∝ Size (in group)

36. (*c*) Weaker bases are better leaving group.

37. (*c*) Carbocation Stability

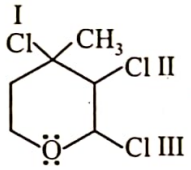


leaving group ability is $Br^{\ominus} > Cl^{\ominus}$

over all reaction order $r_1 > r_3 > r_2$

38. (*c*) On the basis of carbocation stability.

39. (*b*)



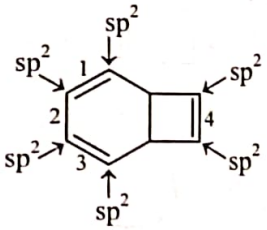
$\xrightarrow{AgNO_3}$ $S_N1$ Reaction

Reaction Intermediate — Carbocation :

III will give most stable carbocation ⇒ stabilised by +M effect of —Ö—

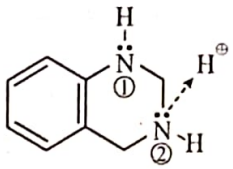
I will give stable carbocation → 3° carbocation so order of reaction = III > I > II.

40. (*c*)



41. (*a*) Stability of carbocation = 3° > 2° > 1° (a) has 3° carbocation.

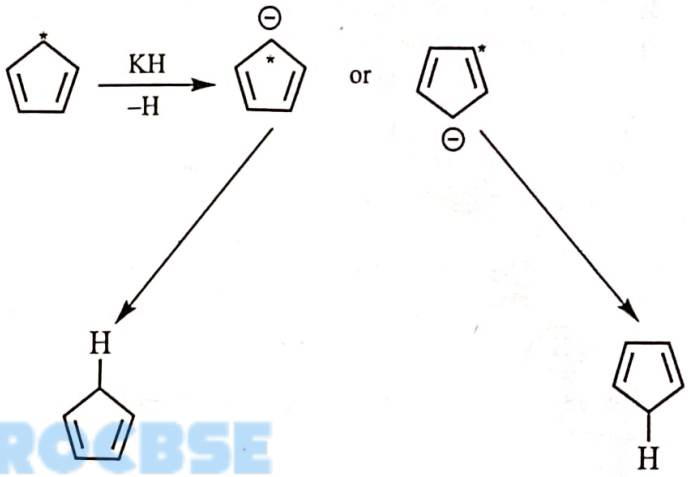
42. (*b*)



Lone pair N–1 is delocalised with benzene ring but N–2 is not so N–2 is more basic and protonation will occur at N–2.

43. (*d*) **Hint:** Look for stability of conjugate base using general concept of organic chemistry.

44. (*d*)



So correct options are (*b*) and (*c*) both.

45. (*c*) Heat of combustion ∝ No. of carbon atom.

46. (*d*) Remove most acidic proton and look for its stability.
In option IV negative charge is stabilized by 3 carbonyl group so is most acidic.
So correct option (*d*) ii < iii < i < iv.

47. (*c*) After dehydrating carbocation is formed and the species with most stable carbocation will be most easily dehydrated.
II upon dehydration forms aromatic compound the most stable.
I Antiaromatic 5-membered Ring Least Stable
III Non Aromatic 7 membered ring.
So order of dehydration ii > iii > i.

48. (*d*) In (*a*) and (*c*) Lone pair undergo resonance in b and d → d is 2° Amine.

49. (*b*) In option 'b' it is 2° carbocation in first and i carbocation is second.

50. (*d*) Alkenes with highest Numebr of $\alpha_H$-are most stable option'd' has 8-$\alpha_H$.

51. (*a*) Option II has –R group so most acidic option III and I has electron releasing groups among I and III, I is more acidic as in III there is +R of oxygen atom.
So correct order iii < i < iv < ii

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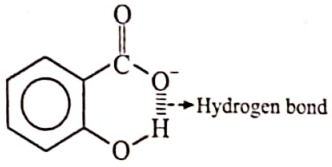
positive charge is isolated so most basic. Between A and B, B has higher number of resonating structure so is more stable so correct order is C > A > B.

53. (*a*) Half life of carbocation ∝ stability of carbocation in option 'a' cyclopropyl ring shows dancing resonance due to ring strain so is most stable among all other carbocations.

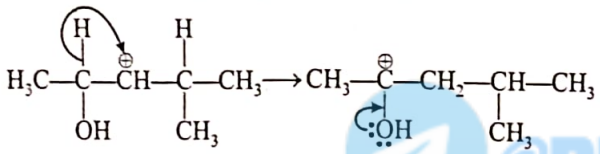
54. (*d*) Negative charge on electronegative atom is more stable. So correct order of stability p > r > q.

55. (*a*) Electron density ∝ no. of $\alpha_H$ with respect to phenyl ring 'a' has maximum that is 3-$\alpha_H$ (b) 2$\alpha_H$ (c) has 1$\alpha_H$ and (d) has zero $\alpha_H$.

56. (*c*) Orthohydroxy benzoic acid is most acidic as conjugate base is stabilized by hydrogen bond.




57. (*d*) H at $C_2$ as after megration carbocation is stabilized by resonance by lone pair of –OH group.

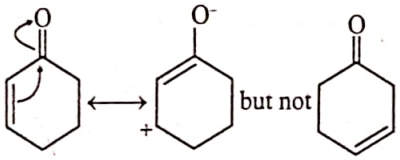


## Exercise-3 (JEE Advanced Level)

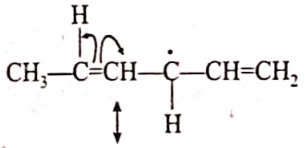
1. (*a,b,d*) Based on definition of inductive effect.

2. (*a,c,d*) Fact based on the concept of Resonance.

3. (*b,d*)

(*a*)  but not 

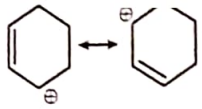
Resonance occurs in conjugation.

(*b*) 

$CH_3—\dot{C}H—CH=CH—CH=CH_2$

are Resonating structure.

(*c*) $CH_3—\overset{O}{\overset{\|}{C}}—CH_2—CH_3$ & $CH_2=\underset{OH}{\underset{|}{CH}}—CH_2—CH_3$ are Tautomers

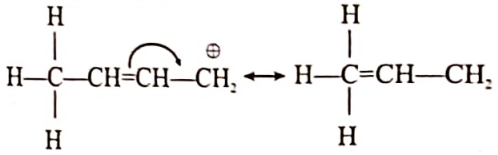
(*d*) 

4. (*b,d*)

(*a*) Conjugated systems are more stable than non-conjugated system so option 'a' incorrect.

(*b*) $CH_2=CH—CH=CH_2—CH_2$ will show higher number of Resonating structure than $\dot{C}H_2—\overset{CH_2}{\overset{\|}{C}}—CH=CH_2$

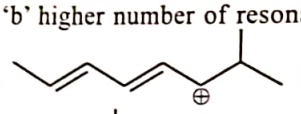
(*c*) $CH_2=\underset{CH_3}{\underset{|}{C}}—\overset{\oplus}{C}H_2$ hyperconjugation and resonance are not in same direction but in $CH_3—C=CH—\overset{\oplus}{C}H_2$ resonance and hyperconjugation in same direction.

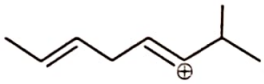


So incorrect stability order.

(*d*) In $CH_3—\overset{O}{\overset{\|}{C}}—O^-$ negative charge is delocalised on two electronegative atoms. So it is correct.

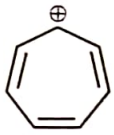
5. (*b,c,d*)

(*b*) In option 'b' higher number of resonating structure possible  more stable

 π-bonds not in conjugation.

(*c*) $CH_3—\ddot{O}—\overset{\oplus}{C}H—CH_3$ more stable due to Resonance.

(*d*) Higher no0 of Resonating structure is more stable.

(*a*) Antiaromatic  Aromatice

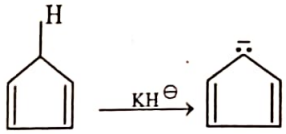
6. (*c,f*)

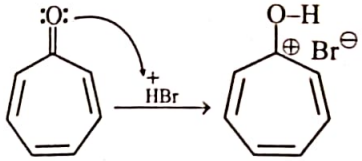
Electron donating groups are $—\ddot{O}—COCH_3$, $—\ddot{N}HCOCH_3$

7. (*b,d,e*)

Lone pair in conjugation will undergo Resonance so correct options are *b, d, e*.

8. (*a,b*)

(*a*)  Aromatic 6πe⁻

(*b*) 

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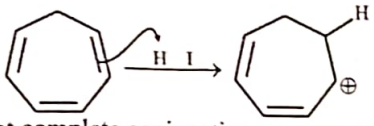
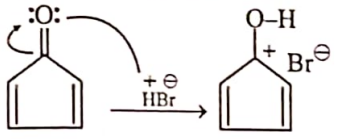
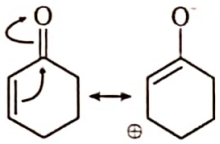
Aromatic $6\pi e^-$

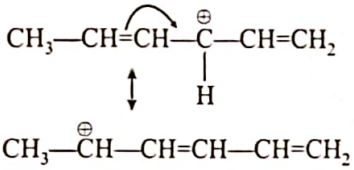
(*c*) 

Not complete conjugation non aromatic

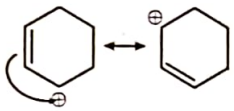
(*d*) 

Anti-Aromatic $4\pi e^-$ system

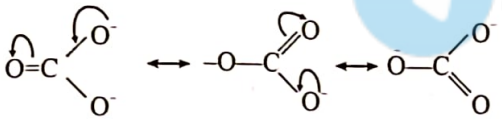
**9.** (*b,d*) 

(*b*) 

(*c*) Shown structure are tautomers.

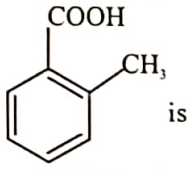
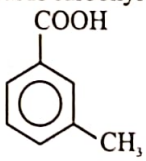
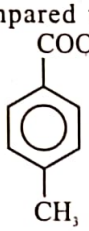
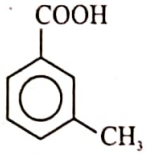
(*d*) 

**10.** (*a,c*) All cononical forms do not always contribute equally to the resonance hybrid.

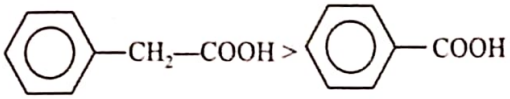
(*c*) 

All are equivalent resonating structures so all C—O bond lenghts are equal.

**11.** (*a,c*) $K_a \propto$ Acidic strength.

(*a*)  is more acidic than due to steric inhibition of resonance due to which electron from Aromatic ring will resonate less towards carbonyl compared to benzoic Acid. Among  and  due to presence of CH3 group at para position in is less acidec than  so given order is correct.

(*b*) Acetic acid has $Pk_a = 4.74$.



$Pk_a = 4.31$ $Pk_a = 4.19$

**12.** (*a,b,c,d*)

(*a*) $PK_a$ of carbonic Acid is 6.37.
$PK_a$ of $CH_3COOH$ is 4.74.

(*b*) Boiling points of acids are higher than Alcohol due to higher extent of hydrogen bonding in Acids.

(*c*) Chloroacetic Acid is stronger acid than acetic acid due to presence of –I group i.e. chlorine.

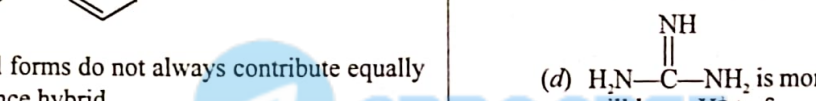
(*d*) In phenol conjugate base is Resonance stabilized.

**13.** (*a,c,d*) Less stable conjugate base will be stronger base so correct options are *a, c, d.*

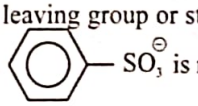
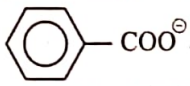
**14.** (*c,d*) In option 'a' lone pair of N is in conjugation so protonation will not occur and reaction not feasible.

(*b*) In Aniline Lone pair will undergo conjugation so will not be protonated and infisible reaction.

(*c*) After reaction a stable aromatic product is obtain from anti-aromatic so feasible reaction.

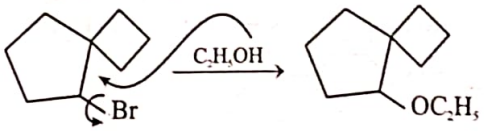
(*d*)  is more basic than  so it will loose $H^+$ to form feasible product is formed.

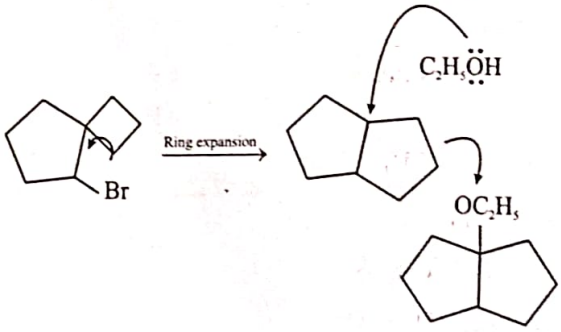
**15.** (*a,b*)

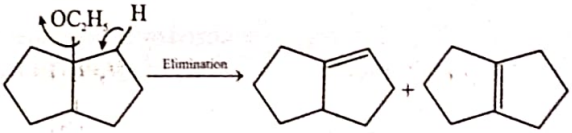
Weaker bases are good leaving group or stable Anion are good leaving group.  is more stable than  . So it is correct. $CF_3—SO_3^-$ is more stable than $CCl_3SO_3^-$ so it is good leaving group.

**16.** (*a,b,c,d*)

All are possible products.

(*a*) 

(*b*) 

(*c*) 

**17.**(*a*,*b*,*d*)

*a*, *b* and *d* are Resonance stabilized as carbocation is in conjugation with π-bond. (Allyl carbocation is stabilized by Resonance)

**18.** (*c*) In option *a*, *b* and *c* after rearrangement more stable carbocation is formed on rearrangement so ΔG will decrease.

**19.** (*d*) In case of singlet conbene there is presence of one vacant orbital and there is possibility of backbonding and get stabilized.

Extent of backbonding is more in $:CF_2$ than $:CCl_2$ due to 2P-2P overlap in $:CF_2$ and 2P-3P overlap in $:CCl_2$.

Triplet $:CH_2$ is more stable than singlet $:CH_2$ as per hunds rule.

(*c*) Singlet $CH_2$ has planar geometry as two electrons will present in one orbital and from trigonal planar geometry.

(*d*) electron affinity of $\dot{C}H_3$ radical is greater as, after getting one electron octet is completed.

**20.**(*a*,*c*) $H_a$ is abstracted more rapidly as it forms stable radical.

$H_b$ is more acidic than $H_c$ as $H_b$ is in conjugation with π-bond and is resonance stabilized.

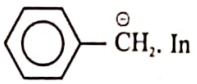
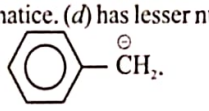
**21.**(*b*,*c*,*d*)

(*a*) Due to resonance, second is more stable.

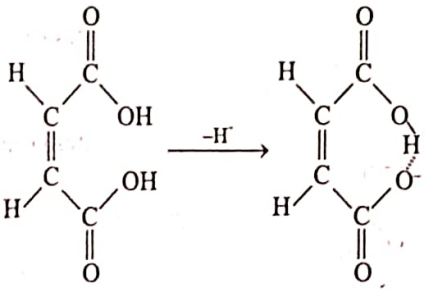
(*b*) First has 5αH, Second has 4αH.

(*c*) Oxygen is more electronegative.

(*d*) Second is not resonance stabilized.

**22.** (*b*) Only *a* and *b* are more stable than  . In '*a*' $-NO_2$ group is –R and '*b*' negative charge delocalize to oxygen a more electronegative atom. (*c*) is Anti-aromatice. (*d*) has lesser number of Resonating structure than  .

**23.**(*a*,*c*)

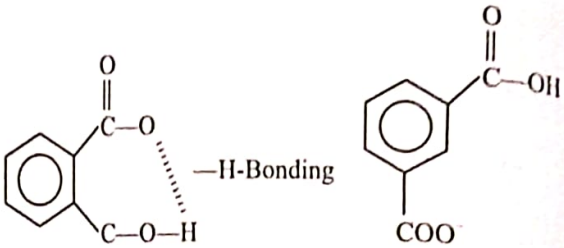


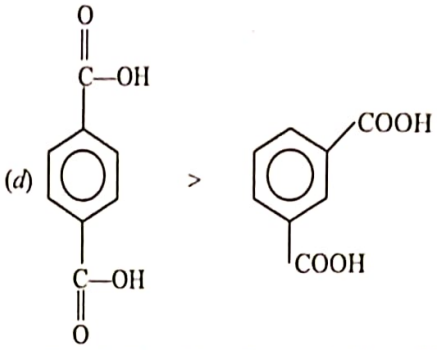
(*a*)

Negative change stabilized by H-bond so is more acidic than fumaric acid. So $PK_1$ of Maclic < $PK_1$ of fumaric acid.

(*b*) $PK_{a_2}$ of Maclic Acid is 6.5 and that of fumaric acid is 4.5.

(*c*) Phthalic acid is stronger acid than isophthalic acid due to Intramolecular H-bonding in phthalate anion.



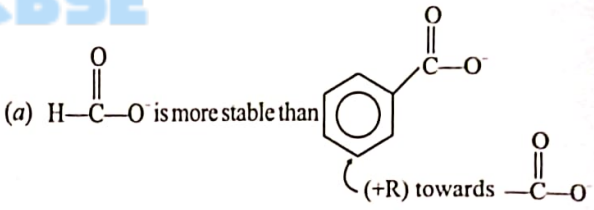



(*d*)

Terepthalic acid Isoterepthalic acid

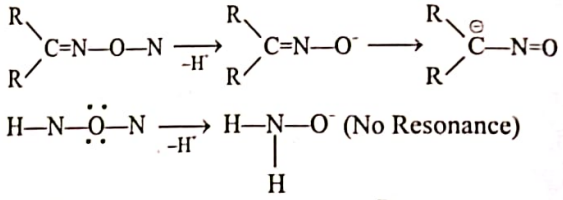
Due to –R Nature in Terepthalic acid so Terepthalic is more acidic.

**24.** (*a*,*b*,*c*)

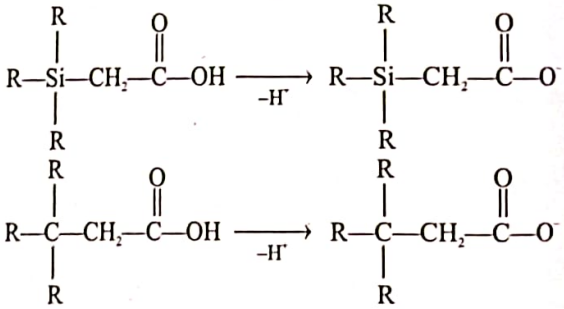



so HCOOH is more acid.

(*b*) In oxime oxide ion is Resonance stabilized




(*c*)




Si is less electronegative than carbon. So, $R_3SiCH_2COOH$ is less acidic than $R_3C\ CH_2COOH$.

(*d*) Highly branched carboxylic acids are less acidic than unbranched acids.

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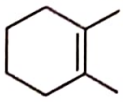
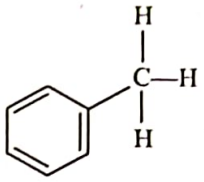
**25.** In 3,4,5-trinitroanilive due to the steric crowding of bulky Nitrogroup, the nitrogroup in para position rotates out of plane hence it can't take part in Resonance with phenyl ring. This phenomenon is called steric inhibition of Resonance (SIR). This phenomenon affects the basicity of compound. $NO_2$ being –I and –R withdraw electron towards itself. But due to SIR lone pair of Nitrogen cannot come towards ring and is more available for donation and is more basic.

There is no steric inhibition of Resonance in 4-cyano-3, 5-dinitro aniline due to linear –C≡N group. So correct options are *a*, *c*, *d*.

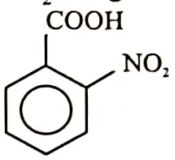
**26.** (*d*) Number of hyperconjugation structure is directly proportional to number of $\alpha_H$. (d)  has maximum number of $\alpha_H = 10$.

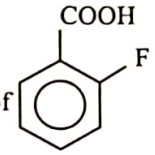
**27.** (*c*) In '*c*' there is no α–H so hyperconjugation not possible.

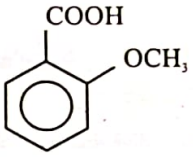
**28.** (*d*)  will show hyperconjugation.

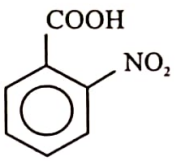
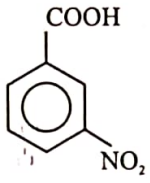
**29.** (*a*) $K_a \propto$ Acidic strength.

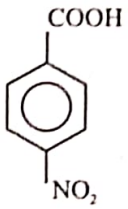
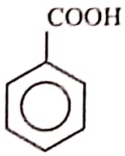
$NO_2$ being strong electron withdrawing group makes

 most acidic.

In case of  'F' being smallest will produce lowest orthoeffect and will be less acidic than bromo and chloro derivative. So correct order is I>II>III>IV

**30.** (*d*)  will be strongest acid due to highest orthoeffect.

**31.** (*d*)  (i)  (ii)

 (iii)  (iv)

Correct order of Acidity is I>III>II>IV.

In (i) $-NO_2$ group is more closer as compared to (iii) so will withdraw more election and make the molecules more acidic.

**32.** (*d*) Stability of carbonion follows 1°>2°>3°. So correct option is (d).

**33.** (*d*) Stability of free radical ∝+I/+R and inversely proportional to –I or –R.

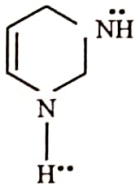
In ph—$CH_2$ → phenyl stabilize the radical by resonance so is most stable.

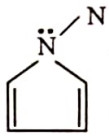
In ph—$CH_2$—$\dot{C}H_2$. Ph act as –I so is less stable than $CH_3$—$CH_2$—$\dot{C}H_2$. So, correct order is III>II>I>IV.

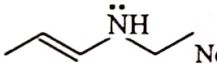
**34.** (*a*) Stability of carbocation ∝+R. with increase in steric crowding at orthoposition extent of Resonance decreases. So, correct order is I>II>III.

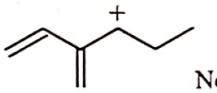
**35.** (*b*) → 6-π Electron and Aromatic

4π–$e^-$ and Anti-Aromatic

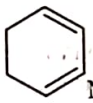
 6π $e^-$ and Aromatic

 6π $e^-$ and Aromatic

**36.** (*c*)  Non-Aromatic, Resonance

 Non-Aromatic, Resonance

Aromatic, Resonance

 Nón-Aromatice, Resonance

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37. (*a*) A→q, r, s; B→q, r; C→p, r; D→p, r

$CH_3—O—\overset{\oplus}{C}H_2$ < $CH_2—NH—\overset{\oplus}{C}H_2$

Less electronegative of N.

$CH_3—O—CH_2—\overset{\oplus}{C}H_2$ < $CH_2—\ddot{O}—\overset{\oplus}{C}H_2$

No Resonance Resonance occur

(*c*) due to steric crowding in second there is no resonance and hence is less stable.

38. (*d*) A→s; B→q; C→r; D→p

39. [4] Electrophiles are electron accepting specis So, $CO_2$, $SnCl_2$, $Fecl_3$ and $BF_3$ are electrophile $NH_4^+$ is not electrophile because there is no any vacant orbital for accepting electron.

40. [4] $\overset{+}{N}O_2$, $:CCl_2$, $AlCl_3$, $SO_3$

↑

Incomplete octet

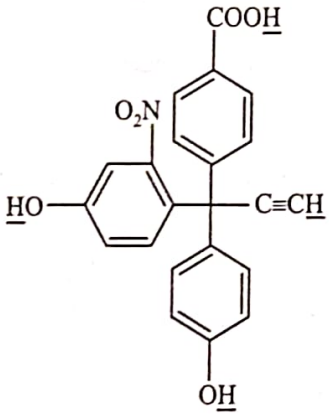
41. [6] No. of Hyperconjugative structure = no. of $\alpha_H$

$(CH_3)_2\overset{\oplus}{C}H$ has 6 $\alpha_H$ so 6 hyperconjugative structure.

42. [6] 6-conjugated electrons.

43. [4] Number of mole of $NaNH_2$ Required = No. of Acidic proton.

Here acidic protons are indicated below



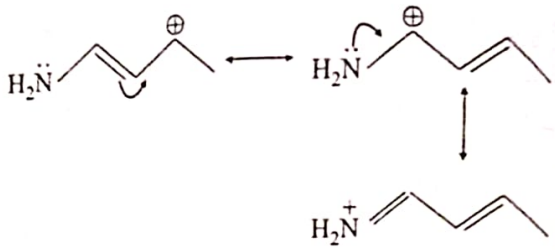

44. [2] HCOOH and $ClCH_2COOH$ are stronger than $CH_3COOH$.

45. [3] $-NO_2$, –CHO and $\overset{+}{C}H_2$

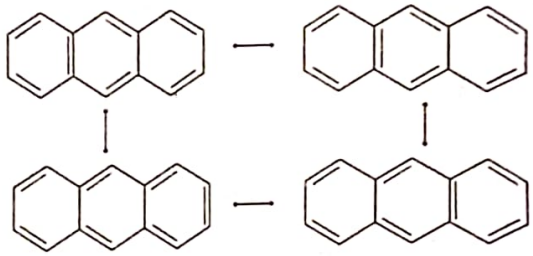
46. [6] Compounds having No any α-H will not show Hyperconjugation.

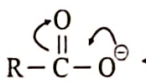
47. [9] No. of H.C. structure = No. of $\alpha_H$. There are $9\alpha_H$.

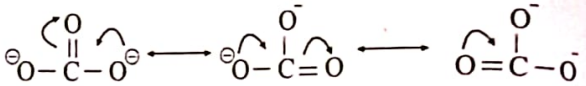
48. [3] 3 Resonance structure can be drawn.

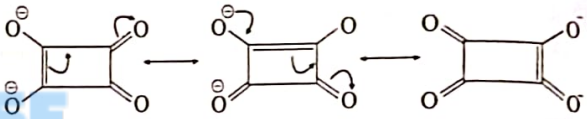


49. [4]

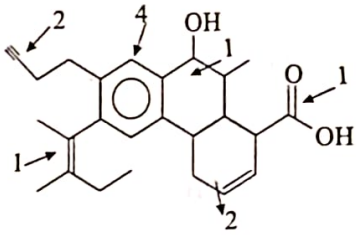


50. [3]  equivalent structure

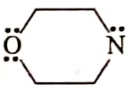




51. [11] DU = Number of double bond + Ring



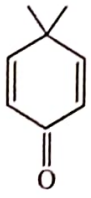

52. [4] '*b*', '*d*', '*e*', '*g*'

53. [3]  N – H = 3 =There are three lone pairs of $e^-$ in the given molecule.

## Exercise-4 (Past Year Questions)

### JEE MAIN

1. (*d*) Least Resonance stabilized



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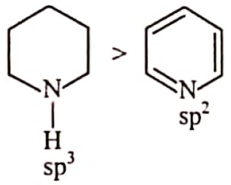
2. (*b*) Basicity depends upon electron density → which is also directly proportional to number of N-atoms basic order increase with +*i* group.

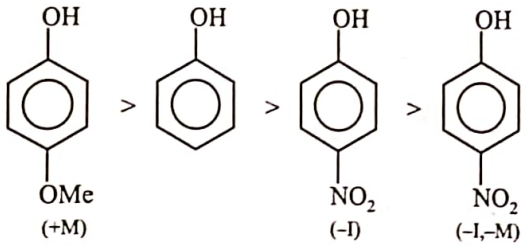
3. (*d*) $\underset{sp}{CH} \equiv \underset{sp}{CH} > \underset{(+I)}{CH_3} - \underset{sp}{C} \equiv CH > CH_2 = \underset{sp}{CH_2}$

S–char. α E.N. α acid strength.

4. (*b*) 

Basic strength of amine increases with electron donating capacity.

5. (*d*) In option 'd' only vander waals interaction will operate but in other option H– Bonding dipole-dipole/ion dipole interaction will occur.

6. (*d*) 

$Pka \propto \frac{1}{ka}$

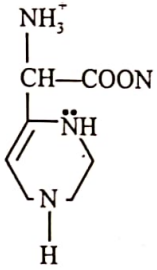
ka ∝ acidity ∝ EWG

7. (*c*) $CH(CN)_3$

Due to high E.N. of $CN^-$.

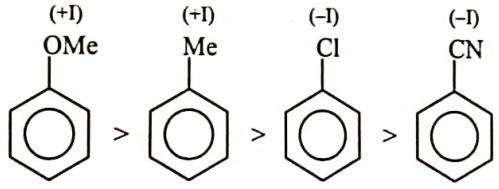
8. (*c*) In strongly acidic medium Lone pair of nitrogen which are localized will undergo protonation

so correct structiure is



9. (*d*) Due to (–I) effect

$-NO_2 > -CN > -F > -Cl$

10. (*c*) 

+I ⇒ E.D.G.

–I ⇒ E.W.G.

11. (*b*) $Pk_b \propto \frac{1}{\text{Basic Nature}}$ Basic Nature $\propto \frac{1}{-R}$

$Pk_b \propto -R$

$-NO_2$ is – R so least basic

so correct order is ii < iv < i < iii

12. (*a*) $\bar{O}H > CH_3-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{C}}-\bar{O} > CH_3-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\bar{O} > H_2O$

Nucleophilicity reduces with l.p. donating tendency of oxygen reduced.

13. (*c*) $Pk_b \propto \frac{1}{\text{Basicity}} \propto \frac{1}{k_b}$

$k_b \Rightarrow B > A > C$

$Pk_b \Rightarrow C > A > B$

14. (*a*) C > B > A

(*c*) ⇒ –ve charge is in conjugation with (=) bond and $NO_2$.

15. (*c*) $\underset{\underset{CH_3}{|}}{\overset{CH_3(+I)}{|}}{-C-CH_3} > \underset{(+I)}{\bar{C}H_3} > \underset{sp^2}{\bar{C}H_2-CH=CH_2} > \underset{sp}{\bar{C} \equiv CH} > \underset{(-I)}{\overset{\ominus}{C}N}$

16. (*c*) Heat of combustion $\propto \frac{1}{\text{stability}}$ C > B > A

17. (*a*) $\ddot{Cl}-CH=CH-\ddot{N}O_2$

Conjugation

Double Bond char. ↑ B.L. ↓

18. (*c*) Do it yourself

19. (*b*) Benzene contains $6sp^2$ carbon and on each carbon there are 3–σ bonds means $3sp^2$ hybrid orbitals per carbon atom so total $sp^2$ hybrid orbitals = 6 × 3 = 18

20. (*a*) Proton with carboxylic acid will be most acidic –OHc ⇒ is at para position with respect to $-NO_2$ so is most acidic after 'b'

21. (*a*) In option 'd' the negative change is stabilized by resonance with two oxygen atom.

22. (*b*) In 'b' lone pair of chlorine is in conjugation with double bond, so will undergo resonance and bear partial double bond between Cl—C so has shortest bond length

23. (*a*) use concept of aromaticity and localization and delocalization of lone pair of $e^-$ in 'b' lone pair is delocalized and makes the ring aromatic

so correct order is B < A < D < C

D.B. char. ↑ B.L. ↓

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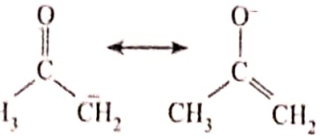
**24.** (*a*) Acidity of H depends on the stability of conjugate base

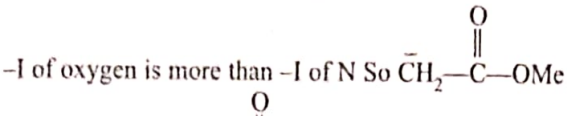
Ph—C—CH—C—Ph is most stable

So is most acidic in option



in option (A) $CH_3$ $\bar{C}H_2$ $CH_3$ $CH_2$

only +I of $CH_3$ group operates to increase electron density and it is further increased by resonance from done pair of oxygen and nitrogen.

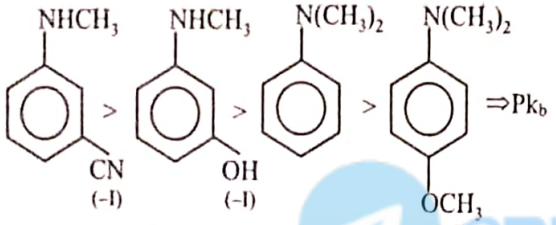


–I of oxygen is more than –I of N So $\bar{C}H_2$—C(=O)—OMe

more stable than $\bar{C}H_2$—O—$NMe_2$

So correct order is D < C < A < B

**25.** (*a*)



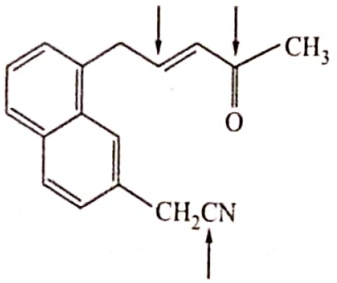
$NHCH_3$ > $NHCH_3$ > $N(CH_3)_2$ > $N(CH_3)_2$ ⇒ $Pk_b$

CN (–I) OH (–I) $OCH_3$ (+I)

$Pk_b \propto \frac{I}{k_b} \propto (-I)$

$k_b \propto$ Basicity. $\propto$ (+I)

**26.** [3]



**27.** (*d*) More stable anion less nucleophilic charged species are more nucleophilic

$\bar{N}h_2 > NH_3$

**28.** (*c*)

: Non-planar heterocyclic compound

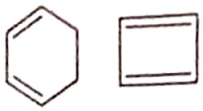
: Bicyclo Compound

: Spiro Compound

: Aromatic Compound

## JEE ADVANCED

**29.** (*b,c*)



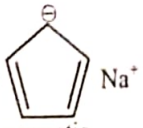
Antiaromatic ($4\pi e^{\ominus}$)

so unstable.

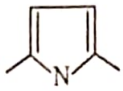
**30.** (*a,b,c,d*)

(P) ⇒ $AlCl_4^-$

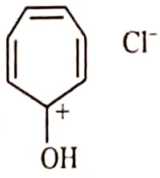
(Aromatic)

(Q) ⇒  $Na^+$

aromatic

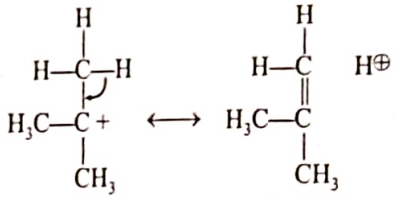
(R) ⇒ 

aromatic

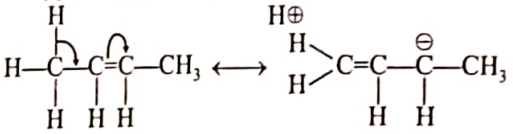
(S) ⇒  $Cl^-$

aromatic

**31.** (*a*)



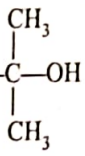
In tert butylcation, carbon bearing positive charge has one vacant p-orbital hence it is σ–p (empty onjugation or hyperconjugation)



In 2-butence, hyperconjugation is between σ–π* orbital.

**32.** (*a,c,d*) We need to draw all the structures of compound given. D.U. = 0. Indicates no any double bond. The alcohol contain –OH

So here are all the possible structures

(*a*) 

Tertiary butanol or 2-methyl propan-2-ol

(*c*) OH and OH

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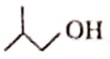
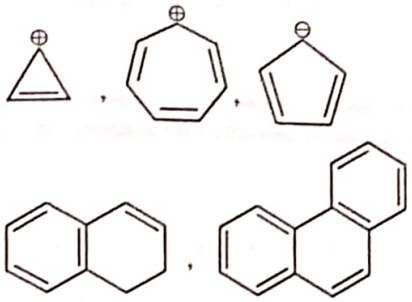
(*d*)  2-methyl propan-1-ol and iso butyl alcohol

So correct answers are (a, c, d)

**33.** [5] Aromatic $\Rightarrow (4n+2)\pi e^{\odot}$



**34.** (d) Correct order of basicity of the given compounds is iv > i > ii> iii

In compound iv when lone pair of electron on N-atom is donated to suitable acid conjugate acid is stabilized by resonance by two $NH_2$ group

In (I) conjugate acid is stabilized by resonance by one $NH_2$ group

In (III) conjugate acid is stabilized by resonance by one NH group

**35.** (*d*) Acidity $\propto$ stability of conjugate base among I and II, I is more acidic as triple bond has higher s-character.

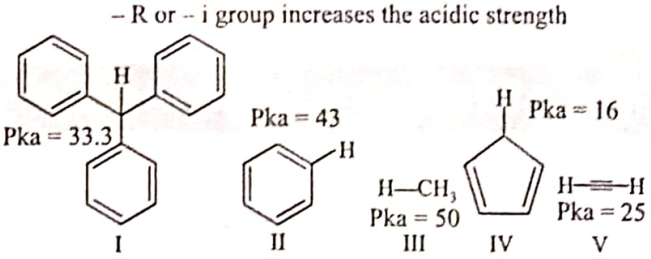
So I > II acidic

between III and IV, III will be more acidic because —COOH group is attached with $sp^2$ carbon where as in IV COOH group is attached with $sp^3$ carbon so III is more acidic than IV

**36.**(*a,b,c*) The conjugate base of compound I is stabilized by delocalization as negative change is in conjugation withe the phenyl rings.

Conjugate base of compound IV is aromatic and is most stable

– R or – i group increases the acidic strength




**37.** (*) IV is strongest base due to SIR effect.

III is weakest base due to – M group of three –NO2 present at ortho and para position.

II is more basic than I due to +I group present. IV is approximately 40,000 times more basic than III.

Where as I and III have little difference in basic strength

so correct options are c and d

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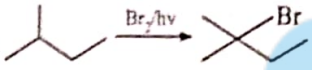
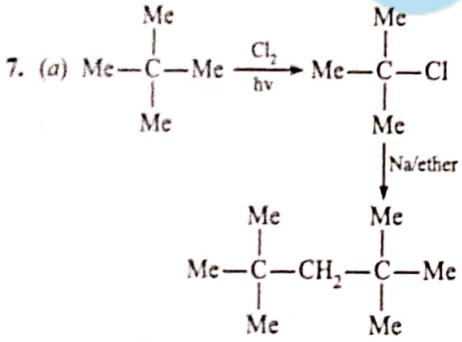
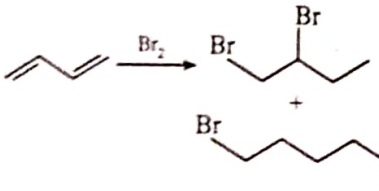
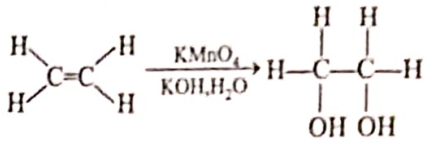
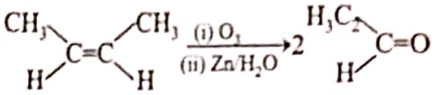
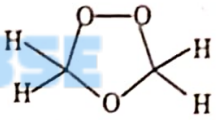
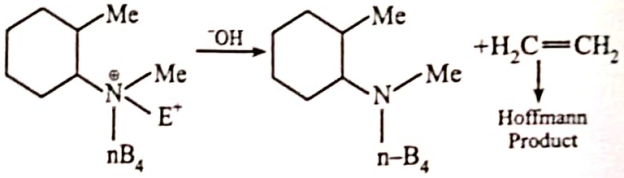
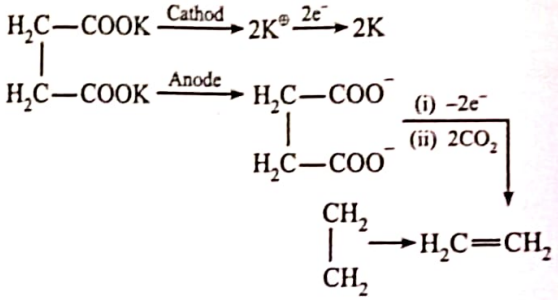
CHAPTER

# 15 Hydrocarbon

## Exercise-1 (Topicwise)

1. (*c*) Unsymmetrical alkane.
   Coery-Hons Synthesis
2. (*d*) $C_7H_{16}(C_nH_{2n+2})$
3. (*a*) Boiling Point $\alpha$ Van Der Walls force.
4. (*d*) Ester on reduction with $LiAlH_4$ produces alcohol.
5. (*b*) neopentane have all equivalent H-atoms. Hence will give only one monosubstituted alkyl Halide.
6. (*d*) 
   Reactivity of 3°H will be highest for Photocatalyzed Halogenation.
7. (*a*) 
8. (*d*) Anti addition to trans alkene gives meso compound and bromine gives trans addition.
9. (*c*) 
   Since formed carbocation can rearrange and will given 1, 2 & 1, 4 both product.
10. (*b*) $CH_2{=}CH_2 \xrightarrow{HOCl} CH_2(Cl)-CH_2(OH) \xrightarrow{aq\ NaHCO_3}$ $CH_2(OH)-CH_2(OH)$ Glycol
11. (*d*) $C_2H_5I + alc.KOH \rightarrow C_2H_4 + KI + H_2O$
12. (*a*) 
13. (*b*) 
14. (*b*) $CH_3-CH_2-CH{=}CH_2 \xrightarrow{HBr} CH_3-CH_2-\underset{Br}{CH}-CH_3$ 2-Bromo butane
15. (*c*)  Ozonide
16. (*b*) Markovnikov's rule can not be applied for symmetrical alkene.
17. (*d*) $CH_3CH_2-OH \xrightarrow[433K]{Conc.H_2SO_4} CH_2=CH_2$ (Intramolecular dehydration)
18. (*b*) 
19. (*c*) 
20. (*d*) Nehydration of Alcohol can be done by (i) conc. $H_2SO_4$ and Heat (ii) $H_3PO_4$ (conc.) and Heat

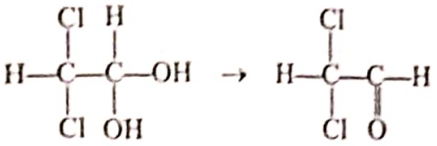
**21.** (*d*) $H_3C-HC=CH-CH_3 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} 2H_3C-CHO$

**22.** (*d*) $H_3C-CHO \longrightarrow m \cdot m = 44$

Hence symmetrical alkene should be

α Butene $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} 2CH_3CHO$

**23.** (*c*) $H-C\equiv C-H + HOCl \rightarrow C(H)=C(H)\rightarrow HOCl$ (Cl, OH)



Unstable

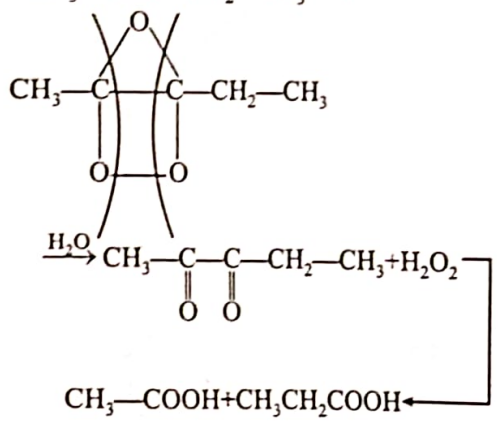
With acetylene it gives dichloroacetaldehyde.

**24.** (*c*) $2CH\equiv CH + 5O_2 \rightarrow 4CO_2 + 2H_2O\ \Delta H = -1300$ KJ

The combustion of acetylene is highly exothermic and the heat produced during the combustion can be used for welding purposes in the form of oxy acetylene flame.

**25.** (*a*) $3\ CH\equiv CH \xrightarrow[Fetube]{Red\ hot}$ Benzene

**26.** (*c*) $CH_3-C\equiv C-CH_2-CH_3 \xrightarrow{O_3}$



$\xrightarrow{H_2O} CH_3-C(=O)-C(=O)-CH_2-CH_3 + H_2O_2$

$\rightarrow CH_3-COOH + CH_3CH_2COOH$

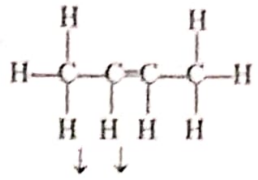
**27.** (*c*) $CH\equiv CH \xrightarrow[H_2SO_4]{H_2O/Hg^{2+}} H_3C-\underset{(x)}{C(=O)}-H \xrightarrow{LiAlH_4}$

Cyclo octatatraene

$CH_3-\underset{(y)}{CH_2}-OH \xrightarrow{P_4/Br_2} \underset{(z)}{C_2H_5Br}$

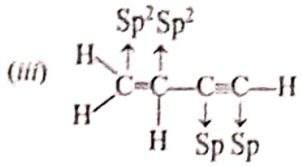
**28.** (*a*) $CH_3-CH_2-C\equiv CH \xrightarrow[H_2SO_4]{Hg_4SO_4,60^\circ C} CH_3-CH_2-C(=O)-CH_3$

**29.** (*a*) (*ii*)



$Sp^2\ Sp^2$

(*iii*)

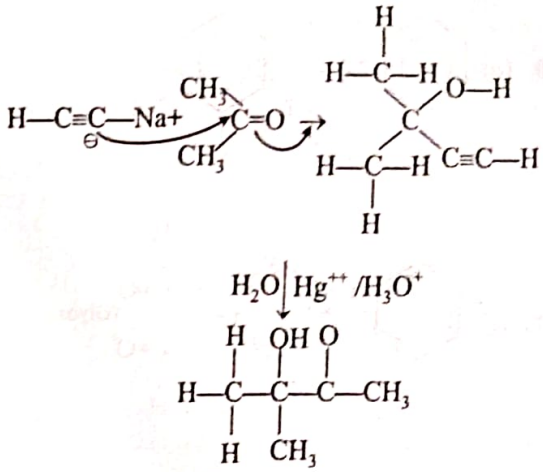


**30.** (*a*) $3CH\equiv CH \xrightarrow[AlCl_3]{Hot\ Cu\ tube}$ Benzene

**31.** (*b*) $H-C(=O)-OH \xleftarrow[(ii)H_2O_2\ (Oxidative\ ozonolysis)]{(i)\ O_3}$ Formic acid

$H-C\equiv C-H \xrightarrow[(ii)Zn/H_2O\ (reductive\ ozonolysis)]{(i)\ O_3} H-C(=O)-C(=O)-H$ Glyoxal

**32.** (*c*) $H-C\equiv C-H \xrightarrow{NaNH_2} H-C\equiv C^{\ominus}\ {}^{\oplus}Na + NH_3$



**33.** (*b*) $CH_3-\underset{CH_3}{C}=CH_2 + HCl \rightarrow CH_3-\underset{CH_3}{\overset{Cl}{C}}-CH_3$

$CH=CH + HCl \rightarrow CH_2=CH-Cl \xrightarrow{HCl} CH_3-CHCl_2$

**34.** (*b*) $CH\equiv CH$ Hydrogen connected to carbon in sp hybridization is acidic Hydrogen.

$CH_2=CH_2$ less acidic.

So order will be $\rightarrow CH\equiv CH > CH_2\equiv CH_2 > CH_3-CH_3$.

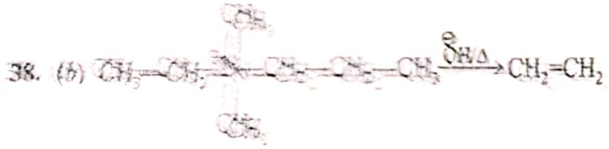
**35.** (*d*) All kinds of unsaturated hydrocarbans will give chlorination.

@PROCBSE

36. (c)

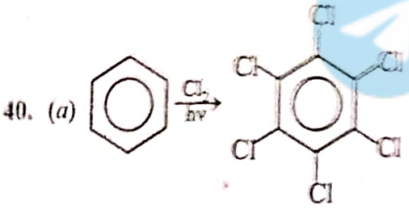
37. (c)

38. (b)

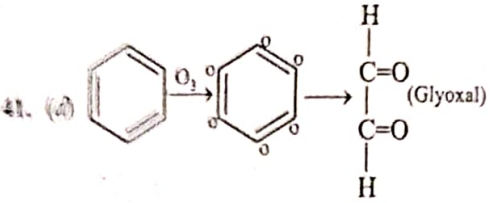


39. (c)

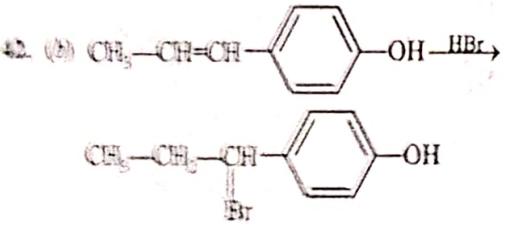
40. (a)



41. (d)



42. (b)

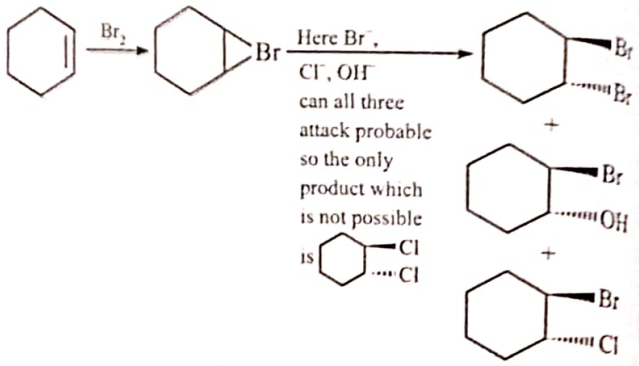


43. (c)

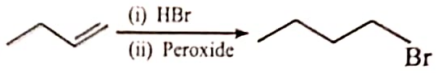
(Anti Markovnkoff Product)

44. (a) Since H—Cl and H—I do not break in radicals. Hence they don't give Anti Markovnikov's product.

45. (d)



46. (d)

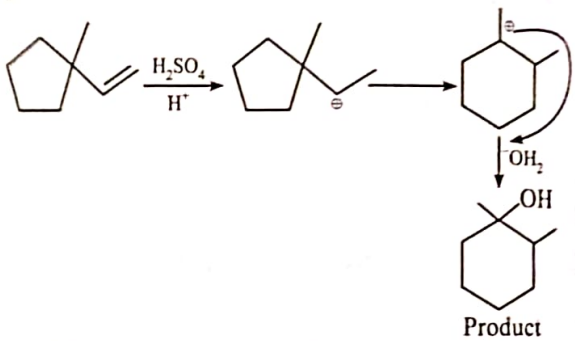


Anti Markovnibovs Product proceed through free radical mechanism.

47. (a) Acid catalyzed Hydration is $SN^1$ mechanism. Hence Reactivity order follow the stability of formed carbocation during the course of reaction. Hence

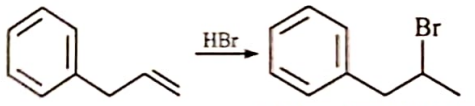
$(CH_3)_2CH=CH_2 > H_3C-CH=CH_2 > H_2C=CH_2$

48. (c)



Product

49. (b)



Markovnikov's Addition

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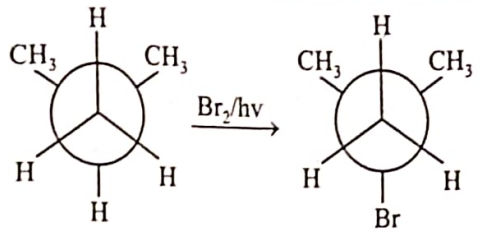
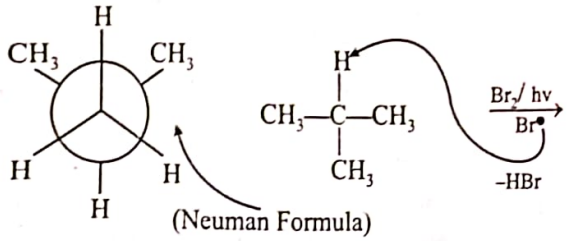
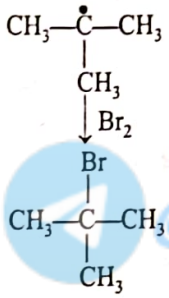
## Exercise-2 (Learning Plus)

**1.** (*b*)

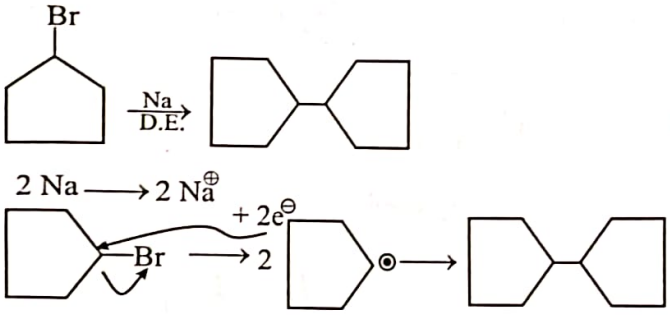



$Br_2 \longrightarrow 2\,Br^{\bullet}$

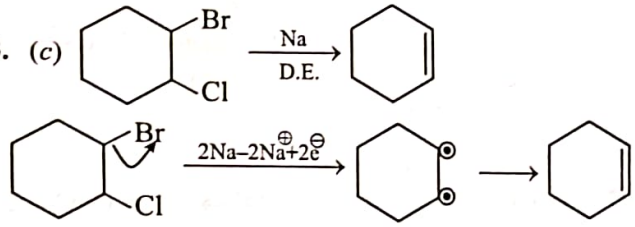


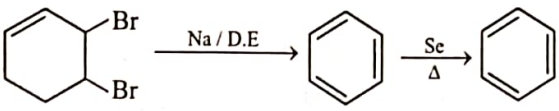

**2.** (*a*) Mechanism



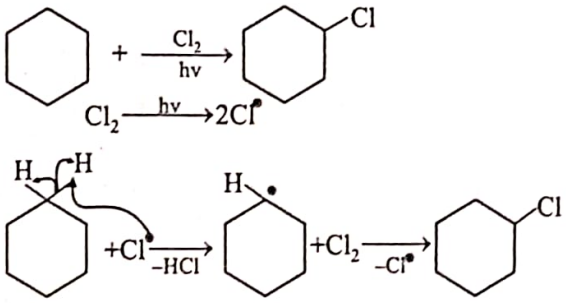

**3.** (*c*)



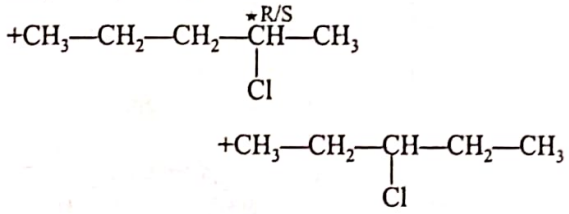

**4.** (*b*)



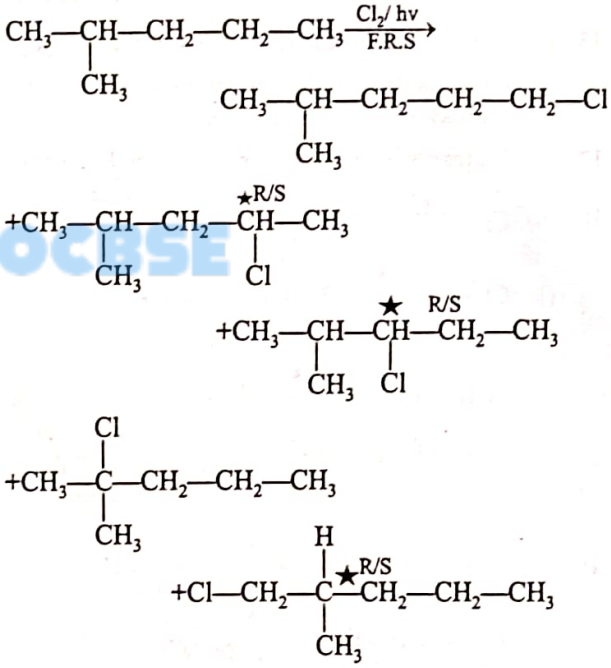

**5.** (*a*) Mechanism




**6.** (*c*) $CH_3—CH_2—CH_2—CH_2—CH_3 \xrightarrow[h\nu]{Cl_2} CH_3—CH_2—CH_2—CH_2—CH_2—Cl$

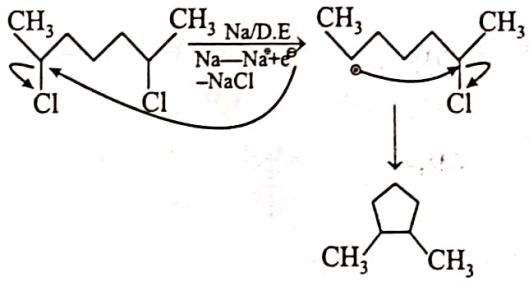



**7.** (*c*)




Total no. of monochloro product = 7

**8.** (*d*)




**9.** (*d*) $CH_3—Cl + CH_3—CH_2—Cl \xrightarrow[D.E]{Na} CH_3—CH_3 + CH_3—CH_2—CH_2—CH_3 + CH_3—\underset{CH_3}{\underset{|}{CH}}—CH_3$

**10.** (*c*)

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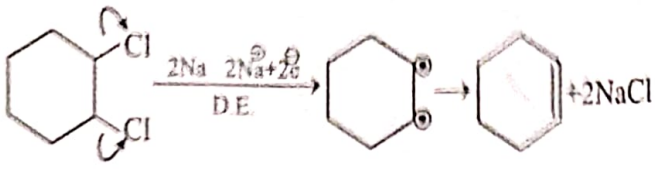
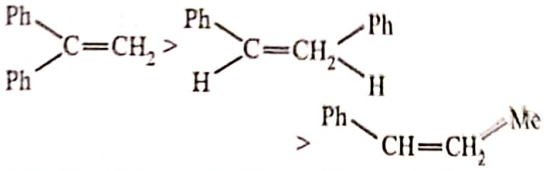


11. (*a*) Stability depends on hyperconjugation which further depends on total number of αH.

12. (*c*) Dipole moment is a vector quantity. In trans 1, 2-Dichloroethene, all the vector cancel each other

13. (*d*) Heat of hydrogenation $\propto \frac{1}{\text{Stability of alkene or crowding across } \pi \text{ bond}}$

14. (*c*) Stability of Alkene depends upon substituents and Hyperconjugation effects. Hence



15. (*c*) Stability α No. of α-Hydorgen for hyperconjugation.

16. (*a*) Electrophilicity of Double Bond α Stability of formed Carbocation α no. of α-H atom.

17. (*d*) No rearrangement in oxymercuration Demercuration.

18. (*a*) $CF_3-\overset{\curvearrowleft}{CH}=CH_2 \xrightarrow{H^+Cl^-} CF_3-CH_2-CH_2-Cl$

(*b*) $Cl-\overset{\curvearrowleft}{CH}=CH_2 \xrightarrow{HCl} Cl-\underset{Cl}{CH}-CH_3$

$CH_3-O-\overset{\curvearrowleft}{CH}=CH_2 \xrightarrow{HCl}$

(*c*) $CH-O-\underset{Cl}{CH}-CH_3$

19. (*b*) $CCl_3CH=CH_2 \xrightarrow{Cl_2+H_2O}$

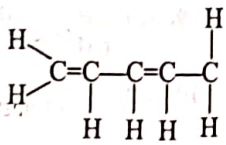
$CCl_3-\overset{\oplus}{\underset{Cl}{CH}}-CH_2 \xrightarrow{H_2\ddot{O}:} CCl_3-\underset{OH}{CH}-\underset{Cl}{CH}$

20. (*b*) $CH_2=CH-CH_2-CH=CH_2 \xrightarrow{NBS}$

$CH_2=CH-\underset{\bullet}{CH}-CH=CH_2$

Resonance

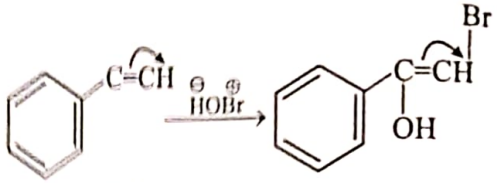
$CH_2=CH-CH=CH-CH_2-Br \xleftarrow{NBS}$

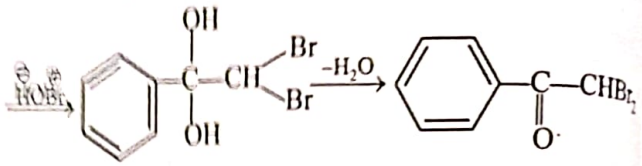


21. (*d*) $CH_3-C\equiv CH \xrightarrow{HBr} CH_3-\underset{Br}{C}=CH_2$

$CH_3-\underset{Br}{C}=CH_2 \xrightarrow{HBr} CH_3-\underset{Br}{\overset{Br}{C}}-CH_3$

22. (*a*)



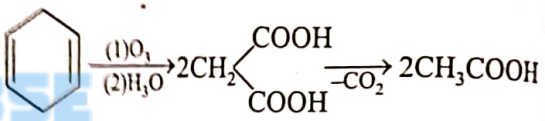


23. (*b*) When double bond and triple bond is in the conjugation then triple bond is more reactive due to more stable carbocation.

$CH_2=CH-\overset{\oplus}{C}=CH_2 \longleftrightarrow \overset{\oplus}{C}H_2-CH=C=CH_2$

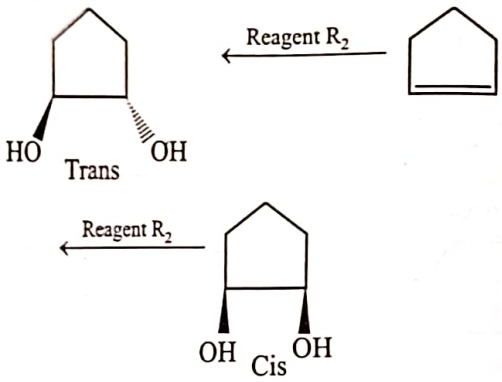
more stable

24. (*a*)



25. (*c*) $CH_3-CH_2-C\equiv CH \xrightarrow{KMnO_4/H^\oplus} CH_3CH_2COOH + CO_2 + H_2O$

26. (*b*)



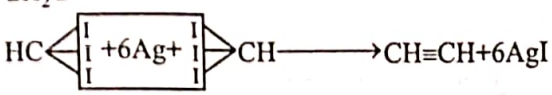
Syn addition with Baeyer's Reagent, So cis will form meso compound however peroxy acid in anti addition gives enantiomer.

27. (*a*) $CH_3-\underset{Cl}{\overset{Cl}{C}}-\underset{Cl}{\overset{Cl}{C}}-CH_3 \xrightarrow{Zn\ dust} CH_3-C\equiv C-CH_3$

(But -2-yne)

28. (*c*) $\underset{CH-COONa}{\overset{CH-COONa}{\|}} \xrightarrow[(aq.)]{Electrolysis} \underset{CH}{\overset{CH}{\|}} + 2CO_2 + 2NaOH +$

29. (*d*) $2H_2O$

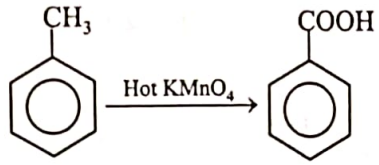



30. (*c*) sp Hybridisation of alkyne and also intermediate form is less stable.

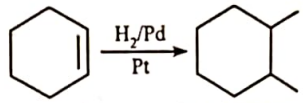
31. (*a*) $CaC_2 + H_3O^+ \longrightarrow Ca(OH)_2 + C_2H_2$ (Acetylene)

32. (*b*) Due to relatively compact $e^-$ cloud of alkynl, there are less reactive to electrophilic addition than alkenes.

33. (*c*)



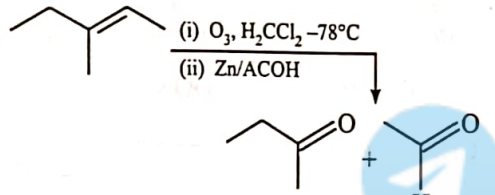

34. (*c*)



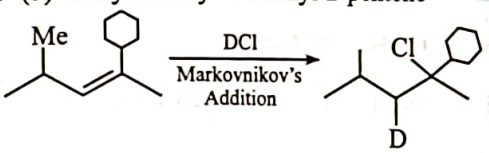

Syn Addition Product

Hydrogenation of Alkene in presence of Metal is a syn addition.

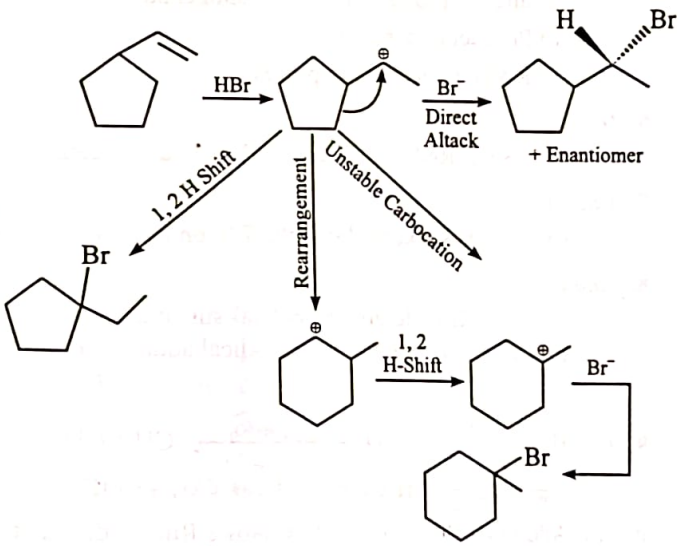
35. (*c*)

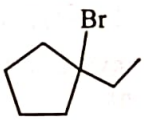
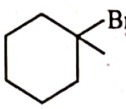



36. (*b*) 2-Cyclohexyl-4-methyl-2-pentene

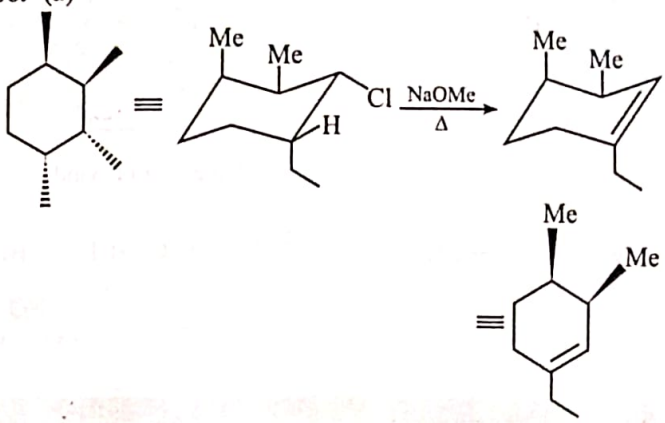



37. (*d*)

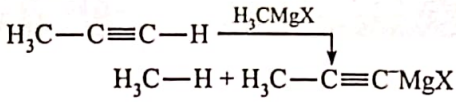



Hence  and  will be major product.

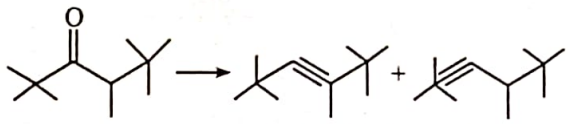
38. (*a*)




39. (*d*) $H_3CMgX$ will take acidic proton from alkyne



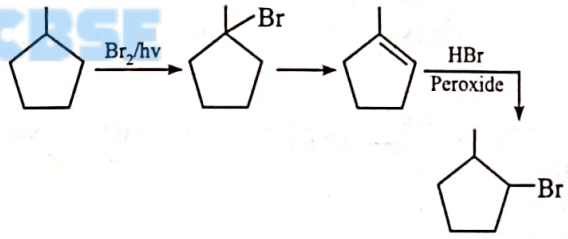

40. (*c*) If we do retro synthesis




(not possible since carbon don't have beyond four valencies)

Hence correct answer will be four.

41. (*c*)



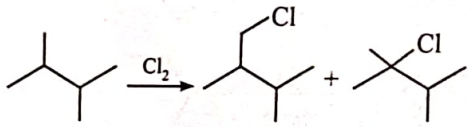

42. (*d*) All of the given reaction give the elimation product of Alkyne.

43. (*c*) Activating groups accelerate the electrophilic susbtitution reaction.

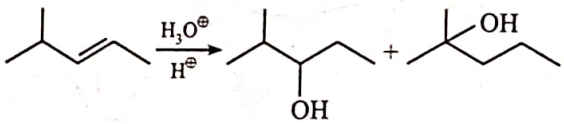
Hence order of rate I > III > IV > II

TM of F > +M of Cl

44. (*c*)

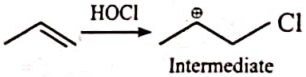



45. (*a*)

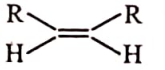



Alkene other than ethene can rearrange to given 2° & 3° alcohols.

46. (*d*)




Intermediate

47. (*a*) Most unstable alkene will go for hydrogenation first, hence correct answer will be 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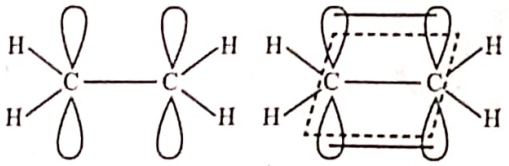
**48.** (*a*)



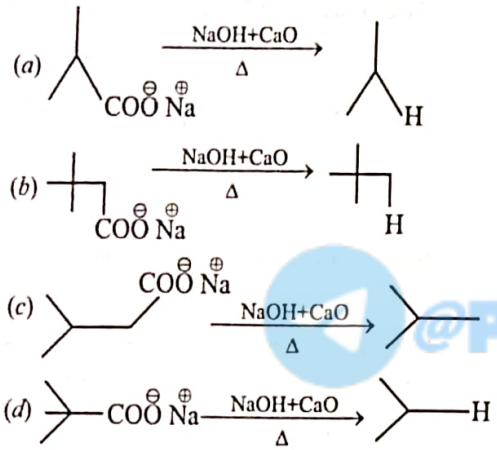
Plane ± to IT bond.

**49.** (*b*) $H_3C-CH\equiv CH_2 + NOCl \longrightarrow H_3C-\underset{Cl}{\underset{|}{CH}}-\underset{NO}{\underset{|}{CH_2}}$

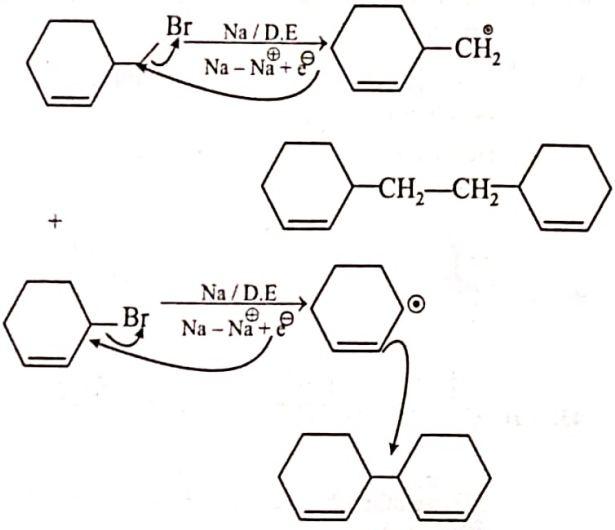
Markovnikov's Product

## Exercise-3 (JEE Advanced Level)

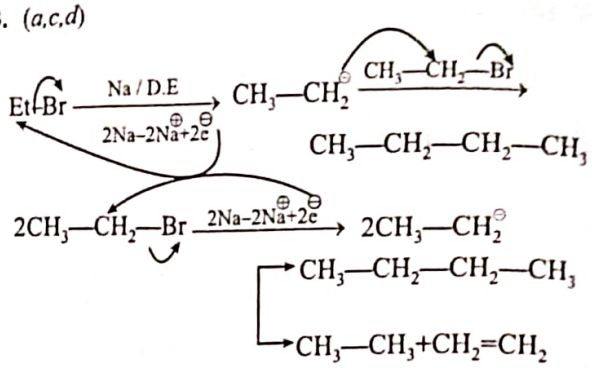
**1.** (*c,d*)



**2.** (*a,b*)

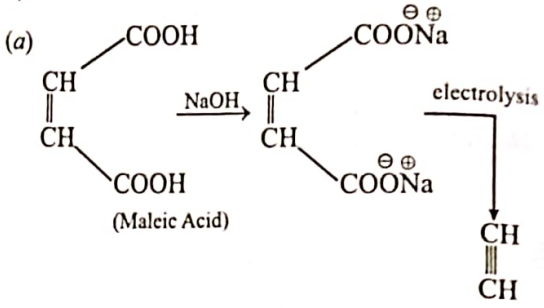


**3.** (*a,c,d*)



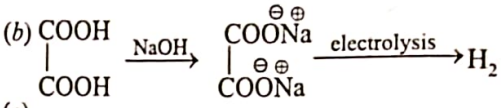
**4.** (*a,b,c*)

(*a*)

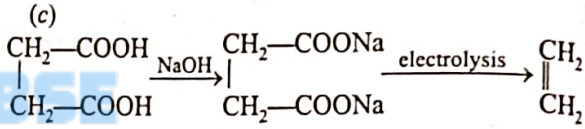


(Maleic Acid)

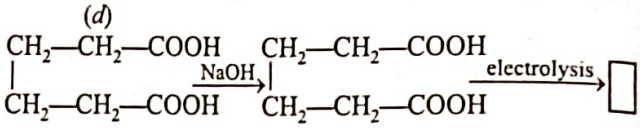
(*b*)



(*c*)



(*d*)



**5.** (*a,b,c,d*)

(*a*) and (*b*) Product of dehydrohalogenation.

(*c*) Product of dehalogenation.

(*d*) Kolbe's Electrolysis process.

**6.** (*a,b,c*)

X is Birch Reducing agent and Y is lindlar's catalyst.

**7.** (*a,c,d*)

(*a,c,d*) gives alkenes by cyclic T.S. on heating.

**8.** (*a,b,c*)

(*a,b,c*) will undergo free radical subsitution reaction however D will perform free radical addition reaction.

**9.** (*b*) $H-C\equiv C-H + H_2O \xrightarrow{H_2SO_4, HgSO_4} CH_2=\underset{OH}{\underset{|}{CH}} \underset{\text{Tautomerise}}{\rightleftharpoons} CH_3CH=O \xrightarrow{[O]} CH_3-COOH$

**10.** (*c*) Addition of $H_2O$ by Markovnikov's Rule and by anti-Markovnikov's Rule. (Hydroboration oxidation).

**11.** (*c*) Addition of $H_2O$ by Markovnikov's Rule (By oxymercuration demercuration).

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**12.** (*d*) $\xrightarrow[Zn+H_2O]{O_3}$ $OHC-(CH_2)_3-CHO$

**13.** (*a*) $(CH_3)_2C=C(CH_3)_2 \xrightarrow[Zn+H_2O]{O_3} 2(CH_3)_2C=O$

**14.** (*a*) $CH_3-CH=CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3 \xrightarrow[Zn+H_2O]{O_3}$

$CH_3CHO + CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CHO$

$X(C_4H_9Br) \xrightarrow{\text{alc. KOH}} Y \xrightarrow{Br_2} Z$ (Dibromide)

$\xrightarrow{NaNH_2} W$ (gas) $\xrightarrow[\text{Solution}]{AgNO_3}$ White ppt

**15.** (*d*) 15 ml Hydrocarbon required 357 ml air (21% $O_2$)

357 ml air contains = $\frac{357\times21}{100} = 75$ ml $O_2$

Ratio of Hydrocarbon and oxygen

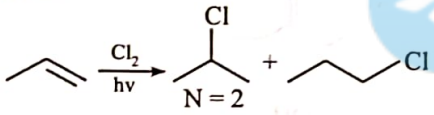
15 : 75 = 1 : 5

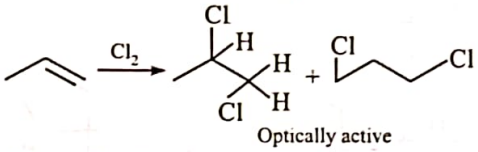
Now for combustion of $C_3H_6$ (A)

$C_3H_6 + SO_2 \rightarrow 3CO_2 + 2H_2O$

Hence correct option will be (d).

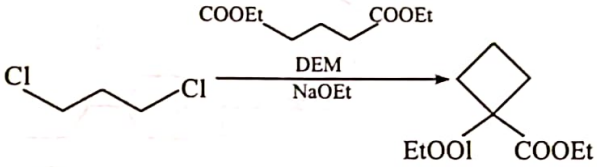
**16.** (*a*) $C_3H_6 \equiv$ will have 2 monochloro Product

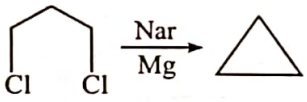


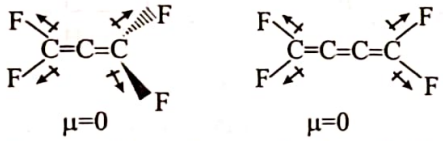
**17.** (*d*) 

Optically active

Hence M = 2

**18.** (*a*) 

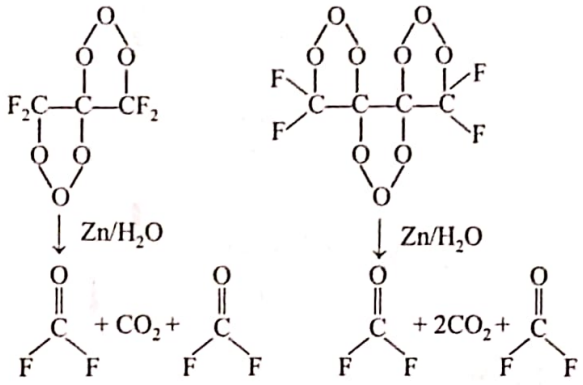
**19.** (*a*) 

**20.** (*d*) 

μ=0 μ=0

Hence neither of them will show dipole moment.

**21.** (*d*) Both are optically inactive hence neither of them will be resolvable.

**22.** (*a*) 

Hence both of them will give same product on reductive ozonolysis.

**23.** (*a*)

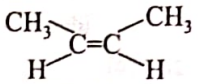
**Regioselectivity :** → Out of two positional isomers, one isomer is preferred.

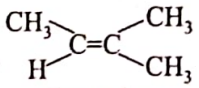
**Stereoselectivity :** → When one stereoisomer of the reactant has selected one stereoisomer of the products

**Chemoselectivity :** → The reagent selects only one functional group /multiple bond/ substituents

**Rearrangement :** → For increasing the stability carbocation undergo for rearrangement.

**24.** (*a*)

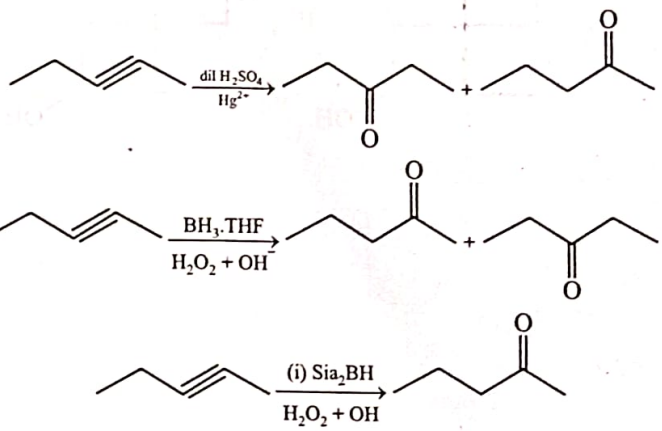
(*a*) $CH_3-C\equiv C-CH_3 \xrightarrow{H_2/Pd/Baso_4}$  Cis-2-butene

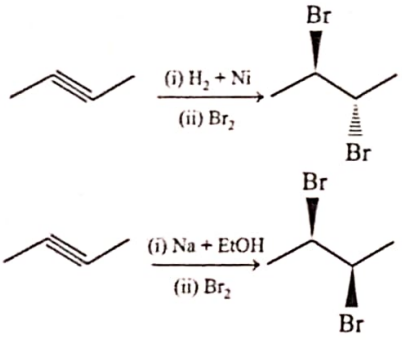
(*b*) $CH_3-C-C-CH_3 \xrightarrow{Na/NH_3/(liq.)}$  Trans-2-butene

(*c*) $CH_3-C\equiv C-CH_3 \xrightarrow{NaNH_2/\Delta} CH_3-CH_2-C\equiv CH$ (1-butyne)

(*d*) $CH_3-CH_2-C\equiv CH \xrightarrow{\text{alc KOH}/\Delta} CH_3-C\equiv C-CH_2$ (2-butyne)

**25.** (*c*)






**26.** [6]

$CH_3—Cl + 2Na + Cl—CH_3 \rightleftharpoons CH_3–CH_3$

$CH_3CH_2—Cl + 2Na + Cl—CH_2CH_3 \rightleftharpoons CH_3CH_2CH_2CH_3$

$CH_3CH_2CH_2—Cl + 2Na + Cl—CH_2CH_2CH_3 \rightleftharpoons CH_3CH_2CH_2CH_2CH_2CH_3$

$CH_3—Cl + 2Na + Cl—CH_2CH_3 \rightleftharpoons CH_3CH_2CH_3$

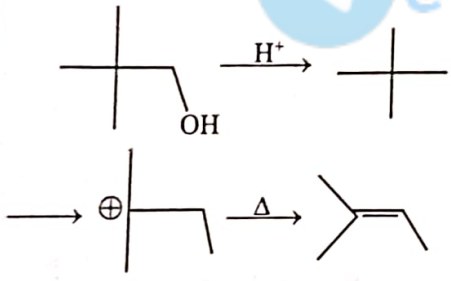
$CH_3CH_2—Cl + 2Na + Cl—CH_2CH_2CH_3 \rightleftharpoons CH_3CH_2CH_2CH_2CH_3$

$CH_3—Cl + 2Na + Cl—CH_2CH_2CH_3 \rightleftharpoons CH_3CH_2CH_2CH_3$

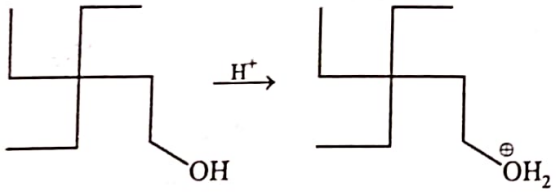
**27.** [4] On reaction with 2-chloro 3-methyl pentane gives non-identical four products.

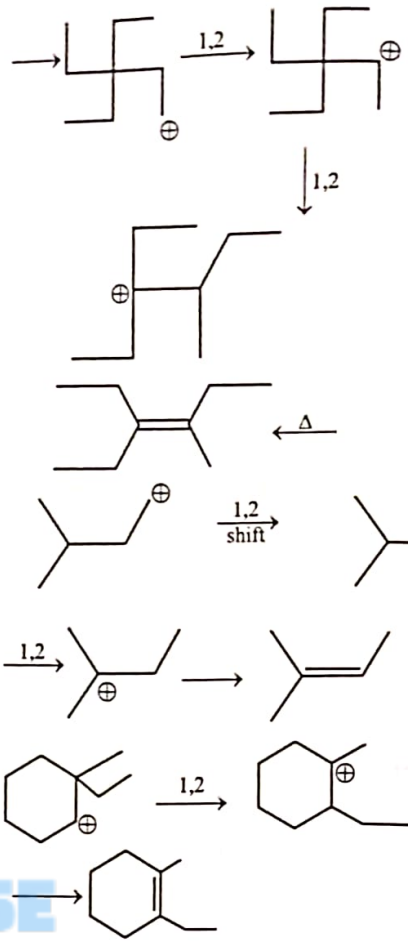
**28.** [4]

(*a*)



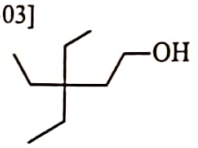

(*b*)



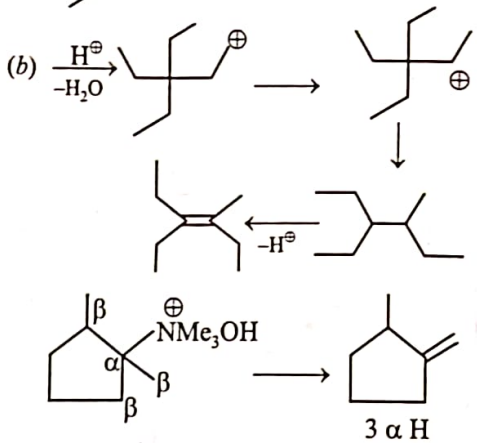




**29.** [0303]

(*a*)




(*b*)

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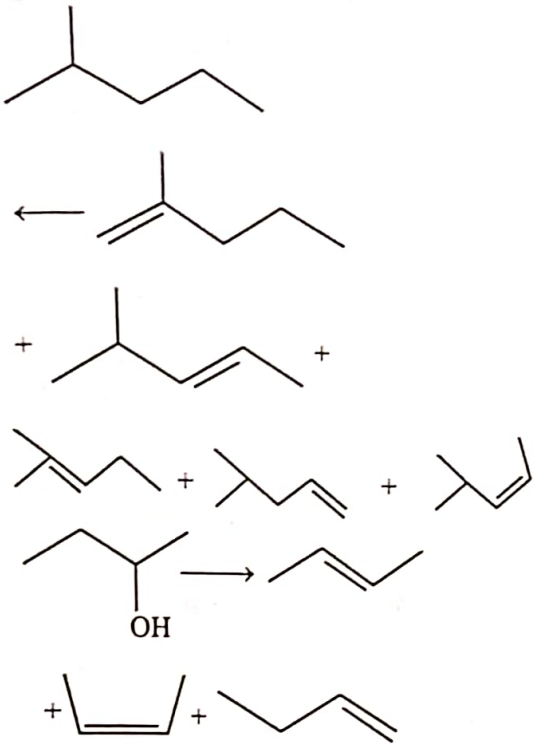
**30.** [0503]

(*a*)



(*b*)

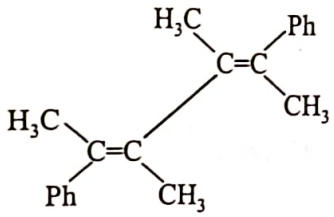




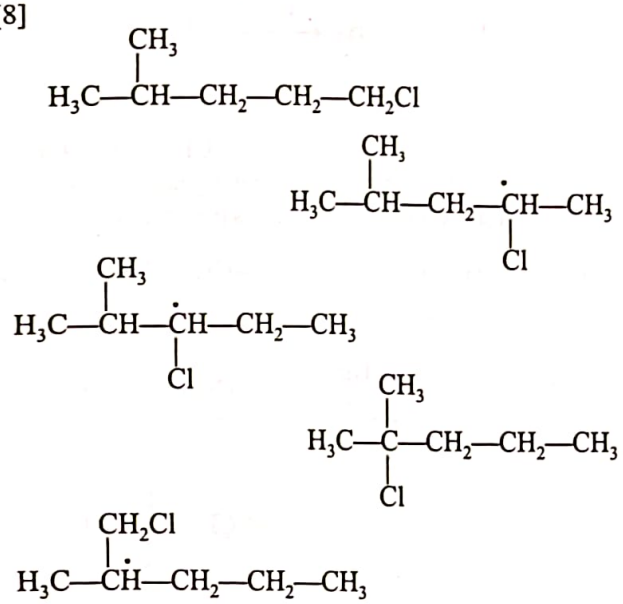


**31.** [3] More will be stability of carbocation more will be reactivity of substrate from which carbocation formed.

**32.** [1]



**33.** [8]



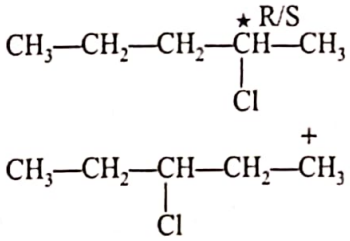
Compounds which have chiral carbon here exist in two forms: d and 1.

**34.** [4] Number of monochoro products are 6(P)

And distillation fractions are 4 (Q)

**35.** [4] $CH_3—CH_2—CH_2—CH_2—CH_3 \xrightarrow[hv]{Cl_2} CH_3—CH_2—CH_2—CH_2—CH_2—Cl$

+





**36.** [5]

$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \longrightarrow nCO_2 + (n+1)H_2O$$

$$C_3H_{8(g)} + \frac{3\times3+1}{2}O_2 \longrightarrow 3CO_2 + 4H_2O$$

1 Mole 5 Mole 3 Mole 4 Mole

**37.** [4]

$$H_3C—Cl + H_3C—CH_2—Cl \xrightarrow[\text{Dry Ether}]{Na} H_3C—CH_3$$

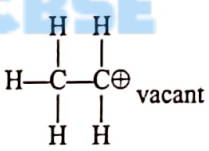
$$+ H_3C—CH_2—CH_2—CH_3$$

$$+ H_3C—CH_2—CH_3$$

$$+ H_2C=CH_2$$

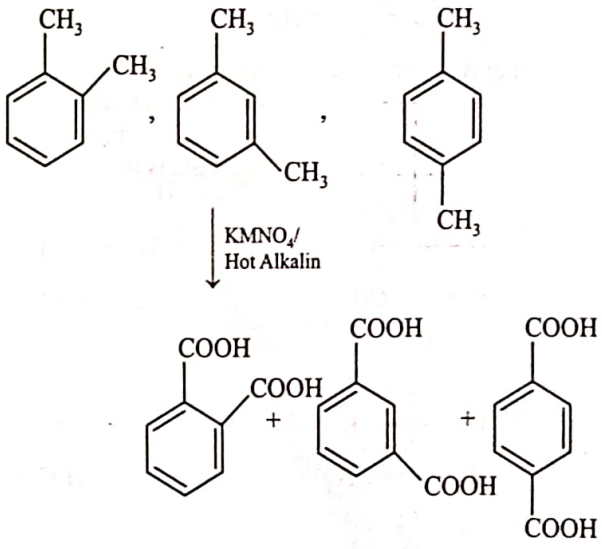
Hence total product will be (4).

**38.** [3]



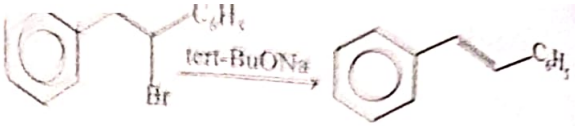
α—H=3 available for hyperconzugation

**39.** [3]

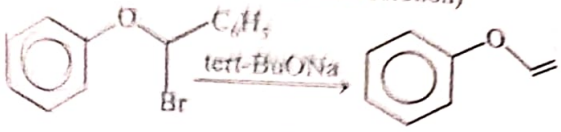


Dicarboxylic Acid

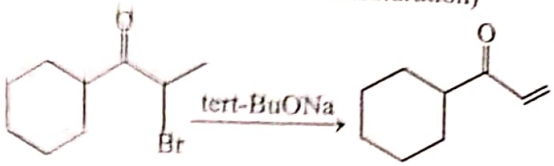
Hence correct answer will be 3.



(It decolorises bromine solution)

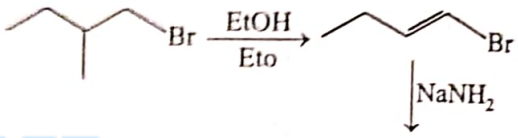


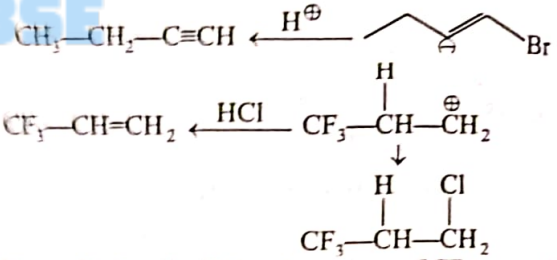
(It decolorises bromine solution due to unsaturation)



(It decolorises bromine solution due to unsaturation)

3. (d) Formation of allyl radical will promote a hydrogen to replace easily.

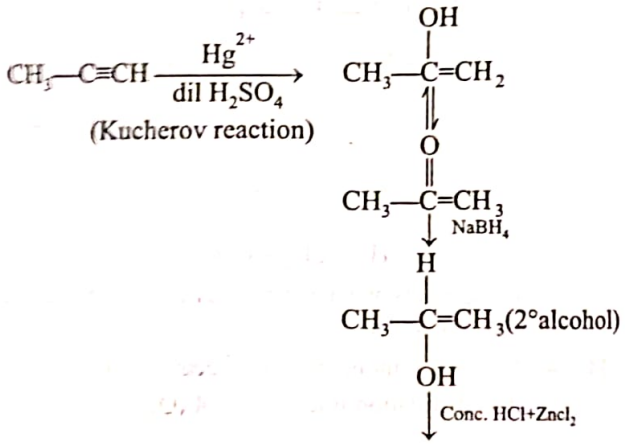
4. (d) 



5. (d) 

Due to higher $e^-$ withdrawing nature of $CF_3$ group. It follow anti markovnikov's product.

6. (b) $CH_3-C\equiv CH \xrightarrow[\Delta]{Ag_2O} CH_3-C\equiv C-Ag\downarrow$ (ppt)



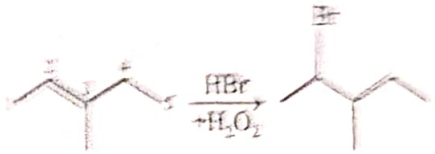
(Kucherov reaction)

Turbidity within 5 min.
(Lucas Reagent Test)

## Exercise–4
## (Past Year Questions)

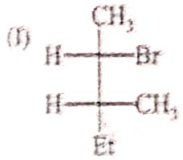
### MAIN

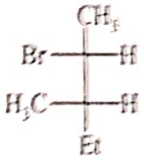
11. (a)

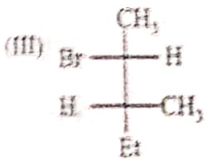


3-methyl pent-2-ene

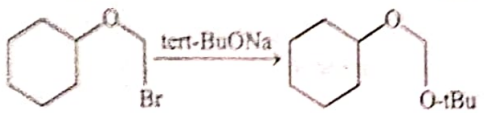
Anti markovnikov product (4 stereo isomers possible due to 2 chiral centre as molecule is nonsymmetric)

(I) 

(II) 

(III) 

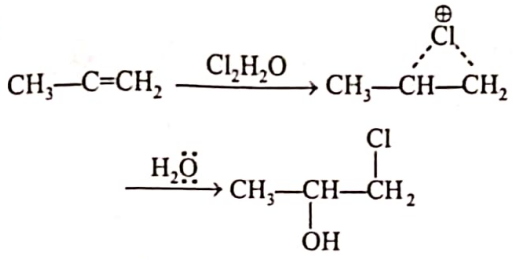
(IV)

2. (a) 

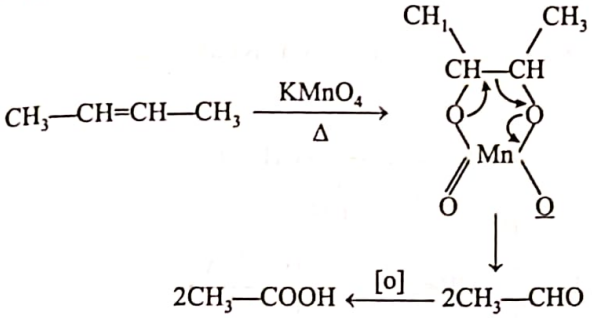
(Fails to decolorise the colour of bromine)

@PROCBSE

7. (*b*)



8. (*c*)



9. (*b*) $\log\frac{-d[A]}{dt} = \log\frac{d[B]}{dt} + 0.3010$

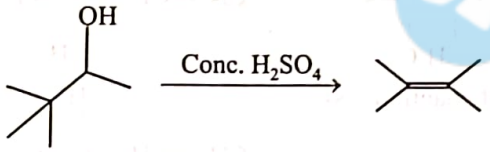
$$\frac{-d[A]}{dt} = 2\times\frac{d(B)}{dt} \Rightarrow \frac{1}{2}\times\frac{-d[A]}{dt} = \frac{d[B]}{dt}$$

$2A \rightarrow B$

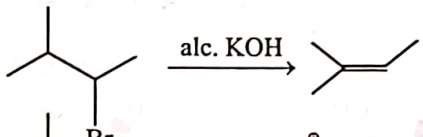
$2C_2H_4 \rightarrow C_4H_8$

10. (*d*) C gives Hoffmann product.

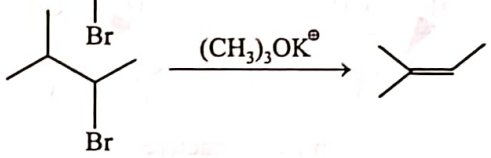
(*a*)



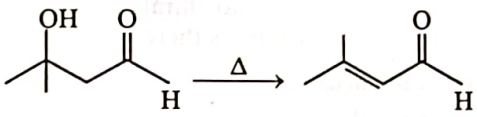
(*b*)



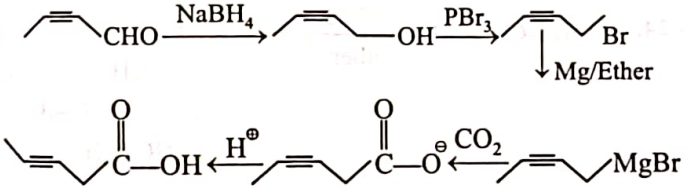
(*c*)



(*d*)

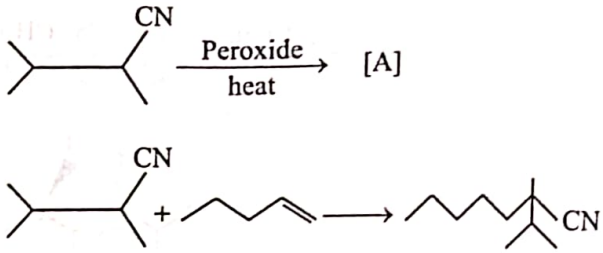


11. (*a*)



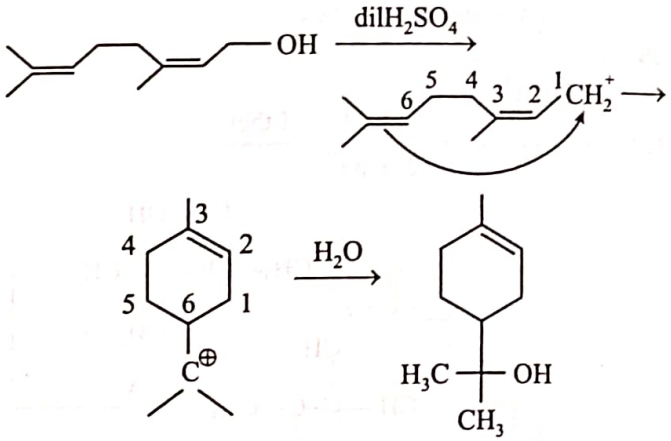
12. (*c*) Vinyl halides and aryl halides do not give Friedel craft's reaction.

13. (*b*)

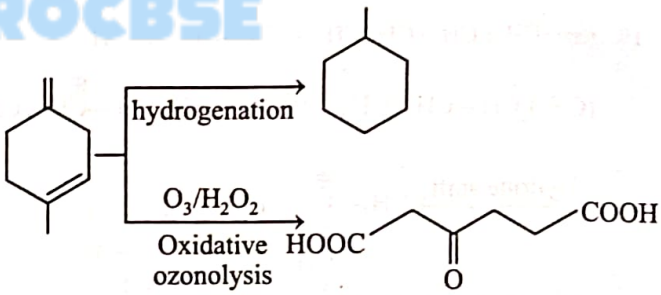


[A] Would be more stable radical and undergoes radical addition to form [B].

14. (*c*)

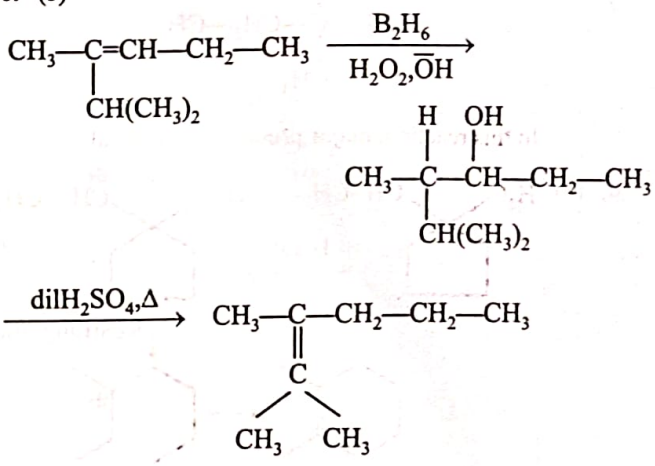


15. (*a*)



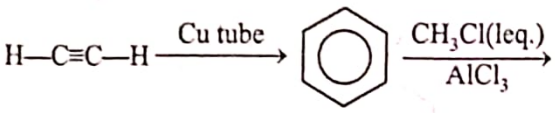
3-Oxohexanedioic acid

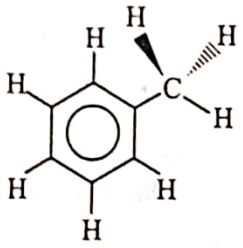
16. (*c*)



@PROCBSE

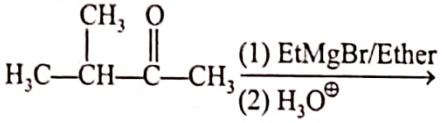
**17.** [13.00 to 13.00]

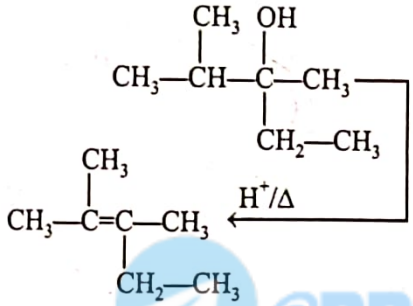


Number of atoms in one plane = 13.

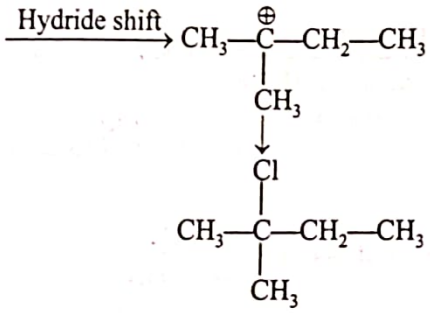
**18.** (*b*)



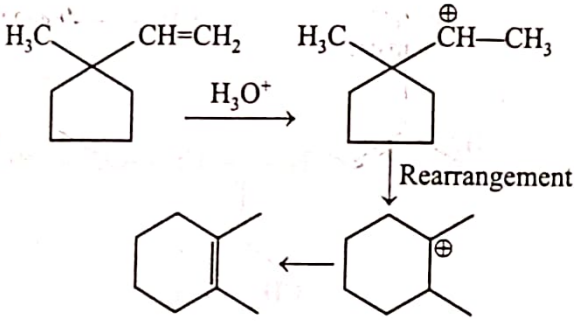

**19.** (*a*) $(CH_3)_2CH—CH=CH_2 \xrightarrow{HCl} (CH_3)_2CH—$

$(CH_3)_2CH—CH_2=CH_2 \xrightarrow{HCl} (CH_3)_2CH—\overset{\oplus}{C}H—CH_3$

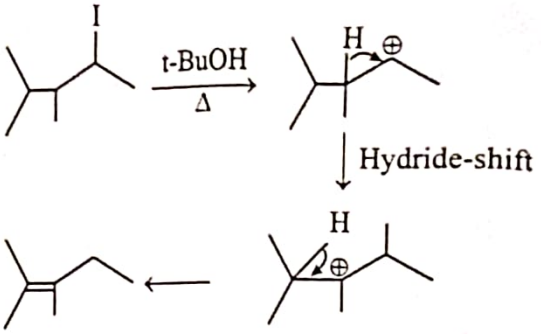



In this reaction, major product is not chiral.

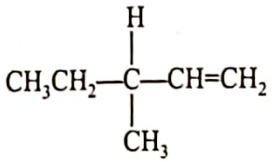
**20.** (*c*)

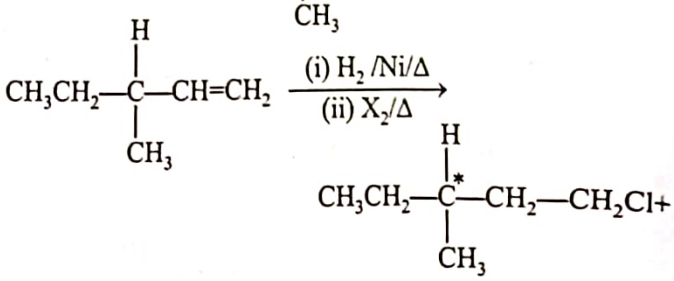



**21.** (*c*)

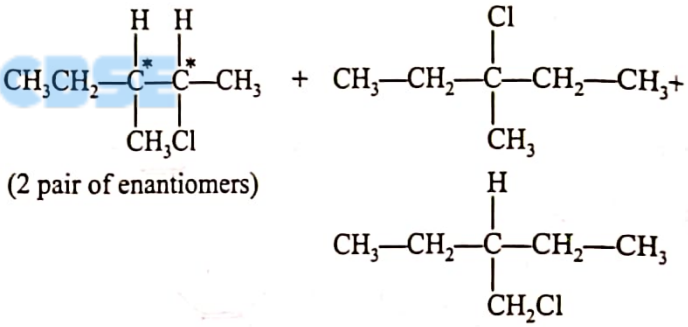



**22.** [8] The simplest optically active alkene is



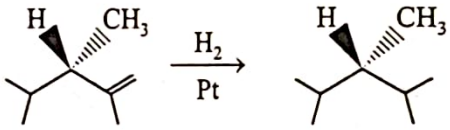

(1 pair of enantiomers)




(2 pair of enantiomers)



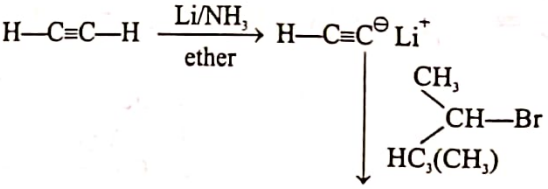

**23.** (*a*)

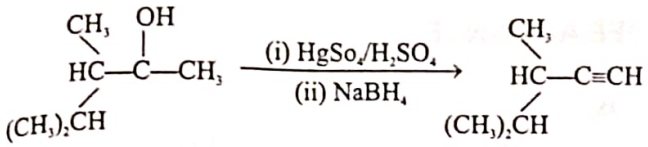


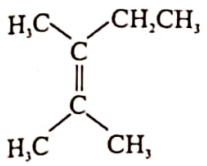

Optically inactive as no chiral centre is there

Optically inactive as no chiral centre is there

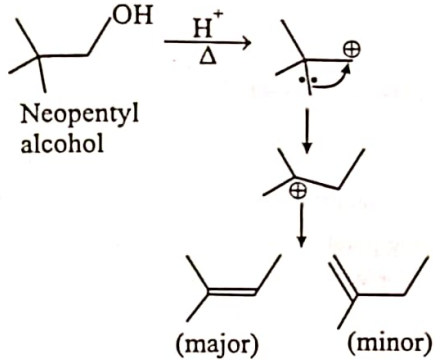
**24.** (*c*)



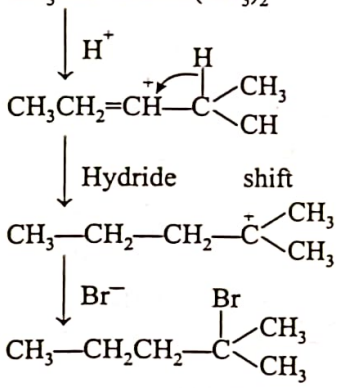


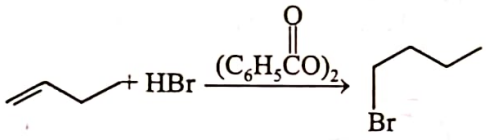
25. (*d*)

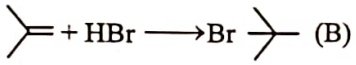


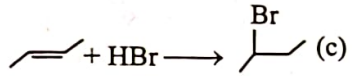

26. (*d*) $CH_3CH{=}CHCH(CH_3)_2$




27. (*c*)



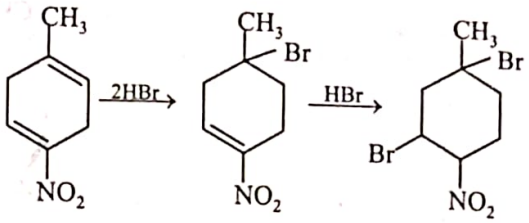




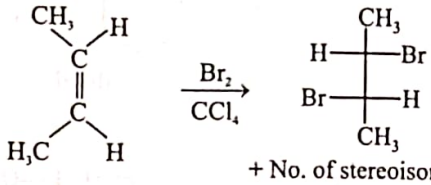
Boiling point decreases with branching

∴ order of B.P. is A > C > B

28. (*a*)

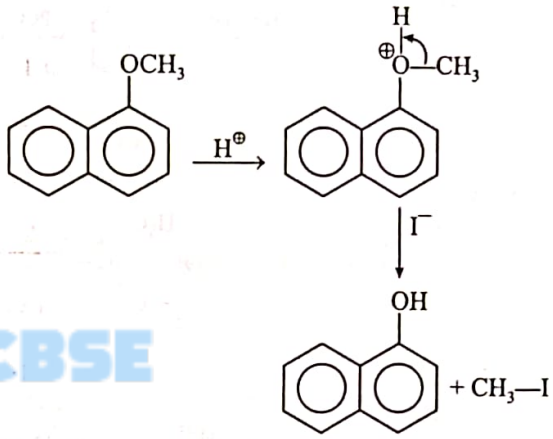


29. [2]



+ No. of stereoisomer = 2

30. (*d*)



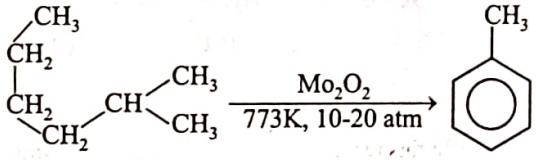
31. (*b*) Alc. KOH causes elimination

$Pd/BaSO_4$ - Lindlar's catalyst

BHC is obtained by the addition reaction of $Cl_2$ with benzene in presence of U.V.

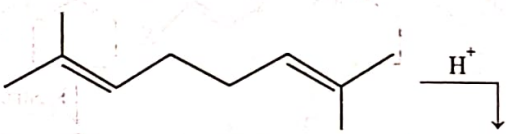
Thin film of polyacetylene can be used as electrode in batteries.

32. (*d*)

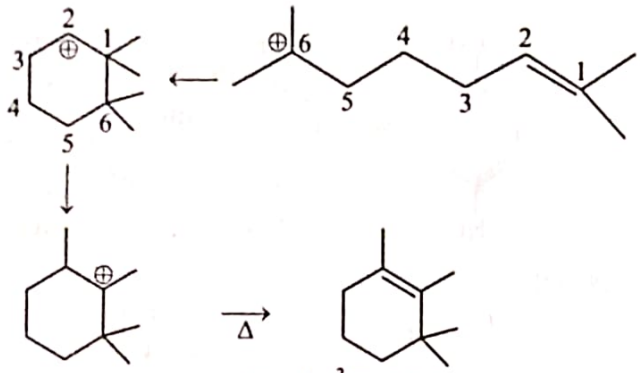


$Mo_2O_3$ at 773K temperature and 10-20-atm pressure is aromatising agent.

33. [2]

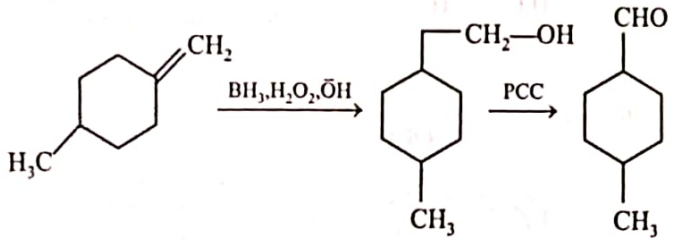


@PROCBSE

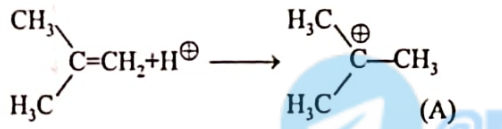


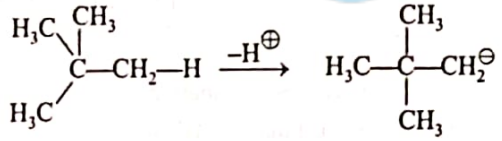
Total $sp^2$ Hybridised C-atom

**34.** (*c*)

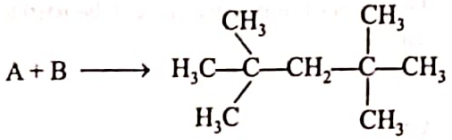


**35.** (*b*)

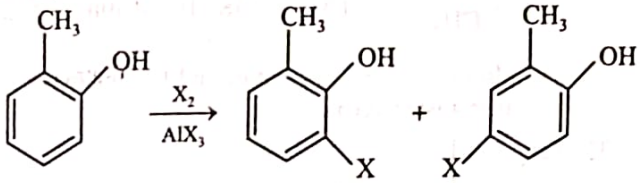




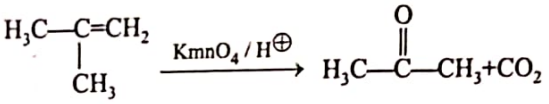
(B)



**36.** (*c*)

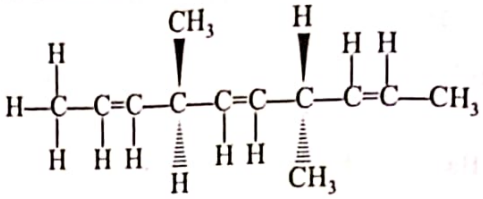


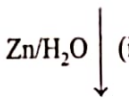
**37.** (*d*)

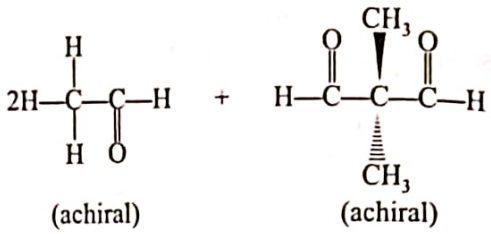


## JEE ADVANCED

**38.** (*a*)







(achiral) (achiral)

**39.** (*b*) Boiling point α Van Der Waals forces

Van Der Waals forces α sufrace area

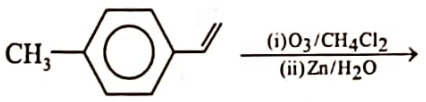
$$\text{Surface Area } \alpha \frac{1}{\text{Branching}}$$

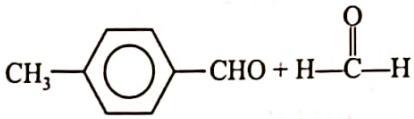
Hence branching leads to lesser surface area which results less boiling point.

Hence B.P. order will be III > II > I.

**40.** (*a,c*)

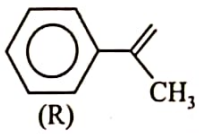
(*a*)



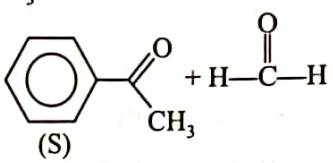


(Q)

given Cannizzaro reaction but no haloform



(R)



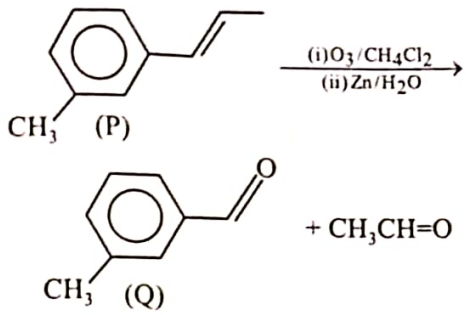
(S)

No Cannizzaro but give haloform

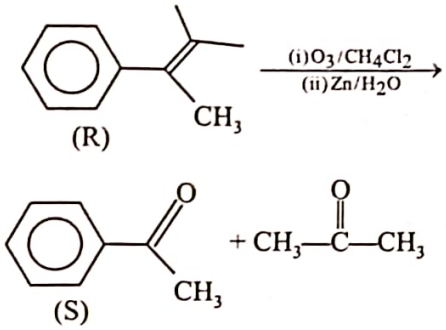
(*b*) Product of ozonolysis of R is having 9 carbon

@PROCBSE

(*c*)



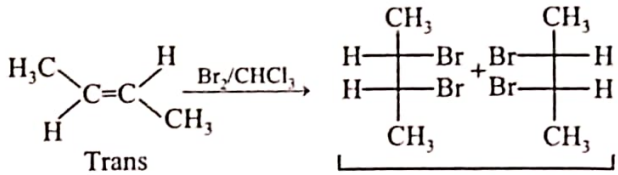

Given Cannizzaro reaction
but no haloform




No Cannizzaro
but give haloform

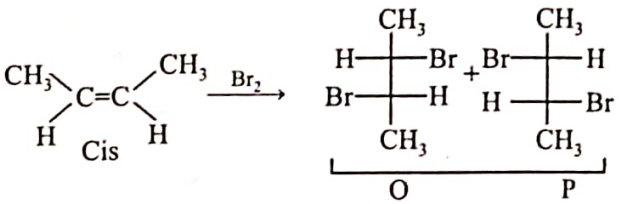
(*d*) Product of ozonolysis of R is having 9 carbon

**41.** (*b,c*)




Trans

Meso (M and N)




Cis

O P

Pair of enantiomers

C is anti Racemic

M and N are meso (identical)

O and P are pair of enantiomers

(*b*) Bromination proceeds through anti addition.

(*c*) (M and O) and (N and P) are two pairs of diastereomer.

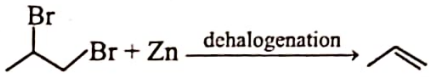
**42.** (*b,c*)

$CH_3—CH_2—CH_2—CO_2Na + H_2O \xrightarrow{\text{electrolysis}}$

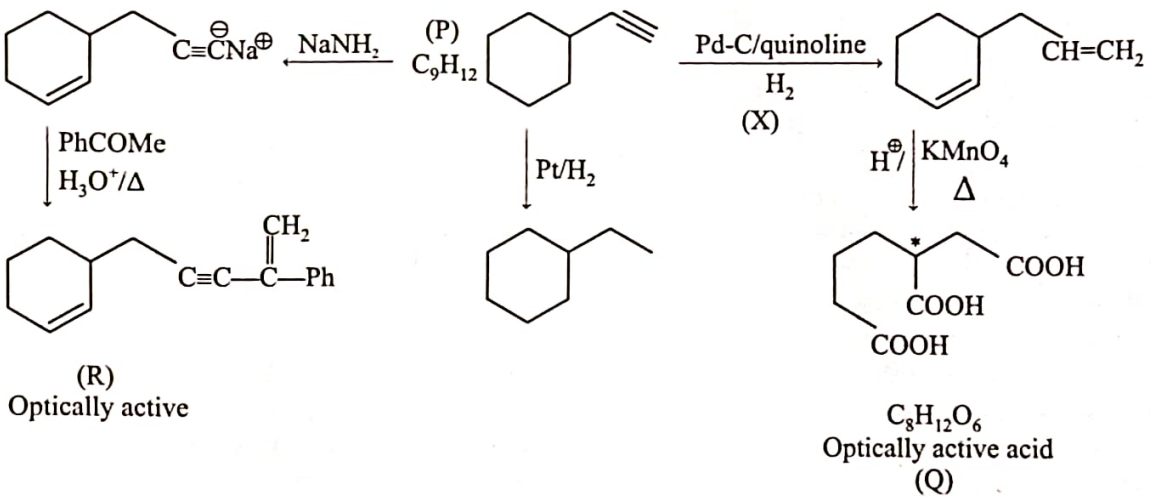
n-hexane

$CH_3—CH_2—CH_2—CO_2Na \xrightarrow[\Delta]{NaOH+CaO} CH_3—CH_2—CH_3$

$CH_3—CH_2—CH_2—Cl + Zn \rightarrow CH_3—CH_2—CH_2—ZnCl \xrightarrow{\text{dil.HCl}} CH_3CH_2CH_3$

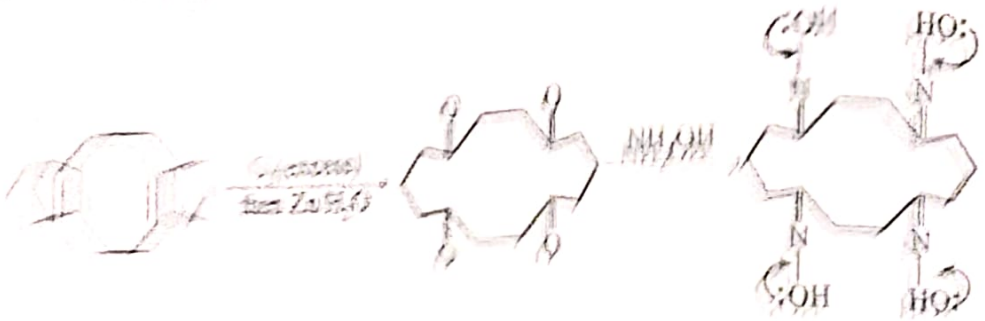



**43.** (*b,c*)




(R)
Optically active

$C_8H_{12}O_6$
Optically active acid
(Q)

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CHAPTER

# 16 Environmental Chemistry

## Exercise-1 (Topicwise)

1. (*c*) Domestic waste contains biodegradable pollutants.
2. (*a*) Tocoferol / Vit E → It has high concentration of antioxidants thus it produces resistance against carcinogens.
3. (*b*) DDT is an organochlorine compound used as an insecticide in agricultural fields. The major disadvantage of DDT is bioaccumulation or not easily degrading nature, due to fat solubility it is accumulated in living tissues.
4. (*c*) World environment day is celebrated on 5$^{th}$ June.
5. (*c*) Photo chemical oxidants are the secondary pollutants that are formed when primery pollutants like hydrocarbons, oxides of sulphur, oxides of nitrogen react with sunlight. Nitrous oxide, PAN and acrolein contribute to the formation of the photochemical smog.
6. (*c*) Freon typically contain hydrocarbon, chlorine or bromine. Thus freons are types of chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs) and conected compounds.
7. (*b*) Buring of fossil fuels is the main source of pollution of $SO_2$.
8. (*c*) $CO_2$ is not a kind of pollutant. It is produced naturally by humans and animals as a product of metabolic activities. It is accountable for sustaining the temperature of the atmosphere and plants use $CO_2$ as an essential element at the time of photosynthesis.
9. (*d*) Secondary air pollutants are produced photochemically from primary pollutants and are called photochemical oxidants. These occur in photochemical smog. PAN is a major constituent of photochemical smog.
10. (*c*) $H_2$ does not have a direct effect on climate. Thus $H_2$ gas is not a greenhouse gas.
11. (*c*) The reaction of $SO_2$ with water molecules present in atmosphere form sulphuric acid which gets precipitated through acid rain. This precipitation can cause decolouring, cracks and decomposition of the monument.
12. (*a*) Acid rain is a result of a chemical reaction occuring in a localized area where the concentration of the oxides of nitrogen and sulphur are high.
13. (*c*) The ozone layer protects us from harmful ultraviolet rays of the sun.
14. (*d*) The greenhouse effect can be defined as the capture and re-radiation of the infrared by the tropospheric gases like $O_3$, $CO_2$, water vapours and dust.
15. (*b*) UV radiations cause skin cancer.
16. (*c*) Greenhouse effect is related to the global warming caused by the accumulation of greenhouse gases like $CO_2$, $CH_4$ etc.
17. (*b*) CFCs reach the stratosphere and break when exposed to the UV rays. This releases the reactive chlorine which further reacts with ozone to form simple oxygen.
18. (*a*) UV rays cause photochemical splitting of $O_2$ molecules in the stratosphere. The nascent oxygent (O) combines with molecular oxgyen to form ozone.
19. (*b*) Hydroelectric power plants are powered by water. This is a clean fuel source, as it won't pollute the air like other power plants that burn fossil fuels such as coal or natural gas.
20. (*a*) CO is harmful to human beings. This is because the haemoglobin found in the human blood has a very great affinity for CO as compared to $O_2$. Thus it competes with $O_2$ for haemoglobin.
21. (*d*) Abundance of the nutrients like phosphate, nitrates results in the enrichment of an ecosystem like pond, lake and cause algal bloom.
22. (*a*) Intake of water contaminated with nitrates can cause the blue baby syndrome, (the $O_2$ carrying capacity of the blood decreases gradually).
23. (*b*) Minimata desease is caused by mercury poisoning in water. It affects the patient's brain, causing, impairment of functions like speech, thoughts, vision etc.

24. (*d*) Methaemoglobinemia caused by the decreased ability of blood to carry vital $O_2$ around the body. One of the most common causes is nitrate in drinking water.

25. (*a*) BOD is biochemical oxygen demand.

26. (*d*) DDT is an insecticide used in agriculture. It is non-biodegradable in nature. It cannot degrade with the help of microbes. So, it starts accumulation and accumulated in the form of ultimate consumer of food chain.

27. (*a*) Chlorinated hydrocarbon is synthetic chemicals used as insecticides. They are stored in the body as fat reserves and hence remain in the environment for long period of time after application.

28. (*b*) Noise pollution is measured in decibels.

29. (*a*) Biodegradable pollutant is sewage.

30. (*c*) CO is present in highest concentration in exhaust emission.

31. (*a*) The best method to determine the quantity of organic pollutants in water is by measuring BOD (the amount of $O_2$ dissolved by aerobic bacteria growing on the organic material present in a water sample at a specific temperature over a specific time period).

32. (*a*) Dental fluorosis (also termed mottled enamel) is an extremely comman disorder, characterized by hypomineralization of tooth enamel caused by ingestion of excessive fluoride during enamel formation.

33. (*a*) Chlorine treatment of water is a process of adding chlorine to drinking water and to kill parasites, bacteria and viruses.

34. (*a*) Pollution is rising due to due to automobiles and industries.

35. (*a*) The sudden mass. death of water animals mainly fishes occure due to the depletion of oxygen in the water body. Thus more likely in case of Eutrophic lake.

36. (*d*) Discharge of untreated sewage into streams which contains high quantity of nutrients induce the disproportionate growth of algae causing eutrophication which results into depletion of dissolved oxygen in the water.

37. (*a*) Jaundice is caused by the intake of contaminated water.

38. (*b*) The pollutants chlorofluorocarbons are the major source of air pollution contributed by aerosols.

39. (*d*) Acid rain is caused by large-scale emission of acidic gases into the atmosphere from thermal power plants, industries, and automobiles. The most common ones are $SO_2$, $NO_2$ etc.

40. (*a*) The basic component of smog is $O_3$ and PAN.

41. (*a*) Discharge of untreated sewage, surface run-off water from agricultural fields. and industrial wastes leads to eutrophication and explosive growth of planktons and blue green alga. This increases turbidity and reduces levels of dissolved oxygen.

42. (*b*) Waste which cannot be decayed as well as decomposed by the enviroment acts as a source of pollution. Thus non biodegradable chemicals are the most harmful type of environment pollutants.

43. (*d*) Freons contains chlorine atoms, these chlorine atoms makes chemical bonding with ozone and thus results in ozone destroy and generates ozone-hole in space.

44. (*c*) The $CO_2$ content in atmospheric air is about 0.034%

45. (*c*) In coming year, skin related disorder will become, very comman due to depletion of ozone layer, thinning of the ozone layer increases the number of UV radiations reaching the earth's surface.

46. (*c*) The term biomagnification refers to the increase in the concentration of nondegradable pollutants as they pass through food chain.

47. (*a*) CO exerts a harmful effect because it is respiratory inhibitor,

48. (*d*) $CO_2$ and water vapours are not direct atmospheric pollutant. $CO_2$ is required by plants to carry on photosynthesis

49. (*c*) The concentration of DDT increases, as it moves or travels along the food chains. Its concentration is maximum in top trophic level. This process is called biomagnification

50. (*a*) Lead (Pb) is the most hazardous metal pollutants of automobile exhausts. It affects central nervous system and distorts the red blood corpuscles.

## Exercise-2 (Learning Plus)

1. (*a*) $CO_2$ absorbs and stores the energy near the earth's surface when incoming solar radiation is reflected back to the space. If there is no $CO_2$ then there would be no absorption of solar energy and hence the temperature of earth's surface would be less than the present level.

2. (*b*) Gas responsible for bhopal tragedy of 1984 was Methyl isocyanate.

3. (*a*) Greenhouse affect refers to warming of earth's surface.

4. (*a*) The main source of aerosols is the emission of jet planes, where fluorocarbon ($CF_2Cl_2$) are used.

5. (*c*) The minimata disease in Japan was caused through the pollution of water by mercury.

6. (*b*) Excess amounts of fluoride ions in drinking water can cause dental fluorosis, skeletal fluorosis, bone damage etc.

7. (*b*) The greater an organism's trophic level in a food chain, the higher the level of concentration of DDT in its body. Thus higher concentration is it to be seen in top carnivores.

8. (*c*) Biomagnification happens because a toxic substance accumulated by an organism cannot be metabolized or

excreted, and it is passed on to the next higher trophic level. In an aquatic ecosystem, birds are at the highest trophic level.

9. (*c*) Lichens are very good pollution indicators as they do not grow in $SO_2$ polluted environments.

10. (*a*) Eutrophication is caused by increase nutrient concentration leading to plentiful growth of simple plant life.

11. (*a*) Increase asthamatic attacks in certain seasons are related to inhalation of seasonal pollen.

12. (*d*) $SO_2$ pollution is indicated by usnea (lichens) as lichens do not grow in $SO_2$ environment.

13. (*c*) The food chain with the organism given in question is phytoplankton → Crab → Eel → sea gull.
where the top carnivore of the fourth trophic level will have the highest concentration of DDT.

14. (*b*) Dissolved $O_2$ is also needed for the decomposition of oragnic matter. That is why fish die in wath bodies polluted by sewage due to reduction in dissolved $O_2$.

15. (*c*) When huge amount of sewage is dumped into a river the BOD will increase.

16. (*b*) Fluorine water pollutant causes. brittleness of teeth.

17. (*c*) Measurement of the rate of $O_2$ in unit volume of water over a period of time is done to find out Biochemical Oxygen demand.

18. (*a*) Sewage discharge in water bodies leads to organic loading or occurence of excess organic matter inside water body. Its putrefaction decreases dissolved oxygen leading to competition for oxygen among fishes.

19. (*b*) Eutrophication causes reduction in dissolved oxygen.

20. (*c*) Photochemical smog occus in day time whereas the classical smog occus in early morning hours.

21. (*c*) Chlorine nitrate which is found in the stratosphere cause break down of the ozone layer.

22. (*a*) Main cause of ozone decay is CFCs.

23. (*c*) It is beneficial because ozone cuts out the ultraviolet radiation of the sun.

24. (*d*) In Antartica ozone depletion is caused due to the formation of chlorine nitrate.

25. (*b*) Ozone layer protects human beings on earth from the harmful effect of ultraviolet radiations coming from sun.

26. (*d*) CFCs are called freon.

27. (*a*) CFCs are chemically most stable, colourless, odourless and harmless gases.

28. (*b*) $Cl^\bullet$ obtained from CFCs reacts with $O_3$ present in stratosphere.

29. (*b*) During the Antarctic winter, when polar stratospheric clouds become widespread, chemical reactions convert less reactive forms of chlorine into large amounts of highly reactive forms.

30. (*b*) Eutrophication is a source of water pollution. It occurs when water is high in nutrients.

31. (*c*) The global warning of the earth's surface is mainly due to deforestation rather than reforestation.

32. (*a*) Hydrocarbon are primary precursors whicle $O_3$, PAN are secondary precursors of photochemical smog.

33. (*c*) Photochemical smog can be reduced by using catalytic. convertor in the automobiles and plantation of certain plants like pinus, juniperus, vitis etc.

34. (*c*) CFCs gets broken by the action of UV radiation coming from the sun and produced $Cl^\bullet$ radicals

$$CF_2Cl_2 \xrightarrow{UV} CF_2Cl^\bullet + Cl^\bullet$$

35. (*a*) Photochemical smog has high concentration of oxidising agents rather than reducing agents and is, therefore, called as reducing smog.

36. (*d*) Soor particles have diameter of about 5nm, $H_2SO_4$ fog particles have size of 500-1000 nm, fly ash particles have diameter of $5 \times 10^5$ nm
So, (*a*), (*b*) and (*c*) are correct.

## Exercise-3 (JEE Advanced Level)

1. (*a,c,d*) Photochemical smog causes headache, corrosion of painted surface, extensive damge to plant life.

2. (*c,d*) $NO_2$ is more harmful as compared to NO. Acid rain contains mainly dissolved carbon dioxide, sulphur dioxide and nitrgen oxides.

3. (*a,b,c*) Particles have a diameter of about 5mm. $H_2SO_4$ fog particles have size of 500 – 1000 nm.

4. (*a,c*) Sodium chlorate and sodium arsenate are weedicides while DDT and BHC are pesticides.

5. (*a,d*) The main cause of ozone depletion is the presence of chlorine containing gases and oxides of nitrogen.

6. (*a,b,c*) Incineration, dumping and anaerobic digestion by microbes are three important methods used for the disposal of sludge.

7. (*a,b,c*) Cadmium, lead and mercury are toxic metals.

8. (*a,b,c,d*) Carbon dioxide ($CO_2$), methane ($CH_4$), Nitrous oxide ($N_2O$), ozone ($O_3$) and water vapours are greenhouse gases present in earth's atmosphere.

9. (*d*) Gaseous CFCs slowly rise into the stratosphere, are broken down by strong ultraviolet radiation, release chlorine atoms and then react with ozone molecules.

10. (*d*) Ozone is preffered over chlorine gas because it avoids the unpleasant smell and taste of chlorine gas.

11. (*d*) (*a*) The dark colour of ozone is due to intense absorption of red light.

(*b*) Oxides of nitrogen and CFCs are major causes of ozone layer depletion.
(*c*) Ozone oxidises dry iodine to $I_4O_9$.
(*d*) $2KOH + 5O_3 \rightarrow 2KO_3 + 5O_2 + H_2O$

12. (*d*) DDT, aldrin and dieldrin are non-biodegradable pesticides.
13. (*d*) Sodium arsenite, sodium chlorate and triazine all are herbicides.
14. (*d*) BHC, malathion and chlorinated hydrocarbon are insecticides.
15. (*a*)

| | |
|---|---|
| Acid rain | Oxides of nitrogen, oxides of sulphur |
| Green house effect | Carbon dioxide |
| Ozone hole | Oxides of nitrogen, CFCs |
| Eutrophication | Phosphate fertilizer i.e. plant nutrient (excess). |

16. (*c*)

| | |
|---|---|
| Classical smog | $SO_2$ |
| Photochemical smog | $NO_2$ |
| Particulate Pollutants | Bacteria, Smoke, $Fe_3O_4$ |
| Gaseous pollutants | $SO_2$, $NO_2$ |

17. (*d*)

| | |
|---|---|
| Peroxy acetyl | Photochemical smog |
| Polycyclic | Carcinogens Molecules |
| Hg compounds | Minamata disease nitrate |
| IR active | Global warming aromatic hydrocarbons |

## Exercise-4 (Past Year Questions)

### JEE MAIN

1. (*c*) Concentration of nitrate is maximum.
2. (*b*) $\left[3Ca_3(PO_4)_2.Ca(OH)_2\right] + \underset{\substack{\text{(drinking water upto 1 ppm)}}}{2F^-} \longrightarrow$

$$\underset{\text{(Harder teeth enamel)}}{\left[3Ca_3(PO_4)_2.CaF_2\right]} + 2OH^-$$

3. (*c*) $CH_4$ is not present in stratosphere.
4. (*a*) $N_2$ Molecule has minimum role in the formation of photochemical smog. While $CH_2 = O$, $O_3$ and NO has major role. When fossil fuels are burnt, a variety of pollutants are emitted. Two of them are hydrocarbons (unburnt) and NO. When these pollutants build upto high levels, a chain reaction occurs from their interaction with sunlight. The reactions involved in the formation of photochemical smog are as follows:

$$NO_2(g) \xrightarrow{h\nu} NO(g) + O(g)$$
$$O(g) + O_2(g) \rightleftharpoons O_3(g)$$
$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$$

$O_3$ reats with unburnt hydrocarbons to produce chemicals such as formaldehyde, acrolein and PAN.

$$3CH_4 + 2O_3 \rightarrow 3CH_2 = O + 3H_2O$$

$\underset{\text{Acrolein}}{CH_2CHCHO}$ $\qquad$ $CH_3COONO_2$ (PAN)

5. (*c*) $SO_2$ gas causes stiffness of flower buds.
6. (*a*) In ozone layer the wavelength of U.V radiation is 300 – 340 nm.
7. (*d*) Freons (CFC's) are not common components of photochemical smog.
8. (*b*) Acid rain reacts with marble. Hence, the Taj Mahal which is made up of marble is discoloured.
9. (*a*) The upper stratosphere conists of ozone ($O_3$), which protect us from harmful ultraviolet (UV) radiations coming from sun.
10. (*c*)
11. (*d*) Excessive release of $CO_2$ into the atmosphere results in global warming.
12. (*d*) Nitrogen oxides and hydrocarbons (unburnt fuel) are primary pollutant that leads to photochemical smog.
13. (*c*) It's a fact, the layer of atmosphere between 10km to 50km above sea level is called as stratosphere.
14. (*c*) Troposphere is the lowest region of atmosphere bounded by Earth beneath and the stratosphere above where most of the clouds form and where life form exists.
15. (*a*) Photochemical smog occurs in warm ( sunlight) and has high concentration of oxidising agent therefore it is called photochemical smog/oxidising smog.
16. (*c*) Fact based
17. (*b*) The permissible level in ppm unit is
    Fe = 0.2
    Mn = 0.05
    Cu = 3
    Zn = 5
    Mn is higher
18. (*c*) In cold water, dissolved oxygen can reach a concentration upto 10 ppm.
19. (*a*) More polluted water has high biological oxygen demand.
20. (*b*) $CO_2$, $H_2O$ vapours CFC'S and $O_3$ are green house gases.
21. (*c*) The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called **Biochemical Oxygen Demand (BOD)**.

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22. (*b*) Aqueous $NaHCO_3$ is used as the first aid. $NaHCO_3$ is basic and not harmful.

23. (*a*) $CF_2Cl_2(g) \xrightarrow{UV} \dot{Cl}(g) + \dot{C}F_2Cl(g)$ ...(i)

$\dot{Cl}(g) + O_3(g) \longrightarrow \underset{\text{Clorine Monoxide}}{\dot{C}lO(g)} + O_2(g)$ ...(ii)

$\dot{C}lO(g) + O(g) \longrightarrow \dot{Cl}(g) + O_2(g)$ ...(iii)

Here, chlorine monoxide is formed in 2nd step so, statement (i) is not true.

24. (*c*) Respiratory ailments in animals may be due to viral infections or bacterial pneumonia etc; which can due to acid rain which contains harmful chemicals acid rain is also harmless for trees.

25. (*d*) From the thermal power plant $CO_2$ is released which causes acid rain.

26. (*c*) Distillation under reduced pressure is done for separating glycerol in soap industries.

27. (*d*) Seldane is an anti-histamine drug that has inhibitory action on histamine receptor.

28. (*d*) **Calcination:** Heating of concentrated ore in the absence (or limited supply of air). Mainly used for carbonate ores.

Metal carbonate $\xrightarrow{\Delta}$ Metal oxide + $CO_2$

(Causes global Warming)

**Roasting:** Heating of concentrated ore in a regular supply of air. Mainly used for sulphide ores.

Metal sulphide + $O_2 \rightarrow$ Metal oxide + $SO_2$

(Causes acid rain)

29. (*c*) Eutrophication is a process by which environment gain nutrients for habitats and cause algal bloom, harmful algal produce toxins that are harmful and create dead zones for fish and polluting the environment.

30. (*a*) Fluoride ($F^-$) ion conc upto 1 ppm makes the enamel of teeth much harder by converting

$\underset{\text{hydroxyapatite}}{[3(Ca_3(PO_4)_2.Ca(OH)_2]} \xrightarrow{F^-} \underset{\text{Fluorapatite (harder)}}{[3(Ca_3(PO_4)_2.CaF_2]}$

31. (*c*) BOD is the biological oxygen demand. The cleanest water sample will have a BOD value equal to 3 ppm as clean water could have a BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

32. (*c*) BOD values of clean water (A) is less than 5 ppm

So, $A < 5$

BOD values of polluted water (B) is greater than 17 ppm

So, $B > 17$

33. (*c*) Ozone in the stratosphere is a product of UV radiations acting on dioxygen ($O_2$) molecules.

Ozone's primary role is to absorb the Sun's UV rays, so protecting the Earth from their destructive effects. The ozone layer's job is to protect the Earth from the sun's harmful UV rays while also keeping it warm.

$O_2(g) \xrightarrow{UV} O(g) + O(g)$

$O(g) + O_2(g) \underset{}{\overset{UV}{\rightleftharpoons}} O_3(g)$

34. (*d*) In stratosphere CFCs get broken down by powerful UV radiations releasing $Cl^{\bullet}$

$CF_2Cl_2(g) \xrightarrow{uv} Cl^{\bullet}(g) + {}^{\bullet}CF_2Cl(g)$

Once in the atmosphere, CFCs slowly rise to the stratosphere, where ultraviolet radiation breaks them down, releasing chlorine atoms that can deplete ozone.

35. (*d*) $CH_4$, $O_3$ and $H_2O$ causes global warming in tropospheric level. $N_2$ does not cause global warming.

36. (*a*) Eutrophication of water body results in loss of Biodiversity.

37. (*c*) Clean water have BOD less than 5 ppm while highly polluted water has BOD greater or equal to 17 ppm. So, assertion is correct.

BOD is measure of oxygen required to oxidise only bio-degradable organic matter. So, reason is false.

38. (*d*) Sodium arsenite is a herbicide.

39. (*a*) Antiseptic dettol is mixture of chloroxylenol and terpineol.

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# 17 Purification, Quantitative and Qualitative Analysis of Organic Compounds

## Exercise-1 (Topicwise)

1. (*d*) In sublimation a solid substance changes into vapour form directly.
2. (*c*) Boiling point of methanol & acetone is very close to each other.
3. (*c*) Turpentine oil is a mixture of volatile & non volatile components so it is not mixed with water easily.
4. (*d*) Chromatography is used for the purification of organic compound whose phase is either solid, liquid or gas.
5. (*b*) Steam distillation is applied for seperation of those compounds which are steam volatile & insoluble in water.
6. (*d*) Distillation is not used in evaporation process.
7. (*b*) Same for the purification, we use steam distillation.
8. (*d*) Chromatography is used for the seperation of small sample of mixture of plant pigments & dye stuffs.
9. (*d*) Azeotropic mixture has constant or same boiling point.
10. (*c*) It is placed between the flask containing the mixture & adapter in fractional distillation.
11. (*b*) It can be purified by vaccum distillation.
12. (*d*) It is purified by sublimation.
13. (*c*) Hydrazine does not contain C so does not give blue colour.
14. (*a*) $NH_2—\overset{\overset{S}{\|}}{C}—NH_2 + Na \longrightarrow \underset{\text{Sodium thiocynate}}{NaCNS}$

    $NaCNS + FeCl_3 \longrightarrow [Fe(SCN)]_3Cl + NaCl$
15. (*b*) Thiourea will give red colour with $FeCl_3$.
16. (*d*) This test is used for presence of halogens, nitrogen & sulphur.
17. (*c*) It does not contain C.
18. (*b*) When covalently bonded $N_2$ is fused with Na it transforms into NaCN.
19. (*c*) Sodium extract + $CH_3COOH \rightarrow (CH_3COO)_2Pb$
    $\downarrow$
    Black ppt
    (PbS)

    It shows the presence of sulphur in organic compound.
20. (*a*) $Fe_4[Fe(CN)_6]_3$ is responsible for the blue colour.
21. (*d*) The Beilstein test is a simple qualitative chemical test for organic halides.
22. (*a*) 0.233 g of $BaSO_4$ contains $\frac{32 \times 0.233}{23.3}$ g of S

    $$\% S = \frac{32 \times 0.233 \times 100}{233 \times 0.32} = 10\%$$
23. (*a*) In Duma's method $N_2$ gas is collected over nitrometer.
24. (*b*) $CuSO_4$ acts as catalytic agent.
25. (*c*) Nitrogen can be estimated by both kjeldahl's & duma's method.
26. (*b*) Nitrogen present in the organic compound is estimated as $NH_3$ in kjeldahl's method.
27. (*c*) $X + NaOH \longrightarrow NH_3 + HCl \longrightarrow NH_4Cl$

    Amount of $NH_3$ generated $= \frac{200 \times 1}{1000} \times 14 \text{ gm} = 2.8\text{gm}$

    $$\% N = \frac{2.8}{6} \times 100 = 46.67\%$$
28. (*a*) $Mg(ClO_4)_2$ & anhydrous $CaCl_2$ used as dessicant because they are distinct water absorbents.
29. (*c*) 0.17 g of $NH_3$ has $= \frac{14}{17} \times 0.17 = 0{\cdot}14$ g of N

    $$\% N = \frac{\text{mass of N}}{\text{mass of compound}} \times 100$$

    $$= \frac{0.14}{28} \times 100 = 50\%$$

## Exercise-2 (Learning Plus)

1. (*b*) $$\% C = \frac{\text{molar mass of C} \times \text{mass of } CO_2 \text{ (given)}}{\text{mass of compound} \times \text{mass of } CO_2}$$
2. (*a*) Benzoic acid reacts with base & become benzoate which is soluble in water & other polar solvent.
3. (*a*) Differential extraction based upon different solubilities.

4. (*b*) $\%\ H = \dfrac{2 \times \text{mass of water}}{1.8 \times \text{mass of compound}} \times 100$

5. (*d*) Column chromatography is used to isolate a single chemical compound from a mixture.

6. (*b*) Silical gel has high specific surface area which allows to adsorb water readily.

7. (*b*) Aniline can be purified by steam distillation since it boils at a temp. of 371.5K in the presence of steam.

8. (*b*) It is used for steam volatile compounds.

9. (*b*) In steam distillation, the vapour pressure of the volatile compound is less than the atmospheric pressure.

10. (*b*) It is used for separation of compound.

11. (*a*) Boiling point of a liquid increases with increase in pressure.

12. (*d*) P is estimated as $(NH_4)_3PO_4 \cdot 12MoO_3$ & $Mg_2P_2O_7$.

13. (*c*) In vaccum distillation liquids are distilled by avoiding decomposition at low temp.

14. (*b*) Naphthalene is volatile and benzoic acid is non-volatile. Hence they can be separated by sublimation.

15. (*c*) The red colour is due to formation of $Fe(CNS)_3$.

16. (*a*) In Carius method Cl is converted into AgCl.

17. (*b*) $Na + C + N + S \xrightarrow{\Delta} NaSCN$

18. (*b*) Fractional crystallization is the one where we can differentiate solids having different solubilities.

19. (*b*) Sulphanilic acid contains N, S & C which give NaSCN. NaSCN with $FeCl_3$ gives red blood colour due to formation of $Fe(SCN)_3$.

20. (*d*) Kjeldahl's method is not applicable for compounds containing N as nitro, azo & compounds having N are directly attached to the ring.

21. (*a*) Chromatography is the most useful & latest technique of separation & purification of organic compounds.

22. (*a*) $\%\ C = \dfrac{12}{44} \times \dfrac{\text{mass of } CO_2\,(gm)}{\text{mass of compound}\,(gm)} \times 100$

$$\%\ H = \frac{2}{18} \times \frac{\text{mass of } H_2O\,(gm)}{\text{mass of compound}\,(gm)} \times 100$$

23. (*b*) $\%\ P = \dfrac{62}{222} \times \dfrac{\text{mass of } Mg_2P_2O_7\,(gm)}{\text{mass of compound}} \times 100$

24. (*a*) This indicates presence of S.

25. (*b*) Yellow colour indicated the presence of P.

26. (*a*) $NaCN + S_8 \xrightarrow{\Delta} NaSCN + Fets\ A$

$NaSCN + Fe^{+3} \longrightarrow Fe(SCN)_3$

Blood red colour

27. (*d*) $\dfrac{\text{mass of organic compound}}{\text{mass of AgBr}} = \dfrac{\text{molar mass of organic compound}}{\text{molar mass of AgBr}}$

$$\text{molar mass of organic compound} = \frac{0.15 \times 188}{0.12}$$

$$\%\ \text{of bromine in organic compound} = \frac{\text{molar mass of Br}}{\text{molar mass of organic compound}} \times 100$$

$$= \frac{80 \times 0.12}{0.15 \times 188} \times 100$$

$$= 34.04\%$$

28. (*b*) Miscible mixture of $C_6H_6 + CHCl_3$, can be separated by distillation method due to the significant difference in their boiling points, boiling point of Benzene is 80.1°C while that of chloroform is 61.2°C.

29. (*c*) O-nitrophenol is a weaker acid than $H_2CO_3$.

30. (*c*) It is an example of partition chromatography.

31. (*b*) Substances are distributed between a stationary phase & a mobile phase.

32. (*b*) Addition of $K_2SO_4$ increases the boiling point of $H_2SO_4$.

33. (*d*) Hydrogen of the organic compound ignites to form $H_2O$.

34. (*c*)

| Element | % | Atomic ratio | Simplest ratio |
|---|---|---|---|
| C | 64.3 | 64.3/12=5.4 | 3 |
| H | 7.2 | 7.2/1 = 7.2 | 4 |
| O | 28.5 | 28.5/16 = 1.8 | 1 |

The empirical formula of compound is $C_3H_4O$.

35. (*c*) Amines are basic in nature. Therefore, aniline reacts with HCl to form a salt which dissolves in water. Hence, aniline can be separated from benzene by using HCl.

## Exercise-3 (JEE Advanced Level)

1. (*a*) Hair contains amino acids which upon fusion with soda-lime (NaOH + CaO) evolve $NH_3$.

2. (*d*) Anhydrous $CuSO_4$ turns blue in presence of water.

$CuSO_4 + 5H_2O \longrightarrow CuSO_4.5H_2O$

White Blue

3. (*b*) The first distillation product of petroleum is petroleum ether. Note that natural gases comes out first during extraction of petroleum.

4. (*c*) Paper chromatography is a liquid-liquid partition chromatography.

5. (*d*) Aryl halides are not hydrolyzed by alcoholic silver nitrate.

6. (*d*) $C = \frac{92.3}{12} = \frac{7.69}{7.69} = 1$

$H = \frac{7.69}{7.69} = 1$

Empirical formula → $C_1H_1$

X is alkyne so it is $C_2H_2$

Y is Benzene → $C_6H_6$

7. (*b*) $CCl_4 + CuO \longrightarrow CuCl_2 + CO_2$

8. (*d*) $C : H : O = \frac{24}{12} : \frac{4}{1} : \frac{32}{16} = 1:2:1$

9. (*c*) In kjeldahl's method all the nitrogen is converted into $(NH_4)_2SO_4.K_2SO_4$.

10. (*a*) In estimation of C, organic compound is heated with CuO.

11. (*b,c,d*)

Thiourea ($NH_2CSNH_2$) contains both N and S. Hence depending upon the amount of Na used, all the three, that is $Na_2S$, NaCN and NaCNS are formed but not $Na_2SO_4$.

12. (*b*) It gives red colour when it has both N & S.

13. (*b*) $\%C = \frac{12}{44} \times \frac{\text{wt. of } CO_2}{\text{wt. of O.C.}} \times 100$

$= \frac{12}{44} \times \frac{0.147}{0.2} \times 100 = 20$

$\%H = \frac{2}{18} \times \frac{\text{wt. of } H_2O}{\text{wt. of O.C.}} \times 100$

$= \frac{2}{18} \times \frac{0.12}{0.2} \times 100 = 6.7$

$\%O = 100 - (20 + 6.7) = 73.3$

14. (*a,b,d*)

Fractional distillation is used for seperating liquids that have difference in boiling point less than 25K.

15. (*a,b,c*)

Steam distillation is used to seperate substance which are steam volatile and immiscible with water.

16. (*a,b,d*)

Urea, acetamide give this test.

17. (*a,b,d*)

As $FeCl_3$ is added as a reagent and not formed during the reaction.

18. (*b,c,d*)

$CuCl_2$, $C_6H_5N$ and thiourea give this test.

19. (*a,b,d*)

Kjeldahl's method is not applicable to nitro or diazo groups.

20. (*b,c*)

In crystallization solute should be more soluble in the hot solvent than in the cold.

21. (*a,d*)

Distillation is used when substance has non volatile impurities & boils without decomposition.

22. (*a,b,c,d*)

All are the correct for differential extraction.

23. (*a,b,c*)

Chromatography is used to seperate compounds having similar boiling points.

24. (*a,b*)

N and S are confirmed

25. (*c*) Br can be identify by red. brown fumes.

26. (*a,c,d*)

In lassaigne's test substance is not converted into soluble ionic compound.

27. (*b*) $\%\ Cl = \frac{35.5}{108.5 + 35.5} \times \frac{0.287}{0.189} \times 100 = 37.56\%$

28. (*c*) $\%S = \frac{32}{233} \times \frac{0.934}{0.243} \times 100$

$= 52.78\%$

29. (*d*) All of the above

30. (*a*) Fact based

31. (*a*) Fact based

32. (*a*) Fact based

33. [4] Kjeldahl's method is not used for nitro & diazo combounds.

34. [5] N, S, Cl, Br I can be detected.

35. [7] $C : H : O = \frac{69.4}{12} : \frac{4.9}{1} : \frac{25.7}{16}$

$= 5.78 : 4.9 : 1.61$

$= 7 : 6 : 2$

$C_7H_6O_2$ → Empirical formula

36. [5] Empirical formula is $C_6H_5NO_2$

37. [3] Naphthalene, benzoic acid and anthracene are purified by sublimation.

38. [6] $\%Cl = \frac{35.5}{143.5} \times \frac{5}{2.561} \times 100 = 47.9$

| Element | % | Atomic ratio | Simplest ratio |
|---|---|---|---|
| C | 49 | 49/12=4.1 | 3 |
| H | 2.72 | 2.72/1 = 2.72 | 2 |
| Cl | 47.9 | 47.9/35.5 = 1.35 | 1 |

$C_3H_2Cl$ is the empirical formula.

n = 147/73.5 = 2

The molecular formula is $C_6H_4Cl_2$.

# Exercise-4 (Past Year Questions)

## JEE MAIN

1. (*d*) Aq. $NaHCO_3$ is used to distinquish between phenol & benzoic acid. It does not react with phenol.

2. (*a*) At atmospheric pressure the boiling point of glycerol is 290°C so it decomposes before it distills.

3. (*d*) Kjeldahl's method is not applicable for nitro, diazo & compounds containing N in the ring.

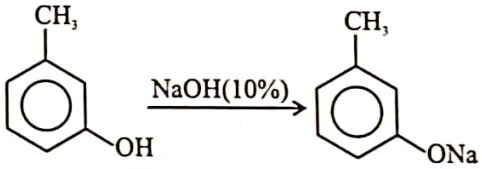
4. (*b*) Benzaldehyde is an absorbate, alumina is an adsorbent (stationary phase) and acetonitrile is mobile phase.

5. (*a*) Due to higher density of dichloromethane than water DCM would be the lower layer and water will form the upper layer in the separating funnel.

6. (*b*) $H_2O$ : Sugar – Recrystallisation
$H_2O$ : Aniline - Separation by steam distillation
$H_2O$ : Toluene - Differential extraction

7. (*c*) $\left[C_xH_yN_z\right] \xrightarrow[\text{method}]{\text{Dumas}} 6CO_2 + 4H_2O + N_2$
Moles of $CO_2 = 6$, so moles of C = 6
Moles of $H_2O = 4$, so moles of H = 8
Moles of $N_2 = 1$, so moles of N = 2
$\therefore$ Formula is $C_6H_8N_2$.

8. (*c*) Phenols (e.g. m-cresol), being weak acid, are soluble in dil NaOH but insoluble in $NaHCO_3$.




9. (*d*) In column chromatograph; a solid adsorbent is packed on a column and a solution containing number of solute particles is allowed to flow down the column. The solute molecules get adsorbed on the surface of adsorbent and move through column at different rates based on differential adsorption of the substances on the solid phase.

10. (*c*) In chromatography, $R_f$ represents retardation factor.

$$R_f = \frac{\text{Distance moved by the substance from baseline}}{\text{Distance moved by the solvent from baseline}}$$

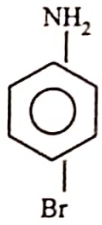
$\therefore$ Higher $R_f$ value means lower adsorption.

11. (*b*) Phosphorus is detected in the form of yellow ppt of ammonium phosphate molybdate on reaction with ammonium molybdate.

12. (*b*) Liquid having lower boiling point comes out first in fractional distillation. Simple distillation can't be used as boiling point difference is very small.

13. (*a*) Kjeldahl method is not applicable four nitro or diazo groups present in the ring, as nitrogen atom can't be converted to ammonium sulfate under the reaction conditions.

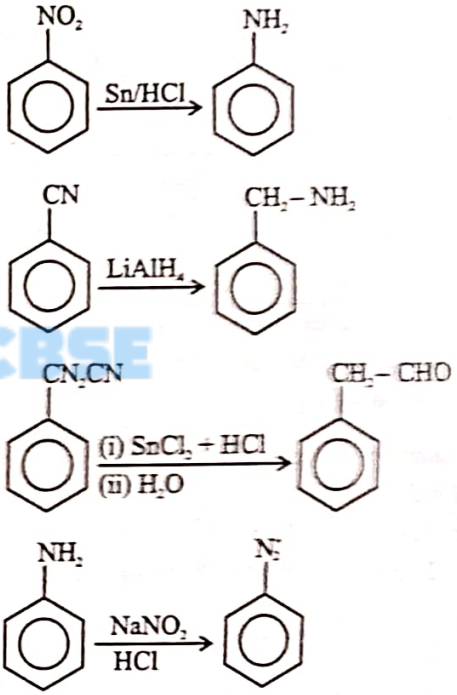
14. (*a*) Given, 0.172g organic compound showed presence of 0.08g Bromine or 172g organic compound will show 80g Bromine which matches exactly with option (*a*).




Contain 80 g Bromine
M.W. = 172

15. (*c*) Distillation under reduced pressure is done for separating glycerol in soap industries.

16. (*c*)




Nitro and diazo compounds do not give Kjeldahl estimation of Nitrogen.

17. [50] Mass of organic compound = 1.6 gm
Mass of AgBr = 1.88 gm

$$\text{Moles of Br} = \text{Moles of AgBr} = \frac{1.88}{188} = 0.01$$

Mass of Br = 0.01 × 80 = 0.80 gm

$$\% \text{ of Br} = \frac{0.80 \times 100}{1.60} = 50\%$$

18. (*b*) Lucas reagent - Conc. $HCl/ZnCl_2$
Dumas method - $CuO/CO_2$
Kjeldahl's method - $H_2SO_4$
Hinsberg test - $C_6H_5SO_2Cl$/aq. KOH

19. (*d*) More polar compound will come out first. The order of polarity in the given compound is
acetophenone > benzanilide > aniline.

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**20.** [4] $R_f = \dfrac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$

On chromatogram distance travelled by compound A is $\rightarrow$ 2 cm

Distance travelled by solvent = 5 cm

So, $R_f = \dfrac{2}{5} = 4 \times 10^{-1} = 0.4$

**21.** (*d*) Fact based

**22.** (*b*) A suitable method for separating a mixture of aniline and chloroform would be steam distillation. Steam distillation is the process used to separate aromatic compound from a mixture because of their temperature sensitivity. Therefore, steam distillation is an ideal method for their separation.

**23.** (*c*) A $\rightarrow$ r; B $\rightarrow$ p; C $\rightarrow$ s; D $\rightarrow$ q

| | |
|---|---|
| Lassaigne's Test | N, S, P and Halogen |
| Cu (II) Oxide | Carbon |
| $AgNOO_3$ | Halogen specifically. |
| Sodium fusion extract given black precipitate with acetic acid and lead acetate ($CH_3COOH/(CH_3COO)_2$ Pb) | Sulphur |

**24.** [19] Given; Aqueous tension at 287 K = 14 mm of Hg.

Hence actual pressure = (758 – 14)

= 744 mm of Hg.

Volume of nitrogen at STP $= \dfrac{273 \times 744 \times 30}{287 \times 760}$

V = 27.938 mL.

$\because$ 22400 mL of $N_2$ at STP weighs = 28 gm.

$\therefore$ 27.94 mL of $N_2$ at STP weighs = 28 gm.

$\left(\dfrac{28}{22400} \times 27.94\right)$ gm

= 0.0349 gm

Hence % of Nitrogen $= \left(\dfrac{0.0349}{0.1840} \times 100\right)$

= 18.97 %

Rond off. Answer = 19%

**25.** [3] $R_f$ = retardation factor

$$R_f = \frac{\text{Distance travelled by the substance from reference line (cm)}}{\text{Distance travelled by the solvent from reference line (cm)}}$$

$R_f$ value of different compounds are different.

**26.** (*b*) Kjeldahl method is not applicable to compounds containing nitrogen in nitrogroup, azo groups and nitrogen present in the ring (e.g Pyridine) as nitrogen of these compounds does not change to ammonium sulphate under these conditions.

**27.** (*d*) In chromotography technique, the purification of a compound is independent of the physical state of the pure compound.

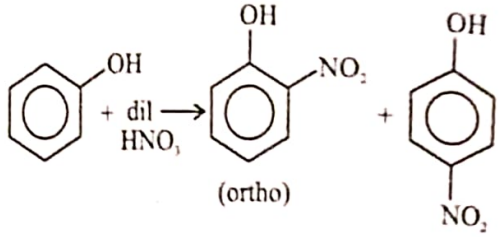
**28.** [14] Weight of organic compound = 0.2 g

mass of $N_2$ (g) evolved $= \dfrac{22.4 \times 10^{-3}}{22.4} \times 28$

$= 28 \times 10^{-3}$ g

% of N $= \dfrac{28 \times 10^{-3}}{0.2} \times 100 = 14$

**29.** (*c*)




(ortho) (para)

Para product has higher boiling point than ortho as intermolecular H-bond is possible in former, where as intramolecular H-bond is possible in ortho product.

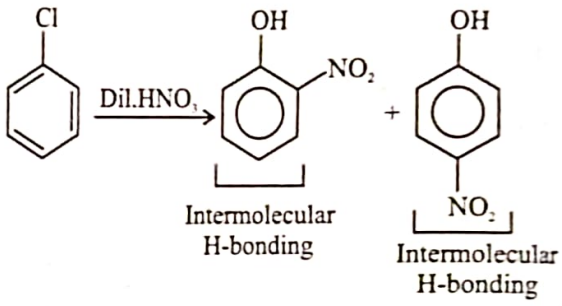
Steam distillation can separate them as ortho product is steam volatile.

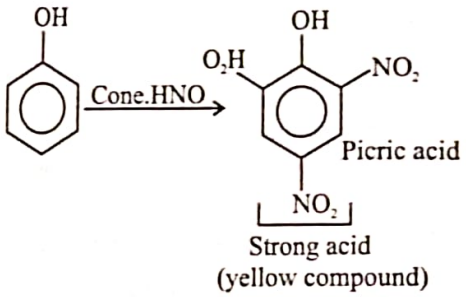
**30.** (*d*) Benzoic acid and napthalene can be effectively separated by crystallization. Benzoic acid is soluble in hot water whereas napthalene is insoluble.

Hence assertion is incorrect but reason is correct

**31.** (*a*) Both Statement I & Statement II are correct.

**32.** [7]




**33.** [40] Wt. of organic compound = 0.25 g

Mass of Cl $= \dfrac{35.5}{143.5} \times 0.4$ g

Mass % of Cl in the organic compound

$= \dfrac{35.5 \times 0.4}{143.5 \times 0.25} \times 100$

= 39.58%

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**34.** (*c*) The function of mobile phase is to carry the components present on TLC.

**35.** (*d*) $Na_2S + Na_2[(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NO_S]$

**36.** [34] O.C $\rightarrow$ AgBr

0.5g 0.4g

$$\text{moles of Br} = \text{moles of AgBr} = \frac{0.4}{188}$$

$$\%Br = \frac{\frac{0.4}{188} \times 80}{0.5} \times 100$$

$= 34.04\%$

**37.** [64] meq of $H_2SO_4$ used by $NH_3 = 12.5 \times 1 \times 2 = 25$ % of N in the compound $= \frac{25 \times 10^{-3} \times 14 \times 100}{0.55} = 63.6$

$\approx 64\%$

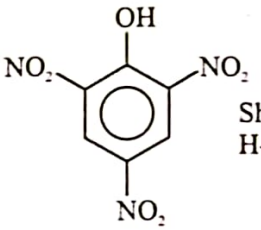
**38.** [56] $\%N = \frac{1.4(N_1V_1)}{\text{mass of organic compound}}$

$$\%N = \frac{1.4(2.5 \times 2 \times 2)}{0.25} = 56$$

**39.** [2] $\frac{R_{f_A}}{R_{f_B}} = \frac{\frac{2.08}{3.25}}{\frac{1.05}{3.25}} = \frac{2.08}{1.05} = 2$

**40.** (*d*)

Shows inter molecular H-bonding



Shows intra molecular H-bonding

Solvent polarity has been related to $R_f$ value of nitro compounds.

100 mg p-nitrophenol and picric acid have different $R_f$ value on silica gel plate.

∴ Preparative TLC is best to separate 100 mg of para nitrophenol and picric acid.

**41.** (*a*) Mass of organic compound = 0.45 gm

Mass of AgBr obtained = 0.36 gm

$$\therefore \text{Moles of AgBr} = \frac{0.36}{188}$$

$$\therefore \text{Mass of Bromine} = \frac{0.36}{188} \times 80 = 0.1532\text{gm}$$

$$\%\text{ Br in compound} = \frac{0.1532}{0.45} \times 100 = 34.04\%$$

**42.** (*d*) (A) Chloroform + Aniline $\rightarrow$ Distillation

(B) Benzoic acid + Napthalene $\rightarrow$ Crystallisation

(C) Water + Aniline $\rightarrow$ Steam distillation

(D) Napthalene + Sodium chloride $\rightarrow$ Sublimation

**43.** (*a*) Thin layer chromoatography (TLC) is another type of adsorption chromatography which involves separation of substance of a mixture over a thin layer of an adsorbent coated on glass plate.

A thin layer (about 0.2 mm thick) of an adsorbent at (silica gel) or (Alumina) is spread over a glass plate of suitable size. Hence Assertion (A) is correct and Reason (R) is correct explanation of (A)

**44.** [22] V = 22.7 ml, T = 280 K

$P_{total} = 759$ mmHg

$P_{N_2} = 759 - 14.2 = 744.8$ mmHg

$$n_{N_2} = \frac{744.8 \times 22.78}{760 \times 1000 \times 0.082 \times 280} = 0.00097$$

$W_{Nitrogen} = 0.02716$

$$\%\,N = \frac{0.02716}{0.125} \times 1000 = 21.728$$

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